Chapter 1

The classical theory

Every number computed in the following chapters is a coefficient in front of a local functional, and its sign and normalisation are inherited from three classical data: the odd pairing on the fields, the BV action and the linear differential Q [C07, §4.3]. The sources supply the ten-dimensional action [ESW, (31)] and its dimensional reduction [ESW, Prop 2.20, §10.3], while the pairing on the antifields stays implicit in the phrase "natural (-1)-shifted symplectic structure" [ESW, Def 1.34]. This chapter fixes all three data for the reduction to Euclidean $\mathbb{R}^4$, writes the interaction vertex by vertex, and shows that the classical master equation determines the relative signs of the pairing.

1.1 Spaces and the dimensional reduction

Definition 1.1 (Spaces)**.** The ten-dimensional space is $V_{10} = V_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, where $V_{\mathbb{R}} = \mathbb{R}^{10}$ carries its Euclidean form and the orthogonal decomposition $V_{\mathbb{R}} = W_{\mathbb{R}} \oplus W_{\mathbb{R}}^{\perp}$ of [ESW, §2.2, §2.6]: $W_{\mathbb{R}} = \mathbb{R}^4$ is spacetime and $W_{\mathbb{R}}^{\perp}$ consists of the six transverse directions. With $V = W_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ the complexification of $\mathbb{R}^4$,

$$V_{10} = V \oplus \big(W_{\mathbb{R}}^{\perp} \otimes_{\mathbb{R}} \mathbb{C}\big),$$

and from here on $W_{\mathbb{R}}^{\perp}$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ denote these complexifications. The nondegenerate symmetric form $(-,-)$ of V_{10} is the complex-bilinear extension of the Euclidean form. The letter W alone denotes the multiplicity space of [ESW, §10].

The gauge Lie algebra $\mathfrak{g}$ is complex, with bracket $[-,-]$ and a nondegenerate invariant symmetric pairing $(-,-)$ [ESW, §3.1], chosen positive definite on the compact real form, so that $\langle e, Qe\rangle$ is negative definite on the real slice of [C07, §6.6] (Remark 1.10). The coupling constant sits in $(-,-)$. The inverse pairing is the element $(-,-)^{-1} \in \mathfrak{g} \otimes \mathfrak{g}$.

The two sixteen-dimensional semi-spin representations Σ and Σ^* of $\mathfrak{so}(V_{10})$ form the $\mathbb{Z}/2$-graded Clifford module $\Sigma \oplus \Sigma^*$ with Clifford action ρ and pairing $(-,-)\colon \Sigma \otimes \Sigma^* \to \mathbb{C}$ [ESW, §2.1]:

$$\rho(v_1)\rho(v_2) + \rho(v_2)\rho(v_1) = 2(v_1, v_2), \qquad (\rho(v)Q_1, Q_2) = (Q_1, \rho(v)Q_2), \qquad v, v_1, v_2 \in V_{10}.$$

The Γ-pairing $\Gamma\colon \mathrm{Sym}^2(\Sigma) \to V_{10}$, $\Gamma\colon \mathrm{Sym}^2(\Sigma^*) \to V_{10}$ is defined by [ESW, (28)]

$$(\rho(v)Q_1, Q_2) = (v, \Gamma(Q_1, Q_2)).$$

The map $q\colon \wedge^{\bullet} V_{10} \to \mathrm{Cl}(V_{10})$ is antisymmetrisation, $q(v_1 \wedge v_2) = v_1 v_2 - (v_1, v_2)$ [ESW, (29)]. Applied to a two-form, such as F_A, the symbol ρ means $\rho \circ q$, as in $\rho(F_A)$ [ESW, (33)].

The exterior derivative d acts along $\mathbb{R}^4$, and every field is constant along $W_{\mathbb{R}}^{\perp}$. The operator d^* is the formal adjoint of d for the Euclidean metric, dvol is the Euclidean volume density of $\mathbb{R}^4$, and $\not{d} = \rho \circ d$ is the Dirac operator from Σ-valued to Σ^*-valued functions [ESW, §2.1]; the same formula on Σ^*-valued functions is written $\not{d}|_{\Sigma^*}$.

The four-dimensional content of Σ is [ESW, §10, §10.3]

$$\Sigma \cong S_+ \otimes W \oplus S_- \otimes W^*, \qquad \dim W = 4, \qquad W \cong S_+^6 \text{ as a representation of } \mathrm{Spin}(6;\mathbb{C}) \cong SL(4,\mathbb{C}),$$

with $S_\pm$ the two-dimensional semi-spin representations of $\mathrm{Spin}(4,\mathbb{C})$.

Proposition 1.2 (Four-dimensional chirality)**.** *Fix an orientation of V. For an oriented orthonormal basis $v_1,\dots,v_4$ of V the element $q(v_1\wedge v_2\wedge v_3\wedge v_4) = v_1v_2v_3v_4$ of $\mathrm{Cl}(V_{10})$ depends on the orientation alone. The operator $\rho(v_1v_2v_3v_4)$ squares to 1 on Σ and on Σ^*, with eigenvalues $+1$ and -1 each of multiplicity 8. Writing*

$$S_+ \otimes W = \{+1 \text{ eigenspace on } \Sigma\}, \qquad S_- \otimes W^* = \{-1 \text{ eigenspace on } \Sigma\},$$
$$S_- \otimes W = \{-1 \text{ eigenspace on } \Sigma^*\}, \qquad S_+ \otimes W^* = \{+1 \text{ eigenspace on } \Sigma^*\},$$

one has, for $v \in V$ and $w \in W_{\mathbb{R}}^{\perp}$,

$$\rho(v)\colon S_+ \otimes W \to S_- \otimes W, \qquad \rho(w)\colon S_+ \otimes W \to S_+ \otimes W^*,$$

and correspondingly on the other summands: $\rho(v)$ exchanges the four-dimensional chiralities and $\rho(w)$ preserves them.

Proof. The element $v_1v_2v_3v_4$ is the antisymmetrisation q of the oriented unit element of $\wedge^4 V$, hence depends on the orientation alone. By the Clifford relation the v_i anticommute pairwise and square to 1, hence

$$(v_1v_2v_3v_4)^2 = (-1)^{3+2+1}\, v_1^2v_2^2v_3^2v_4^2 = 1.$$

Moving v_1 through $v_1v_2v_3v_4$ meets v_2, v_3, v_4 once each, so $v_1\,(v_1v_2v_3v_4) = -(v_1v_2v_3v_4)\,v_1$; the same holds for each v_i, hence for every $v \in V$ by linearity. For $w \in W_{\mathbb{R}}^{\perp}$, w anticommutes with all four v_i, so $w\,(v_1v_2v_3v_4) = (+1)(v_1v_2v_3v_4)\,w$. Since $\rho(v)$ maps $\Sigma \to \Sigma^*$, the first relation reads $\rho(v)\rho(v_1v_2v_3v_4)|_\Sigma = -\rho(v_1v_2v_3v_4)|_{\Sigma^*}\rho(v)$, so $\rho(v)$ maps the $+1$ eigenspace on Σ to the -1 eigenspace on Σ^*; the second relation gives $\rho(w)$ mapping the $+1$ eigenspace on Σ to the $+1$ eigenspace on Σ^*. For the multiplicities, take a unit vector $w \in W_{\mathbb{R}}^{\perp}$. The even element v_1w satisfies $(v_1w)^2 = -v_1^2w^2 = -1$ and $v_1w\,(v_1v_2v_3v_4) = -(v_1v_2v_3v_4)\,v_1w$, so $\rho(v_1w)$ is an invertible endomorphism of Σ (and of Σ^*) exchanging the two eigenspaces of $\rho(v_1v_2v_3v_4)$; hence each eigenspace has dimension $16/2 = 8$. □

Remark 1.3. The names of the eigenspaces depend on the orientation of V; the opposite orientation exchanges $S_+ \leftrightarrow S_-$ in all four of them. This orientation is fixed once; Chapter 2 uses it to name the supercharges.

1.2 Fields, degrees and parities

Definition 1.4 (Fields)**.** The BRST fields of [ESW, Def 3.2], reduced along $W_{\mathbb{R}}^{\perp}$ [ESW, Def 1.55, Prop 1.56, Prop 2.20], form

$$F_{\text{gauge}} = \Omega^0(\mathbb{R}^4;\mathfrak{g}[1]) \oplus \Omega^0(\mathbb{R}^4;V_{10}^\vee \otimes \mathfrak{g}) \oplus \Omega^0(\mathbb{R}^4;\Pi\Sigma \otimes \mathfrak{g}),$$

with sections (c, A, λ). The space of BV fields is $\mathscr{E}$ [C07, §4.1],

$$\mathscr{E} = F_{\text{gauge}} \oplus F_{\text{gauge}}^![-1] \qquad \text{[ESW, Rem 1.36]},$$

with antifields (c^*, A^*, λ^*); duals are identified through $(-,-)$ on $\mathfrak{g}$, the metric of V_{10}, the pairing $\Sigma\otimes\Sigma^* \to \mathbb{C}$ and dvol. The component of A along V is the connection $A|_V$, the component along $W_{\mathbb{R}}^{\perp}$ is $A|_{W_{\mathbb{R}}^{\perp}}$, and $\iota_w A = A(w)$ for $w \in W_{\mathbb{R}}^{\perp}$ are the six scalars. The compactly supported elements of $\mathscr{E}$ form $\mathscr{E}_c$ ($\mathcal{E}_c$ in [ESW, §1.1]).

The degrees are those of [ESW, Rem 1.33]: the ghost c is a section in degree -1, which forces A^* in degree $+1$ and c^* in degree $+2$. Signs follow the Koszul rule for the total parity, the sum of the $\mathbb{Z}$-degree and the $\mathbb{Z}/2$-degree modulo 2 [ESW, §1.1].

field	fibre	$\mathbb{Z}$-degree	$\mathbb{Z}/2$-degree	total parity	$\mathrm{Spin}(4)\times\mathrm{Spin}(6)$ content
c	$\mathfrak{g}$	-1	even	odd	scalar
A	$V_{10}^{\vee}\otimes\mathfrak{g}$	0	even	even	$A\vert_V$ vector of Spin(4), $A\vert_{W_{\mathbb{R}}^{\perp}}$ vector of Spin(6)
λ	$\Pi\Sigma\otimes\mathfrak{g}$	0	odd	odd	$S_+\otimes W\oplus S_-\otimes W^*$
A^*	$V_{10}\otimes\mathfrak{g}$	$+1$	even	odd	as A
λ^*	$\Pi\Sigma^*\otimes\mathfrak{g}$	$+1$	odd	even	$S_-\otimes W\oplus S_+\otimes W^*$
c^*	$\mathfrak{g}$	$+2$	even	even	scalar

The BV fields of the reduced theory. Coordinate functionals on $\mathscr{E}$ carry the opposite $\mathbb{Z}$-degree.

1.3 Pairing and bracket

Definition 1.5 (Odd pairing)**.** The odd pairing $\langle\,,\rangle$ on $\mathscr{E}$ [C07, Def 4.3.1; ESW, Def 1.13, Def 1.16] is nonzero exactly between a field and its antifield. For each of the three pairs, call the member of even total parity the even member: A, λ^*, c^*; their partners A^*, λ, c are odd. For a section e_0 in the slot of an even member and a section e_1 in the slot of its partner,

$$\langle e_0, e_1\rangle = \int \mathrm{dvol}\,(e_0, e_1), \qquad \langle e_1, e_0\rangle = -\langle e_0, e_1\rangle,$$

where (e_0, e_1) is the fibre pairing of Definition 1.4; for the pair (λ^*, λ) it is (e_1, e_0) with $(-,-)\colon \Sigma\otimes\Sigma^*\to\mathbb{C}$ tensored with $(-,-)$ on $\mathfrak{g}$.

Definition 1.6 (Bracket and the kernel of the identity)**.** The kernel K_0 of the identity [CG2, §7.2.3] has nonzero components exactly in the slot pairs

$$A\otimes A^*, \quad A^*\otimes A, \quad \lambda^*\otimes\lambda, \quad \lambda\otimes\lambda^*, \quad c^*\otimes c, \quad c\otimes c^*,$$

each equal to $-\delta(x-y)$ times the inverse of the fibre pairing: $(-,-)^{-1}$ tensored with the inverse metric of V_{10} for A, A^*, with the canonical element of $\Sigma\otimes\Sigma^*$ for λ, λ^*, and with 1 for c, c^*; here $x, y\in\mathbb{R}^4$ are the two positions and $\delta(x-y)$ is the delta distribution of the diagonal, always written with its argument. The bracket is the odd Poisson bracket of [C07, Lemma 4.3.2], defined for f local and g arbitrary; it is given by the formula of [C07, §10.1] for the manifold a point,

$$\{f, g\} = \Delta_0(fg) - \Delta_0(f)g - (-1)^{|f|}f\Delta_0(g), \qquad \Delta_0 = -\partial_{K_0},$$

with ∂_ϕ the order-two operator of [C07, §6.4] and f local. It satisfies

$$\{g, f\} = (-1)^{|f||g|}\{f, g\},$$

the graded symmetry of [ESW, Def 1.15].

Remark 1.7. The overall sign of $\langle\,,\rangle$ is a convention: reversing it reverses K_0 and the bracket, and leaves the master equation $\{S, S\} = 0$ invariant. The relative signs of the three pairs are fixed by the master equation (Proposition 1.14).

1.4 The action

Definition 1.8 (Action)**.** The action is the dimensional reduction of [ESW, (31)]:

$$S = \int_{\mathbb{R}^4} \mathrm{dvol}\Big(-\tfrac{1}{4}(F_A, F_A) + \tfrac{1}{2}(\lambda,\, \not{d}_A \lambda) + (d_A c, A^*) + ([c,\lambda], \lambda^*) + \tfrac{1}{2}([c,c], c^*)\Big), \tag{1.1}$$

with

$$F_A = dA + \tfrac{1}{2}[A \wedge A], \qquad d_A = d + [A, -], \qquad \not{d}_A = \rho \circ d_A.$$

Here A is regarded as a one-form on V_{10} constant along $W_{\mathbb{R}}^{\perp}$; d acts along $\mathbb{R}^4$, and d_A, $\not{d}_A$ use all ten components of A. The pairing (F_A, F_A) is the Hodge pairing on $\wedge^2 V_{10}^{\vee}$ tensored with $(-,-)$ on $\mathfrak{g}$, normalised by

$$\big((v_1,-)\wedge(v_2,-),\, (v_1,-)\wedge(v_2,-)\big) = (v_1,v_1)(v_2,v_2) - (v_1,v_2)^2, \qquad v_1, v_2 \in V_{10}.$$

Every pairing in (1.1) is read with its left entry first, as printed in [ESW, (31)]. In [ESW] the full action is S and the interaction is I; in [C07] the letter S denotes the interaction. Here S is always the full action and I the interaction.

Splitting $A = A|_V + A|_{W_{\mathbb{R}}^{\perp}}$,

$$(F_A, F_A) = (F_{A|_V}, F_{A|_V}) + \big(d_{A|_V} A|_{W_{\mathbb{R}}^{\perp}},\, d_{A|_V} A|_{W_{\mathbb{R}}^{\perp}}\big) + \big(\tfrac{1}{2}[A|_{W_{\mathbb{R}}^{\perp}} \wedge A|_{W_{\mathbb{R}}^{\perp}}],\, \tfrac{1}{2}[A|_{W_{\mathbb{R}}^{\perp}} \wedge A|_{W_{\mathbb{R}}^{\perp}}]\big),$$

$$\not{d}_A \lambda = \not{d}\lambda + \rho([A|_V, \lambda]) + \rho([A|_{W_{\mathbb{R}}^{\perp}}, \lambda]), \qquad (d_A c, A^*) = (dc, A^*|_V) + ([A,c], A^*),$$

where $\rho([A,\lambda])$ is Clifford multiplication by the $V_{10}^{\vee}$ factor of A (through the metric) together with $[-,-]$ on the $\mathfrak{g}$ factors, i.e. $(\not{d}_A - \not{d})\lambda$.

Proposition 1.9 (Normalisation of the gauge kinetic term)**.** *For $Q_1 \in \Sigma$ put*

$$S^{(1)}(Q_1) = \int \mathrm{dvol}\Big(-(\Gamma(Q_1,\lambda), A^*) + \tfrac{1}{2}(\rho(F_A)Q_1, \lambda^*)\Big) \qquad [\textit{ESW, (33)}].$$

For abelian $\mathfrak{g}$ the bracket $\{S, S^{(1)}(Q_1)\}$ is an affine function of the coefficient of the Hodge pairing (F_A, F_A) in (1.1)*, and the identity $\{S, S^{(1)}(Q_1)\} = 0$ of [ESW, Lemma 3.4] fixes this coefficient to $-\frac{1}{4}$. The coefficients $+\frac{1}{4}$ and $-\frac{1}{2}$ (the reading of (F_A, F_A) as a sum over ordered pairs of directions, twice the Hodge pairing) give a nonzero bracket.*

Proof. For abelian $\mathfrak{g}$ the bracket is affine in the coefficient of (F_A, F_A): this coefficient times (dA, dA) meets $-(\Gamma(Q_1,\lambda), A^*)$ through the (A, A^*) pair, $\frac{1}{2}(\lambda, \not{d}\lambda)$ meets $\frac{1}{2}(\rho(dA)Q_1, \lambda^*)$ through the (λ^*, λ) pair, and (dc, A^*) meets $\frac{1}{2}(\rho(dA)Q_1, \lambda^*)$ in $(\rho(d\,dc)Q_1, \lambda^*) = 0$. With the coefficient $-\frac{1}{4}$ the bracket vanishes for every $Q_1 \in \Sigma$ and every $\mathfrak{g}$: this is [ESW, Lemma 3.4] in ten dimensions, pulled back along the reduction [ESW, Prop 2.20], and for $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$ it holds on every element of a basis of Σ (Remark 1.15); the part of this bracket quadratic in the fields is the abelian bracket. With the coefficient $+\frac{1}{4}$ the abelian bracket is a nonzero local functional at an element $Q_1 \in \Sigma$ (Remark 1.15). An affine function that vanishes at $-\frac{1}{4}$ and is nonzero at $+\frac{1}{4}$ has $-\frac{1}{4}$ as its unique zero; in particular it is nonzero at $-\frac{1}{2}$. □

The free part of S is

$$\tfrac{1}{2}\langle e, Qe\rangle = \int \mathrm{dvol}\Big(-\tfrac{1}{4}(dA, dA) + \tfrac{1}{2}(\lambda, \not{d}\lambda) + (dc, A^*)\Big) \qquad [\text{C07, §1, §4.3}], \tag{1.2}$$

and $I = S - \frac{1}{2}\langle e, Qe\rangle$ is the interaction.

Fields and kernels on $\mathbb{R}^4 \times \mathbb{R}^4$ are Fourier transformed as

$$\int \frac{d^4p}{(2\pi)^4} e^{i(p,x)}(\cdots)(p), \qquad \int \frac{d^4p}{(2\pi)^4} e^{i(p,x-y)}(\cdots)(p),$$

with $p \in V$ and $|p|^2 = (p,p)$; thus d becomes $i(p,-)$.

Remark 1.10 (Real slice)**.** The weight is $e^{+S/\hbar}$, with $\langle e, Qe\rangle$ negative definite on the real slice [C07, §6.6]. For A real with respect to the compact real form of $\mathfrak{g}$,

$$\langle A, QA\rangle = -\tfrac{1}{2}\int \operatorname{dvol}(dA, dA) \le 0,$$

and for a fixed nonzero momentum p the quadratic form on the fibre $V_{10}^\vee$ has eigenvalues, in units of $(-,-)$, equal to $-\frac{1}{2}|p|^2$ with multiplicity 9 and 0 with multiplicity 1; the zero eigenvector is the longitudinal part of $A|_V$, which lies in $\operatorname{Im} Q$.

1.5 The linear differential

Proposition 1.11 (The differential Q)**.** *The linear part of $\{S,-\}$ is the derivation $\{\frac{1}{2}\langle e, Qe\rangle, -\}$ of functionals, induced as in [C07, §6.2] by the operator $Q\colon \mathscr{E} \to \mathscr{E}$ given in the following table, and* (1.2) *holds.*

on the slot of	Q	*lands in the slot of*	*momentum space*
c	$c \mapsto -dc$	$A\|_V$	$-i(p,-)\,c$
A	$A \mapsto -\frac{1}{2}\,d^*dA$	A^*	$-\frac{1}{2}\big(\|p\|^2 A\|_V - (p,-)\,\iota_p A\big)$ *and* $-\frac{1}{2}\|p\|^2 A\|_{W_{\mathbb{R}}^\perp}$
A^*	$A^* \mapsto d^*(A^*\|_V)$	c^*	$-i\,(p, A^*\|_V)$
λ	$\lambda \mapsto \not{d}\lambda$	λ^*	$i\rho(p)\lambda$
λ^*, c^*	0		

The operator Q. On $A|_{W_{\mathbb{R}}^\perp}$, d^*d is the positive Laplacian of each scalar $\iota_w A$; on $A|_V$ it is d^*d on one-forms.

Moreover:

- $Q^2 = 0$*;*
- Q *preserves the pairing:* $\langle Qe_0, e_1\rangle + (-1)^{|e_0|}\langle e_0, Qe_1\rangle = 0$ *for all* $e_0, e_1 \in \mathscr{E}_c$*; equivalently* $(Q\otimes 1 + 1\otimes Q)K_0 = 0$.

Proof. $Q^2 = 0$: the composable pairs of arrows of Q are $c \to A|_V \to A^*$ and $A \to A^* \to c^*$. On the first,

$$Q^2 c = -\tfrac{1}{2}\,d^*d(-dc) = \tfrac{1}{2}\,d^*(d\,dc) = 0;$$

on the second, $d^*(A^*|_V)$ applied to $-\frac{1}{2}d^*dA$ gives $-\frac{1}{2}\,d^*d^*(dA) = 0$, since $d^*d^* = 0$.

Preservation of the pairing is checked on the three pairs; all other pairings vanish. For e_0, e_1 in the A slot (even),

$$\langle Qe_0, e_1\rangle = -\langle e_1, -\tfrac{1}{2}d^*de_0\rangle = \tfrac{1}{2}\int \operatorname{dvol}(de_1, de_0), \qquad \langle e_0, Qe_1\rangle = -\tfrac{1}{2}\int \operatorname{dvol}(de_0, de_1),$$

and the sum vanishes. For e_0 in the A^* slot (odd) and e_1 in the c slot,

$$\langle Qe_0, e_1\rangle = \int \operatorname{dvol}(d^*(e_0|_V), e_1) = \int \operatorname{dvol}(e_0, de_1), \qquad \langle e_0, Qe_1\rangle = -\langle -de_1, e_0\rangle = \int \operatorname{dvol}(de_1, e_0),$$

so $\langle Qe_0, e_1\rangle - \langle e_0, Qe_1\rangle = 0$. For e_0, e_1 in the λ slot (odd), with $\not{d} = \rho \circ d$ and the symmetry $(\rho(v)Q_1, Q_2) = (Q_1, \rho(v)Q_2)$,

$$\langle Qe_0, e_1\rangle - \langle e_0, Qe_1\rangle = \int \mathrm{dvol}\big((e_1,\, \not{d}e_0) + (e_0,\, \not{d}e_1)\big) = 0,$$

by integration by parts. The kernel form is the same statement for the kernel of the identity. □

Remark 1.12 (A sign conflict in the sources)**.** [ESW, Def 1.16] prints the adjointness of the BV differential as $\int \omega(e_0, Q_{BV}e_1) = (-1)^{|e_0|}\int \omega(Q_{BV}e_0, e_1)$, where ω and Q_{BV} are the [ESW] names of $\langle\,,\rangle$ and Q. Combined with the graded antisymmetry $\langle e_1, e_0\rangle = -\langle e_0, e_1\rangle$ of Definition 1.5 for e_0 even and e_1 odd, the printed relation gives, for $e_0 = e_1 = A$ in the A slot,

$$\langle A, QA\rangle = \langle QA, A\rangle = -\langle A, QA\rangle,$$

hence $\langle A, QA\rangle = 0$, which contradicts the kinetic term $\frac{1}{2}\langle A, QA\rangle = -\frac{1}{4}\int \mathrm{dvol}\,(dA, dA)$ of (1.2). The relation of Proposition 1.11, $\langle Qe_0, e_1\rangle + (-1)^{|e_0|}\langle e_0, Qe_1\rangle = 0$, is the one satisfied by the action, and it is used throughout.

1.6 The interaction, vertex by vertex

Expanding (1.1) around (1.2),

$$\begin{aligned} I = &\underbrace{\int \mathrm{dvol}\Big(-\tfrac{1}{2}\big(dA, \tfrac{1}{2}[A\wedge A]\big) + \tfrac{1}{2}\big(\lambda, (\not{d}_A - \not{d})\lambda\big) + ([A,c], A^*) + ([c,\lambda],\lambda^*) + \tfrac{1}{2}([c,c],c^*)\Big)}_{\text{cubic part of } I} \\ &+ \underbrace{\int \mathrm{dvol}\Big(-\tfrac{1}{4}\big(\tfrac{1}{2}[A\wedge A], \tfrac{1}{2}[A\wedge A]\big)\Big)}_{\text{quartic part of } I}. \end{aligned} \tag{1.3}$$

term	component types
$-\frac{1}{2}(dA, \frac{1}{2}[A\wedge A])$	$(A\|_V)^3$; $A\|_V\,(A\|_{W_{\mathbb{R}}^\perp})^2$
$-\frac{1}{4}(\frac{1}{2}[A\wedge A], \frac{1}{2}[A\wedge A])$	$(A\|_V)^4$; $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$; $(A\|_{W_{\mathbb{R}}^\perp})^4$
$\frac{1}{2}(\lambda, (\not{d}_A - \not{d})\lambda)$	$\lambda\lambda\, A\|_V$ (gauge); $\lambda\lambda\, \iota_w A$ (Yukawa)
$([A,c], A^*)$	$A\|_V\, c\, A^*\|_V$; $A\|_{W_{\mathbb{R}}^\perp}\, c\, A^*\|_{W_{\mathbb{R}}^\perp}$
$([c,\lambda],\lambda^*)$	$c\,\lambda\,\lambda^*$
$\frac{1}{2}([c,c],c^*)$	$c\,c\,c^*$

Component types of the vertices. The cubic gauge vertex has the component types $(A|_V)^3$ and $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, because d acts along $\mathbb{R}^4$.

Definition 1.13 (Vertex functions)**.** Number the legs of a vertex $1, \dots, n$, give leg j an incoming momentum $p_j \in V$ and a fibre element written $A(p_j)$, $\lambda(p_j)$, $c(p_j)$, $A^*(p_j)$, $\lambda^*(p_j)$ or $c^*(p_j)$ according to its field. Evaluate the corresponding term of I on $\sum_j r_j$ (fibre element of leg j) $e^{i\langle p_j, x\rangle}$, where $r_j = 1$ for even legs and r_j is an odd element of an auxiliary nilpotent supercommutative ring for odd legs [C07, Lemma 6.4.1, proof]. The vertex function is the coefficient of the product of the r_j of the odd legs, in leg order, in the part multilinear in the legs, with the factor $(2\pi)^4\delta(\sum_j p_j)$ removed.

In the following formulas, $\iota_p A(p_j) \in \mathfrak{g}$ is the contraction of the $V_{10}^\vee$ factor with $p \in V$; $[A(p_1), A(p_2)] \in \mathfrak{g}$ contracts the $V_{10}^\vee$ factors with the metric and brackets the $\mathfrak{g}$ factors; $[A(p_1) \wedge A(p_2)]$ is the $\mathfrak{g}$-valued two-form $(v_1, v_2) \mapsto [A(p_1)(v_1), A(p_2)(v_2)] - [A(p_1)(v_2), A(p_2)(v_1)]$, paired by the Hodge pairing.

The gauge vertices, with legs $1,2,3$ (resp. $1,\dots,4$) all of type A:

$$A\,A\,A: \qquad -\frac{i}{2}\Big[\big(\iota_{p_2-p_1}A(p_3),[A(p_1),A(p_2)]\big)+\big(\iota_{p_3-p_2}A(p_1),[A(p_2),A(p_3)]\big) + \big(\iota_{p_1-p_3}A(p_2),[A(p_3),A(p_1)]\big)\Big], \tag{1.4}$$

$$A\,A\,A\,A: \qquad -\frac{1}{2}\Big[\big([A(p_1)\wedge A(p_2)],[A(p_3)\wedge A(p_4)]\big)+\big([A(p_1)\wedge A(p_3)],[A(p_2)\wedge A(p_4)]\big) + \big([A(p_1)\wedge A(p_4)],[A(p_2)\wedge A(p_3)]\big)\Big]. \tag{1.5}$$

The vertices with fermions, ghosts and antifields:

vertex (legs in order)	vertex function
$\lambda\,\lambda\,A$	$r_1r_2\,(\lambda(p_1),\rho([A(p_3),\lambda(p_2)]))$
$A\,c\,A^*$	$r_2r_3\,([A(p_1),c(p_2)],A^*(p_3))$
$c\,\lambda\,\lambda^*$	$r_1r_2\,([c(p_1),\lambda(p_2)],\lambda^*(p_3))$
$c\,c\,c^*$	$r_1r_2\,([c(p_1),c(p_2)],c^*(p_3))$

Momentum-space vertex functions of I with odd legs (all momenta incoming); these four are independent of the momenta.

Each entry is the multilinear part of the corresponding term of (1.3) on plane waves, including all permutations of identical legs; these are the vertex labels of every graph in the later chapters.

1.7 The classical master equation

Proposition 1.14 (Classical master equation)**.** *For every $\mathfrak{g}$ as in Definition 1.1:*

- $\{S,S\}=0$ *[ESW, Def 1.16];*
- *equivalently,* $\{\frac{1}{2}\langle e,Qe\rangle,\frac{1}{2}\langle e,Qe\rangle\}=0$ *(that is, $Q^2=0$) and* $QI+\frac{1}{2}\{I,I\}=0$.

If moreover $[\mathfrak{g},\mathfrak{g}]\neq 0$, the master equation fixes the relative signs of the pairing and the sign of the ghost term:

- *if the sign of the (λ^*,λ) pairing, or of the (c^*,c) pairing, or of both, is reversed relative to the (A,A^*) pairing in Definition 1.5, then $\{S,S\}\neq 0$;*
- *with the pairing of Definition 1.5 and the sign of the term $\frac{1}{2}([c,c],c^*)$ reversed, $\{S,S\}\neq 0$.*

Proof. Equivalence. Write $S=\frac{1}{2}\langle e,Qe\rangle+I$. Both summands are even, so by graded symmetry $\{I,\frac{1}{2}\langle e,Qe\rangle\}=\{\frac{1}{2}\langle e,Qe\rangle,I\}=QI$ (Proposition 1.11), and

$$\{S,S\}=\{\tfrac{1}{2}\langle e,Qe\rangle,\tfrac{1}{2}\langle e,Qe\rangle\}+2\,QI+\{I,I\}.$$

The first term is Q applied to $\frac{1}{2}\langle e,Qe\rangle$; by the preservation of the pairing in Proposition 1.11, $\langle Qe,Qe\rangle=-(-1)^{|e|}\langle e,Q^2e\rangle$, so this term is a multiple of $\langle e,Q^2e\rangle$ and vanishes by $Q^2=0$. This gives the equivalence of the first two items.

$\{S,S\}=0$ for every $\mathfrak{g}$. The action (1.1) is the BV action of the BRST theory of [ESW, Def 3.2], built as in [ESW, Def 1.34] with the natural pairing of the shifted cotangent bundle; it satisfies $\{S,S\}=0$ for every $\mathfrak{g}$, and the dimensional reduction preserves it [ESW, Def 1.55, Prop 1.56]. The natural pairing differs from Definition 1.5 at most by a sign on each of the three pairs, and these signs are independent of $\mathfrak{g}$. An overall sign leaves $\{S,S\}=0$ invariant. For $\mathfrak{g}=\mathfrak{su}(2)$ the density of $\{S,S\}$ computed with the pairing of Definition 1.5 vanishes as a local functional, together with $2QI+\{I,I\}$, while $QI\neq 0$; for each of the three other relative sign patterns it is nonzero (Remark 1.15). Hence the natural pairing

coincides with Definition 1.5 up to an overall sign, and $\{S,S\}=0$ holds for every $\mathfrak{g}$. The same exact computation gives $\{S,S\}=0$ for $\mathfrak{g}=\mathfrak{su}(3)$.

The relative signs. The action is linear in the antifields, so $\{S,S\}$ has antifield number 0 and 1. Each term of $\{S,S\}$ pairs a derivative along a field with a derivative along its partner and carries the sign of that pair. The derivatives of S along the antifields are the gauge transformations $A\mapsto d_Ac$, $\lambda\mapsto[c,\lambda]$, $c\mapsto\frac{1}{2}[c,c]$.

- Antifield number 0: the gauge variation of $\int \mathrm{dvol}\big(-\frac{1}{4}(F_A,F_A)+\frac{1}{2}(\lambda,\not{d}_A\lambda)\big)$. The A-variation enters through the (A,A^*) pair and the λ-variation through the (λ^*,λ) pair.
- Linear in A^*: $[d_Ac,c]$ through the (A,A^*) pair against $d_A(\frac{1}{2}[c,c])$ through the (c^*,c) pair, related by the Leibniz rule $d_A[c,c]=2[d_Ac,c]$.
- Linear in λ^*: $[c,[c,\lambda]]$ through the (λ^*,λ) pair against $\frac{1}{2}[[c,c],\lambda]$ through the (c^*,c) pair, related by the Jacobi identity $2[c,[c,\lambda]]=[[c,c],\lambda]$.
- Linear in c^*: $[c,[c,c]]=0$.

Consider the part of $\{S,S\}$ with field content (c,λ,λ), no A and no antifield. The (c^*,c) pair contributes nothing to it, since $\frac{1}{2}[c,c]$ contains two ghosts. The (λ^*,λ) pair contributes the variation of $\frac{1}{2}\int\mathrm{dvol}\,(\lambda,\not{d}\lambda)$ along $\lambda\mapsto[c,\lambda]$, and the (A,A^*) pair contributes, up to the sign of that pair, a nonzero multiple of the variation of $\frac{1}{2}\int\mathrm{dvol}\,(\lambda,\rho([A,\lambda]))$ along $A\mapsto dc$, which is a nonzero multiple of

$$\int\mathrm{dvol}\,\big(\lambda,\rho([dc,\lambda])\big).$$

With the pairing of Definition 1.5 these two contributions cancel, because $\{S,S\}=0$; reversing the sign of the (λ^*,λ) pair relative to the (A,A^*) pair therefore turns this part of $\{S,S\}$ into twice the (A,A^*) contribution. This functional is nonzero when $[\mathfrak{g},\mathfrak{g}]\neq0$: choose $X,Y,Z\in\mathfrak{g}$ with $(X,[Z,Y])\neq0$, compactly supported Σ-valued functions s_1,s_2 and a compactly supported function f with $\int\mathrm{dvol}\,(df,\Gamma(s_1,s_2))\neq0$, and put $\lambda=r_1s_1X+r_2s_2Y$, $c=r_3fZ$ with r_1,r_2,r_3 odd as in Definition 1.13. Using $r_2r_3r_1=-r_1r_3r_2$, $(Y,[Z,X])=-(X,[Z,Y])$ and the symmetry of Γ,

$$\int\mathrm{dvol}\,\big(\lambda,\rho([dc,\lambda])\big)=2\,r_1r_3r_2\,(X,[Z,Y])\int\mathrm{dvol}\,(df,\Gamma(s_1,s_2))\neq0.$$

This proves the first of the last two items when the (λ^*,λ) sign is reversed, alone or together with the (c^*,c) sign. Next consider the part with field content (c,c,A^*) and no A. The (λ^*,λ) pair contributes nothing to it. The (A,A^*) pair contributes the variation of $\int\mathrm{dvol}\,([A,c],A^*)$ along $A\mapsto dc$, which is $\int\mathrm{dvol}\,([dc,c],A^*)$ up to the sign of the pair, and the (c^*,c) pair contributes the variation of $\int\mathrm{dvol}\,(dc,A^*)$ along $c\mapsto\frac{1}{2}[c,c]$, which is $\int\mathrm{dvol}\,([dc,c],A^*)$ up to the product of the sign of the pair and the sign of the term $\frac{1}{2}([c,c],c^*)$. They cancel with the data of Definitions 1.5 and 1.8, so reversing the (c^*,c) sign alone, or the sign of $\frac{1}{2}([c,c],c^*)$ alone, turns this part into twice $\pm\int\mathrm{dvol}\,([dc,c],A^*)$. On $c=r_1f_1X+r_2f_2Y$ with $[X,Y]\neq0$,

$$\int\mathrm{dvol}\,([dc,c],A^*)=r_1r_2\int\mathrm{dvol}\,\big([X,Y],\,(d(f_1f_2),A^*|_V)\big),$$

which is nonzero for a suitable compactly supported $A^*|_V$. □

Remark 1.15 (Exact verification of identities of local functionals)**.** Propositions 1.9 and 1.14 use the following exact computation. For $\mathfrak{g}=\mathfrak{su}(2)$ and $\mathfrak{g}=\mathfrak{su}(3)$ fix bases of $\mathfrak{g}$, of V_{10} and of $\Sigma\oplus\Sigma^*$ in which the bracket, the pairings, ρ and Γ have Gaussian-rational entries. The density of each local

functional of this chapter, and of each bracket of two of them, is then a polynomial in the jets of the components of the six fields with Gaussian-rational coefficients, computed in exact arithmetic. A local functional $\int \mathrm{dvol}\, D$ without constant term vanishes on $\mathscr{E}_c$ when all Euler–Lagrange derivatives of D vanish identically: for $e \in \mathscr{E}_c$, after integration by parts with compact support, $\frac{d}{ds} \int \mathrm{dvol}\, D(se)$ is the integral of the Euler–Lagrange derivatives of D at se paired with e (odd components multiplied by odd elements of an auxiliary nilpotent ring, as in Definition 1.13), so $\int \mathrm{dvol}\, D(se)$ is independent of s and equals its value 0 at $s = 0$. Conversely, a nonzero Euler–Lagrange derivative shows that the functional is nonzero. Statements that a bracket vanishes for every element of Σ are established on every element of a basis of Σ; this suffices, because the bracket is linear in that element.

Chapter 2

Superconformal symmetry as an equivariant action

The dilatation of a composite operator, and every other superconformal transformation, acts on the fields of $\mathcal{N} = 4$ super Yang–Mills theory [ESW, Def 2.4, §10.3] up to gauge transformations and equations of motion. In the BV formalism this structure is a Maurer–Cartan element of a Chevalley–Eilenberg complex, an L_∞ action in the sense of [ESW, Def 1.26–1.27], and it is the classical input of the equivariant observables used later. [ESW, (32)–(34), Thm 3.3] construct it for the Poincaré supersymmetry algebra; the special conformal transformations and the special supercharges require Weyl weights, a compensating term with the six scalars, and an R-symmetry rotation in the odd bracket. This chapter realises $\mathfrak{psl}(4|4)$, the complexification of $\mathfrak{psu}(2,2|4)$ [SW, (7)], on Euclidean $\mathbb{R}^4$, fixes the sign conventions of the complex, writes the action $S + S^{(1)} + S^{(2)}$, and proves its master equation for every gauge algebra.

2.1 The algebra $\mathfrak{psl}(4|4)$ in complexified Euclidean form

Throughout, $x \in V$ denotes the position on $\mathbb{R}^4$ and E the Euler vector field $E(x) = x$ of [SW, (3)]; in this chapter E always means this vector field. For a vector field v on V_{10}, $(v, -)$ is its metric dual one-form, L_v the Lie derivative [ESW, (32)], $v(-)$ the derivative along v, and $\operatorname{div} v$ the divergence, $L_v \operatorname{dvol} = (\operatorname{div} v) \operatorname{dvol}$.

Definition 2.1 (Even part)**.** The even part of $\mathfrak{psl}(4|4)$ is the Lie algebra of vector fields on $V_{10} = V \oplus W_{\mathbb{R}}^{\perp}$ spanned by

- translations $x \mapsto v_1$, $v_1 \in V$;
- rotations, the linear vector fields given by elements of $\mathfrak{so}(V)$;
- the dilatation $\Delta = 2E$, the generator $\Delta = \operatorname{diag}(1, -1 \,|\, 0)$ of [SW, (17)], with $\operatorname{ad}_\Delta = -2$ on translations (Remark 2.6);
- special conformal fields $x \mapsto (x,x)v_1 - 2(v_1, x)x$, $v_1 \in V$, the special conformal field of [SW, (3)] along v_1;
- linear vector fields on $W_{\mathbb{R}}^{\perp}$ given by elements of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$,

with the bracket $[v, v'] = v(v') - v'(v)$. The first four kinds are the conformal Killing vector fields of V; together with $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ the even part has dimension $15 + 15 = 30$.

Definition 2.2 (Odd part)**.** The odd part of $\mathfrak{psl}(4|4)$ is $\Pi(\Sigma \oplus \Sigma^*)$. An element $Q_1 \in \Sigma$ is realised by the constant spinor Q_1, and an element $Q_2 \in \Sigma^*$ by the Σ-valued function $x \mapsto \rho(x)Q_2$. Together these span the conformal Killing spinors

$$x \mapsto Q_1 + \rho(x)Q_2, \qquad Q_1 \in \Sigma,\ Q_2 \in \Sigma^*.$$

Supercharge spinors always carry a subscript, $Q_1, Q_2, \ldots$, and each formula states whether they lie in Σ or in Σ^*; the unsubscripted Q is the differential of Chapter 1.

Definition 2.3 (Bracket)**.** On two even elements the bracket of $\mathfrak{psl}(4|4)$ is the bracket of vector fields, $[v, v'] = v(v') - v'(v)$. For an even element v and an odd element,

$$[v,\, Q_1 + \rho(x)Q_2] = v(Q_1 + \rho(x)Q_2) + \tfrac{1}{4}\rho(d(v,-))(Q_1 + \rho(x)Q_2) - \tfrac{1}{8}(\operatorname{div} v)(Q_1 + \rho(x)Q_2), \qquad (2.1)$$

where ρ of the two-form $d(v,-)$ means $\rho \circ q$ [ESW, (29)]; the right-hand side is again a conformal Killing spinor. For two odd elements, with $Q_1, Q_3 \in \Sigma$ and $Q_2, Q_4 \in \Sigma^*$,

$$[Q_1 + \rho(x)Q_2,\ Q_3 + \rho(x)Q_4] = \Big(x \mapsto \Gamma\big(Q_1 + \rho(x)Q_2,\, Q_3 + \rho(x)Q_4\big)\big|_V\Big) + \big(\text{rotation of } W_{\mathbb{R}}^{\perp}\big), \qquad (2.2)$$

where $|_V$ is the orthogonal projection $V_{10} \to V$ and the rotation of $W_{\mathbb{R}}^{\perp}$ is the element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ with matrix elements

$$\big(w, (\text{rotation of } W_{\mathbb{R}}^{\perp})\, w'\big) = (\rho(q(w \wedge w'))Q_1, Q_4) + (\rho(q(w \wedge w'))Q_3, Q_2), \qquad w, w' \in W_{\mathbb{R}}^{\perp}.$$

The $W_{\mathbb{R}}^{\perp}$-components of Γ drop out of the bracket, as in the dimensional reduction of the super-translation algebra [ESW, §2.6, Prop 2.20]; they reappear in $S^{(2)}$ (Definition 2.15). The coefficient 1 in front of the rotation of $W_{\mathbb{R}}^{\perp}$ is fixed by the super-Jacobi identity on triples in $\Sigma \times \Sigma \times \Sigma^*$: each term of the Jacobi sum of three odd elements contains exactly one odd–odd bracket, so the sum is affine in this coefficient; it vanishes at 1 and is nonzero at 0 (Remark 2.18), so 1 is its unique zero.

Proposition 2.4 (Brackets of the named generators)**.**

- *For $Q_1, Q_2 \in \Sigma$: $[Q_1, Q_2] = \Gamma(Q_1, Q_2)|_V$, a translation; this is [ESW, Def 2.5] after reduction.*
- *For $Q_1, Q_2 \in \Sigma^*$: $[Q_1, Q_2] = -\big(x \mapsto (x,x)v_1 - 2(v_1, x)x\big)$ with $v_1 = \Gamma(Q_1, Q_2)|_V$, a special conformal field.*
- *For $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: $[Q_1, Q_2] = \big(x \mapsto \Gamma(Q_1, \rho(x)Q_2)|_V\big) + (\text{rotation of } W_{\mathbb{R}}^{\perp})$ as in Definition 2.3; the dilatation part of the first summand is $\frac{1}{2}(Q_1, Q_2)\,\Delta$, and the rest is a rotation of V.*
- *For a translation $x \mapsto v_1$: $[v_1, Q_1] = 0$ for $Q_1 \in \Sigma$ and $[v_1, Q_2] = \rho(v_1)Q_2 \in \Sigma$ for $Q_2 \in \Sigma^*$.*
- *For a special conformal field $x \mapsto (x,x)v_1 - 2(v_1,x)x$: its bracket with $Q_1 \in \Sigma$ is $\rho(v_1)Q_1 \in \Sigma^*$, and with $Q_2 \in \Sigma^*$ it is 0.*
- $[\Delta, Q_1] = -Q_1$ *for* $Q_1 \in \Sigma$ *and* $[\Delta, Q_2] = +Q_2$ *for* $Q_2 \in \Sigma^*$.

Proof. The first two items follow from (2.2): for $Q_2 = Q_4 = 0$ the rotation of $W_{\mathbb{R}}^{\perp}$ vanishes; for $Q_1 = Q_3 = 0$ it vanishes as well, and for $v \in V$

$$(v, \Gamma(\rho(x)Q_1, \rho(x)Q_2)) = (\rho(x)\rho(v)\rho(x)Q_1, Q_2) = 2(v,x)(x, \Gamma(Q_1, Q_2)) - (x,x)(v, \Gamma(Q_1, Q_2)),$$

using $\rho(x)\rho(v)\rho(x) = 2(v,x)\rho(x) - (x,x)\rho(v)$ from the Clifford relation and $(\rho(x)Q_1, Q_2) = (Q_1, \rho(x)Q_2)$. Projecting to V gives $-\big((x,x)v_1 - 2(v_1,x)x\big)$.

For $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ and $v \in V$, $(v, \Gamma(Q_1, \rho(x)Q_2)) = (\rho(x)\rho(v)Q_1, Q_2)$; the part symmetric under $x \leftrightarrow v$ is $\frac{1}{2}(\rho(x)\rho(v) + \rho(v)\rho(x)) = (x, v)$, so the linear vector field $x \mapsto \Gamma(Q_1, \rho(x)Q_2)|_V$ equals $(Q_1, Q_2)\, E = \frac{1}{2}(Q_1, Q_2)\, \Delta$ plus an antisymmetric linear field, i.e. a rotation.

For the even–odd brackets use (2.1). For $v = 2E$: $(v, -) = d(x, x)$ is closed, $\operatorname{div}(2E) = 8$, $2E(Q_1) = 0$ and $2E(\rho(x)Q_2) = 2\rho(x)Q_2$, so $[\Delta, Q_1] = -Q_1$ and $[\Delta, \rho(x)Q_2] = (2-1)\rho(x)Q_2$. For a translation, $(v, -)$ is constant and $\operatorname{div} v = 0$, so (2.1) reduces to $v_1(\rho(x)Q_2) = \rho(v_1)Q_2$. For $v = (x \mapsto (x, x)v_1 - 2(v_1, x)x)$,

$$d(v, -) = d(x, x) \wedge (v_1, -) - 2\, d(v_1, x) \wedge (x, -) = 4\, (x, -) \wedge (v_1, -),$$
$$\operatorname{div} v = 2(x, v_1) - 2(v_1, x) - 8(v_1, x) = -8(v_1, x),$$

so $\frac{1}{4}\rho(d(v, -)) - \frac{1}{8} \operatorname{div} v = \rho(x)\rho(v_1) - (x, v_1) + (v_1, x) = \rho(x)\rho(v_1)$. On a constant $Q_1 \in \Sigma$ this gives $x \mapsto \rho(x)\rho(v_1)Q_1$, the realisation of $\rho(v_1)Q_1 \in \Sigma^*$. On $x \mapsto \rho(x)Q_2$ it gives

$$\begin{aligned} &\rho\big((x, x)v_1 - 2(v_1, x)x\big)Q_2 + \rho(x)\rho(v_1)\rho(x)Q_2 \\ &\quad = (x, x)\rho(v_1)Q_2 - 2(v_1, x)\rho(x)Q_2 + 2(v_1, x)\rho(x)Q_2 - (x, x)\rho(v_1)Q_2 = 0. \end{aligned}$$

□

subspace	realisation	block of [SW, (17)]	dimension	ad_Δ eigenvalue
translations	$x \mapsto v_1$, $v_1 \in V$	$(2, 1)$	4	-2
$\mathfrak{so}(V)$, part acting on S_+	linear vector field of $\mathfrak{so}(V)$	$(1, 1)$	3	0
$\mathfrak{so}(V)$, part acting on S_-	linear vector field of $\mathfrak{so}(V)$	$(2, 2)$	3	0
dilatation	$\Delta = 2E$	$\operatorname{diag}(1, -1 \mid 0)$	1	0
special conformal	$x \mapsto (x, x)v_1 - 2(v_1, x)x$	$(1, 2)$	4	$+2$
$\mathfrak{so}(W_{\mathbb{R}}^{\perp})$	linear vector fields on $W_{\mathbb{R}}^{\perp}$	$(3, 3)$	15	0
$S_+ \otimes W \subset \Sigma$	constant spinor	$(3, 1)$	8	-1
$S_- \otimes W^* \subset \Sigma$	constant spinor	$(2, 3)$	8	-1
$S_- \otimes W \subset \Sigma^*$	$x \mapsto \rho(x)Q_2$	$(3, 2)$	8	$+1$
$S_+ \otimes W^* \subset \Sigma^*$	$x \mapsto \rho(x)Q_2$	$(1, 3)$	8	$+1$

The ten named subspaces of $\mathfrak{psl}(4|4)$; rows and columns of the blocks are ordered as in [SW, (17)]. The eigenspaces of Proposition 1.2 fix the names; in particular the block $(3, 1)$ of [SW, (17)] is $S_+ \otimes W$.

Proposition 2.5 (Identification with $\mathfrak{psl}(4|4)$)**.** *The bracket of Definition 2.3 has the following properties.*

- *The dimension is* $30|32$*, and the bracket of two elements of the ten named subspaces lies in their span.*
- *The bracket is graded antisymmetric and satisfies the super-Jacobi identity.*
- *The* V*-part of every odd–odd bracket is a conformal Killing vector field of* V*; the rotation of* $W_{\mathbb{R}}^{\perp}$ *is nonzero for* $\Sigma \times \Sigma^*$ *and vanishes for* $\Sigma \times \Sigma$ *and* $\Sigma^* \times \Sigma^*$*.*
- *The block rule of the supermatrix presentation [SW, (17), (19)] holds for every pair of the ten named subspaces: their bracket lies in the block with the row of the first subspace and the column of the second when the column of the first equals the row of the second, plus the block with the row of the second and the column of the first when the column of the second equals the row of the first, and vanishes otherwise.*
- ad_Δ *is diagonal on the named subspaces with the eigenvalues of the table above, which are those of* $\Delta = \operatorname{diag}(1, -1 \mid 0)$ *in [SW, (17)].*
- *The Killing form* $\operatorname{str}(\mathrm{ad}_{x_1}\mathrm{ad}_{x_2})$ *for* $x_1, x_2 \in \mathfrak{psl}(4|4)$*, with* str *the supertrace [SW, after (17)], vanishes identically.*

- *The algebra equals its derived algebra and has trivial centre; its even part is the sum of the two commuting ideals formed by the conformal Killing vector fields of V and by $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, each of dimension* 15 *with nondegenerate Killing form.*

These are the properties by which the algebra is identified with the complexification $\mathfrak{psl}(4|4)$ *of* $\mathfrak{psu}(2,2|4)$ *[SW, (7), (17)].*

Proof. The dimension count is $4+6+1+4+15=30$ and $16+16=32$. Every other item is a bilinear or trilinear identity in the arguments, or a statement about the structure constants; each is established on all ordered pairs and all ordered triples of elements of a basis adapted to the ten named subspaces (Remark 2.18), hence for all arguments. □

Remark 2.6 (Dilatation normalisation)**.** The dilatation is the Δ of [SW, (17)], the vector field $2E$, with ad_{Δ} eigenvalue -2 on translations, $+2$ on special conformal fields, -1 on Σ, $+1$ on Σ^* and 0 on rotations. The generator called Δ in [SW, (3)] is the vector field $-E$, and the bosonic part of the Δ of [SW, (12)] is E. Every cochain component is linear in its arguments, so a component evaluated on $2E$ is -2 times the same component evaluated on $-E$:

$$S^{(1)}(2E) = -2\,S^{(1)}(-E),$$

and a coefficient computed in the normalisation of [SW, (3)] is converted to the normalisation used here by the factor -2.

2.2 The Chevalley–Eilenberg complex and its sign conventions

Definition 2.7 (Cochains)**.** Following [ESW, Def 1.26] with the Lie algebra of loc. cit. taken to be $\mathfrak{psl}(4|4)$,

$$C^{\bullet}(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E})) = \bigoplus_n \mathrm{Hom}(\wedge^n \mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))[-n],$$

with $\mathscr{O}_l(\mathscr{E})$ the local functionals [C07, Def 4.2.2] ($\mathcal{O}_{\mathrm{loc}}$ in [ESW, Def 1.26]). The elements of this complex are multilinear maps f of elements $x_1, x_2, \dots$ of $\mathfrak{psl}(4|4)$ (the notation of [ESW, Def 1.26]; the position x carries no subscript) with the antisymmetry

$$f(\dots, x_i, x_{i+1}, \dots) = (-1)^{|x_i||x_{i+1}|+1} f(\dots, x_{i+1}, x_i, \dots);$$

[ESW, Def 1.26] prints the exponent $|x_1||x_2|+1$ [sic]. Here $|f|$ denotes the degree of f in $C^{\bullet}$, and an element of degree 0 has values $f(x_1, \dots, x_k)$ of parity $k + \sum_i |x_i|$.

Definition 2.8 (Cochain bracket and differential)**.** The cochain bracket is the formula of [ESW, Def 1.26]: $\{f, g\}(x_1, \dots, x_{k+l})$ is the sum over (k,l)-shuffles σ of

$$\mathrm{sgn}(\sigma) \cdot (\text{Koszul sign of } \sigma) \cdot (-1)^{|g|k + \sum_{i=1}^{k} |x_{\sigma(i)}|(l+|g|)} \{f(x_{\sigma(1)}, \dots, x_{\sigma(k)}),\, g(x_{\sigma(k+1)}, \dots, x_{\sigma(k+l)})\},$$

with $\{-,-\}$ the bracket of Definition 1.6 and $|g|$ the degree of g in $C^{\bullet}$. The Chevalley–Eilenberg differential is

$$(d_{CE} f)(x_1, \dots, x_n) = \sum_{i<j} (-1)^{|x_i| \sum_{p<i} |x_p| + |x_j| \sum_{p<j,\, p\neq i} |x_p| + i + j + |f| + 1} f([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_n), \tag{2.3}$$

that is, the formula printed in [ESW, Def 1.26] multiplied by -1.

Lemma 2.9 (The sign of the Chevalley–Eilenberg differential)**.** *For $Q_1, Q_2 \in \Sigma$ and $|f| = 0$, the formula printed in [ESW, Def 1.26] gives*

$$(d_{CE}S^{(1)})(Q_1, Q_2) = -S^{(1)}([Q_1, Q_2]),$$

and (2.3) *gives*

$$(d_{CE}S^{(1)})(Q_1, Q_2) = +S^{(1)}([Q_1, Q_2]);$$

the second is [ESW, Lemma 3.6, item 2]. With the bracket of Definition 1.6 and the functionals [ESW, (33), (34)], the equation of [ESW, Lemma 3.6]

$$\{S, S^{(2)}(Q_1, Q_2)\} + S^{(1)}(\Gamma(Q_1, Q_2)) - \{S^{(1)}(Q_1), S^{(1)}(Q_2)\} = 0$$

holds; this fixes the sign of the term $S^{(1)}(\Gamma(Q_1, Q_2))$, and with this bracket [ESW, proof of Lemma 3.6, item 3] holds term by term. The printed formula is the Chevalley–Eilenberg differential of the opposite algebra; the substitution $x_i \mapsto -x_i$ in every argument, $S^{(k)} \mapsto (-1)^k S^{(k)}$, exchanges the two conventions. Hence (2.3) *keeps [ESW, (32)–(34)] verbatim. Under the printed convention the arity-one functionals are -1 times those of [ESW, (32)–(33)], so the sign of every one-generator component is tied to the choice of d_{CE}.*

Proof. For $n = 2$ the pair $(i, j) = (1, 2)$ is the single term, both sums over p are empty, and the exponent in (2.3) is $1 + 2 + |f| + 1 \equiv |f|$; for $|f| = 0$ this gives $+f([x_1, x_2])$, and the printed formula gives $-f([x_1, x_2])$. The displayed equation of [ESW, Lemma 3.6] is the arity-two component of the master equation on $\Sigma \times \Sigma$ (Lemma 2.11 with $|x_1| = 1$), which holds for every $\mathfrak{g}$ by Theorem 2.20. With $-S^{(1)}(\Gamma(Q_1, Q_2))$ in place of $+S^{(1)}(\Gamma(Q_1, Q_2))$ the left side becomes $-2S^{(1)}(\Gamma(Q_1, Q_2))$, which contains $-2(L_v A, A^*)$ for the translation $v = \Gamma(Q_1, Q_2)|_V$ and is nonzero when $\Gamma(Q_1, Q_2)|_V \neq 0$; with the bracket replaced by its negative the left side becomes $2S^{(1)}(\Gamma(Q_1, Q_2))$, nonzero as well. The expression for $\{S, S^{(2)}(Q_1, Q_2)\}$ in [ESW, proof of Lemma 3.6, item 3] is reproduced term by term with the bracket of Definition 1.6 by the exact evaluation of Remark 2.18. For the last statement: d_{CE} is linear in the bracket of $\mathfrak{psl}(4|4)$, so the printed formula is (2.3) for the opposite bracket $-[-,-]$; the linear map $x_1 \mapsto -x_1$ is an isomorphism from $\mathfrak{psl}(4|4)$ to its opposite, since $-[x_1, x_2] = -[-x_1, -x_2]$, and it transports a cochain of arity k to $(-1)^k$ times itself. □

Definition 2.10 (Action of $\mathfrak{psl}(4|4)$)**.** An action of $\mathfrak{psl}(4|4)$ on $(\mathscr{E}, \langle\,,\rangle, S)$ [ESW, Def 1.27] is an element

$$S_{\mathfrak{psl}(4|4)} = \sum_{k \geq 0} S^{(k)}_{\mathfrak{psl}(4|4)} \in C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$$

of degree 0, with $S^{(0)}_{\mathfrak{psl}(4|4)} = S$, each $S^{(k\geq 1)}_{\mathfrak{psl}(4|4)}(x_1, \ldots, x_k)$ at least quadratic in the fields, and

$$d_{CE}S_{\mathfrak{psl}(4|4)} + \tfrac{1}{2}\{S_{\mathfrak{psl}(4|4)}, S_{\mathfrak{psl}(4|4)}\} = 0. \tag{2.4}$$

Equivalently [ESW, Rem 1.29], $I_{\mathfrak{psl}(4|4)} = I + \sum_{k\geq 1} S^{(k)}_{\mathfrak{psl}(4|4)}$ satisfies

$$(d_{CE} + Q)I_{\mathfrak{psl}(4|4)} + \tfrac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0.$$

We abbreviate $S^{(k)} = S^{(k)}_{\mathfrak{psl}(4|4)}$.

Lemma 2.11 (Component equations)**.** *With the conventions of Definitions 2.7–2.8, the component of* (2.4) *on $(x_1, \ldots, x_n)$ reads as follows. In arity 0, $\{S, S\} = 0$. In arity 1, $\{S, S^{(1)}(x_1)\} = 0$. In arity 2,*

$$\{S, S^{(2)}(x_1, x_2)\} + S^{(1)}([x_1, x_2]) + (-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\} = 0.$$

In arity 3*, with* x_1 *even and* x_2, x_3 *odd,*

$$\{S, S^{(3)}(x_1,x_2,x_3)\} + S^{(2)}([x_1,x_2],x_3) + S^{(2)}([x_1,x_3],x_2) + S^{(2)}([x_2,x_3],x_1) \\ + \{S^{(1)}(x_1), S^{(2)}(x_2,x_3)\} - \{S^{(1)}(x_2), S^{(2)}(x_1,x_3)\} - \{S^{(1)}(x_3), S^{(2)}(x_1,x_2)\} = 0.$$

In arity 3 *with* x_1, x_2, x_3 *odd the component is the same expression with all seven signs* $+$*; this is the cyclic form of [ESW, Lemma 3.7]. In arity* 4*, with all arguments odd,*

$$\{S, S^{(4)}(x_1,\dots,x_4)\} + \sum_{i<j} S^{(3)}([x_i,x_j],\dots) - \sum_i \{S^{(1)}(x_i), S^{(3)}(\dots)\} \\ + \{S^{(2)}(x_1,x_2), S^{(2)}(x_3,x_4)\} + \{S^{(2)}(x_1,x_3), S^{(2)}(x_2,x_4)\} + \{S^{(2)}(x_1,x_4), S^{(2)}(x_2,x_3)\} = 0,$$

where the dots are the remaining arguments in increasing order. For $\mathfrak{psl}(4|4)$ *replaced by the supertranslation algebra these are the four equations of [ESW, Thm 3.3].*

Proof. Every component of $S_{\mathfrak{psl}(4|4)}$ has degree 0 in $C^\bullet$, so $|f| = |g| = 0$ in Definition 2.8, the shuffle sign reduces to $\mathrm{sgn}(\sigma)$ times the Koszul sign times $(-1)^{l\sum_{i\le k}|x_{\sigma(i)}|}$, and the value $S^{(k)}(x_1,\dots,x_k)$ has parity $k + \sum|x_i|$.

Arity 1: d_{CE} of $S^{(0)}$ has no pairs; the shuffles $(0,1)$ and $(1,0)$ give $\{S, S^{(1)}(x_1)\}$ and $\{S^{(1)}(x_1), S\} = \{S, S^{(1)}(x_1)\}$, and $\frac{1}{2}$ of their sum is $\{S, S^{(1)}(x_1)\}$.

Arity 2: by Lemma 2.9, $(d_{CE}S^{(1)})(x_1,x_2) = S^{(1)}([x_1,x_2])$. The shuffles $(0,2)$ and $(2,0)$ give $2\{S, S^{(2)}(x_1,x_2)\}$. The $(1,1)$-shuffle $\sigma = \mathrm{id}$ gives $(-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\}$; the transposition gives $-(-1)^{|x_1||x_2|+|x_2|}\{S^{(1)}(x_2), S^{(1)}(x_1)\}$, and graded symmetry with $|S^{(1)}(x_i)| = 1 + |x_i|$ turns it into

$$-(-1)^{|x_1||x_2|+|x_2|+(1+|x_1|)(1+|x_2|)}\{S^{(1)}(x_1), S^{(1)}(x_2)\} = (-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\}.$$

Half the sum is the stated equation.

Arity 3 with x_1 even, x_2, x_3 odd. In (2.3) the pairs $(1,2)$, $(1,3)$, $(2,3)$ have exponents 4, $|x_3||x_2|+5 = 6$ and 6, so all three $S^{(2)}([x_i,x_j],x_k)$ enter with $+$. The $(1,2)$-shuffles give

$$\{S^{(1)}(x_1), S^{(2)}(x_2,x_3)\} - \{S^{(1)}(x_2), S^{(2)}(x_1,x_3)\} - \{S^{(1)}(x_3), S^{(2)}(x_1,x_2)\},$$

the second sign from $\mathrm{sgn} = -1$, and the third from the Koszul sign $|x_3||x_2| = 1$. The $(2,1)$-shuffles $(12|3)$, $(13|2)$, $(23|1)$ carry the signs $-$, $(-)(-)(-) = -$ and $+$; moving $S^{(1)}$ to the left costs $(-1)^{|S^{(2)}||S^{(1)}|} = +1$ in each case, because $S^{(1)}(x_3)$, $S^{(1)}(x_2)$ are even and $S^{(2)}(x_2,x_3)$ is even. The $(2,1)$-shuffles therefore reproduce the same three terms, and half the total is the stated equation; the $(0,3)$ and $(3,0)$ shuffles give $\{S, S^{(3)}\}$. The remaining patterns follow from the same expansion. □

2.3 The Noether current [ESW, Rem 1.28] in arity one

Definition 2.12 (Arity one)**.** For a conformal Killing vector field v of V, or v the linear vector field of an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, regarded as a vector field on V_{10}:

$$S^{(1)}(v) = \int \mathrm{dvol}\Big(\big(L_v A + \tfrac{1}{4}(\mathrm{div}\, v)\, A|_{W_{\mathbb{R}}^{\perp}},\, A^*\big) - \big(v.\lambda, \lambda^*\big) - \big(v.c,\, c^*\big)\Big), \\ v.\lambda = v(\lambda) + \tfrac{1}{4}\rho(d(v,-))\lambda + \tfrac{3}{8}(\mathrm{div}\, v)\lambda, \tag{2.5}$$

where $v.c = v(c)$ is the derivative of c along v [ESW, (32)]. Here A is a one-form on V_{10} constant along $W_{\mathbb{R}}^{\perp}$ [ESW, Def 1.55, Prop 1.56] and L_v is its Lie derivative on V_{10}: for v on V, $L_v(A|_V)$ is the Lie derivative of the one-form and $L_v(\iota_w A) = v(\iota_w A)$; for v the linear vector field of an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, $\iota_w(L_v A) = \iota_{v(w)}A$.

For $Q_1 \in \Sigma$ [ESW, (33)], reduced:

$$S^{(1)}(Q_1) = \int \mathrm{dvol}\Big(-(\Gamma(Q_1,\lambda), A^*) + \tfrac{1}{2}(\rho(F_A)Q_1, \lambda^*)\Big). \tag{2.6}$$

For $Q_2 \in \Sigma^*$, with conformal Killing spinor $x \mapsto \rho(x)Q_2$:

$$S^{(1)}(Q_2) = \int \mathrm{dvol}\Big(-(\Gamma(\rho(x)Q_2,\lambda), A^*) + \tfrac{1}{2}(\rho(F_A)\rho(x)Q_2, \lambda^*) - (\rho(A|_{W_{\mathbb{R}}^\perp})Q_2, \lambda^*)\Big), \tag{2.7}$$

where $\rho(A|_{W_{\mathbb{R}}^\perp})$ is Clifford multiplication by the $W_{\mathbb{R}}^\perp$-component of A, with its $\mathfrak{g}$ factor kept.

The divergences are $\operatorname{div} v = 0$ for translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$, $\operatorname{div}(2E) = 8$, and $\operatorname{div} v = -8(v_1, x)$ for $v = (x \mapsto (x,x)v_1 - 2(v_1,x)x)$. For $v \in \mathfrak{iso}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^\perp)$, (2.5) is [ESW, (32)], whose "$\mathfrak{so}(V)$ action on Σ" is $\frac{1}{4}\rho(d(v,-))$. Since $\not{d}(\rho(x)Q_2) = (\dim V)\, Q_2 = 4Q_2$ by the Clifford relation, the last term of (2.7) equals $-\frac{1}{4}(\rho(A|_{W_{\mathbb{R}}^\perp})\not{d}(\rho(x)Q_2), \lambda^*)$; in this form one formula covers the whole conformal Killing spinor $x \mapsto Q_1 + \rho(x)Q_2$, the term vanishing for constant spinors.

Example 2.13 (The dilatation)**.** For $\Delta = 2E$: $(2E, -) = d(x,x)$ is closed, so $\rho(d(v,-)) = 0$, and $\operatorname{div}(2E) = 8$. On the one-form $A|_V$, $L_{2E}A|_V = 2E(A|_V) + 2A|_V$ with $E(-)$ acting componentwise, and on the scalars $L_{2E}\iota_w A + \frac{1}{4}\cdot 8\,\iota_w A = 2E(\iota_w A) + 2\iota_w A$. Hence

$$S^{(1)}(\Delta) = \int \mathrm{dvol}\Big(\big(2E(A) + 2A, A^*\big) - \big(2E(\lambda) + 3\lambda, \lambda^*\big) - \big(2E(c), c^*\big)\Big) :$$

in the normalisation of Remark 2.6, A, λ, c carry the weights 2, 3, 0, i.e. the Weyl weights 1, $\frac{3}{2}$, 0 with respect to E.

Proposition 2.14 (The coefficients are forced)**.** *Replace the coefficients $\frac{1}{4}$ (spin), $\frac{3}{8}$ (Weyl weight of λ) and $\frac{1}{4}$ (Weyl weight of $A|_{W_{\mathbb{R}}^\perp}$) in* (2.5)*, and the coefficient -1 of the last term of* (2.7)*, by unknowns. For $\mathfrak{g} = \mathfrak{su}(2)$ the arity-one equation $\{S, S^{(1)}(x_1)\} = 0$ at one element x_1 of each type determines the unknowns uniquely:*

- *a rotation in $\mathfrak{so}(V)$, and an element of $\mathfrak{so}(W_{\mathbb{R}}^\perp)$: spin coefficient $\frac{1}{4}$;*
- *Δ: $(\frac{3}{8}, \frac{1}{4})$ for the two Weyl weights;*
- *a special conformal field: $(\frac{1}{4}, \frac{3}{8}, \frac{1}{4})$;*
- *$Q_2 \in \Sigma^*$: -1.*

With these values the arity-one equation holds for every element of $\mathfrak{psl}(4|4)$ and every $\mathfrak{g}$ (Theorem 2.20). The Weyl weight 1 of $A|_V$ is carried by L_v and the weight 0 of c by v.c.

Proof. For each type the bracket $\{S, S^{(1)}(x_1)\}$ is affine in the unknowns. For $\mathfrak{g} = \mathfrak{su}(2)$ and one element x_1 of each type, its density is a polynomial in the jets of the fields with coefficients affine in the unknowns, and the conditions that all Euler–Lagrange derivatives vanish form a linear system whose unique solution is the one stated (Remark 2.18). For translations, rotations, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ and $Q_1 \in \Sigma$ the equation is [ESW, (32), Lemma 3.4] after reduction. □

The functionals $S^{(1)}(Q_1)$ and $S^{(1)}(Q_2)$ contain no c^*; the c^* terms of $S^{(1)}$ are $-(v.c, c^*)$ for even v.

2.4 Quadratic-antifield terms and the end of the tower

Definition 2.15 (Arity two)**.** $S^{(2)}(x_1, x_2) = 0$ when x_1 or x_2 is even. For Q_1, Q_2 each in Σ or in Σ^*, $S^{(2)}(Q_1, Q_2)$ is [ESW, (34)] evaluated pointwise on the conformal Killing spinors, i.e. every $Q_i \in \Sigma^*$ is replaced by the Σ-valued function $x \mapsto \rho(x)Q_i$ in

$$S^{(2)}(Q_1, Q_2) = \int \mathrm{dvol}\Big(\tfrac{1}{4}\big(\Gamma(Q_1, Q_2), \Gamma(\lambda^*, \lambda^*)\big) - \tfrac{1}{2}(Q_1, \lambda^*)(Q_2, \lambda^*) - \big(\iota_{\Gamma(Q_1,Q_2)}A,\, c^*\big)\Big), \tag{2.8}$$

with $\Gamma(Q_1, Q_2) \in V_{10}$ the full ten-component Γ-pairing, so that the gauge parameter is $\iota_{\Gamma(Q_1,Q_2)}A = \iota_{\Gamma(Q_1,Q_2)|_V}A|_V + \iota_{\Gamma(Q_1,Q_2)|_{W_{\mathbb{R}}^\perp}}A$. For $k \geq 3$, $S^{(k)} = 0$.

The field-dependent gauge parameter $\iota_{\Gamma(Q_1,Q_2)}A$ of the c^* term is:

- $Q_1, Q_2 \in \Sigma$: $\Gamma(Q_1, Q_2)$ constant, as in [ESW, (34)];
- $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: $\Gamma(Q_1, \rho(x)Q_2)$, linear in x;
- $Q_1, Q_2 \in \Sigma^*$: $\Gamma(\rho(x)Q_1, \rho(x)Q_2)$, with V-part $-\big((x,x)v_1 - 2(v_1, x)x\big)$, $v_1 = \Gamma(Q_1, Q_2)|_V$, and $W_{\mathbb{R}}^\perp$-part $-(x,x)\,\Gamma(Q_1, Q_2)|_{W_{\mathbb{R}}^\perp}$ (proof of Proposition 2.4).

Both parts of $S^{(2)}$ are required by the arity-two equation: with $S^{(2)} = 0$ its component on every pair of odd elements of a basis is a nonzero local functional, and with the $W_{\mathbb{R}}^\perp$-components of Γ removed from (2.8) its components on $\Sigma \times \Sigma$ and on $\Sigma \times \Sigma^*$ are nonzero (Remark 2.18).

Proposition 2.16 (The tower stops at arity two)**.** *For every $\mathfrak{g}$ and all arguments, with $S^{(k\geq 3)} = 0$:*

- $\{S^{(2)}(x_1, x_2), S^{(2)}(x_3, x_4)\} = 0$*;*
- *the components of* (2.4) *of arity ≥ 4 hold;*
- *the arity-three components with at most one odd argument hold;*
- *the arity-three components with three odd arguments hold.*

Proof. $S^{(2)}$ contains the fields A, λ^*, c^* and none of their partners A^*, λ, c, and the bracket pairs each field with its partner (Definition 1.6); hence the first item. In arity 4, with $S^{(3)} = S^{(4)} = 0$, Lemma 2.11 leaves the three brackets $\{S^{(2)}, S^{(2)}\}$, which vanish; when an argument is even, every surviving term contains $S^{(2)}$ with an even argument and vanishes. In arity $n \geq 5$ every term contains $S^{(k)}$ with $k \geq 3$ or $\{S^{(k)}, S^{(l)}\}$ with $k + l = n$, hence $k \geq 3$ or $l \geq 3$, and vanishes. In arity 3 with at most one odd argument, every term contains $S^{(2)}$ with an even entry, because among $[x_i, x_j]$ and x_k, and among any two of the arguments, at most one entry is odd; so every term vanishes. In arity 3 with three odd arguments, every $[x_i, x_j]$ is even, so the equation of Lemma 2.11 reduces to the cyclic sum $\sum_{\mathrm{cyc}}\{S^{(1)}(x_1), S^{(2)}(x_2, x_3)\}$. In this bracket, $S^{(2)}$ supplies A and λ^*, whose partners A^* and λ occur in $S^{(1)}(x_1)$ exclusively through the term $-(\Gamma(Q_1, \lambda), A^*)$, with Q_1 replaced by the conformal Killing spinor of x_1; the last term of (2.7) contains A and λ^* and has vanishing bracket with $S^{(2)}$. No derivative acts on the spinors, so the cyclic sum is computed pointwise in x exactly as in the proof of [ESW, Lemma 3.7], with Q_1, Q_2, Q_3 replaced by the values of the three conformal Killing spinors at x; it vanishes by [ESW, Assumption 3.1], which holds for the ten-dimensional $\Sigma = S_+$ by [ESW, Thm 2.10] (list after [ESW, Assumption 3.1]), and the Clifford relation. □

2.5 The master equation

Axiom 2.17 (Poincaré supersymmetry, ten dimensions)**.** [ESW, Thm 3.3, Lemmas 3.4–3.7, (32)–(34)] and [ESW, Prop 2.20]: for every $\mathfrak{g}$, the functionals (2.5) on $\mathfrak{iso}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^{\perp})$, (2.6) and (2.8) on $\Sigma \times \Sigma$ satisfy every component of (2.4) with arguments in the subalgebra generated by translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ and Σ, with d_{CE} the differential (2.3) used in the proof of [ESW, Lemma 3.6].

Remark 2.18 (Exact verification on a basis)**.** The computer-assisted statements of this chapter are exact identities established on a basis. Fix a basis of $\mathfrak{psl}(4|4)$ adapted to the ten named subspaces, with $4+3+3+1+4+15$ even and $8+8+8+8$ odd elements, in which ρ, Γ and the pairing $\Sigma \otimes \Sigma^* \to \mathbb{C}$ have Gaussian-rational entries. The bracket of Definition 2.3 is computed in exact arithmetic on every ordered pair of basis elements; its values are expanded in the basis, and graded antisymmetry, the super-Jacobi identity on every ordered triple, the Killing form and the block rule on every ordered pair, the eigenvalues of ad_{Δ}, the derived algebra, the centre and the Killing forms of the two even ideals are computed from these structure constants. With the coefficient 0 in place of 1 in front of the rotation of $W_{\mathbb{R}}^{\perp}$ in (2.2), the super-Jacobi identity fails. For $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$, with the data of Remark 1.15, every component of (2.4) of arity 1, 2 and 3 is evaluated on every unordered tuple of basis elements, repetitions allowed, with at least two odd elements in arity 3; each is a local functional whose density is a polynomial in the jets of the fields with Gaussian-rational coefficients, and it vanishes when all its Euler–Lagrange derivatives vanish identically (Remark 1.15). All these identities are multilinear in their arguments, and the components of (2.4) are graded antisymmetric under exchange of arguments (Definition 2.7); the identities on basis tuples therefore hold for all arguments.

Lemma 2.19 (Colour reduction)**.** *Consider local functionals that are finite sums of terms, each the product of a polynomial in the jets of the V_{10}-, spinor- and form-components of the fields, independent of $\mathfrak{g}$, and a tree tensor in the $\mathfrak{g}$-components of at most five fields: a multilinear form on $\mathfrak{g}^k$, $k \leq 5$, obtained from copies of $([-,-],-)$ by contraction with the inverse pairing $(-,-)^{-1}$ along a tree, or the pairing $(-,-)$ itself for $k = 2$. If the Euler–Lagrange derivatives of the density of such a functional vanish identically for $\mathfrak{g} = \mathfrak{su}(2)$, they vanish identically for every Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric pairing.*

Proof. By invariance of the pairing, every tree tensor with k legs is a sign times $(b(X_1, \dots, X_{k-1}), X_k)$ for a bracket monomial b of $X_1, \dots, X_{k-1} \in \mathfrak{g}$, each used once. Hence the tree tensors with k legs are the values of a linear map, natural in $\mathfrak{g}$ and equivariant under permutations of the legs, from the space of formal trees modulo antisymmetry and the Jacobi identity, which is the multilinear part of the free Lie algebra on $k-1$ letters, of dimension $(k-2)!$. For $\mathfrak{g} = \mathfrak{su}(2)$ this map has rank 1, 1, 2, 6 for $k = 2, 3, 4, 5$, computed exactly, so it is injective for $k \leq 5$, and averaging a left inverse over the permutations of the legs gives an equivariant left inverse. A functional of the stated form is therefore the evaluation in $\mathfrak{g}$ of a formal expression, a sum of jet polynomials times formal trees. Its Euler–Lagrange derivative along a field, paired with a fixed element of $\mathfrak{g}$ treated as one more leg, is again of this form, because differentiation acts on the jet polynomials and leaves the tree. The equivariant left inverse acts on the trees alone; applied to the vanishing $\mathfrak{su}(2)$ evaluation it shows that the formal Euler–Lagrange derivatives vanish, hence their evaluations vanish for every $\mathfrak{g}$. □

Theorem 2.20 (Master equation of the equivariant action)**.** *For every Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric pairing, $S_{\mathfrak{psl}(4|4)} = S + S^{(1)} + S^{(2)}$, with $S^{(1)}$, $S^{(2)}$ of Definitions 2.12–2.15 and $S^{(k\geq 3)} = 0$, is an action of $\mathfrak{psl}(4|4)$ on $(\mathscr{E}, \langle\,,\rangle, S)$: it satisfies (2.4), and $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ satisfies $(d_{CE} + Q)I_{\mathfrak{psl}(4|4)} + \frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0$.*

Proof. By Lemma 2.11 it suffices to prove the components. Arity 0 is Proposition 1.14. The components of arity ≥ 4, and of arity 3 with at most one or with three odd arguments, hold for every $\mathfrak{g}$ by Proposition

2.16. The remaining components, of arities 1 and 2 and of arity 3 with exactly two odd arguments, vanish for $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$ for all arguments (Remark 2.18). Each of them is of the form of Lemma 2.19: every term of S, $S^{(1)}$, $S^{(2)}$ is a tree tensor in at most four, three and two $\mathfrak{g}$-components respectively, built from $(-,-)$ and $[-,-]$; the bracket of Definition 1.6 contracts one leg of each factor with the inverse pairing and joins two trees into a tree; so every term has at most $4+3-2=5$ legs. By Lemma 2.19 these components vanish for every $\mathfrak{g}$. The equation for $I_{\mathfrak{psl}(4|4)}$ is [ESW, Rem 1.29]. On the subalgebra of Axiom 2.17 the statement is [ESW, Thm 3.3]. □

Remark 2.21 (Physical reading)**.** For an even v and Q_1 odd, $S^{(2)}(v, Q_1) = 0$ and the arity-two equation is

$$S^{(1)}([v, Q_1]) = -\{S^{(1)}(v), S^{(1)}(Q_1)\} :$$

the conformal and R-symmetry transformations, with the Weyl weights of Example 2.13, commute with the supersymmetry transformations off shell. For two odd elements,

$$\{S^{(1)}(Q_1), S^{(1)}(Q_2)\} = S^{(1)}([Q_1, Q_2]) + \{S, S^{(2)}(Q_1, Q_2)\},$$

where $[Q_1, Q_2]$ is a conformal Killing vector field plus an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$; $\{S, S^{(2)}\}$ contains the fermion equation of motion (the $\lambda^*\lambda^*$ terms) and the gauge transformation with field-dependent parameter $\iota_{\Gamma(Q_1,Q_2)}A$ (the c^* term).

Chapter 3

Costello's data with a UV boundary

Costello's renormalisation needs a gauge-fixing operator Q^{GF} satisfying the five conditions of [C07, Def 4.3.3], the heat kernel of $H = [Q, Q^{GF}]$ and the propagator $P(\varepsilon, T) = \int_\varepsilon^T (Q^{GF} \otimes 1) K_t \, dt$ [C07, §6.5]. On the fields of Chapter 1 the Laplacian condition forces a unique local operator, and that operator has $(Q^{GF})^2 \neq 0$; [C07, §5.2] records that Yang–Mills theory in second-order form lies outside the axioms. This chapter adds the antighost $\bar{c}$ and its antifield, defines the gauge-fixed action through the gauge fermion $\int (\bar{c}, d^* A|_V)$, and proves that the resulting operator satisfies all five conditions on Euclidean $\mathbb{R}^4$: the decomposition of condition (5) holds at every nonzero momentum, and the zero modes are treated at scale ∞ as in [C07, Lemma 6.5.1]. It then proves that every functional of the fields of Chapter 1 flows with the propagator of the minimal complex, extends the action of $\mathfrak{psl}(4|4)$ to the antighost, and computes how each generator moves the gauge fixing.

3.1 The antighost and the gauge fermion

Definition 3.1 (The antighost pair)**.** The antighost pair is

$$\mathscr{E}^{\bar{c}} = \Omega^0(\mathbb{R}^4; \mathfrak{g}[-1]) \oplus \Omega^0(\mathbb{R}^4; \mathfrak{g}),$$

with sections $\bar{c}$, the antighost, of degree $+1$, and $\bar{c}^*$, its antifield, of degree 0. Both carry Weyl weight 2. The odd pairing of Definition 1.5 is extended to $\mathscr{E} \oplus \mathscr{E}^{\bar{c}}$ by the pair $(\bar{c}^*, \bar{c})$ with even member $\bar{c}^*$: $\langle e_0, e_1 \rangle = \int \mathrm{dvol}\, (e_0, e_1)$ for e_0 in the slot of $\bar{c}^*$ and e_1 in the slot of $\bar{c}$, and $\langle e_1, e_0 \rangle = -\langle e_0, e_1 \rangle$. The kernel of the identity K_0 acquires the components $-\delta(x-y)(-,-)^{-1}$ in $\bar{c} \otimes \bar{c}^*$ and $\bar{c}^* \otimes \bar{c}$. The action and its linear differential are

$$S' = S - \tfrac{1}{4} \int \mathrm{dvol}\, (\bar{c}^*, \bar{c}^*), \qquad Q' = Q \oplus \big(\bar{c}^* \mapsto -\tfrac{1}{2}\bar{c}^* \text{ in the slot of } \bar{c}\big).$$

field	fibre	$\mathbb{Z}$-degree	total parity	Weyl weight	coordinate ghost number
$\bar{c}$	$\mathfrak{g}$	$+1$	odd	2	-1
$\bar{c}^*$	$\mathfrak{g}$	0	even	2	0

The antighost pair, in the conventions of the table of Chapter 1.

The complex $(\mathscr{E}^{\bar{c}}, \bar{c}^* \mapsto -\frac{1}{2}\bar{c}^*)$ is contractible on every space of sections, so S' describes the theory of Chapter 1. The pair is the reduction of the trivial pair formed by $\bar{c}$ and an auxiliary field b.

Proposition 3.2 (Reduction of the trivial pair)**.** *Let b be a $\mathfrak{g}$-valued field of degree 0 and Weyl weight 2 with antifield b^* of degree 1, paired with even member b. The linear map*

$$e \mapsto e + \big(- e_{\bar{c}} \text{ in the slot of } b^* \big) + \big(e_b \text{ in the slot of } \bar{c}^* \big),$$

with $e_{\bar c}$, e_b the components of e, preserves the pairing, and it maps the action $S + \int \mathrm{dvol}\,(b, \bar c^*)$ to

$$S + \int \mathrm{dvol}\,(b, \bar c^* + b) = S' + \int \mathrm{dvol}\,\big(b + \tfrac{1}{2}\bar c^*,\, b + \tfrac{1}{2}\bar c^*\big).$$

The Gaussian integral over b along $b^* = 0$ gives S'. The extension of the even part of $\mathfrak{psl}(4|4)$ to the trivial pair by $-\int(v.\bar c, \bar c^*) + \int(v.b, b^*)$ and $-\int(u.\bar c, b^*)$ on pairs of odd elements reduces to the extension of Definition 3.28.

Proof. The map is $1 + N_b$ with N_b sending the slot of $\bar c$ to that of b^* and the slot of b to that of $\bar c^*$; $\langle N_b e, e'\rangle + \langle e, N_b e'\rangle = 0$ for all e, e', the two coefficients -1, $+1$ having opposite signs because b is the even member of its pair and $\bar c$ the odd member of its pair, and $\langle N_b e, N_b e'\rangle = 0$. Completing the square, $(b,b) + (b, \bar c^*) = (b + \frac{1}{2}\bar c^*, b + \frac{1}{2}\bar c^*) - \frac{1}{4}(\bar c^*, \bar c^*)$. On the extension, the part of the image of $-\int(v.\bar c, \bar c^*) + \int(v.b, b^*)$ that contains b at $b^* = 0$ is $\int\big((v.\bar c, b) + (v.b, \bar c)\big)$, the integral of a total derivative for fields of weights adding to 4, and the image of $-\int(u.\bar c, b^*)$ at $b^* = 0$ is $\int(u.\bar c, \bar c) = -\int(\bar c, u(\bar c))$ (Remark 3.38). □

Definition 3.3 (Gauge fermion and gauge-fixed action)**.** The gauge fermion is the local functional $\Psi = \int \mathrm{dvol}\,(\bar c, d^*A|_V)$ of degree -1. The linear map Φ_Ψ of $\mathscr{E} \oplus \mathscr{E}^{\bar c}$ is

$$\Phi_\Psi(e) = e + \big(-de_{\bar c} \text{ in the slot of } A^*|_V\big) + \big(d^*(e_A|_V) \text{ in the slot of } \bar c^*\big).$$

The gauge-fixed action is $S_\Psi = S' \circ \Phi_\Psi$:

$$S_\Psi = S\big(c, A, \lambda;\, c^*,\, A^* - d\bar c,\, \lambda^*\big) - \tfrac{1}{4}\int \mathrm{dvol}\,\big(\bar c^* + d^*A|_V,\, \bar c^* + d^*A|_V\big), \tag{3.1}$$

where $A^* - d\bar c$ replaces $A^*|_V$ by $A^*|_V - d\bar c$ and keeps $A^*|_{W_{\mathbb{R}}^\perp}$. Its interaction is $I_\Psi = I(c, A, \lambda; c^*, A^* - d\bar c, \lambda^*)$, and Q denotes from now on the linear differential of S_Ψ. The image under Φ_Ψ of the slots of the fields c, A, λ, $\bar c$ is the subspace

$$A^*|_V = -d\bar c, \qquad \bar c^* = d^*A|_V, \qquad A^*|_{W_{\mathbb{R}}^\perp} = 0, \qquad \lambda^* = 0, \qquad c^* = 0,$$

the Lagrangian defined by the gauge fermion Ψ.

Proposition 3.4 (The canonical transformation)**.** *Φ_Ψ preserves $\langle\,,\rangle$, $\Phi_\Psi^{-1} = 2 - \Phi_\Psi$, and $Q = \Phi_\Psi^{-1} Q' \Phi_\Psi$ is given by the table. Among the maps $e \mapsto e + (a_1 de_{\bar c}$ in $A^*|_V) + (a_2 d^*(e_A|_V)$ in $\bar c^*)$ the pairing is preserved exactly when $a_1 = -a_2$.*

on the slot of	Q	lands in the slot of	momentum space
c	$c \mapsto -dc$	$A\|_V$	$-i(p,-)\,c$
c	$c \mapsto d^*dc$	$\bar c^*$	$\lvert p\rvert^2 c$
A	$A \mapsto -\frac{1}{2}(d^*d + dd^*)A$	A^*	$-\frac{1}{2}\lvert p\rvert^2 A$
A	$A \mapsto -\frac{1}{2}d^*(A\|_V)$	$\bar c$	$\frac{i}{2}\,\iota_p A$
A^*	$A^* \mapsto d^*(A^*\|_V)$	c^*	$-i\,(p, A^*\|_V)$
λ	$\lambda \mapsto \not{d}\lambda$	λ^*	$i\rho(p)\lambda$
$\bar c$	$\bar c \mapsto -d^*d\bar c$	c^*	$-\lvert p\rvert^2 \bar c$
$\bar c^*$	$\bar c^* \mapsto -\frac{1}{2}\bar c^*$	$\bar c$	$-\frac{1}{2}$
$\bar c^*$	$\bar c^* \mapsto -\frac{1}{2}d\bar c^*$	$A^*\|_V$	$-\frac{i}{2}\,p\,\bar c^*$
λ^*, c^*	0		

The linear differential Q of S_Ψ. On $A|_{W_{\mathbb{R}}^\perp}$, $dd^* = 0$ and d^*d is the Laplacian of each scalar; on $A|_V$, $d^*d + dd^*$ is the Hodge Laplacian, the Laplacian of each component.

Proof. $N = \Phi_\Psi - 1$ maps the slots of fields to the slots of antifields, so $N^2 = 0$ and $\langle Ne, Ne'\rangle = 0$. For e in the slot of $\bar c$ and e' in the slot of A,

$$\langle Ne, e'\rangle = \langle -de, e'\rangle = -\langle e', -de\rangle = \int \mathrm{dvol}\,(e', de), \qquad \langle e, Ne'\rangle = \langle e, d^*e'\rangle = -\langle d^*e', e\rangle = -\int \mathrm{dvol}\,(d^*e', e),$$

and $\int (d^*e', e) = \int (e', de)$, so the sum vanishes; with a_1, a_2 in place of $-1, 1$ the sum is $-(a_1+a_2)\int (e', de)$. The other pairings vanish or follow by graded antisymmetry. Hence $\frac{1}{2}\langle \Phi_\Psi e, Q'\Phi_\Psi e\rangle = \frac{1}{2}\langle e, \Phi_\Psi^{-1}Q'\Phi_\Psi e\rangle$, and the table is $(1-N)Q'(1+N)$: for instance $Q'\Phi_\Psi c = -dc$ and $(1-N)(-dc) = -dc - d^*(-dc)$ in the slots of $A|_V$ and $\bar c^*$; $Q'\Phi_\Psi A = -\frac{1}{2}d^*dA + d^*(A|_V)\cdot(-\frac{1}{2})$ in the slots of A^* and $\bar c$, and $(1-N)$ adds $-N(-\frac{1}{2}d^*A|_V) = -\frac{1}{2}dd^*A|_V$ in the slot of $A^*|_V$. The remaining rows are computed in the same way. □

3.2 The gauge-fixing operator

Definition 3.5 (Gauge-fixing operator)**.** Q^{GF} is the odd operator of degree -1 on $\mathscr{E} \oplus \mathscr{E}^{\bar e}$ given by the table; it is zero on the slots of the fields c, A, λ, $\bar c$.

on the slot of	Q^{GF}	lands in the slot of	momentum space
A^*	$A^* \mapsto -2A^*$ (all ten components, through the metric of V_{10})	A	-2
λ^*	$\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$	λ	$-i\rho(p)\lambda^*$
$\bar c^*$	$\bar c^* \mapsto \bar c^*$	c	1
c^*	$c^* \mapsto -c^*$	$\bar c$	-1

The gauge-fixing operator Q^{GF}: it maps the slot of each antifield to the slot of a field.

Proposition 3.6 (The Laplacian)**.** $H := [Q, Q^{GF}] = QQ^{GF} + Q^{GF}Q = d^*d + dd^*$ *on every component of every slot of* $\mathscr{E} \oplus \mathscr{E}^{\bar e}$*; in momentum space* $H = |p|^2$*. Moreover* $[Q^{GF}, H] = 0$ *and* $[Q, H] = 0$*.*

Proof. Slot by slot, with the two tables:

$$\begin{aligned}
c:&\quad Q^{GF}Qc = Q^{GF}(-dc) + Q^{GF}(d^*dc \text{ in } \bar c^*) = d^*dc, \qquad QQ^{GF}c = 0;\\
A:&\quad Q^{GF}QA = Q^{GF}\big(-\tfrac{1}{2}(d^*d+dd^*)A\big) + Q^{GF}\big(-\tfrac{1}{2}d^*A|_V \text{ in } \bar c\big) = (d^*d+dd^*)A, \qquad QQ^{GF}A = 0;\\
\bar c:&\quad Q^{GF}Q\bar c = Q^{GF}(-d^*d\bar c \text{ in } c^*) = d^*d\bar c, \qquad QQ^{GF}\bar c = 0;\\
A^*:&\quad QQ^{GF}A^* = Q(-2A^*) = (d^*d+dd^*)A^* + d^*(A^*|_V) \text{ in } \bar c, \qquad Q^{GF}QA^* = Q^{GF}\big(d^*(A^*|_V) \text{ in } c^*\big) = -d^*(A^*|_V) \text{ in}\\
\bar c^*:&\quad QQ^{GF}\bar c^* = Q(\bar c^* \text{ in } c) = -d\bar c^* \text{ in } A|_V + d^*d\bar c^*, \qquad Q^{GF}Q\bar c^* = Q^{GF}\big(-\tfrac{1}{2}d\bar c^* \text{ in } A^*|_V\big) = d\bar c^* \text{ in } A|_V;\\
c^*:&\quad QQ^{GF}c^* = Q(-c^* \text{ in } \bar c) = d^*dc^*, \qquad Q^{GF}Qc^* = 0,
\end{aligned}$$

and $Q^{GF}Q\lambda = -\not{d}|_{\Sigma^*}\not{d}\lambda$, $QQ^{GF}\lambda^* = -\not{d}\,\not{d}|_{\Sigma^*}\lambda^*$, which on a plane wave $e^{i(p,x)}$ are $\rho(p)\rho(p) = (p,p) = |p|^2$ by the Clifford relation, while $QQ^{GF}\lambda = 0 = Q^{GF}Q\lambda^*$. The cross terms in the slots of $\bar c$ and of $A|_V$ cancel. H is a constant-coefficient scalar operator, so it commutes with Q and Q^{GF}. □

Theorem 3.7 (Existence and uniqueness)**.** *Call an operator admissible if it is odd of degree* -1*, a sum of constant-coefficient differential operators of order at most* 1 *between the slots, covariant under* $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$*, and homogeneous of Weyl weight* 2*: each component from a slot of Weyl weight* w *to a slot of Weyl weight* w' *carries* $w' - w + 2$ *derivatives, so that no dimensionful constant enters.*

- *(i) On* $\mathscr{E}$ *with the operator* Q *of Chapter 1 there is exactly one admissible operator* G *with* $[Q, G] = |p|^2$*; it is* $A^* \mapsto -2A^*$*,* $A \mapsto -d^*(A|_V)$ *(slot of* c*),* $c^* \mapsto dc^*$ *(slot of* $A^*|_V$*),* $\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$*, and* $G^2A^* = 2d^*(A^*|_V)$*,* $G^2c^* = -2dc^*$*.*

- *(ii) Replace $-\frac{1}{4}$ in S' by $\frac{1}{2}\alpha$, $\alpha \neq 0$, and let Q_α be the linear differential of the corresponding S_Ψ. The admissible operators G with $[Q_\alpha, G] = |p|^2$ and with the adjointness of Proposition 3.8 form, for every α, exactly one operator; every component of G^2 is a multiple of $(2\alpha+1)^2/(4\alpha)$, and $G^2 = 0$ exactly at $\alpha = -\frac{1}{2}$, where $G = Q^{GF}$.*

Proof. An admissible operator is a combination, with unknown coefficients, of the components allowed by degree, weight and covariance: from a scalar to a scalar with no derivative, from a scalar to a one-form through d, from a one-form to a scalar through d^*, between one-forms or between the scalars of $W_{\mathbb{R}}^{\perp}$ through the identity, and from Σ^* to Σ through $\not{d}|_{\Sigma^*}$; the weights of Definition 3.1 and of Chapter 2 leave, on $\mathscr{E} \oplus \mathscr{E}^{\bar{c}}$, the nine components $A \to c$, $\bar{c}^* \to c$, $A^* \to A$, $A^*|_V \to \bar{c}^*$, $\bar{c} \to A|_V$, $c^* \to A^*|_V$, $c^* \to \bar{c}$, $\lambda^* \to \lambda$ and $A^*|_{W_{\mathbb{R}}^{\perp}} \to A|_{W_{\mathbb{R}}^{\perp}}$. $[Q_\alpha, G] = |p|^2$ and the adjointness are linear in the coefficients; their solution is a single operator for every α, with the coefficients of $A \to c$, $A^*|_V \to \bar{c}^*$, $\bar{c} \to A|_V$ and $c^* \to A^*|_V$ proportional to $2\alpha + 1$. The square of this operator is computed component by component (Remark 3.38). On $\mathscr{E}$ the same computation leaves four components, fixed by the slots of c, of the transverse part of A, of c^* and of λ, as in Proposition 3.6. □

Proposition 3.8 (Adjointness)**.** *For all $e_0, e_1 \in (\mathscr{E} \oplus \mathscr{E}^{\bar{c}})_c$,*

$$\langle Q^{GF} e_0, e_1\rangle = (-1)^{|e_0|}\langle e_0, Q^{GF} e_1\rangle, \qquad \langle Q e_0, e_1\rangle = -(-1)^{|e_0|}\langle e_0, Q e_1\rangle;$$

equivalently $(Q^{GF} \otimes 1 - 1 \otimes Q^{GF})K_0 = 0$ and $(Q \otimes 1 + 1 \otimes Q)K_0 = 0$.

Proof. The relation for Q follows from Proposition 1.11 for Q' and Proposition 3.4. For Q^{GF} the pairing vanishes outside three cases. For e_0, e_1 in the slot of A^* (odd), $\langle Q^{GF} e_0, e_1\rangle = -2\int(e_0, e_1)$ and $\langle e_0, Q^{GF} e_1\rangle = -\langle -2e_1, e_0\rangle = 2\int(e_1, e_0)$. For e_0 in the slot of $\bar{c}^*$ and e_1 in the slot of c^* (both even),

$$\langle Q^{GF} e_0, e_1\rangle = \langle e_0 \text{ in } c,\, e_1\rangle = -\int \mathrm{dvol}\,(e_1, e_0), \qquad \langle e_0, Q^{GF} e_1\rangle = \langle e_0, -e_1 \text{ in } \bar{c}\rangle = -\int \mathrm{dvol}\,(e_0, e_1).$$

For e_0, e_1 in the slot of λ^* (even), with the symmetry of $\rho(v)$ and integration by parts, $\langle Q^{GF} e_0, e_1\rangle = \int \mathrm{dvol}\,(\not{d} e_0, e_1)$ and $\langle e_0, Q^{GF} e_1\rangle = -\int \mathrm{dvol}\,(\not{d} e_1, e_0) = \int \mathrm{dvol}\,(\not{d} e_0, e_1)$. In each case $\langle Q^{GF} e_0, e_1\rangle = (-1)^{|e_0|}\langle e_0, Q^{GF} e_1\rangle$. □

Remark 3.9 (Names of the adjointness relations in the sources)**.** [C07, §4.3] calls Q "self adjoint" and [C07, Def 4.3.3(1)] calls Q^{GF} "skew self adjoint"; [C07, §1, §1.1, §2.1] and [CG2, §7.2.1] use the opposite names. The relations of Proposition 3.8 fix the content, and condition (1) of [C07, Def 4.3.3] is read as: Q^{GF} has the adjointness opposite to that of Q. The printed adjointness relation of [ESW, Def 1.16] is discussed in Remark 1.12.

Theorem 3.10 (The five conditions)**.** *Q^{GF} satisfies the conditions of [C07, Def 4.3.3] on $\mathbb{R}^4$:*

- *(1) the adjointness opposite to that of Q (Proposition 3.8);*
- *(2) every component has order 0 or 1;*
- *(3) $(Q^{GF})^2 = 0$;*
- *(4) $H = d^*d + dd^*$ on every component, a generalised Laplacian with scalar symbol (Proposition 3.6);*
- *(5) at every momentum $p \neq 0$, $\operatorname{Ker} H(p) = 0$ and the fibre of $\mathscr{E} \oplus \mathscr{E}^{\bar{c}}$ is $\operatorname{Im} Q(p) \oplus \operatorname{Im} Q^{GF}(p)$, each summand of dimension 28 times $\dim \mathfrak{g}$; moreover $\operatorname{Ker} Q^{GF}(p) = \operatorname{Im} Q^{GF}(p)$ is the span of the slots of c, A, λ, $\bar{c}$, a Lagrangian subspace, and $\operatorname{Ker} Q(p) = \operatorname{Im} Q(p)$.*

On the real slice of the bosonic part of $\operatorname{Im} Q^{GF}$*, A real for the compact form of* $\mathfrak{g}$*,*

$$\tfrac{1}{2}\langle A, QA\rangle = -\tfrac{1}{4}\int \operatorname{dvol}\big((dA, dA) + (d^*A|_V, d^*A|_V)\big) = -\tfrac{1}{4}\int \operatorname{dvol}(A, HA),$$

negative definite on all ten components at every $p \neq 0$*.*

Proof. (2) is read off the table. (3): Q^{GF} maps antifield slots to field slots and vanishes on field slots. (5): at $p \neq 0$, $H(p) = |p|^2$ is invertible, so every e equals $Q\big(Q^{GF}e/|p|^2\big) + Q^{GF}\big(Qe/|p|^2\big)$; if $e = Qa = Q^{GF}b$ then $|p|^2 e = QQ^{GF}Qa + Q^{GF}QQ^{GF}b = QQ^{GF}e + Q^{GF}Qe = Q(Q^{GF})^2 b + Q^{GF}Q^2 a = 0$. With $Q^2 = 0 = (Q^{GF})^2$, $\operatorname{Im} \subset \operatorname{Ker}$ for both, and the projections $QQ^{GF}/|p|^2$, $Q^{GF}Q/|p|^2$ give equality. The image of $Q^{GF}(p)$ contains every field slot: $A^* \mapsto -2A^*$, $\bar{c}^* \mapsto \bar{c}^*$, $c^* \mapsto -c^*$ are invertible and $\rho(p)$ is invertible for $p \neq 0$; these slots carry half of the dimension and pair to zero among themselves. The ranks are computed in Remark 3.38. The quadratic form: the part of (3.1) without antifields and without ghosts is $-\frac{1}{4}\int(dA, dA) - \frac{1}{4}\int(d^*A|_V, d^*A|_V)$, and $(dA, dA) + (d^*A|_V, d^*A|_V)$ integrates to $\int(A, HA)$. □

Remark 3.11 (Feynman gauge)**.** The term $-\frac{1}{4}\int(d^*A|_V, d^*A|_V)$ of (3.1) is the Feynman-gauge term, and the $A \otimes A$ component of the propagator (Proposition 3.17) is the inverse of the second variation of $\frac{1}{4}\int(A, HA)$. Theorem 3.7 (ii) shows that the coefficient $-\frac{1}{4}$ of $(\bar{c}^*, \bar{c}^*)$ in S', equivalently Feynman gauge, is the value at which a square-zero admissible operator exists.

3.3 The fields of Chapter 1 inside the gauge-fixed theory

The Costello frame is the presentation by S', Q' and $Q_C^{GF} := \Phi_\Psi Q^{GF} \Phi_\Psi^{-1}$; the gauge-fixed frame is the presentation by S_Ψ, Q, Q^{GF}. Let $\mathrm{pr}\colon \mathscr{E} \oplus \mathscr{E}^{\bar{c}} \to \mathscr{E}$ forget the components $\bar{c}$, $\bar{c}^*$, let ι be the inclusion, and put $j_\Psi = \mathrm{pr} \circ \Phi_\Psi$:

$$j_\Psi(c, A, \lambda, \bar{c};\, c^*, A^*, \lambda^*, \bar{c}^*) = (c, A, \lambda;\, c^*,\, A^* - d\bar{c},\, \lambda^*).$$

For a linear map $X\colon \mathscr{E} \oplus \mathscr{E}^{\bar{c}} \to \mathscr{E}$, $X^\dagger\colon \mathscr{E} \to \mathscr{E} \oplus \mathscr{E}^{\bar{c}}$ is defined by $\langle Xe', e\rangle = \langle e', X^\dagger e\rangle$.

Theorem 3.12 (The minimal projection)**.**

- *(i)* $j_\Psi Q = Q_{\min} j_\Psi$*, with* $Q_{\min}$ *the operator of Proposition 1.11;* $j_\Psi j_\Psi^\dagger = 1$*, hence* $(j_\Psi \otimes j_\Psi)K_t = K_t$ *for* $0 \le t \le \infty$*, the right side the kernel on* $\mathscr{E}$*.*

- *(ii)* $j_\Psi Q^{GF} j_\Psi^\dagger$ *is the operator G of Theorem 3.7 (i); hence* $(j_\Psi \otimes j_\Psi)P(\varepsilon, T)$ *is the kernel with the components* $A \otimes A$*,* $\lambda \otimes \lambda$ *of Proposition 3.17 and*

  $$c \otimes A^*|_V,\ A^*|_V \otimes c: \qquad -(-,-)^{-1}\, d_x\, \frac{e^{-|x-y|^2/4T} - e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}, \qquad i\,p\, \frac{e^{-\varepsilon|p|^2} - e^{-T|p|^2}}{|p|^2} \text{ times } -(-,-)^{-1}.$$

- *(iii)* $I_\Psi = I \circ j_\Psi$*, and for every functional J on* $\mathscr{E}$*, every t and* $0 < \varepsilon < T \le \infty$*,*

  $$W\big(P(\varepsilon, T), J \circ j_\Psi\big) = W\big((j_\Psi \otimes j_\Psi)P(\varepsilon, T), J\big) \circ j_\Psi, \quad Q(J \circ j_\Psi) = (Q_{\min} J) \circ j_\Psi, \quad \Delta_t(J \circ j_\Psi) = (\Delta_t J) \circ j_\Psi.$$

- *(iv) In the Costello frame,* $(Q_C^{GF})^2 = 0$*,* $[Q', Q_C^{GF}] = |p|^2$*, and* $\mathrm{pr}\, Q_C^{GF}\, \iota = G$*. The component of* Q_C^{GF} *from the slot of* $\bar{c}$ *to that of* $\bar{c}^*$ *is* $-2d^*d$*, of order two, and* $G^2 = -\mathrm{pr}\, Q_C^{GF}\, \pi_{\bar{c}}\, Q_C^{GF}\, \iota$*, with* $\pi_{\bar{c}}$ *the projection to* $\mathscr{E}^{\bar{c}}$*.*

Proof. (i) $\operatorname{pr} Q' = Q_{\min} \operatorname{pr}$ because Q' is the direct sum of $Q_{\min}$ and an operator on $\mathscr{E}^{\bar{c}}$; with $Q = \Phi_\Psi^{-1} Q' \Phi_\Psi$, $j_\Psi Q = \operatorname{pr} Q' \Phi_\Psi = Q_{\min} j_\Psi$. Since Φ_Ψ preserves the pairing, $\Phi_\Psi^\dagger = \Phi_\Psi^{-1}$, and $\operatorname{pr}^\dagger = \iota$, so $j_\Psi j_\Psi^\dagger = \operatorname{pr} \Phi_\Psi \Phi_\Psi^{-1} \iota = 1$; the kernel $(X \otimes Y) K_0$ represents $XY^\dagger$, and K_t represents $e^{-t|p|^2}$, a scalar. (ii) $j_\Psi Q^{GF} j_\Psi^\dagger = \operatorname{pr} Q_C^{GF} \iota$. For $e \in \mathscr{E}$, $\Phi_\Psi^{-1} e = e - d^*(e_A|_V)$ in the slot of $\bar{c}^*$; Q^{GF} of it is $-2e_{A^*}$ in A, $-\not{d}|_{\Sigma^*} e_{\lambda^*}$ in λ, $-e_{c^*}$ in $\bar{c}$ and $-d^*(e_A|_V)$ in c; Φ_Ψ adds de_{c^*} in $A^*|_V$ and $-2d^*(e_{A^*}|_V)$ in $\bar{c}^*$, and pr keeps $-2e_{A^*}$, $-\not{d}|_{\Sigma^*} e_{\lambda^*}$, $-d^*(e_A|_V)$, de_{c^*}. The kernel follows as in Proposition 3.17. (iii) I contains no $\bar{c}$, $\bar{c}^*$, so $I \circ \Phi_\Psi = I \circ j_\Psi$. For a linear map j and a kernel ϕ, $\partial_\phi(J \circ j) = (\partial_{(j\otimes j)\phi} J) \circ j$ by the definition of ∂_ϕ [C07, §6.4] on products of linear functionals; the flow, Q and Δ_t are built from ∂_ϕ, Q and K_t. (iv) Conjugation by Φ_Ψ gives the first two statements; the component $\bar{c} \to \bar{c}^*$ is $-N Q^{GF} N$ restricted to the slot of $\bar{c}$, $\bar{c} \mapsto -d\bar{c} \mapsto 2d\bar{c} \mapsto -2d^* d\bar{c}$; and $0 = \operatorname{pr}(Q_C^{GF})^2 \iota = G^2 + \operatorname{pr} Q_C^{GF} \pi_{\bar{c}} Q_C^{GF} \iota$. □

Corollary 3.13 (Functionals of the fields of Chapter 1)**.** *Every graph of the flow with vertices $J \circ j_\Psi$, J a functional on $\mathscr{E}$, has the value of the graph with vertices J and edges $(j_\Psi \otimes j_\Psi) P(\varepsilon, T)$, composed with j_Ψ; the special edges $K_\varepsilon - K_0$ and the brackets $\{-,-\}_t$ behave in the same way. In particular the one-loop counterterm of I_Ψ is $I^{CT}[\varepsilon] \circ j_\Psi$, the defect vertex of $I_\Psi - I^{CT}[\varepsilon] \circ j_\Psi$ is the defect vertex of $I - I^{CT}[\varepsilon]$ composed with j_Ψ, and a finite counterterm F on $\mathscr{E}$ solves the master equation for I exactly when $F \circ j_\Psi$ solves it for I_Ψ. The vertices of I_Ψ contain $\bar{c}$ through $A^*|_V - d\bar{c}$, and the components of the propagator with a leg $\bar{c}$ are $\bar{c} \otimes c$ and $c \otimes \bar{c}$.*

Proof. Theorem 3.12 (iii), applied term by term to the graph expansion of Proposition 3.35 and to the defect vertex of Definition 3.21. □

3.4 Heat kernel and propagator

Definition 3.14 (Heat kernel)**.** For $t > 0$, $K_t \in (\mathscr{E} \oplus \mathscr{E}^{\bar{c}})^{\otimes 2}$ is the kernel with the components of K_0 (Definitions 1.6, 3.1) with $\delta(x-y)$ replaced by $(4\pi t)^{-2} e^{-|x-y|^2/4t}$; in momentum space this factor is $e^{-t|p|^2}$. Then $K_t \star e = e^{-tH} e$ [C07, §6.5; C07, §15.2]. The kernel $L_t = (Q^{GF} \otimes 1) K_t$ represents $Q^{GF} e^{-tH}$ [C07, §6.5]. On $\mathbb{R}^4$, $K_\infty := \lim_{t\to\infty} K_t = 0$, pointwise in $x - y$, and in momentum space for every $p \neq 0$.

Proposition 3.15 (Heat-kernel identities)**.**

$$(Q \otimes 1 + 1 \otimes Q) K_t = 0, \qquad \frac{d}{dt} K_t + (Q \otimes 1 + 1 \otimes Q) L_t = 0 \qquad [\textit{C07, §6.5}].$$

Proof. $K_t = (e^{-tH} \otimes 1) K_0$, $[Q, H] = 0$ and $(Q \otimes 1 + 1 \otimes Q) K_0 = 0$ give the first identity. Odd operators on different tensor factors anticommute, and $(1 \otimes Q) K_t = -(Q \otimes 1) K_t$, so $(Q \otimes 1 + 1 \otimes Q) L_t = \big((QQ^{GF} + Q^{GF} Q) \otimes 1\big) K_t = (H \otimes 1) K_t = -\frac{d}{dt} K_t$. □

Definition 3.16 (Propagator)**.**

$$P(\varepsilon, \infty) = \int_\varepsilon^\infty L_t \, dt = \int_\varepsilon^\infty (Q^{GF} \otimes 1) K_t \, dt \qquad [\text{C07, §6.5}], \qquad \varepsilon > 0.$$

The component in a row labelled by two slots in the table below is the value of $\partial_{P(\varepsilon,\infty)}$ [C07, §6.4] on the product of a coordinate functional of the first slot at x and one of the second slot at y, in this order.

Proposition 3.17 (Components of the propagator)**.** *The scalar factor of K_t integrates to*

$$\int_\varepsilon^\infty (4\pi t)^{-2} e^{-|x-y|^2/4t} \, dt = \frac{1 - e^{-|x-y|^2/4\varepsilon}}{4\pi^2 |x-y|^2}, \qquad \int_\varepsilon^\infty e^{-t|p|^2} \, dt = \frac{e^{-\varepsilon|p|^2}}{|p|^2},$$

and the nonzero components of $P(\varepsilon, \infty)$ are:

component	*fibre factor*	*position space*	*momentum space*
$A \otimes A$	$2(-,-)^{-1} \otimes (\textit{inverse metric of } V_{10})$	$\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$c \otimes \bar{c}$	$-(-,-)^{-1}$	$\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$\bar{c} \otimes c$	$(-,-)^{-1}$	$\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$\lambda \otimes \lambda$	$(-,-)^{-1}$	$\not{d}_x\lvert_{\Sigma^*}\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$i\rho(p)\lvert_{\Sigma^*\to\Sigma}\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$

The propagator $P(\varepsilon,\infty)$; the kernel $\rho(p)|_{\Sigma^*\to\Sigma}$ is read as an element of $\Sigma\otimes\Sigma$ through the pairing $\Sigma\otimes\Sigma^*\to\mathbb{C}$.

Every other component vanishes; in particular every component with a leg in the slot of an antifield vanishes. $P(\varepsilon,\infty)$ is even and graded symmetric: exchanging the two factors together with $x \leftrightarrow y$ ($p \mapsto -p$) multiplies each component by the Koszul sign of the two slots.

Proof. $\frac{d}{dt}e^{-|x-y|^2/4t} = \frac{|x-y|^2}{4t^2}e^{-|x-y|^2/4t}$ gives the scalar factor. With Q^{GF} acting on the first factor: the $A^*\otimes A$ component $-(4\pi t)^{-2}e^{-|x-y|^2/4t}$ of K_t goes by $A^*\mapsto -2A^*$ to $+2$ times the scalar in $A\otimes A$; the $\bar{c}^*\otimes\bar{c}$ component goes by $\bar{c}^*\mapsto\bar{c}^*$ to -1 times the scalar in $c\otimes\bar{c}$; the $c^*\otimes c$ component goes by $c^*\mapsto -c^*$ to $+1$ times the scalar in $\bar{c}\otimes c$; the $\lambda^*\otimes\lambda$ component goes to $\not{d}_x|_{\Sigma^*}$ of the scalar. Graded symmetry: $c\otimes\bar{c}$ and $\bar{c}\otimes c$ have the Koszul sign -1 and opposite fibre factors; for $\lambda\otimes\lambda$, $\rho(-p)=-\rho(p)$. □

Proposition 3.18 (The two UV-boundary kernels)**.**

$$(Q\otimes 1 + 1\otimes Q)P(\varepsilon,\infty) = K_\varepsilon, \tag{3.2}$$

$$\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -\varepsilon L_\varepsilon. \tag{3.3}$$

Proof. By Proposition 3.15, $(Q\otimes 1+1\otimes Q)P(\varepsilon,\infty) = -\int_\varepsilon^\infty \frac{d}{dt}K_t\,dt = K_\varepsilon - K_\infty = K_\varepsilon$; this is [C07, Lemma 10.2.1] with the upper scale at ∞. The second identity is the fundamental theorem of calculus. □

Remark 3.19 (Two objects at $t=\varepsilon$)**.** By (3.3), L_ε governs the ε-dependence of every graph: $\varepsilon\frac{d}{d\varepsilon}$ of the value of a graph is minus the sum over its edges of the graph with that edge carrying $\varepsilon L_\varepsilon$, so the singular coefficients sit on L_ε edges. By (3.2), K_ε governs the action of Q (Theorem 3.23): the obstruction to the quantum master equation and the Q-variation of observables sit on edges carrying K_ε, or $K_\varepsilon - K_0$, and on the counterterm vertices.

3.5 The UV-boundary identity

Remark 3.20 (Functionals on $\mathbb{R}^4$)**.** On $\mathbb{R}^4$ the operators ∂_{K_t} and $\partial_{P(\varepsilon,\infty)}$ [C07, §6.4] and the brackets below act on functionals of compactly supported sections. The interaction, its counterterms and their flows are taken modulo constants [CG2, §7.2.6]; observables keep their constant terms. A local functional $\int \mathrm{dvol}\,(\text{density})$ is a density modulo total derivatives, evaluated graph by graph with compactly supported legs. If a contracted Taylor component is an integrable kernel in the two contracted positions, its contraction with K_t is bounded by $(4\pi t)^{-2}$ times the integral of its absolute value and tends to 0 as $t\to\infty$; this applies to local functionals $\int\omega\,(\text{density})$ with ω a compactly supported four-form [CG2, Lemma 10.1.1.3], to local functionals at least quadratic in the fields, and to brackets of two such. For a linear functional $\int \mathrm{dvol}\,(c^*, X)$, $X\in\mathfrak{g}$, the contraction integrates K_t over one position, with value 1 for every t; these are the zero-mode classes of Theorem 3.25.

Definition 3.21 (Scale-ε Laplacian and bracket)**.**

$$\Delta_\varepsilon = -\partial_{K_\varepsilon} \quad \text{[C07, §10.1]}, \qquad \{f,g\}_\varepsilon = \Delta_\varepsilon(fg) - \Delta_\varepsilon(f)g - (-1)^{|f|} f\Delta_\varepsilon(g) \quad \text{[CG2, §7.2.8]}.$$

At scale ∞, $K_\infty := 0$, $\Delta_\infty := 0$ and $\{-,-\}_\infty = 0$ on every functional; on the functionals of Remark 3.20 these are also the limits $t \to \infty$ of Δ_t and $\{-,-\}_t$.

Definition 3.22 (Renormalisation group flow)**.** For an even $\phi \in \mathrm{Sym}^2(\mathscr{E} \oplus \mathscr{E}^{\bar c})$ and an even functional J, at least cubic modulo $\hbar$,

$$W(\phi, J) = \hbar \log\left(e^{\hbar\partial_\phi} e^{J/\hbar}\right) \qquad \text{[CG2, §7.2.7]},$$

which is $\Gamma(\phi, J)$ of [C07, Def 6.6.1].

Theorem 3.23 (UV-boundary identity)**.** *On functionals with values in $C^\bullet(\mathfrak{psl}(4|4))$ (Chapter 2), as formal power series in $\hbar$,*

$$(d_{CE} + Q)\, e^{\hbar\partial_{P(\varepsilon,\infty)}} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\, (d_{CE} + Q + \hbar\Delta_\varepsilon). \tag{3.4}$$

Proof. By [C07, §6.4], $[\partial_\phi, Q] = \partial_{Q\phi}$ with Q acting on $\mathscr{E} \otimes \mathscr{E}$ as $Q \otimes 1 + 1 \otimes Q$; with (3.2), $Q\,\partial_{P(\varepsilon,\infty)} = \partial_{P(\varepsilon,\infty)} Q + \Delta_\varepsilon$. The operators ∂_ϕ commute [C07, §6.4], so $Q\,\partial^n_{P(\varepsilon,\infty)} = \partial^n_{P(\varepsilon,\infty)} Q + n\,\partial^{n-1}_{P(\varepsilon,\infty)}\Delta_\varepsilon$; summing the exponential series gives (3.4) for Q, and d_{CE} acts on the arguments of the cochains and commutes with $\partial_{P(\varepsilon,\infty)}$. □

Corollary 3.24 (Master equations at the two ends)**.** *For an even functional J with values in $C^\bullet(\mathfrak{psl}(4|4))[[\hbar]]$, at least cubic modulo $\hbar$ when the arguments of the cochains are counted as fields [CG2, §13.2.2],*

$$(d_{CE} + Q + \hbar\Delta_\varepsilon) e^{J/\hbar} = 0 \iff (d_{CE} + Q)\, e^{W(P(\varepsilon,\infty),J)/\hbar} = 0,$$

and the two sides are equivalent to $(d_{CE}+Q)J + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J = 0$ and $(d_{CE}+Q)\,W(P(\varepsilon,\infty),J) = 0$.

Proof. Apply (3.4) to $e^{J/\hbar}$; $e^{\hbar\partial_{P(\varepsilon,\infty)}}$ is invertible. This is [C07, Lemma 10.2.2] with the second scale at ∞. The product rule of Definition 3.21 gives $\Delta_\varepsilon e^{J/\hbar} = e^{J/\hbar}\big(\hbar^{-1}\Delta_\varepsilon J + \frac{1}{2}\hbar^{-2}\{J,J\}_\varepsilon\big)$. □

3.6 Acyclicity at scale infinity

For a kernel ϕ write $\phi\star$ for the operator it represents [C07, §6.3]; $P(\varepsilon,\infty)\star = \int_\varepsilon^\infty Q^{GF} e^{-tH} dt$, and $P(0,\infty)\star$ is the limit $\varepsilon \to 0$. A Schwartz section is moment-free when its Fourier transform vanishes to infinite order at $p = 0$.

Theorem 3.25 (Acyclicity at scale ∞)**.** *For every compactly supported α the integral $P(0,\infty)\star\alpha = \int_0^\infty Q^{GF} e^{-tH}\alpha\, dt$ converges to a smooth section, bounded with its first derivatives by a multiple of $(1+|x|)^{-2}$, and*

$$Q\big(P(0,\infty)\star\alpha\big) + P(0,\infty)\star(Q\alpha) = \alpha, \qquad \big(P(0,\infty)\star\big)^2\alpha = 0. \tag{3.5}$$

Moreover:

- *On moment-free Schwartz sections, Q, Q^{GF}, e^{-tH} and $P(0,\infty)\star$ act continuously, (3.5) holds, $QP(0,\infty)\star$ and $P(0,\infty)\star Q$ are complementary projections onto $\mathrm{Im}\,Q$ and $\mathrm{Im}\,Q^{GF}$, and $P(0,\infty)\star$ restricted to $\mathrm{Im}\,Q$ is the inverse of the isomorphism $Q\colon \mathrm{Im}\,Q^{GF} \to \mathrm{Im}\,Q$, the statement of [C07, Lemma 6.5.1] with $\mathrm{Ker}\,H = 0$.*

- *Zero-mode classes. Let e_0 be a section in the slot of c, of $A|_{W^\perp_{\mathbb{R}}}$ or of λ with polynomial components and $Qe_0 = 0$: a constant ghost $X \in \mathfrak{g}$, harmonic polynomial scalars, or a polynomial solution of $\not{d}e_0 = 0$. The linear functional $e \mapsto \langle e_0, e\rangle$ on compactly supported sections is Q-closed and Q-exact for no $e_0 \neq 0$, and it vanishes on moment-free sections; for X it is $-\int \mathrm{dvol}\,(c^*, X)$. The polynomial classes of $\mathscr{E} \oplus \mathscr{E}^{\bar{c}}$ are the images under Φ_Ψ^{-1} of the polynomial classes of $\mathscr{E}$.*

Proof. Compactly supported sections. Each component of $e^{-tH}\alpha$ is the convolution of a component of α with $(4\pi t)^{-2}e^{-|x|^2/4t}$, so $e^{-tH}\alpha$ and its first derivatives are bounded by $(4\pi t)^{-2}$ times $\int \mathrm{dvol}\,|\alpha|$ and $\int \mathrm{dvol}\,|d\alpha|$, and by the suprema of $|\alpha|$ and $|d\alpha|$ for $t \leq 1$. Since Q^{GF} has constant coefficients and order at most 1, $\sup_{x\in\mathbb{R}^4} |Q^{GF}e^{-tH}\alpha|$ is integrable on $(0,\infty)$; the same bounds for the derivatives allow Q to pass under the integral. At a point x with $\mathrm{dist}(x, \mathrm{supp}\,\alpha) \geq 1$, $|e^{-tH}\alpha(x)| \leq (4\pi t)^{-2}e^{-\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2/4t}\int \mathrm{dvol}\,|\alpha|$, and

$$\int_0^\infty (4\pi t)^{-2}e^{-\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2/4t}\,dt = \frac{1}{4\pi^2\,\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2},$$

which gives the bound by a multiple of $(1+|x|)^{-2}$; the first derivatives obey the same bound. Since $[Q,H]=0$, $Q^{GF}e^{-tH}Q\alpha = Q^{GF}Qe^{-tH}\alpha$, and

$$\begin{aligned} Q\,P(0,\infty)\star\alpha + P(0,\infty)\star Q\alpha &= \int_0^\infty (QQ^{GF} + Q^{GF}Q)e^{-tH}\alpha\,dt = \int_0^\infty He^{-tH}\alpha\,dt \\ &= -\int_0^\infty \frac{d}{dt}e^{-tH}\alpha\,dt = \alpha - \lim_{t\to\infty} e^{-tH}\alpha. \end{aligned}$$

The limit vanishes because $|e^{-tH}\alpha| \leq (4\pi t)^{-2}\int \mathrm{dvol}\,|\alpha|$; this is $K_\infty = 0$. Since Q^{GF} commutes with H, $(P(0,\infty)\star)^2\alpha = (Q^{GF})^2\big(\int_0^\infty e^{-tH}dt\big)^2\alpha = 0$.

Moment-free sections. In momentum space Q and Q^{GF} act by multiplication with the matrices of polynomials in p of Proposition 1.11 and Definition 3.5, e^{-tH} by $e^{-t|p|^2}$, and $P(0,\infty)\star$ by the matrix of Q^{GF} times $\int_0^\infty e^{-t|p|^2}dt = |p|^{-2}$. The first three preserve Schwartz functions of p that vanish to infinite order at $p=0$. For such a function h and every n, Taylor's formula at $p=0$ bounds every derivative $\partial^\beta h$ on $|p|\leq 1$ by $|p|^n$ times a multiple of the supremum of the derivatives of h of order $n+|\beta|$ on the unit ball, while every derivative of order k of $|p|^{-2}$ is bounded by a multiple of $|p|^{-2-k}$; by the Leibniz rule every derivative of order k of $|p|^{-2}h$ is bounded on $0<|p|\leq 1$ by $|p|^{n-2-k}$ times continuous seminorms of h. Hence $|p|^{-2}h$ extends smoothly by 0 at $p=0$ and vanishes there to infinite order; on $|p|\geq 1$ the derivatives of $|p|^{-2}$ are bounded, so rapid decay and the seminorm bounds persist. This proves stability and continuity. For $p\neq 0$ the matrices satisfy, with $P = Q^{GF}(p)|p|^{-2}$,

$$QP + PQ = H\,|p|^{-2} = 1, \qquad P^2 = (Q^{GF})^2|p|^{-4} = 0, \qquad PQP = P, \qquad QPQ = Q$$

by Proposition 3.6 and $(Q^{GF})^2 = 0$, and both sides of (3.5) vanish to infinite order at $p=0$; this gives (3.5) on moment-free Schwartz sections. Hence QP and PQ are complementary idempotents with images $\mathrm{Im}\,Q$ and $\mathrm{Im}\,Q^{GF}$, PQ is the identity on $\mathrm{Im}\,Q^{GF}$ and QP the identity on $\mathrm{Im}\,Q$, which is [C07, Lemma 6.5.1] with $\mathrm{Ker}\,H = 0$.

Zero-mode classes. The table of Proposition 3.4 gives $Qe_0 = -de_0$ on the ghost slot, $Qe_0 = -\frac{1}{2}d^*de_0$ on $A|_{W^\perp_{\mathbb{R}}}$ and $Qe_0 = \not{d}e_0$ on λ, which vanish exactly for the three kinds of e_0 listed. For $e \in \mathscr{E}_c$ the pairing $\langle e_0, e\rangle$ is defined, and integration by parts with compact support gives $\langle e_0, Qe\rangle = -(-1)^{|e_0|}\langle Qe_0, e\rangle = 0$, as in Proposition 1.11; so the functional is Q-closed. A Q-exact linear functional on $\mathscr{E}_c$ is, up to sign, $G\circ Q$ for a linear functional G. The operator Q vanishes on the slots of c^*, λ^* and $A^*|_{W^\perp_{\mathbb{R}}}$, so $G\circ Q$ is independent of the components of e in these slots, while $\langle e_0, e\rangle$ is the pairing of e_0 with the component of e in the partner slot of e_0, which is one of these three, and is a nonzero functional of it for $e_0 \neq 0$. Hence $\langle e_0, -\rangle$ is Q-exact for no $e_0 \neq 0$. Since e_0 has polynomial components, $\langle e_0, e\rangle$ is a finite combination

of moments of e; it extends continuously to Schwartz sections and vanishes on moment-free ones. For $e_0 = X$ in the ghost slot, Definition 1.5 gives $\langle X, e\rangle = -\langle e, X\rangle = -\int \mathrm{dvol}\,(e_{c^*}, X)$, with e_{c^*} the component of e in the slot of c^*. Polynomial sections of bounded degree form a finite-dimensional space. The listed e_0 lie in slots on which Φ_Ψ is the identity and Q agrees with the operator of Proposition 1.11 (Proposition 3.4), so the argument applies to them with the gauge-fixed Q. On polynomial sections Φ_Ψ intertwines Q with Q', the direct sum of the operator of Proposition 1.11 and of $\bar{c}^* \mapsto -\frac{1}{2}\bar{c}^*$, an isomorphism from the slot of $\bar{c}^*$ to that of $\bar{c}$; the second summand has zero cohomology, which gives the last statement. □

The derivation Q of [C07, §6.2] acts on an n-linear graded-symmetric functional F of parity $|F|$ by $QF = (-1)^{|F|}F \circ \big(\sum_{j=1}^{n} 1^{\otimes(j-1)} \otimes Q \otimes 1^{\otimes(n-j)}\big)$, with the Koszul rule for odd operators on tensor factors.

Corollary 3.26 (Functionals at scale ∞)**.** *Let F be a continuous graded-symmetric n-linear functional, $n \geq 1$, on moment-free Schwartz sections with $F \circ \big(\sum_j 1^{\otimes(j-1)} \otimes Q \otimes 1^{\otimes(n-j)}\big) = 0$. Then*

$$F = \Big(F \circ \frac{1}{n}\sum_{i=1}^{n} 1^{\otimes(i-1)} \otimes P(0,\infty)\star \otimes 1^{\otimes(n-i)}\Big) \circ \Big(\sum_{j=1}^{n} 1^{\otimes(j-1)} \otimes Q \otimes 1^{\otimes(n-j)}\Big),$$

and F is $(-1)^{|F|+1}$ times Q of a continuous graded-symmetric functional of parity $|F|+1$; the complex of continuous polynomial functionals without constant term on moment-free Schwartz sections is acyclic for Q.

Proof. By Theorem 3.25, $P(0,\infty)\star$ acts continuously on moment-free Schwartz sections, so the composite functional is continuous; averaging over the n factors keeps it graded symmetric, and $P(0,\infty)\star$ is odd, which gives the parity. Odd operators acting on different tensor factors anticommute, so the graded commutator of Q on the j-th factor with $P(0,\infty)\star$ on the i-th factor vanishes for $i \neq j$, and for $i = j$ it is the identity of the i-th factor by (3.5). Summing over i, j and dividing by n, the graded commutator of the tensor-product differential with the averaged operator $\frac{1}{n}\sum_i 1^{\otimes(i-1)} \otimes P(0,\infty)\star \otimes 1^{\otimes(n-i)}$ is $\frac{1}{n}\sum_{i=1}^{n} \mathrm{id} = \mathrm{id}$. Composing with F and using $F \circ \big(\sum_j 1^{\otimes(j-1)} \otimes Q \otimes 1^{\otimes(n-j)}\big) = 0$ gives the displayed identity, and the action of Q on functionals of parity $|F|+1$ gives the last statement. □

Remark 3.27 (Where the content of an operator lies)**.** At scale ∞ the differential on functionals is $d_{CE} + Q$. On moment-free Schwartz sections every Q-closed functional without constant term is Q-exact (Corollary 3.26); on compactly supported sections Q has the zero-mode classes of Theorem 3.25 and, in the slot of $A|_V$, the classes $e \mapsto \langle e_0, e\rangle$ of the one-forms $e_0 = (a,x)(b,-)$ with $(a,b) = 0$, for which $Qe_0 = 0$ (here $d^*e_0 = 0$, so $\Phi_\Psi e_0 = e_0$) and $de_0 = (a,-) \wedge (b,-) \neq 0$. The content of an operator is the local functional $O(\varepsilon)$ at the UV end whose flow to scale ∞ has an $\varepsilon \to 0$ limit; the singular part of $O(\varepsilon)$ is fixed by this requirement and the finite part is the scheme freedom (Chapter 5).

3.7 The superconformal action on the antighost

Definition 3.28 (Equivariant extension)**.** For a conformal Killing vector field v of V (Definition 2.1) put $v.\bar{c} = v(\bar{c}) + \frac{1}{2}(\mathrm{div}\, v)\,\bar{c}$ and $v.\bar{c}^* = v(\bar{c}^*) + \frac{1}{2}(\mathrm{div}\, v)\,\bar{c}^*$, the action on fields of Weyl weight 2. In the Costello frame,

$$S^{(1)\prime}(v) = S^{(1)}(v) - \int \mathrm{dvol}\,(v.\bar{c}, \bar{c}^*), \qquad S^{(1)\prime}(x) = S^{(1)}(x) \text{ for } x \in \mathfrak{so}(W_{\mathbb{R}}^{\perp}) \oplus \Sigma \oplus \Sigma^*,$$

$$S^{(2)\prime}(x_1, x_2) = S^{(2)}(x_1, x_2) - \int \mathrm{dvol}\,\big(\bar{c},\, u_{12}(\bar{c})\big), \qquad u_{12} = \big(x \mapsto \Gamma(\epsilon_1(x), \epsilon_2(x))|_V\big),$$

for odd x_1, x_2 with conformal Killing spinors ϵ_1, ϵ_2 (Definition 2.2), $u_{12}(\bar c)$ the derivative of $\bar c$ along the conformal Killing field u_{12}, $S^{(2)\prime} = 0$ when an argument is even, and $S^{(k)\prime} = 0$ for $k \geq 3$. In the gauge-fixed frame $S^{(k)}_\Psi := S^{(k)\prime} \circ \Phi_\Psi$:

$$S^{(1)}_\Psi(v) = S^{(1)}(v) \circ j_\Psi - \int \mathrm{dvol}\,\big(v.\bar c,\, \bar c^* + d^*A|_V\big), \qquad S^{(1)}_\Psi(x) = S^{(1)}(x) \circ j_\Psi \quad (x \in \mathfrak{so}(W^\perp_{\mathbb{R}}) \oplus \Sigma \oplus \Sigma^*),$$
$$S^{(2)}_\Psi(x_1, x_2) = S^{(2)}(x_1, x_2) \circ j_\Psi - \int \mathrm{dvol}\,\big(\bar c,\, u_{12}(\bar c)\big). \tag{3.6}$$

Theorem 3.29 (Master equation of the extended equivariant action)**.** *For every Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric pairing, $S' + S^{(1)\prime} + S^{(2)\prime}$ and $S_\Psi + S^{(1)}_\Psi + S^{(2)}_\Psi$ satisfy the equivariant master equation* (2.4)*. The coefficient -1 of $\int(\bar c, u_{12}(\bar c))$ is the unique value for which the arity-two component on a pair of odd elements with $u_{12} \neq 0$ holds.*

Proof. Write $S' + S^{(1)\prime} + S^{(2)\prime} = S_{\mathfrak{psl}(4|4)} + S^{\bar c}$, with $S_{\mathfrak{psl}(4|4)}$ of Theorem 2.20 and $S^{\bar c}$ the terms in $\bar c$, $\bar c^*$. The two summands involve disjoint pairs of fields, so their bracket vanishes and (2.4) splits; the first summand satisfies it by Theorem 2.20. For $S^{\bar c}$, the bracket with $-\frac14\int(\bar c^*, \bar c^*)$ acts on functionals as the derivation induced by $\bar c \mapsto -\frac12 \bar c^*$ (Proposition 1.11), and the bracket with $-\int(v.\bar c, \bar c^*)$ as the derivation induced by $v.$ on $\bar c$, $\bar c^*$ (as for c in Lemma 9.16 (i)). Arity one: $\{-\frac14\int(\bar c^*, \bar c^*), -\int(v.\bar c, \bar c^*)\} = \frac12\int(v.\bar c^*, \bar c^*) = \frac14\int\big(v(\bar c^*, \bar c^*) + (\operatorname{div} v)(\bar c^*, \bar c^*)\big) = 0$. Arity two, two even arguments: $v \mapsto v.$ is a representation of the conformal Killing fields, $[v., w.] = [v, w].$, the divergence terms forming the cocycle of the weight. Arity two, two odd arguments: $\{-\frac14\int(\bar c^*, \bar c^*), -\int(\bar c, u(\bar c))\} = \frac12\int(\bar c^*, u(\bar c)) - \frac12\int(\bar c, u(\bar c^*)) = \int(u.\bar c, \bar c^*)$ after integration by parts, which cancels $S^{(1)\prime}([x_1, x_2])$ restricted to $\bar c$, $\bar c^*$, the rotation of $W^\perp_{\mathbb{R}}$ in $[x_1, x_2]$ acting trivially on $\bar c$; the coefficient $+1$ in place of -1 doubles this term. Arity three, one even and two odd arguments: $\{-\int(v.\bar c, \bar c^*), -\int(\bar c, u(\bar c))\} = \int(\bar c, [v, u](\bar c))$ up to a total derivative, and $u_{[v,x_2],x_3} + u_{x_2,[v,x_3]} = [v, u_{23}]$ by the super-Jacobi identity; with three odd arguments $S^{(1)\prime}$ has no term in $\bar c$, $\bar c^*$. Arity four: $S^{(2)\prime}$ contains no $\bar c^*$, so $\{S^{(2)\prime}, S^{(2)\prime}\} = 0$. Finally Φ_Ψ preserves the pairing and involves no generator of $\mathfrak{psl}(4|4)$, so $F \mapsto F \circ \Phi_\Psi$ preserves the bracket and commutes with d_{CE}. The computer-assisted verification is Remark 3.38. □

For a quadratic functional F, $F.$ denotes the linear operator with $\{F, \ell\} = \ell \circ F.$ for every coordinate functional ℓ; for $F = \frac12\langle e, Qe\rangle$ it is Q. For a generator x, $x.$ is the operator of the quadratic part of $S^{(1)}_\Psi(x)$; for even x it is $\Phi^{-1}_\Psi \circ v. \circ \Phi_\Psi$, with $v.$ of Definition 7.4 and of Definition 3.28. The part of $S^{(1)}_\Psi(x)$ without antifields is written $s_x\Psi$.

Proposition 3.30 (The variations of the gauge fermion)**.**

- *(i) For a conformal Killing field v,*

$$s_v\Psi = -\int \mathrm{dvol}\,\big((v.A|_V, d\bar c) + (v.\bar c, d^*A|_V)\big)$$

is the variation of $-\Psi$ along $v.$, and

$$s_v\Psi = -\tfrac12\int \mathrm{dvol}\,\big(\bar c,\, \iota_{\operatorname{grad}(\operatorname{div} v)}A|_V\big).$$

It vanishes for translations, rotations of V and the dilatation; for the special conformal field of v_1, $\operatorname{div} v = -8(v_1, x)$ and $s_v\Psi = 4\int \mathrm{dvol}\,(\bar c, \iota_{v_1}A|_V)$. For $x \in \mathfrak{so}(W^\perp_{\mathbb{R}})$, $s_x\Psi = 0$.

- *(ii) For an odd element with conformal Killing spinor ϵ, $s_\epsilon\Psi = \int \mathrm{dvol}\,\big(\Gamma(\epsilon, \lambda)|_V, d\bar c\big)$, the image of $-\int(\Gamma(\epsilon, \lambda), A^*)$ under $A^* \mapsto A^* - d\bar c$; its operator sends $\lambda \mapsto -d^*\big(\Gamma(\epsilon, \lambda)|_V\big)$ in the slot of $\bar c^*$ and $\bar c \mapsto \rho(d\bar c)\epsilon$ in the slot of λ^*.*

- *(iii) The arity-two vertex of the antighost is* $-\int \mathrm{dvol}\,\big(\bar c, u_{12}(\bar c)\big)$*; for* $x_1, x_2 \in \Sigma$ *it is*

$$-\int \mathrm{dvol}\,\big(\bar c, \iota_{\Gamma(x_1,x_2)|_V} d\bar c\big),$$

and it vanishes when $\Gamma(\epsilon_1,\epsilon_2)|_V = 0$*, in particular on* (ξ,ξ) *for* $\xi \in S_+ \otimes W$ *of rank one.*

Proof. (i) By (3.6) the antifield-free part of $S^{(1)}_\Psi(v)$ comes from $(v.A, A^* - d\bar c)$ and $-(v.\bar c, \bar c^* + d^*A|_V)$. For a conformal Killing field, $d^*(L_vA|_V) = v(d^*A|_V) + \frac12(\mathrm{div}\,v)d^*A|_V + \frac12\iota_{\mathrm{grad}(\mathrm{div}\,v)}A|_V$ (proof of Proposition 4.35), and $\int(v.\bar c, d^*A|_V) = -\int\big(\bar c, v(d^*A|_V) + \frac12(\mathrm{div}\,v)d^*A|_V\big)$ for the weight 2; the sum of $\int(L_vA|_V, d\bar c) = \int(d^*L_vA|_V, \bar c)$ and this term is $\frac12\int(\bar c, \iota_{\mathrm{grad}(\mathrm{div}\,v)}A|_V)$. For $v = (x,x)v_1 - 2(v_1,x)x$, $\mathrm{grad}(\mathrm{div}\,v) = -8v_1$. (ii) $S^{(1)}(\epsilon)$ contains $-(\Gamma(\epsilon,\lambda), A^*)$, and $A^* \mapsto A^* - d\bar c$ gives $(\Gamma(\epsilon,\lambda)|_V, d\bar c)$; the operator follows from the definition of F. (Remark 3.38). (iii) (3.6); $\Gamma(\xi,\xi) = 0$ for $\xi = u \otimes z$ (Lemma 8.21). □

Proposition 3.31 (The generators on the gauge fixing)**.** *For every generator* x*:* $[x., Q] = 0$*, and* $[x., Q^{GF}]$*,* $(P(\varepsilon,T)\star)^2$ *and the line kernel* $P(\varepsilon,T)\star\, x.\, P(\varepsilon,T)\star$ *are as follows, with* $G = \int_\varepsilon^T e^{-tH}dt$*.*

- *(i) For every* x*,* $(P(\varepsilon,T)\star)^2 = (Q^{GF})^2G^2 = 0$ *and*

$$P(\varepsilon,T)\star\, x.\, P(\varepsilon,T)\star = P(\varepsilon,T)\star\,[x., Q^{GF}]\,G = P(\varepsilon,T)\star\,(s_x\Psi).\,P(\varepsilon,T)\star\,.$$

- *(ii) Translations, rotations of* V*,* $\mathfrak{so}(W^\perp_{\mathbb{R}})$*:* $[x., Q^{GF}] = 0$*,* $[x., H] = 0$*, and the line kernel vanishes.*
- *(iii) Dilatation:* $[x., Q^{GF}] = -4Q^{GF}$ *and* $[x., H] = -4H$ *on every slot, and the line kernel vanishes.*
- *(iv) Conformal Killing field* v*:*

$$[v., Q^{GF}] = -\tfrac12(\mathrm{div}\,v)\,Q^{GF} + \tfrac14\rho\big(d(\mathrm{div}\,v)\big)\big|_{\lambda^*\to\lambda} + \big[(s_v\Psi)., Q^{GF}\big];$$

for the special conformal field of v_1 *the last term sends* $A \mapsto -4\iota_{v_1}A$ *(slot of* c*),* $\bar c \mapsto -8(v_1,-)\bar c$ *(slot of* $A|_V$*),* $A^*|_V \mapsto -8\iota_{v_1}A^*$ *(slot of* $\bar c^*$*) and* $c^* \mapsto 4(v_1,-)c^*$ *(slot of* $A^*|_V$*), and the line kernel joins a leg* $A|_V$ *to a leg* c *through the vertex* $4\int(\bar c, \iota_{v_1}A|_V)$*.*

- *(v) Odd* x *with spinor* ϵ*:* $[x., Q^{GF}]$ *is the sum of the commutator of Proposition 7.11, with the components* $A^* \to \lambda$ *and* $\lambda^* \to A$*, and of* $[(s_\epsilon\Psi)., Q^{GF}]$*, with the components* $\lambda \to c$*,* $\bar c \to \lambda$*,* $\lambda^* \to \bar c^*$*,* $c^* \to \lambda^*$*; the line kernel joins a leg* λ *to a leg* c *through the vertex* $s_\epsilon\Psi$*. It is nonzero.*

Proof. $[x., Q] = 0$ is the quadratic part of the arity-one equation of Theorem 3.29. (i) G commutes with Q^{GF}, so $Q^{GF}G\,Q^{GF} = 0$; writing $x.\,Q^{GF} = [x., Q^{GF}] + (-1)^{|x|}Q^{GF}x.$ gives the first identity. In the gauge-fixed frame $x. = \Phi_\Psi^{-1}v.\Phi_\Psi = v. + [v., N]$ for even x, with $N = \Phi_\Psi - 1$, and $[v., N] = (s_v\Psi).$ maps field slots to antifield slots; for odd x likewise. The part of $x.$ preserving field and antifield slots maps $\mathrm{Im}\,Q^{GF}$, the field slots, to itself, and $Q^{GF}G$ annihilates them; this gives the second identity. (ii), (iii): these $x.$ preserve the field slots, $s_x\Psi = 0$ by Proposition 3.30, and on each component of Q^{GF} the weights of Lemma 7.5 and Definition 3.28 give, for $v = 2E$, $(2E+2)(-2) + 2(2E+6) = 8$ on $A^* \mapsto -2A^*$, $2E - (2E+4) = -4$ on $\bar c^* \mapsto \bar c^*$, $(2E+4)(-1) + (2E+8) = 4 = -4\cdot(-1)$ on $c^* \mapsto -c^*$, and -4 on $\lambda^* \mapsto -\not d|_{\Sigma^*}\lambda^*$ as in Proposition 7.10. (iv) On the components of Q^{GF} the weights give $-\frac12\,\mathrm{div}\,v$ for $A^* \to A$, $\bar c^* \to c$, $c^* \to \bar c$, and on $\lambda^* \to \lambda$ the conformal covariance of the Dirac operator from weight $\frac32$ to $\frac52$ leaves $\frac12(\mathrm{div}\,v)\not d|_{\Sigma^*} + \frac14\rho(d(\mathrm{div}\,v))$; the operator of $4\int(\bar c, \iota_{v_1}A|_V)$ is $\bar c \mapsto -4(v_1,-)\bar c$ in $A^*|_V$ and $A \mapsto 4\iota_{v_1}A$ in $\bar c^*$, and its commutator with Q^{GF} has the listed components. (v) Proposition 3.30 (ii) and the same computation. The verification on generic elements is Remark 3.38. □

Proposition 3.32 (Two descriptions of the same graphs)**.** *Let $P_C(\varepsilon,T) = (\Phi_\Psi \otimes \Phi_\Psi)P(\varepsilon,T)$, the propagator of Q_C^{GF}. For every functional J on $\mathscr{E} \oplus \mathscr{E}^{\bar c}$,*

$$W\big(P_C(\varepsilon,T), J\big) \circ \Phi_\Psi = W\big(P(\varepsilon,T), J \circ \Phi_\Psi\big).$$

In the Costello frame the vertices are S', $S^{(1)\prime}$, $S^{(2)\prime}$, and a vertex $S^{(1)\prime}(x)$ on a line contributes $P_C \star [x., Q_C^{GF}]\, G$, the variation kernel of the gauge fixing along x; in the gauge-fixed frame the same line is $P\star(s_x\Psi).P\star$, the vertex $s_x\Psi$ joined to two edges $P(\varepsilon,T)$. The two contributions agree, $P_C\star[x., Q_C^{GF}]\, G = \Phi_\Psi\big(P \star (s_x\Psi).P \star\big)\Phi_\Psi^{-1}$.

Proof. The first identity is Theorem 3.12 (iii) with Φ_Ψ in place of j_Ψ, Φ_Ψ being invertible. The line kernels: Proposition 3.31 (i) in both frames and $Q_C^{GF} = \Phi_\Psi Q^{GF} \Phi_\Psi^{-1}$, $x.' = \Phi_\Psi x.\Phi_\Psi^{-1}$, G commuting with Φ_Ψ. □

Proposition 3.33 (Supersymmetry and special conformal transformations move every gauge fixing)**.** *Let $\Psi' = \int \mathrm{dvol}\,(\bar c, \mathcal{G})$ with $\mathcal{G}$ a local, $\mathfrak{g}$-equivariant, $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$-invariant function of the fields of Weyl weight 2 and ghost number 0 whose linear part fixes the gauge, i.e. pairs nontrivially with dc. Then the linear part of $\mathcal{G}$ is $\kappa\, d^*A|_V$, $\kappa \neq 0$, and for every action of Σ, Σ^* and the special conformal fields on $\bar c$ the part of $s_x\Psi'$ bilinear in $\bar c$ and the fields is nonzero.*

Proof. A Lorentz scalar of weight 2 linear in the fields is built from $A|_V$ with one derivative, $d^*A|_V$; a term linear in λ needs a constant spinor, a term linear in $A|_{W_{\mathbb{R}}^\perp}$ an invariant vector of $W_{\mathbb{R}}^\perp$, and c has ghost number 1. The variations of $\bar c$ of the right weight and ghost number are at least quadratic in the fields for odd x, and for the special conformal field they are $v(\bar c) + \frac{w}{4}(\mathrm{div}\, v)\bar c$. Hence the bilinear part of $s_x\Psi'$ is $-\kappa \int (\bar c, d^*\Gamma(\epsilon,\lambda)|_V) \neq 0$ for odd x, and $-\kappa \int \big(\bar c, \frac{1}{2}\iota_{\mathrm{grad}(\mathrm{div}\, v)} A|_V + \frac{w-2}{4}(\mathrm{div}\, v)\, d^*A|_V\big) \neq 0$ for the special conformal field, as in Proposition 3.30. □

Remark 3.34 (The gauge parameter)**.** With $-\frac14 + \frac12 u$ in place of $-\frac14$, S_Ψ changes by $u\, G_\alpha$, $G_\alpha = \frac12 \int (\bar c^* + d^*A|_V, \bar c^* + d^*A|_V)$, and

$$G_\alpha = -\big(Q + \{I_\Psi, -\}\big) \int \mathrm{dvol}\,\big(\bar c^* + d^*A|_V,\, \bar c\big),$$

the image under Φ_Ψ of $\{\int (\bar c^*, \bar c), -\frac14 \int (\bar c^*, \bar c^*)\} = -\frac12 \int (\bar c^*, \bar c^*)$, the count of $\bar c^*$ minus $\bar c$ (Lemma 4.21). Treating $u\, G_\alpha$ as a formal quadratic vertex, with Q, Q^{GF} and H those of $u = 0$, the u-derivative of a family of one-loop quantisations is s-closed with the exact leading part above, so the cohomology of $\Lambda^{(0)}$ and the induced action of $\mathfrak{psl}(4|4)$ change by an isomorphism, up to the coupling and outer directions, which leave the spectrum of the dilatation and the twisted classes unchanged (Theorems 8.20 (iii), 8.23). At $u \neq 0$, with Q^{GF} fixed, H acquires the non-scalar symbol $|p|^2 - 2u\, p\, \iota_p$ on $A|_V$, and the axioms single out $u = 0$.

3.8 The graph expansion and the one-loop formula

Proposition 3.35 (Graph expansion)**.** *For $J = \sum_{i\geq 0} \hbar^i J_i$ as in Definition 3.22,*

$$W(P(\varepsilon,\infty), J) = \sum_\gamma \hbar^{b_1(\gamma) + \sum_v \deg v} \frac{1}{|\operatorname{Aut}\gamma|}\, w_\gamma \qquad [C07,\ §15.4],$$

the sum over connected graphs γ with legs, with vertex set $V(\gamma)$, edge set $E(\gamma)$ and first Betti number $b_1(\gamma) = |E(\gamma)| - |V(\gamma)| + 1$, each vertex v labelled by a Taylor component of $J_{\deg v}$ (vertices of degree 0 at least trivalent), each edge by $P(\varepsilon,\infty)$, each leg by an argument; $|\operatorname{Aut}\gamma|$ counts the automorphisms fixing the legs. Two contracted coordinate functionals, brought adjacent by Koszul reordering, are replaced by the component of $P(\varepsilon,\infty)$ in the same order. At order $\hbar^1$ either $b_1(\gamma) = 1$ and all vertices have degree 0, or $b_1(\gamma) = 0$ and exactly one vertex has degree 1.

Proof. Write $\phi = P(\varepsilon,\infty) = \sum \phi' \otimes \phi''$. Expanding the exponentials, a term of $e^{\hbar\partial_\phi} e^{J/\hbar}$ is a set of vertices, each a factor $\hbar^{\deg v - 1} J_{\deg v}$, with $|E(\gamma)|$ pairs of half-edges contracted by $\hbar\partial_\phi$; the factor $\frac{1}{2}$ in $\partial_\phi = \frac{1}{2}\sum \partial_{\phi''}\partial_{\phi'}$ [C07, §6.4] and the factorials of the exponentials combine to $1/|\operatorname{Aut}\gamma|$, and on two coordinate functionals ∂_ϕ returns the component of $P(\varepsilon,\infty)$ in their order. The logarithm keeps the connected graphs, and the overall factor $\hbar$ gives the power

$$\hbar^{1+|E(\gamma)|-|V(\gamma)|+\sum_v \deg v} = \hbar^{b_1(\gamma)+\sum_v \deg v}.$$

The last statement is $b_1(\gamma) + \sum_v \deg v = 1$. □

For the one-loop formula let $\mathscr{E}$ be a finite-dimensional super vector space, as when the manifold of [C07, §6.6, §10.1] is a point, and let $\phi = \sum \phi' \otimes \phi'' \in \operatorname{Sym}^2 \mathscr{E}$ be even with complex coefficients [C07, §6.4]. Let J be a quadratic functional on $\mathscr{E}$, even as a whole, whose coefficients lie in a supercommutative algebra: the complex numbers, or an auxiliary nilpotent ring with odd elements as in [C07, Lemma 6.4.1, proof]. The one-loop operator of (ϕ, J) is the linear map

$$e \mapsto \sum \phi' \, (-1)^{|\phi''|} \, \partial_{\phi''}\partial_e J,$$

i.e. ϕ composed with the parity operator $(-1)^{|\cdot|}$ composed with the second derivative of J, the right-hand slot differentiated first, with the coefficients written to the right and extended linearly over them. STr is the supertrace on such linear maps [CG2, §13.7.2]: on the map $e \mapsto e_0\,\ell(e)$, with $e_0 \in \mathscr{E}$ homogeneous and ℓ a linear functional with values in the coefficients, it is $(-1)^{|e_0|}\ell(e_0)$.

Proposition 3.36 (One-loop formula)**.** *As formal power series in J,*

$$\hbar \log\left(e^{\hbar\partial_\phi} e^{J/\hbar}\right)\Big|_{e=0} = \hbar \sum_{n\geq 1} \frac{1}{2n} \operatorname{STr}\big((\textit{one-loop operator})^n\big) = -\frac{\hbar}{2} \operatorname{STr} \log\big(1 - \textit{one-loop operator}\big). \quad (3.7)$$

Proof. Two identities for an even quadratic functional F with coefficients as above. The second derivative $\partial_e\partial_{e'}F$ is a coefficient of parity $|e| + |e'|$ with $\partial_e\partial_{e'}F = (-1)^{|e||e'|}\partial_{e'}\partial_e F$; since ϕ is even, $|\phi'| = |\phi''|$ in every term.

First, $\frac{1}{2}$STr of the one-loop operator of F is $\partial_\phi F$: by the rank-one formula,

$$\tfrac{1}{2}\operatorname{STr} = \tfrac{1}{2}\sum (-1)^{|\phi'|+|\phi''|}\,\partial_{\phi''}\partial_{\phi'}F = \tfrac{1}{2}\sum \partial_{\phi''}\partial_{\phi'}F = \partial_\phi F.$$

Second, since ∂_ϕ is of order two, $\partial_\phi(F_1F_2) - (\partial_\phi F_1)F_2 - F_1(\partial_\phi F_2)$ is the sum of the two terms with one derivative on each factor; for F_1 even, the graded symmetry $\sum \phi' \otimes \phi'' = \sum (-1)^{|\phi'||\phi''|}\phi'' \otimes \phi'$ makes the two terms equal, so this sum is $\sum (\partial_{\phi''}F_1)(\partial_{\phi'}F_2)$. It is a derivation in each argument; hence, by induction on m,

$$\partial_\phi(F^m) = m\,(\partial_\phi F)\,F^{m-1} + \tfrac{1}{2}m(m-1)\,F^{m-2}\sum (\partial_{\phi''}F)(\partial_{\phi'}F),$$

$$\partial_\phi e^{F/\hbar} = e^{F/\hbar}\Big(\hbar^{-1}\partial_\phi F + \tfrac{1}{2}\hbar^{-2}\sum (\partial_{\phi''}F)(\partial_{\phi'}F)\Big).$$

The one-loop operator of the quadratic functional $\frac{1}{2}\sum (\partial_{\phi''}F)(\partial_{\phi'}F)$ is the square of the one-loop operator of F. Indeed, the Leibniz rule for the linear functionals $\partial_{\phi''}F$ and $\partial_{\phi'}F$ gives two terms for the second derivative, which coincide after the graded symmetry of ϕ and the exchange of the two coefficients, so that

$$\partial_e\partial_{e'}\Big(\tfrac{1}{2}\sum (\partial_{\phi''}F)(\partial_{\phi'}F)\Big) = \sum (-1)^{|e'||\phi''|}\,(\partial_e\partial_{\phi''}F)\,(\partial_{e'}\partial_{\phi'}F).$$

Writing $\phi = \sum_k \phi'_k \otimes \phi''_k = \sum_l \phi'_l \otimes \phi''_l$ for two copies of the same sum, the one-loop operator of this functional sends e to

$$\sum_{l,k} \phi'_l (-1)^{|\phi''_l|}(-1)^{|e||\phi''_k|}(\partial_{\phi''_l}\partial_{\phi''_k}F)(\partial_e\partial_{\phi'_k}F) = \sum_{l,k} \phi'_l (-1)^{|\phi''_l|}(\partial_{\phi''_l}\partial_{\phi'_k}F)(-1)^{|\phi''_k|}(\partial_{\phi''_k}\partial_e F),$$

where the graded symmetry of ϕ in the index k contributes $(-1)^{|\phi''_k|}$ and $\partial_e\partial_{\phi''_k}F = (-1)^{|e||\phi''_k|}\partial_{\phi''_k}\partial_e F$; the right side is the one-loop operator of F applied twice to e.

The flow in an auxiliary formal parameter s. Let J_s be the quadratic functional determined, as a formal power series in s and in J, by

$$\frac{d}{ds}J_s = \tfrac12 \sum (\partial_{\phi''}J_s)(\partial_{\phi'}J_s), \qquad J_0 = J.$$

By the formula for $\partial_\phi e^{F/\hbar}$ with $F = J_s$, the functional $\exp\big(\int_0^s \partial_\phi J_{s'}\,ds'\big)e^{J_s/\hbar}$ satisfies $\frac{d}{ds}(\cdots) = \hbar\partial_\phi(\cdots)$, as does $e^{s\hbar\partial_\phi}e^{J/\hbar}$; both equal $e^{J/\hbar}$ at $s=0$, and this equation determines the coefficient of each power of s from the previous one, so they coincide. At order m in J the powers s^k that occur have $k \le m$, because ∂_ϕ lowers the polynomial degree by two and J^m has degree $2m$; so $s=1$ may be substituted. At the origin $e^{J_1/\hbar}$ equals 1, hence

$$\hbar \log\big(e^{\hbar\partial_\phi}e^{J/\hbar}\big)\Big|_{e=0} = \hbar\int_0^1 \partial_\phi J_s\,ds.$$

The one-loop operator of J_s depends linearly on J_s, so by the second identity its derivative in s is its square; its value at $s=0$ is the one-loop operator of J, and the unique formal solution is $\sum_{n\ge1} s^{n-1}(\text{one-loop operator of } J)^n$. By the first identity,

$$\partial_\phi J_s = \tfrac12 \sum_{n\ge1} s^{n-1}\,\mathrm{STr}\big((\text{one-loop operator})^n\big),$$

$$\hbar\int_0^1 \partial_\phi J_s\,ds = \hbar\sum_{n\ge1}\frac{1}{2n}\,\mathrm{STr}\big((\text{one-loop operator})^n\big),$$

which is (3.7), with $\mathrm{STr}\log(1-\text{one-loop operator}) = -\sum_{n\ge1}\frac1n \mathrm{STr}\big((\text{one-loop operator})^n\big)$. The term $n=1$ is $\hbar\,\partial_\phi J$ by the first identity. In the graph expansion with vertices J the graphs without legs and with bivalent vertices are the cycles, and a cycle with n vertices has $2n$ automorphisms. □

Remark 3.37 (Use in the field theory)**.** For the one-loop part of $W(P(\varepsilon,\infty), I_\Psi)$ at a background, J is the part of I_Ψ quadratic in the fluctuation and $\phi = P(\varepsilon,\infty)$; the supertrace produces the signs of the loops of λ and of the ghost line c–$\bar c$. By Corollary 3.13, for vertices built from the fields of Chapter 1 the ghost line c–$\bar c$ with the vertex $([A,c], -d\bar c)$ gives the value of the line c–$A^*|_V$ of $(j_\Psi \otimes j_\Psi)P(\varepsilon,\infty)$ with the vertex $([A,c], A^*)$.

Remark 3.38 (Exact verification)**.** The computer-assisted statements of this chapter are exact. In momentum space Q, Q', Q^{GF}, Φ_Ψ, j_Ψ and K_0 are matrices of polynomials in p over $\mathbb{Q}(i)$ in the realisation of Chapter 2, and every identity of Propositions 3.4, 3.6, 3.8, 3.17, Theorems 3.7, 3.10, 3.12 is a matrix identity for symbolic p; the ranks of (5) are computed at a rational p. The local functionals of Theorem 3.29 and Propositions 3.30, 3.31, and the bracket of Definition 1.6, are evaluated on jets with Euler–Lagrange derivatives as in Remark 1.15: the linear differential of S' and of S_Ψ agrees with Q' and Q on every coordinate functional; the components of (2.4) of arities one and two for the abelian theory, on translations, rotations, the dilatation, two special conformal fields, $\mathfrak{so}(W^\perp_{\mathbb{R}})$ and generic elements of Σ and Σ^* (all pairs of types), vanish in both frames; with the coefficient 0 in place of -1 in $S^{(2)\prime}$ the four components on pairs of odd elements are nonzero; the sector of $\bar c$, $\bar c^*$ satisfies the components of arities one to four for every pair of conformal Killing fields; and $\{S_\Psi, S_\Psi\} = 0$ for $\mathfrak{g} = \mathfrak{su}(2)$ in the gauge and ghost sector, with $A^* - d\bar c$ replaced by $A^* + d\bar c$ or $\bar c^* + d^*A|_V$ by $\bar c^* - d^*A|_V$ giving nonzero brackets.

Remark 3.39 (Two-component form)**.** In the components of Definition 8.1, $d^*A|_V = -\partial_m A_m$, $\partial^2 = \partial_m\partial_m$, and with $(Qe)_X$ the component of Qe in the slot of X:

$(Qe)_{A_m} = -\partial_m c$, $\quad (Qe)_{\bar c^*} = -\partial^2 c$, $\quad (Qe)_{A^*_m} = \tfrac12\partial^2 A_m - \tfrac12\partial_m\bar c^*$, $\quad (Qe)_{A^*_{ij}} = \tfrac12\partial^2 A_{ij}$,
$(Qe)_{\bar c} = \tfrac12\partial_m A_m - \tfrac12\bar c^*$, $\quad (Qe)_{c^*} = -\partial_m A^*_m + \partial^2\bar c$, $\quad (Qe)_{\bar\lambda^{*\dot\alpha i}} = -i\,\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i$, $\quad (Qe)_{\lambda^{*\alpha}{}_i} = i\,\partial_m\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}$,
$(Q^{GF}e)_{A_m} = -2A^*_m$, $\quad (Q^{GF}e)_{A_{ij}} = -2A^*_{ij}$, $\quad (Q^{GF}e)_c = \bar c^*$, $\quad (Q^{GF}e)_{\bar c} = -c^*$,
$(Q^{GF}e)_{\bar\lambda_{\dot\alpha i}} = -i\,\partial_m\lambda^{*\alpha}{}_i\sigma^m_{\alpha\dot\alpha}$, $\quad (Q^{GF}e)_{\lambda_\alpha{}^i} = i\,\sigma^m_{\alpha\dot\alpha}\partial_m\bar\lambda^{*\dot\alpha i}$.

The gauge-fixed action (3.1) contains

$$S_\Psi \supset \int d^4x\Big\{ -\tfrac18(F_{mn},F_{mn}) - \tfrac14(\partial_m A_m,\partial_n A_n) - (D_m c,\partial_m\bar c) + (D_m c, A^*_m) - \tfrac14(\bar c^*,\bar c^*) + \tfrac12(\bar c^*,\partial_m A_m)\Big\},$$

with $D_m = \partial_m + [A_m,-]$; for $\mathfrak{su}(N)$ with $(X,Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY)$, $A_m = -igA^a_m T^a$, $c = -igc^aT^a$, $\bar c = -ig\bar c^aT^a$, $\operatorname{tr}(T^aT^b) = \tfrac12\delta^{ab}$, the first three terms of $S_\Psi/\hbar$ are $-\tfrac14F^a_{mn}F^a_{mn} - \tfrac12(\partial_m A^a_m)^2 - 2(D_m c)^a\partial_m\bar c^a$: the first two are the Wick image of Srednicki's $-\tfrac14F^{a\mu\nu}F^a_{\mu\nu} - \tfrac12(\partial^\mu A^a_\mu)^2$ in Feynman gauge, $\xi = 1$, and the third is the ghost term in this normalisation of c and $\bar c$. The propagator components, each times $e^{-\varepsilon|p|^2}/|p|^2$, are

$$\langle A_m A_n\rangle :\ 2(-,-)^{-1}\delta_{mn}, \qquad \langle c\,\bar c\rangle :\ -(-,-)^{-1}, \qquad \langle\bar c\,c\rangle :\ (-,-)^{-1}, \qquad \langle\lambda_\alpha{}^i\bar\lambda_{\dot\beta j}\rangle :\ -(-,-)^{-1}p_m\sigma^m_{\alpha\dot\beta}\delta^i_j.$$

With the conformal Killing spinor $\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ and $\Gamma(\epsilon_1,\epsilon_2)|^m_V = i\big(\bar\epsilon_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{1\alpha}{}^i + \bar\epsilon_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{2\alpha}{}^i\big)$, the vertices of Proposition 3.30 are

$$s_\epsilon\Psi = i\int d^4x\,\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_\alpha{}^i + \bar\epsilon_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i,\ \partial_m\bar c\big), \qquad s_K\Psi = 4\int d^4x\,(\bar c,\, b_m A_m),$$
$$S^{(2)}_\Psi(x_1,x_2) \supset -\int d^4x\,\big(\bar c,\ \Gamma(\epsilon_1,\epsilon_2)|^m_V\,\partial_m\bar c\big),$$

for the special conformal field K_b, $v^m = x^2b^m - 2(b\cdot x)x^m$. On the Dirac column, $\not d = \gamma^\mu_W\partial_\mu$ with Weinberg's matrices gives Weinberg's form of each line.

Chapter 4

Counterterms at one loop and the quantum master equation at the UV boundary

The one-loop counterterm, the analogue on $\mathbb{R}^4$ at order $\hbar$ of the counterterm of [C07, Thm B] (Remark 4.9), is fixed by the requirement that the order-$\hbar$ part of the flow $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ have a limit as $\varepsilon \to 0$ at generic leg momenta, and it is unique among the translation-invariant and $\mathfrak{g}$-invariant local functionals with coefficients in the span of ε^{-1} and $\log\varepsilon$ (Propositions 4.8 and 4.10). The regulator, however, cuts every edge at the Schwinger time ε and breaks the gauge symmetry, so the one-loop counterterm contains the mass terms $\varepsilon^{-1}\int \mathrm{dvol}\,(A,A)$, whose Q-variation is nonzero. Transported to the UV boundary by Theorem 3.23, the quantum master equation then fails by a finite local functional, and a finite local counterterm has to be chosen to remove it: this is the first renormalisation condition of the theory. This chapter computes $I^{CT}[\varepsilon]$ in every one-loop sector of superficial degree ≥ 0, identifies its $\log\varepsilon$ part as a rescaling of the fields with vanishing component along the coupling, and evaluates the master equation at the UV boundary in the sector with legs $(c, A|_V)$, where the first finite counterterm appears.

4.1 The one-loop counterterm and its boundary form

Throughout this chapter $\mathfrak{g}$ is simple. Then the Killing form $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)$ is a multiple of the invariant pairing, and the ratio $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$, for $X, Y \in \mathfrak{g}$ with $(X,Y) \neq 0$, is a number independent of X and Y. Every one-loop coefficient of this chapter is a rational multiple of

$$\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2\,(X,Y)},$$

except the coefficients of the fourth-order adjoint traces of the quartic graphs, which are rational multiples of $\hbar/16\pi^2$. The upper Schwinger limit is ∞ (Chapter 3).

The value of a translation-invariant functional with k legs on plane waves $e^{i(p_j,x)}$, $j = 1,\dots,k$, with fibre elements as in Definition 1.13, is $(2\pi)^4\delta(p_1+\cdots+p_k)$ times a function of $(p_1,\dots,p_k)$ on the subspace where $p_1+\cdots+p_k = 0$; this function is its plane-wave value. Leg momenta are generic when every leg momentum is nonzero and no sum of the momenta of a nonempty proper subset of the legs vanishes. For $k = 1$ the set of generic configurations is empty; for $k \geq 2$ it is the complement of finitely many proper linear subspaces of the subspace $p_1+\cdots+p_k = 0$, an open dense subset. A functional depending on ε has a limit as $\varepsilon \to 0$ at generic leg momenta when the plane-wave values of its Taylor components converge as $\varepsilon \to 0$ at every generic configuration of the leg momenta. Every limit of a flow in this chapter is a limit in this sense.

Definition 4.1 (One-loop counterterm)**.** For a connected one-particle-irreducible graph γ with legs, $b_1(\gamma) = 1$ and all vertices labelled by I (Proposition 3.35), let $w_\gamma(\varepsilon)$ be its value with every edge labelled by $P(\varepsilon,\infty)$ [C07, §15.4]. Every vertex of γ lies on its cycle and carries a leg, so at generic leg momenta the momenta of distinct edges of γ differ, and by Lemma 4.6, as $\varepsilon \to 0$, the plane-wave value of $w_\gamma(\varepsilon)$ is the sum of multiples of negative integer powers of ε, a multiple of $\log\varepsilon$, and a function of ε with a limit; the first two are polynomials in the leg momenta, and $\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$ denotes the local functional whose plane-wave value is their sum [CG2, §13.7.2]. The one-loop counterterm is

$$I^{CT}[\varepsilon] = \hbar \sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|}\,\mathrm{Sing}_\varepsilon\, w_\gamma(\varepsilon), \tag{4.1}$$

the sum over the connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$ and at least two legs, modulo constants. Its coefficients lie in the span of ε^{-1} and $\log\varepsilon$ (Proposition 4.20), the one-loop part of the renormalisation scheme $\mathscr{A}_{<0}$ [C07, Def 1.4.1; CG2, §13.7.2].

The ε-dependence of a graph is carried by the lower ends of its Schwinger integrals. By (3.3), $\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -L_\varepsilon$ with $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$.

Proposition 4.2 (Boundary form of the scale derivative)**.** *Let γ be a graph with edge set $E(\gamma)$ [C07, §15.5], and let $w_\gamma(\varepsilon)$ be its value with every edge labelled by $P(\varepsilon,\infty)$. Then*

$$\varepsilon\frac{d}{d\varepsilon}w_\gamma(\varepsilon) = \sum_{e\in E(\gamma)} w_\gamma\big(\textit{edge } e \textit{ labelled by } -\varepsilon L_\varepsilon,\ \textit{the other edges by } P(\varepsilon,\infty)\big). \tag{4.2}$$

In the Schwinger form $w_\gamma(\varepsilon) = \int_{[\varepsilon,\infty)^{E(\gamma)}} f_\gamma \prod_e dt_e$ of [C07, §15.5], with the edge e carrying L_{t_e}, the e-th term of (4.2) *is*

$$-\varepsilon \int_{[\varepsilon,\infty)^{E(\gamma)\setminus\{e\}}} f_\gamma\big|_{t_e=\varepsilon} \prod_{e'\neq e} dt_{e'}.$$

Proof. The value w_γ is multilinear in the kernels labelling the edges, so the Leibniz rule gives

$$\frac{d}{d\varepsilon}w_\gamma = \sum_e w_\gamma\Big(\text{edge } e \text{ labelled by } \frac{d}{d\varepsilon}P(\varepsilon,\infty)\Big),$$

and $\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -L_\varepsilon$ by (3.3). In the Schwinger form, $\frac{d}{d\varepsilon}\int_\varepsilon^\infty(\cdots)\,dt_e = -(\cdots)|_{t_e=\varepsilon}$ for each edge. □

Corollary 4.3 (Singular part from the boundary)**.** *At fixed leg momenta, if the right-hand side of* (4.2) *equals $-a\,\varepsilon^{-1} + b$ plus a term bounded by a multiple of ε, with a, b independent of ε, then*

$$w_\gamma(\varepsilon) = a\,\varepsilon^{-1} + b\log\varepsilon + (\textit{a function of } \varepsilon \textit{ with a limit}), \qquad \mathrm{Sing}_\varepsilon w_\gamma(\varepsilon) = a\,\varepsilon^{-1} + b\log\varepsilon.$$

Proof. $\frac{d}{d\varepsilon}\big(w_\gamma - a\varepsilon^{-1} - b\log\varepsilon\big) = \varepsilon^{-1}\big(\varepsilon\frac{d}{d\varepsilon}w_\gamma + a\varepsilon^{-1} - b\big)$ is ε^{-1} times a term bounded by a multiple of ε, hence bounded near $\varepsilon = 0$, so $w_\gamma - a\varepsilon^{-1} - b\log\varepsilon$ has a limit and differs from it by a term bounded by a multiple of ε. □

[figure] Boundary graphs. (a), (b): the two terms of (4.2) for the bubble with two vertices $A|_V(A|_{W_\mathbb{R}^\perp})^2$ and external $A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp}$; the shaded edge is pinned at $t=\varepsilon$ and carries $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$. (c)–(e): the same mechanism for a triangle with one insertion of a composite operator.

The singular coefficients of every graph are therefore computed from graphs with one edge pinned at $t = \varepsilon$. For one-loop graphs with at most two edges, the Schwinger integrals reduce to four elementary ones after a Gaussian integral over the loop momentum.

Lemma 4.4 (Gaussian integral over the loop momentum)**.** *Let $t_1, t_2 > 0$, $p \in V$, and let the numerator be a polynomial function of $p_1 \in V$ of degree at most two. Then*

$$\int \frac{d^4p_1}{(2\pi)^4}\, e^{-t_1|p_1|^2 - t_2|p_1-p|^2}\,(\textit{numerator}) \\ = \frac{e^{-t_1t_2|p|^2/(t_1+t_2)}}{16\pi^2(t_1+t_2)^2}\Big[\textit{numerator} - \frac{1}{4(t_1+t_2)}\big(\textit{positive Laplacian of the numerator}\big)\Big]_{p_1=\frac{t_2}{t_1+t_2}p},$$

the positive Laplacian, minus the sum of the second derivatives along an orthonormal basis, being taken in the variable p_1. For one edge, $\int \frac{d^4p_1}{(2\pi)^4} e^{-t|p_1|^2} = (16\pi^2t^2)^{-1}$.

Proof. Completing the square and integrating the Gaussian,

$$t_1|p_1|^2 + t_2|p_1-p|^2 = (t_1+t_2)\Big|p_1 - \tfrac{t_2}{t_1+t_2}p\Big|^2 + \tfrac{t_1t_2}{t_1+t_2}|p|^2, \\ \int \frac{d^4p_1}{(2\pi)^4} e^{-(t_1+t_2)|p_1|^2} = \frac{\pi^2}{(2\pi)^4(t_1+t_2)^2} = \frac{1}{16\pi^2(t_1+t_2)^2}.$$

The normalised Gaussian average in the shifted variable has vanishing odd moments, and its second moment sends $(v,p_1)(v',p_1)$ to $(v,v')/(2(t_1+t_2))$ for $v, v' \in V$. A polynomial of degree at most two is its value at the centre plus a linear term plus its quadratic part; the average of the quadratic part is $\frac{1}{4(t_1+t_2)}$ times the sum of the second derivatives along an orthonormal basis, i.e. $-\frac{1}{4(t_1+t_2)}$ times the positive Laplacian of the numerator in p_1. □

Lemma 4.5 (Elementary Schwinger integrals)**.** *For $p \neq 0$, modulo functions of ε with a limit as $\varepsilon \to 0$ ($\mathscr{A}_{\geq 0}$ of [C07, Def 1.4.1]),*

$$\int_\varepsilon^\infty \frac{dt}{t^2} = \varepsilon^{-1}, \\ \iint_{[\varepsilon,\infty)^2} \frac{e^{-t_1t_2|p|^2/(t_1+t_2)}}{(t_1+t_2)^3}\, dt_1dt_2 = \tfrac{1}{4}\varepsilon^{-1} + \tfrac{1}{6}|p|^2\log\varepsilon, \\ \iint_{[\varepsilon,\infty)^2} \frac{e^{-t_1t_2|p|^2/(t_1+t_2)}}{(t_1+t_2)^2}\, dt_1dt_2 = -\log\varepsilon, \\ \iint_{[\varepsilon,\infty)^2} \frac{t_1t_2\, e^{-t_1t_2|p|^2/(t_1+t_2)}}{(t_1+t_2)^4}\, dt_1dt_2 = -\tfrac{1}{6}\log\varepsilon.$$

Proof. The integrands are symmetric in t_1, t_2, so by Proposition 4.2 each $\varepsilon\frac{d}{d\varepsilon}$ equals -2ε times the integral over $t \in [\varepsilon,\infty)$ of the integrand at $t_1 = \varepsilon$, $t_2 = t$. On the pinned edge the exponent satisfies $0 \leq \frac{\varepsilon t|p|^2}{\varepsilon+t} \leq \varepsilon|p|^2$, so $e^{-\varepsilon t|p|^2/(\varepsilon+t)}$ equals $1 - \frac{\varepsilon t|p|^2}{\varepsilon+t}$ plus a remainder between 0 and $\frac{1}{2}\varepsilon^2|p|^4$, uniformly in t. With

$$\int_\varepsilon^\infty \frac{dt}{(\varepsilon+t)^2} = \frac{1}{2\varepsilon}, \qquad \int_\varepsilon^\infty \frac{dt}{(\varepsilon+t)^3} = \frac{1}{8\varepsilon^2}, \\ \int_\varepsilon^\infty \frac{t\,dt}{(\varepsilon+t)^4} = \int_\varepsilon^\infty \Big(\frac{1}{(\varepsilon+t)^3} - \frac{\varepsilon}{(\varepsilon+t)^4}\Big)dt = \frac{1}{8\varepsilon^2} - \frac{1}{24\varepsilon^2} = \frac{1}{12\varepsilon^2},$$

the three pinned integrals are, up to terms bounded by a multiple of ε,

$$2\varepsilon\int_\varepsilon^\infty \frac{e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^3}dt = \frac{1}{4\varepsilon} - \frac{|p|^2}{6}, \\ 2\varepsilon\int_\varepsilon^\infty \frac{e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^2}dt = 1, \qquad 2\varepsilon\int_\varepsilon^\infty \frac{\varepsilon t\, e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^4}dt = \frac{1}{6}.$$

Corollary 4.3 with (a,b) equal to $(\frac14, \frac16|p|^2)$, $(0,-1)$, $(0,-\frac16)$ gives the three expansions. The one-edge integral is exact. □

Lemma 4.6 (Singular part of a one-loop graph)**.** *Let γ be a connected one-particle-irreducible graph with $b_1(\gamma)=1$, so that every vertex lies on the cycle and $|E(\gamma)|=|V(\gamma)|$. Let its vertex functions be polynomials in the momenta, up to a factor depending on the leg momenta alone (such as the Fourier transform of a four-form at minus the total leg momentum, Chapter 5), and let its numerator, the product of the vertex functions and of the polynomial factors of the edge components of $P(\varepsilon,\infty)$ (Proposition 3.17), be homogeneous in the loop momentum and the leg momenta; for vertices labelled by I and legs in fixed slots this holds by Lemma 4.18. Its superficial degree is the degree of the numerator plus $4-2|E(\gamma)|$. Let the leg momenta be such that the momenta of any two distinct edges differ by a nonzero vector. Then, at these leg momenta and as $\varepsilon\to0$,*

$$w_\gamma(\varepsilon) = \sum_{1\le k\le \frac12(\textit{superficial degree})} \varepsilon^{-k}\, a_k + b\,\log\varepsilon + (\textit{a function of } \varepsilon \textit{ with a limit}),$$

where a_k and b are polynomials in the leg momenta, a_k homogeneous of degree (superficial degree)$-2k$ and b homogeneous of degree equal to the superficial degree, and $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tends to 0. In particular the sum of the terms with ε^{-k} and $\log\varepsilon$ is the plane-wave value of a local functional, $\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$, and it vanishes when the superficial degree is negative.

Proof. Let the edge $e\in E(\gamma)$ carry the momentum p_1-q_e, with p_1 the loop momentum and q_e a sum of leg momenta, so that $q_e\ne q_f$ for $e\ne f$. By Proposition 3.17, $w_\gamma(\varepsilon)$ is the integral over p_1, with the measure $d^4p_1/(2\pi)^4$, of the numerator times $\prod_e e^{-\varepsilon|p_1-q_e|^2}/|p_1-q_e|^2$, and $e^{-\varepsilon|k|^2}/|k|^2=\int_\varepsilon^\infty e^{-t|k|^2}dt$ introduces a Schwinger time t_e for each edge. By Tonelli's theorem,

$$\int\frac{d^4p_1}{(2\pi)^4}\int_{[\varepsilon,\infty)^{E(\gamma)}}\big|\text{numerator}\big|\,e^{-\sum_e t_e|p_1-q_e|^2}\prod_e dt_e=\int\frac{d^4p_1}{(2\pi)^4}\,\big|\text{numerator}\big|\prod_e\frac{e^{-\varepsilon|p_1-q_e|^2}}{|p_1-q_e|^2},$$

and the right side is finite: the points $p_1=q_e$ are distinct, near each of them exactly one factor is singular, $|p_1-q_e|^{-2}$ being integrable near $p_1=q_e$ in four dimensions, and the Gaussian factors bound the polynomial numerator at large $|p_1|$. By Fubini's theorem, $w_\gamma(\varepsilon)$ is the integral over $t\in[\varepsilon,\infty)^{E(\gamma)}$ of the integral over p_1, with the measure $\prod_e dt_e\,d^4p_1/(2\pi)^4$, of $\exp\big(-\sum_e t_e|p_1-q_e|^2\big)$ times the numerator. Write $\tau=\sum_e t_e$. Completing the square,

$$\sum_e t_e|p_1-q_e|^2=\tau\Big|p_1-\sum_e\frac{t_e}{\tau}\,q_e\Big|^2+\tau\cdot\frac12\sum_{e,f}\frac{t_e}{\tau}\frac{t_f}{\tau}|q_e-q_f|^2,$$

and the Gaussian integral over p_1, with the moments of Lemma 4.4 (each contracted pair of loop momenta gives $(2\tau)^{-1}$ times the metric), gives

$$w_\gamma(\varepsilon)=\sum_{j\ge0}\int_{[\varepsilon,\infty)^{E(\gamma)}}\frac{(\text{numerator})_j}{16\pi^2\,\tau^{2+j}}\,\exp\Big(-\tau\cdot\frac12\sum_{e,f}\frac{t_e}{\tau}\frac{t_f}{\tau}|q_e-q_f|^2\Big)\prod_e dt_e,$$

where $(\text{numerator})_j$, the part with j contracted pairs, is a polynomial in the ratios t_e/τ and the leg momenta, homogeneous of degree (degree of the numerator)$-2j$, and zero when $2j$ exceeds the degree of the numerator. Put $m_j=j+2-|E(\gamma)|$; then $(\text{numerator})_j\ne0$ implies $2m_j\le$ (superficial degree). The ratios t_e/τ range over the simplex $\sum_e t_e/\tau=1$, and $\prod_e dt_e$ is $\tau^{|E(\gamma)|-1}d\tau$ times the Lebesgue measure of the simplex. Since $\sum_{e\ne f}t_et_f/\tau^2=1-\sum_e t_e^2/\tau^2\ge1-\max_e t_e/\tau$, the factor multiplying τ in the exponent is at least one half of the smallest of the numbers $|q_e-q_f|^2$, $e\ne f$, times $1-\max_e t_e/\tau$, and $\tau(1-\max_e t_e/\tau)$ is the sum of the Schwinger times other than the largest.

The region $\tau \geq 1$. The integrand of the j-th term is bounded by a multiple of τ^{-2-j} times the exponential of minus a positive multiple of the sum of the Schwinger times other than the largest. Integrating these times against the exponential, and the largest time, which is at least $\tau/|E(\gamma)| \geq 1/|E(\gamma)|$, against τ^{-2-j}, gives a bound independent of ε; by dominated convergence this part has a limit as $\varepsilon \to 0$.

The region $\tau \leq 1$. If $m_j < 0$, the integrand times $\tau^{|E(\gamma)|-1}$ is bounded by a multiple of τ^{-1-m_j}, which is integrable, and the part has a limit. If $m_j \geq 0$, expand the exponential to order m_j. The remainder is bounded by $1/(m_j+1)!$ times the (m_j+1)-th power of the exponent, so its contribution is bounded by a multiple of $\tau^{-2-j+m_j+1}\tau^{|E(\gamma)|-1} = \tau^0$ and has a limit. The term of order $i \leq m_j$ is the integral, over the part of the simplex where $\min_e t_e/\tau \geq \varepsilon$, of $(16\pi^2\, i!)^{-1}(\text{numerator})_j$ times the i-th power of minus the factor multiplying τ in the exponent, times

$$\int_{\varepsilon/\min_e(t_e/\tau)}^{1} \tau^{-1-(m_j-i)}\, d\tau = \begin{cases} \dfrac{\varepsilon^{-(m_j-i)}\big(\min_e t_e/\tau\big)^{m_j-i} - 1}{m_j - i}, & i < m_j, \\ \log\big(\min_e t_e/\tau\big) - \log\varepsilon, & i = m_j. \end{cases}$$

The part of the simplex where $\min_e t_e/\tau < \varepsilon$ has measure at most a multiple of ε, the function $(\min_e t_e/\tau)^{m_j-i}$ is at most ε^{m_j-i} there, and $\log(\min_e t_e/\tau)$ is integrable on the simplex. Hence the term of order $i < m_j$ is $\varepsilon^{-(m_j-i)}$ times an integral over the whole simplex plus a function with a limit, and the term of order m_j is $-\log\varepsilon$ times an integral over the whole simplex plus a function with a limit. These integrals over the simplex, of polynomials in the ratios and the leg momenta times $(\min_e t_e/\tau)^{m_j-i}$, are polynomials in the leg momenta, homogeneous of degree (degree of the numerator)$-2j+2i$, which equals (superficial degree)$-2(m_j-i)$. Collecting the powers $k = m_j - i$ gives the expansion, with $1 \leq k \leq m_j \leq \frac{1}{2}$(superficial degree); for negative superficial degree every m_j with $(\text{numerator})_j \neq 0$ is negative, and the singular part vanishes.

Finally, $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tends to 0. That function is the sum, over j and i, of the following pieces; in the second and third item the simplex integrand of the term of order i is $(16\pi^2\, i!)^{-1}(\text{numerator})_j$ times the i-th power of minus the factor multiplying τ in the exponent, a polynomial in the ratios and the leg momenta.

- The integrals over $t \in [\varepsilon,\infty)^{E(\gamma)}$ restricted to $\tau \geq 1$, those restricted to $\tau \leq 1$ with $m_j < 0$, and the remainders of the expansion of the exponential on $\tau \leq 1$. By Proposition 4.2, $\varepsilon\frac{d}{d\varepsilon}$ of each is minus ε times its integrals over the faces $t_e = \varepsilon$; by the bounds above these face integrals are bounded on $\tau \geq 1$ and bounded by a multiple of $1 + |\log\varepsilon|$ on $\tau \leq 1$, the integrand being at most a multiple of $\tau^{1-|E(\gamma)|}$ there.

- For the term of order $i < m_j$, with $k = m_j - i$: $-k^{-1}\varepsilon^{-k}h(\varepsilon)$, where $h(\varepsilon)$ is the integral of the simplex integrand times $(\min_e t_e/\tau)^k$ over the part of the simplex where $\min_e t_e/\tau < \varepsilon$; and $-k^{-1}$ times the integral of the simplex integrand over the part where $\min_e t_e/\tau \geq \varepsilon$, which comes from the -1 in the integral over τ.

- For the term of order m_j: $h(\varepsilon)\log\varepsilon$, where $h(\varepsilon)$ is the integral of the simplex integrand over the part where $\min_e t_e/\tau < \varepsilon$; and the integral of the simplex integrand times $\log(\min_e t_e/\tau)$ over the part where $\min_e t_e/\tau \geq \varepsilon$.

The part of the simplex where $\min_e t_e/\tau < s$ is the union over e of the parts where $t_e/\tau < s$. For a function continuous on the part where $\min_e t_e/\tau \geq s_0$, its integral over the part where $\min_e t_e/\tau \geq s$, $s \geq s_0$, is therefore differentiable in s, with derivative minus a sum of integrals of the function over the faces where $\min_e t_e/\tau = s$, each face having $(|E(\gamma)|-2)$-dimensional measure at most that of a face of the simplex; the same holds, with the opposite sign and $s_0 = 0$, for the integral of a function continuous on the simplex over the part where $\min_e t_e/\tau < s$, and that integral is at most the maximum

of the absolute value of the function there times a multiple of s. In the second item the integrand of h is at most a multiple of ε^k where $\min_e t_e/\tau < \varepsilon$, so $h(\varepsilon)$ and $\varepsilon\frac{dh}{d\varepsilon}$ are bounded by multiples of ε^{k+1}, and $\varepsilon\frac{d}{d\varepsilon}\big(\varepsilon^{-k}h(\varepsilon)\big) = -k\varepsilon^{-k}h(\varepsilon) + \varepsilon^{1-k}\frac{dh}{d\varepsilon}$ is bounded by a multiple of ε; $\varepsilon\frac{d}{d\varepsilon}$ of the integral over the part where $\min_e t_e/\tau \geq \varepsilon$ is bounded by a multiple of ε. In the third item $h(\varepsilon)$ and $\varepsilon\frac{dh}{d\varepsilon}$ are bounded by multiples of ε, so $\varepsilon\frac{d}{d\varepsilon}\big(h(\varepsilon)\log\varepsilon\big) = h(\varepsilon) + \varepsilon\frac{dh}{d\varepsilon}\log\varepsilon$ is bounded by a multiple of $\varepsilon(1 + |\log\varepsilon|)$; and $\log(\min_e t_e/\tau) = \log\varepsilon$ on the faces where $\min_e t_e/\tau = \varepsilon$, so $\varepsilon\frac{d}{d\varepsilon}$ of the integral over the part where $\min_e t_e/\tau \geq \varepsilon$ is bounded by a multiple of $\varepsilon|\log\varepsilon|$. Every piece has $\varepsilon\frac{d}{d\varepsilon}$ tending to 0. □

Lemma 4.7 (One-loop graphs as trees with a one-loop vertex)**.** *Consider the graphs of Proposition 3.35, with labelled legs, whose vertices are labelled by Taylor components of I, except at most one vertex of valence at least 1, the marked vertex, labelled by a Taylor component of a further functional (in Chapter 5, the operator O). For a connected graph Γ with $b_1(\Gamma) = 1$, let γ be its cycle together with the vertices on the cycle, the remaining half-edges at these vertices being the legs of γ, and let T be the tree obtained from Γ by contracting γ to one vertex s, whose half-edges are the legs of γ.*

- *γ is connected and one-particle-irreducible with $b_1(\gamma) = 1$. The automorphisms of T fixing the legs are the identity, and so are those of every tree with the given legs whose vertices are labelled as above except one distinguished vertex s. Restriction to γ is a bijection from the automorphisms of Γ fixing the legs to the automorphisms of γ fixing the legs of γ. The value w_Γ is w_T with s labelled by the multilinear form w_γ in the half-edges of s.*

- *The sum of $w_\Gamma/|\operatorname{Aut}\Gamma|$ over the connected graphs Γ with $b_1(\Gamma) = 1$ and the given legs equals the sum of $w_T/|\operatorname{Aut}T| = w_T$ over the trees T with the given legs and one vertex s, labelled by the Taylor component, with legs the half-edges of s, of the sum of $w_\gamma/|\operatorname{Aut}\gamma|$ over the connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$.*

- *A connected one-particle-irreducible graph with $b_1 = 1$, one leg and all vertices labelled by I is a cubic vertex of I with one edge from the vertex to itself, and its value is 0 before any integration over positions or momenta.*

Proof. First item. A connected graph with $b_1 = 1$ contains exactly one cycle. Every vertex of γ lies on it, so removing an edge of γ leaves γ connected, and $b_1(\gamma) = |E(\gamma)| - |V(\gamma)| + 1 = 1$. Contracting the cycle lowers b_1 by one, so T is a tree.

Let T be such a tree, with distinguished vertex s. An automorphism α of T fixing the legs preserves the labels of the vertices, so it fixes s and the marked vertex. For a vertex $v \neq s$, the vertices beyond v are v and the vertices separated from s by v. A vertex beyond v at maximal distance from s has one half-edge towards s and its other half-edges are legs; if it is labelled by I it has at least two legs. Hence the vertices beyond v carry a leg or include the marked vertex. Let $v' \neq v$ be a second vertex different from s, the two being exchanged if v lies beyond v'. If v' lies beyond v and v is labelled by I, then v has, besides its half-edges towards s and towards v', a further half-edge, which is a leg or leads to vertices beyond v and not beyond v' that carry a leg or include the marked vertex; if v' lies beyond v and v is the marked vertex, the vertices beyond v include the marked vertex and those beyond v' exclude it; if neither lies beyond the other, the two sets of vertices beyond them are disjoint, and each carries a leg or includes the marked vertex. In every case v and v' differ in the set of legs carried by the vertices beyond them or in whether these include the marked vertex. Both data are preserved by α, so α fixes every vertex. A tree has no two edges with the same ends and no edge from a vertex to itself, so α fixes every edge, and, with the legs, every half-edge.

An automorphism of Γ fixing the legs maps the cycle to itself and therefore induces an automorphism of T fixing the legs, which is the identity. It fixes every vertex outside the cycle and every half-edge outside the edges of the cycle, the legs of γ included, so it restricts to an automorphism of γ fixing the

legs of γ, and this restriction determines it. Conversely, an automorphism of γ fixing its legs extends by the identity to an automorphism of Γ fixing the legs.

The value w_Γ is the contraction of the tensor product of the vertex functions along the edges of Γ, with the edge rule of Proposition 3.35. The contractions along different edges commute [C07, §6.4], so contracting the edges of the cycle first gives w_γ, a multilinear form in the half-edges of s, and the remaining contractions give w_T with s labelled by w_γ.

Second item. By the first item the half-edges of s are determined by T and its legs, and $\Gamma \mapsto (T, \gamma)$, with the legs of γ identified with the half-edges of s, is a bijection from the isomorphism classes of connected graphs with $b_1 = 1$ and the given legs to the pairs consisting of an isomorphism class of trees T with the given legs and a vertex s, and an isomorphism class of connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$ whose legs are the half-edges of s; the inverse glues γ into T in place of s. With $|\operatorname{Aut}\Gamma| = |\operatorname{Aut}\gamma|$, $|\operatorname{Aut}T| = 1$ and the linearity of w_T in the label of s,

$$\sum_\Gamma \frac{w_\Gamma}{|\operatorname{Aut}\Gamma|} = \sum_T \sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|}\, w_T\big(s \text{ labelled by } w_\gamma\big) = \sum_T \frac{1}{|\operatorname{Aut}T|}\, w_T\Big(s \text{ labelled by } \sum_\gamma \frac{w_\gamma}{|\operatorname{Aut}\gamma|}\Big).$$

Third item. A cycle with n vertices of valence at least three has at least n legs, so one leg forces $n = 1$ and a cubic vertex, two of whose slots are joined by the edge. By Definition 1.13, every cubic vertex function of I, namely $A\,A\,A$ (which contains $(A|_V)^3$ and $A|_V(A|_{W_\mathbb{R}^\perp})^2$), $\lambda\,\lambda\,A$, $A\,c\,A^*$, $c\,\lambda\,\lambda^*$ and $c\,c\,c^*$, is the form $(X_1, [X_2, X_3])$ in the $\mathfrak{g}$-factors X_1, X_2, X_3 of its three legs, up to sign, times a factor independent of them; by the invariance of $(-,-)$ and the antisymmetry of the bracket, $(X_1, [X_2, X_3])$ is totally antisymmetric. The $\mathfrak{g}$-factor of every component of $P(\varepsilon, \infty)$ is a multiple of $(-,-)^{-1}$ (Proposition 3.17), which is symmetric. The contraction of two arguments of a totally antisymmetric form with a symmetric tensor is 0, so the vertex with two of its slots joined is the zero functional, before any integration. □

Proposition 4.8 (Existence of the limit)**.** *The order-$\hbar$ part of $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ has a limit as $\varepsilon \to 0$ at generic leg momenta.*

Proof. By Proposition 3.35 with $J = I - I^{CT}[\varepsilon]$, the order-$\hbar$ part is $\hbar$ times the sum of $w_\Gamma/|\operatorname{Aut}\Gamma|$ over the connected graphs Γ with $b_1(\Gamma) = 1$ and vertices labelled by I, plus $\hbar$ times the sum of $w_T/|\operatorname{Aut}T|$ over the trees T with one vertex s labelled by a Taylor component of $-\hbar^{-1}I^{CT}[\varepsilon]$ and the other vertices labelled by I. By Lemma 4.7 (without marked vertex), $|\operatorname{Aut}T| = 1$ for these trees, and the first sum is the sum over the same trees of w_T with s labelled by the sum of $w_\gamma/|\operatorname{Aut}\gamma|$ over the connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$. A cycle of vertices of valence at least three carries legs, so every such γ has legs; those with one leg have value 0 (Lemma 4.7), and $I^{CT}[\varepsilon]$ has no Taylor component with one leg (Definition 4.1). By (4.1), the order-$\hbar$ part is therefore $\hbar$ times the sum over the trees T whose vertex s has at least two half-edges of w_T with s labelled by

$$\sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|}\Big(w_\gamma(\varepsilon) - \operatorname{Sing}_\varepsilon w_\gamma(\varepsilon)\Big),$$

the sum over the γ whose legs are the half-edges of s. For a fixed number of legs this sum is finite: a tree with r vertices labelled by I and a vertex s with m half-edges has r edges and at least $3r+m-2r = r+m$ legs, and a cycle with m legs has at most m vertices.

Fix generic leg momenta. Each edge of T separates T into two parts, and each part carries a leg: in the part without s, a vertex at maximal distance from s carries at least two legs; in the part with s, the vertex s has, besides the half-edge on the path from s to the edge, a further half-edge, which is a leg or leads to vertices separated from s and from the edge, among which a vertex at maximal distance from s carries at least two legs. Hence the edge carries the sum of the momenta of a nonempty proper subset of the legs, a nonzero momentum k, and its edge factor, a polynomial times $e^{-\varepsilon|k|^2}/|k|^2$ (Proposition

3.17), converges to the polynomial times $|k|^{-2}$. The half-edges of s carry the sums of the momenta of disjoint nonempty sets of legs, one set for each half-edge. Two distinct edges of the cycle of γ cut it into two arcs, each containing a vertex, and every vertex of γ carries a leg of γ; the momenta of the two edges differ by the sum of the momenta entering γ along one arc, the sum of the momenta of a nonempty proper subset of the legs of T, which is nonzero. By Lemma 4.6, with the homogeneity of Lemma 4.18, the plane-wave value of $w_\gamma(\varepsilon) - \mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$ at these momenta has a limit as $\varepsilon \to 0$. The plane-wave value of w_T is the sum, over the components of the internal slots, of the products of the vertex functions of T, of its edge factors and of this value, a finite sum of products of functions of ε with limits. □

Remark 4.9 (Relation with Theorem B)**.** [C07, Thm B] constructs, on a compact manifold with a finite upper scale, a counterterm at all orders in $\hbar$, unique among the local functionals with coefficients in $\mathscr{A}_{<0}$, for which the flow of the interaction minus the counterterm has a limit as $\varepsilon \to 0$ as a functional; at one loop, its part with k legs is the singular part of the order-$\hbar$, k-leg part of the flow of the interaction minus the counterterms with fewer legs, and it is independent of the upper scale [C07, §1.5]. Definition 4.1 and Propositions 4.8 and 4.10 give its analogue on $\mathbb{R}^4$ with upper scale ∞, for the even, local, at least cubic interaction I of (1.3), in the following scope: the order is $\hbar$; the singular parts are computed graph by graph on the one-particle-irreducible graphs, and the one-particle-reducible graphs carry them through their cycles (Lemma 4.7); the limit is the limit of plane-wave values at generic leg momenta; and uniqueness holds among the translation-invariant and $\mathfrak{g}$-invariant local functionals whose coefficients lie in a space of functions of ε containing ε^{-1} and $\log\varepsilon$ in which every function with a limit as $\varepsilon \to 0$ is zero.

Proposition 4.10 (Uniqueness of the one-loop counterterm)**.** *Let I' be a local functional of order $\hbar$, without constant term, invariant under translations and under the adjoint action of $\mathfrak{g}$, whose coefficients lie in a space of functions of ε containing ε^{-1} and $\log\varepsilon$ in which every function with a limit as $\varepsilon \to 0$ is zero, for instance the span of ε^{-1} and $\log\varepsilon$, or a complement $\mathscr{A}_{<0}$ of $\mathscr{A}_{\geq 0}$ containing them [C07, Def 1.4.1]. If the order-$\hbar$ part of $W(P(\varepsilon,\infty), I - I')$ has a limit as $\varepsilon \to 0$ at generic leg momenta, then $I' = I^{CT}[\varepsilon]$.*

Proof. The vertex functions of I and the components of $P(\varepsilon,\infty)$ are translation invariant, and their $\mathfrak{g}$-factors are built from the bracket, $(-,-)$ and $(-,-)^{-1}$; so $I^{CT}[\varepsilon]$, and with it the difference $I^{CT}[\varepsilon] - I'$, is invariant under translations and under the adjoint action of $\mathfrak{g}$, with coefficients in the given space of functions. A translation-invariant local functional with one leg is, modulo total derivatives, the integral of a constant linear form on the fibre of the bundle of fields. The fibre is $\mathfrak{g}$ tensored with a representation of $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$ (Chapter 1), so a $\mathfrak{g}$-invariant linear form on it lies in the tensor product of the invariant linear forms on $\mathfrak{g}$ with the dual of that representation; the invariant linear forms on $\mathfrak{g}$ are, through $(-,-)$, the centre of $\mathfrak{g}$, which is 0 for $\mathfrak{g}$ simple. Hence the difference has no Taylor component with one leg, and by hypothesis none without legs.

The order-$\hbar$ parts of $W(P(\varepsilon,\infty), I - I')$ and $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ differ by $\hbar$ times the sum of the trees with one vertex labelled by a Taylor component of $\hbar^{-1}(I^{CT}[\varepsilon] - I')$ and all other vertices labelled by I (Proposition 3.35). That vertex has at least two half-edges, so, as in the proof of Proposition 4.8, every edge of such a tree carries a nonzero momentum at generic leg momenta. By the hypothesis and Proposition 4.8, the sum of the trees has a limit at generic leg momenta. Suppose the difference is nonzero, and let $k \geq 2$ be the smallest number of legs of its nonzero Taylor components. A tree with $r \geq 1$ vertices labelled by I, each with at least three half-edges, has r edges, and its vertex labelled by the difference has at least k half-edges, so it has at least $k + 3r - 2r = k + r$ legs; the k-leg part of the sum of the trees is therefore the k-leg Taylor component of $\hbar^{-1}(I^{CT}[\varepsilon] - I')$ itself, which then has a limit at generic leg momenta. Its plane-wave value is a finite sum of products of linearly independent coefficient functions of ε with polynomials in $(p_1, \ldots, p_k)$ on the subspace where $p_1 + \cdots + p_k = 0$.

At a generic configuration this sum is a function of ε in the given space with a limit, hence zero, and by linear independence every polynomial vanishes there. The generic configurations are dense in the subspace, so the polynomials vanish on it. A polynomial vanishing on that subspace is a combination of the components of $p_1+\cdots+p_k$ with polynomial coefficients, the plane-wave value of a total derivative; so the k-leg component is zero, a contradiction. $\square$

4.2 Two-point graphs

A quadratic functional is evaluated on plane waves. For two legs of $A|_V$ the background is $A(p)e^{i(p,x)}+A(-p)e^{-i(p,x)}$ with $A(\pm p)\in V^\vee\otimes\mathfrak{g}$, and the plane-wave value of a functional is the coefficient of $(2\pi)^4\delta(0)$ in its part bilinear in $A(p)$ and $A(-p)$; for two odd legs the background is $r_1(\cdot)e^{i(p,x)}+r_2(\cdot)e^{-i(p,x)}$ and the plane-wave value is the coefficient of r_1r_2, as in Definition 1.13. Since d becomes $i(p,-)$ and d^* becomes $-i\,\iota_p$,

$$\begin{gathered}\int \mathrm{dvol}\,(A|_V,A|_V)\mapsto 2\,(A(p),A(-p)),\qquad \int \mathrm{dvol}\,(d^*A|_V,d^*A|_V)\mapsto 2\,(\iota_pA(p),\iota_pA(-p)),\\ \int \mathrm{dvol}\,(dA|_V,dA|_V)\mapsto 2\,\big(|p|^2(A(p),A(-p))-(\iota_pA(p),\iota_pA(-p))\big),\\ \int \mathrm{dvol}\,(dc,A^*)\mapsto i\,(p\,c(-p),A^*(p))\quad\text{for the legs } r_1A^*(p)e^{i(p,x)}+r_2c(-p)e^{-i(p,x)},\end{gathered}\tag{4.3}$$

where $(A(p),A(-p))$ pairs the $V_{10}^\vee$ factors by the metric and the $\mathfrak{g}$ factors by $(-,-)$, $(p\,c(-p),A^*(p))$ is the fibre pairing of $p\otimes c(-p)\in V\otimes\mathfrak{g}$ with $A^*(p)$, and the Hodge normalisation of Definition 1.8 gives $((p,-)\wedge A(p),(p,-)\wedge A(-p))=|p|^2(A(p),A(-p))-(\iota_pA(p),\iota_pA(-p))$.

[figure] The one-loop one-particle-irreducible graphs with external $A|_V,A|_V$; every edge carries $P(\varepsilon,\infty)$. (a), (b): quartic vertex with an $A|_V$, resp. $A|_{W_{\mathbb{R}}^\perp}$, loop. (c): gauge bubble with $A|_V$ lines. (d): bubble with $A|_{W_{\mathbb{R}}^\perp}$ lines. (e): gaugino loop. (f): ghost loop of c–A^* lines.

graph	$1/\lvert\operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	$\frac12$	$\frac32\int\mathrm{dvol}\,(A\|_V,A\|_V)$	0
(b)	$\frac12$	$3\int\mathrm{dvol}\,(A\|_V,A\|_V)$	0
(c)	$\frac12$	$-\frac{9}{16}\int\mathrm{dvol}\,(A\|_V,A\|_V)$	$\frac{19}{24}\int\mathrm{dvol}\,(dA\|_V,dA\|_V)-\frac18\int\mathrm{dvol}\,(d^*A\|_V,d^*A\|_V)$
(d)	$\frac12$	$-\frac34\int\mathrm{dvol}\,(A\|_V,A\|_V)$	$-\frac12\int\mathrm{dvol}\,(dA\|_V,dA\|_V)$
(e)	$\frac12$	$-\int\mathrm{dvol}\,(A\|_V,A\|_V)$	$-\frac43\int\mathrm{dvol}\,(dA\|_V,dA\|_V)$
(f)	1	$\frac{1}{16}\int\mathrm{dvol}\,(A\|_V,A\|_V)$	$\frac{1}{24}\int\mathrm{dvol}\,(dA\|_V,dA\|_V)+\frac18\int\mathrm{dvol}\,(d^*A\|_V,d^*A\|_V)$
total		$\frac94\int\mathrm{dvol}\,(A\|_V,A\|_V)$	$-\int\mathrm{dvol}\,(dA\|_V,dA\|_V)$

Contributions $\hbar\lvert\operatorname{Aut}\gamma\rvert^{-1}\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$ of the graphs with external $A|_V,A|_V$ to $I^{CT}[\varepsilon]$, in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$; the ε^{-1} part multiplies ε^{-1} and the $\log\varepsilon$ part multiplies $\log\varepsilon$. The coefficients of $\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)$ cancel between (c) and (f).

[figure] The one-loop one-particle-irreducible graphs with external $A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp}$: (a), (b) quartic vertex with an $A|_V$, resp. $A|_{W_{\mathbb{R}}^\perp}$, loop; (c) bubble with one $A|_V$ and one $A|_{W_{\mathbb{R}}^\perp}$ line; (d) gaugino loop with two vertices $\lambda\lambda A|_{W_{\mathbb{R}}^\perp}$. (e): the gaugino loop with external $A|_V,A|_{W_{\mathbb{R}}^\perp}$.

graph	$1/\lvert\operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	$\frac{1}{2}$	$2\int \operatorname{dvol}(A\vert_{W_{\mathbb{R}}^{\perp}}, A\vert_{W_{\mathbb{R}}^{\perp}})$	0
(b)	$\frac{1}{2}$	$\frac{5}{2}\int \operatorname{dvol}(A\vert_{W_{\mathbb{R}}^{\perp}}, A\vert_{W_{\mathbb{R}}^{\perp}})$	0
(c)	1	$-\frac{1}{4}\int \operatorname{dvol}(A\vert_{W_{\mathbb{R}}^{\perp}}, A\vert_{W_{\mathbb{R}}^{\perp}})$	$\int \operatorname{dvol}(dA\vert_{W_{\mathbb{R}}^{\perp}}, dA\vert_{W_{\mathbb{R}}^{\perp}})$
(d)	$\frac{1}{2}$	$-2\int \operatorname{dvol}(A\vert_{W_{\mathbb{R}}^{\perp}}, A\vert_{W_{\mathbb{R}}^{\perp}})$	$-2\int \operatorname{dvol}(dA\vert_{W_{\mathbb{R}}^{\perp}}, dA\vert_{W_{\mathbb{R}}^{\perp}})$
total		$\frac{9}{4}\int \operatorname{dvol}(A\vert_{W_{\mathbb{R}}^{\perp}}, A\vert_{W_{\mathbb{R}}^{\perp}})$	$-\int \operatorname{dvol}(dA\vert_{W_{\mathbb{R}}^{\perp}}, dA\vert_{W_{\mathbb{R}}^{\perp}})$
(e)	$\frac{1}{2}$	0	0

Contributions of the graphs with external $A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}}$ (rows (a)–(d)) and $A|_V, A|_{W_{\mathbb{R}}^{\perp}}$ (row (e)) to $I^{CT}[\varepsilon]$, in the units of the previous table.

[figure] The one-loop one-particle-irreducible graphs with external λ, λ: (a) with an $A|_V$ line, (b) with an $A|_{W_{\mathbb{R}}^{\perp}}$ line; and with external $c, A^*|_V$: (c) with one $A|_V$ line and one c–A^* line.

graph	$1/\lvert\operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	1	0	$\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
(b)	1	0	$3\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
total λ, λ		0	$4\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
(c)	1	0	$-\int \operatorname{dvol}(dc, A^*)$

Contributions of the graphs with external λ, λ and $c, A^*|_V$ to $I^{CT}[\varepsilon]$, in the units of the previous tables.

Five representative graphs are computed below in five steps: the amplitude, from the one-loop formula (Proposition 3.36), the vertex functions of Definition 1.13 and the propagator components of Proposition 3.17; the numerator algebra; the Schwinger integral, by Lemma 4.4; the ε-expansion, by Lemma 4.5; and the local part, by (4.3). For two legs, the order-n term of (3.7) enters through its part bilinear in the legs: for $n = 2$,

$$\frac{\hbar}{4}\mathrm{STr}\big((\text{one-loop operator})^2\big)\Big|_{\text{bilinear}} = \frac{\hbar}{2}\mathrm{STr}\big((\text{part linear in leg 1})(\text{part linear in leg 2})\big),$$

and for $n = 1$ it is $\frac{\hbar}{2}\mathrm{STr}$ of the part of the one-loop operator bilinear in the legs. In the bubbles the two edges carry the momenta p_1 and $p_1 - p$, with Schwinger times t_1 and t_2:

$$\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2} = \int_\varepsilon^\infty e^{-t_1|p_1|^2}dt_1, \qquad \frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2} = \int_\varepsilon^\infty e^{-t_2|p_1-p|^2}dt_2.$$

The colour trace of the loop is $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}) = \frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{(X,Y)}(X_1, X_2)$ for the $\mathfrak{g}$-components X_1, X_2 of the two legs, and the factor (X_1, X_2) is part of the pairings $(A(p), A(-p))$ and $(\iota_p A(p), \iota_p A(-p))$.

Example 4.11 (Gauge bubble, graph (c) of the $A|_V$ two-point function)**.** Amplitude. The two vertices are (1.4) at the incoming momenta $(p, -p_1, p_1 - p)$ and $(-p, p_1, p - p_1)$; each edge carries the $A|_V \otimes A|_V$ component $2(-,-)^{-1} \otimes$ (inverse metric of V) of $P(\varepsilon, \infty)$, times $\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2}$, resp. $\frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2}$. The plane-wave value of $\hbar|\operatorname{Aut}\gamma|^{-1}w_\gamma(\varepsilon)$ is

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{(X,Y)}\int\frac{d^4p_1}{(2\pi)^4}\,\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2}\,\frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2}\,(\text{numerator}).$$

Numerator algebra. Contracting the two vertex functions along the two edges (the exchange of the edges is the automorphism counted by $1/|\operatorname{Aut}\gamma| = \frac{1}{2}$),

$$\begin{aligned}\text{numerator} = &-\big(|p_1|^2 + |p_1 - p|^2 + 4|p|^2\big)(A(p), A(-p)) + 2(\iota_p A(p), \iota_p A(-p))\\ &- 5\big[(\iota_{p_1} A(p), \iota_{p_1 - p} A(-p)) + (\iota_{p_1 - p} A(p), \iota_{p_1} A(-p))\big].\end{aligned}$$

At $p_1 = \frac{t_2}{t_1+t_2}p$, $|p_1|^2 + |p_1 - p|^2 = \big(1 - \frac{2t_1t_2}{(t_1+t_2)^2}\big)|p|^2$, and each of the two bracketed pairings equals $-\frac{t_1t_2}{(t_1+t_2)^2}(\iota_p A(p), \iota_p A(-p))$; the positive Laplacian in p_1 of the numerator is $36\,(A(p), A(-p))$.

Schwinger integral. By Lemma 4.4 the plane-wave value is

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\iint_{[\varepsilon,\infty)^2}\frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\Big[-\frac{9\,(A(p),A(-p))}{t_1+t_2}+\Big(\frac{2t_1t_2}{(t_1+t_2)^2}-5\Big)|p|^2(A(p),A(-p)) \\ +\Big(\frac{10t_1t_2}{(t_1+t_2)^2}+2\Big)(\iota_pA(p),\iota_pA(-p))\Big]dt_1dt_2.$$

ε-expansion. By Lemma 4.5 the double integral of the bracket is, modulo $\mathscr{A}_{\geq 0}$,

$$-9\Big(\tfrac{1}{4}\varepsilon^{-1}+\tfrac{1}{6}|p|^2\log\varepsilon\Big)(A(p),A(-p))+\Big(-\tfrac{1}{3}+5\Big)|p|^2\log\varepsilon\,(A(p),A(-p)) \\ +\Big(-\tfrac{10}{6}-2\Big)\log\varepsilon\,(\iota_pA(p),\iota_pA(-p)) \\ =-\tfrac{9}{4}\varepsilon^{-1}(A(p),A(-p))+\tfrac{19}{6}|p|^2\log\varepsilon\,(A(p),A(-p))-\tfrac{11}{3}\log\varepsilon\,(\iota_pA(p),\iota_pA(-p)).$$

Local part. The plane-wave coefficients are $\frac{1}{2}\cdot(-\frac{9}{4}) = -\frac{9}{8}$, $\frac{1}{2}\cdot\frac{19}{6} = \frac{19}{12}$, $\frac{1}{2}\cdot(-\frac{11}{3}) = -\frac{11}{6}$. By (4.3) these are the plane-wave coefficients of

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[-\tfrac{9}{16}\varepsilon^{-1}\int\mathrm{dvol}\,(A|_V,A|_V) \\ +\log\varepsilon\Big(\tfrac{19}{24}\int\mathrm{dvol}\,(dA|_V,dA|_V)-\tfrac{1}{8}\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)\Big)\Big],$$

since $2\cdot(-\frac{9}{16}) = -\frac{9}{8}$, $2\cdot\frac{19}{24} = \frac{19}{12}$ and $2\big(-\frac{19}{24}-\frac{1}{8}\big) = -\frac{11}{6}$. This is row (c) of the first table.

Example 4.12 (Gaugino loop, graph (e) of the $A|_V$ two-point function)**.** Amplitude. Two vertices $\lambda\lambda A|_V$ (Definition 1.13); each edge carries the $\lambda\otimes\lambda$ component $(-,-)^{-1}\otimes i\rho(\cdot)|_{\Sigma^*\to\Sigma}$ of $P(\varepsilon,\infty)$ at the momentum p_1, resp. p_1-p; the parity operator in STr gives the sign of the loop of odd fields. The prefactor and the scalar propagator factors are those of Example 4.11.

Numerator algebra. The trace over Σ of the products of Clifford multiplications, evaluated with $\rho(v)\rho(v')+\rho(v')\rho(v) = 2(v,v')$, gives

$$\text{numerator} = -16\big[(p_1,p_1-p)(A(p),A(-p))-(\iota_{p_1}A(p),\iota_{p_1-p}A(-p))-(\iota_{p_1-p}A(p),\iota_{p_1}A(-p))\big],$$

with value $\frac{16t_1t_2}{(t_1+t_2)^2}\big(|p|^2(A(p),A(-p))-2(\iota_pA(p),\iota_pA(-p))\big)$ at the centre and positive Laplacian $64\,(A(p),A(-p))$.

Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\iint_{[\varepsilon,\infty)^2}\frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\Big[-\frac{16\,(A(p),A(-p))}{t_1+t_2}+\frac{16t_1t_2}{(t_1+t_2)^2}|p|^2(A(p),A(-p)) \\ -\frac{32t_1t_2}{(t_1+t_2)^2}(\iota_pA(p),\iota_pA(-p))\Big]dt_1dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, the double integral of the bracket is $-4\varepsilon^{-1}(A(p),A(-p))-\frac{16}{3}\log\varepsilon\big(|p|^2(A(p),A(-p))-(\iota_pA(p),\iota_pA(-p))\big)$, the $|p|^2$ terms being $-16\cdot\frac{1}{6}-16\cdot\frac{1}{6}$.

Local part. $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big[-\varepsilon^{-1}\int\mathrm{dvol}\,(A|_V,A|_V)-\frac{4}{3}\log\varepsilon\int\mathrm{dvol}\,(dA|_V,dA|_V)\big]$: the $\log\varepsilon$ part of the gaugino loop is transverse.

Example 4.13 (Ghost loop, graph (f) of the $A|_V$ two-point function)**.** Amplitude. Two vertices $A|_V\, c\, A^*|_V$ (Definition 1.13); the edges carry the components $c \otimes A^*|_V$ and $A^*|_V \otimes c$ of $P(\varepsilon,\infty)$, equal to $-(-,-)^{-1}\, i$ (edge momentum) times the scalar factor (Proposition 3.17); the parity operator gives the sign of the loop of odd fields.

Numerator algebra. The two orientations of the ghost line contribute

$$\text{numerator} = (\iota_{p_1} A(p), \iota_{p_1-p} A(-p)) + (\iota_{p_1-p} A(p), \iota_{p_1} A(-p)),$$

with value $-\frac{2t_1t_2}{(t_1+t_2)^2}(\iota_p A(p), \iota_p A(-p))$ at the centre and positive Laplacian $-4\,(A(p), A(-p))$.

Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)} \iint_{[\varepsilon,\infty)^2} \frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2} \Big[\frac{(A(p), A(-p))}{t_1+t_2} - \frac{2t_1t_2}{(t_1+t_2)^2} (\iota_p A(p), \iota_p A(-p)) \Big] dt_1 dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, the double integral of the bracket is $\frac{1}{4}\varepsilon^{-1}(A(p), A(-p)) + \log\varepsilon\big(\frac{1}{6}|p|^2 (A(p), A(-p)) + \frac{1}{3}(\iota_p A(p), \iota_p A(-p))\big)$.

Local part.

$$\frac{\hbar\, \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)} \Big[\tfrac{1}{16}\varepsilon^{-1} \int \operatorname{dvol}(A|_V, A|_V) + \log\varepsilon \Big(\tfrac{1}{24} \int \operatorname{dvol}(dA|_V, dA|_V) + \tfrac{1}{8} \int \operatorname{dvol}(d^*A|_V, d^*A|_V) \Big) \Big],$$

since $2\cdot\frac{1}{16} = \frac{1}{2}\cdot\frac{1}{4}$, $2\cdot\frac{1}{24} = \frac{1}{2}\cdot\frac{1}{6}$ and $2\big(-\frac{1}{24}+\frac{1}{8}\big) = \frac{1}{2}\cdot\frac{1}{3}$. The longitudinal coefficient $\frac{1}{8}$ cancels the $-\frac{1}{8}$ of Example 4.11.

Example 4.14 (Seagull, graph (a) of the $A|_V$ two-point function)**.** Amplitude. One quartic vertex (1.5) with two legs and one $A|_V$ edge closed into a loop; the term $n = 1$ of (3.7).

Numerator algebra. Contracting two legs of (1.5) with $2(-,-)^{-1} \otimes$ (inverse metric of V) and summing over the pairs of legs gives the numerator $6\,(A(p), A(-p))$, independent of p_1.

Schwinger integral. By Lemma 4.4 with one edge,

$$\frac{\hbar}{2}\,\frac{\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)} \int_\varepsilon^\infty \frac{6\,(A(p), A(-p))}{t^2}\, dt.$$

ε-expansion. The integral equals $6\,\varepsilon^{-1}(A(p), A(-p))$ exactly; its $\varepsilon\frac{d}{d\varepsilon}$ is the graph with the edge pinned at $t = \varepsilon$, $-\varepsilon\cdot 6\,\varepsilon^{-2}(A(p), A(-p))$, as in Proposition 4.2.

Local part. $\frac{\hbar\, \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)}\, \frac{3}{2}\, \varepsilon^{-1} \int \operatorname{dvol}(A|_V, A|_V)$, a mass term with vanishing $\log\varepsilon$ part.

Example 4.15 (The graph with external $c, A^*|_V$)**.** Amplitude. Legs $r_1 A^*(p) e^{i(p,x)}$ and $r_2 c(-p) e^{-i(p,x)}$; two vertices $A|_V\, c\, A^*|_V$; one $A|_V$ edge and one c–A^* edge (graph (c) of the third figure).

Numerator algebra. The coefficient of $r_1 r_2$ is $2i\,(p\,c(-p), A^*(p))$, independent of p_1.

Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)} \iint_{[\varepsilon,\infty)^2} \frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\, 2i\,(p\,c(-p), A^*(p))\, dt_1 dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, $-2i\log\varepsilon\,(p\,c(-p), A^*(p))$.

Local part. The plane-wave coefficient is $\frac{1}{2}\cdot(-2i) = -i$, and the plane-wave value of $\int \operatorname{dvol}(dc, A^*)$ is $i\,(p\,c(-p), A^*(p))$ by (4.3); the contribution is $\frac{\hbar\, \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)} \big(-\log\varepsilon\big) \int \operatorname{dvol}(dc, A^*)$.

Theorem 4.16 (The quadratic counterterm)**.** *The part of $I^{CT}[\varepsilon]$ quadratic in the fields is*

$$\begin{aligned}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big\{&\tfrac{9}{4}\,\varepsilon^{-1}\Big[\int \mathrm{dvol}\,(A|_V,A|_V)+\int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})\Big]\\&+\log\varepsilon\Big[-\int \mathrm{dvol}\,(dA|_V,dA|_V)-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp},dA|_{W_{\mathbb{R}}^\perp})\\&\qquad+4\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)-\int \mathrm{dvol}\,(dc,A^*)\Big]\Big\}.\end{aligned}\tag{4.4}$$

The coefficient of $\int \mathrm{dvol}\,(d^*A|_V,d^*A|_V)$ is 0, and every quadratic local functional mixing $A|_V$ with $A|_{W_{\mathbb{R}}^\perp}$, or c with $A^*|_{W_{\mathbb{R}}^\perp}$, has coefficient 0. In terms of the ten-component field A,

$$\begin{aligned}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big\{&\tfrac{9}{4}\,\varepsilon^{-1}\int \mathrm{dvol}\,(A,A)\\&+\log\varepsilon\Big[-\int \mathrm{dvol}\,(dA,dA)+4\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)-\int \mathrm{dvol}\,(dc,A^*)\Big]\Big\}.\end{aligned}$$

Proof. $I^{CT}[\varepsilon]$ is even and of degree 0, so by the degrees of Chapter 1 its quadratic part has legs (A,A), (λ,λ), (c,A^*) or (c,λ^*). The regulator is covariant under $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$, since $H=d^*d+dd^*$ and Q^{GF} commute with it, so $I^{CT}[\varepsilon]$ is invariant. The invariant pairings of a vector of V with a vector of $W_{\mathbb{R}}^\perp$, of a scalar with a spinor, and of a scalar with a vector of $W_{\mathbb{R}}^\perp$ vanish; this gives the vanishing mixings and the vanishing of the sectors (c,λ^*) and $(c,A^*|_{W_{\mathbb{R}}^\perp})$. The remaining sectors are the sums of the three tables, whose rows are all one-particle-irreducible one-loop graphs with these legs; the representative rows are derived in Examples 4.11–4.15. The sums are

$$\begin{aligned}A|_V:&\quad \tfrac{3}{2}+3-\tfrac{9}{16}-\tfrac{3}{4}-1+\tfrac{1}{16}=\tfrac{9}{4},\qquad \tfrac{19}{24}-\tfrac{1}{2}-\tfrac{4}{3}+\tfrac{1}{24}=-1,\qquad -\tfrac{1}{8}+\tfrac{1}{8}=0;\\ A|_{W_{\mathbb{R}}^\perp}:&\quad 2+\tfrac{5}{2}-\tfrac{1}{4}-2=\tfrac{9}{4},\qquad 1-2=-1;\qquad \lambda:\quad 1+3=4;\qquad (c,A^*):\quad -1.\end{aligned}$$

The ten-component form uses $(dA,dA)=(dA|_V,dA|_V)+(dA|_{W_{\mathbb{R}}^\perp},dA|_{W_{\mathbb{R}}^\perp})$, the cross term vanishing by the orthogonality of V and $W_{\mathbb{R}}^\perp$. □

Remark 4.17 (Two properties of the quadratic counterterm)**.** The per-graph entries of the $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ sectors differ, and their totals agree: (4.4) is a functional of the ten-component field A. The $\log\varepsilon$ part is transverse, the longitudinal coefficients $-\frac{1}{8}$ of the gauge bubble and $+\frac{1}{8}$ of the ghost loop cancelling, and it is a sum of multiples of the three terms of the free action (1.2):

$$\begin{aligned}&-\int \mathrm{dvol}\,(dA,dA)+4\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)-\int \mathrm{dvol}\,(dc,A^*)\\&\quad=4\Big(-\tfrac{1}{4}\int \mathrm{dvol}\,(dA,dA)\Big)+8\Big(\tfrac{1}{2}\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)\Big)-1\cdot\int \mathrm{dvol}\,(dc,A^*).\end{aligned}$$

The ε^{-1} part is a mass term for all ten components of A.

4.3 Cubic and quartic graphs

The cubic and quartic parts of $I^{CT}[\varepsilon]$ are organised by the Weyl weights of Example 2.13: A has weight 1, λ weight $\frac{3}{2}$, c weight 0, and the antifields A^*,λ^*,c^* the complementary weights $3,\frac{5}{2},4$, so that each pairing of a field with its antifield has weight 4. The weight of a local functional $\int \mathrm{dvol}\,(\cdots)$ is the sum of the weights of its fields plus the number of its derivatives, and every term of S has weight 4. The superficial degree of a graph is 4 minus the sum of the weights of its legs: 2 for (A,A), 1 for (λ,λ), (c,A^*), $(A|_V,A|_V,A|_V)$, $(A|_V,A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$, and 0 for the other cubic and the quartic leg sets below.

Lemma 4.18 (Weights of the singular coefficients)**.** *Let γ be a one-loop one-particle-irreducible graph with vertices labelled by I. Its superficial degree in the sense of Lemma 4.6 is* 4 *minus the sum of the weights of its legs. The coefficient of ε^{-k} in its singular part is a local functional of weight $4-2k$, and the coefficient of $\log\varepsilon$ a local functional of weight* 4*; in particular the coefficient of ε^{-1} has weight* 2*.*

Proof. In momentum space, a vertex with fields of total weight $4-k$ is a polynomial of degree k in the momenta, since every term of S has weight 4; an edge component of $P(\varepsilon,\infty)$ between slots of total weight $2+k$ is a polynomial of degree k times $\int_\varepsilon^\infty e^{-t_e|\text{momentum}|^2}dt_e$ (Proposition 3.17: degree 0 for $A\otimes A$, 1 for $c\otimes A^*$ and $\lambda\otimes\lambda$). Each internal half-edge contributes its weight once to a vertex and once to an edge, so, for legs in fixed slots and every assignment of slots to the internal half-edges, the numerator of γ is homogeneous of degree $4|V(\gamma)|-2|E(\gamma)|$ minus the sum of the weights of the legs, in the loop momentum p_1 and the leg momenta, which is the homogeneity required in Lemma 4.6; with $|E(\gamma)|=|V(\gamma)|$ at one loop, its degree plus $4-2|E(\gamma)|$ is 4 minus the sum of the weights of the legs. By Lemma 4.6 the coefficient of ε^{-k} is a polynomial in the leg momenta, homogeneous of degree (superficial degree)$-2k$, and that of $\log\varepsilon$ is homogeneous of degree equal to the superficial degree. The weight of such a local coefficient is the sum of the leg weights, 4−(superficial degree), plus its degree in the momenta, which gives $4-2k$ and 4. □

Lemma 4.19 (Local functionals of weight at most four)**.** *Let a local functional of degree* 0*, without constant term, have weight at most* 4 *and be invariant under* $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$ *and the adjoint action of $\mathfrak{g}$. If it is even, it is a linear combination of: in weight* 2*, $\int\mathrm{dvol}\,(A|_V,A|_V)$ and $\int\mathrm{dvol}\,(A|_{W_\mathbb{R}^\perp},A|_{W_\mathbb{R}^\perp})$; in weight* 4*, functionals whose monomials have one of the field contents*

$$
\begin{array}{ll}
(A|_V)^2 \text{ and } (A|_{W_\mathbb{R}^\perp})^2 \text{ with two derivatives}; & \lambda\lambda \text{ and } c\,A^*|_V \text{ with one derivative};\\
(A|_V)^3 \text{ and } A|_V(A|_{W_\mathbb{R}^\perp})^2 \text{ with one derivative}; & \lambda\lambda A|_V,\ \lambda\lambda A|_{W_\mathbb{R}^\perp},\\
c\,A|_V\,A^*|_V,\ c\,A|_{W_\mathbb{R}^\perp}A^*|_{W_\mathbb{R}^\perp},\ c\,\lambda\,\lambda^*,\ c\,c\,c^*; & (A|_V)^4,\ (A|_V)^2(A|_{W_\mathbb{R}^\perp})^2,\ (A|_{W_\mathbb{R}^\perp})^4,
\end{array}
$$

In weights 1 *and* 3 *there are none.*

Proof. As functionals, c has degree $+1$, A^* and λ^* degree -1, c^* degree -2, and A,λ degree 0 (Chapter 1). Degree 0 requires the number of c to equal the number of A^* and λ^* plus twice the number of c^*, and weight at most 4 allows at most one antifield. With c^* (weight 4): $c\,c\,c^*$. With A^* (weight 3): one c and weight at most 1 more, i.e. $c\,A^*$ with one derivative or $c\,A\,A^*$; invariance pairs $A^*|_V$ with the derivative or with $A|_V$, and $A^*|_{W_\mathbb{R}^\perp}$ with $A|_{W_\mathbb{R}^\perp}$. With λ^* (weight $\frac{5}{2}$): one c and, by $\mathrm{Spin}(4,\mathbb{C})$, one spinor λ. Without antifields there is no c; the monomials are built from A, λ and derivatives, with an even number of spinors. $\mathrm{Spin}(6,\mathbb{C})$ admits invariants of an even number of vectors of $W_\mathbb{R}^\perp$ (the invariant volume form of $W_\mathbb{R}^\perp$ needs six) and $\mathrm{Spin}(4,\mathbb{C})$ of an even number of vector indices from $A|_V$ and the derivatives. In weight 2 this leaves $(A|_V)^2$, $(A|_{W_\mathbb{R}^\perp})^2$; in weight 4 the listed contents, a single A with three derivatives being a total derivative; in weight 1 the single A, without invariant; in weight 3 the contents A^3, $A\,A$ with one derivative, $c\,A^*$, each with an odd number of vector indices of V or of $W_\mathbb{R}^\perp$, and $\lambda\lambda$, whose components lie in $S_+\otimes S_-$ (a vector of V) or in $W\otimes W$, $W^*\otimes W^*$ (without $SL(4,\mathbb{C})$-invariant); all of these have vanishing invariants. □

Proposition 4.20 (Structure of the one-loop counterterm)**.** *$I^{CT}[\varepsilon]$ is the sum of ε^{-1} times the mass term of Theorem 4.16 and $\log\varepsilon$ times a local functional of weight* 4 *with the field contents of Lemma 4.19. Its $\log\varepsilon$ part is therefore determined by the one-particle-irreducible one-loop graphs with the leg sets $(A|_V,A|_V)$, $(A|_{W_\mathbb{R}^\perp},A|_{W_\mathbb{R}^\perp})$, (λ,λ), $(c,A^*|_V)$, the eight cubic sets $(A|_V,A|_V,A|_V)$, $(A|_V,A|_{W_\mathbb{R}^\perp},A|_{W_\mathbb{R}^\perp})$, $(\lambda,\lambda,A|_V)$, $(\lambda,\lambda,A|_{W_\mathbb{R}^\perp})$, $(c,A|_V,A^*|_V)$, $(c,A|_{W_\mathbb{R}^\perp},A^*|_{W_\mathbb{R}^\perp})$, (c,λ,λ^*), (c,c,c^*), and the three quartic sets $(A|_V)^4$, $(A|_V)^2(A|_{W_\mathbb{R}^\perp})^2$, $(A|_{W_\mathbb{R}^\perp})^4$.*

Proof. By Lemmas 4.6 and 4.18, the singular part of a one-particle-irreducible one-loop graph with vertices labelled by I consists of ε^{-k} times a local functional of weight $4-2k$, $1 \le k \le \frac{1}{2}$(superficial degree), and $\log\varepsilon$ times a local functional of weight 4, and it vanishes when the superficial degree is negative. The slots of every vertex of I have degrees adding to 0, and so have the two slots joined by every nonzero component of $P(\varepsilon,\infty)$ (Proposition 3.17: $A\otimes A$, $c\otimes A^*|_V$, $A^*|_V\otimes c$, $\lambda\otimes\lambda$); hence the legs of every graph have degrees adding to 0. With the degrees of the proof of Lemma 4.19, a set of n legs with degrees adding to 0 has weight at least n: every leg other than c has weight at least 1, every c has weight 0 and degree $+1$, and the antifields exceed weight 1 by $\frac{3}{2}$ per unit of negative degree or more (A^*: $3 = 1+2$, λ^*: $\frac{5}{2} = 1+\frac{3}{2}$, c^*: $4 = 1+\frac{3}{2}\cdot 2$), so the weight is at least $n-(\text{number of } c)+\frac{3}{2}(\text{number of } c) \ge n$. The graphs with one leg have value 0 (Lemma 4.7). Hence the superficial degree, 4 minus the weight of the legs, is at most 2, and negative for five or more legs; by Lemma 4.6 no power ε^{-k} with $k \ge 2$ occurs, and the coefficient of ε^{-1} has weight 2. $I^{CT}[\varepsilon]$ is invariant under $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$ and the adjoint action of $\mathfrak{g}$ (proof of Theorem 4.16), so by Lemma 4.19 the coefficient of ε^{-1} is a combination of the two mass terms, computed in Theorem 4.16, and the coefficient of $\log\varepsilon$ has the listed contents. □

4.3.1 Cubic graphs

Every cubic graph below has vanishing coefficient of ε^{-1}, in accordance with Lemma 4.18. Its $\log\varepsilon$ coefficient, as a multilinear form in the three legs, is a rational multiple of the tree vertex function of Definition 1.13 with the same legs; the tables give these multiples, in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$. The tree vertices are: the $(A|_V)^3$ and $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ parts of $\int \mathrm{dvol}\big(-\frac{1}{2}(dA,\frac{1}{2}[A\wedge A])\big)$; the $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ parts of $\int \mathrm{dvol}\,\frac{1}{2}(\lambda,(\not{d}_A-\not{d})\lambda)$; the $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ parts of $\int \mathrm{dvol}\,([A,c],A^*)$; $\int \mathrm{dvol}\,([c,\lambda],\lambda^*)$; $\int \mathrm{dvol}\,\frac{1}{2}([c,c],c^*)$.

[figure] One-loop one-particle-irreducible graphs with external $(A|_V, A|_V, A|_V)$: triangles with (a) $A|_V$, (b) $A|_{W_{\mathbb{R}}^\perp}$, (c) λ lines, (d), (e) the two orientations of the c–A^* loop; bubbles with one quartic vertex and (f)–(h) $A|_V$, (i)–(k) $A|_{W_{\mathbb{R}}^\perp}$ lines, one for each leg at the cubic vertex.

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$	$A\|_V$, $A\|_V$, $A\|_V$	1	1	$\frac{13}{4}$
(b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	2
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	λ, λ, λ	1	1	$\frac{16}{3}$
(d), (e)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	c–A^*, c–A^*, c–A^*	2	1	$-\frac{1}{12}$
(f)–(h)	$(A\|_V)^3$, $(A\|_V)^4$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$	$-\frac{9}{2}$
(i)–(k)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	$\frac{1}{2}$	0
total					6

The $\log\varepsilon$ coefficients of the graphs with external $(A|_V, A|_V, A|_V)$, as multiples of the tree vertex.

[figure] One-loop one-particle-irreducible graphs with external $(A|_V, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$: triangles (a)–(c); bubbles with one quartic vertex (d)–(g).

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$\frac{3}{2}$
(b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	1
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ	1	1	8
(d)	$(A\|_V)^3$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	1	$\frac{1}{2}$	0
(e)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	$\frac{1}{2}$	0
(f), (g)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$-\frac{9}{2}$
total					6

The $\log\varepsilon$ coefficients of the graphs with external $(A|_V, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, as multiples of the tree vertex.

[figure] One-loop one-particle-irreducible graphs with external $(\lambda, \lambda, A|_V)$.

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$A\|_V$, $A\|_V$, λ	1	1	3
(b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, λ	1	1	3
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$A\|_V$, λ, λ	1	1	1
(d)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_{W_{\mathbb{R}}^\perp}$, λ, λ	1	1	3
total					10

The $\log\varepsilon$ coefficients of the graphs with external $(\lambda, \lambda, A|_V)$, as multiples of the tree vertex; the tree vertex pairs the two four-dimensional chiralities $S_+ \otimes W$ and $S_- \otimes W^*$ of λ.

[figure] One-loop one-particle-irreducible graphs with external $(\lambda, \lambda, A|_{W_{\mathbb{R}}^\perp})$; (a), (b) differ by the leg attached to the vertex $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$.

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a), (b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, λ	2	1	2
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_V$, λ, λ	1	1	4
(d)	$\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_{W_{\mathbb{R}}^\perp}$, λ, λ	1	1	4
total					10

The $\log\varepsilon$ coefficients of the graphs with external $(\lambda, \lambda, A|_{W_{\mathbb{R}}^\perp})$, as multiples of the tree vertex. The tree vertex has two structures, both λ in $S_+ \otimes W$ or both in $S_- \otimes W^*$ (Proposition 1.2); the total is 10 times the tree in each of them.

[figure] One-loop one-particle-irreducible graphs with external $(c, A|_V, A^*|_V)$.

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	$A\|_V$, $A\|_V$, c–A^*	1	1	$\frac{3}{4}$
(b)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	$A\|_V$, c–A^*, c–A^*	1	1	$\frac{1}{4}$
total					1

The $\log\varepsilon$ coefficients of the graphs with external $(c, A|_V, A^*|_V)$, as multiples of the tree vertex.

[figure] The one-loop one-particle-irreducible graph with external $(c, A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$.

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V\, c\, A^*\|_V$, $A\|_{W_{\mathbb{R}}^\perp}\, c\, A^*\|_{W_{\mathbb{R}}^\perp}$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, c–A^*	1	1	1
total					1

The $\log\varepsilon$ coefficient of the graph with external $(c, A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$, as a multiple of the tree vertex.

[figure] The one-loop one-particle-irreducible graph with external (c, λ, λ^*).

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$c\,\lambda\,\lambda^*$, $A\|_V\, c\, A^*\|_V$, $\lambda\lambda A\|_V$	$A\|_V$, c–A^*, λ	1	1	1
total					1

The $\log\varepsilon$ coefficient of the graph with external (c, λ, λ^*), as a multiple of the tree vertex, for λ in either chirality.

[figure] The one-loop one-particle-irreducible graph with external (c, c, c^*).

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$c\,c\,c^*$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	$A\|_V$, c–A^*, c–A^*	1	1	1
total					1

The $\log\varepsilon$ coefficient of the graph with external (c, c, c^*), as a multiple of the tree vertex.

4.3.2 Quartic graphs

For the quartic sets leg $j \in \{1,2,3,4\}$ carries a decomposable fibre element $(v_j, -) \otimes X_j$ with $X_j \in \mathfrak{g}$, $v_j \in V$ for a leg $A|_V$ and $v_j \in W_{\mathbb{R}}^{\perp}$ for a leg $A|_{W_{\mathbb{R}}^{\perp}}$; in the set $(A|_V)^2(A|_{W_{\mathbb{R}}^{\perp}})^2$ legs $1,2$ are $A|_V$ and legs $3,4$ are $A|_{W_{\mathbb{R}}^{\perp}}$. A multilinear form is determined by its values on such legs, and by Lemma 4.18 the $\log\varepsilon$ coefficients are independent of the momenta. They are written in the basis of products of a colour factor and a Lorentz factor:

- colour factors: the pairing $([X_1,X_2],[X_3,X_4])$ of the bracket of the colour elements of legs $1,2$ with the bracket of those of legs $3,4$; the pairing $([X_1,X_4],[X_2,X_3])$; and the averaged trace $\frac{1}{24}\sum_{\text{orderings}} \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}\mathrm{ad}_{X_3}\mathrm{ad}_{X_4})$, the fourth-order adjoint trace averaged over the 24 orderings of $X_1,\dots,X_4$. The third pairing $([X_1,X_3],[X_2,X_4])$ is the sum of the first two, by the Jacobi identity and the invariance of $(-,-)$.
- Lorentz factors: $(v_1,v_2)(v_3,v_4)$, $(v_1,v_3)(v_2,v_4)$, $(v_1,v_4)(v_2,v_3)$, products of metric pairings.

The two bracket pairings carry the unit $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$, the averaged trace the unit $\hbar/16\pi^2$. On such legs, by the Hodge normalisation of Definition 1.8,

$$\begin{aligned}&([A(p_1)\wedge A(p_2)],[A(p_3)\wedge A(p_4)])\\ &\quad=([X_1,X_2],[X_3,X_4])\big((v_1,v_3)(v_2,v_4)-(v_1,v_4)(v_2,v_3)\big),\end{aligned}$$

and similarly for the other two terms of (1.5). With the Jacobi identity in the form

$$([X_1,X_3],[X_2,X_4])=([X_1,X_2],[X_3,X_4])+([X_1,X_4],[X_2,X_3]),$$

the tree vertex function (1.5) is

$$\begin{aligned}-\tfrac{1}{2}\Big[&([X_1,X_2],[X_3,X_4])\big((v_1,v_2)(v_3,v_4)+(v_1,v_3)(v_2,v_4)-2(v_1,v_4)(v_2,v_3)\big)\\ &+([X_1,X_4],[X_2,X_3])\big(2(v_1,v_2)(v_3,v_4)-(v_1,v_3)(v_2,v_4)-(v_1,v_4)(v_2,v_3)\big)\Big],\end{aligned}\tag{4.5}$$

and for $(A|_V)^2(A|_{W_{\mathbb{R}}^{\perp}})^2$, where $(v_1,v_3)=(v_1,v_4)=0$, it is

$$-\tfrac{1}{2}\big[([X_1,X_2],[X_3,X_4])+2([X_1,X_4],[X_2,X_3])\big](v_1,v_2)(v_3,v_4).$$

[figure] Box graphs with external $(A|_V)^4$. Columns: the three cyclic orders of the legs. Rows: loops of $A|_V$ (a)–(c), $A|_{W_{\mathbb{R}}^{\perp}}$ (d)–(f), λ (g)–(i), and c–A^* in the two orientations (j)–(o).

[figure] Triangle graphs (one quartic vertex) with external $(A|_V)^4$: loops of $A|_V$ (a)–(f) and of $A|_{W_{\mathbb{R}}^{\perp}}$ (g)–(l).

[figure] Bubble graphs (two quartic vertices) with external $(A|_V)^4$: $A|_V$ lines (a)–(c), $A|_{W_{\mathbb{R}}^{\perp}}$ lines (d)–(f).

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$
box (a)–(c)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$	$A\|_V$, $A\|_V$, $A\|_V$, $A\|_V$	3	1
box (d)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$	$A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$	3	1
box (g)–(i)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	λ, λ, λ, λ	3	1
box (j)–(o)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	c–A^*, c–A^*, c–A^*, c–A^*	6	1
triangle (a)–(f)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^4$	$A\|_V$, $A\|_V$, $A\|_V$	6	1
triangle (g)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^{\perp}})^2$	$A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$	6	1
bubble (a)–(c)	$(A\|_V)^4$, $(A\|_V)^4$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$
bubble (d)–(f)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^{\perp}})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^{\perp}})^2$	$A\|_{W_{\mathbb{R}}^{\perp}}$, $A\|_{W_{\mathbb{R}}^{\perp}}$	3	$\frac{1}{2}$

The classes of one-loop graphs with external $(A|_V)^4$; the panel letters refer to the box, triangle and bubble figures of this set.

colour	Lorentz	box (a)–(c)	box (d)–(f)	box (g)–(i)	box (j)–(o)	triangle (a)–(f)	triangle (g)–(l)	bubble (a)–(c)	bubble (d)–(f)	total	8×tree
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_2)(v_3,v_4)$	$\frac{5}{12}$	0	$-\frac{8}{3}$	0	$-\frac{13}{4}$	-2	$\frac{5}{2}$	1	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_3)(v_2,v_4)$	$\frac{5}{12}$	0	$-\frac{8}{3}$	0	$-\frac{13}{4}$	-2	$\frac{5}{2}$	1	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{5}{6}$	0	$\frac{16}{3}$	0	$\frac{13}{2}$	4	-5	-2	8	8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_2)(v_3,v_4)$	$\frac{5}{6}$	0	$-\frac{16}{3}$	0	$-\frac{13}{2}$	-4	5	2	-8	-8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_3)(v_2,v_4)$	$-\frac{5}{12}$	0	$\frac{8}{3}$	0	$\frac{13}{4}$	2	$-\frac{5}{2}$	-1	4	4
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{5}{12}$	0	$\frac{8}{3}$	0	$\frac{13}{4}$	2	$-\frac{5}{2}$	-1	4	4
averaged trace	$(v_1,v_2)(v_3,v_4)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0
averaged trace	$(v_1,v_3)(v_2,v_4)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0
averaged trace	$(v_1,v_4)(v_2,v_3)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0

The $\log\varepsilon$ coefficients of the graphs with external $(A|_V)^4$ in the basis of colour and Lorentz factors, summed over each class, with the total and 8 times the tree vertex (4.5). Rows with a bracket pairing are in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$, rows with the averaged trace in units of $\hbar/16\pi^2$.

[figure] Box graphs with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$; (g)–(i): gaugino loops in the three cyclic orders of the legs.

[figure] Triangle graphs (one quartic vertex) with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$.

[figure] Bubble graphs (two quartic vertices) with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$.

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$	$([X_1,X_2],[X_3,X_4])\times(v_1,v_2)(v_3,v_4)$	$([X_1,X_4],[X_2,X_3])\times(v_1,v_2)(v_3,v_4)$	averaged trace $\times(v_1,v_2)(v_3,v_4)$
box (a), (b)	$(A\|_V)^3$, $(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{1}{8}$	$\frac{1}{4}$	$-\frac{3}{2}$
box (c), (f)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	0	0	0
box (d), (e)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{1}{6}$	$\frac{1}{3}$	-2
box (g)–(i)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ, λ	3	1	-4	-8	0
triangle (a)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_V$	1	1	$-\frac{3}{4}$	$-\frac{3}{2}$	9
triangle (b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$-\frac{5}{6}$	$-\frac{5}{3}$	10
triangle (c)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^4$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$-\frac{1}{8}$	$-\frac{1}{4}$	$\frac{3}{2}$
triangle (d), (i)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	5	1	$-\frac{5}{24}$	$-\frac{5}{12}$	4
triangle (e)–(h)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	4	1	$-\frac{5}{4}$	$-\frac{5}{2}$	-3
bubble (a)	$(A\|_V)^4$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	1	$\frac{1}{2}$	$\frac{1}{2}$	1	-6
bubble (b)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	$\frac{1}{2}$	$\frac{5}{6}$	$\frac{5}{3}$	-10
bubble (c), (d)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{13}{6}$	$\frac{13}{3}$	-2
total					-4	-8	0
8×tree					-4	-8	0

The $\log\varepsilon$ coefficients of the graphs with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$ (legs $1,2$ along V, legs $3,4$ along $W_{\mathbb{R}}^\perp$), summed over each class, with the total and 8 times the tree vertex; units as in the previous table.

[figure] Box graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$: alternating $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ lines (a)–(f), gaugino loops (g)–(i).

[figure] Triangle graphs (one quartic vertex) with external $(A|_{W_{\mathbb{R}}^\perp})^4$: quartic vertex $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$ (a)–(f) and $(A|_{W_{\mathbb{R}}^\perp})^4$ (g)–(l).

[figure] Bubble graphs (two quartic vertices) with external $(A|_{W_{\mathbb{R}}^\perp})^4$: $A|_V$ lines (a)–(c), $A|_{W_{\mathbb{R}}^\perp}$ lines (d)–(f).

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$
box (a)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
box (g)–(i)	$\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ, λ	3	1
triangle (a)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
triangle (g)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
bubble (a)–(c)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$
bubble (d)–(f)	$(A\|_{W_{\mathbb{R}}^\perp})^4$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	$\frac{1}{2}$

The classes of one-loop graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$.

colour	Lorentz	box (a)–(f)	box (g)–(i)	triangle (a)–(f)	triangle (g)–(l)	bubble (a)–(c)	bubble (d)–(f)	total	8×tree
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_2)(v_3,v_4)$	$\frac{1}{6}$	$-\frac{16}{3}$	$-\frac{1}{3}$	-2	$\frac{2}{3}$	$\frac{17}{6}$	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_3)(v_2,v_4)$	$\frac{1}{6}$	$-\frac{16}{3}$	$-\frac{1}{3}$	-2	$\frac{2}{3}$	$\frac{17}{6}$	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{1}{3}$	$\frac{32}{3}$	$\frac{2}{3}$	4	$-\frac{4}{3}$	$-\frac{17}{3}$	8	8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_2)(v_3,v_4)$	$\frac{1}{3}$	$-\frac{32}{3}$	$-\frac{2}{3}$	-4	$\frac{4}{3}$	$\frac{17}{3}$	-8	-8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_3)(v_2,v_4)$	$-\frac{1}{6}$	$\frac{16}{3}$	$\frac{1}{3}$	2	$-\frac{2}{3}$	$-\frac{17}{6}$	4	4
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{1}{6}$	$\frac{16}{3}$	$\frac{1}{3}$	2	$-\frac{2}{3}$	$-\frac{17}{6}$	4	4
averaged trace	$(v_1,v_2)(v_3,v_4)$	-2	16	4	0	-8	-10	0	0
averaged trace	$(v_1,v_3)(v_2,v_4)$	-2	16	4	0	-8	-10	0	0
averaged trace	$(v_1,v_4)(v_2,v_3)$	-2	16	4	0	-8	-10	0	0

The $\log\varepsilon$ coefficients of the graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$ in the basis of colour and Lorentz factors, with the total and 8 times the tree vertex (4.5); units as in the $(A|_V)^4$ table.

In each quartic set the averaged traces cancel in the total, so that the total is a combination of the two bracket pairings, and the total equals 8 times the tree vertex.

4.3.3 The logarithmic part as a rescaling of the fields

Write $\int(A,A^*)$, $\int(\lambda^*,\lambda)$, $\int(c^*,c)$ for the local functionals $e\mapsto\langle e_A,e_{A^*}\rangle$, $\langle e_{\lambda^*},e_\lambda\rangle$, $\langle e_{c^*},e_c\rangle$, the pairing of Definition 1.5 applied to the components of the field e in the two slots of a pair, the even member first; the first uses all ten components of A. For a monomial in the fields, let $n_A,n_{A^*},n_\lambda,n_{\lambda^*},n_c,n_{c^*}$ denote the numbers of its factors of each kind.

Lemma 4.21 (The bracket as a field count)**.** *For every monomial F in the fields,*

$$\Big\{\int(A,A^*),F\Big\}=(n_A-n_{A^*})F,\qquad \Big\{\int(\lambda^*,\lambda),F\Big\}=(n_{\lambda^*}-n_\lambda)F,\qquad \Big\{\int(c^*,c),F\Big\}=(n_{c^*}-n_c)F.$$

Hence, on every monomial of S,

$$\Big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),\,F\Big\}=\big(2(n_A-n_{A^*})+4(n_\lambda-n_{\lambda^*})+(n_c-n_{c^*})\big)F.$$

Proof. By Definition 1.6, the kernel K_0 pairs each even member A,λ^*,c^* with its odd partner A^*,λ,c with the coefficient $-\delta(x-y)$ times the inverse fibre pairing, which is the structure $K_0=-\sum(x_i\otimes\xi_i+\xi_i\otimes x_i)$, $\Delta_0=\sum\partial_{x_i}\partial_{\xi_i}$ of [C07, §10.1] for the manifold a point, in the notation of loc. cit.: x_i an even coordinate and ξ_i its odd partner, applied pointwise at each position. For one pair, $x_i\xi_i$ is odd, and

$$\begin{aligned}\{x_i\xi_i,x_i\}&=\Delta_0(x_i^2\xi_i)-\Delta_0(x_i\xi_i)\,x_i+x_i\xi_i\,\Delta_0(x_i)=2x_i-x_i=x_i,\\ \{x_i\xi_i,\xi_i\}&=\Delta_0(x_i\xi_i\xi_i)-\Delta_0(x_i\xi_i)\,\xi_i+x_i\xi_i\,\Delta_0(\xi_i)=-\xi_i.\end{aligned}$$

The bracket with a fixed functional is a derivation, because Δ_0 is a second-order operator, so $\{x_i\xi_i,-\}=x_i\partial_{x_i}-\xi_i\partial_{\xi_i}$ counts x_i minus ξ_i. The three functionals of the lemma are the sums of $x_i\xi_i$ over the components, with x_i the even member, which gives the three identities; the combination follows by linearity. □

Theorem 4.22 (The logarithmic part of the one-loop counterterm)**.** *The $\log\varepsilon$ part of $I^{CT}[\varepsilon]$ is*

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\log\varepsilon\,\big(Q+\{I,-\}\big)\Big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big).\tag{4.6}$$

Equivalently, it multiplies each monomial of S by $2(n_A-n_{A^*})+4(n_\lambda-n_{\lambda^*})+(n_c-n_{c^*})$: the bracket acts as a weighted field count with weights 2 for A, 4 for λ, 1 for c, and the opposite weights for the antifields.

Proof. The functional $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ is odd. By Proposition 1.11, Q applied to it equals its bracket with $\frac12\langle e,Qe\rangle$, so

$$\begin{aligned}\big(Q+\{I,-\}\big)\Big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big)&=\Big\{S,\,2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big\}\\&=\Big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),\,S\Big\}\end{aligned}$$

by the graded symmetry of Definition 1.6, S being even. By Lemma 4.21, this bracket multiplies the terms of S by

$$\begin{aligned}&-\tfrac14\int\mathrm{dvol}\,(dA,dA):\ 4,\qquad \tfrac12\int\mathrm{dvol}\,(\lambda,\not d\lambda):\ 8,\qquad \int\mathrm{dvol}\,(dc,A^*):\ 1-2=-1,\\&\int\mathrm{dvol}\,\big(-\tfrac12(dA,\tfrac12[A\wedge A])\big):\ 6,\qquad \int\mathrm{dvol}\,\tfrac12(\lambda,(\not d_A-\not d)\lambda):\ 8+2=10,\\&\int\mathrm{dvol}\,([A,c],A^*):\ 2+1-2=1,\qquad \int\mathrm{dvol}\,([c,\lambda],\lambda^*):\ 1+4-4=1,\\&\int\mathrm{dvol}\,\tfrac12([c,c],c^*):\ 2-1=1,\qquad \int\mathrm{dvol}\,\big(-\tfrac14(\tfrac12[A\wedge A],\tfrac12[A\wedge A])\big):\ 8.\end{aligned}$$

These are the computed multiples: the quadratic ones by Theorem 4.16 and Remark 4.17 (with the coefficient 0 of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$, absent from S), the cubic ones $6, 6, 10, 10, 1, 1, 1, 1$ by the totals of the eight cubic tables, and the quartic ones $8, 8, 8$ by the totals of the three quartic tables. In the cubic and quartic sets the computed coefficient equals the multiple of the tree vertex as a multilinear form, including the two chirality structures of $(\lambda, \lambda, A|_{W_{\mathbb{R}}^\perp})$ and of (c, λ, λ^*). By Proposition 4.20 these leg sets carry the whole $\log\varepsilon$ part of $I^{CT}[\varepsilon]$, so the two local functionals coincide. □

Corollary 4.23 (The coupling counterterm vanishes at one loop)**.** *Suppose the* $\log\varepsilon$ *part of* $I^{CT}[\varepsilon]$ *is written as* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon$ *times the sum of* $\{S,-\}$ *of an element of the span of* $\int(A|_V, A^*|_V)$, $\int(A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$, $\int(c^*, c)$ *and the two chiral parts of* $\int(\lambda^*, \lambda)$*, and a multiple of* S*. Then the multiple of* S *is* 0 *and the element is* $2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$*. The same holds with* S *replaced by its classical part, the terms of* S *without antifields. Since the coupling constant sits in* $(-,-)$ *and enters* S *as an overall factor, the one-loop* $\log\varepsilon$ *counterterm is a rescaling of the fields with vanishing component along the coupling.*

Proof. By Lemma 4.21, $\{S,-\}$ of an element of the span multiplies each monomial of S by a sum of field counts weighted by the coefficients of the element, additive over the factors of the monomial, and a multiple of S adds the same number to every monomial. The terms of S built from $A|_V$ alone contain 2, 3, 4 factors $A|_V$, with computed multiples 4, 6, 8: an additive weight times $2, 3, 4$ plus a common constant fits $4, 6, 8$ with weight 2 and constant $3\cdot 4 - 2\cdot 6 = 0$. With the constant 0: the kinetic terms of $A|_{W_{\mathbb{R}}^\perp}$ and λ give the weight 2 of $A|_{W_{\mathbb{R}}^\perp}$ and the sum 8 of the two chiral weights of λ; the vertex $\lambda\lambda A|_{W_{\mathbb{R}}^\perp}$, whose two structures have both λ of the same chirality, gives each chiral weight $(10-2)/2 = 4$; the ghost kinetic term $\int \mathrm{dvol}\,(dc, A^*)$ gives the weight $-1+2 = 1$ of c; the remaining eight multiples are then reproduced. For the classical part: by Lemma 4.21 every term of S containing an antifield has $n_c - n_{c^*} = 1$ and every classical term has $n_c = n_{c^*} = 0$, so S minus its classical part equals $\{S, -\int(c^*, c)\}$, an element of the span; replacing S by its classical part shifts the element by a multiple of $\int(c^*, c)$ and leaves the multiple unchanged. □

Remark 4.24 (The mass term has nonzero BV variation)**.** The ε^{-1} part of $I^{CT}[\varepsilon]$, the mass term $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{4}\varepsilon^{-1}\int \mathrm{dvol}\,(A, A)$, has nonzero variation under $Q + \{I,-\}$. Indeed $\{I, \int \mathrm{dvol}\,(A, A)\}$ pairs the A^* of $\int \mathrm{dvol}\,([A, c], A^*)$ with A and is a multiple of $\int \mathrm{dvol}\,([A, c], A) = -\int \mathrm{dvol}\,(c, [A, A]) = 0$, the bracket $[A, A]$ contracting the symmetric metric with the antisymmetric bracket of $\mathfrak{g}$. By Proposition 1.11, Q inserts $Qc = -dc$ into the slot of $A|_V$, so

$$\big(Q + \{I,-\}\big)\Big(\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{4}\varepsilon^{-1}\int \mathrm{dvol}\,(A, A)\Big) = -\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{2}\varepsilon^{-1}\int \mathrm{dvol}\,(dc, A|_V).$$

This is the breaking of the gauge symmetry by the regulator. The next section shows how the master equation at the UV boundary absorbs it and what it leaves behind.

4.4 The quantum master equation at the UV boundary

Definition 4.25 (Defect vertex)**.** For an even functional J with values in $\mathbb{C}[[\hbar]]$, at least cubic modulo $\hbar$, the defect vertex at scale ε is the odd functional

$$QJ + \tfrac{1}{2}\{J, J\}_\varepsilon + \hbar\Delta_\varepsilon J,$$

with Δ_ε and $\{-,-\}_\varepsilon$ of Definition 3.21. By Corollary 3.24, its vanishing is the quantum master equation at scale ε.

Proposition 4.26 (The Q-variation at scale ∞ is the flow of the defect)**.** *On functionals without superconformal ghosts,*

$$Q\,W(P(\varepsilon,\infty),J) = e^{-W(P(\varepsilon,\infty),J)/\hbar}\, e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big(QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)\,e^{J/\hbar}\Big). \tag{4.7}$$

Thus $Q\,W(P(\varepsilon,\infty),J)$ is the sum of the connected graphs with exactly one vertex labelled by the defect vertex, all other vertices by J, and all edges by $P(\varepsilon,\infty)$, with the weights of Proposition 3.35 [C07, §15.4].

Proof. For an even J, the product rule of Definition 3.21 gives $(Q+\hbar\Delta_\varepsilon)e^{J/\hbar} = \hbar^{-1}\big(QJ+\frac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)e^{J/\hbar}$, as in the proof of Corollary 3.24. Theorem 3.23 without superconformal ghosts gives

$$Q\,e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(Q+\hbar\Delta_\varepsilon\big)e^{J/\hbar} = \hbar^{-1}e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big(QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)\,e^{J/\hbar}\Big).$$

The left side is $Q\,e^{W(P(\varepsilon,\infty),J)/\hbar} = \hbar^{-1}\big(Q\,W(P(\varepsilon,\infty),J)\big)e^{W(P(\varepsilon,\infty),J)/\hbar}$, Q being a derivation; multiplying by $\hbar\,e^{-W(P(\varepsilon,\infty),J)/\hbar}$ gives (4.7). Expanding $e^{\hbar\partial_{P(\varepsilon,\infty)}}$ into graphs as in Proposition 3.35, the right side of (4.7) is the sum of the graphs with exactly one vertex labelled by the defect vertex; the factor $e^{-W(P(\varepsilon,\infty),J)/\hbar}$, the exponential of the connected graphs of J, removes the components without that vertex, which leaves the connected graphs. □

With $\Delta_\infty = 0$ and $\{-,-\}_\infty = 0$, the quantum master equation at scale ∞ for the renormalised flow is $Q\lim_{\varepsilon\to0} W(P(\varepsilon,\infty),J) = 0$ (Corollary 3.24). Its failure at order $\hbar$ is the order-$\hbar$ part of $\lim_{\varepsilon\to0} Q\,W(P(\varepsilon,\infty),I-I^{CT}[\varepsilon])$. By (4.7) it is a sum of graphs with one defect vertex, among them trees of I attached to that vertex; it is the flow to scale ∞ of an effective defect vertex, the one-loop obstruction. On a compact manifold with a finite upper scale the obstruction is the local functional of [C07, Prop 12.0.3], closed and independent of ε by [C07, Lemma 12.0.4]; on $\mathbb{R}^4$ with upper scale ∞ these properties are part of Step 2. In the sector with legs $(c, A|_V)$ no tree of I contributes at order $\hbar$, and the limit computed below is itself local.

Proposition 4.27 (The defect vertex at one loop)**.** *For $J = I - I^{CT}[\varepsilon]$, with I the classical interaction* (1.3)*,*

$$QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J = \tfrac12\big(\{I,I\}_\varepsilon-\{I,I\}\big)+\hbar\Delta_\varepsilon I-Q\,I^{CT}[\varepsilon]-\{I,I^{CT}[\varepsilon]\}_\varepsilon+(\textit{terms of order }\hbar^2). \tag{4.8}$$

The first term is the bracket built from $\Delta_\varepsilon-\Delta_0 = -\partial_{K_\varepsilon-K_0}$, i.e. two vertices of I joined by one special edge carrying

$$K_\varepsilon - K_0 = -\int_0^\varepsilon (H\otimes1)K_t\,dt,$$

which in momentum space is $e^{-\varepsilon|p|^2}-1$ times the components of K_0; its Schwinger time lies in $[0,\varepsilon]$.

Proof. Expanding $QJ+\frac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J$ with $J = I - I^{CT}[\varepsilon]$ and $I^{CT}[\varepsilon]$ of order $\hbar$: at order $\hbar^0$, $QI+\frac12\{I,I\}_\varepsilon$; at order $\hbar$, $-QI^{CT}[\varepsilon]-\frac12\big(\{I,I^{CT}[\varepsilon]\}_\varepsilon+\{I^{CT}[\varepsilon],I\}_\varepsilon\big)+\hbar\Delta_\varepsilon I$, where the two brackets of even functionals coincide; the rest is $\frac12\{I^{CT}[\varepsilon],I^{CT}[\varepsilon]\}_\varepsilon-\hbar\Delta_\varepsilon I^{CT}[\varepsilon]$, of order $\hbar^2$. The classical master equation (Proposition 1.14) gives $QI = -\frac12\{I,I\}$, with $\{-,-\}$ built from Δ_0 (Definition 1.6). From $K_t = (e^{-tH}\otimes1)K_0$, $\frac{d}{dt}K_t = -(H\otimes1)K_t$, and integration from 0 to ε gives the special edge; in momentum space e^{-tH} is $e^{-t|p|^2}$. □

At order $\hbar$, the graphs of (4.7) with two legs are: the term $-QI^{CT}[\varepsilon]$ as a vertex; the bubbles formed by the two vertices of $\frac12(\{I,I\}_\varepsilon-\{I,I\})$, joined by the special edge and one edge $P(\varepsilon,\infty)$; and one-particle-reducible graphs in which a loop closes at a single cubic vertex. The term $\{I,I^{CT}[\varepsilon]\}_\varepsilon$ has at least three legs.

Lemma 4.28 (Two-leg sectors of degree one)**.** *An invariant local functional of degree* 1 *with two legs and weight at most* 4 *is a combination of* $\int \mathrm{dvol}\,(dc, A|_V)$ *(weight* 2*) and* $\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$ *(weight* 4*). In particular the sectors with legs* $(c, A|_{W_{\mathbb{R}}^{\perp}})$ *and* (c, λ) *vanish.*

Proof. Degree 1 with two legs requires one c and one leg of degree 0, A or λ. The invariant pairings of a scalar with a spinor and with a vector of $W_{\mathbb{R}}^{\perp}$ vanish. With $A|_V$: one derivative in weight 2, $\int \mathrm{dvol}\,(dc, A|_V)$; three derivatives in weight 4, where $\int \mathrm{dvol}\,(dc, d^*dA|_V) = \int \mathrm{dvol}\,(d\,dc, dA|_V) = 0$ and $\int \mathrm{dvol}\,(dc, dd^*A|_V) = \int \mathrm{dvol}\,(d^*dc, d^*A|_V)$. □

[figure] The one-loop graphs of the defect vertex with external $c, A|_V$: the shaded edge is the special edge $K_\varepsilon - K_0$ of $\frac{1}{2}(\{I,I\}_\varepsilon - \{I,I\})$, the other edge carries $P(\varepsilon,\infty)$. Special edge between the slots (a) $A|_V$, $A^*|_V$ with an $A|_V$ line; (b) $A|_V$, $A^*|_V$ with a c–A^* line; (c) $A|_{W_{\mathbb{R}}^{\perp}}$, $A^*|_{W_{\mathbb{R}}^{\perp}}$ with an $A|_{W_{\mathbb{R}}^{\perp}}$ line; (d) λ^*, λ with a λ line; (e) c^*, c with a c–A^* line.

[figure] The one-particle-reducible graphs of the defect vertex with external $c, A|_V$. Row 1: from $\frac{1}{2}(\{I,I\}_\varepsilon - \{I,I\})$, with an edge $P(\varepsilon,\infty)$ closing at one cubic vertex. Row 2: from $\hbar\Delta_\varepsilon I$, a special edge K_ε closing at one cubic vertex. Each closed loop at a cubic vertex carries the colour factor $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X) = 0$.

Lemma 4.29 (Graphs with one special edge)**.** *The plane-wave value of each bubble of the first figure is* ε^{-1} *times a multiple of* $\int \mathrm{dvol}\,(dc, A|_V)$ *plus a multiple of* $\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$ *plus a term bounded by a multiple of* ε*, with vanishing coefficient of* $\log\varepsilon$*.*

Proof. The special edge carries $-\int_0^\varepsilon (\cdots) e^{-t_1|p_1|^2} dt_1$ and the other edge $\int_\varepsilon^\infty (\cdots) e^{-t_2|p_1-p|^2} dt_2$. After the Gaussian integral over p_1 (Lemma 4.4, extended to the polynomial numerators of degree three of these graphs by the same moments), the integrand is $(t_1+t_2)^{-2} e^{-t_1t_2|p|^2/(t_1+t_2)}$ times a sum of terms $(t_1+t_2)^{-m}$, $m \geq 0$, with polynomial coefficients in $\frac{t_1}{t_1+t_2}$ and p. On $[0,\varepsilon]\times[\varepsilon,\infty)$ the exponent satisfies $0 \leq \frac{t_1t_2|p|^2}{t_1+t_2} \leq t_1|p|^2 \leq \varepsilon|p|^2$, so its exponential is a power series in $\varepsilon|p|^2$ with uniform remainders. Substituting εt_1, εt_2 for t_1, t_2 maps the region to $[0,1]\times[1,\infty)$, on which $(t_1+t_2)^{-2-m}$ is integrable, and each term becomes an integer power of ε times a convergent integral. Hence the value is a series in integer powers of ε; by Lemma 4.28 the coefficients of ε^{-1} and ε^0 are the two listed functionals. □

graph	special edge $K_\varepsilon - K_0$	edge $P(\varepsilon,\infty)$	coefficient of $\varepsilon^{-1}\int \mathrm{dvol}\,(dc, A\|_V)$	coefficient of $\int \mathrm{dvol}\,(d^*dc, d^*A\|_V)$
(a)	$A\|_V$–$A^*\|_V$	$A\|_V$	$-\frac{9}{4}$	$-\frac{5}{8}$
(b)	$A\|_V$–$A^*\|_V$	c–A^*	$\frac{1}{8}$	$-\frac{1}{12}$
(c)	$A\|_{W_{\mathbb{R}}^{\perp}}$–$A^*\|_{W_{\mathbb{R}}^{\perp}}$	$A\|_{W_{\mathbb{R}}^{\perp}}$	$-\frac{9}{2}$	$-\frac{5}{4}$
(d)	λ^*–λ	λ	2	$-\frac{4}{3}$
(e)	c^*–c	c–A^*	$\frac{1}{8}$	$-\frac{1}{12}$
$-QI^{CT}[\varepsilon]$			$\frac{9}{2}$	0
total			0	$-\frac{27}{8}$

The one-loop part of $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ with legs $(c, A|_V)$, graph by graph, in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$. Each bubble has weight 1: the $\frac{1}{2}$ of $\frac{1}{2}\{I,I\}$ cancels against the two assignments of the two vertices. Every entry has vanishing $\log\varepsilon$ coefficient.

Theorem 4.30 (The one-loop obstruction in the sector $(c, A|_V)$)**.** *The part of* $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ *of order* $\hbar$ *with legs* $(c, A|_V)$ *has the following* ε*-expansion: the special-edge bubbles contribute* $-\frac{9}{2}\,\varepsilon^{-1}$ *times* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} \int \mathrm{dvol}\,(dc, A|_V)$*, the vertex* $-QI^{CT}[\varepsilon]$ *contributes* $+\frac{9}{2}\,\varepsilon^{-1}$ *times the same functional, the coefficient of* $\log\varepsilon$ *is* 0*, and the limit is*

$$-\frac{27}{8}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} \int \mathrm{dvol}\,(d^*dc, d^*A|_V) = \frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,Q \int \mathrm{dvol}\,(d^*A|_V, d^*A|_V). \tag{4.9}$$

The sectors $(c, A|_{W_{\mathbb{R}}^{\perp}})$ *and* (c, λ) *of the one-loop obstruction vanish.*

Proof. By Proposition 4.26 and (4.8), the order-$\hbar$ part with two legs is the sum of the graphs listed before Lemma 4.28. The one-particle-reducible graphs vanish: a loop closing at one cubic vertex contracts two legs of a vertex of the form $([X,Y],Z)$ with $(-,-)^{-1}$ and gives $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X) = 0$ or the contraction of the antisymmetric bracket with the symmetric $(-,-)^{-1}$, both zero. The same argument gives $\Delta_\varepsilon I = 0$: K_ε pairs a field with its antifield, the vertices of I containing an antifield are $([A,c],A^*)$, $([c,\lambda],\lambda^*)$, $\frac{1}{2}([c,c],c^*)$, and each self-contraction is a trace of ad. The five bubbles are given in the table; their ε^{-1} coefficients sum to $-\frac{9}{4}+\frac{1}{8}-\frac{9}{2}+2+\frac{1}{8} = -\frac{9}{2}$, and by Lemma 4.29 they have vanishing $\log\varepsilon$ coefficient. By Theorem 4.16 and Remark 4.24, $-QI^{CT}[\varepsilon]$ restricted to $(c,A|_V)$ is $-\frac{9}{4}\cdot(-2) = \frac{9}{2}$ times $\varepsilon^{-1}\int \mathrm{dvol}\,(dc, A|_V)$ in the unit; its $\log\varepsilon$ part restricted to $(c,A|_V)$ vanishes, because $Q\int \mathrm{dvol}\,(dA,dA)$ contains $d\,dc = 0$, $Q\int \mathrm{dvol}\,(dc,A^*)$ restricted to $(c,A|_V)$ is $-\frac{1}{2}\int \mathrm{dvol}\,(dc, d^*dA|_V) = -\frac{1}{2}\int \mathrm{dvol}\,(d\,dc, dA|_V) = 0$, and the coefficient of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ is 0. The finite parts sum to $-\frac{5}{8}-\frac{1}{12}-\frac{5}{4}-\frac{4}{3}-\frac{1}{12} = -\frac{81}{24} = -\frac{27}{8}$. Finally $Q\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) = 2\int \mathrm{dvol}\,(d^*A|_V, d^*(-dc)) = -2\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$, so $-\frac{27}{8} = \frac{27}{16}\cdot(-2)$. The last statement is Lemma 4.28. □

Remark 4.31 (The obstruction as the longitudinal part of the two-point function)**.** Q preserves the polynomial degree of functionals, so the part of $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ with legs $(c,A|_V)$ at order $\hbar$ is Q applied to the order-$\hbar$ quadratic part of $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$: the renormalised two-point function, whose limit as $\varepsilon\to 0$ at nonzero momentum exists by Theorem 4.16 and Proposition 4.8. Q annihilates its (c,A^*) part on $(c,A|_V)$, since ι_p of $|p|^2A|_V - p\,\iota_p A|_V$ is 0, and its transverse $A|_V$ part, which depends on $A|_V$ through $dA|_V$. Hence the cancellation of ε^{-1} and the vanishing of $\log\varepsilon$ in Theorem 4.30 follow from Theorem 4.16, and the obstruction is Q of the longitudinal part of the renormalised $A|_V$ two-point function at scale ∞. A longitudinal quadratic functional of $A|_V$ is $\int \mathrm{dvol}\,(d^*A|_V, G\,d^*A|_V)$ for a translation-invariant operator G on functions, and Q sends it to $-2\int \mathrm{dvol}\,(d^*dc, G\,d^*A|_V)$; comparing with (4.9) shows that G is the constant $\frac{27}{16}$ in the unit, so the longitudinal part is the local functional

$$\frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V).$$

Equivalently, Q applied to the longitudinal parts of the graphs of the $A|_V$ two-point function, with every edge cut off at ε, sums to $-\frac{9}{2}\varepsilon^{-1}$ and $-\frac{27}{8}$ times the functionals of (4.9); there the gauge bubble and the ghost loop carry separate $\log(\varepsilon|p|^2)$ and $\log 2$ terms in their longitudinal parts, which cancel in their sum. Theorem 4.30 is the evaluation of the same quantity through the UV-boundary identity, on graphs with one special edge confined to $[0,\varepsilon]$.

Theorem 4.32 (The finite counterterm in the sector $(c,A|_V)$)**.** *Replace the renormalised interaction* $I - I^{CT}[\varepsilon]$ *by*

$$I - \frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) - I^{CT}[\varepsilon]. \tag{4.10}$$

Then $I^{CT}[\varepsilon]$ *at one loop is unchanged, the order-*$\hbar$ *part of the flow has a limit at generic leg momenta, and the part with legs* $(c,A|_V)$ *of the one-loop obstruction vanishes. The added term is local, even, of degree* 0 *and weight* 4*, independent of* ε*, and invariant under translations,* $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$*, the adjoint action of* $\mathfrak{g}$ *and the dilatation. Among the quadratic functionals of* $A|_V$ *of weight* 4 *and degree* 0*, it is unique up to adding multiples of* $\int \mathrm{dvol}\,(dA|_V, dA|_V)$*, which is* Q*-closed.*

Proof. The added term is of order $\hbar$, so at order $\hbar$ it enters the flow as a tree vertex, with a limit as $\varepsilon\to 0$, and leaves the one-loop singular parts unchanged. In (4.8) it adds $-Q$ of itself at order $\hbar$, and $-\{I,\cdot\}_\varepsilon$ of itself, which has at least three legs. By Theorem 4.30 and $Q\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) = -2\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$, the $(c,A|_V)$ part becomes $-\frac{27}{8}+\frac{27}{16}\cdot 2 = 0$ in the unit. Invariance under the dilatation holds because the weight is 4. For uniqueness: by Lemma 4.19, the quadratic functionals

of $A|_V$ of weight 4 are spanned by $\int \mathrm{dvol}\,(dA|_V, dA|_V)$ and $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$; the first is Q-closed, since $d\,dc = 0$, and the second has Q-variation $-2\int \mathrm{dvol}\,(d^*dc, d^*A|_V) \neq 0$, so the coefficient $\frac{27}{16}$ of the second is forced. □

Remark 4.33 (Renormalisation conditions belong to the definition of the theory)**.** $I^{CT}[\varepsilon]$ is purely singular, and unique among the translation-invariant and $\mathfrak{g}$-invariant local functionals of Proposition 4.10 (the one-loop analogue of [C07, Thm B], in the scope of Remark 4.9); the finite term of (4.10) is a finite change of the interaction, $I \mapsto I - \frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$, of the kind that lifts a theory across a Q-exact obstruction in [C07, Lemma 12.0.4]. The order-$\hbar$ part of the regulated flow with $I - I^{CT}[\varepsilon]$ has a finite limit at generic leg momenta (Proposition 4.8) and fails the master equation at scale ∞ by a finite local amount; the master equation forces the coefficient $\frac{27}{16}$ and leaves the Q-closed finite counterterms, such as multiples of $\int \mathrm{dvol}\,(dA, dA)$, free. This choice is a renormalisation condition, and it is part of the definition of the quantum theory: every observable $O[\infty]$ is built from the flow of the interaction, and changes with it. In this program it is the first instance of that statement; the same mechanism recurs for the equivariant counterterms of the superconformal ghosts.

Remark 4.34 (The cubic sector of the obstruction)**.** The one-loop obstruction also has components with three legs: $(c, A|_V, A|_V)$, $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, (c, λ, λ) and $(c, c, A^*|_V)$. They are sums of triangles and quartic-vertex bubbles with one special edge, of the vertex $-\{I, I^{CT}[\varepsilon]\}_\varepsilon$, and of one-particle-reducible graphs in which $-QI^{CT}[\varepsilon]$ is joined to I by an edge $P(\varepsilon, \infty)$. The finite term of (4.10) changes them by $-\{I, \cdot\}$ of itself, a nonzero functional built from $\int \mathrm{dvol}\,([A, c], A^*)$. The full one-loop master equation, i.e. the vanishing of the obstruction in all sectors after a finite counterterm, is the subject of Step 2.

Proposition 4.35 (The finite counterterm under special conformal transformations)**.** *Let v be the special conformal field $x \mapsto (x,x)v_1 - 2(v_1, x)x$, $v_1 \in V$ (Definition 2.1), acting on $A|_V$ by the Lie derivative L_v (Definition 2.12). The derivative of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ along $A|_V \mapsto L_vA|_V$ is*

$$2\int \mathrm{dvol}\,(d^*A|_V, d^*L_vA|_V) = -8\int \mathrm{dvol}\,(d^*A|_V, \iota_{v_1}A|_V) \neq 0,$$

while for translations, rotations and the dilatation $2E$ the derivative vanishes.

Proof. For a conformal Killing field v of V, the symmetric part of the derivative of $(v, -)$ is $\frac{1}{4}\operatorname{div} v$ times the metric; taking one more divergence, d^*d applied to the components of v is $\frac{1}{2}\operatorname{grad}(\operatorname{div} v)$. The Lie derivative of the one-form $A|_V$ is $v(A|_V) + A|_V \circ (\text{derivative of } v)$, and the divergence of this expression, with the two identities above, is

$$d^*(L_vA|_V) = v(d^*A|_V) + \tfrac{1}{2}(\operatorname{div} v)\, d^*A|_V + \tfrac{1}{2}\,\iota_{\operatorname{grad}(\operatorname{div} v)}A|_V.$$

Hence

$$\begin{aligned}2\int \mathrm{dvol}\,(d^*A|_V, d^*L_vA|_V) = &\int \mathrm{dvol}\, v\big((d^*A|_V, d^*A|_V)\big) + \int \mathrm{dvol}\,(\operatorname{div} v)(d^*A|_V, d^*A|_V)\\ &+ \int \mathrm{dvol}\,(d^*A|_V, \iota_{\operatorname{grad}(\operatorname{div} v)}A|_V),\end{aligned}$$

and integration by parts cancels the first two terms, since $\int \mathrm{dvol}\, v(\cdot) = -\int \mathrm{dvol}\,(\operatorname{div} v)(\cdot)$. For the special conformal field, $\operatorname{div} v = -8(v_1, x)$ (Chapter 2) and $\operatorname{grad}(\operatorname{div} v) = -8v_1$, which gives the stated value. For translations and rotations $\operatorname{div} v = 0$, and for $2E$, $\operatorname{div} v = 8$ is constant. □

The finite counterterm of Theorem 4.32 is the Feynman-gauge longitudinal term, and Proposition 4.35 shows that it breaks the special conformal symmetry. In Step 2 the components of the equivariant counterterms along the special conformal generators have to absorb this variation, or exhibit it as part of an obstruction.

Remark 4.36 (Method of computation)**.** Each per-graph coefficient of this chapter is an exact rational number. It is obtained in an explicit realisation of V_{10}, Σ and $\mathfrak{g}$ over $\mathbb{Q}(i)$, with the loop momentum integrated as a Gaussian (Lemma 4.4) and the ε-expansion taken on the graphs with one pinned edge (Proposition 4.2), in exact rational arithmetic; for the two-point graphs the polarisations and momenta are symbolic. For the cubic and quartic sets each multilinear form is determined by exact linear algebra on a basis of the $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$-invariant multilinear forms of the given legs, including the parity-odd ones, from evaluations at integer polarisations and momenta drawn uniformly from a finite set of integers. A nonzero polynomial vanishes at such a point with probability at most its total degree divided by the size of the set (Schwartz–Zippel), so the sampled system has full rank with high probability; full rank is then established exactly, and the solution is exact. The colour factors are reduced exactly for a general simple $\mathfrak{g}$, and the resulting rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$ are the same in the explicit realisations $\mathfrak{g}=\mathfrak{su}(2)$ and $\mathfrak{g}=\mathfrak{su}(3)$. The sector $(c, A|_V)$ of the obstruction is also given by Q of the renormalised two-point function (Remark 4.31).

Chapter 5

Local operators: from functionals to local $O(\varepsilon)$

The insertion of a composite operator into a one-loop graph diverges as $\varepsilon \to 0$, and Costello's construction asks for a family $O(\varepsilon)$ whose flow to scale ∞ has a limit. At scale ∞, however, $K_\infty = 0$ and $P(0,\infty)$ is a contracting homotopy for Q on moment-free Schwartz sections (Chapter 3), so a general functional family $O(\varepsilon)$ produces the zero class on moment-free Schwartz sections. The content of an operator is therefore the requirement that $O(\varepsilon)$ be local, $O(\varepsilon) = O - \partial_\delta I^{CT}[\varepsilon]$ with coefficients in $\mathrm{span}\{\varepsilon^{-1}, \log\varepsilon\}$: locality and the existence of the limit fix the singular part, and the finite part is the scheme. This chapter sets up $O(\varepsilon)$ and $O[\infty]$, proves $O(\varepsilon) = O$ for single letters at one loop, and computes $O(\varepsilon)$ for the scalar bilinears graph by graph.

5.1 Observables as first-order deformations

Definition 5.1 (Background field and the family $I + \delta O$)**.** A background field is a compactly supported four-form $\omega \in \Omega^4_c(\mathbb{R}^4)$ [CG2, Lemma 10.1.1.3]. A local operator is a local functional

$$O = \int \omega\,(\text{density}),$$

linear in ω, whose density is a polynomial in the fields and their derivatives at a point; the Weyl weight w_O of O is the sum of the Weyl weights of the fields in the density (c: 0, A: 1, λ: $\frac{3}{2}$, A^*: 3, λ^*: $\frac{5}{2}$, c^*: 4) plus the number of derivatives. For O of cohomological degree i, δ is a square-zero parameter of degree $-i$ [CG2, Def 8.3.1.1], and $I + \delta O$ is a family of interactions over $\mathbb{C}[\delta]/\delta^2$; by [CG2, after Def 8.3.1.3] it satisfies the axioms of a family of theories over $\mathbb{C}[\delta]/\delta^2$ except locality of the flowed functionals. The scalar bilinears and the letters $\int \omega\,(X, \iota_w A)$, $\int \omega\,(X, \iota_v(A|_V))$, $\int \omega\,(\zeta, \lambda)$ have $i = 0$; the ghost letter $\int \omega\,(X, c)$ has $i = 1$, and then δ has degree -1.

The square-zero parameter δ is written without argument; the delta distribution is always written with its argument, as in $\delta(x - y)$.

Definition 5.2 (The local family and the observable at scale ∞)**.** Let $I^{CT}[\varepsilon]$ be the one-loop counterterm of I (Definition 4.1; the analogue of [C07, Thm B], Remark 4.9), and write the counterterm of the family $I + \delta O$ as $I^{CT}[\varepsilon] + \delta\,\partial_\delta I^{CT}[\varepsilon]$. Put

$$O(\varepsilon) := O - \partial_\delta I^{CT}[\varepsilon], \qquad O[\infty] := \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I - I^{CT}[\varepsilon] + \delta\, O(\varepsilon)\big), \tag{5.1}$$

with W the flow of Definition 3.22 [CG2, §7.2.7]. With the global ghosts of $\mathfrak{psl}(4|4)$, I is replaced by the equivariant interaction $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ [ESW, Rem 1.29] (Definition 2.10) and $I^{CT}[\varepsilon]$ by

the counterterm of the family $I_{\mathfrak{psl}(4|4)} + \delta O$ over $C^\bullet(\mathfrak{psl}(4|4))$:

$$O(\varepsilon) := O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon], \qquad O[\infty] := \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \delta\, O(\varepsilon)\big).$$

Both are cochains on $\mathfrak{psl}(4|4)$; their components of arity 0 are the functionals (5.1). This chapter computes the components of arity 0; the components with arguments in $\mathfrak{psl}(4|4)$ involve the counterterms of $I_{\mathfrak{psl}(4|4)}$ along the ghosts; Chapter 6 treats the dilatation through the scaling covariance, and the ghost components are part of Step 2 (Chapter 7).

At scale ∞ the differential on observables is $(d_{CE} + Q)O[\infty]$, since the scale-∞ structure is defined by the kernel $K_\infty := 0$, so that $\Delta_\infty = 0$ and $\{-,-\}_\infty = 0$ (Definition 3.21) [CG2, Def 8.3.1.3, Prop 13.4.0.1].

[C07, Thm B] is stated for interactions that are at least cubic modulo $\hbar$, and δO is quadratic for the operators below. The construction of [C07, Thm B] is applied graph by graph:

Proposition 5.3 (Counterterms of the family)**.** *Let O be a homogeneous local operator. At order $\hbar$ there is a unique local functional $\partial_\delta I^{CT}[\varepsilon]$ with coefficients in the span of the negative integer powers of ε and of $\log\varepsilon$, a subspace of $\mathscr{A}_{<0}$ [C07, Def 1.4.1; CG2, §13.7.2], such that the limit* (5.1) *exists at generic leg momenta. It is the singular part of the sum of the one-loop graphs in which the vertex O lies on the cycle, each evaluated with $P(\varepsilon,\infty)$; for $w_O \le 3$ its coefficients lie in* $\operatorname{span}\{\varepsilon^{-1}, \log\varepsilon\}$.

Proof. By Proposition 3.35, the order-$\hbar$ part of $\partial_\delta W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon] + \delta O)$ is a sum over connected graphs with exactly one vertex O (since $\delta^2 = 0$) and either one cycle and vertices I, or no cycle and one vertex $-I^{CT}[\varepsilon]$. Every vertex of I is at least trivalent and O occurs once, so for a fixed number of legs the sum is finite. Three classes occur.

- O lies off the cycle. Removing the edges that join O to the rest leaves a one-loop graph without O; the corresponding tree graph with the vertex $-I^{CT}[\varepsilon]$ in its place cancels its singular part by Lemma 4.7, with O as the marked vertex, and Definition 4.1, and the trees joining O converge to trees with $P(0,\infty)$, because $P(\varepsilon,\infty) - P(0,\infty) = -\int_0^\varepsilon L_t\, dt$ is bounded by a multiple of ε at fixed nonzero momentum.

- O lies on the cycle. The vertex function of O is a polynomial in the momenta times the Fourier transform of ω at minus the total momentum of its legs (Lemma 5.8), and the numerator is homogeneous, O being homogeneous, so by Lemma 4.6 the singular part of such a graph is a local functional with coefficients in the span of the negative powers of ε and of $\log\varepsilon$, and the tree with the vertex $-\delta\,\partial_\delta I^{CT}[\varepsilon]$ in place of the cycle removes it (Lemma 4.7, with O as the marked vertex). Its superficial degree is w_O minus the sum of the Weyl weights of the legs (proof of Proposition 5.13), at most w_O, so for $w_O \le 3$ the powers ε^{-k}, $k \ge 2$, are absent.

- Trees with the vertex $-I^{CT}[\varepsilon]$ and with O: already paired in the first class.

Uniqueness: two such counterterms differ by a local functional with coefficients in $\mathscr{A}_{<0}$ whose tree-level flow has a limit. The part of that flow with as many legs as the component of the difference with the fewest legs is that component itself, which therefore has a limit and vanishes; by induction on the number of legs, as in Proposition 4.10, the difference vanishes. □

Proposition 5.4 (Local representatives)**.** *Let O be a homogeneous local operator.*

- *Two local families $O(\varepsilon)$, $O'(\varepsilon)$ for which the limit* (5.1) *exists have the same singular part.*

- *Adding to $O(\varepsilon)$ an ε-independent local functional of order $\hbar$ adds to $O[\infty]$, at order $\hbar$, the flow of that functional to scale ∞ through trees with vertices I.*

- *Let F be a continuous polynomial functional of moment-free Schwartz sections, without constant term, with $QF = 0$. Then F is Q-exact (Corollary 3.26). On compactly supported sections the Q-closed functionals carry in addition nonzero classes, among them the zero-mode classes of Theorem 3.25 and Remark 3.27, finitely many in each polynomial degree and infinitely many in all.*

Consequently a statement about $O[\infty]$ has content when it is written in the form

$$(d_{CE} + Q)\, O[\infty] = (\textit{a local functional})[\infty],$$

and, for an operator whose observable is a cocycle (Remark 5.6), an anomalous dimension is read off in the cohomology of local functionals.

Proof. The first item is the uniqueness part of Proposition 5.3. The second follows from linearity of (5.1) in the δ-component: at order $\hbar$ the added functional enters through its tree-level flow. The third item is Corollary 3.26, which rests on the homotopy identity $Q\,P(0,\infty)\star + P(0,\infty)\star\, Q = \mathrm{id}$ of Theorem 3.25 on moment-free Schwartz sections; the classes on compactly supported sections are those described in Theorem 3.25 and Remark 3.27. Every observable $O[\infty]$ that extends to a continuous functional on moment-free Schwartz sections, with $(d_{CE}+Q)O[\infty] = 0$ there, therefore lies in the image of Q on such functionals; what it records about O is the local functional $O(\varepsilon)$, through the first two items. □

Remark 5.5 (Where the content of an operator lies)**.** By Proposition 5.4, the singular part of $O(\varepsilon)$ is uniquely fixed and the finite part is the scheme freedom; this is the statement of Remark 3.27 for operators. In the rest of this chapter $\partial_\delta I^{CT}[\varepsilon]$ is computed; its $\log\varepsilon$ part is the input of the anomalous dimensions in Chapter 6.

Remark 5.6 (The cocycle condition)**.** Reading the dilatation component of Chapter 6 as the anomalous dimension of a cohomology class requires $O[\infty]$ to be a cocycle at arity zero,

$$Q\,O[\infty] = 0 \qquad \text{at order } \hbar,$$

the part linear in δ of the quantum master equation of the family $I + \delta O$ at scale ∞. For O with $(Q + \{I, -\})O = 0$, Proposition 4.26 applied to $J = I - I^{CT}[\varepsilon] + \delta O(\varepsilon)$ expresses $Q\,O[\infty]$ at order $\hbar$ as the limit of the graphs with one vertex labelled by the part of the defect vertex linear in δ: the special edge $K_\varepsilon - K_0$ between I and O, from $\{I, O\}_\varepsilon - \{I, O\}$; the vertices $-Q\,\partial_\delta I^{CT}[\varepsilon] - \{I, \partial_\delta I^{CT}[\varepsilon]\}_\varepsilon$ and $-\{I^{CT}[\varepsilon], O\}_\varepsilon$; the self-contraction $\hbar\Delta_\varepsilon O$, zero for operators without antifields; and the trees joining O to the action defect of Proposition 4.27, whose cubic sector is open (Remark 4.34). For the scalar bilinears the local output of Weyl weight 2, ghost degree 1 and colour singlet includes functionals of the type $\int \omega\,(dc, A|_V)$. A nonzero result of this kind is removed by a finite local counterterm of Weyl weight w_O added to $O(\varepsilon)$, for instance a multiple of $\int \omega\,(A|_V, A|_V)$, since

$$Q \int \omega\,(A|_V, A|_V) = -2 \int \omega\,(dc, A|_V).$$

Such a counterterm is independent of ε and of order $\hbar$; by Proposition 5.4 it adds to $O[\infty]$ the flow of a local functional of the same Weyl weight, and it leaves the $\log\varepsilon$ coefficients of this chapter and the dilatation coefficients of Chapter 6 unchanged. The evaluation of these graphs is part of Step 2 (Chapter 7).

5.2 Tools for one-loop insertions

The gauge algebra $\mathfrak{g}$ is simple. The inverse pairing $(-,-)^{-1} \in \mathfrak{g} \otimes \mathfrak{g}$ is symmetric, and a bilinear form on $\mathfrak{g}$ can be evaluated on it.

Lemma 5.7 (Colour contractions)**.** *For $X_1, X_2 \in \mathfrak{g}$:*

- *the bracket $[-,-]$ evaluated on $(-,-)^{-1}$ is 0;*
- *for every linear map of $\mathfrak{g}$, the bilinear form $(Z, Z') \mapsto (Z,$ the map applied to $Z')$ evaluated on $(-,-)^{-1}$ is the trace $\mathrm{tr}_{\mathfrak{g}}$ of the map; in particular $(-,-)$ evaluated on $(-,-)^{-1}$ is $\dim \mathfrak{g}$, and $(X_1, [-,-])$ and $(-, [X_1,-])$ evaluated on $(-,-)^{-1}$ are 0;*
- *the bilinear form $([X_1,-],[X_2,-])$ evaluated on $(-,-)^{-1}$ is $-\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$;*
- *for all $X, Y \in \mathfrak{g}$ with $(X,Y) \neq 0$, the ratio $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(X,Y)$ is the same number, and $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}) = \frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{(X,Y)}\,(X_1, X_2)$.*

Proof. The bracket is antisymmetric and $(-,-)^{-1}$ is symmetric. The inverse pairing is characterised by the property that contracting its second factor with $(Z,-)$ returns Z; hence the bilinear form of a linear map, evaluated on $(-,-)^{-1}$, is the trace of the map. By invariance, $(X_1, [Z, Z']) = ([X_1, Z], Z') = -(Z, [X_1, Z'])$, which is the case of the map $-\mathrm{ad}_{X_1}$, of trace 0 because ad_{X_1} is antisymmetric for $(-,-)$, and $(-, [X_1,-])$ is the case of ad_{X_1}; and $([X_1, Z], [X_2, Z']) = -(Z, [X_1, [X_2, Z']])$ is the case of $-\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}$. For simple $\mathfrak{g}$ every invariant symmetric bilinear form is a multiple of $(-,-)$, which gives the last item. □

In the following, the number

$$\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2\,(X,Y)}$$

is the unit of every one-loop coefficient; it is negative when $(-,-)$ is positive definite on the compact real form.

Lemma 5.8 (Vertex functions on decomposable legs)**.** *Give each leg of a vertex a fibre element $(v,-)\otimes Z$ with $v \in V_{10}$ and $Z \in \mathfrak{g}$; by linearity this determines every vertex function. The quartic vertex function* (1.5) *with legs $(v_j,-)\otimes Z_j$, $j = 1,\dots,4$, equals*

$$\begin{aligned} -\frac{1}{2}\Big[&\big((v_1,v_3)(v_2,v_4) - (v_1,v_4)(v_2,v_3)\big)\big([Z_1,Z_2],[Z_3,Z_4]\big) \\ &+ \big((v_1,v_2)(v_3,v_4) - (v_1,v_4)(v_3,v_2)\big)\big([Z_1,Z_3],[Z_2,Z_4]\big) \\ &+ \big((v_1,v_2)(v_4,v_3) - (v_1,v_3)(v_4,v_2)\big)\big([Z_1,Z_4],[Z_2,Z_3]\big)\Big], \end{aligned} \tag{5.2}$$

the three terms belonging to the pairings (12|34), (13|24), (14|23) *of the legs. The cubic vertex function* (1.4) *with two legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$, $w, w' \in W_{\mathbb{R}}^{\perp}$, of incoming momenta p, p', and one leg $(v,-)\otimes Z''$, $v \in V$, equals*

$$-\frac{i}{2}\,(v, p'-p)\,(w,w')\,\big(Z'', [Z, Z']\big); \tag{5.3}$$

The vertex function of $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ on legs $(v,-)\otimes Z$, $(v',-)\otimes Z'$ is

$$\big[(w_1,v)(w_2,v') + (w_1,v')(w_2,v)\big](Z,Z'),$$

times the Fourier transform of ω at minus the total momentum of the legs.

Proof. With $A_j = (v_j,-)\otimes Z_j$, the two-form $[A_1 \wedge A_2]$ of Chapter 1 is $(v_1,-)\wedge(v_2,-)\otimes[Z_1,Z_2]$, because $[A_1\wedge A_2](e,e') = (v_1,e)(v_2,e')[Z_1,Z_2] - (v_1,e')(v_2,e)[Z_1,Z_2]$, and likewise for every pair of legs. Polarising the normalisation of the Hodge pairing in Definition 1.8 gives

$$\big((v_1,-)\wedge(v_2,-),(v_3,-)\wedge(v_4,-)\big) = (v_1,v_3)(v_2,v_4) - (v_1,v_4)(v_2,v_3),$$

and likewise for the other two pairings, which gives (5.2). In (1.4), $\iota_p A(p_j) = (v_j, p) Z_j$ and $[A(p_i), A(p_j)] = (v_i, v_j)[Z_i, Z_j]$. With legs $1, 2$ along $W_{\mathbb{R}}^{\perp}$ and leg 3 along V, the momenta lie in $V \perp W_{\mathbb{R}}^{\perp}$, so $(v_1, p) = (v_2, p) = 0$ and $(v_2, v_3) = (v_3, v_1) = 0$; the second and third terms of (1.4) vanish and the first is $-\frac{i}{2}(v_3, p_2 - p_1)(v_1, v_2)(Z_3, [Z_1, Z_2])$, which is (5.3). The last item is the multilinear part of $(\iota_{w_1} A, \iota_{w_2} A)$ on two plane waves (Definition 1.13). □

Lemma 5.9 (Insertion on a cycle)**.** *Expand around a background consisting of two plane waves with fibre elements $A(p_1)$, $A(p_2)$, as in Remark 3.37. The part of the order-$\hbar$ term of $\partial_\delta W(P(\varepsilon, \infty), I + \delta O)$ that is bilinear in $A(p_1)$, $A(p_2)$ and comes from cycles through the vertex O whose edges all lie in the A slot is $\hbar$ times the sum of*

- $\frac{1}{2}$ *times each bubble: the cycle through O and one quartic vertex carrying both external legs;*
- 1 *times each triangle: the cycle through O, a cubic vertex carrying $A(p_1)$ and a cubic vertex carrying $A(p_2)$, traversed in one orientation.*

In each cycle the internal legs are contracted along the edges with the $A \otimes A$ component of $P(\varepsilon, \infty)$ (Proposition 3.17): an edge of momentum p replaces the fibre elements $(v, -) \otimes Z$, $(v', -) \otimes Z'$ at its two ends by 2 times the inverse metric in $v \otimes v'$ and $(-, -)^{-1}$ in $Z \otimes Z'$, times $\int_\varepsilon^\infty e^{-t|p|^2} dt = e^{-\varepsilon|p|^2} |p|^{-2}$, and the loop momentum p is integrated with $\int d^4p/(2\pi)^4$.

Proof. On the A slots the parity operator and the supertrace of Proposition 3.36 are trivial, and the one-loop operator is $P(\varepsilon, \infty)$ composed with the second derivative of $I + \delta O$ in the fluctuation. The δ-derivative of $\frac{1}{2n}\mathrm{STr}$ of its n-th power is $\frac{1}{2}\mathrm{STr}$ of the product of $n - 1$ factors built from I and one factor built from O. The second derivative of O is its vertex function with both legs internal, since O is quadratic. For $n = 2$ the factor built from I is bilinear in the two external legs through the quartic vertex: by Definition 1.13 the part of the quartic term of I quadratic in the fluctuation and bilinear in $A(p_1)$, $A(p_2)$ is one half of (1.5) with two internal legs, whose second derivative is (1.5) with two legs open; this gives the weight $\frac{1}{2}$. For $n = 3$ the two factors built from I each carry one external leg through a cubic vertex; the two orders of these factors give equal traces, since all second derivatives and $P(\varepsilon, \infty)$ are symmetric and the trace is invariant under transposition and cyclic permutation, so $\frac{1}{2}$ times their sum is one triangle with weight 1. □

The singular parts are read off from the following lemma, which also locates them on the lower end $t = \varepsilon$ of the Schwinger integrals (Remark 3.19).

Lemma 5.10 (Logarithmic coefficient from the UV boundary)**.** *Let γ be a one-loop graph with edge set $E(\gamma)$ [C07, §15.5], $|E(\gamma)| \geq 2$, whose value is*

$$F_\gamma(\varepsilon) = \int_{[\varepsilon,\infty)^{E(\gamma)}} \prod_{e \in E(\gamma)} dt_e \int \frac{d^4p}{(2\pi)^4} \big(\textit{numerator of } \gamma\big)(p) \, \exp\Big(- \sum_{e \in E(\gamma)} t_e |p + p_e|^2 \Big) \qquad [\textit{C07, §15.4}],$$

with generic $p_e \in V$ and a numerator that is a polynomial in p and the p_e, homogeneous of degree $2|E(\gamma)| - 4$; let $\big(\textit{numerator of } \gamma\big)_{\mathrm{top}}$ be its part of degree $2|E(\gamma)| - 4$ in p, and write

$$\textit{(average)} = \frac{1}{2\pi^2} \int_{|p|=1} \big(\textit{numerator of } \gamma\big)_{\mathrm{top}}(p) \, \mathrm{darea}(p)$$

for its average over the unit sphere of V, of area $2\pi^2$. Then

$$\begin{aligned} \varepsilon \frac{dF_\gamma}{d\varepsilon} &= - \sum_{e \in E(\gamma)} \big(\varepsilon \cdot \textit{the value of } \gamma \textit{ with } t_e = \varepsilon \textit{ fixed}\big), \\ \lim_{\varepsilon \to 0} \varepsilon \frac{dF_\gamma}{d\varepsilon} &= - \frac{\textit{(average)}}{16\pi^2}, \end{aligned} \tag{5.4}$$

and each of the $|E(\gamma)|$ terms of the sum tends to $\frac{1}{|E(\gamma)|}\cdot\frac{(average)}{16\pi^2}$. The coefficient of $\log\varepsilon$ in $F_\gamma(\varepsilon)$ is $-\frac{(average)}{16\pi^2}$, and each edge carries the fraction $\frac{1}{|E(\gamma)|}$ of it.

Proof. The first identity is Proposition 4.2. Fix an edge e. As in the proof of Lemma 4.6, with $\tau=\varepsilon+\sum_{e'\neq e}t_{e'}$, the Gaussian integral over p writes the value of γ with $t_e=\varepsilon$ fixed as

$$\sum_{j=0}^{|E(\gamma)|-2}\int_{[\varepsilon,\infty)^{E(\gamma)\setminus e}}\frac{(\text{numerator})_j}{16\pi^2\,\tau^{2+j}}\exp\Big(-\tau\cdot\frac12\sum_{e',e''}\frac{t_{e'}}{\tau}\frac{t_{e''}}{\tau}|p_{e'}-p_{e''}|^2\Big)\prod_{e'\neq e}dt_{e'},$$

the exponent being ≥ 0. The term $j=|E(\gamma)|-2$ contracts all loop momenta of the top-degree part, so in polar coordinates

$$\begin{aligned}\int\frac{d^4p}{(2\pi)^4}\big(\text{numerator of }\gamma\big)_{\text{top}}(p)\,e^{-\tau|p|^2}&=(\text{average})\cdot\frac{2\pi^2}{(2\pi)^4}\int_0^\infty|p|^{2|E(\gamma)|-1}e^{-\tau|p|^2}\,d|p|\\&=\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2\,\tau^{|E(\gamma)|}},\end{aligned}$$

i.e. $(\text{numerator})_{|E(\gamma)|-2}=(|E(\gamma)|-1)!\,(\text{average})$, a constant. Substituting $t_{e'}\mapsto\varepsilon t_{e'}$ for $e'\neq e$, ε times this term becomes an integral over $[1,\infty)^{E(\gamma)\setminus e}$ of $\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2}\big(1+\sum_{e'\neq e}t_{e'}\big)^{-|E(\gamma)|}$ times an exponential of ε times a nonnegative function, which is at most 1 and tends to 1. The function $\big(1+\sum_{e'\neq e}t_{e'}\big)^{-|E(\gamma)|}$ is integrable on $[1,\infty)^{|E(\gamma)|-1}$, so by dominated convergence the limit is

$$\begin{aligned}\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2}\int_{[1,\infty)^{E(\gamma)\setminus e}}\Big(1+\sum_{e'\neq e}t_{e'}\Big)^{-|E(\gamma)|}\prod_{e'\neq e}dt_{e'}&=\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2\,|E(\gamma)|!}\\&=\frac{1}{|E(\gamma)|}\cdot\frac{(\text{average})}{16\pi^2}.\end{aligned}$$

The terms $j\leq|E(\gamma)|-3$, of lower degree in p, are bounded separately. Their integrands are bounded by a multiple of τ^{-2-j} times the exponential; on $\tau\geq1$ the integral is bounded as in the proof of Lemma 4.6, and on $\tau\leq1$ it is bounded by a multiple of $\int_0^1(\varepsilon+\rho)^{|E(\gamma)|-4-j}\,d\rho$, with $|E(\gamma)|-4-j\geq-1$, hence by a multiple of $1+|\log\varepsilon|$. Multiplied by ε these terms tend to 0. This proves the limit of each of the $|E(\gamma)|$ terms, and summing over the edges gives the limit in (5.4). Finally, by Lemma 4.6 with superficial degree 0, $F_\gamma(\varepsilon)$ is a multiple of $\log\varepsilon$ plus a function g with a limit and with $\varepsilon\frac{dg}{d\varepsilon}\to0$; hence the multiple is $\lim\varepsilon\,dF_\gamma/d\varepsilon$. □

For $|E(\gamma)|=2$ and numerator 1 the lemma gives the coefficient $-\frac{1}{16\pi^2}$ of $\log\varepsilon$ in

$$\int\frac{d^4p}{(2\pi)^4}\frac{e^{-\varepsilon|p|^2}}{|p|^2}\frac{e^{-\varepsilon|p+p_1|^2}}{|p+p_1|^2};$$

in position space this is the singular part $-\frac{\log\varepsilon}{16\pi^2}\,\delta(x-y)$ of the square of $\frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}$.

5.3 Single letters

Proposition 5.11 (Letters)**.** *Let $X\in\mathfrak{g}$, $w\in W_{\mathbb{R}}^{\perp}$, $v\in V$ and $\zeta\in\Sigma^*\otimes\mathfrak{g}$. For each of the letters*

$$\int\omega\,(X,\iota_wA),\qquad\int\omega\,(X,\iota_v(A|_V)),\qquad\int\omega\,(\zeta,\lambda),\qquad\int\omega\,(X,c),$$

one has $\partial_\delta I^{CT}[\varepsilon]=0$ at one loop, hence

$$O(\varepsilon)=O.\tag{5.5}$$

Proof. The vertex O is univalent, and a cycle passes through every one of its vertices along two edges; hence O lies off the cycle in every one-loop graph, and by Proposition 5.3 $\partial_\delta I^{CT}[\varepsilon]$, the singular part of the graphs with O on the cycle, is 0. The graphs with O off the cycle consist of a one-loop graph with the vertices of I, joined to O by a tree. If the one-loop graph has one external leg, it is a cubic vertex of I with two of its legs joined by an edge; every cubic vertex of I has the form $(Z_1,[Z_2,Z_3])$ in its colour factors (Chapter 1), and joining two of its legs evaluates $(Z_1,[-,-])$ or $(-,[Z_2,-])$ on $(-,-)^{-1}$, which is 0 by Lemma 5.7. If the one-loop graph has more external legs, its singular part is cancelled by the tree with the vertex $-I^{CT}[\varepsilon]$ (first class in the proof of Proposition 5.3). □

5.4 Scalar bilinears

Definition 5.12 (Scalar bilinears and the symmetric square of $W_{\mathbb{R}}^{\perp}$)**.** For $w_1,w_2\in W_{\mathbb{R}}^{\perp}$,

$$O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A),$$

a gauge-invariant local operator of Weyl weight $w_O=2$, symmetric and bilinear in (w_1,w_2), hence linear in $w_1w_2\in\mathrm{Sym}^2W_{\mathbb{R}}^{\perp}$. Under $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, with $\mathrm{Spin}(6;\mathbb{C})\cong SL(4,\mathbb{C})$ [ESW, §10.3], the symmetric square of the six-dimensional space $W_{\mathbb{R}}^{\perp}$ is the sum of the traceless symmetric tensors, of dimension 20, and the line spanned by the inverse metric, of dimension 1. Correspondingly

$$\begin{aligned}\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)=\Big[\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)\\ -\tfrac{1}{6}(w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})\Big]\\ +\tfrac{1}{6}(w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}}),\end{aligned}\tag{5.6}$$

where $(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})$ contracts the $W_{\mathbb{R}}^{\perp}$ factors with the metric and the $\mathfrak{g}$ factors with $(-,-)$. The bracket is the traceless part: replacing w_1w_2 by the inverse metric of $W_{\mathbb{R}}^{\perp}$ turns (w_1,w_2) into 6 and the bracket into $(1-\frac{1}{6}\cdot 6)\int\omega(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})=0$. The last term is the trace part, a multiple of $\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})$.

Proposition 5.13 (The graphs)**.** *The one-loop graphs with the vertex O of Definition 5.12 on the cycle and superficial degree ≥ 0 are the six graphs of the figure:*

- *(a) the vacuum graph: one $A|_{W_{\mathbb{R}}^{\perp}}$ edge from O to itself, degree 2;*
- *(b) one external $A|_V$: a cycle of two $A|_{W_{\mathbb{R}}^{\perp}}$ edges through a cubic vertex, degree 1;*
- *(c), (d) two external $A|_V$: the bubble with the quartic vertex of type $(A|_V)^2(A|_{W_{\mathbb{R}}^{\perp}})^2$, and the triangle with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^{\perp}})^2$ and three $A|_{W_{\mathbb{R}}^{\perp}}$ edges, degree 0;*
- *(e), (f) two external $A|_{W_{\mathbb{R}}^{\perp}}$: the bubble with the quartic vertex of type $(A|_{W_{\mathbb{R}}^{\perp}})^4$, and the triangle with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^{\perp}})^2$, edges $A|_{W_{\mathbb{R}}^{\perp}}$, $A|_V$, $A|_{W_{\mathbb{R}}^{\perp}}$, degree 0.*

[figure] The one-loop graphs (a)–(f) with one vertex $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ (double line) on the cycle and superficial degree ≥ 0; dashed lines are $A|_{W_{\mathbb{R}}^{\perp}}$, wavy lines $A|_V$, every edge carries $P(\varepsilon,\infty)$.

Proof. All vertices of I have Weyl weight 4 (fields plus derivatives), so a one-loop graph with one vertex O has superficial degree w_O minus the sum of the Weyl weights of its external legs, i.e. 2 minus that sum. Its external legs have total cohomological degree 0; an external c (weight 0) needs an antifield (weight $\geq \frac{5}{2}$) to compensate its degree, and two λ have weight 3. A single external λ, of weight $\frac{3}{2}$, is excluded: every vertex of I and of O contains an even number of factors λ, λ^*, and every edge carrying the $\lambda \otimes \lambda$ component of $P(\varepsilon, \infty)$ joins two of them, so the number of external legs λ, λ^* is even; equivalently, the element -1 of $\mathrm{Spin}(4, \mathbb{C})$, which acts by -1 on $S_\pm$ and trivially on V, preserves the action and the regulator and changes the sign of every functional with an odd number of spinor legs. The external legs of degree ≥ 0 graphs are therefore none, one A, or two A. The element -1 of $SO(W_{\mathbb{R}}^\perp)$ lies in the image of $\mathrm{Spin}(6; \mathbb{C})$ and is a symmetry of the action and of the regulator (H and Q^{GF} commute with $\mathrm{Spin}(6; \mathbb{C})$); it fixes $w_1 w_2$ and changes the sign of every functional with an odd number of $A|_{W_{\mathbb{R}}^\perp}$ legs, so the singular parts with one $A|_{W_{\mathbb{R}}^\perp}$ leg, or with legs $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, vanish. Both edges at O are $A|_{W_{\mathbb{R}}^\perp}$ edges. A cycle that enters a Yukawa vertex along $A|_{W_{\mathbb{R}}^\perp}$ continues along λ and needs an external λ; a cycle that enters the vertex $([A, c], A^*)$ along $A|_{W_{\mathbb{R}}^\perp}$ leaves it along c or along $A^*|_{W_{\mathbb{R}}^\perp}$; $P(\varepsilon, \infty)$ has no component in the slot of $A^*|_{W_{\mathbb{R}}^\perp}$ (Proposition 3.17), and if the cycle leaves along c, through the $c \otimes A^*|_V$ component, the leg $A^*|_{W_{\mathbb{R}}^\perp}$ of that vertex is external, of weight 3, which is excluded. Hence the cycle passes through cubic vertices of type $A|_V (A|_{W_{\mathbb{R}}^\perp})^2$ and quartic vertices, with at most two external legs: one cubic vertex gives (b), one quartic vertex gives (c) or (e), and two cubic vertices give (d) or (f), according to whether the edge between them is $A|_{W_{\mathbb{R}}^\perp}$ or $A|_V$. □

Proposition 5.14 (Singular parts graph by graph)**.** *The singular parts of the graphs (a)–(f) are:*

graph	$\varepsilon^{-1} \int \omega$	$\int \omega(\iota_{w_1} A, \iota_{w_2} A)$	$(w_1, w_2) \int \omega(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$(w_1, w_2) \int \omega(A\|_V, A\|_V)$
(a)	$\frac{2\hbar \dim \mathfrak{g}\,(w_1, w_2)}{16\pi^2}$	0	0	0
(b)	0	0	0	0
(c)	0	0	0	-2
(d)	0	0	0	2
(e)	0	2	-2	0
(f)	0	2	0	0
total	$\frac{2\hbar \dim \mathfrak{g}\,(w_1, w_2)}{16\pi^2}$	4	-2	0

Singular parts of the graphs of the figure. The first column is the coefficient of $\varepsilon^{-1} \int \omega$; the other entries are coefficients of $\frac{\hbar\, \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X, Y)} \log \varepsilon$ times the functional at the head of the column.

Proof. External fibre elements are taken decomposable, which suffices by linearity; the external momenta are p_1, p_2 and ω absorbs $-p_1 - p_2$. By Lemma 5.10, the $\log \varepsilon$ coefficient of a graph of degree 0 depends on the top-degree part of its numerator in the loop momentum p, so in numerators the external momenta are set to 0. The local functionals of the table have the following vertex functions (Lemma 5.8), the first two on legs $(w_3, -) \otimes X_1$, $(w_4, -) \otimes X_2$ and the third on legs $(v_3, -) \otimes X_1$, $(v_4, -) \otimes X_2$, with $w_3, w_4 \in W_{\mathbb{R}}^\perp$, $v_3, v_4 \in V$, $X_1, X_2 \in \mathfrak{g}$:

$$\begin{aligned} \int \omega\,(\iota_{w_1} A, \iota_{w_2} A) : &\quad \big[(w_1, w_3)(w_2, w_4) + (w_1, w_4)(w_2, w_3)\big](X_1, X_2), \\ (w_1, w_2) \int \omega\,(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}) : &\quad 2(w_1, w_2)(w_3, w_4)(X_1, X_2), \\ (w_1, w_2) \int \omega\,(A|_V, A|_V) : &\quad 2(w_1, w_2)(v_3, v_4)(X_1, X_2). \end{aligned}$$

The graphs are computed in the examples below; (a) and (b) are as follows.

Graph (a) is the term $n=1$ of Proposition 3.36, $\hbar\,\partial_{P(\varepsilon,\infty)}O$. The $A\otimes A$ component of $P(\varepsilon,\infty)$ at coincident points is $2(-,-)^{-1}\otimes$ (inverse metric of V_{10}) $\cdot\,\frac{1}{16\pi^2\varepsilon}$ (Proposition 3.17); contracting $\iota_{w_1}\otimes\iota_{w_2}$ with the inverse metric gives (w_1,w_2), and $(-,-)$ on $(-,-)^{-1}$ gives $\dim\mathfrak{g}$ (Lemma 5.7):

$$\hbar\,\partial_{P(\varepsilon,\infty)}O=\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\,\varepsilon}\int\omega.$$

Graph (b): by (5.3) the cubic vertex with external leg $(v_3,-)\otimes X_1$ and internal legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$ is $-\frac{i}{2}(v_3,p'-p)(w,w')(X_1,[Z,Z'])$. Both edges end at O, whose vertex function is symmetric in its two legs, so the colour factor is $(X_1,[-,-])$ evaluated on $(-,-)^{-1}$, which is 0 by Lemma 5.7. $\square$

Example 5.15 (Graph (e): the bubble with external $A|_{W_{\mathbb{R}}^{\perp}}$)**.** Take external legs $(w_3,-)\otimes X_1$, $(w_4,-)\otimes X_2$ and internal legs $(w_1,-)\otimes Z_1$, $(w_2,-)\otimes Z_2$ at the quartic vertex. The three pairings of (5.2), with legs $1,2$ external and $3,4$ internal, are

$$\begin{aligned}
(12|34):&\quad\big[(w_3,w_1)(w_4,w_2)-(w_3,w_2)(w_4,w_1)\big]\big([X_1,X_2],[Z_1,Z_2]\big),\\
(13|24):&\quad\big[(w_3,w_4)(w_1,w_2)-(w_3,w_2)(w_1,w_4)\big]\big([X_1,Z_1],[X_2,Z_2]\big),\\
(14|23):&\quad\big[(w_3,w_4)(w_2,w_1)-(w_3,w_1)(w_2,w_4)\big]\big([X_1,Z_2],[X_2,Z_1]\big).
\end{aligned}$$

The two edges join O to the internal legs. By Lemma 5.9 each edge contributes the factor 2, the inverse metric and $(-,-)^{-1}$; contracting the vertex function of O through the two edges, the internal legs of the quartic vertex receive $(w_1,-)\otimes(w_2,-)+(w_2,-)\otimes(w_1,-)$ tensored with $(-,-)^{-1}$ in $Z_1\otimes Z_2$, and the quartic vertex function is symmetric in legs $3,4$. By Lemma 5.7, $([X_1,X_2],[Z_1,Z_2])$ gives 0 and the other two colour factors give $-\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_{X_1}\operatorname{ad}_{X_2})$, so the quartic vertex evaluated on $(w_1,-)\otimes(w_2,-)\otimes(-,-)^{-1}$ is

$$-\frac{1}{2}\Big(-\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_{X_1}\operatorname{ad}_{X_2})\Big)\Big[2(w_1,w_2)(w_3,w_4)-(w_1,w_4)(w_2,w_3)-(w_1,w_3)(w_2,w_4)\Big].$$

With the bubble weight $\frac{1}{2}$ and the edge factor 2^2, and the factor 2 from the two terms of the vertex function of O,

$$\begin{aligned}
\text{(e)}=\frac{\hbar}{2}\cdot 2^2\cdot 2\cdot\frac{1}{2}\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_{X_1}\operatorname{ad}_{X_2})\Big[2(w_1,w_2)(w_3,w_4)-(w_1,w_3)(w_2,w_4)-(w_1,w_4)(w_2,w_3)\Big]\\
\cdot\int\frac{d^4p}{(2\pi)^4}\frac{e^{-\varepsilon|p|^2}}{|p|^2}\frac{e^{-\varepsilon|p+p_1+p_2|^2}}{|p+p_1+p_2|^2}.
\end{aligned}$$

Lemma 5.10 with two edges and numerator 1 replaces the integral by $-\frac{1}{16\pi^2}\log\varepsilon$ in the singular part, and Lemma 5.7 replaces $\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_{X_1}\operatorname{ad}_{X_2})$ by $\frac{\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y)}{(X,Y)}(X_1,X_2)$:

$$\begin{aligned}
\text{singular part of (e)}&=\frac{\hbar\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\\
&\quad\cdot\Big[2\big((w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)\big)-4(w_1,w_2)(w_3,w_4)\Big](X_1,X_2)\\
&=\frac{\hbar\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\,\Big[2\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-2(w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})\Big],
\end{aligned}$$

the second line read off from the vertex functions listed in the proof of Proposition 5.14.

Example 5.16 (Graph (f): the triangle with external $A|_{W_{\mathbb{R}}^{\perp}}$)**.** Route the loop momentum p from O to the vertex carrying $(w_3,-)\otimes X_1$, then along the $A|_V$ edge to the vertex carrying $(w_4,-)\otimes X_2$, then back to O. By (5.3), with the external momenta set to 0 in the numerator, the two cubic vertices are

$$\begin{aligned}
\text{first vertex:}&\quad-\tfrac{i}{2}\,(v,0-p)\,(w,w_3)\,\big(Z'',[Z,X_1]\big)=\tfrac{i}{2}\,(v,p)\,(w,w_3)\,\big(Z'',[Z,X_1]\big),\\
\text{second vertex:}&\quad-\tfrac{i}{2}\,(v',0+p)\,(w',w_4)\,\big(Z''',[Z',X_2]\big),
\end{aligned}$$

where $(w,-)\otimes Z$ and $(w',-)\otimes Z'$ are the internal $A|_{W_{\mathbb{R}}^{\perp}}$ legs (incoming momenta p and $-p$) and $(v,-)\otimes Z''$, $(v',-)\otimes Z'''$ the two ends of the $A|_V$ edge. The edges give:

- the $A|_V$ edge: 2, $(v,p)(v',p)\mapsto|p|^2$, and $(Z'',[Z,X_1])(Z''',[Z',X_2])\mapsto([Z,X_1],[Z',X_2])$;
- the two $A|_{W_{\mathbb{R}}^{\perp}}$ edges at O: the factor 2^2; the metric contractions of $(w,w_3)(w',w_4)$ with the vertex function of O, which give $(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)$; and $(-,-)^{-1}$ in $Z\otimes Z'$, so that $([Z,X_1],[Z',X_2])$ becomes $-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$ by Lemma 5.7.

With the triangle weight 1 and $\frac{i}{2}\cdot\big(-\frac{i}{2}\big)=\frac{1}{4}$, the numerator of the graph is

$$\begin{aligned}\big(\text{numerator of (f)}\big)(p) &= 2^3\cdot\frac{1}{4}\,|p|^2\cdot\Big(-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})\Big)\Big[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)\Big]\\ &\quad+(\text{lower degree in } p),\end{aligned}$$

so its average in the sense of Lemma 5.10 is $-2\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)]$, and the lemma with three edges gives

$$\begin{aligned}\text{singular part of (f)} &= \hbar\cdot\frac{2\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}\big[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)\big]\log\varepsilon\\ &= 2\,\frac{\hbar\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\int\omega\,(\iota_{w_1}A,\iota_{w_2}A).\end{aligned}$$

Example 5.17 (Graphs (c) and (d): the cancellation of $(A|_V,A|_V)$)**.** Take external legs $(v_3,-)\otimes X_1$, $(v_4,-)\otimes X_2$ with $v_3,v_4\in V$.

Graph (c). In (5.2) with legs $1,2$ external and $3,4$ internal along w_1,w_2, the pairing $(12|34)$ carries $(v_3,w_1)(v_4,w_2)-(v_3,w_2)(v_4,w_1)=0$, since $V\perp W_{\mathbb{R}}^{\perp}$, and the pairings $(13|24)$, $(14|23)$ carry $(v_3,v_4)(w_1,w_2)$. By Lemma 5.7 the quartic vertex evaluated on $(w_1,-)\otimes(w_2,-)\otimes(-,-)^{-1}$ is $-\frac{1}{2}\cdot2\cdot(-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}))(v_3,v_4)(w_1,w_2)$, and

$$\text{(c)}=\frac{\hbar}{2}\cdot2^2\cdot2\cdot\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})(v_3,v_4)(w_1,w_2)\int\frac{d^4p}{(2\pi)^4}\frac{e^{-\varepsilon|p|^2}}{|p|^2}\frac{e^{-\varepsilon|p+p_1+p_2|^2}}{|p+p_1+p_2|^2},$$

$$\begin{aligned}\text{singular part of (c)} &= -\frac{4\hbar\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}(v_3,v_4)(w_1,w_2)\log\varepsilon\\ &= -2\,\frac{\hbar\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\,(w_1,w_2)\int\omega\,(A|_V,A|_V).\end{aligned}$$

Graph (d). Route p from O to the vertex carrying $(v_3,-)\otimes X_1$, to the vertex carrying $(v_4,-)\otimes X_2$, and back to O, all three edges along $W_{\mathbb{R}}^{\perp}$. In (5.3) the external leg is the V-leg and the internal legs have incoming momenta p and $-p$:

$$-\tfrac{i}{2}\,(v_3,-p-p)\,(w,w')\,\big(X_1,[Z,Z']\big)=i\,(v_3,p)\,(w,w')\,\big(X_1,[Z,Z']\big),$$

and likewise $i\,(v_4,p)(w'',w''')(X_2,[Z'',Z'''])$ at the second vertex. The metric contractions along the cycle give (w,w''') from the middle edge and then $(w_1,w_2)+(w_2,w_1)=2(w_1,w_2)$ at O. The colour factors, contracted along the middle edge, give $(X_2,[[X_1,Z],Z'''])=([X_2,[X_1,Z]],Z''')$, and the two edges at O contract $Z\otimes Z'''$ with $(-,-)^{-1}$, which gives $\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_2}\mathrm{ad}_{X_1})$ by Lemma 5.7. With the edge factor 2^3, the average $\frac{1}{4}(v_3,v_4)$ of $(v_3,p)(v_4,p)/|p|^2$ over the unit sphere of V and $i\cdot i=-1$, the average of the top-degree part of the numerator is

$$2^3\cdot(-1)\cdot\tfrac{1}{4}(v_3,v_4)\cdot2(w_1,w_2)\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})=-4\,(v_3,v_4)(w_1,w_2)\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}),$$

$$\text{singular part of (d)} = +\frac{4\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}(v_3,v_4)(w_1,w_2)\log\varepsilon$$
$$= 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\,(w_1,w_2)\int\omega\,(A|_V,A|_V).$$

The two singular parts cancel, and the coefficient of $(w_1,w_2)\int\omega\,(A|_V,A|_V)$ in $\partial_\delta I^{CT}[\varepsilon]$ is 0.

Theorem 5.18 ($O(\varepsilon)$ for the scalar bilinears)**.** *For $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, at one loop,*

$$\partial_\delta I^{CT}[\varepsilon] = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-2(w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})\Big] + \frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\,\varepsilon}\int\omega. \tag{5.7}$$

For the traceless part (the bracket in (5.6)*, and every traceless combination),*

$$O(\varepsilon) = \Big(1-4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big)\,O, \tag{5.8}$$

and for the trace part $O=\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$,

$$O(\varepsilon) = \Big(1+8\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big)\,O - \frac{12\,\hbar\dim\mathfrak{g}}{16\pi^2\,\varepsilon}\int\omega. \tag{5.9}$$

Proof. By Proposition 5.3, $\partial_\delta I^{CT}[\varepsilon]$ is the sum of the singular parts of the graphs (a)–(f), which is the last row of the table of Proposition 5.14; every other set of external legs gives 0 (proof of Proposition 5.13). This is (5.7). The map $w_1w_2\mapsto\partial_\delta I^{CT}[\varepsilon]$ is linear on $\mathrm{Sym}^2W_{\mathbb{R}}^\perp$. On a traceless combination the terms with the factor (w_1,w_2) add up to zero, leaving 4 times the operator; for the bracket of (5.6) explicitly,

$$\Big[4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-2(w_1,w_2)\,(\mathrm{trace})\Big]-\tfrac{1}{6}(w_1,w_2)\Big[4\,(\mathrm{trace})-12\,(\mathrm{trace})\Big]$$
$$=4\Big[\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-\tfrac{1}{6}(w_1,w_2)\,(\mathrm{trace})\Big],$$

where (trace) stands for $\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$ and the vacuum terms cancel in the same way: $2(w_1,w_2)-\frac{1}{6}(w_1,w_2)\cdot12=0$. For the trace, replace w_1w_2 by the inverse metric of $W_{\mathbb{R}}^\perp$: (w_1,w_2) becomes $\dim W_{\mathbb{R}}^\perp=6$ and $\int\omega(\iota_{w_1}A,\iota_{w_2}A)$ becomes (trace), so the bracket of (5.7) becomes $4\,(\mathrm{trace})-12\,(\mathrm{trace})=-8\,(\mathrm{trace})$ and the vacuum term becomes $\frac{12\hbar\dim\mathfrak{g}}{16\pi^2\varepsilon}\int\omega$. Then $O(\varepsilon)=O-\partial_\delta I^{CT}[\varepsilon]$ gives (5.8) and (5.9). □

Remark 5.19 (Method of computation)**.** The entries of the table of Proposition 5.14 are exact rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$. In explicit realisations of V_{10}, of the Clifford module and of $\mathfrak{g}=\mathfrak{su}(2)$ and $\mathfrak{g}=\mathfrak{su}(3)$, the Schwinger integrals of each graph are integrated in closed form in exact rational arithmetic and the singular part is read off from the exact ε-expansion; the multiples are the same for both gauge algebras, and the colour factors are reduced for a general simple $\mathfrak{g}$ by Lemma 5.7. The graphs (a)–(f) are all one-loop graphs with one vertex O on the cycle and superficial degree ≥ 0 (Proposition 5.13).

Chapter 6

The dilatation component and anomalous dimensions

The dilatation is the simplest superconformal component of the BV transformation of an operator. The classical theory is invariant under the rescaling of the fields by their Weyl weights, and the regulator breaks this invariance through ε alone: the rescaling maps $P(\varepsilon,\infty)$ to $P(s^2\varepsilon,\infty)$. Consequently the scale dependence of $O[\infty]$ is carried by the $\log\varepsilon$ parts of $I^{CT}[\varepsilon]$ and of $\partial_\delta I^{CT}[\varepsilon]$, and by Theorem 4.22 (the logarithmic part of the one-loop counterterm) the part coming from the action is a field redefinition. This chapter proves these statements, computes the resulting coefficients for the scalar bilinears and the single letters, shows how the boundary graphs with an L_ε edge produce them, and states the identification with the equivariant dilatation cochain as the task of Step 2, together with the graphs that Step 2 must evaluate.

6.1 Scaling covariance of the regulator

Definition 6.1 (Weyl scaling)**.** The Weyl weights of the slots are

slot	c	A	λ	A^*	λ^*	c^*
Weyl weight	0	1	$\frac{3}{2}$	3	$\frac{5}{2}$	4

Weyl weights; a field and its antifield have weights adding up to 4. They are one half of the weights $0,2,3$ of c, A, λ read off from $S^{(1)}(\Delta)$ in Example 2.13.

For a real number $s>0$ and a section e in a slot,

$$(\sigma_s e)(x) = s^{-(\text{Weyl weight of the slot})}\, e(s^{-1}x),$$

extended to $\mathscr{E}\otimes\mathscr{E}$ as $\sigma_s\otimes\sigma_s$ and to functionals by $(\sigma_s F)(e)=F(\sigma_s^{-1}e)$. For a local operator $O=\int\omega\,(\text{density})$ of Weyl weight w_O (Definition 5.1) we write

$$(E+w_O)\,O=\int\omega\,(E+w_O)(\text{density}),$$

where E is the Euler vector field of [SW, (3)] acting on the position dependence of the density.

Lemma 6.2 (Scaling of the data)**.**

- $\langle\sigma_s e_0,\sigma_s e_1\rangle=\langle e_0,e_1\rangle$ *and* $(\sigma_s\otimes\sigma_s)K_t=K_{s^2t}$.
- $\sigma_s Q=Q\sigma_s$, $\sigma_s Q^{GF}=s^2Q^{GF}\sigma_s$ *and* $\sigma_s H=s^2H\sigma_s$; *on functionals* σ_s *commutes with* Q *and with the bracket.*
- $\sigma_s S=S$ *and* $\sigma_s I=I$. *A local functional* $\int\mathrm{dvol}\,(\textit{density})$ *whose density has Weyl weight* k *(fields plus derivatives) satisfies* $\sigma_s\int\mathrm{dvol}\,(\textit{density})=s^{k-4}\int\mathrm{dvol}\,(\textit{density})$; *for example* $\sigma_s\int\mathrm{dvol}\,(A,A)=s^{-2}\int\mathrm{dvol}\,(A,A)$.

- *For a local operator O, $\sigma_s O$ is O with ω replaced by s^{w_O} times the pullback of ω along $x \mapsto s^{-1}x$, and $s\frac{d}{ds}\big|_{s=1}\sigma_s O = (E + w_O)\,O$.*

Proof. For the pairing substitute $x = sy$, $\mathrm{dvol}(x) = s^4\mathrm{dvol}(y)$; the two weights add up to 4. The scalar factor of K_t (Definition 3.14) becomes $s^{-4}(4\pi t)^{-2}e^{-|x-y|^2/4s^2t} = (4\pi s^2 t)^{-2}e^{-|x-y|^2/4s^2t}$. An operator with j derivatives from a slot of Weyl weight k to a slot of Weyl weight k' is multiplied by $s^{k+j-k'}$ under conjugation by σ_s. For the arrows of Q (Proposition 1.11) the exponent is $0+1-1$, $1+2-3$, $3+1-4$, $\frac{3}{2}+1-\frac{5}{2}$, all 0; for the arrows of Q^{GF} (Definition 3.5) it is $3+0-1$, $1+1-0$, $4+1-3$, $\frac{5}{2}+1-\frac{3}{2}$, all 2; hence also for $H = [Q, Q^{GF}]$. Every term of (1.1) has a density of Weyl weight 4: for example $1+1+2$ for (dA, dA), $\frac{3}{2}+\frac{3}{2}+1$ for $(\lambda, \not{d}\lambda)$, $0+1+3$ for (dc, A^*) and $([A,c], A^*)$, $0+\frac{3}{2}+\frac{5}{2}$, $0+0+4$, and 4 for the cubic and quartic gauge terms. For a density of Weyl weight k, $(\text{density})(\sigma_s^{-1}e)(x) = s^k(\text{density})(e)(sx)$, and $\int \mathrm{dvol}(x)\, s^k(\text{density})(sx) = s^{k-4}\int \mathrm{dvol}\,(\text{density})$. With ω in place of dvol the same substitution moves the factor s^{w_O} and the dilation onto ω, and $s\frac{d}{ds}\big|_{s=1}s^{w_O}(\text{density})(sx) = (w_O+E)(\text{density})(x)$. □

Proposition 6.3 (Scaling covariance)**.** *For every $s > 0$ and $\varepsilon > 0$,*

$$(\sigma_s \otimes \sigma_s)\, P(\varepsilon, \infty) = P(s^2\varepsilon, \infty), \qquad \sigma_s\, \partial_{P(\varepsilon,\infty)} = \partial_{P(s^2\varepsilon,\infty)}\, \sigma_s, \tag{6.1}$$

and for every even functional J as in Definition 3.22,

$$\sigma_s\, W\big(P(\varepsilon,\infty), J\big) = W\big(P(s^2\varepsilon,\infty), \sigma_s J\big). \tag{6.2}$$

Proof. By Lemma 6.2, $(\sigma_s \otimes \sigma_s)L_t = (\sigma_s Q^{GF} \otimes \sigma_s)K_t = s^2(Q^{GF} \otimes 1)K_{s^2t} = s^2 L_{s^2t}$, and

$$(\sigma_s \otimes \sigma_s)\, P(\varepsilon,\infty) = \int_\varepsilon^\infty s^2 L_{s^2t}\, dt = \int_{s^2\varepsilon}^\infty L_t\, dt = P(s^2\varepsilon,\infty).$$

For linear functionals ℓ_1, ℓ_2, the operator ∂_ϕ of [C07, §6.4] gives $\partial_\phi\big((\ell_1\circ\sigma_s^{-1})(\ell_2\circ\sigma_s^{-1})\big) = (\ell_1\otimes\ell_2)\big((\sigma_s^{-1}\otimes\sigma_s^{-1})\phi\big)$; since ∂_ϕ is determined by its values on products of two linear functionals and σ_s is an even algebra automorphism of functionals, $\partial_\phi\,\sigma_s = \sigma_s\,\partial_{(\sigma_s^{-1}\otimes\sigma_s^{-1})\phi}$. With $\phi = P(s^2\varepsilon,\infty)$ and the first identity for s^{-1} this is the second identity in (6.1). Then $\sigma_s e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar} = e^{\hbar\partial_{P(s^2\varepsilon,\infty)}}e^{\sigma_s J/\hbar}$, and taking $\hbar\log$ gives (6.2). □

Lemma 6.4 (Covariance of the counterterms)**.** *For a local family $J = I + \delta O$ with O homogeneous, let $J^{CT}[\varepsilon] = I^{CT}[\varepsilon] + \delta\,\partial_\delta I^{CT}[\varepsilon]$ be its one-loop counterterm (Proposition 5.3). Then the counterterm of the family $\sigma_s J = I + \delta\,\sigma_s O$ at the cutoff $s^2\varepsilon$ is*

$$(\sigma_s J)^{CT}[s^2\varepsilon] = \sigma_s\big(J^{CT}[\varepsilon]\big) + 2\log s\;\sigma_s\big(\textit{coefficient of } \log\varepsilon \textit{ in } J^{CT}[\varepsilon]\big). \tag{6.3}$$

In particular σ_s maps the coefficient of ε^{-1} in $I^{CT}[\varepsilon]$ to s^{-2} times itself and fixes the coefficient of $\log\varepsilon$.

Proof. At one loop W is affine in the counterterm, which enters through its tree-level flow. By (6.2), $W(P(s^2\varepsilon,\infty), \sigma_s J - \sigma_s J^{CT}[\varepsilon]) = \sigma_s W(P(\varepsilon,\infty), J - J^{CT}[\varepsilon])$ has a limit as $\varepsilon \to 0$, and so has $W(P(s^2\varepsilon,\infty), \sigma_s J - (\sigma_s J)^{CT}[s^2\varepsilon])$. The difference of the two counterterms is a local functional whose tree-level flow has a limit, and its part without δ is invariant under translations and under the adjoint action of $\mathfrak{g}$, which σ_s preserves; so, as in the uniqueness parts of Propositions 4.10 and 5.3, their singular parts in ε coincide. Write $(\sigma_s J)^{CT}[\varepsilon']$ as ε'^{-1} times one local functional plus $\log\varepsilon'$ times another. At $\varepsilon' = s^2\varepsilon$, $\log\varepsilon' = \log\varepsilon + 2\log s$ and $\varepsilon'^{-1} = s^{-2}\varepsilon^{-1}$; matching the coefficients of ε^{-1} and $\log\varepsilon$ with those of $\sigma_s J^{CT}[\varepsilon]$ gives (6.3), the constant $2\log s$ times the $\log\varepsilon$ coefficient being the remainder. For $J = I$, $\sigma_s I = I$ and the uniqueness of Proposition 4.10 give the last statement; it agrees with Lemma 6.2: the coefficient of ε^{-1} is a multiple of $\int \mathrm{dvol}\,(A|_V, A|_V) + \int \mathrm{dvol}\,(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ (Theorem 4.16), of density weight 2. □

6.2 The Callan–Symanzik form

Definition 6.5 (Tree insertion)**.** For local functionals F and X of either parity, the tree insertion of F into $X[\infty]$ is

$$\lim_{\varepsilon\to 0}\Big(\hbar\, e^{-W(P(\varepsilon,\infty),I)/\hbar}\, e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(\hbar^{-1}F\,\hbar^{-1}X\, e^{I/\hbar}\big)\Big)_{\text{connected trees}},$$

the sum of the connected trees with one vertex F, one vertex X, any number of vertices I and edges $P(0,\infty)$, with the weights of Proposition 3.35, the vertex functions of F and X being multiplied in this order. For odd F and odd X this order fixes the sign; the opposite order multiplies the insertion by -1. For an even F of order $\hbar$ and $X=O$, at order $\hbar$ it is

$$\lim_{\varepsilon\to 0}\partial_\delta\Big[W\big(P(\varepsilon,\infty),I-I^{CT}[\varepsilon]+\delta O(\varepsilon)+F\big)-W\big(P(\varepsilon,\infty),I-I^{CT}[\varepsilon]+\delta O(\varepsilon)\big)\Big].$$

For a local functional F of order $\hbar$, $F[\infty]$ denotes at order $\hbar$ the construction (5.1) applied to F: the trees with one vertex F and vertices I.

Theorem 6.6 (Callan–Symanzik form)**.** *For a homogeneous local operator O, modulo $\hbar^2$,*

$$\begin{aligned} s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) = {}& \big((E+w_O)\,O\big)[\infty]+2\,\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty]\\ &+2\,\big(\textit{tree insertion of the coefficient of } \log\varepsilon \textit{ in } I^{CT}[\varepsilon] \textit{ into } O[\infty]\big).\end{aligned}\tag{6.4}$$

The coefficients of ε^{-1} in $I^{CT}[\varepsilon]$ and in $\partial_\delta I^{CT}[\varepsilon]$ are absorbed by the scaling of the cutoff, and the scale dependence of $O[\infty]$ is carried by the $\log\varepsilon$ parts.

Proof. Put $J(\varepsilon)=I-I^{CT}[\varepsilon]+\delta O(\varepsilon)$. Since σ_s is independent of ε and commutes with ∂_δ, (6.2) gives

$$\sigma_s\big(O[\infty]\big)=\lim_{\varepsilon\to 0}\partial_\delta W\big(P(s^2\varepsilon,\infty),\sigma_s J(\varepsilon)\big).$$

By $\sigma_s I=I$ and (6.3), with $\varepsilon'=s^2\varepsilon$,

$$\begin{aligned}\sigma_s J(\varepsilon)=\ &\Big(I-I^{CT}[\varepsilon']+\delta\,(\sigma_s O)(\varepsilon')\Big)\\ &+2\log s\,\sigma_s\big(\text{coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon]+\delta\,\partial_\delta I^{CT}[\varepsilon]\big),\end{aligned}$$

where $(\sigma_s O)(\varepsilon')$ is the local family of Definition 5.2 for the operator $\sigma_s O$ at the cutoff ε', by (6.3). The second term is of order $\hbar$, so modulo $\hbar^2$ it enters linearly and through trees. As $\varepsilon\to 0$, the first bracket gives $(\sigma_s O)[\infty]$, and the second gives $2\log s$ times the observable of the δ-part plus the tree insertion of the δ-independent part, both for $\sigma_s O$. Hence, modulo $\hbar^2$,

$$\begin{aligned}\sigma_s\big(O[\infty]\big)=(\sigma_s O)[\infty]+2\log s\Big[&\sigma_s\big(\text{coefficient of } \log\varepsilon \text{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty]\\ &+\big(\text{tree insertion of the coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon]\\ &\qquad\text{into } (\sigma_s O)[\infty]\big)\Big].\end{aligned}$$

The construction $O\mapsto O[\infty]$ is linear in ω, so $s\frac{d}{ds}\big|_{s=1}(\sigma_s O)[\infty]=((E+w_O)O)[\infty]$ by Lemma 6.2; the derivative of $2\log s$ is 2 at $s=1$. This is (6.4). The ε^{-1} coefficients are absorbed into $I^{CT}[\varepsilon']$ and $\partial_\delta I^{CT}[\varepsilon']$ by (6.3). □

6.3 The action counterterm as a field redefinition

By Theorem 4.22 (the logarithmic part of the one-loop counterterm), the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ is $(Q+\{I,-\})$-exact:

$$\begin{aligned}&\text{coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon]\\ &\quad=\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,(Q+\{I,-\})\Big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big).\end{aligned}\tag{6.5}$$

Here $\int(A,A^*)$, $\int(\lambda^*,\lambda)$, $\int(c^*,c)$ are the local functionals of Chapter 4, the pairing of Definition 1.5 applied to the two slots of a pair with the even member first; we call $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ the field-counting functional. The quadratic part of (6.5) is $\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(-\int\mathrm{dvol}\,(dA,dA)+4\int\mathrm{dvol}\,(\lambda,\not d\lambda)-\int\mathrm{dvol}\,(dc,A^*)\big)$ (Theorem 4.16).

Lemma 6.7 (The field-counting functional)**.** *The derivation* $\big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),-\big\}$ *multiplies the coordinate functionals of* c, A, λ, A^*, λ^*, c^* *by* 1, 2, 4, -2, -4, -1, *hence a monomial by the corresponding weighted count of its fields. On the operators of Chapter 5 it acts as*

$$\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)\mapsto 4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A),\qquad\int\omega\,(X,\iota_wA)\mapsto 2\,(\cdot),$$
$$\int\omega\,(X,\iota_v(A|_V))\mapsto 2\,(\cdot),\qquad\int\omega\,(\zeta,\lambda)\mapsto 4\,(\cdot),\qquad\int\omega\,(X,c)\mapsto(\cdot),$$

where $(\cdot)$ *is the operator on the left.*

Proof. This is Lemma 4.21 applied to the monomials of the operators: two factors A, one A, one A, one λ, one c, and no antifields. □

Lemma 6.8 (Insertion of a BV-exact functional)**.** *Let* F *be an odd local functional and* O *a homogeneous local operator. At order* $\hbar^0$*,*

$$\begin{aligned}&\textit{tree insertion of } (Q+\{I,-\})F \textit{ into } O[\infty]=Q\big(\textit{tree insertion of } F \textit{ into } O[\infty]\big)-\{F,O\}[\infty]\\ &\qquad\qquad+\textit{tree insertion of } F \textit{ into } \big((Q+\{I,-\})O\big)[\infty].\end{aligned}\tag{6.6}$$

Proof. Put $J=I+\delta O$. By Theorem 3.23, $Q\,e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(F\,e^{J/\hbar}\big)=e^{\hbar\partial_{P(\varepsilon,\infty)}}(Q+\hbar\Delta_\varepsilon)\big(F\,e^{J/\hbar}\big)$. By the product rule of Definition 3.21 and $|F|$ odd,

$$\begin{aligned}(Q+\hbar\Delta_\varepsilon)\big(F\,e^{J/\hbar}\big)=\big(&QF+\{J,F\}_\varepsilon+\hbar\Delta_\varepsilon F\big)e^{J/\hbar}\\&-F\,\hbar^{-1}\big(QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)e^{J/\hbar}.\end{aligned}$$

Divide both sides by $e^{W(P(\varepsilon,\infty),J)/\hbar}$. On the left, Q acts on the product $e^{W(P(\varepsilon,\infty),J)/\hbar}\cdot e^{-W(P(\varepsilon,\infty),J)/\hbar}e^{\hbar\partial_{P(\varepsilon,\infty)}}(Fe^{J/\hbar})$, and the term in which it acts on $e^{W(P,J)/\hbar}$ equals, by Corollary 3.24, the disconnected part of the last term on the right; the two cancel. In the remaining terms take the part linear in δ, connected, at order $\hbar^0$. The term $\hbar\Delta_\varepsilon F$ is of order $\hbar$. For even O the δ-part of $\{J,F\}_\varepsilon$ is $\{O,F\}_\varepsilon=\{F,O\}_\varepsilon$. For odd O the parameter δ is odd, and moving δ to the left through F, Q and the bracket gives $\{\delta O,F\}_\varepsilon=-\delta\{O,F\}_\varepsilon=\delta\{F,O\}_\varepsilon$, $Q(\delta O)=-\delta\,QO$, $\{I,\delta O\}_\varepsilon=-\delta\{I,O\}_\varepsilon$ and $F\,\delta=-\delta\,F$; the signs agree term by term, and after removing δ the identities below are the same in both cases, with the vertex functions of F placed first as in Definition 6.5. The δ-part of $QJ+\frac12\{J,J\}_\varepsilon$ is $QO+\{I,O\}_\varepsilon$, and its δ-independent part is $QI+\frac12\{I,I\}_\varepsilon$. This gives, with edges $P(\varepsilon,\infty)$,

$$\begin{aligned}Q\big(\text{tree insertion of } F \text{ into } O\big)=\ &\text{tree insertion of } QF+\{I,F\}_\varepsilon \text{ into } O\ +\ \big(\{F,O\}_\varepsilon \text{ flowed}\big)\\&-\text{tree insertion of } F \text{ into } QO+\{I,O\}_\varepsilon\\&-\text{trees with } F,\ O \text{ and } QI+\tfrac12\{I,I\}_\varepsilon.\end{aligned}$$

As $\varepsilon \to 0$, $\{-,-\}_\varepsilon$ tends to $\{-,-\}$ on local functionals, the edges tend to $P(0,\infty)$, and $QI + \frac{1}{2}\{I,I\} = 0$ by Proposition 1.14. Rearranging gives (6.6). □

Corollary 6.9 (The field redefinition)**.** *For a homogeneous local operator O, modulo $\hbar^2$,*

$$\begin{aligned} s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) = {} & \big((E+w_O)O\big)[\infty] \\ & + 2\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] \\ & - 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big\{\textit{field-counting functional}, O\big\}[\infty] \\ & + Q\Big(2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot \textit{tree insertion of the field-counting functional into } O[\infty]\Big) \\ & + 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} \\ & \qquad \cdot \textit{tree insertion of the field-counting functional into } \big((Q+\{I,-\})O\big)[\infty]. \end{aligned} \tag{6.7}$$

Proof. Insert (6.5) into the last term of (6.4) and apply Lemma 6.8 with F the field-counting functional; the factor $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ makes the order-$\hbar^0$ identity (6.6) an identity at order $\hbar$. □

Remark 6.10 (Reading of the three terms)**.** By (6.5) the action counterterm is the change of S under the infinitesimal rescaling of the fields by the weights of Lemma 6.7; on $O[\infty]$ it acts through the bracket $\{-,O\}$ with the field-counting functional, the field redefinition proper, and through the image under Q of a tree. The tree insertion of the field-counting functional into $O[\infty]$ is a nonlocal functional whose image under Q is the scale-∞ observable of a local functional; by Proposition 5.4 such terms are the ones that a statement about classes of local functionals must specify, and the identification with the equivariant dilatation cochain, stated in the last section of this chapter, specifies how this term enters. The last term involves $(Q+\{I,-\})O$ and vanishes for gauge-invariant O.

6.4 The dilatation component

The dilatation is $\Delta = 2E$, the generator $\mathrm{diag}(1,-1\,|\,0)$ of [SW, (17)] (Remark 2.6). By Example 2.13 it acts classically on a field as $2E$ plus twice its Weyl weight, hence on a local operator as $2(E+w_O)$, and by Lemma 6.2 this is $2s\frac{d}{ds}\big|_{s=1}\sigma_s$.

Theorem 6.11 (Dilatation in Callan–Symanzik form)**.** *For a homogeneous local operator O, modulo $\hbar^2$,*

$$\begin{aligned} & 2s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) - Q\Big(4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot \textit{tree insertion of the field-counting functional into } O[\infty]\Big) \\ & \quad = 2\big((E+w_O)O\big)[\infty] \\ & \qquad + 4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] - 4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big\{\textit{field-counting functional}, O\big\}[\infty] \\ & \qquad + \big(\textit{terms built from } (Q+\{I,-\})O\big), \end{aligned} \tag{6.8}$$

the last terms being $4\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ times the tree insertion of the field-counting functional into $\big((Q+\{I,-\})O\big)[\infty]$. When the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$ minus the unit times the bracket with the field-counting functional is a multiple of O, the right side is

$$2\big(E + w_O + \textit{coefficient}\big)\,O[\infty] + \big(\textit{terms built from } (Q+\{I,-\})O\big),$$

with the coefficient equal to twice that multiple; its values are given in the table.

operator	w_O	$\log\varepsilon$ *coefficient of* $\partial_\delta I^{CT}[\varepsilon]$, *per* O	*bracket with the field-counting functional, per* O	*coefficient*
traceless part of $\int\omega(\iota_{w_1}A,\iota_{w_2}A)$	2	4	4	0
$\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$	2	-8	4	-24
$\int\omega\,(X,\iota_wA)$	1	0	2	-4
$\int\omega\,(X,\iota_v(A|_V))$, *modulo terms built from* $(Q+\{I,-\})O$	1	0	2	-4
$\int\omega\,(\zeta,\lambda)$	$\frac{3}{2}$	0	4	-8
$\int\omega\,(X,c)$	0	0	1	-2

The coefficient in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*; the third column in the same units (Theorem 5.18, Proposition 5.11), the fourth from Lemma 6.7, and the coefficient is twice their difference. With the generator* $-E$ *of [SW, (3)], which is* $-\frac{1}{2}$ *times* Δ *(Remark 2.6), the same statement reads* $-\big(E+w_O+\textit{coefficient}\big)O[\infty]$*. The letters are gauge dependent (Feynman gauge); the ghost letter* $\int\omega\,(X,c)$ *has degree* 1.

Proof. Multiply (6.7) by 2 and move the Q-term to the left; this is (6.8). For the scalar bilinears, (5.8) and (5.9) give the $\log\varepsilon$ coefficients 4 and -8 per operator, and Lemma 6.7 gives 4; for the letters the $\log\varepsilon$ coefficient is 0 by (5.5) and the bracket gives $2,2,4,1$. Twice the differences are $2(4-4)=0$, $2(-8-4)=-24$, $2(0-2)=-4$, -4, $2(0-4)=-8$, $2(0-1)=-2$. The operator $(E+w_O)$ acts on $O[\infty]$ through the linear dependence of $O[\infty]$ on ω. □

Remark 6.12 (The letter $A|_V$)**.** The letters are gauge variant, and $(Q+\{I,-\})O$ contributes. For $\int\omega\,(X,\iota_wA)$, $\int\omega\,(\zeta,\lambda)$ and $\int\omega\,(X,c)$ the linear part of Q on the coordinate functional vanishes (Q has no arrow into the slots of $A|_{W_{\mathbb{R}}^\perp}$, λ, c; Proposition 1.11), and $(Q+\{I,-\})O=\{I,O\}$ is quadratic in the fields: the terms built from $(Q+\{I,-\})O$ have at least two legs, and the coefficients -4, -8, -2 multiply O itself. For $\int\omega\,(X,\iota_v(A|_V))$ the arrow $c\mapsto -dc$ of Q gives the linear functional $(Q+\{I,-\})O\ni-\int\omega\,(X,\iota_v(dc))$; joined through the $c\otimes A^*|_V$ component of $P(0,\infty)$ (Proposition 3.17) to a vertex with an $A^*|_V$ leg, it produces linear functionals of $A|_V$ that depend on $A|_V$ through $d^*A|_V$, nonlocal projections onto the longitudinal part. The entry -4 of the table is therefore the coefficient of O modulo terms built from $(Q+\{I,-\})O$; the coefficient on the longitudinal part requires these terms, at order $\hbar^0$ (Proposition 6.18) and at one loop, and its evaluation is part of Step 2.

Remark 6.13 (The Konishi value)**.** For $\mathfrak{g}=\mathfrak{su}(N)$ with $(X,Y)=-\frac{4\hbar}{g^2}\,\mathrm{tr}(XY)$, tr the trace in the defining representation, $e^{S/\hbar}$ is the standard Euclidean Yang–Mills weight with coupling g, and $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)=2N\,\mathrm{tr}(XY)$. Then

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}=-\frac{g^2N}{32\pi^2},\qquad -24\cdot\Big(-\frac{g^2N}{32\pi^2}\Big)=\frac{3g^2N}{4\pi^2},$$

so the trace bilinear has dimension $2+\frac{3g^2N}{4\pi^2}$ at one loop, the one-loop value for the Konishi operator, and the twenty traceless bilinears keep dimension 2.

Corollary 6.14 (Dilatation component)**.** *Assume the identification of Step 2 (last section of this chapter),*

$$\begin{aligned}\big[(d_{CE}+Q)\,O[\infty]\big](\Delta)=2s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big)\\ -Q\Big(4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot\textit{tree insertion of }2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\\ \textit{into }O[\infty]\Big)+\big(\textit{terms built from }(Q+\{I,-\})O\big)\end{aligned}\tag{6.9}$$

modulo $\hbar^2$*, where the terms built from* $(Q+\{I,-\})O$ *are the trees with one vertex* $(Q+\{I,-\})O(\varepsilon)$ *and one vertex* $S^{(1)}(\Delta)$*, at order* $\hbar^0$*, with their one-loop corrections; they vanish for* $(Q+\{I,-\})O=0$*. Then, with* $\Delta=2E$ *of [SW, (17)],*

$$\big[(d_{CE}+Q)\,O[\infty]\big](\Delta)=2\big(E+w_O+\textit{coefficient}\big)\,O[\infty]+\big(\textit{terms built from }(Q+\{I,-\})O\big),\tag{6.10}$$

with the coefficient of the table of Theorem 6.11: 0 *for the traceless scalar bilinears,* $-24\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *for the trace, and* -4, -4, -8, -2 *times* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *for the letters* $\iota_w A$, $A|_V$ *(modulo terms built from* $(Q+\{I,-\})O$*, Remark 6.12),* λ*,* c*. With the generator* $-E$ *of [SW, (3)] the same statement reads* $-\big(E + w_O + \textit{coefficient}\big)O[\infty]$*.*

Proof. Insert (6.9) into (6.8). □

6.5 The UV boundary at work: graphs with an L_ε edge

By (3.3), $s\frac{d}{ds}\big|_{s=1}P(s^2\varepsilon,\infty) = 2\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -2\varepsilon L_\varepsilon$: the scaling acts on every edge as the boundary kernel $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$ at $t=\varepsilon$ (Remark 3.19).

Proposition 6.15 (The coefficients on the boundary)**.** *Let* γ *be a one-loop graph whose value has the expansion* $(\varepsilon^{-1}\textit{-coefficient})\,\varepsilon^{-1} + (\log\varepsilon\textit{-coefficient})\log\varepsilon + (\textit{a function with a limit})$*. Then*

$$2\cdot(\log\varepsilon\textit{-coefficient}) = \lim_{\varepsilon\to 0}\Big(-2\varepsilon\sum_{e\in E(\gamma)}\big(\textit{value of }\gamma\textit{ with }t_e=\varepsilon\textit{, i.e. edge }e\textit{ carrying }L_\varepsilon\big) + 2\varepsilon^{-1}(\varepsilon^{-1}\textit{-coefficient})\Big).$$

The term $2\varepsilon^{-1}(\varepsilon^{-1}\textit{-coefficient})$ *is the scaling* $s\frac{d}{ds}\big|_{s=1}$ *of the counterterm* $-(\varepsilon^{-1}\textit{-coefficient})\,\varepsilon^{-1}$ *under* σ_s *(Lemma 6.4). Hence the terms* $2(\cdot)$ *of* (6.4) *are the limits of the sums of the graphs with one edge carrying* $-2\varepsilon L_\varepsilon$*, the* ε^{-1} *parts cancelling against the scaling of the* ε^{-1} *counterterms; for graphs of degree* 0 *each edge carries the fraction* $1/|E(\gamma)|$ *(Lemma 5.10).*

Proof. By Proposition 4.2, $\varepsilon\frac{d}{d\varepsilon}(\text{value of }\gamma) = -\varepsilon\sum_e(\text{value with } t_e=\varepsilon)$, and $\varepsilon\frac{d}{d\varepsilon}$ of the expansion is $-(\varepsilon^{-1}\text{-coefficient})\varepsilon^{-1} + (\log\varepsilon\text{-coefficient}) + (\text{a function tending to 0})$. The last term tends to 0 by Lemma 4.6, which gives $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tending to 0. Multiply by 2 and add $2\varepsilon^{-1}(\varepsilon^{-1}\text{-coefficient})$. □

[figure] Boundary graphs with one edge carrying $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$ (shaded). (a), (b): the $A|_{W_{\mathbb{R}}^\perp}$ two-point graph with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, each edge pinned in turn. (c)–(e): graph (f) of Chapter 5, the triangle with external $A|_{W_{\mathbb{R}}^\perp}$, each of its three edges pinned in turn.

Example 6.16 (The scalar bilinears on the boundary)**.** For $O = \int\omega\,(\iota_{w_1}A, \iota_{w_2}A)$, the terms on the right of (6.4) other than $((E+w_O)O)[\infty]$ are, in their bilinear part and in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, the following limits of L_ε graphs (Proposition 6.15).

- Graph (e) of Chapter 5: each of its two edges carrying $-2\varepsilon L_\varepsilon$ tends to $2\int\omega(\iota_{w_1}A,\iota_{w_2}A) - 2(w_1,w_2)\int\omega(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$, one half of twice the singular part of Example 5.15.
- Graph (f): each of its three edges (figure, (c)–(e)) tends to $\frac{4}{3}\int\omega(\iota_{w_1}A,\iota_{w_2}A)$, one third of twice the singular part of Example 5.16.
- Graphs (c), (d): each edge of (c) tends to $-2(w_1,w_2)\int\omega(A|_V,A|_V)$ and each edge of (d) to $\frac{4}{3}(w_1,w_2)\int\omega(A|_V,A|_V)$; the sum is 0 (Example 5.17).
- Graph (a): -2ε times its single edge carrying L_ε is $-2\varepsilon^{-1}$ times its ε^{-1}-coefficient, cancelled by the scaling of the vacuum counterterm; the limit is 0.

- The legs: the $A|_{W_{\mathbb{R}}^{\perp}}$ two-point graphs of Theorem 4.16 with an L_ε edge, joined to O by an edge $P(\varepsilon,\infty)$. Their ε^{-1} parts cancel against the scaling of $\frac{9}{4}\varepsilon^{-1}\int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$, and their limit is twice the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^{\perp}}, dA|_{W_{\mathbb{R}}^{\perp}})$ into O. The graphs in which the joining edge carries L_ε tend to 0, since $\varepsilon L_\varepsilon$ tends to 0 at fixed momentum and the renormalised two-point function has a limit.

The tree insertion is computed from the vertex function $2|p|^2(w,w')(Z,Z')$ of $\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^{\perp}}, dA|_{W_{\mathbb{R}}^{\perp}})$ on legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$ of momenta $\pm p$ and the edge $2(-,-)^{-1}\otimes(\text{inverse metric})|p|^{-2}$ of $P(0,\infty)$: each $A|_{W_{\mathbb{R}}^{\perp}}$ leg of O is multiplied by

$$-1\cdot 2|p|^2\cdot\frac{2}{|p|^2}=-4,$$

so the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^{\perp}}, dA|_{W_{\mathbb{R}}^{\perp}})$ into O has bilinear part $-8\,O$. Summing, with the trace written $\int \omega(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$:

contribution	general w_1w_2	traceless part	trace
graph (e), two L_ε edges	$4O-4(w_1,w_2)\,(\text{trace})$	4	-20
graph (f), three L_ε edges	$4O$	4	4
graphs (c), (d)	0	0	0
legs, $2\times(-8)$	$-16O$	-16	-16
total	$-8O-4(w_1,w_2)\,(\text{trace})$	-8	-32
removed Q-term	$8O$	8	8
coefficient	$-4(w_1,w_2)\,(\text{trace})$	0	-24

Contributions to $s\frac{d}{ds}\big|_{s=1}\sigma_s(O[\infty])-((E+w_O)O)[\infty]$ for the scalar bilinears, bilinear parts, in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$; the traceless and trace columns give the multiple of the operator itself.

The row "removed Q-term" is the bilinear part of minus the Q-term of (6.7), the image under Q of twice the unit times the tree insertion of the field-counting functional into $O[\infty]$. Its value follows from the legs row and Lemmas 6.8, 6.7: the bilinear part involves the quadratic part of (6.5) alone, and twice its tree insertion into O equals the Q-term minus twice $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ times the bracket of the field-counting functional with O. In units, $-16\,O=(\text{bilinear part of the }Q\text{-term})-2\cdot 4\,O$, so the Q-term has bilinear part $-8\,O$ and the removed term is $+8\,O$. The coefficient row agrees with Theorem 6.11: $2(4-4)=0$ and $2(-8-4)=-24$.

Example 6.17 (The letter $\iota_w A$ on the boundary)**.** For $O=\int\omega\,(X,\iota_w A)$ there is no graph with O on the cycle (Proposition 5.11), and the boundary contributions are the legs. With the single leg multiplied by -4, the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^{\perp}}, dA|_{W_{\mathbb{R}}^{\perp}})$ into O is $-4O$, and twice it is $-8O$ in units. The bracket with the field-counting functional is $2\,O$, so by the same argument $-8\,O=(\text{linear part of the }Q\text{-term})-2\cdot 2\,O$, the Q-term has linear part $-4\,O$, and the removed term is $+4\,O$. The result is $-8+4=-4$ units, the coefficient $2(0-2)=-4$ of Theorem 6.11.

6.6 The equivariant dilatation cochain: the task of Step 2

The left side of (6.10) is defined by the equivariant construction of Definition 5.2. By Theorem 3.23 and the product rule used in Corollary 3.24, for every even J with values in $C^\bullet(\mathfrak{psl}(4|4))$,

$$(d_{CE}+Q)\,W\big(P(\varepsilon,\infty),J\big)=e^{-W(P(\varepsilon,\infty),J)/\hbar}\,e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big((d_{CE}+Q)J+\tfrac{1}{2}\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)\,e^{J/\hbar}\Big).\quad(6.11)$$

For $J=I_{\mathfrak{psl}(4|4)}-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]+\delta O(\varepsilon)$, applying ∂_δ and $\varepsilon\to 0$ gives $(d_{CE}+Q)O[\infty]$. Every bracket $\{-,-\}_\varepsilon$ and every $\Delta_\varepsilon=-\partial_{K_\varepsilon}$ is an edge carrying K_ε (Remark 3.19). Evaluated on Δ, the component of arity

one contains the vertex $S^{(1)}(\Delta)$ of Example 2.13, or the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ, exactly once. Its terms are:

- the graphs with one vertex $O(\varepsilon)$, one vertex $S^{(1)}(\Delta)$ joined by an edge K_ε, and edges $P(\varepsilon,\infty)$ (figure): one-loop graphs (a)–(u), and trees (v)–(y) in which the vertex is $\partial_\delta I^{CT}[\varepsilon]$;
- the graphs with the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ, and with the finite counterterms that the equivariant master equation requires along Δ;
- the component along Δ of the equivariant master-equation defect $(d_{CE}+Q)J+\frac{1}{2}\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J$ at $\delta=0$, inserted together with $O(\varepsilon)$;
- the graphs with the vertex $QO(\varepsilon)+\{I-I^{CT}[\varepsilon],O(\varepsilon)\}_\varepsilon+\hbar\Delta_\varepsilon O(\varepsilon)$ and one vertex $S^{(1)}(\Delta)$, which at order $\hbar^0$ are the trees of Proposition 6.18 and vanish for $(Q+\{I,-\})O=0$.

[figure] The graphs with one vertex O, one vertex $S^{(1)}(\Delta)$ (zigzag leg Δ) and one edge carrying K_ε (shaded, arrow from the field to the antifield slot); all other edges carry $P(\varepsilon,\infty)$. (a)–(u): one loop; (v)–(y): trees with the vertex $\partial_\delta I^{CT}[\varepsilon]$ (cross).

Proposition 6.18 (Order $\hbar^0$)**.** *At order $\hbar^0$, the component $[(d_{CE}+Q)O[\infty]](\Delta)$ is the sum of $2\big((E+w_O)O\big)[\infty]$ and the trees with one vertex $(Q+\{I,-\})O$ and one vertex $S^{(1)}(\Delta)$. In particular, for O with $(Q+\{I,-\})O=0$, both sides of* (6.9) *equal $2\big((E+w_O)O\big)[\infty]$, and* (6.9) *holds at order $\hbar^0$ for every O, with these trees as its last term.*

Proof. At order $\hbar^0$ the counterterms and the field-redefinition term are absent, and the first term of the right side of (6.9) is $2((E+w_O)O)[\infty]$ by Theorem 6.6. On the left, by (6.11) at order $\hbar^0$, the arity-one component along Δ is the flow of graphs with one vertex given by the part of $(d_{CE}+Q)J+\frac{1}{2}\{J,J\}_\varepsilon$ that is linear in δ or contains $S^{(1)}(\Delta)$. The defect at $\delta=0$ tends to 0 by the master equation of $S_{\mathfrak{psl}(4|4)}$ (Definition 2.10). The vertex $\{S^{(1)}(\Delta),O\}_\varepsilon$, with all other vertices I, tends to the tree flow of the classical bracket of $S^{(1)}(\Delta)$ with O; by Example 2.13 this bracket transforms each field by $2E$ plus twice its Weyl weight, hence O by $2(E+w_O)$. The vertex $QO+\{I,O\}_\varepsilon$, of arity zero, joined to a vertex $S^{(1)}(\Delta)$, gives the trees with one vertex $(Q+\{I,-\})O$ and one vertex $S^{(1)}(\Delta)$ in the limit. For $O=\int\omega\,(X,\iota_v(A|_V))$ these trees are nonzero: $QO=-\int\omega\,(X,\iota_v(dc))$ is linear, and its c leg joined to the $A^*|_V$ leg of $S^{(1)}(\Delta)$ through the $c\otimes A^*|_V$ component of $P(0,\infty)$ gives a linear functional of $A|_V$ through $d^*A|_V$. □

Remark 6.19 (Step 2)**.** Step 2 of the program consists of three computations at order $\hbar$: the component along Δ of the one-loop counterterm of $I_{\mathfrak{psl}(4|4)}$, with the finite counterterms fixed by the equivariant master equation along Δ; the graphs (a)–(y) of the figure together with the defect insertion; and the proof of (6.9) for every homogeneous local operator, with the trees built from $(Q+\{I,-\})O(\varepsilon)$ and $S^{(1)}(\Delta)$ included. Reading the result as the anomalous dimension of a class also needs the cocycle condition $Q\,O[\infty]=0$ of Remark 5.6; a finite local counterterm of Weyl weight w_O that it may require is independent of ε, scales covariantly, and leaves the coefficients of Theorem 6.11 unchanged. The form of (6.9) is dictated by (6.5): the Weyl scaling and the dilatation cochain differ by the field redefinition generated by $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$, whose image under Q in (6.7) is the term subtracted in (6.9). Until that computation, (6.10) is the dilatation component obtained from scaling covariance, with the coefficients of Theorem 6.11.

Remark 6.20 (Method of computation)**.** The limits of Proposition 6.15 are evaluated exactly in explicit realisations of $\mathfrak{g}=\mathfrak{su}(2)$ and $\mathfrak{g}=\mathfrak{su}(3)$: each boundary integral of the Schwinger parametrisation with one edge at $t=\varepsilon$ is computed in closed form in exact rational arithmetic, edge by edge, and each edge of a graph of degree 0 carries the fraction of Lemma 5.10; the resulting rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$ are the same for both gauge algebras.

Chapter 7

Other superconformal components and the road ahead

The dilatation component of Chapter 6 is one of the components $[(d_{CE}+Q)O[\infty]](x_1)$, one for each generator x_1 of $\mathfrak{psl}(4|4)$. How a generator acts on the regulator decides what a one-loop computation along it can produce. Translations, rotations and the R-symmetry commute with Q^{GF} and with $H = [Q, Q^{GF}]$, so every regulated kernel, every counterterm and the maps $O \mapsto O(\varepsilon)$, $O \mapsto O[\infty]$ are covariant, which is why these components can be chosen with no one-loop correction for gauge-invariant O, a choice realised in Step 2; the dilatation, the special conformal transformations, the supercharges and the special supercharges move Q^{GF}, and along them the one-loop output is confined by weight, Lorentz and R-representation, ghost degree and colour to a short list of local structures. This chapter proves the covariance statements and the selection rules, states exactly what relates the special conformal component to the dilatation component, and lists the objects and graphs of Steps 2–5.

7.1 Components along a generator

Definition 7.1 (Observables and their components)**.** Let O be a local functional of degree $|O|$ and δ a square-zero parameter of degree $-|O|$ [CG2, Def 8.3.1.1]. Let $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ be the equivariant interaction of Chapter 2 [ESW, Rem 1.29] and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ its one-loop counterterms, the singular parts, graph by graph, of the flow of $I_{\mathfrak{psl}(4|4)}$ with values in $C^\bullet(\mathfrak{psl}(4|4))$. Then

$$O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon], \qquad O[\infty] = \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \delta\, O(\varepsilon)\big)$$

[CG2, Def 8.3.1.3, Prop 13.4.0.1]. The component of O along $x_1 \in \mathfrak{psl}(4|4)$ is the arity-one part of the cochain $(d_{CE}+Q)O[\infty]$ evaluated on x_1 [ESW, Def 1.26–1.27], written $[(d_{CE}+Q)O[\infty]](x_1)$; as in Chapter 2, generators carry a subscript and x alone is the position. The component is represented by the local functional $\int \omega\, f_{x_1}$, with f_{x_1} a polynomial in the jets of the fields at a point, if

$$[(d_{CE}+Q)O[\infty]](x_1) = \Big(\int \omega\, f_{x_1}\Big)[\infty].$$

The counterterms computed in Step 1 are the arity-zero part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$, i.e. $I^{CT}[\varepsilon]$ and $\partial_\delta I^{CT}[\varepsilon]$ of Chapters 4 and 5; the arity-one counterterms are part of Step 2 (last section of this chapter).

Lemma 7.2 (Components lie in the image of Q)**.** *For every $x_1 \in \mathfrak{psl}(4|4)$, $[(d_{CE}+Q)O[\infty]](x_1)$ equals Q applied to the arity-one part of $O[\infty]$ evaluated on x_1, up to the Koszul sign of the evaluation. In particular it lies in the image of Q.*

Proof. The Chevalley–Eilenberg differential of Chapter 2 [ESW, Def 1.26] is a sum over pairs $i<j$ of arguments of terms $f([x_i,x_j],\dots)$; it maps cochains of arity k to arity $k+1$ and vanishes on cochains of arity zero, which have no pair of arguments. The arity-one part of $(d_{CE}+Q)O[\infty]$ is therefore d_{CE} of the arity-zero part, which is 0, plus Q of the arity-one part, Q acting on the values of the cochain. □

Remark 7.3 (Where the content of a component lies)**.** On moment-free Schwartz sections every Q-closed functional without constant term is Q-exact (Chapter 3); on compactly supported sections Q has in addition nonzero classes, among them the zero-mode classes, finitely many in each polynomial degree. Lemma 7.2 places every component in the image of Q. The content of a component is therefore its local representative $\int\omega\, f_{x_1}$, and the selection rules below are statements about f_{x_1}.

7.2 Translations, rotations and R-symmetry

Definition 7.4 (Action of an even generator on all slots)**.** Let x_1 be an even generator of $\mathfrak{psl}(4|4)$ with vector field v (Chapter 2). The linear operator $e\mapsto v.e$ on $\mathscr{E}$ is given on the slots of the fields by the transformations in $S^{(1)}(x_1)$ (Chapter 2, arity one):

$$v.A=L_vA+\tfrac14(\operatorname{div}v)A|_{W_{\mathbb{R}}^{\perp}},\qquad v.\lambda=v(\lambda)+\tfrac14\rho(d(v,-))\lambda+\tfrac38(\operatorname{div}v)\lambda,\qquad v.c=v(c),$$

and on the slots of the antifields by the condition

$$\langle v.e_0,e_1\rangle+\langle e_0,v.e_1\rangle=0\qquad\text{for all } e_0,e_1\in\mathscr{E}_c.$$

This extends the notation $v.\lambda$, $v.c$ of [ESW, (32)]. With it,

$$S^{(1)}(x_1)=\int\operatorname{dvol}\Big((v.A,A^*)-(v.\lambda,\lambda^*)-(v.c,c^*)\Big),$$

a functional linear in the fields and linear in the antifields, and the derivation $\{S^{(1)}(x_1),-\}$ of functionals is induced, as in [C07, §6.2], by $e\mapsto v.e$ times one sign fixed by the conventions of the bracket (Chapter 1). Every statement below about commutators of $v.$ is insensitive to this sign.

Lemma 7.5 (Weights on the antifields)**.** *For a conformal Killing vector field v of V and e_1 a section in the slot of $A^*|_{W_{\mathbb{R}}^{\perp}}$,*

$$v.e_1=v(e_1)+\tfrac34(\operatorname{div}v)\,e_1.$$

For the dilatation Δ, $v=2E$, on every component,

$$\begin{aligned}&v.A=2E(A)+2A,&&v.\lambda=2E(\lambda)+3\lambda,&&v.c=2E(c),\\&v.A^*=2E(A^*)+6A^*,&&v.\lambda^*=2E(\lambda^*)+5\lambda^*,&&v.c^*=2E(c^*)+8c^*,\end{aligned}$$

with $E(-)$ acting componentwise. Thus $v.$ is $2E(-)$ plus twice the Weyl weight, the weights of $A,\lambda,c,A^*,\lambda^*,c^*$ being $1,\frac32,0,3,\frac52,4$.

Proof. For compactly supported f_1,f_2, integration by parts gives

$$\int\operatorname{dvol}v(f_1)\,f_2=-\int\operatorname{dvol}f_1\,\big(v(f_2)+(\operatorname{div}v)\,f_2\big).$$

For the pair (A,A^*) the condition of Definition 7.4 reads $\int\operatorname{dvol}\big((v.A,A^*)+(A,v.A^*)\big)=0$, with the fibre pairing of Chapter 1, and likewise for the pairs (λ,λ^*) and (c,c^*). On $A|_{W_{\mathbb{R}}^{\perp}}$, $v.f_1=v(f_1)+\frac14(\operatorname{div}v)f_1$, so $\int\operatorname{dvol}(v.f_1,f_2)=-\int\operatorname{dvol}\big(f_1,\,v(f_2)+(1-\frac14)(\operatorname{div}v)f_2\big)$, which gives the first formula. For $v=2E$, $\operatorname{div}(2E)=8$ and $v.A=2E(A)+2A$, $v.\lambda=2E(\lambda)+3\lambda$, $v.c=2E(c)$ (Chapter 2, the dilatation example); the same computation gives $2E(A^*)+(8-2)A^*$, $2E(\lambda^*)+(8-3)\lambda^*$ and $2E(c^*)+8c^*$. □

Proposition 7.6 (Covariance of the regulator)**.** *Let x_1 be a translation, an element of $\mathfrak{so}(V)$ or an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, with vector field v. Then:*

- $\operatorname{div} v = 0$*, and $v.$ is the infinitesimal action of the translation or rotation on every slot, through the spin representation $\frac{1}{4}\rho \circ q$ on Σ and Σ^* [ESW, (29), (32)];*
- $[v., Q] = 0$*, $[v., Q^{GF}] = 0$, $[v., H] = 0$, and $v.$ preserves $\langle\,,\rangle$;*
- *for every $t > 0$,*

$$(v. \otimes 1 + 1 \otimes v.)K_t = 0, \qquad (v. \otimes 1 + 1 \otimes v.)L_t = 0, \qquad (v. \otimes 1 + 1 \otimes v.)P(\varepsilon, \infty) = 0;$$

- *the derivation $\{S^{(1)}(x_1), -\}$ commutes with $\partial_{P(\varepsilon,\infty)}$ and with Δ_ε, and for every even functional J, at least cubic modulo $\hbar$,*

$$\{S^{(1)}(x_1), W(P(\varepsilon,\infty), J)\} = \partial_\delta W\big(P(\varepsilon,\infty),\, J + \delta\{S^{(1)}(x_1), J\}\big), \tag{7.1}$$

with δ a square-zero parameter of degree 0.

Proof. First item: a translation is constant, a rotation of V is the linear vector field of an antisymmetric endomorphism of V, and an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ is linear and antisymmetric on $W_{\mathbb{R}}^{\perp}$; each is divergence free, so the Weyl-weight terms of Definition 7.4 vanish, and the remaining terms are the Lie derivative of the components of A (for $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ the rotation of the index w of $\iota_w A$) and the spin-lifted Lie derivative $v(\lambda)+\frac{1}{4}\rho(d(v,-))\lambda$. These actions preserve dvol and the fibre pairings, so on the antifields the condition of Definition 7.4 returns the same natural actions.

Second item: for a translation, $v.e$ is the constant-coefficient derivative $v_1(e)$ on every slot, which commutes with every constant-coefficient operator. For a rotation of V or of $W_{\mathbb{R}}^{\perp}$, the exterior derivative d, its adjoint d^*, the Clifford multiplication ρ (equivariant for the spin representation, [ESW, §2.1]) and the metric of V_{10} are invariant. The operator Q is built from d, d^* and $\not{d} = \rho \circ d$ (Chapter 1), and Q^{GF} from d, d^*, $\not{d}|_{\Sigma^*}$ and the identification $A^* \mapsto -2A^*$ through the metric of V_{10} (Chapter 3); both commute with $v.$, hence so does $H = QQ^{GF} + Q^{GF}Q$. The pairing is preserved by construction.

Third item: as for Q in Chapter 3, preservation of the pairing by the even operator $v.$ is $(v. \otimes 1 + 1 \otimes v.)K_0 = 0$, and $K_t = (e^{-tH} \otimes 1)K_0$ with $[v., H] = 0$ gives the first identity. Then

$$(v. \otimes 1 + 1 \otimes v.)(Q^{GF} \otimes 1)K_t = ([v., Q^{GF}] \otimes 1)K_t + (Q^{GF} \otimes 1)(v. \otimes 1 + 1 \otimes v.)K_t = 0,$$

and integration over $t \in (\varepsilon, \infty)$ gives the third.

Fourth item: by the computation of $[\partial_\phi, Q] = \partial_{Q\phi}$ in [C07, §6.4], with the even operator $v.$ in place of Q, the commutator of ∂_ϕ with the derivation induced by $v.$ is the operator ∂ of the kernel $(v. \otimes 1 + 1 \otimes v.)\phi$; for $\phi = P(\varepsilon,\infty)$ and $\phi = K_\varepsilon$ this vanishes by the third item. Hence $\{S^{(1)}(x_1), -\}$ commutes with $e^{\hbar\partial_{P(\varepsilon,\infty)}}$, and, being a derivation of the product [C07, Lemma 4.3.2] of even degree ($S^{(1)}(x_1)$ is odd and the bracket is odd),

$$\{S^{(1)}(x_1), e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar}\} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\hbar^{-1}\{S^{(1)}(x_1), J\}\, e^{J/\hbar}\Big).$$

Dividing by $\hbar^{-1}e^{W(P(\varepsilon,\infty),J)/\hbar}$ gives (7.1), whose right side is $e^{-W(P(\varepsilon,\infty),J)/\hbar}e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(\{S^{(1)}(x_1), J\}e^{J/\hbar}\big)$. □

Corollary 7.7 (Invariant counterterms, equivariant operators)**.** *For x_1 as in Proposition 7.6:*

- $\{S^{(1)}(x_1), I\} = 0$ *and* $\{S^{(1)}(x_1), I^{CT}[\varepsilon]\} = 0$*;*
- *for every local functional O, the counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta\{S^{(1)}(x_1), O\}$ equals $\{S^{(1)}(x_1), -\}$ applied to the counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta O$;*

- *consequently the arity-zero parts satisfy*

$$\big(\{S^{(1)}(x_1),O\}\big)(\varepsilon)=\{S^{(1)}(x_1),O(\varepsilon)\},\qquad \big(\{S^{(1)}(x_1),O\}\big)[\infty]=\{S^{(1)}(x_1),O[\infty]\}.$$

Proof. The arity-one equation $\{S,S^{(1)}(x_1)\}=0$ holds for these generators and every $\mathfrak{g}$ (Chapter 2). Its part of polynomial degree two is $\{\frac{1}{2}\langle e,Qe\rangle,S^{(1)}(x_1)\}=0$, since $S^{(1)}(x_1)$ is quadratic and $\{I,S^{(1)}(x_1)\}$ has degree at least three; the remaining part is $\{I,S^{(1)}(x_1)\}=0$, which by graded symmetry is the first identity. The counterterms are the singular parts, order by order and graph by graph, of $W(P(\varepsilon,\infty),J)$, for J equal to I minus the counterterms of lower order and for $J=I-I^{CT}[\varepsilon]+\delta O$ (Chapters 4 and 5). By (7.1), $\{S^{(1)}(x_1),-\}$ maps $W(P(\varepsilon,\infty),J)$ to the linearised flow of $\{S^{(1)}(x_1),J\}$, and it acts on the fields and leaves the coefficient functions of ε untouched, so it commutes with taking the singular part. Induction on the order gives $\{S^{(1)}(x_1),I^{CT}[\varepsilon]\}=0$, and the same argument at first order in δ, with $\{S^{(1)}(x_1),\delta O\}=\delta\{S^{(1)}(x_1),O\}$ (the bracket with the odd functional $S^{(1)}(x_1)$ has even degree), gives the second item. The third follows from the definitions of $O(\varepsilon)$ and $O[\infty]$ and from (7.1) once more; the limit $\varepsilon\to 0$ commutes with the ε-independent derivation. □

Corollary 7.8 (No mixing between representations)**.** *Let $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ with $w_1,w_2\in W_{\mathbb{R}}^{\perp}$, and let w run over an orthonormal basis of $W_{\mathbb{R}}^{\perp}$. The one-loop counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I+\delta O$ is a linear combination of the four functionals*

$$\int\omega\,(\iota_{w_1}A,\iota_{w_2}A),\qquad (w_1,w_2)\sum_w\int\omega\,(\iota_wA,\iota_wA),\qquad (w_1,w_2)\int\omega\,(A|_V,A|_V),\qquad (w_1,w_2)\int\omega,$$

with coefficients independent of w_1,w_2, the first three multiples of $\log\varepsilon$ and the fourth a multiple of ε^{-1}. In particular the traceless part of O is renormalised into itself, and the trace part into the span of $\sum_w\int\omega\,(\iota_wA,\iota_wA)$, $\int\omega\,(A|_V,A|_V)$ and $\int\omega$. The component of the quadratic counterterm $I^{CT}[\varepsilon]$ pairing $A|_V$ with $A|_{W_{\mathbb{R}}^{\perp}}$ vanishes.

Proof. By Corollary 7.7 the map $O\mapsto\partial_\delta I^{CT}[\varepsilon]$ is equivariant for translations, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$; it is also equivariant for the adjoint action of $\mathfrak{g}$, which commutes with Q, Q^{GF}, H and preserves $\langle\,,\rangle$ and I, by the same argument. At one loop the coefficients lie in the span of ε^{-1} and $\log\varepsilon$, and under the scaling of Chapter 6, which multiplies ε in $P(\varepsilon,\infty)$ by the square of the scale factor, ε^{-1} multiplies a density of Weyl weight 0 and $\log\varepsilon$ a density of Weyl weight 2, the weight of $(\iota_{w_1}A,\iota_{w_2}A)$. The densities of weight 0 and 2, of ghost degree 0, invariant under $\mathfrak{g}$, $\mathfrak{so}(V)$ and translations, are the constant 1 (weight 0) and $(A|_V,A|_V)$, $(\iota_wA,\iota_{w'}A)$ with $w,w'\in W_{\mathbb{R}}^{\perp}$ (weight 2): a $\mathfrak{g}$-invariant needs two adjoint fields, a ghost c of degree $+1$ needs an antifield of weight at least $\frac{5}{2}$, and a derivative on a single field leaves it in the adjoint. As a representation of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, $\mathrm{Sym}^2W_{\mathbb{R}}^{\perp}$ is the sum of the irreducible traceless part and the trace line spanned by $\sum_w w\otimes w$; the span of the densities $(\iota_wA,\iota_{w'}A)$ is a copy of $\mathrm{Sym}^2W_{\mathbb{R}}^{\perp}$, and $(A|_V,A|_V)$ and 1 are invariant. By Schur's lemma an equivariant linear map out of $\mathrm{Sym}^2W_{\mathbb{R}}^{\perp}$ is a multiple of the identity on the traceless part and maps the trace line into the invariants, which gives the four functionals. For $I^{CT}[\varepsilon]$, a bilinear invariant of a vector of V and a vector of $W_{\mathbb{R}}^{\perp}$ is 0. □

The coefficients computed in Chapter 5 are

$$\partial_\delta I^{CT}[\varepsilon]=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-2(w_1,w_2)\sum_w\int\omega\,(\iota_wA,\iota_wA)\Big]+\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\varepsilon}\int\omega,$$

with coefficient 0 for $(w_1,w_2)\int\omega\,(A|_V,A|_V)$, and the quadratic $I^{CT}[\varepsilon]$ of Chapter 4 has vanishing $A|_V$–$A|_{W_{\mathbb{R}}^{\perp}}$ component, as Corollary 7.8 requires. Here $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$ is independent of $X,Y\in\mathfrak{g}$ for simple $\mathfrak{g}$.

Remark 7.9 (Internal insertions and the status of these components)**.** For x_1 as in Proposition 7.6, the arity-one part of $O[\infty]$ along x_1 consists of the classical tree with the vertex $S^{(1)}(x_1)$ joined to O, the graphs in which $S^{(1)}(x_1)$ sits on an internal line, and the arity-one counterterms along x_1. On an internal line, $(v.A, A^*)$ is the term of $S^{(1)}(x_1)$ with both legs joined by $P(\varepsilon,\infty)$ (every component of the propagator lies in the slots of A, A^*, c and λ), its A leg to an $A\otimes A$ component and its A^* leg to an $A^*\otimes c$ component. The resulting kernel is that of $P(\varepsilon,\infty)\star\circ v.\circ P(\varepsilon,\infty)\star = v.\circ\big(P(\varepsilon,\infty)\star\big)^2$, and since $[Q^{GF},H]=0$,

$$\big(P(\varepsilon,\infty)\star\big)^2 = (Q^{GF})^2\Big(\int_\varepsilon^\infty e^{-tH}dt\Big)^2, \qquad (Q^{GF})^2A^* = 2\,d^*(A^*|_V)\ \ \text{(in the slot of } c),$$

which is nonzero (Chapter 3). These graphs exist because $(Q^{GF})^2\neq 0$. By Proposition 7.6 every kernel, vertex and counterterm entering them is x_1-covariant, but the internal-insertion graphs still contribute: for a translation $x_1=P_b$, on the part of the scalar bilinears with two legs $A|_{W_{\mathbb{R}}^\perp}$, they give $-\big(\frac13+\theta_P\big)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\{S^{(1)}(P_b),O\}$ (θ_P as in Theorem 9.35), which the finite part of the operators of Proposition 9.36 removes; along $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ the composite kernel vanishes, since $v.$ acts trivially on the slot of c. When these graphs vanish or are removed in this way, for O with $(Q+\{I,-\})O=0$,

$$[(d_{CE}+Q)O[\infty]](x_1) = \big(\{S^{(1)}(x_1),O\}\big)[\infty],$$

the classical transformation of O carried to scale ∞. For general O the right side acquires, already at order $\hbar^0$, the trees with one vertex $(Q+\{I,-\})O$ and one vertex $S^{(1)}(x_1)$, as in Proposition 6.18; for the letter $A|_V$ and a translation they are nonzero. The arity-one counterterms along x_1 and the internal-insertion graphs are evaluated in Chapter 9 along the dilatation and the translations; along $\mathfrak{so}(V)$ they remain to be evaluated.

7.3 The generators that move the regulator

Proposition 7.10 (Dilatation and special conformal transformations)**.** *Let x_1 be the dilatation or a special conformal generator, with vector field v. Then $[v.,Q]=0$, and on the slot of $A^*|_{W_{\mathbb{R}}^\perp}$*

$$[v.,Q^{GF}]\,e_1 = (\operatorname{div} v)\,e_1 \quad (\textit{in the slot of } A|_{W_{\mathbb{R}}^\perp}), \tag{7.2}$$

with $\operatorname{div}(2E)=8$ *and* $\operatorname{div} v=-8(v_1,x)$ *for* $v=\big(x\mapsto(x,x)v_1-2(v_1,x)x\big)$*. For the dilatation, $v=2E$, on every slot*

$$[v.,Q^{GF}] = -4\,Q^{GF}, \qquad [v.,H]=-4\,H, \qquad (v.\otimes 1+1\otimes v.)P(\varepsilon,\infty) = 4\varepsilon L_\varepsilon = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty).$$

Among the even generators, (7.2) *vanishes exactly for those with* $\operatorname{div} v=0$*: translations, rotations and* $\mathfrak{so}(W_{\mathbb{R}}^\perp)$*.*

Proof. $[v.,Q]=0$ is the part of polynomial degree two of the arity-one equation $\{S,S^{(1)}(x_1)\}=0$ (Chapter 2). On the slot of $A^*|_{W_{\mathbb{R}}^\perp}$, $Q^{GF}e_1=-2e_1$ in the slot of $A|_{W_{\mathbb{R}}^\perp}$ (Chapter 3); by Definition 7.4 and Lemma 7.5, $v.$ acts there by $v(-)+\frac14\operatorname{div} v$ and $v(-)+\frac34\operatorname{div} v$ respectively, plus the same rotation of the index for both. Hence

$$[v.,Q^{GF}]e_1 = -2\big(v(e_1)+\tfrac14(\operatorname{div}v)e_1\big)+2\big(v(e_1)+\tfrac34(\operatorname{div}v)e_1\big) = (\operatorname{div}v)\,e_1.$$

The divergences are those of Chapter 2. For $v=2E$ use Lemma 7.5 on each arrow of Q^{GF}, together with the homogeneity relations $[E,d]=-d$, $[E,d^*]=-d^*$ and $[E,\not\partial]=-\not\partial$ on componentwise E (each

lowers the degree of homogeneity in x by one):

$$\begin{aligned}
A^* \mapsto -2A^*: &\quad (2E+2)(-2A^*) + 2(2E+6)A^* = 8A^* = -4(-2A^*),\\
A \mapsto -d^*(A|_V): &\quad -2E\,d^*A + d^*(2E+2)A = 2d^*A + 2d^*A = -4\big(-d^*A\big),\\
c^* \mapsto dc^*: &\quad (2E+6)dc^* - d(2E+8)c^* = 2[E,d]c^* + 6dc^* - 8dc^* = -4\,dc^*,\\
\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*: &\quad -(2E+3)\not{d}\lambda^* + \not{d}(2E+5)\lambda^* = 2\not{d}\lambda^* - 3\not{d}\lambda^* + 5\not{d}\lambda^* = -4\big(-\not{d}\lambda^*\big),
\end{aligned}$$

where in each line $A|_V$, $A^*|_V$ are handled componentwise (Lemma 7.5). Similarly $[2E, d^*d] = -4\,d^*d$ and the weight terms are constants, so $[v., H] = -4H$. Consequently $[v., e^{-tH}] = 4tHe^{-tH} = -4t\frac{d}{dt}e^{-tH}$; since $v.$ preserves the pairing, $(v.\otimes 1 + 1\otimes v.)K_t = ([v., e^{-tH}]\otimes 1)K_0 = -4t\frac{d}{dt}K_t$, and

$$(v.\otimes 1 + 1\otimes v.)L_t = -4L_t - 4t\frac{d}{dt}L_t.$$

Integrating over $t \in (\varepsilon,\infty)$, with $tL_t \to 0$ as $t\to\infty$ for $p \neq 0$,

$$(v.\otimes 1 + 1\otimes v.)P(\varepsilon,\infty) = -4\int_\varepsilon^\infty L_t\,dt - 4\big[tL_t\big]_\varepsilon^\infty + 4\int_\varepsilon^\infty L_t\,dt = 4\varepsilon L_\varepsilon,$$

which equals $-4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty)$ by Chapter 3. The last statement holds because the even generators are the conformal Killing fields and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, and among the conformal Killing fields of Chapter 2 the divergence vanishes exactly on translations and rotations. □

The identity $(v.\otimes 1+1\otimes v.)P(\varepsilon,\infty) = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty)$ is the infinitesimal form of the scaling covariance used in Chapter 6: the dilatation moves the regulator along its own scale.

Proposition 7.11 (Supercharges and special supercharges)**.** *Let $Q_1 \in \Sigma$ be nonzero and write $Q_1.$, in the notation $v.$ of [ESW, (32)], for the odd linear operator on $\mathscr{E}$ that induces, as in [C07, §6.2], the derivation given by the bracket with the quadratic part*

$$\int \mathrm{dvol}\Big(-(\Gamma(Q_1,\lambda),A^*) + \tfrac{1}{2}(\rho(dA)Q_1,\lambda^*)\Big)$$

of $S^{(1)}(Q_1)$. Then:

- *$[Q_1., Q] = 0$, $[Q_1., H] = 0$, and $Q_1.$ preserves $\langle\,,\rangle$;*
- *the graded commutator $[Q_1., Q^{GF}]$ is nonzero; its component from the slot of A^* to the slot of λ is nonzero;*
- *consequently*
$$(Q_1.\otimes 1 + 1\otimes Q_1.)P(\varepsilon,\infty) = \int_\varepsilon^\infty \big([Q_1., Q^{GF}]\otimes 1\big)K_t\,dt \neq 0;$$
- *$S^{(1)}(Q_1)$ contains the cubic term $\frac{1}{2}\big(\rho(\frac{1}{2}[A\wedge A])Q_1,\lambda^*\big)$.*

For $Q_2 \in \Sigma^*$, the corresponding operator contains multiplication by the function $x \mapsto \Gamma(\rho(x)Q_2,-)$, linear in x, and fails to commute with H.

Proof. First item: $[Q_1., Q] = 0$ is the part of polynomial degree two of $\{S, S^{(1)}(Q_1)\} = 0$ (Chapter 2); $Q_1.$ is a differential operator of order at most one with constant coefficients, and H is the scalar Laplacian d^*d on every component (Chapter 3), so they commute; the derivation of a quadratic functional preserves the constant pairing.

Second item. The term $-(\Gamma(Q_1,\lambda),A^*) = -(\rho(A^*)Q_1,\lambda)$ [ESW, (28)] pairs the slots of λ and A^*; it contributes to Q_1. an arrow from λ to A and an arrow from A^* to λ^*, the latter a nonzero multiple of $A^* \mapsto \rho(A^*)Q_1$. The term $\frac{1}{2}(\rho(dA)Q_1,\lambda^*)$ contributes an arrow from A to λ, a nonzero multiple of $A \mapsto \rho(dA)Q_1$, and an arrow from λ^* to A^*. The operator Q^{GF} maps $A^* \mapsto -2A^*$ into the slot of A and $\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$ into the slot of λ (Chapter 3). Hence the component of $[Q_1., Q^{GF}]$ from A^* to λ is the sum of the two compositions $A^* \to A \to \lambda$ and $A^* \to \lambda^* \to \lambda$. On a plane wave $A^* = a\, e^{i(p,x)}$ with $p, a \in V$ real and nonzero, and with $\rho(q(p \wedge a)) = \rho(p)\rho(a) - (p,a)$ [ESW, (29)], these are nonzero multiples, fixed by the conventions, of

$$\big(\rho(p)\rho(a) - (p,a)\big)Q_1\, e^{i(p,x)} \qquad \text{and} \qquad \rho(p)\rho(a)Q_1\, e^{i(p,x)}.$$

Suppose their sum, with the sign of the graded commutator, vanished for all such p, a. When p and a are orthogonal, $(p,a) = 0$, the product $\rho(p)\rho(a)$ is invertible ($\rho(a)\rho(p)\rho(p)\rho(a) = |p|^2|a|^2$), and the two multiples would be opposite; at $a = p$ the sum would then reduce to the first multiple times $-|p|^2Q_1$, which is nonzero. Hence the component is nonzero.

Third item: as in Proposition 7.6, with Q_1. odd, $(1 \otimes Q_1.)(Q^{GF} \otimes 1) = -(Q^{GF} \otimes 1)(1 \otimes Q_1.)$ and $(1 \otimes Q_1.)K_t = -(Q_1. \otimes 1)K_t$ by the first item, so $(Q_1. \otimes 1 + 1 \otimes Q_1.)L_t = \big((Q_1.Q^{GF} + Q^{GF}Q_1.) \otimes 1\big)K_t$. The integral is nonzero because $\int_\varepsilon^\infty e^{-tH}dt$ is $e^{-\varepsilon|p|^2}/|p|^2 \neq 0$ in momentum space and commutes with $[Q_1., Q^{GF}]$.

Fourth item: $F_A = dA + \frac{1}{2}[A \wedge A]$ in $\frac{1}{2}(\rho(F_A)Q_1,\lambda^*)$. For Q_2: the term $-(\Gamma(\rho(x)Q_2,\lambda),A^*)$ of $S^{(1)}(Q_2)$ (Chapter 2) gives the stated multiplication operator, and for a function f_1 linear in x, $d^*d(f_1f_2) - f_1\, d^*df_2 = -2(df_1, df_2)$, which is nonzero for nonconstant f_1. □

Remark 7.12 (Large-distance behaviour of internal insertions)**.** The coefficients of $S^{(1)}(x_1)$ grow polynomially in the position x for the moving generators: linearly for Δ and for $Q_2 \in \Sigma^*$, quadratically for the special conformal fields. On an internal line, an insertion of $(v.A, A^*)$ at a point y is joined to the rest of the graph by an $A \otimes A$ component of $P(\varepsilon,\infty)$, of order $|y|^{-2}$ at large $|y|$, differentiated once by $v.$, and by an $A^* \otimes c$ component, of order $|y|^{-3}$ (Chapter 3). For the special conformal field the integrand is of order $|y|^2 \cdot |y|^{-3} \cdot |y|^{-3} = |y|^{-4}$, the borderline of integrability over $\mathbb{R}^4$; for Δ, and for Q_2 with its λ and A^* legs, it is of order $|y|^{-5}$. The convergence at large $|y|$ of the special conformal internal insertions, with no infrared scale available, is part of Step 2.

7.4 Allowed output structures

Throughout this section $O = \int \omega\, f$ with ω a compactly supported four-form [CG2, Lemma 10.1.1.3] and f a polynomial in the jets of the fields at a point. The Weyl weight of a monomial is the sum of the weights of its fields (A: 1, λ: $\frac{3}{2}$, c: 0, A^*: 3, λ^*: $\frac{5}{2}$, c^*: 4, Lemma 7.5) plus the number of derivatives; its ghost degree is its degree as a functional, the sum of the degrees of its factors (c: $+1$; A, λ: 0; A^*, λ^*: -1; c^*: -2, as in the proof of Lemma 4.19); it is a colour singlet if it is invariant under the adjoint action of $\mathfrak{g}$ on all fields. O has Weyl weight w_O if f is homogeneous of that weight.

Proposition 7.13 (Selection rules)**.** *Let $O = \int \omega\, f$ be a colour singlet of Weyl weight w_O and ghost degree $|O|$, and let x_1 be an odd generator, realised by the conformal Killing spinor $x \mapsto Q_1 + \rho(x)Q_2$ with $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ (Chapter 2), or a special conformal generator, with vector field $v = \big(x \mapsto (x,x)v_1 - 2(v_1,x)x\big)$. Assume that the counterterms of Step 2 are chosen equivariantly for translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ and $\mathfrak{g}$, which Proposition 7.6 permits, and that the component along x_1 is represented by a local functional $\int \omega\, f_{x_1}$. Then f_{x_1} can be chosen with the following properties.*

- *Ghost degree $|O|$, colour singlet.*

- *At a point $x_0 \in V$, f_{x_1} depends on x_1 through its conformal Killing data at x_0: for an odd generator, f_{x_1} is the sum of a term linear in the value $Q_1 + \rho(x_0)Q_2$, of Weyl weight $w_O + \frac{1}{2}$, and a term linear in the Dirac derivative $\not{d}(Q_1 + \rho(x)Q_2) = 4Q_2$, of weight $w_O - \frac{1}{2}$; for a special conformal generator, f_{x_1} is the sum of terms linear in $v(x_0)$, in the first derivative of v at x_0 and in its second derivatives, of weights $w_O + 1$, w_O and $w_O - 1$.*

- *The map $(x_1, O) \mapsto f_{x_1}$ is equivariant for translations, $\mathfrak{so}(V)$ and $\mathfrak{so}(W^\perp_{\mathbb{R}})$, with x_1 transforming by the bracket of $\mathfrak{psl}(4|4)$.*

Proof. Ghost degree: $(d_{CE} + Q)O[\infty]$ has degree $|O| + 1$, and evaluating an arity-one cochain on a generator, whose ghost coordinate has degree 1, lowers the ghost degree by one. Colour and equivariance: the adjoint action of $\mathfrak{g}$ and the action of the covariant subalgebra commute with Q, Q^{GF}, H and preserve the pairing (Proposition 7.6); the classical action $S + S^{(1)} + S^{(2)}$ is invariant under the simultaneous action on fields and on the arguments of the cochains, by the arity-two components of its master equation with one argument in the covariant subalgebra (Chapter 2); with equivariant counterterms every graph sum is equivariant, as in Corollary 7.7. Dependence on x_1: the vertices $S^{(1)}(x_1)$ and $S^{(2)}(x_1, -)$ depend on x_1 through the spinor $Q_1 + \rho(x)Q_2$ and its Dirac derivative $4Q_2$ (the last term of $S^{(1)}(Q_2)$ equals $-\frac{1}{4}(\rho(A|_{W^\perp_{\mathbb{R}}})\not{d}(\rho(x)Q_2), \lambda^*)$, Chapter 2), resp. through v, dv and $\operatorname{div} v$; the spinor has vanishing second derivatives and derivative $\rho(v_2)Q_2 = \frac{1}{4}\rho(v_2)\not{d}(Q_1 + \rho(x)Q_2)$ along $v_2 \in V$, and a conformal Killing field of V is a polynomial of degree two, so the data at a vertex point y are polynomials in $y - x_0$ with coefficients the data at x_0 listed in the statement. Weights: give the conformal Killing data the weights that make each vertex $S^{(1)}(x_1)$, $S^{(2)}(x_1, -)$ homogeneous of the weight 4 of the densities of the action: $-\frac{1}{2}$ to the value of the conformal Killing spinor and $+\frac{1}{2}$ to its Dirac derivative (in $(\Gamma(Q_1, \lambda), A^*)$ the fields have weights $\frac{3}{2} + 3$), and -1, 0, $+1$ to the value, the first and the second derivatives of a conformal Killing field (in $(L_v A, A^*)$ the derivative and the fields have weights $1 + 1 + 3$). Under the scaling of Chapter 6, which multiplies each field by the power of the scale factor given by its Weyl weight and ε in $P(\varepsilon, \infty)$ by the square of the scale factor, every vertex is then homogeneous, and the output has the weight of O; each datum therefore multiplies a density whose weight is w_O minus the weight of the datum, which gives the weights listed. A local term whose weight differs from the stated one by $2k$ carries a coefficient ε^k, possibly times powers of $\log \varepsilon$; for $k > 0$ it vanishes as $\varepsilon \to 0$, and singular coefficients are excluded by the existence of the limit defining $\big(\int \omega\, f_{x_1}\big)[\infty]$. □

Corollary 7.14 (Supercharges on the scalar bilinears)**.** *Let $O = \int \omega\, (\iota_{w_1} A, \iota_{w_2} A)$ ($|O| = 0$, $w_O = 2$) and $Q_1 \in S_+ \otimes W$ (the generators Q^i_α of [SW, (8)]) or $Q_1 \in S_- \otimes W^*$ (the generators $\bar{Q}_{\dot{\alpha} i}$). Under the hypotheses of Proposition 7.13, f_{Q_1} is a colour-singlet density of Weyl weight $\frac{5}{2}$ and ghost degree 0, odd in the fields, linear in Q_1, symmetric in (w_1, w_2), without derivatives, and $\int \omega\, f_{Q_1}$ lies in the span of the following four functionals, grouped in three types:*

- *type $(A|_{W^\perp_{\mathbb{R}}}, \lambda)$ (classical type):*

$$\int \omega \Big[(\iota_{w_1} A, (\rho(w_2)Q_1, \lambda)) + (\iota_{w_2} A, (\rho(w_1)Q_1, \lambda))\Big], \qquad (w_1, w_2) \int \omega\, (\rho(A|_{W^\perp_{\mathbb{R}}})Q_1, \lambda);$$

- *type $(A|_V, \lambda)$ (gauge-variant type): $(w_1, w_2) \int \omega\, (\rho(A|_V)Q_1, \lambda)$;*

- *type (c, λ^*) (BRST type): $(w_1, w_2) \int \omega\, (c, (Q_1, \lambda^*))$.*

Here $(\rho(w)Q_1, \lambda)$, (Q_1, λ^*) use the pairing $\Sigma^* \otimes \Sigma \to \mathbb{C}$ and are $\mathfrak{g}$-valued, paired with the remaining $\mathfrak{g}$-valued field through $(-, -)$, and $\rho(A|_V)$, $\rho(A|_{W^\perp_{\mathbb{R}}})$ are Clifford multiplication by the components of A with their $\mathfrak{g}$ factor kept. In particular, for $(w_1, w_2) = 0$, where O lies in the traceless part, $\int \omega\, f_{Q_1}$ is a multiple of the first functional, which is the type of the classical component $\{S^{(1)}(Q_1), O\}$; the gauge-variant and BRST types occur in the trace part $\sum_w \int \omega\, (\iota_w A, \iota_w A)$.

Proof. By Proposition 7.13, $\not{d} Q_1 = 0$ and f_{Q_1} has weight $\frac{5}{2}$ and ghost degree 0. Field content: a half-integral weight requires an odd number of fields among λ, λ^*, the fields of half-integral weight. With one λ^* (weight $\frac{5}{2}$, degree -1) the rest has weight 0 and degree $+1$: one c. With one λ the rest has weight 1 and degree 0: one A, or one derivative; a single λ with a derivative is in the adjoint of $\mathfrak{g}$, and a colour singlet needs two adjoint fields joined by $(-,-)$, the invariant symmetric form of the simple $\mathfrak{g}$. Three spinor fields have weight at least $\frac{9}{2}$. Hence the types (c, λ^*), $(A|_V, \lambda)$, $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$, without derivatives.

Representations, for $Q_1 \in S_+ \otimes W$ (the case $S_- \otimes W^*$ is the same with $S_\pm$ and W, W^* exchanged). Under $\mathfrak{so}(W_{\mathbb{R}}^\perp) \cong \mathfrak{sl}(W)$ [ESW, §10.3], $\mathrm{End}(W) = W \otimes W^* \cong \mathbb{C} \oplus \wedge^2 W_{\mathbb{R}}^\perp$ (dimensions $1 + 15$), and $\mathrm{Sym}^2 W_{\mathbb{R}}^\perp$ is the sum of its irreducible traceless part, of dimension 20, and the trace line.

- Type (c, λ^*): c is a Lorentz and R-singlet, so the structure is $(c, (B\, Q_1, \lambda^*))$ with B an endomorphism of Σ depending bilinearly and equivariantly on (w_1, w_2). Lorentz invariance pairs Q_1 with the $S_+ \otimes W^*$ component of λ^*, so B restricts to an element of $\mathrm{End}(W)$; by Schur's lemma the traceless part of $\mathrm{Sym}^2 W_{\mathbb{R}}^\perp$, of dimension 20, has no image in $\mathbb{C} \oplus \wedge^2 W_{\mathbb{R}}^\perp$, and the trace line maps to the identity: $B = (w_1, w_2)$ times a constant.

- Type $(A|_V, \lambda)$: $A|_V$ is an R-singlet and a vector of $V \cong S_+ \otimes S_-$; Lorentz invariance pairs $\rho(A|_V) Q_1 \in S_- \otimes W$ with the $S_- \otimes W^*$ component of λ, and the R-dependence is again an equivariant map from $\mathrm{Sym}^2 W_{\mathbb{R}}^\perp$ to $\mathrm{End}(W)$: a multiple of (w_1, w_2).

- Type $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$: Lorentz invariance pairs Q_1 with the $S_+ \otimes W$ component of λ through an odd element of the Clifford algebra of $W_{\mathbb{R}}^\perp$ built from the index w of $\iota_w A$ and from w_1, w_2. Equivariant multilinear expressions in w, w_1, w_2 with values in this Clifford algebra are combinations of products of $\rho(w), \rho(w_1), \rho(w_2)$, of the metric and of the volume element of $W_{\mathbb{R}}^\perp$, and the volume element acts on each of $S_+ \otimes W$, $S_- \otimes W^*$ by a scalar (Chapter 1: Σ is a semi-spin representation and the four-dimensional chirality is fixed on each summand); the odd ones symmetric in (w_1, w_2) reduce, by $\rho(w_1)\rho(w_2) + \rho(w_2)\rho(w_1) = 2(w_1, w_2)$ and $\rho(w)\rho(w_i) + \rho(w_i)\rho(w) = 2(w, w_i)$, to the span of $(w, w_1)\rho(w_2) + (w, w_2)\rho(w_1)$ and $(w_1, w_2)\rho(w)$; contracting w with $\iota_w A$ gives the two functionals listed, using $\sum_w \iota_w A\, \rho(w) = \rho(A|_{W_{\mathbb{R}}^\perp})$. The term $\rho(q(w \wedge w_1 \wedge w_2))$ is antisymmetric in (w_1, w_2).

For $(w_1, w_2) = 0$ every functional with the factor (w_1, w_2) vanishes. The classical component replaces A by its transformation $\Gamma(Q_1, \lambda)$ in O (Chapter 2), and $\iota_w \Gamma(Q_1, \lambda) = (w, \Gamma(Q_1, \lambda)) = (\rho(w) Q_1, \lambda)$ by [ESW, (28)], which gives the first functional. □

Corollary 7.15 (Special supercharges on the scalar bilinears)**.** *Let O be as in Corollary 7.14 and $Q_2 \in S_+ \otimes W^*$ (the generators S_i^α) or $Q_2 \in S_- \otimes W$ (the generators $\bar{S}^{\dot\alpha i}$). Under the hypotheses of Proposition 7.13, $\int \omega\, f_{Q_2}$ lies in the span of the four functionals of Corollary 7.14 with Q_1 replaced by the value $\rho(x) Q_2$ of the conformal Killing spinor, which lies in $S_- \otimes W^*$, resp. $S_+ \otimes W$. The term linear in $\not{d}(\rho(x) Q_2) = 4 Q_2$ is absent: it would require a colour-singlet density of Weyl weight $\frac{3}{2}$ and ghost degree 0, and the one field of weight $\frac{3}{2}$, λ, is in the adjoint of $\mathfrak{g}$ while every field of weight 0 has ghost degree $+1$.*

The coefficients of the four functionals along Q_2 are independent data; they are computed in Step 3 together with those along Q_1.

Corollary 7.16 (Special conformal components)**.** *Let $v(x) = (x, x) v_1 - 2(v_1, x) x$. Its first derivative at x_0 is $\frac{1}{4} \operatorname{div} v(x_0) \cdot \mathrm{id}$ plus an antisymmetric endomorphism of V, with $\frac{1}{4} \operatorname{div} v(x_0) = -2(v_1, x_0)$, and its second derivatives are constant and linear in v_1. Under the hypotheses of Proposition 7.13, for O a scalar bilinear or a single letter $\iota_w A$, $A|_V$, λ, c (with its colour index fixed by an element of $\mathfrak{g}$):*

- *the term linear in the second derivatives of v is absent: it would require a vector density of weight $w_O - 1$, ghost degree $|O|$ and the colour of O; for a scalar bilinear this is a colour-singlet vector of*

weight 1*, and the field of weight* 1 *with a vector index,* $A|_V$*, is in the adjoint; for the letters the weight* $w_O - 1 \in \{0, 0, \frac{1}{2}, -1\}$ *with ghost degree* $|O| \in \{0, 0, 0, 1\}$ *admits no monomial;*

- *the term linear in the first derivative is* $\frac{1}{4} \operatorname{div} v(x_0)$ *times a density of weight* w_O *with the R-representation of* O*, plus a density of weight* w_O *contracted with the antisymmetric part of* $dv(x_0)$*. For a scalar bilinear the latter lies in* $\wedge^2 V$ *and vanishes, since the one candidate* $(A|_V \wedge A|_V)$ *vanishes by the symmetry of* $(-,-)$*; for the letters* $\iota_w A$ *and* c*, which carry no index of* V *or of* $S_\pm$*, it vanishes. For the letter* $\int \omega\,(X, \iota_{v_2}(A|_V))$*,* $v_2 \in V$*, it is the rotation of the vector index,* $\int \omega\,\big(X, \iota_{v_3}(A|_V)\big)$ *with* v_3 *the antisymmetric part of* $dv(x_0)$ *applied to* v_2*; for the letter* $\int \omega\,(\zeta, \lambda)$ *it is the rotation of the spinor index,* $\int \omega\,\big(\zeta, \frac{1}{4}\rho(d(v,-)(x_0))\lambda\big)$ *[ESW, (29)]. Both are present classically, through* $L_v A$ *and* $\frac{1}{4}\rho(d(v,-))\lambda$ *in* $S^{(1)}(v)$ *(Definition 2.12), and are allowed by every selection rule;*

- *for a scalar bilinear with* $(w_1, w_2) = 0$*, the density of weight* 2 *in the first-derivative term is a multiple of* $(\iota_{w_1} A, \iota_{w_2} A)$*; for the trace it lies in the span of* $\sum_w (\iota_w A, \iota_w A)$ *and* $(A|_V, A|_V)$*;*

- *the term linear in* $v(x_0)$ *contracts* $v(x_0)$*, of degree two in* x_0*, with a vector density of weight* $w_O + 1$*.*

Proof. The derivative of v is $dv(x_0)v_2 = 2(x_0, v_2)v_1 - 2(v_1, v_2)x_0 - 2(v_1, x_0)v_2$; its trace is $\operatorname{div} v(x_0) = -8(v_1, x_0)$ and its symmetric traceless part vanishes, which is the conformal Killing equation. The remaining statements are the weight count of Proposition 7.13 with the field weights and degrees listed above, and the enumeration in the proof of Corollary 7.8. □

Remark 7.17 (The special conformal component and the dilatation)**.** The relation between the two components has three parts.

- Classical. At x_0 the one-jet of v is that of the translation by $v(x_0)$, plus a rotation, plus $\frac{1}{8} \operatorname{div} v(x_0)$ times the dilatation $2(x - x_0)$ centred at x_0; correspondingly the Weyl-weight terms of $S^{(1)}(v)$ at x_0, $\frac{1}{4} \operatorname{div} v(x_0)$ times the weight, are $\frac{1}{8} \operatorname{div} v(x_0)$ times those of $S^{(1)}(\Delta)$, twice the weight (Lemma 7.5).

- What the complex gives. The cochain $(d_{CE} + Q)O[\infty]$ has degree $|O| + 1$, and $(d_{CE} + Q)\big((d_{CE} + Q)O[\infty]\big) = 0$. For even x_1, x_2 the differential of Chapter 2 gives $\big(d_{CE}(d_{CE} + Q)O[\infty]\big)(x_1, x_2) = (-1)^{|O|+1}[(d_{CE} + Q)O[\infty]]([x_1, x_2])$, and for the special conformal generator K_{v_1} of [SW, (3)] along $v_1 \in V$ and the translation P_{v_2} of [SW, (3)] along $v_2 \in V$, with the bracket of vector fields of Chapter 2,

$$[K_{v_1}, P_{v_2}] = (v_1, v_2)\,\Delta + \big(x \mapsto 2(v_1, x)v_2 - 2(v_2, x)v_1\big),$$

  the second summand a rotation. The arity-two component of $(d_{CE} + Q)^2 O[\infty] = 0$ on (K_{v_1}, P_{v_2}) therefore reads

$$\begin{aligned}(-1)^{|O|+1}\Big((v_1, v_2)\,[(d_{CE} + Q)O[\infty]](\Delta) + [(d_{CE} + Q)O[\infty]]\big(x \mapsto 2(v_1, x)v_2 - 2(v_2, x)v_1\big)\Big)\\ = -\big(Q(d_{CE} + Q)O[\infty]\big)(K_{v_1}, P_{v_2}),\end{aligned}$$

  which expresses the dilatation component, together with a rotation component, through Q of the arity-two part of $(d_{CE} + Q)O[\infty]$. The special conformal component $[(d_{CE} + Q)O[\infty]](K_{v_1})$ enters through no term.

- The conformal expectation. For O localised at x_0, the conformal algebra suggests that the anomalous part of the special conformal component is $\frac{1}{4} \operatorname{div} v(x_0) = -2(v_1, x_0)$ times the anomalous coefficient of the dilatation component (Chapter 6) times O, i.e. the first-derivative term of Corollary 7.16 with the weight w_O shifted by that coefficient. Its derivation requires the composition of

components as a module structure on observables up to homotopy, the translation of O to x_0 by the components of Remark 7.9, and control of the second derivatives of v and of the large-distance behaviour of Remark 7.12 in the K_ε graphs. These are tasks of Steps 2 and 3.

generator	subspace	ad_Δ	weight of output	regulator	one-loop output for $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$
translations	V	-2	w_O+1	covariant	can be chosen classical (Step 2)
rotations	$\mathfrak{so}(V)$	0	w_O	covariant	can be chosen classical (Step 2)
R-symmetry	$\mathfrak{so}(W_{\mathbb{R}}^{\perp})$	0	w_O	covariant	can be chosen classical (Step 2)
dilatation Δ	$2E$	0	w_O	$-4\varepsilon\frac{d}{d\varepsilon}$	Chapter 6
supercharges	$S_+\otimes W\subset\Sigma$	-1	$w_O+\frac{1}{2}$	moved	Corollary 7.14
supercharges	$S_-\otimes W^*\subset\Sigma$	-1	$w_O+\frac{1}{2}$	moved	Corollary 7.14
special supercharges	$S_+\otimes W^*\subset\Sigma^*$	$+1$	$w_O+\frac{1}{2}$ at $\rho(x)Q_2$	moved	Corollary 7.15
special supercharges	$S_-\otimes W\subset\Sigma^*$	$+1$	$w_O+\frac{1}{2}$ at $\rho(x)Q_2$	moved	Corollary 7.15
special conformal	conformal Killing fields of degree 2	$+2$	$w_O+1,\ w_O,\ w_O-1$	moved	Corollary 7.16

Components along the ten named subspaces of $\mathfrak{psl}(4|4)$; in the names of [SW, (8)] the rows are P, M and $\bar M$, R, Δ, Q, $\bar Q$, S, $\bar S$, K, and the ad_Δ eigenvalues are those of [SW, (17)]. The column "regulator" records the action on $P(\varepsilon,\infty)$: covariant (Proposition 7.6), the infinitesimal change of scale (Proposition 7.10), or moved (Propositions 7.10 and 7.11). For the moving generators the output is a local functional of the listed weight and of ghost degree $|O|$, colour singlet, equivariant in (x_1, O).

7.5 The road ahead

Step 1 has produced the arity-zero part of the one-loop theory: $I^{CT}[\varepsilon]$ through quartic order, the quantum master equation in the sector $(c, A|_V)$ with the finite counterterm

$$\frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$$

of Chapter 4, the local $O(\varepsilon)$ of the letters and scalar bilinears, and their dilatation through the scaling covariance. The next steps are organised by one identity. For an even functional J with values in $C^\bullet(\mathfrak{psl}(4|4))$, the UV-boundary identity of Chapter 3, applied to $e^{J/\hbar}$, shows that $(d_{CE}+Q)W(P(\varepsilon,\infty),J)$ is the sum of the connected graphs with edges $P(\varepsilon,\infty)$, one vertex labelled by

$$(d_{CE}+Q)J+\tfrac{1}{2}\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J, \tag{7.3}$$

and every other vertex labelled by J. For $J=I_{\mathfrak{psl}(4|4)}-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$, the classical equation $(d_{CE}+Q)I_{\mathfrak{psl}(4|4)}+\frac{1}{2}\{I_{\mathfrak{psl}(4|4)},I_{\mathfrak{psl}(4|4)}\}=0$ [ESW, Rem 1.29] turns (7.3) at one loop into

$$\begin{aligned}&\tfrac{1}{2}\Big(\{I_{\mathfrak{psl}(4|4)},I_{\mathfrak{psl}(4|4)}\}_\varepsilon-\{I_{\mathfrak{psl}(4|4)},I_{\mathfrak{psl}(4|4)}\}\Big)+\hbar\Delta_\varepsilon I_{\mathfrak{psl}(4|4)}\\&\quad-(d_{CE}+Q)I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]-\{I_{\mathfrak{psl}(4|4)},I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]\}_\varepsilon,\end{aligned} \tag{7.4}$$

whose first term joins two vertices by a special edge carrying $K_\varepsilon-K_0$ and whose second is a K_ε self-contraction. The $\varepsilon\to0$ limit of the graphs with one vertex (7.4) is the flow to scale ∞ of the one-loop obstruction to the quantum master equation, an effective defect vertex. On a compact manifold with a finite upper scale the obstruction is a local functional, closed for $d_{CE}+Q+\{I_{\mathfrak{psl}(4|4)},-\}$, and a finite local counterterm removes it exactly when it is exact [C07, Prop 12.0.3, Lemma 12.0.4; CG2, §13.3.1]; on $\mathbb{R}^4$ with upper scale ∞ these properties are to be proved in Step 2. In the sector $(c, A|_V)$ of Chapter 4 the limit is itself local.

7.5.1 Step 2: equivariant counterterms and the full one-loop master equation

Objects.

- The cubic sector of the arity-zero obstruction, with external legs $(c, A|_V, A|_V)$, $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, (c, λ, λ) and $(c, c, A^*|_V)$, for the interaction $I - I^{CT}[\varepsilon]$ minus the finite counterterm above; then a local functional cubic in the fields whose image under Q equals it, or the class it defines; then the sectors with more legs.

- For each generator x_1 (translations, $\mathfrak{so}(V)$, Δ, special conformal, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$, Σ, Σ^*): the arity-one counterterm along x_1, the singular part of the one-loop graphs with one vertex $S^{(1)}(x_1)$; and the arity-one obstruction along x_1, the limit of the graphs with one vertex given by the arity-one part of (7.4),

$$\begin{aligned}&\{I, S^{(1)}(x_1)\}_\varepsilon - \{I, S^{(1)}(x_1)\} + \hbar\Delta_\varepsilon S^{(1)}(x_1)\\ &\quad - Q\, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](x_1) - \{I, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](x_1)\}_\varepsilon - \{S^{(1)}(x_1), I^{CT}[\varepsilon]\}_\varepsilon.\end{aligned}$$

  For translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ all these graphs are covariant (Proposition 7.6), and their evaluation realises the choice of Remark 7.9; for the special conformal fields the large-distance convergence of Remark 7.12 is part of the computation.

- The dilatation component $[(d_{CE} + Q)O[\infty]](\Delta)$ computed directly at the UV boundary, for the scalar bilinears, and its identification with the result of the scaling covariance of Chapter 6, including the trees built from $(Q + \{I, -\})O$.

- The cocycle condition $Q\,O[\infty] = 0$ at order $\hbar$ for the letters and the scalar bilinears (Remark 5.6): the graphs with a special edge $K_\varepsilon - K_0$ between I and O, the vertices $-Q\,\partial_\delta I^{CT}[\varepsilon] - \{I, \partial_\delta I^{CT}[\varepsilon]\}_\varepsilon$ and $-\{I^{CT}[\varepsilon], O\}_\varepsilon$, the trees of O with the action defect; and the finite counterterms of weight w_O that it requires.

- Finite counterterms in $C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$ removing the whole one-loop obstruction, or the precise class that obstructs them.

Graphs. In the sector $(c, A|_V)$ the five one-loop graphs with one special edge $K_\varepsilon - K_0$ drawn in Chapter 4 (a bubble with an $A|_V$, an $A|_{W_{\mathbb{R}}^\perp}$, a λ and two c–A^* lines), together with the vertex $-QI^{CT}[\varepsilon]$, give the obstruction of Chapter 4. The cubic sector uses the same vertices with one more external leg: one-loop graphs with three external legs, one special edge $K_\varepsilon - K_0$ from the first term of (7.4) and the other edges $P(\varepsilon, \infty)$ (triangles with three cubic vertices, bubbles with one quartic vertex); the trees of $-\{I, I^{CT}[\varepsilon]\}_\varepsilon$ and of minus the bracket of I with the finite counterterm, with one K_ε edge joining a cubic vertex of I to the quadratic counterterm; and the vertex $-QI^{CT}[\varepsilon]$ restricted to the sector. The direct dilatation component of a scalar bilinear uses the twenty-five graphs (a)–(y) drawn in Chapter 6, with one O vertex, one $S^{(1)}(\Delta)$ vertex and one K_ε edge: twenty-one one-loop graphs (a)–(u), with the K_ε edge on the cycle and the other edges $P(\varepsilon, \infty)$ through cubic and quartic vertices of I (two of them, (t) and (u), with a c–A^* line ending at $S^{(1)}(\Delta)$), and four trees (v)–(y) with the vertex $\partial_\delta I^{CT}[\varepsilon]$; to them are added the graphs with the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ and the insertion of the defect (7.4) along Δ.

7.5.2 Step 3: supersymmetry and special supersymmetry of letters and scalar bilinears

Objects.

- For O a letter or a scalar bilinear and $x_1 = Q_1 \in \Sigma$ or $x_1 = Q_2 \in \Sigma^*$: the local functional representing $[(d_{CE}+Q)O[\infty]](x_1)$. By Corollaries 7.14 and 7.15 this is one number for the traceless bilinear (the multiple of the classical structure) and three numbers for the trace bilinear (on the trace the two classical-type functionals are multiples of $\int \omega\,(\rho(A|_{W_{\mathbb{R}}^\perp})Q_1, \lambda)$, and the gauge-variant and BRST-type functionals are added), for each of Q^i_α, $\bar{Q}_{\dot\alpha i}$, S^α_i, $\bar{S}^{\dot\alpha i}$.

- The arity-two components on pairs of odd generators, which contain $S^{(2)}$ [ESW, (34)] and express the closure of the quantum supersymmetry on O through $(d_{CE}+Q)^2O[\infty]=0$, as in Remark 7.17.

- The Konishi multiplet: the components of the trace bilinear along Σ and Σ^* and the coefficients of its gauge-variant and BRST-type structures.

Graphs. One-loop graphs with one $O(\varepsilon)$ vertex, one vertex of $S^{(1)}(x_1)$ and one K_ε edge, the other edges $P(\varepsilon,\infty)$. The vertices of $S^{(1)}(Q_1)$ are the bilinear vertices $-(\Gamma(Q_1,\lambda),A^*)$ and $\frac{1}{2}(\rho(dA)Q_1,\lambda^*)$ and the cubic vertex $\frac{1}{2}(\rho(\frac{1}{2}[A\wedge A])Q_1,\lambda^*)$; for Q_2 the spinor is $\rho(x)Q_2$ and the vertex $-(\rho(A|_{W_{\mathbb{R}}^\perp})Q_2,\lambda^*)$ is added. Trees with the vertex $\partial_\delta I^{CT}[\varepsilon]$ and with the arity-one counterterm along x_1 of Step 2 complete the list. For a scalar bilinear the external legs are those of Corollary 7.14: (A,λ) and (c,λ^*).

7.5.3 Step 4: all bilinears and covariant jets

Objects. For every gauge-invariant local functional bilinear in the letters $\iota_w A$, F_A, λ and their covariant derivatives d_A applied repeatedly: $\partial_\delta I^{CT}[\varepsilon]$ and $O(\varepsilon)$, a mixing matrix among operators of equal Weyl weight, ghost degree and representation of $\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^\perp)$ (Corollary 7.7); the dilatation component by the scaling covariance of Chapter 6; the other components with the selection rules of this chapter.

Graphs. One-loop one-particle-irreducible graphs with one O vertex whose superficial degree

$$w_O-\sum_{\text{external legs}}(\text{Weyl weight of the leg})$$

is non-negative, the external legs ranging over A, λ, c and the antifields A^*, λ^*, c^* with the weights of Lemma 7.5.

7.5.4 Step 5: two loops

Objects. The counterterms at order $\hbar^2$, the singular parts of the order-$\hbar^2$ terms of $W\big(P(\varepsilon,\infty),J\big)$, with J equal to $I-I^{CT}[\varepsilon]$ minus the finite counterterms of Step 2, in the order of [C07, Thm B]; $\partial_\delta I^{CT}[\varepsilon]$ and $O(\varepsilon)$ at order $\hbar^2$; the two-loop obstruction, the order-$\hbar^2$ part of the graphs with one vertex (7.3), the UV-boundary identity behind (7.3) holding to all orders in $\hbar$; the dilatation at order $\hbar^2$, read from the $\log\varepsilon$ coefficients through the scaling covariance of Chapter 6. The singular ε-functions now include $(\log\varepsilon)^2=\int_\varepsilon^1\int_\varepsilon^1(t_1t_2)^{-1}dt_1dt_2$, an element of the space of [C07, Def 7.1.1].

Graphs. Two-loop graphs with vertices of I and edges $P(\varepsilon,\infty)$; one-loop graphs with one vertex of $I^{CT}[\varepsilon]$ or of a finite counterterm, both of order $\hbar$, which subtract the subdivergences; trees with one vertex of order $\hbar^2$; and, for the obstruction, the same graphs with one vertex (7.3).

Chapter 8

The one-loop realisation of $\mathfrak{psl}(4|4)$: definitions

Chapters 4–6 quantised the action and the local operators with the global ghosts of $\mathfrak{psl}(4|4)$ entering through the classical cochain $S^{(1)} + S^{(2)}$. Realising the superconformal symmetry at one loop means two things: an equivariant effective interaction that solves the quantum master equation with values in $C^\bullet(\mathfrak{psl}(4|4))$, and, on local operators, a quantum differential whose components along the generators are the one-loop corrections of the supercharges, the special supercharges, the dilatation and the special conformal transformations. The special conformal fields grow quadratically at large distance and their insertions produce logarithms of the endpoint of the transfer of [C07], so the objects are defined at a finite endpoint T, their content is proved to be independent of T, and the finite choices on which they depend are separated from the invariant content. This chapter states the definitions at every order in $\hbar$, proves that they are well posed, and defines the conjugations and inner products in which $Q^\dagger = S$ becomes a statement about the one-loop components.

Names. In this chapter the spinor parameters written $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ in Definition 2.2 are written $\xi \in \Sigma$ and $\eta \in \Sigma^*$, and general elements of $\Sigma \oplus \Sigma^*$ are written ζ; the letters $Q_1, S_1, D_1, K_1, P_1, M_1, R_1$ denote the one-loop components of Definition 8.16, and Q alone is the linear BV differential. The local functionals are $\mathscr{O}_l(\mathscr{E})$ [C07, Def 4.2.2], the cochains $C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$ of Definition 2.7, with the Chevalley–Eilenberg differential and the cochain bracket of Definition 2.8; for a functional f, $\mathrm{ad}_f := \{f, -\}$. The translation along $b \in V$ is P_b and the special conformal field is K_b, $x \mapsto (x,x)b - 2(b,x)x$ (Definition 2.1). The unit is

$$\frac{\hbar \,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)}.$$

8.1 Two-component form of the data

Every definition of this chapter is stated index-free and in the two-component notation of Srednicki [Srednicki] (M. Srednicki, Quantum Field Theory, Cambridge University Press, 2007, §§34–36), adapted to Euclidean signature by the Wick map below.

Definition 8.1 (Two-component realisation)**.** The Wick map is $v^0 = -iv^4$, $v^k = v^k$ $(k = 1,2,3)$, so $x^4 = it$ and $\partial_0 = i\partial_4$. The Euclidean matrices

$$\sigma^m_{\alpha\dot\alpha} = (\sigma^1, \sigma^2, \sigma^3, i), \qquad \bar\sigma^{m\dot\alpha\alpha} = (-\sigma^1, -\sigma^2, -\sigma^3, i)$$

satisfy $\sigma^\mu v_\mu = \sigma^m v_m$, $\bar\sigma^\mu v_\mu = \bar\sigma^m v_m$ for Srednicki's $\sigma^\mu = (1, \vec\sigma)$, $\bar\sigma^\mu = (1, -\vec\sigma)$, and $\sigma^m\bar\sigma^n + \sigma^n\bar\sigma^m = -2\delta^{mn}$. Indices are raised and lowered with $\varepsilon^{12} = \varepsilon^{\dot1\dot2} = +1$, $\varepsilon_{12} = \varepsilon_{\dot1\dot2} = -1$, and $\chi\psi = \chi^\alpha\psi_\alpha$, $\bar\chi\bar\psi = \bar\chi_{\dot\alpha}\bar\psi^{\dot\alpha}$. The chirality S_+ carries undotted indices (Srednicki's left-handed $(2,1)$); W carries an

upper index $i = 1, \dots, 4$, W^* a lower one. An element of $\Sigma = S_+ \otimes W \oplus S_- \otimes W^*$ has components $(\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$, an element of $\Sigma^* = S_+ \otimes W^* \oplus S_- \otimes W$ components $(\eta^\alpha{}_i, \bar\eta^{\dot\alpha i})$, and

$$(\xi,\eta) = \xi_\alpha{}^i \eta^\alpha{}_i - \bar\xi_{\dot\alpha i}\bar\eta^{\dot\alpha i}, \qquad (\rho(v)\eta)_\alpha{}^i = -i\,v_m \sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad (\rho(v)\eta)_{\dot\alpha i} = i\,v_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}, \tag{8.1}$$
$$(\rho(v)\xi)^\alpha{}_i = i\,v_m \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}, \qquad (\rho(v)\xi)^{\dot\alpha i} = -i\,v_m \bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i.$$

The six scalar directions $W_{\mathbb R}^\perp$ are described by the 't Hooft blocks $\Sigma^I_{ij} = -\Sigma^I_{ji}$, $I = 1, \dots, 6$: for $a,b,c \in \{1,2,3\}$, $\Sigma^a_{bc} = \epsilon_{abc}$, $\Sigma^a_{b4} = -\Sigma^a_{4b} = \delta_{ab}$, $\Sigma^{a+3}_{bc} = i\,\epsilon_{abc}$, $\Sigma^{a+3}_{b4} = -\Sigma^{a+3}_{4b} = -i\,\delta_{ab}$; they satisfy $\sum_I \Sigma^I_{ij}\Sigma^I_{kl} = 2\varepsilon_{ijkl}$ with $\varepsilon_{1234} = \varepsilon^{1234} = 1$. For $w \in W_{\mathbb R}^\perp$ with components w_I, $w_{ij} = w_I \Sigma^I_{ij}$ and $w^{ij} = \frac12 \varepsilon^{ijkl} w_{kl}$, and

$$(\rho(w)\xi)^\alpha{}_j = \varepsilon^{\alpha\beta} w_{ji} \xi_\beta{}^i, \quad (\rho(w)\xi)^{\dot\alpha j} = \varepsilon^{\dot\alpha\dot\beta} w^{ji} \bar\xi_{\dot\beta i}, \quad (\rho(w)\eta)_\alpha{}^k = -w^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \quad (\rho(w)\eta)_{\dot\beta k} = -w_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}.$$

The Γ-pairing $(\rho(v)\zeta_1, \zeta_2) = (v, \Gamma(\zeta_1,\zeta_2))$ [ESW, (28)] has the components

$$\Gamma(\xi_1,\xi_2)\big|^m_V = i\big(\bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i + \bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i\big), \qquad \Gamma(\eta_1,\eta_2)\big|^m_V = -i\big(\eta_2{}^\alpha{}_i \sigma^m_{\alpha\dot\alpha}\bar\eta_1^{\dot\alpha i} + \eta_1{}^\alpha{}_i \sigma^m_{\alpha\dot\alpha}\bar\eta_2^{\dot\alpha i}\big),$$
$$\Gamma(\xi_1,\xi_2)_{kl} = 2\varepsilon_{jikl}\xi_{1\alpha}{}^j \varepsilon^{\alpha\beta}\xi_{2\beta}{}^i - 2\big(\bar\xi_{1\dot\alpha k}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta l} - \bar\xi_{1\dot\alpha l}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta k}\big),$$
$$\Gamma(\eta_1,\eta_2)^{kl} = -2\varepsilon^{klpj}\varepsilon_{\alpha\beta}\eta_1{}^\beta{}_j \eta_2{}^\alpha{}_p + 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha l}\bar\eta_2^{\dot\beta k} - \varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha k}\bar\eta_2^{\dot\beta l}\big),$$

with $\Gamma_{kl} = \Gamma^I \Sigma^I_{kl}$, $\Gamma^{kl} = \frac12 \varepsilon^{klij}\Gamma_{ij}$ and $(\Gamma, u) = \Gamma^m u_m + \frac14 \Gamma_{ij} u^{ij}$. The conformal Killing spinor $x \mapsto \xi + \rho(x)\eta$ of Definition 2.2 has components

$$\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\,x_m \sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad \bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\,x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}. \tag{8.2}$$

The fields are A_m; the scalars $A_{ij} = A_I \Sigma^I_{ij}$, $A^{ij} = \frac12 \varepsilon^{ijkl} A_{kl}$, with $\iota_w A = \frac14 w^{ij} A_{ij}$; the gaugino $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$; the ghost c; the antifields A^*_m, A^*_{ij}, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^*, with $(\lambda, \lambda^*) = (\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})$. The generators of [SW, (8)] are labelled by the blocks of [SW, (17)]: Q^i_α (component $\xi_\alpha{}^i$), $\bar Q_{\dot\alpha i}$ ($\bar\xi_{\dot\alpha i}$), $S^\alpha{}_i$ ($\eta^\alpha{}_i$), $\bar S^{\dot\alpha i}$ ($\bar\eta^{\dot\alpha i}$); in components K_b is $v^m = x^2 b^m - 2(b\cdot x)x^m$.

Proposition 8.2 (Two-component form of the action and of the propagator)**.** *The formulas of Definition 8.1 realise ρ, $(-,-)$ and Γ of Definition 1.1: they satisfy the Clifford relation on V_{10}, $(\rho(v)\zeta_1,\zeta_2) = (\zeta_1, \rho(v)\zeta_2)$, [ESW, Assumption 3.1], and $\rho(e_1e_2e_3e_4) = +1$ on $S_+ \otimes W$. On the Dirac column $(\psi_\alpha, \bar\chi^{\dot\alpha})$, $\rho(v) = -i\gamma^\mu_S v_\mu$ with Srednicki's γ matrices, which is $\gamma^\mu_W v_\mu$ with Weinberg's ($\gamma_W = -i\gamma_S$); this gives Weinberg's form of every two-component formula below. Up to total derivatives the kinetic part of the action of Definition 1.8 is*

$$S \supset \int d^4x \Big\{ -\tfrac18 (F_{mn}, F_{mn}) - \tfrac{1}{16}(D_m A^{ij}, D_m A_{ij}) + i\big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha} D_m \lambda_\alpha{}^i\big) + (D_m c, A^*_m)\Big\},$$

and for $\mathfrak g = \mathfrak{su}(N)$ with $(X,Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY)$, $A_m = -igA^a_m T^a$, $A_I = -igA^a_I T^a$, $\lambda = -\frac{ig}{\sqrt2}\lambda^a T^a$, $\operatorname{tr}(T^a T^b) = \frac12 \delta^{ab}$,

$$\frac{S}{\hbar} \supset \int d^4x \Big\{ -\tfrac14 F^a_{mn}F^a_{mn} - \tfrac12 (D_m A_I)^a (D_m A_I)^a + i\bar\lambda^a \bar\sigma^m D_m \lambda^a \Big\},$$

the Wick image of Srednicki's Lagrangian $-\frac14 F^{a\mu\nu}F^a_{\mu\nu} - \frac12 D^\mu A_I D_\mu A_I + i\lambda^{\dagger a}\bar\sigma^\mu D_\mu \lambda^a$ [Srednicki, (69.16), (95.77)], with $e^{iS_{\text{Srednicki}}} = e^{S/\hbar}$. The components of the propagator of Definition 8.3 in momentum space are

$$\langle A_m A_n\rangle :\ 2(-,-)^{-1}\delta_{mn}, \qquad \langle A_{ij}A_{kl}\rangle :\ 4(-,-)^{-1}\varepsilon_{ijkl}, \qquad \langle c\, A^*_m\rangle :\ -(-,-)^{-1} i p_m, \qquad \langle \lambda_\alpha{}^i \bar\lambda_{\dot\beta j}\rangle :\ -(-,-)^{-1} p_m \sigma^m_{\alpha}$$

each multiplied by $(e^{-\varepsilon|p|^2} - e^{-T|p|^2})/|p|^2$.

Proof. Each identity is a finite matrix identity, verified by evaluation against the realisation of Chapter 2 on the ten basis vectors of V_{10}; the intertwiner with that realisation is unique up to a common sign. The action is the density of (1.1) rewritten with $(F,F)=\frac12(F_{mn},F_{mn})+\frac14(D_mA^{ij},D_mA_{ij})+\frac{1}{32}([A^{ij},A^{kl}],[A_{ij},A_{kl}])$ and $\frac12\int(\lambda,\not{d}_A\lambda)=\int\{i(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}D_m\lambda_\alpha{}^i)+(\text{Yukawa terms})\}$. The su($N$) form follows with $(X,Y)=2\hbar X^aY^a$. The propagator is Proposition 3.17 with the scalar factor of Definition 8.3. □

8.2 The transfer endpoint

Definition 8.3 (Propagator with an endpoint)**.** For $0<\varepsilon<T\le\infty$,

$$P(\varepsilon,T)=\int_\varepsilon^T(Q^{GF}\otimes1)K_t\,dt,\qquad(Q\otimes1+1\otimes Q)\,P(\varepsilon,T)=K_\varepsilon-K_T \tag{8.3}$$

[C07, Lemma 10.2.1], with $K_\infty=0$ (Definition 3.14), and $\Delta_T=-\partial_{K_T}$, $\{-,-\}_T$ as in Definition 3.21 with T in place of ε. The scalar factor of $P(\varepsilon,T)$ is

$$\int_\varepsilon^T\frac{e^{-|x-y|^2/4t}}{(4\pi t)^2}\,dt=\frac{e^{-|x-y|^2/4T}-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2},\qquad\int_\varepsilon^T e^{-t|p|^2}dt=\frac{e^{-\varepsilon|p|^2}-e^{-T|p|^2}}{|p|^2}.$$

T is the endpoint of the transfer, the scale of the effective interaction of [C07]. For $|x-y|^2\gg T$ every component of $P(\varepsilon,T)$ and of its derivatives decays as $e^{-|x-y|^2/4T}$, and for every T they are bounded by a constant multiple, independent of ε and T, of the corresponding components of $P(0,\infty)$. Here ∂_ϕ is read as in Definition 3.16, its value on the product of a coordinate functional of the first factor and one of the second, in this order, being the corresponding component of ϕ, and a linear operator X on $\mathscr{E}$ acts on functionals as the derivation induced by $\ell\mapsto\ell\circ X$ [C07, §6.2]; then $[\partial_\phi,X]=\partial_{(X\otimes1+1\otimes X)\phi}$ as a graded commutator for kernels and operators of either parity (Lemma 9.15).

Lemma 8.4 (Transfer between endpoints)**.** *For $0<\varepsilon<T<T'\le\infty$, on functionals with values in $C^\bullet(\mathfrak{psl}(4|4))[[\hbar]]$,*

$$(d_{CE}{+}Q{+}\hbar\Delta_T)\,e^{\hbar\partial_{P(\varepsilon,T)}}=e^{\hbar\partial_{P(\varepsilon,T)}}\,(d_{CE}{+}Q{+}\hbar\Delta_\varepsilon),\qquad W\big(P(\varepsilon,T'),J\big)=W\Big(P(T,T'),W\big(P(\varepsilon,T),J\big)\Big). \tag{8.4}$$

At $T=\infty$ the first identity is Theorem 3.23; with (ε,T) replaced by (T,T') it relates the structures at two endpoints [C07, Lemma 10.2.2].

Proof. By [C07, §6.4] and Definition 8.3, $[\partial_\phi,Q]=\partial_{(Q\otimes1+1\otimes Q)\phi}$; with (8.3), $\partial_{P(\varepsilon,T)}Q-Q\,\partial_{P(\varepsilon,T)}=\partial_{K_\varepsilon}-\partial_{K_T}=-\Delta_\varepsilon+\Delta_T$. The operators ∂_ϕ commute, so $Q\,\partial^n_{P(\varepsilon,T)}=\partial^n_{P(\varepsilon,T)}Q+n\,\partial^{n-1}_{P(\varepsilon,T)}(\Delta_\varepsilon-\Delta_T)$, and summing the exponential series gives the first identity; d_{CE} commutes with $\partial_{P(\varepsilon,T)}$. The second follows from $P(\varepsilon,T')=P(\varepsilon,T)+P(T,T')$ and $e^{\hbar\partial_{\phi_1+\phi_2}}=e^{\hbar\partial_{\phi_2}}e^{\hbar\partial_{\phi_1}}$ (Definition 3.22). □

8.3 The quantised equivariant interaction

Definition 8.5 (Equivariant quantisation)**.** Let $I_{\mathfrak{psl}(4|4)}=I+S^{(1)}+S^{(2)}$ be the equivariant interaction (Definition 2.10) and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ its counterterms: graph by graph, the singular parts of the flow, order by order in $\hbar$ with the counterterms of lower order inserted, local cochains with coefficients in the span of the functions $\varepsilon^{-m}\log^k\varepsilon$, $(m,k)\neq(0,0)$ (Definition 4.1 at order $\hbar$, [C07, Thm B] at higher order). A finite counterterm is a local cochain $F_{\mathfrak{psl}(4|4)}=\hbar F_1+\hbar^2F_2+\cdots$ of degree 0 and Weyl weight 4, independent of ε; here the explicit position has weight -1 and an argument x has weight one half of

its ad_Δ-eigenvalue (Remark 2.6), so that a value on $x_1, \dots, x_k$ has weight 4 minus the weights of the arguments (Proposition 7.13). The effective interaction at scale T is

$$I_{\mathfrak{psl}(4|4)}[T] := \lim_{\varepsilon \to 0} W\big(P(\varepsilon, T),\, I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - F_{\mathfrak{psl}(4|4)}\big), \tag{8.5}$$

and $F_{\mathfrak{psl}(4|4)}$ is a quantisation of the equivariant theory if the equivariant quantum master equation holds at scale T [C07, Def 10.3.1], [CG2, Prop 13.3.2.1]:

$$(d_{CE} + Q)\, I_{\mathfrak{psl}(4|4)}[T] + \tfrac{1}{2}\big\{ I_{\mathfrak{psl}(4|4)}[T], I_{\mathfrak{psl}(4|4)}[T] \big\}_T + \hbar \Delta_T\, I_{\mathfrak{psl}(4|4)}[T] = 0. \tag{8.6}$$

A one-loop quantisation is the same statement modulo $\hbar^2$. The one-loop BV structure of the theory is a one-loop quantisation together with the quantum differential of local operators of Definition 8.14.

Proposition 8.6 (Independence of the endpoint)**.**

- *(i) At order $\hbar$ the singular part of every graph with edges $P(\varepsilon, T)$ is independent of $T \in (0, \infty)$, and equal to that at $T = \infty$ in the components whose graphs converge at large distance. At every order in $\hbar$ the counterterms $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ constructed with $P(\varepsilon, T)$ are the same for all $T \in (0, \infty)$ [C07, §1.5].*
- *(ii) For $T < T' < \infty$, $I_{\mathfrak{psl}(4|4)}[T'] = W\big(P(T, T'), I_{\mathfrak{psl}(4|4)}[T]\big)$, and (8.6) holds at T exactly when it holds at T'. At $T' = \infty$ the same holds in every component in which the graph sums converge as $T' \to \infty$.*

Proof. (i) At order $\hbar$, in the Schwinger representation of Lemma 4.6, the coefficients of ε^{-k} and $\log \varepsilon$ come from the region in which the Schwinger times of the cycle sum to at most a fixed $\tau_0 < T$; there the integration domain is independent of T, and the complementary region gives a function with a limit as $\varepsilon \to 0$. At higher order, if $\lim_{\varepsilon \to 0} W(P(\varepsilon, T), J)$ exists, so does $\lim_{\varepsilon \to 0} W(P(\varepsilon, T'), J) = W\big(P(T, T'), \lim_{\varepsilon \to 0} W(P(\varepsilon, T), J)\big)$, the edges $P(T, T')$ being smooth, and the uniqueness of the counterterms [C07, Thm B] identifies the counterterms at T and at T'. (ii) The flow identity is the second identity of (8.4) in the limit $\varepsilon \to 0$. By the first identity with (T, T') in place of (ε, T), $(d_{CE} + Q + \hbar\Delta_{T'}) e^{I[T']/\hbar} = e^{\hbar \partial_{P(T,T')}} (d_{CE} + Q + \hbar \Delta_T) e^{I[T]/\hbar}$, and $e^{\hbar \partial_{P(T,T')}}$ is invertible; (8.6) is the statement $(d_{CE} + Q + \hbar \Delta_T) e^{I[T]/\hbar} = 0$ (Corollary 3.24). □

Definition 8.7 (Defect and obstruction)**.** With $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - F_{\mathfrak{psl}(4|4)}$, the defect vertex is

$$(d_{CE} + Q) J + \tfrac{1}{2}\{J, J\}_\varepsilon + \hbar \Delta_\varepsilon J,$$

and by (8.4) the left side of (8.6) is the $\varepsilon \to 0$ limit of the connected graphs with one defect vertex, other vertices from J and edges $P(\varepsilon, T)$ (Proposition 4.26). With $F_{\mathfrak{psl}(4|4)} = 0$, the one-loop obstruction is the local cochain Ob of degree 1 whose tree-level flow with edges $P(0, T)$ equals the coefficient of $\hbar$ in this limit; it is the equivariant form of Proposition 4.27 and of the anomaly of [CG2, Prop 13.3.2.1], and by Proposition 8.6 it is independent of T.

Theorem 8.8 (One-loop obstruction theory)**.** *Write $s = d_{CE} + Q + \{I_{\mathfrak{psl}(4|4)}, -\}$ on local cochains; $s^2 = 0$ by (2.4). Assume that the obstruction of Definition 8.7 is local.*

- *(i) $s\,\mathrm{Ob} = 0$.*
- *(ii) A finite counterterm $\hbar F_1$ is a one-loop quantisation exactly when $sF_1 = \mathrm{Ob}$.*
- *(iii) The one-loop quantisations form an affine space over the s-closed local cochains of degree 0. Two of them whose difference is $s\phi$, ϕ local of degree -1, differ by the field redefinition generated by ϕ; the differences modulo s-exact cochains are the classes of H^0 of s on local cochains.*

- *(iv) Two classes of H^0 are always present: the coupling direction $S - S^{(2)}$, the derivative at $\mu = 1$ of the solutions $\mu S + S^{(1)} + \mu^{-1} S^{(2)}$ of (2.4), and the outer direction G_B of Definition 8.9.*

Proof. (i) is the equivariant form of [CG2, Prop 13.3.2.1]: with E the left side of (8.6), $(d_{CE} + Q + \hbar\Delta_T)e^{I[T]/\hbar} = \hbar^{-1} E e^{I[T]/\hbar}$, and applying $d_{CE} + Q + \hbar\Delta_T$ once more gives $\big((d_{CE} + Q)E + \{I[T], E\}_T + \hbar\Delta_T E\big)e^{I[T]/\hbar} = 0$, the term E^2 vanishing for the odd E; at order $\hbar$ this is the tree-level flow of s Ob, and the tree-level flow is injective on local cochains by the argument of Proposition 4.10 (lowest number of legs first). (ii) Replacing $F_{\mathfrak{psl}(4|4)} = 0$ by $\hbar F_1$ adds to the defect vertex $-\hbar\big((d_{CE}+Q)F_1 + \{I_{\mathfrak{psl}(4|4)}, F_1\}_\varepsilon\big)$ modulo $\hbar^2$; the special-edge part $\{I_{\mathfrak{psl}(4|4)}, F_1\}_\varepsilon - \{I_{\mathfrak{psl}(4|4)}, F_1\}$ sits on a tree and is multiplied by $e^{-\varepsilon|p|^2} - 1$ at a generic momentum, so it vanishes in the limit; the coefficient of $\hbar$ of the obstruction becomes $\mathrm{Ob} - sF_1$. (iii) By (ii) the difference G of two solutions satisfies $sG = 0$; for $G = s\phi$ the effective interactions differ by the flow of $\hbar s\phi$, the infinitesimal field redefinition generated by ϕ. (iv) The family $\mu S + S^{(1)} + \mu^{-1} S^{(2)}$ solves (2.4) because, by Lemma 2.11, the arity-n component of the Maurer–Cartan equation is multiplied by μ^{2-n} in this family; its derivative at $\mu = 1$ is s-closed. Its class is nonzero: in arity zero $(s\phi)^{(0)} = \{S, \phi^{(0)}\}$; $\{S, -\}$ preserves the degree in the explicit position, so $\{S, \phi^{(0)}\} = S$ requires the same identity for the part of $\phi^{(0)}$ of degree zero in the position, which is translation invariant of degree -1 and Weyl weight 4, a combination of the field-counting functionals (Lemma 4.19), and that identity is excluded by the computation of Corollary 4.23. G_B: Proposition 8.10. □

Definition 8.9 (The outer derivation)**.** B is the even linear map of $\mathfrak{psl}(4|4)$ equal to 0 on the even part, to $+1$ on $S_+ \otimes W \subset \Sigma$ and on $S_- \otimes W \subset \Sigma^*$, and to -1 on $S_- \otimes W^* \subset \Sigma$ and on $S_+ \otimes W^* \subset \Sigma^*$; it is induced by the bracket with $\mathrm{diag}(0_4, 1_4)$ in $\mathfrak{gl}(4|4)$, equivalently minus the bracket with $\mathrm{diag}(1_4, 0_4)$. Two-component: B multiplies $\xi_\alpha{}^i$ and $\bar\eta^{\dot\alpha i}$ by $+1$ and $\bar\xi_{\dot\alpha i}$ and $\eta^\alpha{}_i$ by -1; it counts upper minus lower W indices. The outer direction is

$$G_B := \frac{d}{dt}\Big|_{t=0} \big(e^{tB}\big)^* S_{\mathfrak{psl}(4|4)}, \qquad G_B^{(0)} = 0, \quad G_B^{(1)}(x) = S^{(1)}(Bx), \quad G_B^{(2)}(x_1, x_2) = S^{(2)}(Bx_1, x_2) + S^{(2)}(x_1, Bx_2).$$

Proposition 8.10 (The outer direction)**.** *B is an outer derivation of $\mathfrak{psl}(4|4)$. G_B is s-closed of degree 0 and Weyl weight 4, and its class in H^0 is independent of the class of the coupling direction.*

Proof. On odd x, y with B-values b_x, b_y the derivation rule reads $(b_x + b_y)[x, y] = 0$, and brackets within one B-eigenspace vanish: two elements of Σ, or two of Σ^*, of equal chirality have $\Gamma|_V = 0$ by Definition 8.1, and an element of Σ and one of Σ^* in the same eigenspace have opposite chiralities, so their bracket has the Lorentz type $S_+ \otimes S_-$, absent from $[\Sigma, \Sigma^*] \subset \mathfrak{so}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^\perp) \oplus \mathbb{C}\Delta$ (Proposition 2.4); the even part preserves the eigenspaces, translations mapping Σ^* to Σ and special conformal fields Σ to Σ^* by ρ, which keeps the factor W or W^*. An inner derivation ad_y equal to B would centralise the semisimple even part, and $\Sigma \oplus \Sigma^*$ contains no invariant vector, so $y = 0$, while $B \neq 0$. Pulling back by the automorphisms e^{tB} maps solutions of (2.4) to solutions, so G_B is closed; B commutes with Δ, which fixes degree and weight. Independence: suppose $G_B = a(S - S^{(2)}) + s\phi$ with ϕ local of degree -1. Along a translation, $0 = G_B^{(1)}(P_b) = (s\phi)^{(1)}(P_b)$, so $\{S^{(1)}(P_b), \phi^{(0)}\}$ is $(Q + \{I, -\})$-exact; since $\{S^{(1)}(P_b), -\}$ acts on the explicit polynomial dependence on the position as $-\partial_b$ and $Q + \{I, -\}$ preserves the degree in the position, induction on that degree replaces ϕ by $\phi - s\chi$, with χ local of degree -2 and $\phi^{(0)}$ translation invariant, and $s\phi$ is unchanged; this excludes $\phi^{(0)} = S^{(1)}(\Delta)$, which is invariant under the compact groups and satisfies $\{S^{(1)}(\xi), S^{(1)}(\Delta)\} = S^{(1)}(\xi)$ for all $\xi \in \Sigma$, while $\{S^{(1)}(P_b), S^{(1)}(\Delta)\} = -2S^{(1)}(P_b)$. The part of degree zero in the position of the arity-one component along $\xi \in S_+ \otimes W$ then reads $S^{(1)}(\xi) = \pm\{S^{(1)}(\xi), \phi^{(0)}\}$, since the space of translation-invariant local functionals of degree -2 and Weyl weight $\frac{9}{2}$ is zero; averaging over the compact forms of $\mathrm{Spin}(4) \times SU(4)$ and of the gauge group, $\phi^{(0)}$ is a combination $\sum_f a_f \int (f, f^*)$ of the field-counting functionals of $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, the two chiralities of λ and c, and $\{S^{(1)}(\xi), \phi^{(0)}\}$ multiplies each term of (2.6) by a difference of two a_f. The four linear terms of $S^{(1)}(\xi)$ require four such differences equal to ± 1 whose sum is 0, a contradiction. □

8.4 Observables and the quantum differential

Definition 8.11 (Renormalised local cochains)**.** A scheme for local operators is a linear map ψ of local cochains, preserving the degree and the Weyl weight (counted as in Definition 8.5), independent of ε and linear over $C^\bullet(\mathfrak{psl}(4|4))$. For a local cochain M of degree $|M|$ and a square-zero parameter δ of degree $-|M|$ [CG2, Def 8.3.1.1],

$$M(\varepsilon) := M - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](M) + \hbar\,\psi(M), \qquad M[T] := \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,T),\, I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - F_{\mathfrak{psl}(4|4)} + \delta\, M(\varepsilon)\big),$$

where $\partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](M)$ is the counterterm of the family $I_{\mathfrak{psl}(4|4)} + \delta M$, the singular part of the graphs through the vertex δM (Proposition 5.3). At $T=\infty$, $\psi = 0$ and $F_{\mathfrak{psl}(4|4)} = 0$ this is Definition 7.1; $\psi = 0$ is the minimal scheme. $M \mapsto M[T]$ is linear over $C^\bullet(\mathfrak{psl}(4|4))$.

Definition 8.12 (Quantum differential at scale T)**.** For an observable $O[T]$ at scale T [CG2, Def 8.3.1.3, Prop 13.4.0.1],

$$d_T\, O[T] := (d_{CE} + Q)\, O[T] + \big\{ I_{\mathfrak{psl}(4|4)}[T], O[T] \big\}_T + \hbar \Delta_T\, O[T].$$

At $T=\infty$, $d_\infty = d_{CE} + Q$.

Lemma 8.13 (Flow covariance)**.** *Assume* (8.6). *Then $d_T^2 = 0$; for $T < T' < \infty$, $W\big(P(T,T'), I_{\mathfrak{psl}(4|4)}[T] + \delta M[T]\big) = I_{\mathfrak{psl}(4|4)}[T'] + \delta M[T']$, and the linearised flow intertwines d_T with $d_{T'}$.*

Proof. $e^{(I[T]+\delta M[T])/\hbar} = e^{I[T]/\hbar}(1 + \hbar^{-1}\delta M[T])$ and, by (8.6) and the product rule of Definition 3.21, $(d_{CE} + Q + \hbar\Delta_T)\big(\delta M[T] e^{I[T]/\hbar}\big) = \delta\,(d_T M[T])\, e^{I[T]/\hbar}$ up to the sign of δ; $(d_{CE} + Q + \hbar\Delta_T)^2 = 0$ gives $d_T^2 = 0$. The flow identity is the second identity of (8.4) at δ-linear order in the limit $\varepsilon \to 0$, and the intertwining is the first identity with (T,T') in place of (ε, T). □

Definition 8.14 (Local representative)**.** A local representative of the quantum differential of a local cochain M is a local cochain $L(M)$ of degree $|M|+1$, a formal series in $\hbar$, with

$$d_T\, M[T] = \big(L(M)\big)[T]. \tag{8.7}$$

Theorem 8.15 (Well-posedness of the quantum differential)**.** *Assume* (8.6).

- *(i) If $N[T] = 0$ for a local cochain N, then $N = 0$. Hence $L(M)$ is unique when it exists.*
- *(ii) If* (8.7) *holds at one $T_0 \in (0,\infty)$, it holds with the same $L(M)$ at every $T \in (0,\infty)$, and at $T = \infty$ when the graph sums of $M[T]$ and $L(M)[T]$ converge as $T \to \infty$.*
- *(iii) If $L(M)$ and $L(L(M))$ exist, then $L(L(M)) = 0$.*
- *(iv) $L = d_{CE} + \Lambda$ with Λ linear over $C^\bullet(\mathfrak{psl}(4|4))$, and $L \circ L = 0$ is the Maurer–Cartan equation $d_{CE}\Lambda + \Lambda d_{CE} + \Lambda\Lambda = 0$ for the cochain $\Lambda = \sum_k \Lambda^{(k)}$ with values in the operators on $\mathscr{O}_l(\mathscr{E})[[\hbar]]$, where $(\Lambda\Lambda')(x_1,\dots,x_n)$ is the sum over (k,l)-shuffles σ of $\operatorname{sgn}(\sigma)\cdot$(Koszul sign of σ)$\cdot$ $(-1)^{|\Lambda'|k + \sum_{i=1}^k |x_{\sigma(i)}|(l+|\Lambda'|)}\, \Lambda^{(k)}(x_{\sigma(1)},\dots,x_{\sigma(k)}) \circ \Lambda'^{(l)}(x_{\sigma(k+1)},\dots,x_{\sigma(n)})$, the sign rule of Definition 2.8; the same rule defines the products of field-independent cochains with local cochains, and, with the composition replaced by the product of values and $|\Lambda'|$ read as the total parity (degree plus fermion number modulo 2), the product of any two functional-valued cochains, in particular $e^{J/\hbar}$ in* (8.5)*; this product is associative and graded commutative. Modulo $\hbar$, $L = s$, $\Lambda^{(0)} = Q + \{I, -\}$, $\Lambda^{(1)}(x)O = (-1)^{|O|(1+|x|)}\{S^{(1)}(x), O\}$, $\Lambda^{(2)}(x_1,x_2)O = (-1)^{|O|(|x_1|+|x_2|)}\{S^{(2)}(x_1,x_2), O\}$, $\Lambda^{(k\geq 3)} = 0$, where in these two formulas $|O|$ is the degree of O in the sense of Definition 2.8 (its ghost number), so that $\Lambda^{(1)}(x) = \{S^{(1)}(x), -\}$ on functionals of degree 0 of either parity and $x \mapsto -\Lambda^{(1)}(x)$ is a representation on the cohomology also on odd operators.*

Proof. (i) At lowest order in $\hbar$, $N[T]$ is the tree-level flow of N with vertices $I_{\mathfrak{psl}(4|4)}$ and edges $P(0,T)$. By induction on the arity, a tree with $r \geq 1$ vertices I has at least r more legs than its vertex $N^{(k)}$, so the part with the fewest legs is a Taylor component of $N^{(k)}$, which therefore vanishes; induction on the number of legs (Proposition 4.10) and then on the order in $\hbar$ gives $N = 0$. (ii) By Lemma 8.13, $M[T]$ and $L(M)[T]$ are the linearised flows from T_0 of $M[T_0]$ and $L(M)[T_0]$, and the flow intertwines d_{T_0} with d_T; for $T < T_0$ the flow is inverted. At $T = \infty$ the flow with $P(T_0, \infty)$ is the limit of the convergent graph sums, and the intertwining is the first identity of (8.4) with (T_0, ∞) in place of (ε, T), in which $\Delta_\infty = 0$ (Definition 3.21). (iii) $L(L(M))[T] = d_T(L(M)[T]) = d_T^2 M[T] = 0$, and (i). (iv) For a field-independent cochain c', $(c'M)[T] = c'\,M[T]$ and $d_T(c'M[T]) = (d_{CE}c')M[T] \pm c'\,d_T M[T]$, so $L(c'M) = (d_{CE}c')M \pm c'\,L(M)$; $\Lambda := L - d_{CE}$ is therefore $C^\bullet$-linear and determined by its values on local functionals, and $L \circ L = 0$ expands to the stated equation. Modulo $\hbar$, $M[T]$ is the tree-level flow of M, and by the first identity of (8.4) $d_T M[T]$ is the tree-level flow of the defect vertex $(d_{CE} + Q)M + \{I_{\mathfrak{psl}(4|4)}, M\}_\varepsilon$; the special edges of trees carry $e^{-\varepsilon|p|^2} - 1$ at generic momenta and vanish in the limit, which leaves the tree-level flow of sM, so $L = s$; the signs of the components are those of the cochain bracket of Definition 2.8 with the arity-zero argument O. □

Definition 8.16 (Components)**.** For a local functional O (arity zero) write $L(O) = \sum_k L(O)^{(k)}$ by arity and $L = s + \hbar L_1 + \hbar^2 L_2 + \cdots$. The one-loop components of O are the arity-one values

$$Q_1(O) := \hbar L_1(O)^{(1)}(\xi), \quad \xi \in \Sigma; \qquad S_1(O) := \hbar L_1(O)^{(1)}(\eta), \quad \eta \in \Sigma^*; \qquad D_1(O) := \hbar L_1(O)^{(1)}(\Delta);$$
$$K_1(O) := \hbar L_1(O)^{(1)}(K_b); \qquad P_1(O) := \hbar L_1(O)^{(1)}(P_b); \qquad M_1(O),\ R_1(O):\ \text{the same along } \mathfrak{so}(V),\ \mathfrak{so}(W_{\mathbb{R}}^{\perp}),$$

together with the one-loop BRST correction $\hbar L_1(O)^{(0)}$ and the arity-two values $\hbar L_1(O)^{(2)}(x_1, x_2)$. For $|O| = 0$ the full transformation of O along $x \in \mathfrak{psl}(4|4)$ is $L(O)^{(1)}(x) = \{S^{(1)}(x), O\} + \hbar L_1(O)^{(1)}(x) + O(\hbar^2)$. Two-component: $Q_1(O) = \xi_\alpha{}^i\, Q_1(O)^\alpha{}_i + \bar\xi_{\dot\alpha i}\, \bar Q_1(O)^{\dot\alpha i}$, with the coefficient functionals along Q^i_α and $\bar Q_{\dot\alpha i}$, and likewise $S_1(O) = \eta^\alpha{}_i\, S_1(O)_\alpha{}^i + \bar\eta^{\dot\alpha i}\, \bar S_1(O)_{\dot\alpha i}$.

Corollary 8.17 (The realisation up to homotopy)**.** *Assume* (8.6) *and that the local representatives exist on a space of local functionals stable under* Λ*. In arity one and two the Maurer–Cartan equation of Theorem 8.15 (iv) states, with graded commutators:*

- *(i)* $\Lambda^{(1)}(x)$ *commutes with* $\Lambda^{(0)}$ *for every* x*;*
- *(ii)* $[\Lambda^{(1)}(x_1), \Lambda^{(1)}(x_2)] + \Lambda^{(1)}([x_1, x_2])$ *is a graded commutator of* $\Lambda^{(0)}$ *with an operator built from* $\Lambda^{(2)}(x_1, x_2)$*; modulo* $\hbar$ *this is the identity*

$$\big[\{S^{(1)}(x_1), -\}, \{S^{(1)}(x_2), -\}\big] = -\{S^{(1)}([x_1, x_2]), -\} - \big\{\{S, S^{(2)}(x_1, x_2)\}, -\big\}$$

obtained from Lemma 2.11 and the Jacobi identity of Definition 1.6.

Hence $x \mapsto -\Lambda^{(1)}(x)$ *induces a representation of* $\mathfrak{psl}(4|4)$ *on the cohomology of* $\Lambda^{(0)}$*; it is the quantum realisation of the superconformal symmetry on local operators, and the one-loop components are its order-*$\hbar$ *part.*

Proof. On an arity-zero input the arity-one component of $d_{CE}\Lambda + \Lambda d_{CE} + \Lambda\Lambda$ is $[\Lambda^{(0)}, \Lambda^{(1)}(x)]$, since d_{CE} vanishes on arity zero; the arity-two component contains the term $\Lambda^{(1)}([x_1, x_2])$ from d_{CE}, the products of two $\Lambda^{(1)}$ and the products of $\Lambda^{(0)}$ with $\Lambda^{(2)}$. The equation has the same form at every order in $\hbar$, and its order-$\hbar^0$ part is the displayed identity, which fixes the signs. On cohomology the last term acts as 0. □

Remark 8.18 (What is computed)**.** The one-loop BV structure of Definition 8.5 consists of the solution F_1 of $sF_1 = \mathrm{Ob}$ (Theorem 8.8) and of Λ modulo $\hbar^2$ on local operators. By Theorem 8.15 (ii) every

one-loop component is a local functional independent of the endpoint; it is read off from the $\varepsilon \to 0$ limit of the graphs with one defect vertex of Lemma 8.4 applied to $e^{(J+\delta O(\varepsilon))/\hbar}$, as in (6.11) along Δ. Chapter 6 computed the dilatation coefficients through the scaling covariance of the regulator. On the part of the scalar bilinears with two legs $A|_{W_{\mathbb{R}}^{\perp}}$, Theorem 9.30 gives $D_1(O)$ equal to twice the coefficient of Theorem 6.11 times O plus $-(\frac{1}{3}+\theta)$ units times $\{S^{(1)}(\Delta), O\}$, where θ units is the coefficient of $\int (2E(A|_{W_{\mathbb{R}}^{\perp}}) + 2A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})$ in $\hbar F_1^{(1)}(\Delta)$, a scheme parameter that the two-point part of $sF_1 = \mathrm{Ob}$ leaves free. When the one-loop quantisation and the local representatives exist through arity two along (Δ, P_b), the component of Corollary 8.17 (ii) along (Δ, P_b) gives $\theta = -\frac{1}{3}$ (Theorem 9.33); then $D_1(O)$ vanishes for the traceless scalar bilinears and is -48 units times O for the trace bilinear, the value $2 \times (-24)$ of Theorem 6.11, and (6.9) holds on that part (Corollary 9.34).

8.5 Scheme freedom and invariant content

Proposition 8.19 (Transformation laws)**.** *Under the following changes, with the labels O of the operators fixed, L_1 changes as stated.*

- *(a) Field redefinition: $F_{\mathfrak{psl}(4|4)} \mapsto F_{\mathfrak{psl}(4|4)} + \hbar\, s\phi$, ϕ local of degree -1 and Weyl weight 4: $L_1 \mapsto L_1 + [s, \mathrm{ad}_\phi] = L_1 - \mathrm{ad}_{s\phi}$, i.e. L is conjugated by $1 + \hbar\, \mathrm{ad}_\phi$.*
- *(b) Finite parts of operators: $\psi \mapsto \psi + \psi'$: $L_1 \mapsto L_1 + [s, \psi']$, i.e. L is conjugated by $1 + \hbar\psi'$.*
- *(c) Closed directions: $F_{\mathfrak{psl}(4|4)} \mapsto F_{\mathfrak{psl}(4|4)} + \hbar\, a\, G$ with $sG = 0$: $L_1 \mapsto L_1 - a\, \mathrm{ad}_G$. For the coupling direction $G = S - S^{(2)}$: $L_1^{(0)} \mapsto L_1^{(0)} - a(Q + \{I, -\})$, $L_1^{(1)}$ unchanged, $L_1^{(2)}(x_1, x_2) \mapsto L_1^{(2)}(x_1, x_2) + a\{S^{(2)}(x_1, x_2), -\}$. For $G = G_B$: $L_1^{(0)}$ unchanged and, along a B-eigenvector x with eigenvalue b, $L_1^{(1)}(x) \mapsto L_1^{(1)}(x) - a\, b\, \{S^{(1)}(x), -\}$.*

In particular G_B changes Q_1 by $-a\hbar\{S^{(1)}(\xi), -\}$ for $\xi \in S_+ \otimes W$ and by $+a\hbar\{S^{(1)}(\xi), -\}$ for $\xi \in S_- \otimes W^*$, changes S_1 by $+a\hbar\{S^{(1)}(\eta), -\}$ for $\eta \in S_+ \otimes W^*$ and by $-a\hbar\{S^{(1)}(\eta), -\}$ for $\eta \in S_- \otimes W$, and leaves D_1, K_1, P_1, M_1, R_1 unchanged; two-component, Q_1 along Q^i_α and S_1 along $\bar{S}^{\dot\alpha i}$ (upper W index) change by $-a\hbar$ times the classical transformation, Q_1 along $\bar{Q}_{\dot\alpha i}$ and S_1 along $S^\alpha{}_i$ (lower W index) by $+a\hbar$ times it.

Proof. (a) At order $\hbar$, $O[T]$ changes by minus the tree insertion of $\hbar s\phi$; the equivariant form of Lemma 6.8 gives the change $\hbar\big(s\{\phi, O\} - \{\phi, sO\}\big)$ of the local representative, and $s\{\phi, O\} - \{\phi, sO\} = -\{s\phi, O\}$ for odd ϕ by the derivation rules of the bracket. (b) $O[T]$ changes by $\hbar\psi'(O)[T]$ and $L(O)[T]$ by $\hbar\psi'(L(O))[T]$, while $d_T(\psi'(O)[T]) = (s\psi'(O))[T] + O(\hbar)$. (c) Differentiating the flow along $J \mapsto J - \hbar a G$ gives $d_T \,\mathrm{tree}(G, O) = \{G, O\}[T] + \mathrm{tree}(G, sO)$ for closed even G; the components of ad_G are those of G: $(S - S^{(2)})^{(0)} = S$ with $\{S, -\} = Q + \{I, -\}$, $(S - S^{(2)})^{(1)} = 0$, $(S - S^{(2)})^{(2)} = -S^{(2)}$; $G_B^{(1)}(x) = S^{(1)}(Bx) = b\, S^{(1)}(x)$. The list follows from Definition 8.9. □

Theorem 8.20 (Invariant content)**.**

- *(i) Under (a) and (b) of Proposition 8.19, the cohomology of $\Lambda^{(0)}$ on local functionals and the representation of $\mathfrak{psl}(4|4)$ induced on it by $-\Lambda^{(1)}$ (Corollary 8.17) change by an isomorphism; under the coupling direction they are unchanged.*
- *(ii) Under the outer direction the induced representation is composed with the automorphism $e^{-a\hbar B}$ of $\mathfrak{psl}(4|4)$: the actions of Q^i_α, $\bar{S}^{\dot\alpha i}$ are multiplied by $1 - a\hbar$, those of $\bar{Q}_{\dot\alpha i}$, $S^\alpha{}_i$ by $1 + a\hbar$, and the even generators are unchanged.*
- *(iii) The spectrum of the operator induced by $\Lambda^{(1)}(\Delta)$ on the cohomology classes of a fixed classical Weyl weight (the one-loop anomalous dimensions) is invariant under (a), under (b), under the coupling direction and under the outer direction G_B; under (c) with an arbitrary s-closed G it is*

invariant when every s-closed local cochain of degree 0 and Weyl weight 4 lies in $\mathbb{C}(S - S^{(2)}) + \mathbb{C}\,G_B + \operatorname{Im} s$.

Proof. (i) Conjugation by $1 + \hbar\,\mathrm{ad}_\phi$ or $1 + \hbar\psi'$ maps Λ to the conjugated Maurer–Cartan element and the cohomology isomorphically, intertwining the induced actions. The coupling direction multiplies $\Lambda^{(0)}$ by $1 - a\hbar$ modulo $\hbar^2$ and leaves $\Lambda^{(1)}$ unchanged. (ii) Proposition 8.19 (c) with $G = G_B$, $b = \pm 1$ on the four odd blocks and $b = 0$ on the even part. (iii) Conjugation preserves the spectrum; the coupling direction rescales $\Lambda^{(0)}$ and leaves $\Lambda^{(1)}$ unchanged; for G_B, $G_B^{(0)} = 0$ and $B\Delta = 0$; under the stated hypothesis a closed G is a combination of these and of $s\phi$, which is (a). □

8.6 The supercharge components and the twist

Lemma 8.21 (Evaluation along a square-zero supercharge)**.** *Let $\xi \in \Sigma$ with $[\xi, \xi] = 0$ and let q be a variable of degree 1 and even total parity. The map*

$$r_\xi \colon f \mapsto \sum_{k \geq 0} \frac{(-1)^{k(k-1)/2}}{k!}\, f^{(k)}(\xi, \dots, \xi)\, q^k$$

intertwines d_{CE} with 0, the cochain bracket with the q-linear extension of the bracket of values, and the products of operator-valued and of functional-valued cochains of Theorem 8.15 (iv) with the composition, resp. the product, of their values; hence $r_\xi W(P(\varepsilon, T), J) = W(P(\varepsilon, T), r_\xi J)$. Every $\xi \in S_+ \otimes W$ satisfies $[\xi, \xi] = 0$. For $\xi = u \otimes z$ with $u \in S_+$, $z \in W$ (two-component ${\xi_\alpha}^i = u_\alpha z^i$, $\bar\xi = 0$), $\Gamma(\xi, \xi) = 0$ in all ten components, and

$$S^\xi := r_\xi(S_{\mathfrak{psl}(4|4)})\big|_{q=1} = S + S^{(1)}(\xi) - \tfrac{1}{2} S^{(2)}(\xi, \xi), \qquad S^{(2)}(\xi, \xi) = -\tfrac{1}{2} \int \big(u_\alpha z^i {\lambda^*}^\alpha{}_i,\ u_\beta z^j {\lambda^*}^\beta{}_j\big), \quad (8.8)$$

solves $\{S^\xi, S^\xi\} = 0$, and $c = -\frac{1}{2}$ is the unique value for which $S + S^{(1)}(\xi) + c\, S^{(2)}(\xi, \xi)$ solves it.

Proof. For a (k, l)-shuffle with all arguments equal to the odd ξ, the signs of Definition 2.8 combine to $(-1)^{kl}$, so $\{f, g\}(\xi^{k+l}) = \binom{k+l}{k}(-1)^{kl}\{f(\xi^k), g(\xi^l)\}$, and the coefficients $a_k = (-1)^{k(k-1)/2}/k!$ satisfy $a_k a_l = a_{k+l}\binom{k+l}{k}(-1)^{kl}$; the same shuffle count applies to the product of operator-valued cochains. d_{CE} on $(\xi, \dots, \xi)$ is a sum of terms containing $[\xi, \xi] = 0$. By Lemma 2.11 with $[\xi, \xi] = 0$, $\{S, S^{(2)}(\xi, \xi)\} = \{S^{(1)}(\xi), S^{(1)}(\xi)\}$ and $\{S^{(1)}(\xi), S^{(2)}(\xi, \xi)\} = 0$, so the square of $S + S^{(1)}(\xi) + cS^{(2)}(\xi, \xi)$ is $(2c + 1)\{S^{(1)}(\xi), S^{(1)}(\xi)\}$; this equals $(2c + 1)\{S, S^{(2)}(\xi, \xi)\}$, whose part linear in λ is $-(2c + 1)\int \mathrm{dvol}\,\big((\xi, \lambda^*), (\xi, \not\partial\lambda)\big)$ up to the sign of the pair, with $(\xi, \not\partial\lambda) = i\, u_\alpha z^i \partial_m \bar\lambda_{\dot\alpha i} \bar\sigma^{m\dot\alpha\alpha} \neq 0$. The two-component form of $S^{(2)}(\xi, \xi)$ is (2.8) with $\Gamma(\xi, \xi) = 0$. □

Definition 8.22 (Twisted differentials)**.** For ξ as in Lemma 8.21, the twisted differential on local functionals is

$$Q^\xi := r_\xi(\Lambda)\big|_{q=1} = \Lambda^{(0)} + \Lambda^{(1)}(\xi) - \tfrac{1}{2}\Lambda^{(2)}(\xi, \xi) - \tfrac{1}{6}\Lambda^{(3)}(\xi, \xi, \xi) + \cdots = Q_0^\xi + \hbar\, Q_1^\xi + O(\hbar^2),$$

with $Q_0^\xi = \{S^\xi, -\}$ and $Q_1^\xi = L_1^{(0)} + L_1^{(1)}(\xi) - \frac{1}{2}L_1^{(2)}(\xi, \xi) - \frac{1}{6}L_1^{(3)}(\xi, \xi, \xi) + \cdots$. In the names of Definition 8.16, $\hbar Q_1^\xi(O) = \hbar L_1(O)^{(0)} + Q_1(O) - \frac{1}{2}\hbar L_1(O)^{(2)}(\xi, \xi) - \cdots$.

Theorem 8.23 (The twisted one-loop differential as a consequence)**.** *Assume (8.6), that the local representatives exist on a space of local functionals stable under Λ, and that the sums defining $r_\xi(\Lambda)$ are finite there. Then $(Q^\xi)^2 = 0$; at orders $\hbar^0, \hbar^1$, $(Q_0^\xi)^2 = 0$ and $Q_0^\xi Q_1^\xi + Q_1^\xi Q_0^\xi = 0$, so Q_1^ξ induces a map $[Q_1^\xi]$ on the cohomology of Q_0^ξ. Under (a), (b) of Proposition 8.19, Q_1^ξ changes by a graded commutator with Q_0^ξ; under the coupling direction by $-aQ_0^\xi$ plus such a commutator; under G_B by such a commutator. Hence $[Q_1^\xi]$, its kernel, image and rank are invariant under all of them.*

Proof. By Lemma 8.21, r_ξ maps the Maurer–Cartan equation $d_{CE}\Lambda + \Lambda d_{CE} + \Lambda\Lambda = 0$ to $r_\xi(\Lambda)^2 = 0$. For (a), (b), r_ξ of a conjugation is a conjugation, which changes Q_1^ξ by $[Q_0^\xi, \mathrm{ad}_{r_\xi\phi}]$, resp. $[Q_0^\xi, r_\xi\psi']$. For a closed G, Q_1^ξ changes by $-a\,\mathrm{ad}_{r_\xi(G)}$: $r_\xi(S - S^{(2)})|_{q=1} = S^\xi - S^{(1)}(\xi) + S^{(2)}(\xi,\xi)$ and $r_\xi(G_B)|_{q=1} = S^{(1)}(\xi) - S^{(2)}(\xi,\xi)$; with $R_\xi \in \mathfrak{so}(W_{\mathbb{R}}^\perp)$, $[R_\xi, \xi] = \xi$, the derivation $N := \{S^{(1)}(R_\xi), -\}$ satisfies $NS^\xi = -S^{(1)}(\xi) + S^{(2)}(\xi,\xi)$ by Lemma 2.11, so the two changes are $-aQ_0^\xi$, resp. 0, plus the graded commutator of Q_0^ξ with $\mp aN$. Each of these induces 0 on the cohomology of Q_0^ξ. □

Remark 8.24 (Comparison with the holomorphic twist)**.** The twisted theory appears as a consequence: S^ξ, Q_0^ξ, Q_1^ξ are images of the untwisted equivariant objects under r_ξ, computed with the untwisted Feynman-gauge propagator and the vertices $S^{(1)}(\xi)$, $S^{(2)}(\xi,\xi)$. The one-loop differential of the holomorphic twist [BK] (K. Budzik and J. Kulp, arXiv:2512.07771, (3.64)–(3.70)) is to be compared with the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ, after its transport to the holomorphic letters through the equivalence of [ESW, Thm 10.9] and provided the two quantisations differ by an element of $\mathbb{C}\,S^\xi + \mathrm{Im}\,Q_0^\xi$, which contains $\{S^{(1)}(R_\xi), S^\xi\} = Q_0^\xi S^{(1)}(R_\xi)$ and $r_\xi(G_B)|_{q=1} = -Q_0^\xi S^{(1)}(R_\xi)$; by Theorem 8.23 this class is invariant under the scheme changes listed there, while the individual entries of Q_1 are scheme dependent. Two-component: $S^{(1)}(\xi)$ for ${\xi_\alpha}^i = u_\alpha z^i$ is

$$\int \Big\{ -i\big(\bar\lambda_{\dot\alpha i} z^i \bar\sigma^{m\dot\alpha\alpha} u_\alpha, A_m^*\big) + \tfrac{1}{2}\varepsilon_{jikl} z^j \big((u\lambda^i), A^{*kl}\big) - \tfrac{1}{4}\big(F_{mn}(\sigma^m\bar\sigma^n u)_\alpha, {\lambda^*}^\alpha{}_i z^i\big) - \tfrac{1}{4}\big([A^{ik}, A_{kj} z^j] u_\alpha, {\lambda^*}^\alpha{}_i\big) - \tfrac{i}{2}\big(u^\alpha \sigma^m_{\alpha\dot\alpha} D_m(A$$

index-free $\int \mathrm{dvol}\big(-(\Gamma(\xi,\lambda), A^*) + \frac{1}{2}(\rho(F_A)\xi, \lambda^*)\big)$ (2.6).

8.7 Large distance and the special conformal components

Proposition 8.25 (Where the endpoint may be removed)**.** *A single insertion of $S^{(1)}(x)$ with both legs on edges of a cycle decays at large distance $|y|$ as $|y|^{-6}$ for $x \in \Sigma$ and translations, as $|y|^{-5}$ for $x \in \Sigma^*$, Δ and $\mathfrak{so}(V)$, and as $|y|^{-4}$ for the special conformal fields (Remark 7.12). Consequently the graph sums of the arity-zero and arity-one components along Σ, Σ^*, Δ, the translations, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ converge as $T \to \infty$, and so do the arity-two graphs along (ξ,ξ') and (ξ,η): for two floating odd insertions joined by a λ edge the integrand has degree -1 at large distance in $\mathbb{R}^8$ for (ξ,ξ'), and -1 after the sum over the two orientations for (ξ,η). Since the component of (8.7) along x is built from the components of arity at most one along x, Theorem 8.15 (ii) applies component by component, and Q_1, S_1, D_1, P_1, M_1, R_1 and the twisted Q_1^ξ may be computed at $T = \infty$. Along a special conformal field the graphs with the insertion on a cycle are cut off by $P(\varepsilon, T)$ at $|y|^2 \sim T$ and grow logarithmically in T; the same holds at arity two for two special supercharges η_1, η_2 with $\Gamma(\eta_1,\eta_2)|_V \neq 0$, whose bracket is a special conformal field. Two floating insertions of even generators are adjacent through an $A|_V \otimes A|_V$ edge; the chain has both ends on ghost legs, its degree at large distance in $\mathbb{R}^8$ is $d_1 + d_2 - 2$ for one orientation ($d = 0, 1, 1, 2$ for translations, $\mathfrak{so}(V)$, Δ and the special conformal fields) and one less after the sum over the two orientations. The arity-two graphs along pairs of translations, rotations of V and Δ converge after this sum, and so do those along (η_1,η_2) with $\Gamma(\eta_1,\eta_2)|_V = 0$; those containing a special conformal field are defined at finite T. The coefficient of $\log T$ of a graph with $S^{(1)}(K_b)$ floating on its cycle is $\frac{3}{8\pi^2}\hbar$ times the tree obtained by cutting the insertion out of the cycle and evaluating the freed legs on the constant modes $A|_V = b \otimes X_a$ and $c = X^a$, summed over a basis $\{X_a\}$ of $\mathfrak{g}$ with dual basis $\{X^a\}$ for $(-,-)$. Along Δ, the translations and $\mathfrak{so}(V)$ the insertions on a cycle are composite $A|_V$–c lines as well; their graphs converge at $T = \infty$, single composite graphs have nonlocal Q-images, and the components are read off from the complete sets of composite graphs.*

Proof. On a cycle the terms of $S^{(1)}(x)$ with both legs on components of the propagator are $(v.A, A^*)$ for even x, with $v.A = L_v A + \frac{1}{4}(\mathrm{div}\, v)A|_{W_{\mathbb{R}}^\perp}$ (Definition 7.4), and $-(\Gamma(\epsilon(y),\lambda), A^*)$ for odd x, $\epsilon(y) =$

$\xi + \rho(y)\eta$. The $A^*|_V$ leg meets an $A^*|_V \otimes c$ component and the λ leg a $\lambda \otimes \lambda$ component, both of order $|y|^{-3}$; the A leg meets an $A \otimes A$ component, of order $|y|^{-2}$ (Proposition 3.17). A vector field of degree k enters through the derivative along v, of order $|y|^k$ acting on the $A \otimes A$ component, and through dv and $\operatorname{div} v$, of order $|y|^{k-1}$, so the insertion is of order $|y|^{k-6}$; a Killing spinor of degree g gives $|y|^{g-6}$; at finite T every edge is bounded by a constant multiple of its $T = \infty$ form, uniformly in T, and decays as $e^{-|y|^2/4T}$, so dominated convergence applies in the first cases. For a λ-edge pair the three propagators decay as $|y|^{-3}$ each and the Killing spinors grow as $|y|^{g_1+g_2}$ with $g = 0$ along Σ, $g = 1$ along Σ^*, so the degree in $\mathbb{R}^8$ is $g_1 + g_2 - 1$; for (ξ, η) the two orientations traverse the λ edge in opposite directions and their leading terms cancel. For two floating even insertions the $A \otimes A$ edge between them carries both derivatives and is of order $|y|^{-4}$, the two $A^*|_V \otimes c$ edges are of order $|y|^{-3}$, and the two orientations of the chain enter with opposite signs, so their leading terms cancel. For K_b the product is of order $|y|^{-4}$ with a nonzero average over the sphere; this term comes from the antisymmetric part of dv acting on $A|_V$, while on $A|_{W_{\mathbb{R}}^\perp}$ the leading terms of the derivative along v and of $\frac{1}{4}\operatorname{div} v$ cancel. $\int_{|y|^2<T} |y|^{-4} d^4y$ grows as $\pi^2 \log T$; $T\partial_T$ of the scalar integral of the composite line tends to $-\frac{3}{16\pi^2} b$, and with the fibre factors $2(-,-)^{-1}$ of $A \otimes A$ and $-(-,-)^{-1}$ of $A^*|_V \otimes c$ (Proposition 8.2) it gives the coefficient $\frac{3}{8\pi^2}$ of the cut tree. After the integration over the insertion point the two edges form the kernel of $P(\varepsilon, T)\star\, v.\, P(\varepsilon, T)\star$, with $P(\varepsilon, T)\star = \int_\varepsilon^T Q^{GF} e^{-tH} dt$ the operator represented by $P(\varepsilon, T)$; along Δ it is of order $|p|^{-3}$ at small momentum ($|x-y|^{-1}$ in position space at $T = \infty$), and along translations of order $|p|^{-2}$ (Proposition 9.18 (iv)), so the one-loop integrals with external legs at generic momenta converge. □

Definition 8.26 (The special conformal components)**.** $K_1(O) := \hbar L_1(O)^{(1)}(K_b)$ with the local representative of Definition 8.14 at a finite endpoint T; the arity-two components along (η_1, η_2) and $(K_b, P_{b'})$ are defined in the same way at finite T. The limit $\varepsilon \to 0$ is taken at fixed T; at a fixed ε the graphs of the defect carry a term of order $\varepsilon \log T$ for the scalar bilinears and along (η_1, η_2) with $\Gamma(\eta_1, \eta_2)|_V \neq 0$, so the defect graphs are evaluated at finite T.

Theorem 8.27 (Independence of the endpoint for K_1)**.** *If the local representative $L(O)$ exists at one finite T_0, then $K_1(O)$ is the same local functional at every finite T, and therefore also in the limit $T \to \infty$. The observables keep, in their component along K_b, the term $\frac{3}{8\pi^2}\hbar \log T$ times $\sum_a \partial_{A|_V = b \otimes X_a} \partial_{c = X^a}$ of the order-$\hbar^0$ part of $O[\infty]$, with $\{X_a\}$ a basis of $\mathfrak{g}$ and $\{X^a\}$ the dual basis for $(-,-)$; this tree is nonzero for the scalar bilinears, the term is generated by the flow of Lemma 8.13, its image under Q vanishes for gauge-invariant O, and the local representative is independent of T. The effective interaction $I_{\mathfrak{psl}(4|4)}[T]$ keeps the same term with the tree of $W(P(0,\infty), I)$, which is nonzero in every two-point sector.*

Proof. Theorem 8.15 (ii) applies at every finite T; the local representative is a T-independent local functional, so its limit is trivial. The second statement is Lemma 8.13: $O[T']$ is the linearised flow of $O[T]$, whatever its behaviour as $T' \to \infty$. □

Remark 8.28 (The existence along K)**.** In a graph of the defect along K_b with $S^{(1)}(K_b)$ floating on its cycle, the coefficient of $\log T$ at a fixed ε is $\frac{3}{8\pi^2}\hbar$ times the cut tree of Proposition 8.25; this tree contains the special edge $K_\varepsilon - K_0$ at a generic momentum p and carries the factor $e^{-\varepsilon|p|^2} - 1$. Summed over the graphs, the coefficient is $\frac{3}{8\pi^2}\hbar$ times Q of $\sum_a \partial_{A|_V = b \otimes X_a} \partial_{c = X^a}$ of the order-$\hbar^0$ part of $\partial_\delta W(P(\varepsilon, \infty), I + \delta O)$: it vanishes identically in the two-point sectors of the action and is of order ε, and nonzero, for the scalar bilinears. Along $(\eta_1, \eta_2) \in \Sigma^* \times \Sigma^*$ with $\Gamma(\eta_1, \eta_2)|_V \neq 0$ the coefficient multiplies the first moment of the cut tree and carries the gradient $-2\varepsilon p\, e^{-\varepsilon|p|^2}$ of the special-edge factor. In every case it vanishes as $\varepsilon \to 0$ at fixed T. At a fixed T, the derivative $T\partial_T$ of every graph with the special edge on its cycle is of order ε, because the edge pinned at T confines the loop momentum and the special edge carries $e^{-\varepsilon|q|^2} - 1$; the limit $\varepsilon \to 0$ is therefore the same at every T, and the existence of the local representative at fixed T is the locality of that limit, the same statement as along the other generators. Every insertion of $S^{(1)}(K_b)$ on a cycle is a composite $A|_V$–c

line with kernel $P(\varepsilon,T)\star\, v.\,P(\varepsilon,T)\star = v.\,(P(\varepsilon,T)\star)^2 + [P(\varepsilon,T)\star, v.]\,P(\varepsilon,T)\star$, v the vector field of K_b; the first term exists because $(Q^{GF})^2 \neq 0$, and the existence of $L(O)$ along K_b is a statement about the complete set of composite graphs (Proposition 8.25). Three relations constrain K_1 on the cohomology of $\Lambda^{(0)}$: the component of $L\circ L = 0$ at $(K_b, P_{b'})$, which ties K_1 to D_1, P_1 and M_1 through $[K_b, P_{b'}] = (b,b')\Delta + (\text{rotation})$ (Remark 7.17); the component at (η_1,η_2), which expresses K_1 along $K_{\Gamma(\eta_1,\eta_2)|_V}$ through S_1 and the classical transformations along η_1, η_2; and the relation $K = P^\dagger$ of Corollary 8.41.

8.8 Conjugations

Lemma 8.29 (Conjugation of spinors)**.** *There is a unique antilinear map $\zeta\mapsto\overline{\zeta}$ of $\Sigma\oplus\Sigma^*$ with $\overline{\Sigma}=\Sigma^*$, $\overline{\Sigma^*}=\Sigma$, $\overline{\overline{\zeta}}=\zeta$, $\overline{\rho(v)\zeta}=\rho(\bar v)\overline{\zeta}$ for $v\in V_{10}$, and $(\zeta,\overline{\zeta})>0$ for $\zeta\in\Sigma\setminus 0$. It maps $S_+\otimes W$ to $S_+\otimes W^*$ and $S_-\otimes W^*$ to $S_-\otimes W$, and $\Gamma(\overline{\zeta_3},\overline{\zeta_4})=\overline{\Gamma(\zeta_3,\zeta_4)}$. Two-component:*

$$\overline{\xi} = \big(\eta^\alpha{}_i = \overline{\xi_\alpha{}^i},\ \bar\eta^{\dot\alpha i} = -\overline{\xi_{\dot\alpha i}}\big), \qquad (\xi,\overline{\xi}) = \sum|\xi_\alpha{}^i|^2 + \sum|\bar\xi_{\dot\alpha i}|^2. \tag{8.9}$$

Proof. In the realisation of Chapter 2 the matrices of $\rho(e_M)$ on Σ^* are the complex conjugates of those on Σ, so componentwise conjugation has the properties. Two such maps differ by a linear automorphism of $\Sigma\oplus\Sigma^*$ commuting with every $\rho(v)$, hence by one scalar a on the irreducible Clifford module $\Sigma\oplus\Sigma^*$ (since $\rho(v)$ exchanges Σ and Σ^*, the scalars on the two summands agree); involutivity gives $|a|=1$ and positivity $a>0$, so $a=1$. The two-component form follows from (8.1); the minus sign is that of the pairing. □

Definition 8.30 (Radial conjugation)**.** With the inversion $I(x)=x/(x,x)$ and its pushforward I_*, for an even element v (a conformal Killing field plus an element of $\mathfrak{so}(W_{\mathbb{R}}^\perp)$) and an odd element with Killing spinor $\epsilon(x)=\xi+\rho(x)\eta$,

$$v^\dagger := -I_*\bar v, \qquad \epsilon^\dagger(x) := -\rho(x)\,\overline{\epsilon(I(x))}, \quad \text{i.e.} \quad (\xi+\rho(x)\eta)^\dagger = -\overline{\eta} - \rho(x)\overline{\xi}.$$

Theorem 8.31 (The radial conjugation)**.**

- *(i) On the generators: $P_b^\dagger = -K_{\bar b}$, $K_b^\dagger = -P_{\bar b}$, $\Delta^\dagger = \Delta$, $\omega^\dagger = -\bar\omega$ for rotations of V and of $W_{\mathbb{R}}^\perp$, $\xi^\dagger = -\overline{\xi}\in\Sigma^*$, $\eta^\dagger = -\overline{\eta}\in\Sigma$. Two-component:*

  $$(\xi^\dagger)^\alpha{}_i = -\overline{\xi_\alpha{}^i}, \quad (\xi^\dagger)^{\dot\alpha i} = \overline{\bar\xi_{\dot\alpha i}}, \qquad (\eta^\dagger)_\alpha{}^i = -\overline{\eta^\alpha{}_i}, \quad (\eta^\dagger)_{\dot\alpha i} = \overline{\bar\eta^{\dot\alpha i}},$$

  so for the unit basis elements $(Q^i_\alpha)^\dagger = -S^\alpha{}_i$, $(\bar Q_{\dot\alpha i})^\dagger = \bar S^{\dot\alpha i}$, $(S^\alpha{}_i)^\dagger = -Q^i_\alpha$, $(\bar S^{\dot\alpha i})^\dagger = \bar Q_{\dot\alpha i}$; on $\mathfrak{su}(4)$ the conjugation is the matrix adjoint.

- *(ii) $(X^\dagger)^\dagger = X$ and $[X,Y]^\dagger = [Y^\dagger, X^\dagger]$ for all X, Y. The antilinear automorphism $X\mapsto -X^\dagger$ (even X), $X\mapsto -iX^\dagger$ (odd X) has the fixed real form $\mathfrak{psu}(2,2|4)$ [SW, (7)], with maximal compact even subalgebra $\mathfrak{so}(4)\oplus\mathbb{R}\,i\Delta\oplus\mathfrak{so}(6)$.*

- *(iii) The antilinear involutive anti-automorphisms with the even part of Definition 8.30 are exactly $X\mapsto e^{tB}\big((\pm1)^{|X|}X^\dagger\big)$, $t\in\mathbb{R}$, with B of Definition 8.9. Positivity of the norm, in a representation π with $\pi(X^\natural) = \pi(X)^{\mathrm{adj}}$ for the conjugation $X^\natural$ in question, in which $-\frac12\Delta$ acts as a self-adjoint operator with a positive eigenvalue on a vector annihilated by Σ^* and invariant under $\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^\perp)$, selects the sign +; the realisation on the fields (Theorem 8.33) selects $t=0$.*

Proof. (i) $(I_*u)(x) = (x,x)u(I(x)) - 2(u(I(x)),x)x$, which maps the constant field b to $(x,x)b - 2(b,x)x$, fixes rotations and maps $2E$ to $-2E$. The odd statements are the definition at $\eta=0$ and at $\xi=0$ with Lemma 8.29. (ii) Write $T_I\epsilon(x) = \rho(x)\overline{\epsilon(I(x))}$; $T_I^2 = 1$ by $\rho(x)\rho(I(x)) = 1$. I_* and complex conjugation are automorphisms of vector fields; T_I intertwines the Lie derivative $\mathcal{L}_v$ of (2.1) with $\mathcal{L}_{I_*\bar v}$,

by $\rho(x)\rho(b)\rho(x) = -\rho(K_b(x))$ and the transformation of $\frac{1}{4}\rho(d(v,-)) - \frac{1}{8}\operatorname{div} v$; for the odd–odd bracket, $\Gamma(\rho(x)\zeta_1, \rho(x)\zeta_2) = -(x,x)(r_{x/|x|} \oplus 1)\Gamma(\zeta_1,\zeta_2)$ together with $(I_* u)(x) = (x,x) r_{x/|x|} u(I(x))$, and the rotation of $W^\perp_{\mathbb{R}}$ goes to minus its conjugate. The product of the two signs -1 of the odd conjugates enters the odd–odd bracket, so the identity holds for a factor c' with $c'^2 = 1$ and fails for $c' = i$. The real form: the fixed points of the even part are the real rotations, $i\Delta$, $P_b + K_b$, $i(P_b - K_b)$ for real b and the compact $\mathfrak{so}(W^\perp_{\mathbb{R}})$; $P_b - K_b$ generates a periodic flow and $P_b + K_b$ a hyperbolic one. (iii) The composite of two such maps is the identity on the even part, hence a scalar on each of the two inequivalent irreducible even-modules of the odd part, with product 1, real by involutivity. For $\xi \in S_+ \otimes W$ and a unit vector h annihilated by Σ^*, invariant under the rotations, with Δ-eigenvalue $-2d$, $d > 0$, $\|\pi(\xi)h\|^2 = \langle h, \pi([\xi^\natural, \xi])h\rangle$ is $\pm e^{-t}(\xi,\bar\xi)\, d\, \|h\|^2$, by the dilatation part $\frac{1}{2}(\xi,\eta)\Delta$ of $[\xi,\eta]$ (Proposition 2.4). All identities are verified by exact evaluation on all pairs of basis elements in the realisation of Chapter 2. □

Definition 8.32 (Reflections)**.** For a unit vector $n \in V_{\mathbb{R}}$ with reflection $r_n(x) = x - 2(n,x)n$, $R_n = r_n \oplus 1$ on V_{10}, the reflection conjugation of $\mathfrak{psl}(4|4)$ is $v^{\dagger_n} := -(r_n)_* \bar v$, $(\xi + \rho(x)\eta)^{\dagger_n} = \rho(n)\bar\xi - \rho(x)\rho(n)\bar\eta$, and the reflection of the fields is

$$(\Theta_n A)(x) = R_n \overline{A(r_n x)}, \quad (\Theta_n c)(x) = \overline{c(r_n x)}, \quad (\Theta_n \lambda)(x) = \rho(n)\overline{\lambda(r_n x)},$$
$$(\Theta_n A^*)(x) = -R_n \overline{A^*(r_n x)}, \quad (\Theta_n \lambda^*)(x) = -\rho(n)\overline{\lambda^*(r_n x)}, \quad (\Theta_n c^*)(x) = -\overline{c^*(r_n x)},$$

with the conjugation of Lemma 8.29 on spinors and that of the compact real form on $\mathfrak{g}$. A functional is conjugated by $F^\Theta(e) := F(\Theta e)^{\mathrm{c}}$, where c conjugates coefficients and reverses the order of odd factors. Two-component:

$$(\Theta_n A)_m = (r_n)_{mk}\overline{A_k(r_n x)}, \quad (\Theta_n A)_{ij} = \overline{A^{ij}(r_n x)}, \quad (\Theta_n \lambda)_\alpha{}^i = i\, n_m \sigma^m_{\alpha\dot\alpha}\overline{\lambda_{\dot\alpha i}(r_n x)}, \quad (\Theta_n \bar\lambda)_{\dot\alpha i} = i\, n_m \overline{\lambda_\alpha{}^i(r_n x)}\sigma^m_{\alpha\dot\alpha},$$
$$(\Theta_n \lambda^*)^\alpha{}_i = i\, n_m \overline{\bar\lambda^{*\dot\alpha i}(r_n x)}\bar\sigma^{m\dot\alpha\alpha}, \quad (\Theta_n \bar\lambda^*)^{\dot\alpha i} = i\, n_m \bar\sigma^{m\dot\alpha\alpha}\overline{\lambda^{*\alpha}{}_i(r_n x)},$$

and $(\xi^{\dagger_n})_\alpha{}^i = i\, n_m \sigma^m_{\alpha\dot\alpha}\overline{\bar\xi_{\dot\alpha i}}$, $(\xi^{\dagger_n})_{\dot\alpha i} = i\, n_m \overline{\xi_\alpha{}^i}\sigma^m_{\alpha\dot\alpha}$: the reflection exchanges Q^i_α with combinations of $\bar Q_{\dot\alpha i}$.

Theorem 8.33 (The reflection is a symmetry of the regularised theory)**.**

- *(i)* $S^{\Theta_n} = S$, $S^{(1)}(X)^{\Theta_n} = (-1)^{|X|} S^{(1)}(X^{\dagger_n})$, $S^{(2)}(X,Y)^{\Theta_n} = S^{(2)}(Y^{\dagger_n}, X^{\dagger_n})$ *for all* $X, Y \in \mathfrak{psl}(4|4)$*, pointwise in the densities.*

- *(ii)* Θ_n *is compatible with* K_t, $P(\varepsilon,T)$, Q, Q^{GF}, Δ_t, $\{-,-\}_t$ *and the flow:* $(\partial_{P(\varepsilon,T)} F)^{\Theta_n} = \partial_{P(\varepsilon,T)}(F^{\Theta_n})$, $(\Delta_t F)^{\Theta_n} = (-1)^{|F|}\Delta_t(F^{\Theta_n})$, $\{f,g\}_t^{\Theta_n} = (-1)^{|f|+|g|}\{g^{\Theta_n}, f^{\Theta_n}\}_t$, $(FG)^{\Theta_n} = G^{\Theta_n}F^{\Theta_n}$*, and* $W(P(\varepsilon,T),J)^{\Theta_n} = W(P(\varepsilon,T), J^{\Theta_n})$ *for even* J*.*

- *(iii) At every order in* $\hbar$ *the counterterms are reflection invariant,* $I^{CT}[\varepsilon]^{\Theta_n} = I^{CT}[\varepsilon]$*, and so are the defect vertex and the obstruction of a reflection-invariant interaction; if* F *solves the equation of Theorem 8.8 (ii), so does its conjugate, and the average is a reflection-compatible solution. In the quadratic sector the finite counterterm* $\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ *of Theorem 4.32 is reflection invariant.*

Proof. (i) The phases of the antifields are forced by the terms (dc, A^*), $\frac{1}{2}([c,c],c^*)$, $([c,\lambda],\lambda^*)$ with the order reversal of odd factors; the bosonic terms are invariant because R_n and the conjugation of $\mathfrak{g}$ preserve $(-,-)$ and $[-,-]$, and the Dirac term because $\not d(\rho(n)\overline{\ell(r_n\cdot)}) = -\rho(n)\overline{(\not d\ell)(r_n\cdot)}$. For $S^{(1)}$, the field transformations of (2.5) are the infinitesimal conformal action on weighted fields, intertwined by Θ_n with that of $(r_n)_*\bar v$; for odd X and for $S^{(2)}$ the signs are fixed by evaluation on all odd basis directions. (ii) In momentum space the identities are matrix identities for Q, Q^{GF}, H and the kernels, together with the anti-multiplicativity of $F \mapsto F^\Theta$; the bracket rule follows from the definition of $\{-,-\}_t$ through Δ_t (Definition 3.21). (iii) By (ii) the order-$\hbar^k$ graphs of a reflection-invariant interaction are mapped

to themselves; the singular part is unique (Proposition 4.10 at order $\hbar$, [C07, Thm B] beyond) and the functions $\varepsilon^{-m}\log^k\varepsilon$ are real; induction on k. The listed functional is invariant pointwise. □

Proposition 8.34 (From the hyperplane to the sphere)**.** *Let C be the time $-\frac{\pi}{4}$ flow of $P_n - K_n$, $x \mapsto n(1-(x,x)) + 2(n,x)x$,*

$$C(x) = \frac{2x + \big((x,x) - 1 - 2(n,x)\big)n}{|x+n|^2}.$$

C maps the half space $\{(n,x) > 0\}$ onto the open unit ball, the hyperplane onto the unit sphere, and $C \circ r_n \circ C^{-1} = I$. With $\mathrm{Ad}(C) = \exp\big(\frac{\pi}{4}\,\mathrm{ad}_{P_n - K_n}\big)$,

$$X^\dagger = \mathrm{Ad}(C)\Big(\big(\mathrm{Ad}(C)^{-1}X\big)^{\dagger_n}\Big), \qquad \Theta_I = (C^{-1})^* \circ \Theta_n \circ C^*,$$

where Θ_I is the inversion of the fields with the Weyl weights $1, 0, \frac{3}{2}, 3, \frac{5}{2}, 4$ of $A, c, \lambda, A^*, \lambda^*, c^*$, the spinor factors $-\rho(x)/|x|$ on λ and $\rho(x)/|x|$ on λ^*, and the factors -1 on A^* and c^*, and g^* is the pullback with the Weyl weights; Θ_I satisfies (i) of Theorem 8.33 with $\dagger$ in place of $\dagger_n$, the scalar kinetic density changing by a total derivative; (ii) and (iii) there hold for Θ_n and fail for Θ_I, since K_t, $P(\varepsilon,T)$ and the counterterm $\frac{9}{4}\varepsilon^{-1}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int(A,A)$ change under the inversion, and the one-loop radial conjugation is therefore built from Θ_n (Definition 8.35). Two-component: $(\Theta_I A)_m(x) = (x,x)^{-1}\big(\delta_{mk} - 2x_m x_k/(x,x)\big)\overline{A_k(I(x))}$, $(\Theta_I A)_{ij}(x) = (x,x)^{-1}\overline{A^{ij}(I(x))}$, $(\Theta_I\lambda)_\alpha{}^i(x) = -i(x,x)^{-2}x_m\sigma^m_{\alpha\dot\alpha}\overline{\lambda_{\dot\alpha i}(I(x))}$, $(\Theta_I\bar\lambda)_{\dot\alpha i}(x) = -i(x,x)^{-2}x_m\overline{\lambda_\alpha{}^i(I(x))}\sigma^m_{\alpha\dot\alpha}$, $\underline{(\Theta_I\lambda^*)^\alpha{}_i}(x) = -i(x,x)^{-3}x_m\overline{\bar\lambda^{*\dot\alpha i}(I(x))}\bar\sigma^{m\dot\alpha\alpha}$, $(\Theta_I\bar\lambda^*)^{\dot\alpha i}(x) = -i(x,x)^{-3}x_m\bar\sigma^{m\dot\alpha\alpha}\overline{\lambda^{*\alpha}{}_i(I(x))}$, $(\Theta_I c)(x) = \overline{c(I(x))}$, and $\Theta_I A^*$, $\Theta_I c^*$ as $\Theta_I A$, $\Theta_I c$ with the weights 3, 4 in place of 1, 0 and the factor -1.

Proof. The flow of $P_n - K_n$ is the rotation of S^4 by the angle $2t$ in the plane of n and the axis of stereographic projection; at $t = -\frac{\pi}{4}$ it maps the great sphere orthogonal to n to the equator, so $C \circ r_n \circ C^{-1}$ is the reflection in the equator, the inversion. The identity for $\dagger$ holds on the even part since C is real and $(C \circ r_n \circ C^{-1})_* = I_*$; by Theorem 8.31 (iii) the two sides differ by $e^{tB}(\pm1)^{|X|}$, and evaluation of $\mathrm{Ad}(C)$ as the exponential of $\frac{\pi}{4}\mathrm{ad}_{P_n-K_n}$ in the realisation gives the factor $+1$. The field identity is the chain rule for the conformal factors and the spinor lifts. □

Definition 8.35 (Radial conjugation of renormalised observables)**.** Assume the local representatives along P_n and K_n exist, and put $\delta^\hbar := \Lambda^{(1)}(P_n) - \Lambda^{(1)}(K_n)$. For a renormalised local observable O supported in the open unit ball,

$$O^{\Theta_I,\hbar} := \exp\big(-\tfrac{\pi}{4}\,\delta^\hbar\big)\Big[\big(\exp(\tfrac{\pi}{4}\,\delta^\hbar)\,O\big)^{\Theta_n}\Big],$$

modulo $\Lambda^{(0)}$-exact terms and $O(\hbar^2)$. At order $\hbar^0$ this is O^{Θ_I} (Proposition 8.34); at order $\hbar$ it depends on P_1 and K_1 through $\Lambda^{(1)}(P_n)$ and $\Lambda^{(1)}(K_n)$. The condition $P_1 = 0$ is a renormalisation condition. Proposition 7.6 gives the covariance of the regulator; the translations also enter cycles through the composite kernel $2\big(\partial_b(P(\varepsilon,\infty)\star)^2 \otimes 1\big)K_0$, present because $(Q^{GF})^2 \neq 0$, and on the part of the scalar bilinears with two legs $A|_{W^\perp_{\mathbb{R}}}$ these graphs give $P_1(O) = -(\frac{1}{3} + \theta_P)$ units times $\{S^{(1)}(P_b), O\}$, with θ_P the coefficient in the finite counterterm along P_b of Theorem 9.35. On a family stable under Λ on which $P_1 = a_P\{S^{(1)}(P_b), -\}$ for a number a_P, the finite part $\hbar\psi' = \frac{a_P}{2}\{S^{(1)}(\Delta), -\}$ of the operators of the family (Definition 8.11) gives $P_1 = 0$ there, leaves D_1, M_1, R_1 unchanged and changes the components along K_b, Σ, Σ^* by a_P, $-\frac{a_P}{2}$, $\frac{a_P}{2}$ times the classical transformation (Proposition 9.36); it is compatible with the reflection of Theorem 8.33 exactly when a_P is real, since $\{S^{(1)}(\Delta), -\}$ acts on a local operator of Weyl weight w as $2(E+w)$ (Lemma 9.16 (i)), which commutes with Θ_n. A field redefinition generated by a multiple of $S^{(1)}(\Delta)$ (Proposition 8.19 (a)) changes these components by the same multiples of the classical transformation on every operator, so the condition $P_1 = 0$ is imposed family by family through the finite parts of the operators.

Proposition 8.36 (Radial conjugation of a scalar primary)**.** *Assume moreover $P_1(O) = M_1(O) = R_1(O) = 0$, the first in a scheme with $P_1 = 0$ on the operators considered (Definition 8.35). Let O_x be the density at x of a scalar primary $\int \omega f$ of Weyl weight w_O with $D_1(O) = 2\kappa O$ and $K_1(O) = \frac{1}{4}(\operatorname{div} K_b)\,\kappa O = -2(b,x)\,\kappa O$ for a number κ of order one unit. Then $(O_x)^{\Theta_I,\hbar} = (x,x)^{-(w_O+\kappa)}\,\overline{f}(I(x))$ modulo $\Lambda^{(0)}$-exact terms and $O(\hbar^2)$, with $\overline{f}$ the density with conjugated components.*

Proof. $-\Lambda^{(1)}(v)$ acts on O_x as the classical action of the conformal Killing field v on a density of weight $w_O + \kappa$, since $v \mapsto \frac{1}{4}\operatorname{div} v$ is a 1-cocycle of the conformal Killing fields with values in functions; the finite action of $C^{\pm 1}$ is the pullback with the conformal factor to the power $w_O + \kappa$, Θ_n has conformal factor 1, and the conformal factors of C^{-1}, r_n, C multiply to that of I, $(x,x)^{-1}$. □

8.9 Two-point functions and inner products

Definition 8.37 (Two-point function)**.** For gauge-invariant local operators O_1, O_2 of degree 0 with $(Q + \{I, -\})O_a = 0$, background forms ω_1, ω_2 of disjoint supports, and square-zero parameters t_1, t_2 of degree 0 and of the parities of O_1, O_2,

$$\big\langle O_1, O_2 \big\rangle := \lim_{\varepsilon\to 0} \partial_{t_1}\partial_{t_2} W\big(P(\varepsilon,\infty),\, I - F^{(0)}_{\mathfrak{psl}(4|4)} - I^{CT}[\varepsilon] + t_1 O_1(\varepsilon) + t_2 O_2(\varepsilon)\big)\Big|_{e=0},$$

the coefficient of $t_2 t_1$ at the zero field, i.e. $(-1)^{|O_1||O_2|}$ times the coefficient of $t_1 t_2$ ($|O_a|$ the total parity), which at tree level is the free expectation of $O_1 O_2$ with O_1 on the left, the form in which the Ward identity of Theorem 8.40 holds, with $F^{(0)}_{\mathfrak{psl}(4|4)}$ the arity-zero part of the finite counterterm (at one loop the arity-zero part of a solution $\hbar F_1$ of Theorem 8.8 (ii): in the quadratic sector $\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ plus Q-closed quadratic terms, tied to the cubic terms by the cubic part of the obstruction and fixed, modulo the arity-zero parts of s-closed cochains, by the arity-one components of $sF_1 = \mathrm{Ob}$; through relative order one unit the two-point function of two operators of equal length depends on its quadratic part only) and $O_a(\varepsilon)$ of Definition 8.11; with the global ghosts the same formula with $I_{\mathfrak{psl}(4|4)}$, $F_{\mathfrak{psl}(4|4)}$, $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ and $P(\varepsilon, T)$ in place of $P(\varepsilon,\infty)$ defines the two-point cochain at scale T; its components of arity at most one along Σ, Σ^*, Δ, the translations and the rotations, and of arity two along (ξ,ξ') and (ξ,η), have limits as $T\to\infty$ (Proposition 8.25), while those along K_b and along (η_1,η_2) with $\Gamma(\eta_1,\eta_2)|_V \neq 0$ contain $\log T$ and are used at finite T only. Its tree part is the part of lowest order in the unit, its one-loop part the part of relative order one unit, including the entries between operators of different lengths.

Proposition 8.38 (Properties of the two-point function)**.**

- *(i) Constant terms of $O_a(\varepsilon)$ and Q-exact terms of the flow contribute zero. Assume* (8.6) *modulo $\hbar^2$, and local representatives with $L(O_2)^{(0)} = 0$ modulo $\hbar^2$. Then $\langle L(A)^{(0)}, O_2\rangle = 0$ through relative order one unit for every local A of degree -1 with a local representative, where $L(A)^{(0)} = (Q + \{I,-\})A + \hbar L_1(A)^{(0)}$; the two-point function descends to the cohomology of $\Lambda^{(0)}$.*

- *(ii) Through relative order one unit the limit exists at separated supports, and the vertex integrations converge at large distance; the endpoint $T = \infty$ is safe.*

- *(iii) Scaling identity: assume the finite counterterm has Weyl weight 4 and ψ preserves the Weyl weight, and let D_1 denote the dilatation coefficients of Theorem 6.11, obtained from the scaling covariance of the regulator (on the part of the scalar bilinears with two legs $A|_{W^\perp_{\mathbb{R}}}$, the component of Definition 8.16 along Δ when the one-loop quantisation and the local representatives exist through arity two along (Δ, P_b), Corollary 9.34; the scaling identity is an arity-zero statement and*

holds with the coefficients of Theorem 6.11). For operators of Weyl weights w_1, w_2, with kernel $\langle O_1, O_2\rangle(x,y)$,

$$\big(x\cdot\partial_x + y\cdot\partial_y + w_1 + w_2\big)\langle O_1,O_2\rangle(x,y) = -\tfrac{1}{2}\Big(\langle D_1(O_1), O_2\rangle_{\text{tree}} + \langle O_1, D_1(O_2)\rangle_{\text{tree}}\Big)(x,y),$$

so for Lorentz-scalar operators O_a of common weight w with tree matrix $C|x|^{-2w}$, $C_{ab}|x|^{-2w} = \langle O_a, O_b\rangle_{\text{tree}}$, and $D_1(O_b) = \sum_a (D_1)_{ab}O_a$, the two-point matrix is $|x|^{-2w}\big[C + (\text{one-loop finite part}) - \frac{1}{4}(D_1^T C + C D_1)\log(|x|^2/\ell^2)\big]$, with ℓ the unit of length in which ε is measured.

Proof. (i) $W(P, J + c'\,t_1) = W(P,J) + c'\,t_1$ for a constant c'; Q is induced by a linear operator on fields, so $(QF)|_{e=0} = 0$. The last statement is the arity-zero, $t_1 t_2$, $e = 0$ part of the flow of the defect vertex of $J + t_1 A(\varepsilon) + t_2 O_2(\varepsilon)$: the left side vanishes by the first statement, and the right side is $\langle L(A)^{(0)}, O_2\rangle \pm \langle A, L(O_2)^{(0)}\rangle$. (ii) A collapsing set of vertices containing both operators is excluded by the separation; the one-loop subgraphs without operators are subtracted by $I^{CT}[\varepsilon]$ (Proposition 4.8) and those through O_a by $\partial_\delta I^{CT}[\varepsilon](O_a)$ (Proposition 5.3); at large distance each internal vertex of weight 4 with its edges decays faster than the measure grows, because the weights of A, λ and the antifields are positive. (iii) The scaling covariance $\sigma_\lambda P(\varepsilon,\infty) = P(\lambda^2\varepsilon,\infty)$ (Proposition 6.3) moves the $\log\varepsilon$ counterterms into the dependence on the separation; the counterterm of the action is s-exact (Theorem 4.22) and acts on the two operators as the field counting, which combines with the operator counterterms into D_1 as in Theorem 6.11. With $\langle O_a, O_b\rangle = r^{-2w}(C_{ab} + E_{ab}\log r^2)$, $(r\partial_r + 2w)$ gives $2E_{ab}r^{-2w}$, while $\langle D_1(O_a), O_b\rangle_{\text{tree}} = (D_1^T C)_{ab} r^{-2w}$ and $\langle O_a, D_1(O_b)\rangle_{\text{tree}} = (CD_1)_{ab} r^{-2w}$. □

Definition 8.39 (Inner products)**.** For renormalised observables a, b supported in the half space $\{(n,x) > 0\}$, the reflection inner product is $\langle a,b\rangle_n := \langle a^{\Theta_n}, b\rangle$; for a, b supported in the open unit ball minus the origin, the radial inner product is $\langle a,b\rangle_{\text{rad}} := \langle a^{\Theta_I,\hbar}, b\rangle$ with Definition 8.35. On operators at the origin, the radial inner product is the limit as the supports shrink to the origin, which exists on a family of one classical Weyl weight exactly when the one-loop dilatation matrix of the family is self-adjoint for the tree part; in a scheme with $P_1 = 0$ on the family it equals the two-point coefficient of Proposition 8.38 (iii) with the first family conjugated, at the separation ℓ, and in a scheme with $P_1 = a_P\{S^{(1)}(P_b), -\}$ on a family of Weyl weight w, a_P real, it is $1 - 2a_P w$ times that coefficient; written in a basis O_a it is a Gram matrix, with tree part $\text{Gram}_{\text{tree}}$ and one-loop part $\hbar\,\text{Gram}_1$, so that $\langle a,b\rangle_{\text{rad}} = \overline{a}^T(\text{Gram}_{\text{tree}} + \hbar\,\text{Gram}_1)\,b$ on coefficient columns. Matrices of operators act on coefficient columns, $YO_b = \sum_a Y_{ab}O_a$, and $Y^{\text{adj}} := \text{Gram}_{\text{tree}}^{-1}\overline{Y}^T\text{Gram}_{\text{tree}}$ is the adjoint for the tree inner product. In a reflection-compatible scheme $\langle b,a\rangle_n = \overline{\langle a,b\rangle_n}$ modulo $O(\hbar^2)$, since $W(P(\varepsilon,T),J)^{\Theta_n} = W(P(\varepsilon,T),J^{\Theta_n})$, the value of F^{Θ_n} at the zero field is the conjugate of that of F, $(FG)^{\Theta_n} = G^{\Theta_n}F^{\Theta_n}$ and $\Theta_n^2 = 1$ (Theorem 8.33); with the Ward identity along $P_n - K_n$, which moves $\exp(-\frac{\pi}{4}\delta^\hbar)$ to the second argument, the same holds for $\langle -,-\rangle_{\text{rad}}$, so $\text{Gram}_{\text{tree}}$ and Gram_1 are Hermitian.

Theorem 8.40 (Adjointness)**.** *Assume: the scheme is reflection compatible (the equivariant finite counterterms and the finite parts of the operators are fixed by the conjugation of Theorem 8.33); the local representatives exist on the operators considered; and the two-point function of Definition 8.37 satisfies, for every $Y \in \mathfrak{psl}(4|4)$, the Ward identity $\langle \Lambda^{(1)}(Y)F, G\rangle = -(-1)^{|Y||F|}\langle F, \Lambda^{(1)}(Y)G\rangle$ on the observables considered, where $|F|$, $|Y|$ are total parities. Then for every $X \in \mathfrak{psl}(4|4)$*

$$\big\langle -\Lambda^{(1)}(X)a,\, b\big\rangle_n = \big\langle a,\, -\Lambda^{(1)}(X^{\dagger_n})b\big\rangle_n, \qquad \big\langle -\Lambda^{(1)}(X)a,\, b\big\rangle_{\text{rad}} = \big\langle a,\, -\Lambda^{(1)}(X^{\dagger})b\big\rangle_{\text{rad}},$$

modulo $O(\hbar^2)$. At order $\hbar^0$ both identities hold for the free theory.

Proof. By Theorem 8.33 and the anti-multiplicativity of the conjugation, $\big(\{S^{(1)}(X), a\}\big)^{\Theta_n} = (-1)^{1+|a||X|}\{S^{(1)}(X^{\dagger_n}), a$ reflection compatibility extends this to $\Lambda^{(1)}$ at order $\hbar$. The Ward identity applied with $F = a^{\Theta_n}$ and $Y = X^{\dagger_n}$ gives the first identity. The second follows from the first by Proposition 8.34 and Definition 8.35: conjugation by $\exp(\frac{\pi}{4}\delta^\hbar)$ carries $\Lambda^{(1)}(X)$ to $\Lambda^{(1)}(\text{Ad}(C)^{-1}X)$ modulo $\Lambda^{(0)}$-exact terms by Corollary 8.17. □

Corollary 8.41 (The test $Q^\dagger = S$ in Gram form)**.** *Under the hypotheses of Theorem 8.40, on a family of operators at the origin of one length, closed under the generators considered, let $X_0 + \hbar X_1$ be the matrix of $-\Lambda^{(1)}(X)$ and $Y_0 + \hbar Y_1$ that of $-\Lambda^{(1)}(X^\dagger)$. Then*

$$Y_0 = X_0^{\mathrm{adj}}, \qquad Y_1 = X_1^{\mathrm{adj}} + \big[Y_0,\, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big].$$

In terms of the one-loop components (their matrices on the family, $\hbar X_1$ being minus the matrix of the component along X), for the unit generators of Definition 8.1:

$$\begin{aligned}
&S_1 \text{ along } S^\alpha{}_i = -\big(Q_1 \text{ along } Q^i_\alpha\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(S^\alpha{}_i), -\},\, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],\\
&S_1 \text{ along } \bar S^{\dot\alpha i} = \big(Q_1 \text{ along } \bar Q_{\dot\alpha i}\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(\bar S^{\dot\alpha i}), -\},\, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],\\
&K_1 \text{ along } K_{\bar b} = -\big(P_1 \text{ along } P_b\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(K_{\bar b}), -\},\, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],\\
&D_1 = D_1^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(\Delta), -\},\, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big].
\end{aligned}$$

In a scheme with $P_1 = 0$ on the family (Definition 8.35) the third relation is the statement $K = P^\dagger$: K_1 along $K_{\bar b}$ equals $\hbar\big[\{S^{(1)}(K_{\bar b}), -\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big]$. For $P_1 = a_P\{S^{(1)}(P_b), -\}$ on the family, a_P real, it reads K_1 along $K_{\bar b} = a_P\{S^{(1)}(K_{\bar b}), -\} + \hbar\big[\{S^{(1)}(K_{\bar b}), -\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big]$, since $(\{S^{(1)}(P_b), -\})^{\mathrm{adj}} = -\{S^{(1)}(K_{\bar b}), -\}$, with the radial Gram matrix of Definition 8.39 in that scheme, $1 - 2a_P w$ times the two-point coefficient on Weyl weight w; with the ratio of the two-point coefficients in place of $\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1$ the first term is $-a_P\{S^{(1)}(K_{\bar b}), -\}$, since $\{S^{(1)}(K_b), -\}$ lowers the Weyl weight by one. Under a finite part $a \mapsto (1 + \hbar\psi')a$ of the operators, with ψ' reflection compatible, $\mathrm{Gram} \mapsto \overline{(1+\hbar\psi')}^{\,T}\mathrm{Gram}\,(1 + \hbar\psi')$ and $X \mapsto (1 + \hbar\psi')^{-1}X(1 + \hbar\psi')$ for the matrix of every generator, so $X_1 \mapsto X_1 + [X_0, \psi']$ and $\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1 \mapsto \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1 + \psi' + \psi'^{\mathrm{adj}}$, and the relation is preserved. The shift $F_{\mathfrak{psl}(4|4)} \mapsto F_{\mathfrak{psl}(4|4)} + \hbar a G_B$ changes X_1 by $-aX_0$ and Y_1 by $+aY_0$ for $X = Q^i_\alpha$ (Proposition 8.19); it preserves the relation exactly when $\bar a = -a$, which is also the condition for aG_B to be reflection compatible, B anticommuting with $\dagger_n$: reflection compatibility (Theorem 8.33) fixes the real part of the outer direction, since the arity-one finite counterterm along $S_- \otimes W^*$ must be the Θ_n-image of the one along $S_+ \otimes W$.

Proof. Write the radial inner product as $\langle a, b\rangle_{\mathrm{rad}} = \overline{a}^{\,T}(\mathrm{Gram}_{\mathrm{tree}} + \hbar\,\mathrm{Gram}_1)b$ and the adjointness of Theorem 8.40 as $\overline{X}^{\,T}\mathrm{Gram} = \mathrm{Gram}\,Y$. Order $\hbar^0$ gives $Y_0 = X_0^{\mathrm{adj}}$; order $\hbar$ gives $\overline{X_1}^{\,T}\mathrm{Gram}_{\mathrm{tree}} + \overline{X_0}^{\,T}\mathrm{Gram}_1 = \mathrm{Gram}_{\mathrm{tree}}Y_1 + \mathrm{Gram}_1 Y_0$, i.e. $Y_1 = X_1^{\mathrm{adj}} + Y_0\,\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1 - \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\,Y_0$. The component form uses Theorem 8.31 (i): for $X = Q^i_\alpha$, $X^\dagger = -S^\alpha{}_i$, so $\hbar X_1 = -Q_1$ and $\hbar Y_1 = +S_1$ along the unit generators, and Y_0 is the matrix of $\{S^{(1)}(S^\alpha{}_i), -\}$; the other lines are the same with $(\bar Q_{\dot\alpha i})^\dagger = \bar S^{\dot\alpha i}$, $P_b^\dagger = -K_{\bar b}$, $\Delta^\dagger = \Delta$. The covariance is the computation with $(1 + \hbar\psi')$ inserted, using $[A, B]^{\mathrm{adj}} = [B^{\mathrm{adj}}, A^{\mathrm{adj}}]$. For the outer direction, the change $-aX_0$ requires the change $-\bar a Y_0$ on the right side, while Proposition 8.19 gives $+aY_0$; hence $\bar a = -a$. On a family containing lengths L and $L+1$ the tree norms of the two lengths differ by one power of the unit; in a basis normalised by the tree norms the relation is expanded in the square root of the unit: $Y_{\frac12} = X_{\frac12}^{\mathrm{adj}} + \big[Y_0, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_{\frac12}\big]$ and $Y_1 = X_1^{\mathrm{adj}} + \big[Y_0, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big] + \big[X_{\frac12}^{\mathrm{adj}}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_{\frac12}\big] - \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_{\frac12}\big[Y_0, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_{\frac12}\big]$, with $X_{\frac12}$ the classical length-increasing part and $\mathrm{Gram}_{\frac12}$ the entries between the two lengths of order one half. □

Remark 8.42 (What the test checks)**.** The three ingredients of Corollary 8.41 are computed independently: Q_1 and S_1 from the defect graphs of the equivariant quantum differential, Gram_1 from the two-point graphs of Definition 8.37, and the conjugation from the exact reflection Θ_n transported by Proposition 8.34. Agreement tests the equivariant quantum master equation, the reflection compatibility of the scheme and the definition of K_1 together; the relation for D_1 states that the one-loop dilatation is self-adjoint for the tree inner product on each space of fixed classical weight, where $\{S^{(1)}(\Delta), -\}$ is a multiple of the identity.

8.10 Beyond one loop

Remark 8.43 (The definitions at higher order)**.** Every definition of this chapter is stated at all orders in $\hbar$: the counterterms are the singular parts graph by graph, with the functions $\varepsilon^{-m}\log^k\varepsilon$, $(m,k)\neq(0,0)$, and the counterterms of lower order inserted [C07, Thm B]; the obstruction at order $\hbar^k$ is a local s-cocycle once the lower orders are solved; $L = s+\hbar L_1+\hbar^2 L_2+\cdots$ with $L\circ L=0$; the twisted differential is $r_\xi(\Lambda)$; the scheme acts by conjugation; the inner products and the adjointness are formal series. At two loops the graphs contain one-loop subgraphs with their counterterm vertices, and the defect graphs contain one special edge and one further cycle; the enumeration of graphs and the exact evaluation of their Schwinger integrals are algorithmic tasks of a size that grows with the loop order.

Chapter 9

The cubic obstruction and the dilatation cochain at one loop

Chapter 4 removed the part of the one-loop obstruction with two legs by the finite counterterm $\frac{27}{16}$ units times $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$, and Chapter 6 read the dilatation coefficients off the scaling of the regulator. This chapter computes the part of the obstruction with three legs and solves the equation for the cubic finite counterterm, and then evaluates the equivariant structure of Chapter 8 along the dilatation: the arity-one counterterm, obstruction and finite counterterm along Δ, the component D_1 of Definition 8.16 on the scalar bilinears, and the translation component P_1. Two features carry the content. The obstruction with three legs has a component outside the image of Q on cubic functionals, and the quadratic counterterm of Chapter 4 matches it exactly, which is the Wess–Zumino condition of this sector; and because $(Q^{GF})^2 \neq 0$ in Feynman gauge (Proposition 3.10), $S^{(1)}(\Delta)$ also enters cycles as a composite line from $A|_V$ to c, which adds to D_1 a multiple of the classical transformation that the two-point equations leave free and the action-level arity-two equations along (Δ, P_b) and (Δ, ξ) fix on two legs (Theorem 9.28).

The unit is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$, and every coefficient below is an element of $\mathbb{Q} + \mathbb{Q}\log\frac{3}{2}$ times it. The obstruction and the finite counterterm are written with their factor $\hbar$: a one-loop quantisation is a finite counterterm $\hbar F_1$ with $s(\hbar F_1) = \hbar\,\mathrm{Ob}$, $s = d_{CE} + Q + \{I_{\mathfrak{psl}(4|4)}, -\}$ (Theorem 8.8), and both sides are multiples of the unit. For a cochain f, $f^{(0)}$ is its arity-zero part and $f^{(1)}(x)$ its arity-one part evaluated on $x \in \mathfrak{psl}(4|4)$, as in Definition 8.16. An identity between functionals in a sector, a prescribed list of legs, is the identity between their parts with these legs; for a quadratic functional X the part of $\{I, X\}$ with three legs is the bracket with the cubic part of I in (1.3). Every functional containing λ or λ^* is also given in the two-component form of Definition 8.1 (Remarks 9.13 and 9.29); Weinberg's form follows from $\not{d} = \gamma_W^\mu \partial_\mu$ on the Dirac column (Proposition 8.2).

9.1 The cubic sector of the one-loop obstruction

With the finite counterterm of Theorem 4.32 the obstruction vanishes in the sector $(c, A|_V)$ (Theorem 4.30). The sectors with three legs of Remark 4.34 come next: this section fixes a basis, computes the obstruction class of graphs by class of graphs, and solves the equation for the cubic finite counterterm.

Definition 9.1 (Bases of the three-leg functionals)**.** With $(-,-)$ the pairing of $\mathfrak{g}$ tensored with the metric, d^*d and dd^* acting on the one-form $A|_V$, and d^*d acting on each scalar $\iota_w A$, $w \in W_{\mathbb{R}}^{\perp}$, the

functionals of degree 1 are

$$G_1 = \int \mathrm{dvol}\,([c, A|_V], d^*dA|_V), \qquad G_2 = \int \mathrm{dvol}\,([c, A|_V], dd^*A|_V), \qquad G_3 = \int \mathrm{dvol}\,([c, A|_{W_{\mathbb{R}}^\perp}], d^*dA|_{W_{\mathbb{R}}^\perp}),$$
$$G_4 = \int \mathrm{dvol}\,([c, dc], A^*), \qquad G_5 = \int \mathrm{dvol}\,([c, \lambda], \not{d}\lambda),$$

with the pairings of the terms (dc, A^*) and $\frac{1}{2}(\lambda, \not{d}\lambda)$ of the action in G_4 and G_5, and the functionals of degree 0 are

$$H_1 = \int \mathrm{dvol}\,(dA|_V, [A|_V \wedge A|_V]), \qquad H_2 = \int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, [A|_V, A|_{W_{\mathbb{R}}^\perp}]), \qquad H_3 = \int \mathrm{dvol}\,([A|_V, c], A^*|_V),$$
$$H_4 = \int \mathrm{dvol}\,([A|_{W_{\mathbb{R}}^\perp}, c], A^*|_{W_{\mathbb{R}}^\perp}), \qquad H_5 = \int \mathrm{dvol}\,([c, \lambda], \lambda^*), \qquad H_6 = \int \mathrm{dvol}\,([c, c], c^*),$$

together with $\int \mathrm{dvol}\,(\lambda, \rho([A|_V, \lambda]))$ and $\int \mathrm{dvol}\,(\lambda, \rho([A|_{W_{\mathbb{R}}^\perp}, \lambda]))$, whose sum is twice the cubic term $\frac{1}{2}\int \mathrm{dvol}\,(\lambda, (\not{d}_A - \not{d})\lambda)$ of I. In H_2 the one-form $d(A|_{W_{\mathbb{R}}^\perp})$ and the one-form $[A|_V, A|_{W_{\mathbb{R}}^\perp}]$ are paired with the metrics of V and of $W_{\mathbb{R}}^\perp$.

Lemma 9.2 (Leg sets and colour)**.** *A local functional of degree* 1 *with three legs and Weyl weight at most* 4*, invariant under translations,* $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$ *and the adjoint action of* $\mathfrak{g}$*, has one of the leg sets*

$$(c, A|_V, A|_V), \qquad (c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}), \qquad (c, \lambda, \lambda), \qquad (c, c, A^*|_V),$$

and its colour factor is a multiple of $(X_0, [X_1, X_2])$*,* X_j *the* $\mathfrak{g}$*-factor of leg* j*. Every one-loop colour factor of these sectors reduces to it through*

$$\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y \mathrm{ad}_Z) = \tfrac{1}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{(X,Y)}\,(X, [Y, Z]).$$

Proof. The functional degrees are $+1$ for c, 0 for A and λ, -1 for A^* and λ^*, -2 for c^*, and the Weyl weights are $0, 1, \frac{3}{2}, 3, \frac{5}{2}, 4$ (Definition 6.1). Degree 1 with three legs means

$$n_c - n_{A^*} - n_{\lambda^*} - 2n_{c^*} = 1, \qquad n_c + n_A + n_\lambda + n_{A^*} + n_{\lambda^*} + n_{c^*} = 3,$$

solved by one c with two legs among A, λ, and by two c with one A^* or one λ^*. A single spinor leg ($cA\lambda$, $cc\lambda^*$) is odd under $-1 \in \mathrm{Spin}(4,\mathbb{C})$, and a single vector of $W_{\mathbb{R}}^\perp$ ($cA|_V A|_{W_{\mathbb{R}}^\perp}$, $ccA^*|_{W_{\mathbb{R}}^\perp}$) admits no $\mathrm{Spin}(6,\mathbb{C})$-invariant pairing. The remaining leg sets have the weights $2, 2, 3, 3$. A tree contraction of vertices of I gives $([X_0, X_1], X_2)$. A one-loop colour factor is a trace of products of ad's; the invariance of $(-,-)$ gives $\mathrm{ad}_X^T = -\mathrm{ad}_X$, hence $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y \mathrm{ad}_Z) = -\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Z \mathrm{ad}_Y) = \frac{1}{2}\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_{[Y,Z]})$, which is the stated identity; a bubble with a quartic vertex gives $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_i} \mathrm{ad}_{[X_j, X_k]})$. □

Proposition 9.3 (Basis of the sectors)**.** *Write* $f = ([X_0, X_1], X_2)$*,* p_j *for the momentum of leg* j *(leg* 0 *is the first c),* $a_j \in V$*,* $w_j \in W_{\mathbb{R}}^\perp$*,* $b \in V$ *and* $\lambda_j \in \Sigma$ *for the fibre elements, and take plane-wave values as in Definition 1.13.*

- *(i)* $(c, A|_V, A|_V)$ *has dimension* 2*, with basis* $G_1 = f\big[(|p_2|^2 - |p_1|^2)(a_1, a_2) - (a_1, p_2)(a_2, p_2) + (a_1, p_1)(a_2, p_1)\big]$ *and* $G_2 = f\big[(a_1, p_2)(a_2, p_2) - (a_1, p_1)(a_2, p_1)\big]$*.*
- *(ii)* $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ *has dimension* 1*, with* $G_3 = f\,(w_1, w_2)(|p_2|^2 - |p_1|^2)$*.*
- *(iii)* (c, λ, λ) *is spanned by* $G_5 = -if\,(\lambda_1, \rho(p_0)\lambda_2)$ *and the parity-odd structure* $\int \mathrm{dvol}\,([c, \lambda], \not{d}\rho(v_1v_2v_3v_4)\lambda)$*, of value if* $(\lambda_1, \rho(p_2 - p_1)\rho(v_1v_2v_3v_4)\lambda_2)$*,* $v_1, \ldots, v_4$ *an oriented orthonormal basis of* V *(Proposition 1.2).*

- *(iv)* $(c, c, A^*|_V)$ *has dimension* 1*, with* $G_4 = -if\,(p_2, b)$.
- *(v) The functionals* $\int \mathrm{dvol}\,([c, A|_V], A|_V)$ *and* $\int \mathrm{dvol}\,([c, A|_{W_{\mathbb{R}}^\perp}], A|_{W_{\mathbb{R}}^\perp})$ *of weight* 2 *vanish, and weight* 3 *carries no invariant.*

The parity-odd structure has coefficient 0 *in every functional built from* S, $I^{CT}[\varepsilon]$, $P(\varepsilon, \infty)$, K_ε *and* K_0.

Proof. Relabelling two identical legs leaves a functional unchanged: for two legs A its value is symmetric under their exchange, for two odd legs antisymmetric, since $r_i r_j = -r_j r_i$. The colour factor f is antisymmetric under $1 \leftrightarrow 2$ and under $0 \leftrightarrow 1$, so the kinematic factor is antisymmetric under $(p_1, a_1) \leftrightarrow (p_2, a_2)$ for (c, A, A) and symmetric for (c, λ, λ) and $(c, c, A^*|_V)$. In $(c, A|_V, A|_V)$ at weight 4 it is bilinear in a_1, a_2 and quadratic in p_1, p_2; the antisymmetric parity-even combinations are $(a_1, a_2)(|p_1|^2 - |p_2|^2)$ and $(a_1, p_1)(a_2, p_1) - (a_1, p_2)(a_2, p_2)$, on which (G_1, G_2) has the invertible matrix $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$, from $d^*d \mapsto |p|^2 - p\,(p, -)$ and $dd^* \mapsto p\,(p, -)$ on the second $A|_V$; the parity-odd $(a_1 \wedge a_2 \wedge p_1 \wedge p_2,\, v_1 \wedge v_2 \wedge v_3 \wedge v_4)$ is symmetric. In $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, $\mathrm{Spin}(6, \mathbb{C})$ admits (w_1, w_2) alone. In (c, λ, λ) the invariant bilinears of two elements of Σ and one vector $q \in V$ are $(\lambda_1, \rho(q)\lambda_2)$, symmetric in λ_1, λ_2, and $(\lambda_1, \rho(q)\rho(v_1v_2v_3v_4)\lambda_2)$, antisymmetric (Proposition 1.2); the symmetric combinations are $q = p_1 + p_2 = -p_0$ in the first and $q = p_2 - p_1$ in the second. In $(c, c, A^*|_V)$ the symmetric combination is $(b, p_0 + p_1) = -(b, p_2)$. At weight 2 the factor (a_1, a_2), resp. (w_1, w_2), is symmetric, and at weight 3 the leg sets admit no invariant (Lemma 4.19). Let an element of $\mathrm{Spin}(10, \mathbb{C})$ lift the product of the reflections in one direction of V and one direction of $W_{\mathbb{R}}^\perp$. It preserves the action, Q, Q^{GF} and H, hence every kernel and counterterm, fixes G_5 and maps $\rho(v_1v_2v_3v_4)$ to $-\rho(v_1v_2v_3v_4)$, since v_1 anticommutes with v_2, v_3, v_4 and the direction of $W_{\mathbb{R}}^\perp$ commutes with all four; so the parity-odd coefficient vanishes. □

Proposition 9.4 (The maps Q and $\{I, -\}$). • *(i)* $QH_1 = -2G_1$, $QH_2 = -G_3$, $QH_3 = -\frac{1}{2}G_1 - G_4$, $QH_4 = -\frac{1}{2}G_3$, $QH_5 = G_5$, $QH_6 = 2G_4$, $Q\int \mathrm{dvol}\,(\lambda, \rho([A|_V, \lambda])) = -2G_5$ *and* $Q\int \mathrm{dvol}\,(\lambda, \rho([A|_{W_{\mathbb{R}}^\perp}, \lambda])) = 0$. *The cubic part of* I *is* $-\frac{1}{4}H_1 - \frac{1}{2}H_2 + \frac{1}{2}\int \mathrm{dvol}\,(\lambda, \rho([A, \lambda])) + H_3 + H_4 + H_5 + \frac{1}{2}H_6$*, and its image under* Q *vanishes.*

- *(ii)* $H_1, \ldots, H_6$, $\int \mathrm{dvol}\,(\lambda, \rho([A|_V, \lambda]))$ *and* $\int \mathrm{dvol}\,(\lambda, \rho([A|_{W_{\mathbb{R}}^\perp}, \lambda]))$ *span the parity-even invariant local functionals of degree* 0*, Weyl weight* 4 *and three legs. On their span* Q *has rank* 4 *and image* $\mathrm{span}\{G_1, G_3, G_4, G_5\}$*, and its kernel is spanned by*

$$H_1 - 4H_3 - 2H_6, \qquad H_2 - 2H_4, \qquad \int \mathrm{dvol}\,(\lambda, \rho([A|_V, \lambda])) + 2H_5, \qquad \int \mathrm{dvol}\,(\lambda, \rho([A|_{W_{\mathbb{R}}^\perp}, \lambda])).$$

Every invariant local functional of degree -1 *with three legs has Weyl weight at least* 6*, so these four functionals represent four independent classes of the cohomology of* Q *on local functionals of weight* 4.

- *(iii) In the sectors with three legs, with* $H = d^*d + dd^*$ *the componentwise Laplacian of Proposition 3.10 (the functionals* H_j *carry an index),*

$$\Big\{I, \int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)\Big\} = 2G_2, \qquad \Big\{I, \int \mathrm{dvol}\,(dA|_V, dA|_V)\Big\} = 2G_1, \qquad \Big\{I, \int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})\Big\} =$$
$$\Big\{I, \int \mathrm{dvol}\,(\lambda, \not{d}\lambda)\Big\} = 2G_5, \qquad \Big\{I, \int \mathrm{dvol}\,(dc, A^*)\Big\} = \tfrac{1}{2}G_1 + \tfrac{1}{2}G_3 - G_5,$$
$$\Big\{I, \int \mathrm{dvol}\,(A, A)\Big\} = 0, \qquad \Big\{I, \int \mathrm{dvol}\,(A, HA)\Big\} = 2(G_1 + G_2 + G_3).$$

Proof. (i) Q acts by $F(e+rQe) = F(e) + r\,(QF)(e)$ with r odd [C07, §6.2], that is by the substitutions $A|_V \mapsto A|_V - r\,dc$, $A^* \mapsto A^* - \frac{r}{2} d^* dA$, $c^* \mapsto c^* + r\,d^*(A^*|_V)$, $\lambda^* \mapsto \lambda^* + r\,\not{d}\lambda$ (Proposition 1.11), with r moved to the left. For H_1 the cubic term gives $-2r[A|_V \wedge dc]$, and with $d[A,c] = [dA,c] - [A \wedge dc]$ and $\int \mathrm{dvol}\,(dA, [dA,c]) = 0$,

$$QH_1 = -2\int \mathrm{dvol}\,(dA|_V, [A|_V \wedge dc]) = 2\int \mathrm{dvol}\,([A|_V, c], d^* dA|_V) = -2G_1.$$

For H_3 the substitution in $A|_V$ gives $-\int \mathrm{dvol}\,([dc,c], A^*|_V) = -G_4$, since $[dc,c] = [c,dc]$ for the odd c, and the substitution in $A^*|_V$ gives $\frac{1}{2}\int \mathrm{dvol}\,([A|_V, c], d^* dA|_V) = -\frac{1}{2}G_1$; H_4 receives the second step alone, H_5 gives $\int \mathrm{dvol}\,([c,\lambda],\, \not{d}\lambda)$, and H_6 gives $\int \mathrm{dvol}\,(d[c,c], A^*) = 2G_4$. For the Yukawa-type functionals, Q hits the $A|_V$ slot alone, and with the symmetry of $(\lambda_1, \rho(v)[Y,\lambda_2])$ in λ_1, λ_2,

$$Q\int \mathrm{dvol}\,(\lambda, \rho([A|_V, \lambda])) = \int \mathrm{dvol}\,(\lambda, \rho([dc,\lambda])) = -2\int \mathrm{dvol}\,([c,\lambda],\, \not{d}\lambda) = -2G_5.$$

The cubic part of I follows from $(dA, [A \wedge A]) = (dA|_V, [A|_V \wedge A|_V]) + 2(dA|_{W_{\mathbb{R}}^\perp}, [A|_V, A|_{W_{\mathbb{R}}^\perp}])$, and its image is $(\frac{1}{2}G_1 + \frac{1}{2}G_3) - G_5 + (-\frac{1}{2}G_1 - G_4) - \frac{1}{2}G_3 + G_5 + G_4 = 0$, the cubic part of the classical master equation (Proposition 1.14).

(ii) Degree 0, three legs and weight 4 leave (A,A,A) with one derivative, and (A,λ,λ), (c,A,A^*), (c,λ,λ^*), (c,c,c^*) without derivatives. For $(A|_V)^3$ the parity-even invariant is H_1, and the parity-odd one is the cubic part of the total derivative $(F_A \wedge F_A)$; for $A|_V (A|_{W_{\mathbb{R}}^\perp})^2$ the invariants agree up to integration by parts with H_2; three vectors of $W_{\mathbb{R}}^\perp$ admit no invariant. For (A,λ,λ) the kinematic factor is symmetric in the two λ legs, as in Proposition 9.3: with $a \in V$ this leaves $(\lambda_1, \rho(a)\lambda_2)$, and with $w \in W_{\mathbb{R}}^\perp$, which commutes with $v_1v_2v_3v_4$, it leaves $(\lambda_1, \rho(w)\lambda_2)$ and the parity-odd $(\lambda_1, \rho(w)\rho(v_1v_2v_3v_4)\lambda_2)$. The remaining leg sets give H_3, H_4, H_5 with a parity-odd partner $\int \mathrm{dvol}\,([c,\lambda], \rho(v_1v_2v_3v_4)\lambda^*)$, and H_6. The rank and the kernel follow from (i). Degree -1 with three legs requires an antifield; a single spinor leg or a single vector index of V or of $W_{\mathbb{R}}^\perp$ admits no invariant, and $\lambda^* \otimes \lambda$, $\lambda^* \otimes \lambda^*$ contain no invariant vector or scalar, so the invariant leg sets of degree -1 have weight at least 6, the lowest being (A^*,λ,λ) and $(A^*|_V, A^*|_V, c)$; the image under Q of the functionals of degree -1, weight 4 and three legs is therefore zero, and the four kernel elements represent independent classes.

(iii) In the cubic part of I the antifields enter linearly, with $\partial_{A^*} = -[A,c]$, $\partial_{\lambda^*} = [c,\lambda]$, $\partial_{c^*} = \frac{1}{2}[c,c]$ (left derivatives; the first sign from moving A^* across $[A,c]$). By Definition 1.6, for X a functional of A alone $\{I,X\} = -\frac{d}{dt}\big|_{t=0} X(A + t[A,c])$ with the odd factor to the left, and for X a functional of λ alone $\{I,X\}$ is the derivative of X along $\lambda \mapsto [c,\lambda]$. Integration by parts gives the first three brackets, $\{I, \int(A,A)\} = -2\int \mathrm{dvol}\,([A,c],A) = 0$ and $\{I, \int(A,HA)\} = -2\int \mathrm{dvol}\,([A,c],HA) = 2(G_1+G_2+G_3)$; for λ, $\int \mathrm{dvol}\,([c,\lambda],\, \not{d}\lambda) - \int \mathrm{dvol}\,(\lambda,\, \not{d}[c,\lambda]) = 2G_5$. For $\int \mathrm{dvol}\,(dc, A^*)$ the bracket consists of the parts of Q applied to the cubic part of I coming from the substitutions in $A|_V$ and in c^*: $(\frac{1}{2}G_1 + \frac{1}{2}G_3 - G_5 - G_4) + G_4$. □

Corollary 9.5 (Completions)**.** *In the sectors with three legs, Q of the second entry plus the bracket of I with the first entry vanishes for each of the pairs*

$$\Big(\int (dA|_V, dA|_V),\, H_1\Big), \quad \Big(\int (dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp}),\, 2H_2\Big),$$
$$\Big(\int (\lambda,\, \not{d}\lambda),\, \int (\lambda, \rho([A,\lambda]))\Big), \quad \Big(\int (dc, A^*),\, H_3 + H_4 + H_5 + \tfrac{1}{2}H_6\Big),$$

the quadratic and cubic parts of $\int \mathrm{dvol}\,(F_{A|_V}, F_{A|_V})$, $\int \mathrm{dvol}\,(d_{A|_V} A|_{W_{\mathbb{R}}^\perp}, d_{A|_V} A|_{W_{\mathbb{R}}^\perp})$, $\int \mathrm{dvol}\,(\lambda,\, \not{d}_A\lambda)$ and of the antifield part $\int \mathrm{dvol}\big((d_A c, A^*) + ([c,\lambda],\lambda^*) + \frac{1}{2}([c,c],c^*)\big)$ of S. In the third pair $\int(\lambda, \rho([A,\lambda]))$ may be replaced by $-2H_5$; the two choices differ by the Q-closed $\int(\lambda,\rho([A,\lambda])) + 2H_5$.

Proof. By Proposition 9.4: $-2G_1+2G_1=0$, $-2G_3+2G_3=0$, $-2G_5+2G_5=0$ and $(-\frac{1}{2}G_1-G_4)-\frac{1}{2}G_3+G_5+G_4+\frac{1}{2}G_1+\frac{1}{2}G_3-G_5=0$. The functionals are the stated parts by $F_{A|_V}=dA|_V+\frac{1}{2}[A|_V\wedge A|_V]$, $d_{A|_V}A|_{W_{\mathbb{R}}^\perp}=d(A|_{W_{\mathbb{R}}^\perp})+[A|_V,A|_{W_{\mathbb{R}}^\perp}]$, $\not{d}_A\lambda=\not{d}\lambda+\rho([A,\lambda])$ and $d_Ac=dc+[A,c]$. □

Lemma 9.6 (The sector $(c,c,A|_V)$)**.** *On the sector $(c,c,A|_V)$ of degree 2, $QG_1=QG_3=QG_4=QG_5=0$,*

$$QG_2=\int \mathrm{dvol}\,([c,dc],dd^*A|_V)+\int \mathrm{dvol}\,([c,A|_V],dd^*dc)\neq 0,\qquad \Big\{I,\int \mathrm{dvol}\,(d^*dc,d^*A|_V)\Big\}=QG_2.$$

Proof. $[c,dc]=\frac{1}{2}d[c,c]$ and $d^*d^*=0$ give $QG_1=0$ and $QG_4=0$; Q hits no slot of G_3 and G_5; in G_2 the substitution $A|_V\mapsto A|_V-r\,dc$ hits both $A|_V$. The plane-wave value of QG_2 is $-if\,r_0r_1\big(|p_0|^2(p_0,b)+|p_1|^2(p_1,b)+|p_2|^2(p_2,b)\big)$, nonzero. For the bracket, $\int \mathrm{dvol}\,(d^*dc,d^*A|_V)=Q\big(-\frac{1}{2}\int \mathrm{dvol}\,(d^*A|_V,d^*A|_V)\big)$, and by the Jacobi identity and Proposition 9.4 (i), $\{I,QF\}=-Q\{I,F\}$ in the sector, so the bracket is $-Q\big(-\frac{1}{2}\cdot 2G_2\big)=QG_2$. □

Proposition 9.7 (Graph classes)**.** *By Proposition 4.26 and* (4.8)*, the part with three legs of the order-$\hbar$ term of $Q\,W(P(\varepsilon,\infty),I-I^{CT}[\varepsilon])$ is the sum of the following classes of graphs, the special edge carrying $K_\varepsilon-K_0$ between a field and its antifield ($A|_V$–$A^*|_V$, $A|_{W_{\mathbb{R}}^\perp}$–$A^*|_{W_{\mathbb{R}}^\perp}$, λ–λ^*, c–c^*) and the other edges $P(\varepsilon,\infty)$:*

- *(i) cycle graphs: the special edge on the cycle of a triangle of three cubic vertices or of a bubble of one quartic and one cubic vertex;*
- *(ii) bridge graphs: the special edge joins a cubic vertex with two external legs to a one-loop two-point subgraph;*
- *(iii) the counterterm vertex $-Q\,I^{CT}[\varepsilon]$;*
- *(iv) the counterterm bracket $-\{I,I^{CT}[\varepsilon]\}_\varepsilon$, with the cubic part of I and the quadratic part of $I^{CT}[\varepsilon]$, split into its part with K_0 and its part with $K_\varepsilon-K_0$;*
- *(v) reducible graphs: a defect subgraph with two legs, one of the bubbles (a)–(e) of Theorem 4.30 or the vertex $-Q\,I^{CT}[\varepsilon]$ with two legs, joined by an edge $P(\varepsilon,\infty)$ to a cubic vertex with two external legs.*

Proof. The vertex $\frac{1}{2}(\{I,I\}_\varepsilon-\{I,I\})$ has order $\hbar^0$, so its graphs have one cycle; the other terms of (4.8) have order $\hbar$, so their graphs are trees, and $\hbar\Delta_\varepsilon I=0$ (proof of Theorem 4.30). If the special edge lies on the cycle, a cycle of n vertices of valence at least three carries at least n further half-edges; with three legs, and every tree hanging off the cycle carrying at least two legs, the cycle is a triangle of cubic vertices, a quartic-cubic bubble, or a cubic-cubic bubble with one half-edge joined to a cubic vertex with two legs, which is class (v). If the special edge is a bridge, its tree side has at least two legs: two legs at one cubic vertex and a two-point one-loop side, class (ii), or three legs and a one-point side, which vanishes by Lemma 4.7. The trees are $-Q\,I^{CT}[\varepsilon]$ with three legs, $-Q\,I^{CT}[\varepsilon]$ with two legs joined to a cubic vertex, and $-\{I,I^{CT}[\varepsilon]\}_\varepsilon$; the brackets of the quartic part of I with the quadratic counterterm and of the cubic part of I with the cubic counterterm have four legs. □

The cycle graphs are listed row by row in the following table. A vertex is named by the component types of its term of I (the table after (1.3)), with its external leg in parentheses.

sector	vertices on the cycle	special edge	other edges	graphs	value of the row
$(c, A\|_V, A\|_V)$	$A\|_V cA^*\|_V$ (c), $(A\|_V)^3$, $(A\|_V)^3$	$A^*\|_V$ at the c-vertex – $A\|_V$	$A\|_V$, $A\|_V$	2	$(\frac{103}{144}+\frac{2}{3}\log\frac{3}{2})G_1+(\frac{41}{48}+\frac{1}{4}\log\frac{3}{2})G_2$
	$A\|_V cA^*\|_V$ (c), $(A\|_V)^3$, $A\|_V cA^*\|_V$ $(A\|_V)$	$A\|_V$ at $(A\|_V)^3$ – $A^*\|_V$	$A\|_V$, c–A^*	2	$(\frac{1}{2}+\frac{1}{2}\log\frac{3}{2})G_1$
	$A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ (c), $A\|_V (A\|_{W_R^\perp})^2$, $A\|_V (A\|_{W_R^\perp})^2$	$A^*\|_{W_R^\perp} - A\|_{W_R^\perp}$	$A\|_{W_R^\perp}$, $A\|_{W_R^\perp}$	2	$(\frac{11}{12}+\log\frac{3}{2})G_1+\frac{5}{4}G_2$
	$A\|_V cA^*\|_V$ (c), $A\|_V cA^*\|_V$, $A\|_V cA^*\|_V$	$A\|_V$ at the c-vertex – $A^*\|_V$	c–A^*, c–A^*	2	$-(\frac{11}{288}+\frac{1}{24}\log\frac{3}{2})G_1-(\frac{1}{32}+\frac{1}{8}\log\frac{3}{2})G_2$
	ccc^* (c), $A\|_V cA^*\|_V$, $A\|_V cA^*\|_V$	$c^* - c$	c–A^*, c–A^*	2	$-(\frac{11}{288}+\frac{1}{24}\log\frac{3}{2})G_1-(\frac{1}{32}+\frac{1}{8}\log\frac{3}{2})G_2$
	$c\lambda\lambda^*$ (c), $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$\lambda^* - \lambda$	λ, λ	2	$(\frac{10}{9}+\frac{8}{3}\log\frac{3}{2})G_1+\frac{4}{3}G_2$
	bubble: $A\|_V cA^*\|_V$ (c), $(A\|_V)^4$	$A^*\|_V - A\|_V$	$A\|_V$	1	0
	bubble: $A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ (c), $(A\|_V)^2 (A\|_{W_R^\perp})^2$	$A^*\|_{W_R^\perp} - A\|_{W_R^\perp}$	$A\|_{W_R^\perp}$	1	0
	total			14	$(\frac{11}{4}+\frac{19}{4}\log\frac{3}{2})G_1+\frac{27}{8}G_2$
$(c, A\|_{W_R^\perp}, A\|_{W_R^\perp})$	$A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ (c), $A\|_V (A\|_{W_R^\perp})^2$, $A\|_V (A\|_{W_R^\perp})^2$	$A^*\|_{W_R^\perp} - A\|_{W_R^\perp}$	$A\|_{W_R^\perp}$, $A\|_V$	2	$(\frac{17}{48}+\frac{1}{2}\log\frac{3}{2})G_3$
	$A\|_V cA^*\|_V$ (c), $A\|_V (A\|_{W_R^\perp})^2$, $A\|_V (A\|_{W_R^\perp})^2$	$A^*\|_V - A\|_V$	$A\|_V$, $A\|_{W_R^\perp}$	2	$-(\frac{1}{48}+\frac{1}{4}\log\frac{3}{2})G_3$
	$A\|_V cA^*\|_V$ (c), $A\|_V (A\|_{W_R^\perp})^2$, $A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ $(A\|_{W_R^\perp})$	$A\|_{W_R^\perp} - A^*\|_{W_R^\perp}$	$A\|_V$, c–A^*	2	$(\frac{1}{12}+\frac{1}{2}\log\frac{3}{2})G_3$
	$c\lambda\lambda^*$ (c), $\lambda\lambda A\|_{W_R^\perp}$, $\lambda\lambda A\|_{W_R^\perp}$	$\lambda^* - \lambda$	λ, λ	2	$(\frac{7}{3}+4\log\frac{3}{2})G_3$
	bubble: $A\|_V cA^*\|_V$ (c), $(A\|_V)^2 (A\|_{W_R^\perp})^2$	$A^*\|_V - A\|_V$	$A\|_V$	1	0
	bubble: $A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ (c), $(A\|_{W_R^\perp})^4$	$A^*\|_{W_R^\perp} - A\|_{W_R^\perp}$	$A\|_{W_R^\perp}$	1	0
	total			10	$(\frac{11}{4}+\frac{19}{4}\log\frac{3}{2})G_3$
(c, λ, λ)	$A\|_V cA^*\|_V$ (c), $c\lambda\lambda^*$ (λ), $\lambda\lambda A\|_V$ (λ)	$\lambda^* - \lambda$	$A\|_V$, c–A^*	2	$-\log\frac{3}{2}G_5$
	$A\|_V cA^*\|_V$ (c), $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$A^*\|_V - A\|_V$	$A\|_V$, λ	2	$-(\frac{1}{2}+\log\frac{3}{2})G_5$
	$A\|_{W_R^\perp} cA^*\|_{W_R^\perp}$ (c), $\lambda\lambda A\|_{W_R^\perp}$, $\lambda\lambda A\|_{W_R^\perp}$	$A^*\|_{W_R^\perp} - A\|_{W_R^\perp}$	$A\|_{W_R^\perp}$, λ	2	$-(\frac{3}{2}+3\log\frac{3}{2})G_5$
	$c\lambda\lambda^*$ (c), $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$\lambda^* - \lambda$	λ, $A\|_V$	2	$(\frac{1}{2}-\log\frac{3}{2})G_5$
	$c\lambda\lambda^*$ (c), $\lambda\lambda A\|_{W_R^\perp}$, $\lambda\lambda A\|_{W_R^\perp}$	$\lambda^* - \lambda$	λ, $A\|_{W_R^\perp}$	2	$(\frac{3}{2}-3\log\frac{3}{2})G_5$
	total			10	$-9\log\frac{3}{2}G_5$
$(c, c, A^*\|_V)$	$A\|_V cA^*\|_V$ (c), $A\|_V cA^*\|_V$ (c), $A\|_V cA^*\|_V$ $(A^*\|_V)$	$A\|_V$ at one c-vertex – $A^*\|_V$ at the other	c–A^*, $A\|_V$	2	$\log\frac{3}{2}G_4$
	$A\|_V cA^*\|_V$ (c), ccc^* (c), $A\|_V cA^*\|_V$ $(A^*\|_V)$	$c^* - c$ at the A^*-vertex	$A\|_V$, c–A^*	2	$-(\frac{1}{4}+\frac{1}{2}\log\frac{3}{2})G_4$
	$A\|_V cA^*\|_V$ (c), $A\|_V cA^*\|_V$ (c), $A\|_V cA^*\|_V$ $(A^*\|_V)$	$A^*\|_V$ at a c-vertex – $A\|_V$ at the A^*-vertex	$A\|_V$, c–A^*	2	$(\frac{1}{4}-\frac{1}{2}\log\frac{3}{2})G_4$
	total			6	0

The cycle graphs, row by row, with the $\varepsilon \to 0$ value of each row in units. The two graphs of a triangle row differ by the exchange of the two identical external legs; every graph has weight 1, the $\frac{1}{2}$ of $\frac{1}{2}\{I, I\}$ cancelling the two orders of the vertices joined by the special edge. The coefficients of ε^{-1} and of $\log\varepsilon$ of every row vanish.

Proposition 9.8 (Limits of the bridge and reducible graphs)**.**

- *(i) The sum of the bridge graphs and of the counterterm bracket with $K_\varepsilon - K_0$ tends to 0 as $\varepsilon \to 0$. Separately, the counterterm bracket with $K_\varepsilon - K_0$ tends to $\frac{9}{4}\{I, \int(A, HA)\} = \frac{9}{2}(G_1 + G_2 + G_3)$ units and the bridge graphs tend to the opposite.*

- *(ii) The reducible graphs tend to the tree with one vertex given by the quadratic part of $\hbar\,\mathrm{Ob}^{(0)}$, one cubic vertex of I and one edge $P(0,\infty)$, which is the local functional*

$$\Big(\tfrac{27}{8}G_1 - \tfrac{27}{8}G_2 + \tfrac{27}{8}G_3 - \tfrac{27}{4}G_4 - \tfrac{27}{4}G_5\Big) \text{ units.}$$

Proof. (i) Removing the bridge separates a bridge graph into the cubic vertex and a connected one-loop two-point graph of $W(P(\varepsilon,\infty), I)$; summed with their weights, the bridge graphs are the cross terms of $\frac{1}{2}\{W, W\}$ built with $K_\varepsilon - K_0$ between the cubic part of I and the one-loop two-point part of $W(P(\varepsilon,\infty), I)$. In momentum space $K_\varepsilon - K_0 = (e^{-\varepsilon|p|^2} - 1)K_0$, p the momentum through the bridge, so the bridge graphs and the counterterm bracket with $K_\varepsilon - K_0$ add up to $e^{-\varepsilon|p|^2} - 1$ times the bracket of the cubic part of I with the renormalised two-point function at p. The latter has a limit at $p \neq 0$ (Theorem 4.16, Proposition 4.8), which gives the first statement. In the counterterm bracket alone, the $\log\varepsilon$ terms of $I^{CT}[\varepsilon]$ give $(e^{-\varepsilon|p|^2} - 1)\log\varepsilon \to 0$, and the mass term gives, with $\varepsilon^{-1}(e^{-\varepsilon H} - 1) \to -H$ and Proposition 9.4 (iii),

$$-\tfrac{9}{4}\varepsilon^{-1}\Big\{I, \int \mathrm{dvol}\,\big(A, (e^{-\varepsilon H} - 1)A\big)\Big\} \longrightarrow \tfrac{9}{4}\Big\{I, \int \mathrm{dvol}\,(A, HA)\Big\} = \tfrac{9}{2}(G_1 + G_2 + G_3).$$

(ii) The reducible graphs form the tree with one vertex given by the defect graphs with two legs and one cubic vertex of I, joined by $P(\varepsilon,\infty)$. By Lemma 4.29 and Theorem 4.30, at fixed $p \neq 0$ the defect graphs with two legs equal the quadratic part of $\hbar\,\mathrm{Ob}^{(0)}$ at p plus $O(\varepsilon)$, their coefficients of ε^{-1} cancelling, and the edge carries $e^{-\varepsilon|p|^2}$ times $P(0,\infty)$ at p; both factors converge. The quadratic part of $\hbar\,\mathrm{Ob}^{(0)}$ is $-\frac{27}{8}\int \mathrm{dvol}\,(c, d^*dd^*A|_V)$ units: through the $A|_V$ line the factor $|p|^2$ of d^*d on c, and through the c–A^* line the factor $|p|^2$ of d^*d on $A|_V$, cancels the factor $|p|^{-2}$ of $P(0,\infty)$, so the tree is local, and its evaluation with the components of Proposition 3.17 gives the stated functional. □

Theorem 9.9 (Singular parts). • *(i) Each cycle graph has the value $\varepsilon^{-\omega/2}$ times a first coefficient plus a second coefficient plus $O(\varepsilon\log\varepsilon)$, where, in the scaled Schwinger times $\tau_1\in[0,1]$ of the special edge and $\tau_2,\tau_3\in[1,\infty)$, the two coefficients are the integrals of the Taylor coefficients of order 0 and of order ω in the external momenta of the Gaussian-integrated integrand, $\omega=2$ for (c,A,A) and $\omega=1$ for (c,λ,λ) and $(c,c,A^*|_V)$. These integrals converge, the coefficient of $\log\varepsilon$ vanishes, and the coefficient of ε^{-1} vanishes after the sum over the graphs of a row.*

- *(ii) The counterterm vertex and the counterterm bracket with K_0 are multiples of $\log\varepsilon$, with the coefficients of the second table below, and their sum vanishes in every sector: with the unit times the field-counting functional $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ of Chapter 6, the sum is minus Q of the bracket of I with this functional minus the bracket of I with its image under Q, which is zero.*

- *(iii) In every sector with three legs the coefficients of ε^{-1} and of $\log\varepsilon$ of the order-$\hbar$ term of $Q\,W(P(\varepsilon,\infty),I-I^{CT}[\varepsilon])$ vanish.*

Proof. (i) The special edge carries $-\int_0^\varepsilon|q|^2e^{-t|q|^2}dt$ times the components of K_0 (Proposition 4.27). The superficial degree ω is the total momentum degree of the vertex functions and of the numerators of the edges $P(\varepsilon,\infty)$ for triangles, and that degree plus 2 for bubbles, and after the substitution $t_e=\varepsilon\tau_e$ the Taylor coefficient of order j in the external momenta carries $\varepsilon^{(j-\omega)/2}$. For odd $j<\omega$ it is Lorentz-odd and vanishes; for $j=0$ it is $\varepsilon^{-\omega/2}$ times a functional of weight $4-\omega$: for $\omega=2$ it vanishes after the symmetrisation over the identical legs, and for $\omega=1$ the leg sets carry no invariant of weight 3 (Proposition 9.3 (v)). For a triangle, on $\tau_2,\tau_3\geq R$ the Gaussian integral over the loop momentum gives $(\tau_1+\tau_2+\tau_3)^{-2}$ times monomials of total degree $\omega+2$ in the loop momentum and the external momenta, so the coefficient of order j is bounded by a multiple of $R^{-2-(\omega+2-j)/2}$, and the shell $R\leq\tau_2+\tau_3\leq 2R$ contributes at most a multiple of $R^{-(\omega+2-j)/2}$, summable over dyadic R for $j<\omega+2$; for bubbles the exponent $\tau_1\tau_2|p|^2/(\tau_1+\tau_2)$ is at most $|p|^2$, uniformly in τ_2 (Lemma 4.29). The order $\omega+1$ is Lorentz-odd, and the orders $\omega+2$ and beyond contribute $O(\varepsilon\log\varepsilon)$. (ii) By Theorem 4.22, the cubic part of $I^{CT}[\varepsilon]$ is $\log\varepsilon$ times the bracket of I with the unit times the field-counting functional (the multiples $6,6,10,10,1,1,1,1$ of the cubic vertices), with coefficient 0 of ε^{-1}, and the quadratic $\log\varepsilon$ part is $\log\varepsilon$ times the image under Q of the same functional. With Proposition 9.4 the counterterm vertex gives, vertex by vertex, $-3G_1$ from $(A|_V)^3$ and $+\frac12G_1$ from $A|_VcA^*|_V$; $-3G_3$ and $+\frac12G_3$ from $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ and $A|_{W_{\mathbb{R}}^\perp}cA^*|_{W_{\mathbb{R}}^\perp}$; $+10G_5$ and $-G_5$ from $\lambda\lambda A|_V$ and $c\lambda\lambda^*$; $+G_4$ and $-G_4$ from $A|_VcA^*|_V$ and ccc^*; and the counterterm bracket with K_0 gives $+\frac12G_1+2G_1$, $+\frac12G_3+2G_3$, $-G_5-8G_5$ and $-G_4+G_4$. Their sum is minus Q of the bracket of I with that functional minus the bracket of I with its image under Q, which vanishes by the Jacobi identity and the vanishing image under Q of the cubic part of I. The mass term contributes $\{I,\int(A,A)\}=0$. (iii) follows from (i), (ii) and Proposition 9.8. □

contribution	$(c,A\|_V,A\|_V)$	$(c,A\|_{W_{\mathbb{R}}^\perp},A\|_{W_{\mathbb{R}}^\perp})$	(c,λ,λ)	$(c,c,A^*\|_V)$
counterterm vertex, coefficient of $\log\varepsilon$	$-\frac52G_1$	$-\frac52G_3$	$9G_5$	0
counterterm bracket with K_0, coefficient of $\log\varepsilon$	$\frac52G_1$	$\frac52G_3$	$-9G_5$	0
bridge graphs, limit	$-\frac92(G_1+G_2)$	$-\frac92G_3$	0	0
counterterm bracket with $K_\varepsilon-K_0$, limit	$\frac92(G_1+G_2)$	$\frac92G_3$	0	0
reducible graphs, limit	$\frac{27}{8}(G_1-G_2)$	$\frac{27}{8}G_3$	$-\frac{27}{4}G_5$	$-\frac{27}{4}G_4$

The classes (ii)–(v) of Proposition 9.7, sector by sector, in units.

Theorem 9.10 (The cubic part of the obstruction). *In the sectors with three legs,*

$$\hbar\,\mathrm{Ob}^{(0)}=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\big(\tfrac{11}{4}+\tfrac{19}{4}\log\tfrac32\big)(G_1+G_3)+\tfrac{27}{8}G_2-9\log\tfrac32\,G_5\Big],\tag{9.1}$$

the sum of the totals of the table of cycle graphs. The coefficient of G_4 and that of the parity-odd structure of (c,λ,λ) vanish.

Proof. By Definition 8.7 with the finite counterterm set to zero, the part with three legs of the order-$\hbar$ term of $\lim_{\varepsilon\to 0} QW(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ is the part with three legs of the tree-level flow of $\hbar\,\mathrm{Ob}^{(0)}$: the cubic part of $\hbar\,\mathrm{Ob}^{(0)}$ plus the tree of Proposition 9.8 (ii). By Proposition 9.7 the limit is the sum of the five classes. By Proposition 9.8 the bridge graphs cancel against the counterterm bracket with $K_\varepsilon - K_0$, and the reducible graphs give the tree; by Theorem 9.9 the counterterm vertex and the counterterm bracket with K_0 are multiples of $\log\varepsilon$ that cancel. The cubic part of $\hbar\,\mathrm{Ob}^{(0)}$ is therefore the limit of the cycle graphs, the totals of the table. The parity-odd coefficient vanishes by Proposition 9.3. □

Theorem 9.11 (The cubic finite counterterm)**.** *Write $\hbar\,\mathrm{Ob}^{(0)}$ in the sectors with three legs as $\sum_i \alpha_i G_i$ units and the arity-zero finite counterterm as*

$$\hbar F_1^{(0)} = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{27}{16}\int(d^*A|_V, d^*A|_V) + \beta_1\int(dA|_V, dA|_V) + \beta_2\int(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp}) + \beta_3\int(\lambda, \not{d}\lambda) + \beta_4\int(dc,$$
$$+\sum_{j=1}^{6} h_jH_j + y_V\int(\lambda,\rho([A|_V,\lambda])) + y_W\int(\lambda,\rho([A|_{W_{\mathbb{R}}^\perp},\lambda]))\Big] + (\textit{terms with four or more leg}$$

with numbers β_k, h_j, y_V, y_W; its quadratic part is the general solution of the part with two legs, the finite counterterm of Theorem 4.32 plus the Q-closed quadratic local functionals of degree 0 and Weyl weight 4, which are spanned by the four functionals with the coefficients β_k (Theorems 4.30, 4.32). The part with three legs of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ at arity zero reads

$$\begin{aligned} &G_1 : \alpha_1 = -2h_1 - \tfrac{1}{2}h_3 + 2\beta_1 + \tfrac{1}{2}\beta_4, && G_2 : \alpha_2 = \tfrac{27}{8},\\ &G_3 : \alpha_3 = -h_2 - \tfrac{1}{2}h_4 + 2\beta_2 + \tfrac{1}{2}\beta_4, && G_4 : \alpha_4 = -h_3 + 2h_6,\\ &G_5 : \alpha_5 = h_5 - 2y_V + 2\beta_3 - \beta_4. \end{aligned}$$

It is solvable exactly when $\alpha_2 = \frac{27}{8}$, which holds by (9.1)*, and with* (9.1) *its solutions are, in units,*

$$\begin{aligned} \sum_j h_jH_j + y_V\int(\lambda,\rho([A|_V,\lambda])) + y_W\int(\lambda,\rho([A|_{W_{\mathbb{R}}^\perp},\lambda])) = {}& -\big(\tfrac{11}{8}+\tfrac{19}{8}\log\tfrac{3}{2}\big)H_1 - \big(\tfrac{11}{2}+\tfrac{19}{2}\log\tfrac{3}{2}\big)H_4 - 9\log\tfrac{3}{2}\,H_5\\ &+\beta_1H_1 + 2\beta_2H_2 - 2\beta_3H_5 + \beta_4\big(H_3+H_4+H_5+\tfrac{1}{2}H_6\big)\\ &+\kappa_1(H_1 - 4H_3 - 2H_6) + \kappa_2(H_2 - 2H_4)\\ &+\kappa_3\Big(\int(\lambda,\rho([A|_V,\lambda])) + 2H_5\Big) + \kappa_4\int(\lambda,\rho([A|_{W_{\mathbb{R}}^\perp},\lambda])) \end{aligned}$$

with free numbers $\beta_1,\dots,\beta_4$, $\kappa_1,\dots,\kappa_4$. The directions β_k are the completions of Corollary 9.5, and the directions κ_k are the four Q-cohomology classes of Proposition 9.4 (ii). The choice $\beta_1 = \beta_2 = \frac{1}{2}\alpha_1$, $\beta_3 = \frac{1}{2}\alpha_5$, $\beta_4 = 0$, $\kappa_1 = \kappa_3 = \kappa_4 = 0$, $\kappa_2 = -\alpha_3$ gives a vanishing cubic part, with

$$\hbar F_1^{(0)} = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{27}{16}\int(d^*A|_V, d^*A|_V) + \big(\tfrac{11}{8}+\tfrac{19}{8}\log\tfrac{3}{2}\big)\int(dA, dA) - \tfrac{9}{2}\log\tfrac{3}{2}\int(\lambda, \not{d}\lambda)\Big] \quad (9.2)$$

up to terms with four or more legs, $\int(dA,dA) = \int(dA|_V, dA|_V) + \int(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})$.

Proof. At arity zero $s = Q + \{I, -\}$, and its part with three legs applied to $\hbar F_1^{(0)}$ is Q of the cubic part plus the bracket of I with the quadratic part; Propositions 9.4 (i), (iii) give the five equations. G_2 is reached by $\frac{27}{16}\{I, \int(d^*A|_V, d^*A|_V)\} = \frac{27}{8}G_2$ alone. The other four equations are solved by $h_3 = 2h_6 - \alpha_4$, $h_1 = \beta_1 + \frac{1}{4}\beta_4 - \frac{1}{2}h_6 + \frac{1}{4}\alpha_4 - \frac{1}{2}\alpha_1$, $h_4 = 4\beta_2 + \beta_4 - 2h_2 - 2\alpha_3$, $h_5 = \alpha_5 + 2y_V - 2\beta_3 + \beta_4$, with h_2, h_6, y_V, y_W free; with $\alpha_1 = \alpha_3 = \frac{11}{4} + \frac{19}{4}\log\frac{3}{2}$, $\alpha_4 = 0$, $\alpha_5 = -9\log\frac{3}{2}$ this is the displayed family, with $\kappa_1 = \frac{1}{4}\beta_4 - \frac{1}{2}h_6$, $\kappa_2 = h_2 - 2\beta_2$, $\kappa_3 = y_V$, $\kappa_4 = y_W$. For the last choice $h_1 = -\frac{1}{2}\alpha_1 + \frac{1}{2}\alpha_1 = 0$, $h_2 = \alpha_1 - \alpha_3 = 0$, $h_4 = -2\alpha_3 + 2\alpha_3 = 0$, $h_5 = \alpha_5 - \alpha_5 = 0$ and $h_3 = h_6 = 0$. □

Theorem 9.12 (Wess–Zumino condition)**.** *The part with three legs of* $s\,\mathrm{Ob} = 0$ *(Theorem 8.8 (i)) at arity zero is*

$$Q\big(\textit{cubic part of } \hbar\,\mathrm{Ob}^{(0)}\big) + \big\{I, \textit{quadratic part of } \hbar\,\mathrm{Ob}^{(0)}\big\} = \big(\alpha_2 - \tfrac{27}{8}\big)\,QG_2 \textit{ units},$$

with the coefficient $-\frac{27}{8}$ *of the quadratic part (Theorem 4.30). The coefficients* $\alpha_1, \alpha_3, \alpha_4, \alpha_5$ *enter it with coefficient* 0*, and the computed value* $\alpha_2 = \frac{27}{8}$ *satisfies it. The equation follows from* $Q^2 = 0$*, the vanishing image under* Q *of the cubic part of* I *and of the quadratic part of* $\hbar\,\mathrm{Ob}^{(0)}$*, and Proposition 9.8; it is equivalent to the solvability of Theorem 9.11, since the image of the cubic functionals and of the brackets with the* Q*-closed quadratic counterterms is* $\mathrm{span}\{G_1, G_3, G_4, G_5\}$*.*

Proof. Lemma 9.6 gives $Q\sum_i \alpha_i G_i = \alpha_2 QG_2$ and $\{I, -\frac{27}{8}\int(d^*dc, d^*A|_V)\} = -\frac{27}{8}QG_2$. For the second statement, Q acts on plane-wave values by polynomial substitutions, so Q applied to the limit of Q of the order-$\hbar$ part of $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ is the limit of Q^2 of it, which vanishes. By Theorem 9.10 and Proposition 9.8 the part with three legs of that limit is the cubic part of $\hbar\,\mathrm{Ob}^{(0)}$ plus the tree of its quadratic part with the cubic part of I. By Theorem 3.25, $(Q\otimes 1 + 1\otimes Q)P(0,\infty) = K_0$, so Q applied to that tree is the tree of the images under Q plus the bracket of the two vertices; the quadratic part $-\frac{27}{8}\int(d^*dc, d^*A|_V)$ has the image $\frac{27}{8}\int(d^*dc, d^*dc) = 0$ (c is odd and the pairing symmetric), and the cubic part of I has image 0. What remains is the displayed sum. □

Remark 9.13 (Two-component form)**.** With the components of Definition 8.1, the pairing of $\mathfrak{g}$ written as $(-,-)$ around the components and the factors in the order of the index-free expressions,

$$\begin{aligned}
&\int \mathrm{dvol}\,(\lambda,\, \not{d}\lambda) = 2i\int d^4x\,\big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}\partial_m {\lambda_\alpha}^i\big),\\
&G_5 = i\int d^4x\Big\{\big([c, {\lambda_\alpha}^i], \bar\sigma^{m\dot\alpha\alpha}\partial_m\bar\lambda_{\dot\alpha i}\big) + \big([c,\bar\lambda_{\dot\alpha i}], \bar\sigma^{m\dot\alpha\alpha}\partial_m {\lambda_\alpha}^i\big)\Big\},\\
&H_5 = \int d^4x\Big\{\big([c, {\lambda_\alpha}^i], {\lambda^*}^\alpha{}_i\big) - \big([c,\bar\lambda_{\dot\alpha i}], \bar\lambda^{*\dot\alpha i}\big)\Big\},\\
&\int \mathrm{dvol}\,(\lambda, \rho([A|_V,\lambda])) = i\int d^4x\Big\{\big({\lambda_\alpha}^i, \bar\sigma^{m\dot\alpha\alpha}[A_m,\bar\lambda_{\dot\alpha i}]\big) + \big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}[A_m, {\lambda_\alpha}^i]\big)\Big\},\\
&\int \mathrm{dvol}\,(\lambda, \rho([A|_{W_\mathbb{R}^\perp},\lambda])) = \int d^4x\Big\{\big({\lambda_\alpha}^j, \varepsilon^{\alpha\beta}[A_{ji}, \lambda_\beta{}^i]\big) - \big(\bar\lambda_{\dot\alpha j}, \varepsilon^{\dot\alpha\dot\beta}[A^{ji}, \bar\lambda_{\dot\beta i}]\big)\Big\},
\end{aligned}$$

with $A_{ij} = A_I\Sigma^I_{ij}$ and $A^{ij} = \frac{1}{2}\varepsilon^{ijkl}A_{kl}$; they follow from $(\zeta_1, \rho(v)\zeta_2) = i\,v_m\big({\zeta_{1\alpha}}^i\bar\sigma^{m\dot\alpha\alpha}\bar\zeta_{2\dot\alpha i} + \bar\zeta_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}{\zeta_{2\alpha}}^i\big)$ and $(\zeta_1, \rho(w)\zeta_2) = {\zeta_{1\alpha}}^j\varepsilon^{\alpha\beta}w_{ji}{\zeta_{2\beta}}^i - \bar\zeta_{1\dot\alpha j}\varepsilon^{\dot\alpha\dot\beta}w^{ji}\bar\zeta_{2\dot\beta i}$ for $\zeta_1, \zeta_2 \in \Sigma$, $v \in V$, $w \in W_\mathbb{R}^\perp$, the first identity after an integration by parts. The plane-wave value of G_5 is $f\,p_{0m}\big({\lambda_{1\alpha}}^i\bar\sigma^{m\dot\alpha\alpha}\bar\lambda_{2\dot\alpha i} + \bar\lambda_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}{\lambda_{2\alpha}}^i\big)$, and, with the orientation of Proposition 8.2, the parity-odd structure of (c,λ,λ) is $i\int d^4x\big\{([c,\bar\lambda_{\dot\alpha i}], \bar\sigma^{m\dot\alpha\alpha}\partial_m{\lambda_\alpha}^i) - ([c,{\lambda_\alpha}^i], \bar\sigma^{m\dot\alpha\alpha}\partial_m\bar\lambda_{\dot\alpha i})\big\}$, the difference of the two chiral terms of G_5. In particular the fermionic coefficients of (9.1) and (9.2) read $-9\log\frac{3}{2}$ units times the two-component G_5 and $-9i\log\frac{3}{2}$ units times $\int d^4x\,(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}\partial_m{\lambda_\alpha}^i)$. On the Dirac column $({\lambda_\alpha}^i, \bar\lambda^{\dot\alpha}{}_i)$, $\rho(v) = \gamma_W^\mu v_\mu$ gives Weinberg's form of each line.

Remark 9.14 (Method of computation)**.** The value of each cycle graph is an exact element of $\mathbb{Q} + \mathbb{Q}\log\frac{3}{2}$: the coefficient of ε^0 of Theorem 9.9 reduces to master integrals over the scaled Schwinger region, evaluated in closed form, and each multilinear form is decomposed on the basis of Proposition 9.3 by exact evaluation on integer plane waves in explicit realisations of V_{10}, Σ over $\mathbb{Q}(i)$ and $\mathfrak{g} = \mathfrak{su}(2)$, as in Remark 4.36; the colour reduction of Lemma 9.2 makes the result valid for every simple $\mathfrak{g}$. The rows are organised by the census of Proposition 9.7; the complete order-$\hbar$ sums at finite ε in the sectors $(c, A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ and $(c, c, A^*|_V)$ have the limits α_3, α_4 and the reducible tree.

9.2 Closed forms along the dilatation

In the construction every component along Δ is the coefficient of ε^0 of the defect cycles of the defect vertex of Lemma 8.4, with $S^{(1)}(\Delta)$ placed on the cycle at the special edge or as a line (Proposition 9.25 (ii), Theorem 11.44). The closed forms of this section serve two purposes: they give the line placement, the composite kernel, and they give a second evaluation of the same flow, through Q of the renormalised amplitudes, whose agreement with the defect cycles is a theorem of the construction (Proposition 9.25 (iii), Theorem 10.38). Along Δ the generator $S^{(1)}(\Delta)$ is quadratic (Example 2.13), and the regulator moves under it through its two endpoints (Proposition 7.10 at $T=\infty$, Proposition 9.21 at finite T). The graphs with one vertex $S^{(1)}(\Delta)$ therefore have closed forms at finite ε and T. In this section and the next two, $v=2E$, so that $v.$ is the operator of Lemma 7.5 on all slots, and $P(\varepsilon,T)\star=\int_\varepsilon^T Q^{GF}e^{-tH}dt$ is the operator represented by $P(\varepsilon,T)$ (Definition 8.3). For an operator X on $\mathscr{E}$ the kernel $(X\otimes 1)K_0$ represents X, so $P(\varepsilon,T)=(P(\varepsilon,T)\star\otimes 1)K_0$. The operator ∂_ϕ is read as in Definition 3.16: on the product of a coordinate functional of the first factor and one of the second, in this order, it is the corresponding component of ϕ. A linear operator X on $\mathscr{E}$ acts on functionals as the derivation induced by $\ell\mapsto\ell\circ X$ on linear functionals ℓ [C07, §6.2], as Q does, and XF denotes this derivation applied to F. For a functional F and an even J, the linearised flow of F at J is $e^{-W(P(\varepsilon,T),J)/\hbar}e^{\hbar\partial_{P(\varepsilon,T)}}\big(F\,e^{J/\hbar}\big)$; for even F it is $\partial_\delta W(P(\varepsilon,T),J+\delta F)$ with δ a square-zero parameter of degree 0, as in (7.1).

Lemma 9.15 (Operators and kernels)**.** *For a kernel $\phi\in\mathscr{E}\otimes\mathscr{E}$ and a linear operator X on $\mathscr{E}$, of either parity,*

$$[\partial_\phi,X]=\partial_{(X\otimes 1+1\otimes X)\phi} \tag{9.3}$$

as a graded commutator of operators on functionals; for $X=Q$ this is [C07, §6.4]. If X is even and preserves $\langle\,,\rangle$, then $(1\otimes X)K_0=-(X\otimes 1)K_0$, and $(1\otimes Q^{GF})K_0=(Q^{GF}\otimes 1)K_0$; the same holds with any function of H in place of Q^{GF}.

Proof. Both sides of (9.3) are second-order operators with constant coefficients, determined by their values on products $\ell_1\ell_2$ of linear functionals. On such a product ∂_ϕ gives the constant $(\ell_1\otimes\ell_2)(\phi)$, on which X vanishes, and $X(\ell_1\ell_2)=(\ell_1\circ X)\ell_2+(-1)^{|X||\ell_1|}\ell_1(\ell_2\circ X)$, so the left side gives $(\ell_1\otimes\ell_2)\big((X\otimes 1+1\otimes X)\phi\big)$, the sign being the Koszul sign of X passing ℓ_1. Preservation of the pairing by an even X is $(X\otimes 1+1\otimes X)K_0=0$, as in Proposition 7.6; for Q^{GF} the identity is Proposition 3.8, and H is symmetric for the pairing. □

Lemma 9.16 (Brackets with $S^{(1)}(\Delta)$)**.**

- *(i) For every functional F, $\{S^{(1)}(\Delta),F\}=v.F$: the sign of Definition 7.4 is $+1$. For a local functional $\int\mathrm{dvol}\,(\text{density})$ of Weyl weight k (fields plus derivatives) $v.$ multiplies it by $2(k-4)$, and for a local operator O of Weyl weight w_O, $v.O=2(E+w_O)O$ (Proposition 6.18).*
- *(ii) For $t>0$, $\{S^{(1)}(\Delta),F\}_t=(e^{-tH}v.)F$.*
- *(iii) $Q\,S^{(1)}(\Delta)=0$, $\{I,S^{(1)}(\Delta)\}=0$ and $\partial_{P(\varepsilon,T)}S^{(1)}(\Delta)=0$.*
- *(iv) For every functional F, $\partial_{P(\varepsilon,T)}\big(S^{(1)}(\Delta)F\big)-S^{(1)}(\Delta)\,\partial_{P(\varepsilon,T)}F=\big(P(\varepsilon,T)\star v.\big)F$.*

Proof. (i) By Definition 1.6 and the proof of Lemma 4.21, the bracket is at each position that of [C07, §10.1] for a point, with $\Delta_0=\sum\partial_{x_i}\partial_{\xi_i}$ over the even members x_i and their odd partners ξ_i. For a number a,

$$\{a\,x_i\xi_i,x_i\}=\Delta_0(a\,x_i^2\xi_i)-\Delta_0(a\,x_i\xi_i)\,x_i=a\,x_i,\qquad \{a\,x_i\xi_i,\xi_i\}=-a\,\xi_i.$$

The terms of $S^{(1)}(\Delta)$ (Example 2.13) are of this form: $(v.A,A^*)$ with the even member first, and $-(v.\lambda,\lambda^*)$, $-(v.c,c^*)$ with the even members λ^*, c^* paired against the image of the partner. With the action of $v.$ on the antifields of Definition 7.4, the bracket sends every coordinate functional ℓ to $\ell\circ v.$,

and it is a derivation [C07, Lemma 4.3.2]. Since $[E, \partial_m] = -\partial_m$, $v.$ multiplies a monomial of Weyl weight k by $2E + 2k$, and $\int \mathrm{dvol}\, E(\cdot) = -4 \int \mathrm{dvol}\,(\cdot)$. (ii) $\{-,-\}_t$ is built from $K_t = (e^{-tH} \otimes 1)K_0$ (Definition 3.21), which replaces the contracted leg of $S^{(1)}(\Delta)$ by its image under e^{-tH}. (iii) $Q\, S^{(1)}(\Delta) = \{\frac{1}{2}\langle e, Qe\rangle, S^{(1)}(\Delta)\}$ (Proposition 1.11) is, up to sign, $v.$ of $\frac{1}{2}\langle e, Qe\rangle$, which vanishes by $[v., Q] = 0$ and the preservation of the pairing (Proposition 7.10); $\{I, S^{(1)}(\Delta)\}$ is, up to sign, $v.I = 0$, every density of I having Weyl weight 4; and $P(\varepsilon, T)$ has no component in the slot pairs $A \otimes A^*$, $\lambda \otimes \lambda^*$, $c \otimes c^*$ of $S^{(1)}(\Delta)$ (Proposition 3.17). (iv) The left side contracts one leg of $S^{(1)}(\Delta)$ with one leg of F through $P(\varepsilon, T)$; it is a derivation of F, and on a linear functional ℓ it gives $\ell \circ (P(\varepsilon, T) \star v.)$, the antifield leg of $S^{(1)}(\Delta)$ being contracted and its field leg carrying $v.$. □

Definition 9.17 (The composite kernel)**.** For $0 < \varepsilon < T \leq \infty$ the composite kernel is $\big(P(\varepsilon, T) \star v. \otimes 1 + 1 \otimes P(\varepsilon, T) \star v.\big)P(\varepsilon, T)$: the two legs of $S^{(1)}(\Delta)$ joined, in both orders, to two edges $P(\varepsilon, T)$.

Proposition 9.18 (The composite kernel)**.** *Write $P = P(\varepsilon, T)$.*

- *(i) The composite kernel equals $2\big(P\star\, v.\, P\star \otimes 1\big)K_0$, the kernel of the even operator $2P\star\, v.\, P\star = v.\,(P\star)^2 + (P\star)^2\, v.$, with $(P\star)^2 = (Q^{GF})^2\big(\int_\varepsilon^T e^{-tH} dt\big)^2$; it has degree -1.*
- *(ii) Its components lie in $c \otimes A|_V$ and $A|_V \otimes c$: in a graph it joins a leg c to a leg $A|_V$. It is nonzero because $(Q^{GF})^2 A^* = 2\, d^*(A^*|_V)$ and $(Q^{GF})^2 c^* = -2\, dc^*$ (Proposition 3.10).*
- *(iii) $Q\, P\star + P\star\, Q = e^{-\varepsilon H} - e^{-TH}$ and $Q\,(P\star\, v.\, P\star) = (P\star\, v.\, P\star)\, Q$. Hence $(Q \otimes 1 + 1 \otimes Q)$ annihilates the composite kernel, and ∂ of the composite kernel anticommutes with Q.*
- *(iv) At $T = \infty$, in momentum space, $(P\star)^2 = (Q^{GF})^2 e^{-2\varepsilon|p|^2}|p|^{-4}$; on the slot of $A^*|_V$ it is of order $|p|^{-3}$ as $p \to 0$, and so is the composite kernel, $v.$ acting as an Euler operator plus constants. At finite T, $(P\star)^2$ is bounded by a multiple of $T^2|p|$.*

Proof. (i) $(P\star v.\otimes 1)P = (P\star v.P\star\otimes 1)K_0$. For the second term, the Koszul sign of two odd operators on different factors and Lemma 9.15 give

$$(1\otimes P\star v.)(P\star\otimes 1)K_0 = -(P\star\otimes 1)(1\otimes Q^{GF})\Big(1\otimes \int_\varepsilon^T e^{-tH} dt\Big)(1\otimes v.)K_0 = (P\star\otimes 1)(v.\, P\star\otimes 1)K_0 = (P\star v.P\star\otimes 1)K_0.$$

By Proposition 7.10, $[v., Q^{GF}] = -4Q^{GF}$ and $[v., H] = -4H$, so for a function of H the commutator with $v.$ is $-4H$ times its derivative in H, and $[v., P\star]$ is Q^{GF} times a function of H. Since Q^{GF} commutes with H, $P \star [v., P\star] - [v., P\star]P\star$ is $(Q^{GF})^2$ times the commutator of two functions of H, which vanishes; this is $2P \star v.P \star -v.(P\star)^2 - (P\star)^2 v.$. The degree is that of K_0 minus 2, $P\star$ having degree -1 and $v.$ degree 0. (ii) $(Q^{GF})^2$ vanishes on the slots of c, A, λ, λ^*, and $v.$ preserves the slots; the components of K_0 in $A^* \otimes A$ and $c^* \otimes c$ go to $c \otimes A|_V$ and $A|_V \otimes c$. (iii) $QQ^{GF} + Q^{GF}Q = H$ and $H \int_\varepsilon^T e^{-tH} dt = e^{-\varepsilon H} - e^{-TH}$ give the first identity. With $[v., Q] = 0$ it gives

$$Q\, P\star v.P\star - P\star v.P\star\, Q = (e^{-\varepsilon H} - e^{-TH})\, v.\, P\star - P\star\, v.\,(e^{-\varepsilon H} - e^{-TH}),$$

and moving $v.$ to the right by the rule of (i), the right side is $-4Q^{GF}$ times $(e^{-\varepsilon H} - e^{-TH})\frac{d}{dH}\big(H \int_\varepsilon^T e^{-tH} dt\big) - H \int_\varepsilon^T e^{-tH} dt\, \frac{d}{dH}(e^{-\varepsilon H} - e^{-TH})$, which vanishes because $H \int_\varepsilon^T e^{-tH} dt = e^{-\varepsilon H} - e^{-TH}$. For the even operator $P\star v.P\star$, $(Q \otimes 1 + 1 \otimes Q)(P\star v.P\star\otimes 1)K_0 = \big((Q\, P\star v.P\star - P\star v.P\star\, Q)\otimes 1\big)K_0 = 0$, and (9.3) gives the last statement. (iv) In momentum space $\int_\varepsilon^\infty e^{-t|p|^2} dt = e^{-\varepsilon|p|^2}|p|^{-2}$, and on $A^*|_V$, $(Q^{GF})^2$ is $-2i\,\iota_p$, of order $|p|$; at finite T the integral is bounded by T. □

Proposition 9.19 (Graphs with one vertex $S^{(1)}(\Delta)$)**.** *Write $P = P(\varepsilon, T)$, let J be even and at least cubic modulo $\hbar$, and $W = W(P, J)$. As operators on functionals, with $S^{(1)}(\Delta)$ acting by multiplication,*

$$e^{\hbar\partial_P}\, S^{(1)}(\Delta) = \Big(S^{(1)}(\Delta) + \hbar\, P\star\, v. + \tfrac{\hbar^2}{2}\,\partial_{2(P\star v.P\star\otimes 1)K_0}\Big)\, e^{\hbar\partial_P}, \tag{9.4}$$

and the linearised flow of $S^{(1)}(\Delta)$ at J is

$$\begin{aligned} e^{-W/\hbar}e^{\hbar\partial_P}\big(S^{(1)}(\Delta)\, e^{J/\hbar}\big) &= e^{-W/\hbar}\Big(S^{(1)}(\Delta) + \hbar\, P\star\, v. + \tfrac{\hbar^2}{2}\,\partial_{2(P\star v.P\star\otimes 1)K_0}\Big)e^{W/\hbar} \\ &= S^{(1)}(\Delta) + (P\star\, v.)W + \tfrac{\hbar}{2}\,\partial_{2(P\star v.P\star\otimes 1)K_0}W + \tfrac{1}{4}\Big(\partial_{2(P\star v.P\star\otimes 1)K_0}(W^2) - 2W\,\partial_{2(P\star v.P\star\otimes 1)K_0} \end{aligned} \tag{9.5}$$

The four terms are the vertex $S^{(1)}(\Delta)$ alone, $S^{(1)}(\Delta)$ joined by one edge P to each leg of W in turn, two legs of W closed by the composite kernel, and two copies of W joined by it.

Proof. As operators, $e^{\hbar\partial_P}S^{(1)}(\Delta)e^{-\hbar\partial_P} = \sum_n \frac{\hbar^n}{n!}\mathrm{ad}^n_{\partial_P}\big(S^{(1)}(\Delta)\big)$. By Lemma 9.16 (iii), (iv), $\mathrm{ad}_{\partial_P}(S^{(1)}(\Delta)) = P\star v.$, and by (9.3) and Proposition 9.18 (i), $\mathrm{ad}_{\partial_P}(P\star v.)$ is ∂ of the composite kernel, a second-order operator with constant coefficients which commutes with ∂_P; the series stops, which is (9.4). Insert (9.4) with $e^{\hbar\partial_P}e^{J/\hbar} = e^{W/\hbar}$. The derivation gives $\hbar(P\star v.)e^{W/\hbar} = \big((P\star v.)W\big)e^{W/\hbar}$, and the product rule of Definition 3.21 for the odd second-order operator gives $\partial(e^{W/\hbar}) = \big(\hbar^{-1}\partial W + \frac{1}{2}\hbar^{-2}(\partial(W^2) - 2W\,\partial W)\big)e^{W/\hbar}$. □

Proposition 9.20 (The image under Q)**.** *In the setting of Proposition 9.19,*

$$Q\,(P\star\, v.) + (P\star\, v.)\,Q = \{S^{(1)}(\Delta), -\}_\varepsilon - \{S^{(1)}(\Delta), -\}_T, \tag{9.6}$$

and, with $[-, QW]$ the graded commutator with the multiplication by QW,

$$Q\Big(e^{-W/\hbar}e^{\hbar\partial_P}\big(S^{(1)}(\Delta)e^{J/\hbar}\big)\Big) = \{S^{(1)}(\Delta), W\}_\varepsilon - \{S^{(1)}(\Delta), W\}_T - \hbar^{-1}e^{-W/\hbar}\Big[S^{(1)}(\Delta) + \hbar\, P\star v. + \tfrac{\hbar^2}{2}\,\partial_{2(P\star v.P\star\otimes 1)K_0},\, QW \tag{9.7}$$

The last term consists of $(P\star v.)(QW)$, $\frac{\hbar}{2}\,\partial(QW)$ and $\frac{1}{2}\big(\partial((QW)W) - (\partial\, QW)W + (QW)\,\partial W\big)$, with ∂ the operator of the composite kernel: the graphs of (9.5) *with QW in place of one copy of W, with the signs reversed.*

Proof. The graded commutator of the derivations induced by two odd operators is induced by their anticommutator, and $Q\,P\star v. + P\star v.\,Q = (e^{-\varepsilon H} - e^{-TH})v.$ by Proposition 9.18 (iii) and $[v., Q] = 0$; Lemma 9.16 (ii) gives (9.6). The operator $S^{(1)}(\Delta) + \hbar P\star v. + \frac{\hbar^2}{2}\partial$ in the bracket of (9.7) is odd, and its graded commutator with Q is $\hbar\big(\{S^{(1)}(\Delta), -\}_\varepsilon - \{S^{(1)}(\Delta), -\}_T\big)$, by $Q\,S^{(1)}(\Delta) = 0$, (9.6) and Proposition 9.18 (iii). Q is a derivation and $Qe^{W/\hbar} = \hbar^{-1}(QW)e^{W/\hbar}$, so Q of $e^{-W/\hbar}$ times this operator applied to $e^{W/\hbar}$ consists of $-\hbar^{-1}(QW)$ times the linearised flow, of $e^{-W/\hbar}\hbar\big(\{S^{(1)}(\Delta), -\}_\varepsilon - \{S^{(1)}(\Delta), -\}_T\big)e^{W/\hbar} = \{S^{(1)}(\Delta), W\}_\varepsilon - \{S^{(1)}(\Delta), W\}_T$, and of minus $\hbar^{-1}e^{-W/\hbar}$ times the operator applied to $(QW)e^{W/\hbar}$. Moving the multiplication by QW to the left through the odd operator produces the graded commutator and $-(QW)$ times the linearised flow, which cancels the first term. The expansion of the commutator uses $[S^{(1)}(\Delta), QW] = 0$ for two odd functionals and the product rule for ∂. □

Proposition 9.21 (Scaling Ward identity)**.** *For $0 < \varepsilon < T \leq \infty$,*

$$(v.\otimes 1 + 1\otimes v.)\,P(\varepsilon, T) = -4\Big(\varepsilon\frac{\partial}{\partial\varepsilon} + T\frac{\partial}{\partial T}\Big)P(\varepsilon, T), \tag{9.8}$$

and for every even J, with δ a square-zero parameter of degree 0 and the derivatives in ε, T acting on $P(\varepsilon, T)$,

$$\{S^{(1)}(\Delta), W(P(\varepsilon, T), J)\} = \partial_\delta W\big(P(\varepsilon, T), J + \delta\{S^{(1)}(\Delta), J\}\big) + 4\Big(\varepsilon\frac{\partial}{\partial\varepsilon} + T\frac{\partial}{\partial T}\Big)W\big(P(\varepsilon, T), J\big). \tag{9.9}$$

Proof. By Proposition 7.10, $(v.\otimes 1+1\otimes v.)L_t = -4L_t - 4t\frac{d}{dt}L_t$; integration over (ε, T) gives $-4[tL_t]_\varepsilon^T = 4\varepsilon L_\varepsilon - 4TL_T$, and $\partial_\varepsilon P(\varepsilon,T) = -L_\varepsilon$, $\partial_T P(\varepsilon,T) = L_T$. By (9.3) with $X = v.$, $v.\partial_P = \partial_P v. - \partial_{(v.\otimes 1+1\otimes v.)P}$, and the operators ∂_ϕ commute, so $v.\, e^{\hbar\partial_P} = e^{\hbar\partial_P} v. + 4\big(\varepsilon\partial_\varepsilon + T\partial_T\big)e^{\hbar\partial_P}$. Apply this to $e^{J/\hbar}$, use $v.\, e^{J/\hbar} = \hbar^{-1}(v.J)e^{J/\hbar}$ and Lemma 9.16 (i), and divide by $\hbar^{-1}e^{W/\hbar}$. □

Corollary 9.22 (Callan–Symanzik form at a finite endpoint)**.** *Let O be a homogeneous local operator, $J = I - I^{CT}[\varepsilon] - \hbar F_1^{(0)} + \delta\, O(\varepsilon)$ with $\hbar F_1^{(0)}$ of Weyl weight 4 and $O(\varepsilon)$ of Definition 8.11 with $\psi = 0$. Modulo $\hbar^2$,*

$$\{S^{(1)}(\Delta), O[T]\} - 4T\frac{\partial}{\partial T}O[T] = \big(\{S^{(1)}(\Delta), O\}\big)[T] + 4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[T] \\ + 4\big(\textit{tree insertion at scale } T \textit{ of the coefficient of } \log\varepsilon \textit{ in } I^{CT}[\varepsilon] \textit{ into } O[T]\big).$$

At $T = \infty$ this is twice Theorem 6.6.

Proof. By (9.9), $\{S^{(1)}(\Delta), -\}$ minus $4\varepsilon\frac{d}{d\varepsilon} + 4T\partial_T$ of the flow, with the total derivative in ε, equals $\partial_\delta W\big(P, J + \delta(v.J - 4\varepsilon\frac{dJ}{d\varepsilon})\big)$. Here $v.I = 0$ and $v.\hbar F_1^{(0)} = 0$ (Lemma 9.16 (i)). At one loop $I^{CT}[\varepsilon]$ is ε^{-1} times a functional of weight 2, multiplied by -4 under $v.$, plus $\log\varepsilon$ times a functional of weight 4, so $(v. - 4\varepsilon\frac{d}{d\varepsilon})I^{CT}[\varepsilon] = -4$ times the coefficient of $\log\varepsilon$; by Lemma 6.4 the same computation gives $(v. - 4\varepsilon\frac{d}{d\varepsilon})O(\varepsilon) = (v.O)(\varepsilon) + 4$ times the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$, up to constants, which drop out of the flow (Remark 3.20). The flow has a limit as $\varepsilon \to 0$ and $\varepsilon\frac{d}{d\varepsilon}$ of it tends to 0, its dependence on ε at one loop being a sum of terms $\varepsilon^a \log^b\varepsilon$ with $a > 0$ (Lemma 4.6); take ∂_δ and the limit. At $T = \infty$, $v. = 2s\frac{d}{ds}\big|_{s=1}\sigma_s$ on functionals (Definition 6.1). □

9.3 The arity-one equation along the dilatation

Definition 9.23 (Arity-one data along Δ)**.** Let $\hbar F_1^{(0)}$ be an arity-zero finite counterterm containing the finite counterterm of Theorem 4.32, $J = I - I^{CT}[\varepsilon] - \hbar F_1^{(0)}$, and let $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ and $\hbar F_1^{(1)}(\Delta)$ be the arity-one parts along Δ of the counterterms and of the finite counterterm of Definition 8.5, evaluated as in Chapter 2, so that the arity-one part of $I_{\mathfrak{psl}(4|4)}$ along Δ is $S^{(1)}(\Delta)$. The arity-one part along Δ of $W\big(P(\varepsilon,T), I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \hbar F_1\big)$ is the linearised flow at J of the arity-one vertex

$$S^{(1)}(\Delta) - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta) - \hbar F_1^{(1)}(\Delta).$$

The arity-one part along Δ of the obstruction of Definition 8.7, $\hbar\,\mathrm{Ob}^{(1)}(\Delta)$, is, since the vertex $\hbar F_1^{(0)}$ adds $-\{\hbar F_1^{(0)}, S^{(1)}(\Delta)\}_\varepsilon$, whose limit is $\{S^{(1)}(\Delta), \hbar F_1^{(0)}\} = 0$ (Lemma 9.16 (i)), the local functional of degree 0 whose tree-level flow with edges $P(0,\infty)$ is the order-$\hbar$ part of the $\varepsilon \to 0$ limit of Q applied to the linearised flow at J of $S^{(1)}(\Delta) - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$, with edges $P(\varepsilon,\infty)$. By Lemma 8.4 this limit is the flow of the arity-one defect vertex

$$\{I, S^{(1)}(\Delta)\}_\varepsilon - \big\{I^{CT}[\varepsilon] + \hbar F_1^{(0)}, S^{(1)}(\Delta)\big\}_\varepsilon - Q\, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta) - \big\{I, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)\big\}_\varepsilon$$

modulo constants and $\hbar^2$, the arity-one part along Δ of (7.4) with the finite counterterm included.

On the line $\mathbb{C}\Delta$, $d_{CE} = 0$, and $\{S^{(1)}(\Delta), \hbar F_1^{(0)}\} = 0$ by Lemma 9.16 (i). By Theorem 8.8 (ii) and Proposition 8.25, the equivariant quantum master equation holds at arity one along Δ exactly when

$$(Q + \{I, -\})\,\hbar F_1^{(1)}(\Delta) = \hbar\,\mathrm{Ob}^{(1)}(\Delta). \tag{9.10}$$

Lemma 9.24 (Components of s and the Wess–Zumino identities)**.** *Write $\Lambda^{(0)} = Q + \{I, -\}$ on local functionals and $S(x) = S^{(1)}(x)$, in the normalisation of Lemma 2.11. For a local cochain F of degree 0 and a local cochain G of degree 1,*

$$\begin{aligned}
(sF)^{(1)}(x) &= \Lambda^{(0)}F^{(1)}(x) + \{F^{(0)}, S(x)\},\\
(sF)^{(2)}(x,y) &= \Lambda^{(0)}F^{(2)}(x,y) + F^{(1)}([x,y]) + (-1)^{|x|}\big(\{F^{(1)}(x), S(y)\} + \{S(x), F^{(1)}(y)\}\big) + \{F^{(0)}, S^{(2)}(x,y)\},\\
(sG)^{(1)}(x) &= \Lambda^{(0)}G^{(1)}(x) + \{G^{(0)}, S(x)\},\\
(sG)^{(2)}(x,y) &= \Lambda^{(0)}G^{(2)}(x,y) - G^{(1)}([x,y]) + (-1)^{|x|}\{G^{(1)}(x), S(y)\} - \{S(x), G^{(1)}(y)\} + \{G^{(0)}, S^{(2)}(x,y)\}.
\end{aligned} \tag{9.11}$$

Hence the one-loop equation $sF_1 = \mathrm{Ob}$ of Theorem 8.8 (ii) reads, through arity two,

$$\begin{aligned}
&\Lambda^{(0)}F_1^{(0)} = \mathrm{Ob}^{(0)}, \qquad \Lambda^{(0)}F_1^{(1)}(x) + \{F_1^{(0)}, S(x)\} = \mathrm{Ob}^{(1)}(x),\\
&\Lambda^{(0)}F_1^{(2)}(x,y) + F_1^{(1)}([x,y]) + (-1)^{|x|}\big(\{F_1^{(1)}(x), S(y)\} + \{S(x), F_1^{(1)}(y)\}\big) + \{F_1^{(0)}, S^{(2)}(x,y)\} = \mathrm{Ob}^{(2)}(x,y),
\end{aligned} \tag{9.12}$$

and $s\,\mathrm{Ob} = 0$ (Theorem 8.8 (i)) gives the Wess–Zumino identities

$$\begin{aligned}
\Lambda^{(0)}\mathrm{Ob}^{(1)}(x) &= -\{\mathrm{Ob}^{(0)}, S(x)\},\\
\Lambda^{(0)}\mathrm{Ob}^{(2)}(x,y) &= \mathrm{Ob}^{(1)}([x,y]) - (-1)^{|x|}\{\mathrm{Ob}^{(1)}(x), S(y)\} + \{S(x), \mathrm{Ob}^{(1)}(y)\} - \{\mathrm{Ob}^{(0)}, S^{(2)}(x,y)\}.
\end{aligned}$$

If $F_1^{(0)}$ and $F_1^{(1)}$ solve the equations of arity zero and one, the arity-two residual

$$\mathcal{R}^{(2)}(x,y) := \mathrm{Ob}^{(2)}(x,y) - F_1^{(1)}([x,y]) - (-1)^{|x|}\big(\{F_1^{(1)}(x), S(y)\} + \{S(x), F_1^{(1)}(y)\}\big) - \{F_1^{(0)}, S^{(2)}(x,y)\}$$

is $\Lambda^{(0)}$-closed, and the arity-two equation along (x,y) is solvable exactly when its class in the cohomology $H^{-1}(\Lambda^{(0)})$ of the local functionals with its quantum numbers vanishes.

Proof. The formulas for sF are the derivatives of the component formulas of Lemma 2.11 along $S_{\mathfrak{psl}(4|4)} + tF$, since $\frac{d}{dt}\frac{1}{2}\{f,f\} = \{f, \dot f\}$ for even f. For the bracket of Definition 1.6, $\{f,g\} = \Delta_0(fg) - \Delta_0(f)g - (-1)^{|f|}f\Delta_0(g)$, a derivation $\mathcal{D}$ of degree 1 that commutes with Δ_0 up to sign ($\mathcal{D} = Q$, $\{I,-\}$, $\Lambda^{(0)}$, or $\{F,-\}$ with $|F| = 0$) satisfies

$$\mathcal{D}\{a,b\} = -\{\mathcal{D}a, b\} - (-1)^{|a|}\{a, \mathcal{D}b\},$$

as in the proof of Lemma 10.15. The components of sG are $\Lambda^{(0)}G^{(2)} + \alpha G^{(1)}([x,y]) + \beta\{G^{(1)}(x), S(y)\} + \gamma\{S(x), G^{(1)}(y)\} + \delta\{G^{(0)}, S^{(2)}(x,y)\}$ and $\Lambda^{(0)}G^{(1)}(x) + \delta_1\{G^{(0)}, S(x)\}$ with signs fixed by $s^2 = 0$ on the exact cochains $G = sF$. For $F = F^{(0)}$, $\Lambda^{(0)}\{F^{(0)}, S(x)\} = -\{\Lambda^{(0)}F^{(0)}, S(x)\}$ since $\Lambda^{(0)}S(x) = 0$, so $\delta_1 = 1$. For $F = F^{(1)}$, with $|S(x)| \equiv 1 + |x|$, $\Lambda^{(0)}\big(F^{(1)}([x,y]) + (-1)^{|x|}(\{F^{(1)}(x), S(y)\} + \{S(x), F^{(1)}(y)\})\big) = \Lambda^{(0)}F^{(1)}([x,y]) - (-1)^{|x|}\{\Lambda^{(0)}F^{(1)}(x), S(y)\} + \{S(x), \Lambda^{(0)}F^{(1)}(y)\}$, so $\alpha = \gamma = -1$ and $\beta = (-1)^{|x|}$. For $F = F^{(0)}$ in arity two, $\Lambda^{(0)}S^{(2)}(x,y) = -S([x,y]) - (-1)^{|x|}\{S(x), S(y)\}$ (Lemma 2.11) and the derivation rule for $\{F^{(0)}, -\}$ leave $(\delta - 1)\{\Lambda^{(0)}F^{(0)}, S^{(2)}(x,y)\}$, so $\delta = 1$. The Wess–Zumino identities are $(s\,\mathrm{Ob})^{(1)} = (s\,\mathrm{Ob})^{(2)} = 0$. For $G = \mathrm{Ob} - sF_1$, whose components of arity zero and one vanish, $(sG)^{(2)} = \Lambda^{(0)}G^{(2)} = \Lambda^{(0)}\mathcal{R}^{(2)}$ vanishes, and the arity-two equation is $\Lambda^{(0)}F_1^{(2)}(x,y) = \mathcal{R}^{(2)}(x,y)$. □

Proposition 9.25 (Arity-one counterterm and obstruction along Δ)**.** • *(i) At one loop, $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ is the singular part of the composite cycles $\frac{\hbar}{2}\,\partial_{2(P\star v.P\star\otimes 1)K_0}W(P(\varepsilon,\infty), I)_{\mathrm{tree}}$, $P = P(\varepsilon,\infty)$, in which the composite kernel closes two legs of a tree of I; it is $\log\varepsilon$ times a local functional, with no coefficient of ε^{-m}, $m \geq 1$. Every other one-loop graph with one vertex $S^{(1)}(\Delta)$ has a limit as $\varepsilon \to 0$ at generic momenta.*

- *(ii) (Local formula.) Assume that $\hbar F_1^{(0)}$ solves the arity-zero part of $s(\hbar F_1) = \hbar \operatorname{Ob}$ in the sectors with at most n legs. A defect cycle along Δ is a cycle all of whose vertices lie on it, with exactly one special edge, exactly one vertex $S^{(1)}(\Delta)$ and the other vertices from I, the vertex $S^{(1)}(\Delta)$ placed on a line, its term $(v.A|_V, A^*|_V)$ with both legs on edges P, which is the composite kernel, or at the special edge, with one leg on it. Each defect cycle $\mathcal{C}$ with external legs at generic momenta is $\sum_{m\geq 1} \varepsilon^{-m} a_m(\mathcal{C}) + a_0(\mathcal{C}) + o(1)$ with local coefficients and no term in $\log\varepsilon$, and in the sectors with n legs*

$$\hbar \operatorname{Ob}^{(1)}(\Delta)\big|_{n \text{ legs}} = \sum_{\mathcal{C} \text{ with } n \text{ legs}} a_0(\mathcal{C}), \qquad \sum_{\mathcal{C} \text{ with two legs}} a_1(\mathcal{C}) = \tfrac{9}{4} \text{ units} \cdot \Big\{ \int \operatorname{dvol}(A,A), S^{(1)}(\Delta) \Big\}, \tag{9.13}$$

the other pole sums vanishing. This is the definition of the numbers of the obstruction.

- *(iii) (Second evaluation.) Assume moreover that the $\varepsilon \to 0$ limit of Q of the composite cycles minus $(Q + \{I, -\}) I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ is the tree-level flow of a local functional, the composite part. Then, in the sectors with at most n legs,*

$$\hbar \operatorname{Ob}^{(1)}(\Delta) = 4\big(\text{coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon]\big) + (\text{composite part}) = (Q + \{I,-\})\Big(4 \frac{\hbar \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2 (X,Y)} \Big(2 \int (A, A^*) - \tag{9.14}$$

Proof. (i) By (9.5) at $T = \infty$, the one-loop graphs with one vertex $S^{(1)}(\Delta)$ are $(P \star v.)$ applied to the one-loop part of $W(P,J)$, the composite cycles, and the graphs joining the tree part and the one-loop part of $W(P,J)$ by the composite kernel. The one-loop part of $W(P,J)$ has a limit at generic momenta (Proposition 4.8), and the first and third classes attach to its legs an edge P or the composite kernel at generic momenta, so they have limits; the counterterm, the singular part of the graphs (Definition 8.5), is that of the composite cycles. A coefficient of ε^{-m} of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ is a colour-singlet local functional of degree -1 and Weyl weight $4 - 2m$, the position at weight -1 and at most linear in it; a colour singlet of degree -1 contains an antifield of weight at least $\frac{5}{2}$ and a second adjoint field of weight at least 1, or c^* and c, so its total weight is at least $\frac{5}{2}$, above $4 - 2m \leq 2$.

(ii) By Lemma 8.4 the obstruction is read from the $\varepsilon \to 0$ limit of the connected graphs with one vertex from the defect vertex of Definition 9.23, the other vertices from $I - I^{CT}[\varepsilon] - \hbar F_1^{(0)}$ and $S^{(1)}(\Delta)$, edges $P(\varepsilon, \infty)$. At order $\hbar$ a graph has one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Proposition 3.35). The vertices of order $\hbar^0$ of the defect vertex are special-edge parts of brackets, so the special edge lies on the cycle or on a tree. The graphs are: the defect cycles; the reducible graphs, a defect cycle with fewer legs or a one-loop subgraph of the arity-zero defect joined by edges P to trees of I and $S^{(1)}(\Delta)$; the graphs whose special edge is a bridge; the trees with one counterterm vertex; and the constant $\hbar \Delta_\varepsilon S^{(1)}(\Delta)$. A bridge at a generic momentum p separates a one-loop subgraph G; with the special-edge part of the counterterm bracket across the same edge, its counterterm partner, the sum is $(e^{-\varepsilon|p|^2} - 1)(G - \operatorname{Sing} G)$, $e^{-\varepsilon|p|^2} - 1$ times a convergent graph, which tends to 0; the partners carrying $\log\varepsilon$ give $(e^{-\varepsilon|p|^2} - 1)\log\varepsilon \to 0$. By induction on the number of legs the reducible graphs tend to the trees of the components of the obstruction with fewer legs, the arity-zero two-leg subgraphs being Theorem 4.30. A defect cycle is a sum of simplex integrals over the region where the Schwinger coordinate of the special edge is the smallest, the domains with the cutoff shapes $1/\min_{f \neq \sigma} \alpha_f$ and $1/\min_f \alpha_f$ differing there, so the terms in $\log\varepsilon$ cancel (Lemma 10.24 and the argument of Theorem 11.11, which apply to cycles of vertices of I with the composite kernel of order $|p|^{-3}$, Proposition 9.18 (iv)). The left side has a limit and the defect cycles carry no $\log\varepsilon$, so the terms in $\log\varepsilon$ of the counterterm trees cancel among themselves, their special-edge parts tending to 0; the only pole of the counterterm trees is that of $-\{I^{CT}[\varepsilon]^{(0)}, S^{(1)}(\Delta)\}_\varepsilon$, with $I^{CT}[\varepsilon]^{(0)}$ the counterterm of Theorem 4.16, by (i). The cancellation of the poles gives the pole identity, and the tree-level flow of the part with n legs of the obstruction is

$\sum a_0(\mathcal{C})$ plus the trees of the components with fewer legs; the injectivity of the tree-level flow (Theorem 8.15 (i)) gives (9.13). The vertex $\hbar F_1^{(0)}$ adds $-\{\hbar F_1^{(0)}, S^{(1)}(\Delta)\}_\varepsilon$, whose limit $\{S^{(1)}(\Delta), \hbar F_1^{(0)}\}$ vanishes (Lemma 9.16 (i)).

(iii) Apply (9.7) at $T=\infty$, where $\{-,-\}_\infty = 0$. At order $\hbar$: $\{S^{(1)}(\Delta), W\}_\varepsilon$ differs from $\{S^{(1)}(\Delta), W\}$ by a special edge on a leg, carrying $e^{-\varepsilon|p|^2}-1$ at generic momenta, and $\{S^{(1)}(\Delta), W\} = v.W$ tends by (9.9), as in the proof of Corollary 9.22, to the tree-level flow of 4 times the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$. QW is the flow of the arity-zero defect (Proposition 4.26); its tree part carries a special edge on a tree and tends to 0, and its one-loop part tends to 0 by the arity-zero equation, so $(P\star v.)(QW)$ and the graphs joining QW and W tend to 0, while $-\frac{\hbar}{2}\partial(QW_{\text{tree}}) = Q\big(\frac{\hbar}{2}\partial W_{\text{tree}}\big)$ by Proposition 9.18 (iii). Q of the linearised flow of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ tends to the tree-level flow of its image under $Q+\{I,-\}$, as in Lemma 6.8. Both (ii) and (iii) compute the tree-level flow of the same local functional, which is unique (Theorem 8.15 (i)). The second form is (6.5). □

Theorem 9.26 (The quadratic sector along Δ)**.** *In the sectors $(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$, $(A|_V, A^*|_V)$, (λ,λ^*) and (c,c^*) of degree -1 and the corresponding sectors (A,A), (λ,λ), $(c, A^*|_V)$ of degree 0, with x the position, write the sector parts of $S^{(1)}(\Delta)$ as*

$$S^{(1)}(\Delta)_\perp = \int \big(2E(A|_{W_\mathbb{R}^\perp})+2A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp}\big), \qquad S^{(1)}(\Delta)_\lambda = -\int \big(2E(\lambda)+3\lambda, \lambda^*\big), \qquad S^{(1)}(\Delta)_V = \int \big(2E(A|_V)+2A$$

whose sum is the quadratic part of $S^{(1)}(\Delta)$; each is Q-closed.

- *(i) The quadratic part of the arity-one counterterm is*

  $$I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta) = \frac{\hbar\,\mathrm{tr}_\mathfrak{g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[\int (d(\iota_x A|_V), A^*|_V) - \int(\iota_x dc, c^*) - \int(\rho(x)\not{d}\lambda, \lambda^*)\Big] = \frac{\hbar\,\mathrm{tr}_\mathfrak{g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log$$

  Its coefficient of ε^{-1} vanishes, and so does its part in the sector $(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$.

- *(ii) The quadratic part of the arity-one obstruction, the sum* (9.13) *over the defect cycles with two legs, is*

  $$\hbar\,\mathrm{Ob}^{(1)}(\Delta) = \frac{\hbar\,\mathrm{tr}_\mathfrak{g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[-\tfrac{11}{3}\int(dA,dA) + \tfrac{47}{3}\int(\lambda,\not{d}\lambda) - \tfrac{14}{3}\int(dc,A^*)\Big], \tag{9.15}$$

  $\int(dA,dA) = \int(dA|_V, dA|_V) + \int(dA|_{W_\mathbb{R}^\perp}, dA|_{W_\mathbb{R}^\perp})$*, with vanishing coefficient of $\int(d^*A|_V, d^*A|_V)$ and of every functional with an explicit factor x. The defect cycles with $S^{(1)}(\Delta)$ at the special edge give $\big[-4\int(dA,dA)+16\int(\lambda,\not{d}\lambda)-4\int(dc,A^*)\big]$ units and those with $S^{(1)}(\Delta)$ on a line give $\big[\frac{1}{3}\int(dA,dA)-\frac{1}{3}\int(\lambda,\not{d}\lambda)-\frac{2}{3}\int(dc,A^*)\big]$ units: placement class by placement class these are the first summand and the composite part of* (9.14)*.*

- *(iii) The quadratic part of* (9.10) *is solved exactly by*

  $$\begin{aligned}\hbar F_1^{(1)}(\Delta) = {}& \frac{\hbar\,\mathrm{tr}_\mathfrak{g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{22}{3}\int(A,A^*) - \tfrac{47}{3}\int(\lambda^*,\lambda) - \tfrac{8}{3}\int(c^*,c)\Big] + \textit{units}\cdot\Big(\theta\,S^{(1)}(\Delta)_\perp + \theta_\lambda\,S^{(1)}(\Delta)_\lambda + \theta_V\,\\ &+ \textit{units}\cdot Q\Big(\kappa_V\int(\iota_x A|_V, c^*) + \kappa_\lambda\int(\rho(x)\lambda^*,\lambda^*)\Big),\end{aligned} \tag{9.16}$$

  with free numbers $\theta,\theta_\lambda,\theta_V,\kappa_V,\kappa_\lambda$. The first bracket is the unique translation-invariant solution; it is 4 units times $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ plus $\big[-\frac{2}{3}\int(A,A^*)+\frac{1}{3}\int(\lambda^*,\lambda)+\frac{4}{3}\int(c^*,c)\big]$ units. On the reflection-even functionals the free directions are all the Q-closed quadratic local functionals of degree -1, Weyl weight 4 and degree at most one in x (Proposition 11.24); modulo

Q-exact functionals, the class of $\hbar F_1^{(1)}(\Delta)$ minus the first bracket is $\theta[S^{(1)}(\Delta)_\perp]+\theta_\lambda[S^{(1)}(\Delta)_\lambda]+\theta_V[S^{(1)}(\Delta)_V]$ units. In the sector $(A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$ the solutions are

$$\hbar F_1^{(1)}(\Delta)=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{22}{3}\int(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})+\theta\int\big(2E(A|_{W_{\mathbb{R}}^\perp})+2A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp}\big)\Big],\qquad \theta\ \textit{a number.} \tag{9.17}$$

The numbers $\theta,\theta_\lambda,\theta_V$ are fixed by the arity-two equations to $-\frac{1}{3}$ (Theorem 9.28).

Proof. (i) By Proposition 9.25 (i), the counterterm is the singular part of the composite cycles with two legs, in which the composite kernel closes two legs of a tree of I with four legs; their exact evaluation gives the stated multiple of $\log\varepsilon$, with vanishing coefficient of ε^{-1}, and the graphs with $S^{(1)}(\Delta)$ joined by an edge to the one-loop two-point subgraphs and to the vertex $-I^{CT}[\varepsilon]$ have opposite singular parts. In the sector $(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})$ the exact evaluation of the composite cycle gives the coefficient 0 of $\log\varepsilon$. The second form follows from the substitutions of Proposition 1.11: $Q\int(\iota_xA|_V,c^*)=-\int(\iota_xdc,c^*)+\int(d(\iota_xA|_V),A^*|_V)$, and $-\frac{1}{2}Q\int(\rho(x)\lambda^*,\lambda^*)=-\int(\rho(x)\not{d}\lambda,\lambda^*)$ by the symmetry of $(\rho(x)\zeta_1,\zeta_2)$. (ii) By Proposition 9.25 (ii) with $n=2$, whose hypothesis is Theorem 4.32: the defect cycles with two legs have one or two vertices of I, the vertex $S^{(1)}(\Delta)$ at the special edge or on a line, and are evaluated exactly by their simplex forms, with the position realised by a momentum injected at $S^{(1)}(\Delta)$; the cycles with $S^{(1)}(\Delta)$ at the special edge and those with $S^{(1)}(\Delta)$ on a line give the two stated brackets, and the coefficients of the functionals with an explicit factor x cancel; by Proposition 11.10 (i) this is the vanishing of the quadratic part of $\mathrm{Ob}^{(1)}(P_b)$, which the defect cycles along P_b give directly (Theorem 9.35 (i)). By Proposition 9.25 (iii) the same functional is 4 times the quadratic part of Theorem 4.16 plus the image under Q of the renormalised composite two-point functional of Remark 9.27; the two evaluations agree sector by sector. (iii) By Proposition 1.11, $Q\int(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})=-\frac{1}{2}\int(dA|_{W_{\mathbb{R}}^\perp},dA|_{W_{\mathbb{R}}^\perp})$, $Q\int(A|_V,A^*|_V)=-\frac{1}{2}\int(dA|_V,dA|_V)-\int(dc,A^*)$, $Q\int(\lambda^*,\lambda)=-\int(\lambda,\not{d}\lambda)$ and $Q\int(c^*,c)=-\int(dc,A^*)$. These four functionals span the parity-even translation-invariant local functionals of degree -1 and Weyl weight 4 in the four sectors, and their images are linearly independent, which gives existence and uniqueness of the translation-invariant solution; Q of the first bracket is (9.15). The sector parts are Q-closed because $v.$ commutes with Q and Q preserves the sectors $(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})$ and (λ,λ^*) and mixes $(A|_V,A^*|_V)$ with (c,c^*) only; Proposition 11.24 gives the closed and the exact directions. In the sector $(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})$ the functionals of degree at most one in x are spanned by $\int(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})$ and $\int(E(A|_{W_{\mathbb{R}}^\perp}),A^*|_{W_{\mathbb{R}}^\perp})$, and $Q\int(E(A|_{W_{\mathbb{R}}^\perp}),A^*|_{W_{\mathbb{R}}^\perp})=\frac{1}{2}\int(dA|_{W_{\mathbb{R}}^\perp},dA|_{W_{\mathbb{R}}^\perp})$; the equation fixes the difference of the two coefficients to $\frac{22}{3}$. □

Remark 9.27 (The renormalised composite two-point functional)**.** In the second evaluation of Proposition 9.25 (iii) the composite part enters through the following functional. The quadratic part of the composite cycles minus the quadratic part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ has a limit as $\varepsilon\to0$ at nonzero momentum, the renormalised composite two-point functional. In the sector $(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})$ it is the local functional

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{5}{3}\int\big(2E(A|_{W_{\mathbb{R}}^\perp})+2A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp}\big)-\tfrac{2}{3}\int(A|_{W_{\mathbb{R}}^\perp},A^*|_{W_{\mathbb{R}}^\perp})\Big],$$

and in the sectors $(A|_V,A^*|_V)$, (λ,λ^*), (c,c^*) it contains $\log|p|^2$ of the momentum, in $(A|_V,A^*|_V)$ together with $(p,a)(p,b)|p|^{-2}$ for polarisations a,b. Its image under Q is local in every sector, equal to the composite part of Theorem 9.26 (ii): the logarithms and the ratios $(p,a)(p,b)|p|^{-2}$ cancel in the image. The translation-invariant local functional with the same image is $\big[-\frac{2}{3}\int(A,A^*)+\frac{1}{3}\int(\lambda^*,\lambda)+\frac{4}{3}\int(c^*,c)\big]$ units, the second summand of the solution of Theorem 9.26 (iii).

Theorem 9.28 (The sector coefficients along Δ from the arity-two equations)**.** *Let $\hbar F_1^{(0)}$ contain the finite counterterm of Theorem 4.32, and let the arity-one equations along Δ, P_b and $\xi \in \Sigma$ be solved on two legs by* (9.16)*, by $\hbar F_1^{(1)}(P_b)$ of Theorem 9.35 (i), with the sector coefficients $\theta_P, \theta_{P,\lambda}, \theta_{P,V}$, and by $\hbar F_1^{(1)}(\xi)$ of Proposition 11.50 (i), with its free coefficients. The obstructions of arity two on two legs are the sums of the coefficients of ε^0 of the cycles with exactly one special edge and either two vertices $S^{(1)}(x)$, $S^{(1)}(y)$ or one vertex $S^{(2)}(x,y)$ on the cycle, the other vertices from I, as in Proposition 9.25 (ii).*

- *(i) With $[\Delta, P_b] = -2P_b$ (Remark 2.6) and $S^{(2)}(\Delta, P_b) = 0$ (Definition 2.15), the two-leg part of* (9.12) *along (Δ, P_b) is*

$$Q\, F_1^{(2)}(\Delta, P_b) + 2\ \mathit{units}\cdot \Big(\theta\, S^{(1)}(P_b)_\perp + \theta_\lambda\, S^{(1)}(P_b)_\lambda + \theta_V\, S^{(1)}(P_b)_V\Big) = \mathrm{Ob}^{(2)}(\Delta, P_b) + Q(\mathit{local}), \tag{9.18}$$

  with the sector parts $S^{(1)}(P_b)_d$ of Definition 11.1; the coefficients along P_b drop out. The classes $[S^{(1)}(P_b)_\perp]$, $[S^{(1)}(P_b)_\lambda]$, $[S^{(1)}(P_b)_V]$ are linearly independent in the cohomology of Q on the quadratic local functionals of degree -1 and Weyl weight 5, and

$$\big[\hbar\, \mathrm{Ob}^{(2)}(\Delta, P_b)\big] = -\tfrac{2}{3}\ \mathit{units}\cdot \Big([S^{(1)}(P_b)_\perp] + [S^{(1)}(P_b)_\lambda] + [S^{(1)}(P_b)_V]\Big),$$

  with no position part and no pole. Hence the arity-two equation along (Δ, P_b) is solvable on two legs exactly when $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$.

- *(ii) With $[\Delta, \xi] = -\xi$, for ξ of either chirality, the two-leg part of* (9.12) *along (Δ, ξ) is solvable exactly when $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, for every value of the free coefficients along ξ. The obstruction is*

$$\hbar\, \mathrm{Ob}^{(2)}(\Delta, \xi) = \mathit{units}\cdot \Big[8(T_1{+}T_2) - \tfrac{26}{3}(T_3{+}T_4) + \log\tfrac{4}{3}\, Q(\psi_1{+}\psi_2)\Big], \qquad \psi_1 = \int x^\mu \big(\Gamma(\xi,\lambda)^\mu, c^*\big), \quad \psi_2 = \int x^\mu \big(A^*_\mu$$

  with $T_1, \dots, T_4$ the terms of Table 11.6 and $\Gamma(\xi,\lambda)^\mu$ the component of $\Gamma(\xi,\lambda)|_V$ along e_μ, and at these values $\hbar F_1^{(2)}(\Delta, \xi) = \log\frac{4}{3}\ \mathit{units}\cdot(\psi_1 + \psi_2)$ plus Q-closed functionals.

- *(iii) At $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ the quadratic part of $\hbar F_1^{(1)}(\Delta)$ in the sector $(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$ is $\big[\frac{20}{3}\int (A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp}) - \frac{2}{3}\int (E(A|_{W_\mathbb{R}^\perp}), A^*|_{W_\mathbb{R}^\perp})\big]$ units, and the class of $\hbar F_1^{(1)}(\Delta)$ minus the translation-invariant solution is $-\frac{1}{3}$ units times $[S^{(1)}(\Delta)]$: the uniform coefficient along Δ is $-\frac{1}{3}$, independently of the coefficients along P_b, Σ and Σ^* and of the field redefinition $\phi^{(0)} = c\,\mathit{units}\cdot S^{(1)}(\Delta)$, which shifts the coefficient of $S^{(1)}(x)$ in $\hbar F_1^{(1)}(x)$ by $c\lambda_x$ units, $\lambda_\Delta = 0$.*

Proof. (i) In (9.12) at (Δ, P_b) the term $\{F_1^{(0)}, S^{(2)}(\Delta, P_b)\}$ vanishes and the coefficients enter through $F_1^{(1)}(-2P_b)$, $\{F_1^{(1)}(\Delta), S^{(1)}(P_b)\}$ and $\{S^{(1)}(\Delta), F_1^{(1)}(P_b)\}$. The sector parts of $S^{(1)}(\Delta)$ and of $S^{(1)}(P_b)$ in different sectors have no common legs, and within one sector $\{S^{(1)}(\Delta)_d, S^{(1)}(P_b)_d\} = v.S^{(1)}(P_b)_d = 2S^{(1)}(P_b)_d$, $S^{(1)}(P_b)_d$ having Weyl weight 5 (Lemma 9.16 (i)). The terms are therefore $2\theta_d\, S^{(1)}(P_b)_d$ units from $\{F_1^{(1)}(\Delta), S^{(1)}(P_b)\}$, $2\theta_{P,d}\, S^{(1)}(P_b)_d$ units from $\{S^{(1)}(\Delta), F_1^{(1)}(P_b)\}$ and $-2\theta_{P,d}\, S^{(1)}(P_b)_d$ units from $F_1^{(1)}(-2P_b)$. The translation-invariant solution of (9.16) is a combination of field-counting functionals, whose brackets with $S^{(1)}(P_b)$ vanish, and the κ-directions are Q-exact, with Q-exact brackets. Independence: Q preserves the three sectors $(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$, (λ, λ^*) and $(A|_V, A^*|_V) \oplus (c, c^*)$ of the quadratic functionals of degree -1. In the first there is no functional of degree -2, $\int f(x)(A^*|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$ vanishing for the odd A^* and the symmetric fibre pairing, so $[S^{(1)}(P_b)_\perp] \neq 0$. In the second the translation-invariant image is spanned by $Q\int(\rho(b)\lambda^*, \lambda^*) = 2\int(\partial_b\lambda, \lambda^*) + 2\int(\rho(b\wedge\partial)\lambda, \lambda^*)$, not a multiple of

$S^{(1)}(P_b)_\lambda$. In the third it is spanned by $Q\int(\iota_b A|_V, c^*)$, which differs from $S^{(1)}(P_b)_V$ by the nonzero $\int(\iota_b dA|_V, A^*|_V)$ up to sign. The class of the obstruction is the sum of the coefficients of ε^0 of the cycles with the vertices $S^{(1)}(\Delta)$, $S^{(1)}(P_b)$ and one special edge, computed exactly on complete bases of the two-leg functionals with these quantum numbers; the residual of Lemma 9.24 is then $(-\frac{2}{3}-2\theta_d)$ units times $[S^{(1)}(P_b)_d]$ in the sector d. (ii) The same computation along (Δ,ξ), with $S^{(2)}(\Delta,\xi)=0$: the coefficients along ξ enter through $F_1^{(1)}(-\xi)$ and $\{S^{(1)}(\Delta), F_1^{(1)}(\xi)\}$, which cancel, $S^{(1)}(\xi)$ having Weyl weight $\frac{9}{2}$, and the sector coefficients along Δ through $\{F_1^{(1)}(\Delta), S^{(1)}(\xi)\}$, whose sector parts act on the legs of $T_1,\dots,T_4$; the cohomology on two legs at Weyl weight $\frac{9}{2}$ and position degree at most one is computed on complete bases, and the residual class vanishes exactly at the stated values. (iii) Insert $\theta=-\frac{1}{3}$ into (9.17). The field redefinition generated by $\phi^{(0)}$ changes $F_1^{(1)}(x)$ by $\{\phi^{(0)}, S^{(1)}(x)\}$, which is $c\lambda_x$ units times $S^{(1)}(x)$ modulo exact terms (Lemma 2.11, $S^{(2)}(\Delta,x)=0$). □

Remark 9.29 (Two-component forms along Δ)**.** With the components of Definition 8.1, $(\sigma^n\bar\sigma^m)_\alpha{}^\beta = \sigma^n_{\alpha\dot\alpha}\bar\sigma^{m\dot\alpha\beta}$ and $(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} = \bar\sigma^{m\dot\beta\alpha}\sigma^n_{\alpha\dot\alpha}$,

$$\int(\lambda^*,\lambda) = \int d^4x\Big\{\big(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i\big) - \big(\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}\big)\Big\}, \qquad -\int(2E(\lambda)+3\lambda,\lambda^*) = -\int d^4x\Big\{\big((2E+3)\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i\big) - \big((2E+3$$

$$\int(\rho(x)\not\partial\lambda,\lambda^*) = \int d^4x\Big\{-x_n(\sigma^n\bar\sigma^m)_\alpha{}^\beta\big(\partial_m\lambda_\beta{}^i, \lambda^{*\alpha}{}_i\big) + x_n\big(\partial_m\bar\lambda_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha}, \bar\lambda^{*\dot\alpha i}\big)\Big\}, \qquad \int(\rho(x)\lambda^*,\lambda^*) = -2i\int d^$$

and $\int(\lambda,\not\partial\lambda)$ as in Remark 9.13. Thus (9.15) contains $\frac{94}{3}\,i$ units times $\int d^4x\,(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i)$, the translation-invariant $\hbar F_1^{(1)}(\Delta)$ contains $-\frac{47}{3}$ units times $\int d^4x\{(\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i)-(\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i})\}$, and Weinberg's forms follow from $\rho(v)=\gamma_W^\mu v_\mu$ on the Dirac column. The sector parts of $S^{(1)}(P_b)$ in Theorem 9.28 are $\frac{1}{4}\int d^4x\,(b_m\partial_m A_{ij}, A^{*ij})$, $-\int d^4x\big\{(b_m\partial_m\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i) - (b_m\partial_m\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i})\big\}$ and $\int d^4x\big\{(b_m\partial_m A_n, A_n^*) - (b_m\partial_m c, c^*)\big\}$, and the solution (9.16) at $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$, $\kappa_V=\kappa_\lambda=0$, is

$$\hbar F_1^{(1)}(\Delta) = \text{ units}\int d^4x\Big\{\big((\tfrac{22}{3}-\tfrac{1}{3}(2x_n\partial_n+2))A_m, A_m^*\big) + \tfrac{1}{4}\big((\tfrac{22}{3}-\tfrac{1}{3}(2x_n\partial_n+2))A_{ij}, A^{*ij}\big) - \big((\tfrac{8}{3}-\tfrac{2}{3}x_n\partial_n)c, c^*\big)$$
$$-\big((\tfrac{47}{3}-\tfrac{1}{3}(2x_n\partial_n+3))\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i\big) + \big((\tfrac{47}{3}-\tfrac{1}{3}(2x_n\partial_n+3))\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}\big)\Big\}.$$

For $\xi=(\xi_\alpha{}^i, 0)$ the functionals of Theorem 9.28 (ii) are $\psi_1 = i\int d^4x\,x_m\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i, c^*\big)$ and $\psi_2 = \int d^4x\,x_m\big(A_m^*, \xi_\alpha{}^i\lambda^{*\alpha}{}_i\big)$, and $T_1,\dots,T_4$ are those of Remark 11.68.

9.4 The dilatation component of the scalar bilinears

For $O=\int\omega\,(\iota_{w_1}A, \iota_{w_2}A)$ (Definition 5.12), $(Q+\{I,-\})O=0$. By Lemma 8.4, applied to the exponential of the equivariant interaction with the vertex $\delta\,O(\varepsilon)$ added, the arity-one part along Δ of $d_\infty O[\infty]$ is the $\varepsilon\to 0$ limit of the graphs with one vertex given by the part of the defect vertex linear in δ and in the arity-one vertex of Definition 9.23, together with the products of $O(\varepsilon)$ with the arity-zero and arity-one defects. The local representative along Δ is $L(O)^{(1)}(\Delta) = \{S^{(1)}(\Delta), O\} + D_1(O)$ (Definitions 8.14, 8.16); by Proposition 8.25 and Theorem 8.15 (ii) it is computed at $T=\infty$, and by the master formula of Theorem 11.44 (iii) $D_1(O)$ is the sum over the pairs of letters of O of the coefficients of ε^0 of their defect cycles along Δ, minus the bracket of O with $\hbar F_1^{(1)}(\Delta)$.

Theorem 9.30 (The component along Δ)**.** *Let $\hbar F_1^{(0)}$ contain the finite counterterm of Theorem 4.32, let $\hbar F_1^{(1)}(\Delta)$ solve the quadratic part of* (9.10)*, with the part* (9.17) *in the sector $(A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$, and let $\psi=0$. For a scalar bilinear O the arity-one counterterm of O along Δ vanishes, and on the part with*

two legs $A|_{W_{\mathbb{R}}^{\perp}}$ the local representative along Δ exists and is given by the master formula of Theorem 11.44 (iii),

$$D_1(O) = \Upsilon_\Delta(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}}) - \big\{\hbar F_1^{(1)}(\Delta), O\big\},$$

with the two-body kernel Υ_Δ, the coefficient of ε^0 of the defect cycles of the pair of letters of O along Δ, equal to $\frac{44}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ units on the traceless part and $-\frac{100}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ units on the trace. Hence

$$\begin{aligned} D_1(O) &= -\big(\tfrac{1}{3}+\theta\big)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\{S^{(1)}(\Delta), O\} && \text{for the traceless part,}\\ D_1(O) &= -48\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O - \big(\tfrac{1}{3}+\theta\big)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\{S^{(1)}(\Delta), O\} && \text{for the trace,} \end{aligned} \tag{9.19}$$

$\{S^{(1)}(\Delta), O\} = 2(E+2)O$: twice the coefficient of Theorem 6.11 times O, plus $-(\frac{1}{3}+\theta)$ units times the classical transformation. At the value $\theta = -\frac{1}{3}$ fixed by the arity-two equations (Theorem 9.28),

$$D_1(O) = 0 \quad \text{(traceless part)}, \qquad D_1(O) = -48\ \text{units}\cdot O = \frac{3g^2N}{2\pi^2}\,O \quad \text{(trace, } \mathfrak{g} = \mathfrak{su}(N)),$$

and these values are the scheme-independent content: they are unchanged by every free direction of the action-side solution compatible with the arity-two equations, by the finite parts ψ of Lemma 10.37 and by the finite counterterms (Theorem 8.20 (iii)). The contributions are those of the following table.

contribution, part with two legs $A\|_{W_{\mathbb{R}}^{\perp}}$	traceless part	trace
defect cycles of the pair along Δ, Υ_Δ	$\frac{44}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$	$-\frac{100}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$
$-\{\hbar F_1^{(1)}(\Delta), O\}$	$-\frac{44}{3}\,O - \theta\,\{S^{(1)}(\Delta), O\}$	$-\frac{44}{3}\,O - \theta\,\{S^{(1)}(\Delta), O\}$
$-\{I^{CT}_{\mathfrak{psl}(4\|4)}[\varepsilon]^{(1)}(\Delta), O\}$, arity-one counterterm of O	0	0
$D_1(O)$	$-\big(\frac{1}{3}+\theta\big)\{S^{(1)}(\Delta), O\}$	$-48\,O - \big(\frac{1}{3}+\theta\big)\{S^{(1)}(\Delta), O\}$

The component $D_1(O)$ of a scalar bilinear on the part with two legs $A|_{W_{\mathbb{R}}^{\perp}}$, in units; for the trace, O stands for $\int \omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$. At the point of the density the first row is $\frac{40}{3}\,O$ and $-\frac{104}{3}\,O$. By Corollary 11.56 it equals 4 times the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$ (Theorem 5.18), $16\,O$ and $-32\,O$, plus the composite pair kernel $-\frac{4}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$, an identity of two evaluations of the same flow.

Proof. By Theorem 11.44 (iii) at $T = \infty$, which applies along Δ (Proposition 8.25), $D_1(O)$ is the two-body kernel of the single pair of letters of O minus the bracket of O with $\hbar F_1^{(1)}(\Delta)$. The two-body kernel on the outputs $A|_{W_{\mathbb{R}}^{\perp}} A|_{W_{\mathbb{R}}^{\perp}}$ is the coefficient of ε^0 of the defect cycles of the pair: cycles through the two letters with one special edge, vertices of I and the vertex $S^{(1)}(\Delta)$ at the special edge, with one leg on it and the other on an edge P or at a letter, or on a line, the composite kernel; they carry no $\log\varepsilon$ (Theorem 11.45 (i)), their poles cancel against the poles of the counterterm trees ((11.7)), and their exact evaluation, with the position realised by a momentum injected at $S^{(1)}(\Delta)$, gives the first row, whose position part along b is $2\,\Upsilon_{P_b} = -\frac{2}{3}$ units times $\partial_b O$ (Theorem 9.35 (ii)). The graphs with a special edge on a tree tend to 0 together with the special-edge parts of their counterterm partners, and the graphs whose cycle passes through vertices of the action alone tend to 0 by the arity-zero equation, whose two-leg part vanishes in the sector $(c, A|_V)$ (Theorem 4.32), and by (9.10) on two legs. The one-body term is computed by Lemma 4.21 and Lemma 9.16 (i): $-\frac{22}{3}\cdot 2\,O - \theta\{S^{(1)}(\Delta), O\}$ units from (9.17), and 0 from the counterterm, whose quadratic part contains no $A^*|_{W_{\mathbb{R}}^{\perp}}$ (Theorem 9.26 (i)). The arity-one counterterm of O along Δ is a colour-singlet local functional of degree -1 and Weyl weight 2, with terms linear in the position carrying densities of weight 3 (the weight counting in the proof of Proposition 7.13); every colour singlet of degree -1 contains an antifield and a second adjoint field and has weight at least 4, so it vanishes. The sum is the last row. By Theorem 5.18 and Lemma 6.7, 4

times the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$ minus $16\,O$ is twice the coefficient of Theorem 6.11 times O: $2\cdot 0$ for the traceless part and $2\cdot(-24)$ for the trace. The invariance statement: by Theorem 9.28 the coefficient θ is the same in every solution of the arity-two equations, the inner and outer directions leave the uniform coefficient along Δ unchanged, $\lambda_\Delta = B_\Delta = 0$, and Lemma 10.37 and Theorem 8.20 (iii) give the rest. □

Proposition 9.31 (The composite triangle)**.** *Let O be a scalar bilinear, and consider the one-loop graph of the arity-one part of $O[\infty]$ along Δ whose cycle contains O, a cubic vertex of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, the vertex $([A,c],A^*)$ and the composite kernel from the leg $A|_V$ of the first to the leg c of the second; its external legs are $A|_{W_{\mathbb{R}}^\perp}$, of momentum p_1, and $A^*|_{W_{\mathbb{R}}^\perp}$, of momentum p_2, with $p_1+p_2+k=0$ for the momentum k of ω. In the form $\alpha(p_1)\delta(p_1+p_2+k)+\beta(p_1)^\mu\frac{\partial}{\partial p_2^\mu}\delta(p_1+p_2+k)$, with p_1 held fixed and the value of O taken with the polarisation of $A^*|_{W_{\mathbb{R}}^\perp}$ in place of its second leg, the value of the graph is $\alpha=0$ and*

$$\beta(p_1)=8\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,(\textit{value of }O)\,\Big(\frac{p_2}{|p_2|^2}+16\pi^2|p_1|^2\int\frac{d^4l}{(2\pi)^4}\,\frac{l}{|l|^4\,|l+p_2|^2\,|l-p_1|^2}\Big).$$

Its image under Q, a functional of two legs $A|_{W_{\mathbb{R}}^\perp}$, is the local functional

$$Q\,(\textit{composite triangle})=-2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\{S^{(1)}(\Delta),O\}. \tag{9.20}$$

Proof. The composite line is longitudinal, so by the Ward identity at the vertex $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ the graph is a bubble term plus $|p_1|^2$ times the triangle vector integral, with $\alpha=0$; its two coefficients are 8 and 8, the values fixed by Remark 9.32. The term with $\frac{\partial}{\partial p_2}\delta$ carries the explicit position of the density. Exchanging the two legs turns (α,β) into $\big(\alpha(p_2)-\frac{\partial\beta^\mu}{\partial p_1^\mu}(p_2),\,\beta(p_2)\big)$, with $p_2=-p_1-k$, by $g(p_2)\,\partial\delta = g(-p_1-k)\,\partial\delta-(\partial g)(-p_1-k)\,\delta$, and a functional of two legs of the same field depends on its value through the sum of the value and its exchange. Q replaces the leg $A^*|_{W_{\mathbb{R}}^\perp}$ by $-\frac12 d^*dA|_{W_{\mathbb{R}}^\perp}$ (Proposition 1.11), which multiplies the value by $-\frac12|p_2|^2$ and, by the same rule, turns $(0,\beta)$ into $\big((p_2,\beta),-\frac12|p_2|^2\beta\big)$. The vector integral is odd under the exchange of its two momenta, by $l\mapsto -l$, and its divergence in p_1 at fixed k is $2|p_1|^{-2}|p_2|^{-2}$, by $\partial_\nu(l^\nu|l|^{-4})=2\pi^2\delta^4(l)$. In the sum over the two orderings the terms containing the integral cancel: the coefficient of $\partial\delta$ is $-4p_2-4p_1=4k$ and that of δ is 8, in units times the value of O. The classical transformation $\{S^{(1)}(\Delta),O\}=2(E+2)O$ has the value -4 and $-2k$ in the same form, by $\int\mathrm{dvol}\,e^{i(k,x)}2\big((x,\partial)+2\big)=2\int\mathrm{dvol}\,\big(-2-i(k,x)\big)e^{i(k,x)}$; the ratio is -2. □

Remark 9.32 (Two evaluations)**.** The table of Theorem 9.30 is the master formula: the defect cycles of Lemma 8.4. A second evaluation, through the closed forms of Section 9.2, applies Q to the arity-one part of $O[\infty]$ along Δ (Lemma 7.2); its graphs are the tree of O with $S^{(1)}(\Delta)$, the graphs with $S^{(1)}(\Delta)$ on a leg of the one-loop two-point function joined to O, the composite triangle, the renormalised composite two-point functional on a leg of O, and the trees with the vertices $-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)-\hbar F_1^{(1)}(\Delta)$. With (9.20) it gives (9.19). The two evaluations agree in detail on the terms beyond the scaling of Corollary 9.22: in the first, the cycles through O with a special edge give, beyond 4 times the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]$, the terms $-\frac43 O-\frac13\{S^{(1)}(\Delta),O\}$, and the tree insertion into O of the composite part $\frac13\int(dA|_{W_{\mathbb{R}}^\perp},dA|_{W_{\mathbb{R}}^\perp})$ of Theorem 9.26 (ii) gives $\frac83 O$; in the second, the composite triangle gives $-2\{S^{(1)}(\Delta),O\}$, and the composite two-point functional on a leg of O gives, by Lemma 6.8 and Remark 9.27, $\frac83 O+\frac53\{S^{(1)}(\Delta),O\}-\frac43 O$. Both sums are $\frac43 O-\frac13\{S^{(1)}(\Delta),O\}$ units, and this agreement determines both coefficients 8 in the value of the composite triangle.

Theorem 9.33 (The multiple of the classical transformation)**.** *Let the family of scalar bilinears consist of* $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ *for all compactly supported four-forms* ω *and* $w_1, w_2 \in W_{\mathbb{R}}^{\perp}$*. Assume:*

- *(i) a one-loop quantisation through arity two along* Δ *and* P_b*:* (8.6) *modulo* $\hbar^2$ *in its components of arity at most two along these generators, with* $\hbar F_1^{(1)}(\Delta)$ *of degree at most one in* x *in the sector* $(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})$*;*
- *(ii) local representatives through arity two along* (Δ, P_b) *(Definition 8.14), with* $\psi = 0$*, on a space of local functionals stable under* Λ *that contains the family.*

Then $\theta = -\frac{1}{3}$*: on the part with two legs* $A|_{W_{\mathbb{R}}^{\perp}}$ *the coefficient of* $\{S^{(1)}(\Delta), O\}$ *in* $D_1(O)$ *vanishes. This is the operator-level form of Theorem 9.28 (i): the value computed there from the action-level arity-two equation along* (Δ, P_b) *is reproduced by the arity-two component of* $L \circ L = 0$ *on the scalar bilinears, a cross-check of that computation.*

Proof. By Theorem 8.15 (iii), (iv), $L \circ L = 0$ on that space, and its component along (Δ, P_b) is Corollary 8.17 (ii): applied to O in the family, $[\Lambda^{(1)}(\Delta), \Lambda^{(1)}(P_b)]O + \Lambda^{(1)}([\Delta, P_b])O$ is a graded commutator of $\Lambda^{(0)}$ with an operator built from $\Lambda^{(2)}(\Delta, P_b)$, applied to O. Here $[\Delta, P_b] = -2P_b$ (Remark 2.6), and $S^{(2)}(\Delta, P_b) = 0$ (Definition 2.15), so that operator has order $\hbar$, and with $sO = 0$ the right side at order $\hbar$ is s applied to the local functional obtained from O by its order-$\hbar$ part, the arity-two output of O.

Left side. On the part with two legs $A|_{W_{\mathbb{R}}^{\perp}}$, by Theorems 9.30 and 9.35, the order-$\hbar$ parts are $a\{S^{(1)}(\Delta), -\} + n$ along Δ and $a_P\{S^{(1)}(P_b), -\}$ along P_b, with $a = -(\frac{1}{3} + \theta)$ units, $a_P = -(\frac{1}{3} + \theta_P)$ units and n the multiplication by 0 or -48 units (the part of $\hbar F_1^{(1)}(P_b)$ in the sector $(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})$ has the form of Theorem 9.35, $\int(\partial_b(A|_{W_{\mathbb{R}}^{\perp}}), A^*|_{W_{\mathbb{R}}^{\perp}})$ being, up to integration by parts, the invariant local functional of degree -1 in that sector with the datum b and density weight 5); the classical operators $\{S^{(1)}(\Delta), -\}$, $\{S^{(1)}(P_b), -\}$ preserve the field content of a functional, so these parts compose among themselves. With $[\{S^{(1)}(\Delta), -\}, \{S^{(1)}(P_b), -\}] = 2\{S^{(1)}(P_b), -\}$, the order-$\hbar^0$ part of the same component, and $[n, \{S^{(1)}(P_b), -\}] = 0$, the order-$\hbar$ part of the left side is

$$(a + a_P) \cdot 2\{S^{(1)}(P_b), O\} - 2a_P\{S^{(1)}(P_b), O\} = 2a\{S^{(1)}(P_b), O\} = 2a \int \omega\, \partial_b(\iota_{w_1}A, \iota_{w_2}A).$$

Right side. The arity-two output of O is a colour-singlet local functional of degree -1; by the weight counting in the proof of Proposition 7.13, with the data $2x_0$, 2 of Δ and b of P_b at a point x_0, its density at x_0 is a density of Weyl weight 3 plus x_0 times densities of Weyl weight 4. A colour singlet of degree -1 contains an antifield and a second adjoint field, so its density has weight at least 4: the first part vanishes, and the densities of weight 4, built from (A^*, A), (λ^*, λ) and (c^*, c), carry no derivatives. Q substitutes fields by derivatives of fields, and $\{I, -\}$ pairs the antifields of the arity-two output with the fields of I and differentiates the vertices of I alone, so s of the arity-two output is linear in x_0. The left side is independent of x_0, and nonzero for $a \neq 0$. Hence $a = 0$. □

Corollary 9.34 (The dilatation coefficients as components)**.** *At the values* $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ *of Theorem 9.28, which are also those of Theorem 9.33 under its hypotheses, on the part with two legs* $A|_{W_{\mathbb{R}}^{\perp}}$ *of the scalar bilinears:*

- *(i)* $\hbar F_1^{(1)}(\Delta)$ *in the sector* $(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})$ *is* $\left[\frac{20}{3}\int(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}}) - \frac{2}{3}\int(E(A|_{W_{\mathbb{R}}^{\perp}}), A^*|_{W_{\mathbb{R}}^{\perp}})\right]$ *units;*
- *(ii)* $D_1(O) = 0$ *for the twenty traceless bilinears and* $D_1(O) = -48$ *units times* O *for the trace, twice the coefficients* 0 *and* -24 *units of Theorem 6.11, and the identification* (6.9) *holds on this part;*

- *(iii) when the part of $D_1(O)$ with legs $(A|_V, A|_V)$ and $(c, A^*|_V)$ lies in the image of $\Lambda^{(0)}$, the one-loop anomalous dimensions are 0 for the traceless bilinears and -24 units for the trace, $\frac{3g^2N}{4\pi^2}$ for $\mathfrak{g} = \mathfrak{su}(N)$ (Remark 6.13).*

For every θ, the coefficients of O in $D_1(O)$ for the trace and for the traceless part differ by -48 units.

Proof. Insert $\theta = -\frac{1}{3}$ into (9.17) and (9.19). Then $L(O)^{(1)}(\Delta) = 2(E + 2 + \text{coefficient})O$ on this part, which is (6.10) there; with Theorem 6.11 this is (6.9). The last statement is (9.19). □

9.5 Translations

For a translation P_b, $b \in V$, $v.$ is the derivative ∂_b on every slot, and it commutes with Q, Q^{GF} and H (Proposition 7.6). Propositions 9.19 and 9.20 hold with ∂_b in place of $v.$ and (9.9) without the scaling term; with $P = P(\varepsilon, \infty)$ the composite kernel is $\big(P\star\,\partial_b\otimes 1 + 1\otimes P\star\,\partial_b\big)P = 2\big(\partial_b(P\star)^2\otimes 1\big)K_0$, nonzero because $(Q^{GF})^2 \neq 0$.

Theorem 9.35 (The translation component)**.** *Let $\hbar F_1^{(0)}$ contain the finite counterterm of Theorem 4.32 and let $\psi = 0$.*

- *(i) The quadratic part of the arity-one obstruction along P_b, the sum of the coefficients of ε^0 of the defect cycles with two legs and the vertex $S^{(1)}(P_b)$ at the special edge or on a line, vanishes. The quadratic part of the arity-one equation along P_b is therefore solved exactly by*

$$\hbar F_1^{(1)}(P_b) = \text{units}\cdot\Big(\theta_P\, S^{(1)}(P_b)_\perp + \theta_{P,\lambda}\, S^{(1)}(P_b)_\lambda + \theta_{P,V}\, S^{(1)}(P_b)_V\Big) + Q\Big(\kappa'_V \int (\iota_b A|_V, c^*) + \kappa'_\lambda \int (\rho(b)\lambda^*, \lambda^*)\Big),$$

  with the sector parts of Definition 11.1 and free numbers $\theta_P, \theta_{P,\lambda}, \theta_{P,V}, \kappa'_V, \kappa'_\lambda$; in the sector $(A|_{W_\mathbb{R}^\perp}, A^*|_{W_\mathbb{R}^\perp})$ it is θ_P units times $\int(\partial_b(A|_{W_\mathbb{R}^\perp}), A^*|_{W_\mathbb{R}^\perp})$.

- *(ii) For a scalar bilinear O, on the part with two legs $A|_{W_\mathbb{R}^\perp}$, the master formula of Theorem 11.44 (iii) gives $P_1(O) = \Upsilon_{P_b}(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp}) - \{\hbar F_1^{(1)}(P_b), O\}$ with $\Upsilon_{P_b} = -\frac{1}{3}$ units times $\{S^{(1)}(P_b), O\}$:*

$$P_1(O) = -\big(\tfrac{1}{3}+\theta_P\big)\frac{\hbar\,\mathrm{tr}_\mathfrak{g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\{S^{(1)}(P_b), O\}, \qquad \{S^{(1)}(P_b), O\} = \int \omega\,\partial_b(\iota_{w_1}A, \iota_{w_2}A). \quad (9.21)$$

  The term $-\frac{1}{3}$ units is the limit of the cycles through O that contain $S^{(1)}(P_b)$ as the composite kernel and one special edge.

- *(iii) The arity-two equations along (ξ, P_b), $\xi \in \Sigma$, give $\theta_P = \theta_{P,\lambda} = \theta_{P,V} =: t_P$, and those along (ξ_+, ξ_-), $\xi_\pm$ of the two chiralities, give $t_P = t_{\xi_+} + t_{\xi_-} - \frac{27}{4} - 2\log\frac{3}{2}$ (Theorem 11.51). In the translation-covariant scheme of Definition 11.23, $t_P = \theta = -\frac{1}{3}$, and $P_1(O) = 0$.*

Proof. (i) The defect cycles along P_b are those of Proposition 9.25 (ii) with ∂_b in place of $v.$; every edge, vertex and counterterm commutes with ∂_b (Corollary 7.7), so the cycles with $S^{(1)}(P_b)$ at the special edge are the translations of the arity-zero special-edge cycles, and the sum over the two-leg cycles, evaluated exactly, vanishes, in agreement with the position part of Theorem 9.26 (ii) (Proposition 11.10 (i)). The closed and exact directions are the position parts of those of Theorem 9.26 (iii) (Proposition 11.10). (ii) As in Theorem 9.30, with ∂_b in place of $v.$: the graphs with $S^{(1)}(P_b)$ on an edge or on a leg reproduce the flow of the classical term $\{S^{(1)}(P_b), O\}$, the remaining graphs are the cycles through O with the composite kernel and one special edge, whose limit on this part is $-\frac{1}{3}$ units times $\{S^{(1)}(P_b), O\}$, and $-\{\hbar F_1^{(1)}(P_b), O\}$ gives $-\theta_P$ units times it. In the second evaluation of Remark 9.32 the composite

triangle along P_b, symmetrised over the two legs $A|_{W_{\mathbb{R}}^{\perp}}$, gives -2 units and the composite two-point functional along P_b on a leg of O gives $\frac{5}{3}$ units times $\{S^{(1)}(P_b), O\}$, with the same sum. (iii) Theorem 11.51; in the translation-covariant scheme the position part of $\hbar F_1^{(1)}(\Delta)$ along b is $2\hbar F_1^{(1)}(P_b)$, so $\theta_P = \theta$, and Theorem 9.28 gives $\theta = -\frac{1}{3}$. □

Proposition 9.36 (A finite part of the operators along the dilatation)**.** *Let a be a real multiple of the unit and change the finite part of the operators by $\hbar\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$ (Definition 8.11) on a family of local functionals stable under Λ. Then for every eigenvector x of ad_Δ with eigenvalue λ_x (Remark 2.6) the component of the operators along x changes by*

$$\tfrac{a}{2}\,\lambda_x\,\{S^{(1)}(x), -\}:$$

by $-a$ along translations, by $+a$ along special conformal fields, by $-\frac{a}{2}$ along Σ, by $+\frac{a}{2}$ along Σ^*, and by 0 along Δ, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$. The arity-zero part $\Lambda^{(0)}$ is unchanged, ψ' is compatible with the reflection of Theorem 8.33, and the spectrum of Theorem 8.20 (iii) is unchanged. With $a = -(\frac{1}{3} + \theta_P)$ units, ψ' gives $P_1 = 0$ on the part with two legs $A|_{W_{\mathbb{R}}^{\perp}}$ of the scalar bilinears, leaves D_1, M_1, R_1 unchanged and changes the components along K_b, Σ and Σ^* by $-(\frac{1}{3} + \theta_P)$, $\frac{1}{2}(\frac{1}{3} + \theta_P)$ and $-\frac{1}{2}(\frac{1}{3} + \theta_P)$ units times the classical transformation.

Proof. By Proposition 8.19 (b), L_1 changes by $[s, \psi']$. The map ψ' is linear over $C^\bullet(\mathfrak{psl}(4|4))$ and preserves the degree and the Weyl weight, so it commutes with d_{CE}; its commutator with $\Lambda^{(0)} = Q + \{I, -\}$ is $\frac{a}{2}\{(Q+\{I,-\})S^{(1)}(\Delta), -\} = 0$ (Lemma 9.16 (iii)). Along x the change is $\frac{a}{2}\big[\{S^{(1)}(x), -\}, \{S^{(1)}(\Delta), -\}\big] = -\frac{a}{2}\{S^{(1)}([x, \Delta]), -\}$ by Corollary 8.17 (ii) at order $\hbar^0$, with $S^{(2)}(x, \Delta) = 0$ (Definition 2.15), and $[x, \Delta] = -\lambda_x x$. By Theorem 8.33, $S^{(1)}(\Delta)$ is mapped to $-S^{(1)}(\Delta)$ by the reflection, $\Delta^{\dagger_n} = -\Delta$, and the bracket rule gives $\big(\{S^{(1)}(\Delta), F\}\big)^{\Theta_n} = \{S^{(1)}(\Delta), F^{\Theta_n}\}$, so ψ' commutes with the reflection for real a. The spectrum is invariant under (b) by Theorem 8.20 (iii). The last statement is Theorem 9.35 with $\lambda_x = -2, +2, -1, +1$. □

Remark 9.37 (Endpoint and large distance)**.** Along Δ and the translations the insertion of $S^{(1)}(x)$ on a cycle decays as $|y|^{-5}$ and $|y|^{-6}$ (Proposition 8.25): for Δ the factor $2y$ of the vector field meets the $A \otimes A$ edge, of order $|y|^{-2}$, through $(2(y, \partial_y) + 2)|y - a|^{-2} = -2|y|^{-2} + O(|y|^{-3})$, and the $c \otimes A^*|_V$ edge, of order $|y|^{-3}$. The composite kernel at $T = \infty$ is of order $|p|^{-3}$ at small momentum, $|x - y|^{-1}$ in position space (Proposition 9.18 (iv)), and the one-loop graphs with external legs at generic momenta converge; the components of this chapter are therefore evaluated at $T = \infty$, and by Theorem 8.15 (ii) the local representatives are the same at every finite T. At finite T, with $P = P(\varepsilon, T)$ and the operator $(e^{-\varepsilon H} - e^{-TH})^{-1} P \star\, P \star v.P\star$, bounded at finite T and of order $|p|^{-4}$ on c^* at $T = \infty$, the composite insertion is a homotopy:

$$P \star v.P\star = Q\,\big((e^{-\varepsilon H} - e^{-TH})^{-1} P \star\, P \star v.P\star\big) + \big((e^{-\varepsilon H} - e^{-TH})^{-1} P \star\, P \star v.P\star\big)\,Q,$$

by Proposition 9.18 (iii). The composite triangle of Proposition 9.31 has a local image under Q at $T = \infty$, so this homotopy enters none of the computed components.

Remark 9.38 (Method of computation)**.** Every number of this chapter along Δ and P_b is a coefficient of ε^0 of defect cycles, evaluated in two independent ways: exactly, in the simplex form of Lemma 10.24 and Theorem 11.45 over the region where the Schwinger coordinate of the special edge is the smallest, with the position realised by a momentum injected at the vertex $S^{(1)}(x)$; and in momentum space, with the special edge as the factor $e^{-\varepsilon|p|^2} - 1$, Feynman parameters for the edges P and the special Schwinger time integrated last. The cohomology classes of the obstructions of arity one and two are computed on complete bases of the quadratic local functionals with the given quantum numbers, fixed by representation counts, and the equations are solved by exact linear algebra. The second evaluation

of Proposition 9.25 (iii) and Remark 9.27 evaluates the composite cycles and the graphs with $S^{(1)}(\Delta)$ on a leg directly as $\varepsilon \to 0$ limits in a Feynman-parameter representation, with E acting on Fourier components as $-(4 + (p, \partial_p))$; it agrees with the defect cycles in every sector and placement class by placement class. On the scalar bilinears the agreement of Remark 9.32 fixes the two coefficients of the composite triangle. The colour factors are reduced for a general simple $\mathfrak{g}$ and checked in the realisations $\mathfrak{g} = \mathfrak{su}(2)$, $\mathfrak{g} = \mathfrak{su}(3)$ and $\mathfrak{g} = \mathfrak{gl}(3)$.

Remark 9.39 (The equations that fix the remaining numbers)**.** The numbers left free above enter the next equations of the one-loop structure. The coefficients β_k, κ_k of Theorem 9.11 enter the part of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ with four legs and the three- and four-leg parts of the arity-one equations along Σ. The differences $\theta - \theta_V$ and $\theta_\lambda - \theta_V$ enter the cubic part of (9.10) through the brackets of the cubic part of I with the sector parts of $S^{(1)}(\Delta)$ (Proposition 11.25 (iii)), and the two-leg arity-two equations along (Δ, P_b) and (Δ, ξ) fix all three to $-\frac{1}{3}$ (Theorem 9.28). The uniform coefficients along P_b, Σ, Σ^* and K_b are related by the arity-two equations of Theorem 11.51 and are fixed up to the inner direction generated by $S^{(1)}(\Delta)$ and the outer direction G_B; the special conformal coefficients and the four second-order directions of $\hbar F_1^{(1)}(K_b)$ enter the arity-two equations along $(K_b, P_{b'})$ and (η, η'), at a finite endpoint T. The trace bilinear has the further outputs $(A|_V, A|_V)$ and $(c, A^*|_V)$, the rotations of V carry the composite kernel $2\big(v.(P(\varepsilon,\infty)\star)^2 \otimes 1\big)K_0$ with v a rotation and enter M_1, and the letters, with $(Q + \{I, -\})O \neq 0$, carry composite loops through the vertex $\{I, O\}$; these are given by the same classes of graphs. Chapter 12 states the equations of every arity along every generator together (Lemma 12.6, Theorem 12.29).

Chapter 10

The one-loop counterterms of all local operators

Chapters 5, 6 and 9 computed the one-loop counterterm and the dilatation component of the scalar bilinears graph by graph. A general local operator has letters of every slot with any number of derivatives, so its one-loop graphs form an infinite family and a graph-by-graph treatment reaches no end. At one loop, however, the cycle of a graph meets a local operator in exactly two of its letters and a tree meets it in one, so the counterterm of every local operator is a sum over pairs of letters of a universal pair kernel, and every one-loop component of Definition 8.16 with a local representative is a sum of a two-body and a one-body operator. This chapter computes the pair kernels in closed form, bounds their outputs by Weyl weight and degree, organises them by the BRST covariance of the logarithmic coefficient, evaluates them at the lowest weight and on the planar sectors whose spectrum is that of Beisert's one-loop dilatation operator, and writes the dilatation component D_1 and the translation component P_1 as the pair kernels plus composite terms, which it evaluates on the planar scalar sector.

The unit is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, equal to $-\frac{g^2N}{32\pi^2}$ for $\mathfrak{g}=\mathfrak{su}(N)$ (Remark 6.13). Edges carry $P(\varepsilon,T)$ of Definition 8.3; every counterterm and every component below is independent of the endpoint T of the transfer, and is evaluated at $T=\infty$ where the graph sums converge (Proposition 8.25). On arity zero the classical differential is $Q+\{I,-\}$, the restriction of s of Theorem 8.8. The field-counting functional is $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ (Chapter 6); by Lemma 6.7 its bracket multiplies the coordinate functionals of c, A, λ, A^*, λ^*, c^* by 1, 2, 4, -2, -4, -1, and by Theorem 4.22 the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ is the unit times $(Q+\{I,-\})$ of it. Every functional containing λ or λ^*, and every main formula, is given in the two-component form of Definition 8.1 in Remark 10.50.

10.1 The counterterm map of a local functional

Definition 10.1 (Letters, pair weight and pair degree)**.** A letter is a factor $\partial^\alpha\varphi$ of a monomial of the density of a local functional (Definition 5.1): φ is a coordinate functional of one slot of $\mathscr{E}$ (Definition 3.16), the component of c, $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, λ, A^*, λ^* or c^* along an element of its fibre, and ∂^α is a product of $|\alpha|$ derivatives along V. The Weyl weight $w_{\partial^\alpha\varphi}$ of the letter is the Weyl weight of its slot (Definition 6.1) plus $|\alpha|$, and its degree is the degree of φ: $+1$, 0, 0, -1, -1, -2 for c, A, λ, A^*, λ^*, c^* (proof of Lemma 9.2). A monomial local functional is

$$M=\int\omega\,X_1X_2\cdots X_n,$$

with letters $X_1,\dots,X_n$ at one point, $\omega\in\Omega^4_c(\mathbb{R}^4)$, or $\omega=\mathrm{dvol}$ for a translation-invariant M, and the fibre and colour factors of the letters contracted with constant tensors. For two letters X_a, X_b the pair

weight is $w_{X_a} + w_{X_b}$ and the pair degree is the sum of their degrees. The single letters of Proposition 5.11 are the monomials with one letter.

For a homogeneous local functional M of degree $|M|$ and a square-zero parameter δ of degree $-|M|$ [CG2, Def 8.3.1.1], $\partial_\delta I^{CT}[\varepsilon](M)$ is the δ-component of the one-loop counterterm of the family $I + \delta M$ (Proposition 5.3, Definition 8.11):

$$\partial_\delta I^{CT}[\varepsilon](M) = \hbar \sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|} \operatorname{Sing}_\varepsilon w_\gamma(\varepsilon), \tag{10.1}$$

the sum over the connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$ whose cycle contains the vertex M, all other vertices labelled by I and every edge by $P(\varepsilon, T)$, with $\operatorname{Sing}_\varepsilon$ of Definition 4.1. With $I_{\mathfrak{psl}(4|4)}$ in place of I the same construction gives the counterterm of a local cochain (Definition 8.5), whose arity-zero part is (10.1).

Proposition 10.2 (Structure of the counterterm map)**.** *For a homogeneous local functional M of Weyl weight w_M:*

- *(i) $\partial_\delta I^{CT}[\varepsilon](M)$ is a combination of ε^{-m}, $1 \le m \le \frac{1}{2} w_M$, and $\log\varepsilon$, with local coefficients; at one loop no power of $\log\varepsilon$ beyond the first and no product $\varepsilon^{-m}\log\varepsilon$ occurs.*
- *(ii) The coefficient of ε^{-m} is local of Weyl weight $w_M - 2m$, the coefficient of $\log\varepsilon$ is local of Weyl weight w_M, and ω is untouched.*
- *(iii) The singular part of every graph of (10.1) is the same for every $T \in (0, \infty]$; at $T < \infty$ it is defined at every configuration of the leg momenta, at $T = \infty$ at generic ones. Hence $\partial_\delta I^{CT}[\varepsilon](M)$ is independent of T.*
- *(iv) $M \mapsto \partial_\delta I^{CT}[\varepsilon](M)$ is linear, linear in ω, preserves the degree and the parity, and is covariant under translations, $\operatorname{Spin}(4,\mathbb{C}) \times \operatorname{Spin}(6,\mathbb{C})$ and the adjoint action of $\mathfrak{g}$.*
- *(v) With $v = 2E$ and $v.$ of Lemma 9.16 (i), $v.M = 2(E + w_M)M$ for $M = \int \omega\,(\cdots)$,*

$$\Big(v. - 4\varepsilon\frac{d}{d\varepsilon}\Big)\partial_\delta I^{CT}[\varepsilon](M) = \partial_\delta I^{CT}[\varepsilon](v.M) - 4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](M)\big). \tag{10.2}$$

Proof. (i), (ii). By Lemma 4.6 each graph is $\sum_{1\le k\le D/2} \varepsilon^{-k} a_k + b\log\varepsilon$ plus a function with a limit, D its superficial degree. In the Schwinger form the singular terms come from the integral of τ^{-1-k}, $k \ge 1$, or of τ^{-1} over the total Schwinger time $\tau \in [\varepsilon/\min_e \alpha_e, 1]$ (Theorem 10.22), which produces ε^{-k} and $\log\varepsilon$ alone. By Lemma 10.11, D is the pair weight of the two letters of M on the cycle minus the Weyl weight of the outputs, so the coefficient of ε^{-m} has momentum degree $D - 2m$ and output weight equal to the pair weight minus $2m$, and the coefficient of $\log\varepsilon$ has output weight equal to the pair weight; the other letters are untouched (Theorem 10.4). The output weight is nonnegative, so m is at most half the pair weight, which is at most $\frac{1}{2} w_M$.

(iii). With edges $P(\varepsilon, T)$ the Schwinger times run over $[\varepsilon, T]^{E(\gamma)}$. For $\tau_0 < \min(T, 1)$, on $\{\tau \le \tau_0\}$ the domain is independent of T and the proof of Lemma 4.6 gives all terms ε^{-k} and $\log\varepsilon$. On $\{\tau \ge \tau_0\}$ the integrand after the Gaussian loop integral is bounded by a multiple of τ^{-2-j}; at $T < \infty$ the domain is compact in τ, and at $T = \infty$ the factor $e^{-\tau\cdot(\text{momentum form})}$ of Definition 10.21, the momentum form being at least $\frac{1}{2}\min_{e\ne f} |q_e - q_f|^2 (1 - \max_e \alpha_e)$, gives convergence at generic momenta. This part has a limit by dominated convergence, and its $\varepsilon\frac{d}{d\varepsilon}$, the faces $t_e = \varepsilon$ of Proposition 4.2, is $O(\varepsilon)$. This is Proposition 8.6 (i) for the vertex M.

(iv). The vertex functions and $P(\varepsilon,T)$ are translation invariant and covariant (proof of Theorem 4.16); the singular part is a polynomial in the leg momenta times the Fourier transform of ω at minus the total momentum. Every vertex and every component of $P(\varepsilon,T)$ has degree 0 and is even.

(v). By Lemma 6.4, $(\sigma_s M)^{CT}[s^2\varepsilon] = \sigma_s\big(\partial_\delta I^{CT}[\varepsilon](M)\big) + 2\log s\,\sigma_s\big(\text{coefficient of } \log\varepsilon\big)$. Apply $s\frac{d}{ds}\big|_{s=1}$, use $2s\frac{d}{ds}\big|_{s=1}\sigma_s = v.$ on functionals (proof of Corollary 9.22) and multiply by 2:

$$\partial_\delta I^{CT}[\varepsilon](v.M) + 4\varepsilon\frac{d}{d\varepsilon}\partial_\delta I^{CT}[\varepsilon](M) = v.\,\partial_\delta I^{CT}[\varepsilon](M) + 4\big(\text{coefficient of } \log\varepsilon \text{ in } \partial_\delta I^{CT}[\varepsilon](M)\big).$$

□

10.2 Two letters on the cycle

Definition 10.3 (The dressed propagator and the pair kernel)**.** Let e be a field configuration, compactly supported, with its odd components multiplied by odd elements of an auxiliary nilpotent ring (Definition 1.13), and let $P = P(\varepsilon,T)$. The dressed propagator at e is the kernel

$$\sum_{k\geq 0}\Big(\big(\text{one-loop operator}\big)^k P\star\otimes 1\Big)K_0,$$

with the one-loop operator of P and of the part of $I(e+f)$ quadratic in the fluctuation f (Proposition 3.36) and $P\star$ the operator represented by P (Definition 8.3); it is a formal power series in e, even and graded symmetric. Its term with k factors of the one-loop operator is the sum of the chains of k vertices of I, each entered and left through two of its slots and joined by $k+1$ edges P, the other slots evaluated at e. For two letters $X_a = \partial^\alpha\varphi_a$ and $X_b = \partial^\beta\varphi_b$ the pair kernel is

$$\partial_\delta I^{CT}[\varepsilon](X_aX_b) := \hbar\,\mathrm{Sing}_\varepsilon\Big[\partial_x^\alpha\partial_y^\beta\Big(\text{value of } \partial_{\text{dressed propagator}} \text{ on } \varphi_a(x)\varphi_b(y)\Big)\Big]_{y=x}, \tag{10.3}$$

with ∂_ϕ of [C07, §6.4] read as in Definition 3.16, $\mathrm{Sing}_\varepsilon$ applied to each Taylor component in e, and e then replaced by the fields. It is a local density of the fields at x, the outputs, with coefficients in the span of ε^{-m} and $\log\varepsilon$, and with values in the tensor product of the fibres and colour spaces of the two letters, whose factors are left open; the letters of a monomial other than X_a, X_b are its spectators.

Theorem 10.4 (Decomposition)**.** *Let $M = \int\omega\,X_1X_2\cdots X_n$ be a monomial local functional.*

- *(i) The order-$\hbar$, δ-linear part of $W\big(P(\varepsilon,T), I - I^{CT}[\varepsilon] + \delta M(\varepsilon)\big)$ is the sum of (A) the graphs whose cycle passes through M, (B) the graphs with M on a tree and a cycle of vertices of I, (C) the trees with one vertex $-I^{CT}[\varepsilon]$ and one vertex M, and (D) the trees with one vertex $-\partial_\delta I^{CT}[\varepsilon](M)$. A cycle passes through M along exactly two of its half-edges, two letters X_a, X_b with $a\neq b$, and the graphs (A) are the trees of I attached to the legs of $\hbar\,\partial_{\text{dressed propagator}}M$, evaluated at the legs.*
- *(ii) (B)+(C) has a limit as $\varepsilon\to 0$, at generic leg momenta for $T=\infty$ and at all leg momenta for $T<\infty$; the graphs with M on a tree contribute nothing to $\partial_\delta I^{CT}[\varepsilon](M)$.*
- *(iii) With the Koszul sign $(-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)}$ of moving X_b next to X_a, $|\cdot|$ the total parity, and $\widehat{X_b}$ omitted,*

$$\partial_\delta I^{CT}[\varepsilon](M) = \sum_{1\leq a<b\leq n}(-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)}\int\omega\,X_1\cdots X_{a-1}\,\partial_\delta I^{CT}[\varepsilon](X_aX_b)\,X_{a+1}\cdots\widehat{X_b}\cdots X_n : \tag{10.4}$$

a sum over the unordered pairs of letters of the pair kernel times the spectators, without one-body terms. For $n=2$, $\partial_\delta I^{CT}[\varepsilon](\int\omega\,X_aX_b) = \int\omega\,\partial_\delta I^{CT}[\varepsilon](X_aX_b)$.

- *(iv) The term $k=0$ of the pair kernel, $\hbar\,\partial_x^\alpha\partial_y^\beta$ of the component of $P(\varepsilon,\infty)$ in the slots of φ_a, φ_b at $y=x$, is a pure power $\varepsilon^{-(w_{X_a}+w_{X_b})/2}$ times a number (Lemma 10.6): it has no $\log\varepsilon$ and no finite part.*

Proof. (i) By Proposition 3.35 at order $\hbar$, the δ-linear part consists of the connected graphs with one vertex M and either one cycle and vertices I, or no cycle and one vertex of $-\hbar^{-1}I^{CT}[\varepsilon]$ or of $-\hbar^{-1}\partial_\delta I^{CT}[\varepsilon](M)$: these are (A)–(D). A cycle passes through a vertex along two half-edges, which for M are two letters. For (A), by Lemma 4.7 the graph is a tree whose vertex s is labelled by the one-particle-irreducible cycle through M; the sum over these cycles is the δ-linear part of (3.7) for the family $I+\delta M$ at the configuration e of the half-edges of s. With the one-loop operators of I and of δM at e, $\frac{\hbar}{2}\sum_{n\geq1}\frac{1}{n}\mathrm{STr}\big((\text{one-loop operator of } I+\text{one-loop operator of }\delta M)^n\big)$ has the δ-linear part

$$\frac{\hbar}{2}\,\mathrm{STr}\Big(\big(\text{one-loop operator of }\delta M\big)\big(1-\text{one-loop operator of }I\big)^{-1}\Big),$$

by cyclicity of the supertrace for even operators. By the first identity in the proof of Proposition 3.36, $\frac{1}{2}\mathrm{STr}$ of the one-loop operator of a quadratic functional with an even graded-symmetric kernel ϕ is ∂_ϕ of the functional; here ϕ is the dressed propagator and the quadratic functional is the part of δM quadratic in the fluctuation at e. Hence the cycles through M sum to $\delta\,\hbar\,\partial_{\text{dressed propagator}}M$ at e, the operator ∂ being even.

(ii) For (B), removing the edge that joins the cycle to the part containing M leaves a one-particle-irreducible cycle γ' of vertices I (Lemma 4.7); the trees (C) with $-I^{CT}[\varepsilon]$ in place of γ' remove its singular part (Definition 4.1). At generic leg momenta every edge of the trees carries a nonzero momentum and two distinct edges of γ' carry different momenta: M carries the Fourier transform of ω at minus the total momentum of its legs, momentum is conserved at every other vertex, an edge of a tree separates a part without M that carries a leg, and for two edges of γ' the arc between them that avoids the half-edge to M contains a vertex, the difference of the two momenta being the sum of the leg momenta entering along this arc. Then Lemma 4.6 applies to γ' minus its singular part, and the tree edges converge to $P(0,\infty)$ at nonzero momentum (Proposition 4.8). Cycles γ' with one half-edge vanish (Lemma 4.7). At $T<\infty$ no genericity is needed (Proposition 10.2 (iii)).

(iii) (A)+(D) has a limit exactly when (D) removes the singular part of the vertex s (Lemma 4.6, Proposition 5.3), which is local; by (i) and the uniqueness of the counterterm, $\partial_\delta I^{CT}[\varepsilon](M)=\hbar\,\mathrm{Sing}_\varepsilon\,\partial_{\text{dressed propagator}}M$ with e replaced by the fields. The operator $\partial_\phi=\frac{1}{2}\sum\partial_{\phi''}\partial_{\phi'}$ is of order two: on $X_1\cdots X_n$ it is the sum over the pairs $a<b$ of the term in which X_b is brought next to X_a by Koszul reordering and the adjacent pair is replaced by the component of ϕ in the same order (edge rule of Proposition 3.35), the derivatives of the two letters acting on the two arguments. Each Taylor component in e is a finite sum of cycles, which is (10.3).

(iv) Lemma 10.6. □

Lemma 10.5 (Sign of the pair kernels)**.** *The pair kernel is $\hbar\,\mathrm{Sing}_\varepsilon$ of the value of $\partial_{\text{dressed propagator}}$ on $X_a(x)X_b(y)$ at $y=x$, read with the first factor in the slot of X_a and the Koszul signs of the vertices of I in the order of the chain from X_a to X_b. In the second-derivative form of Proposition 3.36, with the second left derivative of M in place of that of δM,*

$$\partial_j\partial_l(\delta M)=(-1)^{|M|(|j|+|l|)}\,\delta\,\partial_j\partial_l M,$$

so that for $M=\int\omega\,X_aX_b$ the supertrace value with the second derivative of M is $(-1)^{|X_a|+|X_b|}$ times the pair kernel; this sign differs from 1 exactly for the pairs with one odd letter.

Proof. The first statement is (10.3) with the edge rule of Proposition 3.35. For the second, δ, of the parity of M, is moved to the left through the two left derivatives; for $M=\int\omega\,X_aX_b$ the derivatives hit the slots of X_a and X_b, so $|j|+|l|=|X_a|+|X_b|=|M|$ and $(-1)^{|M|^2}=(-1)^{|X_a|+|X_b|}$. □

Lemma 10.6 (The self-contraction)**.** *The scalar factor of $P(\varepsilon,\infty)$ (Proposition 3.17) has the Taylor series*

$$\frac{1-e^{-|x|^2/4\varepsilon}}{4\pi^2|x|^2}=\sum_{m\geq 1}\frac{(-1)^{m+1}}{16\pi^2\,m!\,\varepsilon^m}\Big(\frac{|x|^2}{4}\Big)^{m-1}.$$

For a multi-index γ with $|\gamma|=2(m-1)$ its derivative ∂^γ at $x=0$ is

$$\frac{(-1)^{m+1}}{16\pi^2\,m\,2^{m-1}\,\varepsilon^m}\sum_{\text{pairings of the indices of }\gamma}\prod\delta,$$

and it vanishes for $|\gamma|$ odd. Every component of $P(\varepsilon,\infty)$ is a constant-coefficient differential operator of order w_1+w_2-2, w_1, w_2 the Weyl weights of its two slots, applied to the scalar factor, times its fibre factor: $2(-,-)^{-1}\otimes$ (inverse metric of V_{10}) for $A\otimes A$, $-(-,-)^{-1}d$ for $c\otimes A^*|_V$, and the Dirac operator on $\lambda\otimes\lambda$. With $\partial_x^\alpha\partial_y^\beta f(x-y)|_{y=x}=(-1)^{|\beta|}(\partial^{\alpha+\beta}f)(0)$, the term $k=0$ of $\partial_\delta I^{CT}[\varepsilon](\partial^\alpha\varphi_a\,\partial^\beta\varphi_b)$ is $(-1)^{|\beta|}\hbar$ times the fibre factor times $\partial^{\alpha+\beta}$ of that operator applied to the scalar factor at 0: a single power $\varepsilon^{-(w_{X_a}+w_{X_b})/2}$, zero when $w_{X_a}+w_{X_b}$ is odd. At $T<\infty$ the additional term $-\frac{1-e^{-|x|^2/4T}}{4\pi^2|x|^2}$ of the scalar factor (Definition 8.3) is regular in ε.

Proof. $1-e^{-z}=\sum_{m\geq1}(-1)^{m+1}z^m/m!$ with $z=|x|^2/4\varepsilon$ gives the series. For $|\gamma|=2(m-1)$, $\partial^\gamma(|x|^2)^{m-1}$ at 0 is $2^{m-1}(m-1)!$ times the sum over the pairings of the indices of γ of the products of Kronecker deltas, and $\frac{(m-1)!\,2^{m-1}}{4^{m-1}m!}=\frac{1}{m\,2^{m-1}}$. The scalar factor is entire in $|x|^2$, so each Taylor coefficient is a single power of ε, and the total order of differentiation $w_{X_a}+w_{X_b}-2$ fixes $m=\frac12(w_{X_a}+w_{X_b})$. □

Example 10.7 (The vacuum term of the scalar bilinears)**.** For $M=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ the term $k=0$ gives $\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\varepsilon}\int\omega$, the graph (a) of Proposition 5.14: the $A\otimes A$ component at $x=0$ is $2(-,-)^{-1}\otimes$ (inverse metric) times $\frac{1}{16\pi^2\varepsilon}$, and $(-,-)$ evaluated on $(-,-)^{-1}$ is $\dim\mathfrak{g}$ (Lemma 5.7).

Lemma 10.8 (Colour structure of the pair kernels)**.** *Number the output fields of a chain from X_a to X_b as $Y_1,\dots,Y_r$, Y_1 at the vertex nearest X_a, and write Y_j also for the colour factor of the j-th output.*

- *(i) For a simple $\mathfrak{g}$, the factor in the colour spaces of the two letters of every Taylor component of the dressed propagator is a combination of the elements $(1\otimes\mathrm{ad}_{Y_r}\cdots\mathrm{ad}_{Y_1})(-,-)^{-1}$, with factors $\mathrm{ad}_{[Y_j,Y_{j+1}]}$ from quartic vertices, the first factor in the colour slot of X_a; $r=0$ for $k=0$.*
- *(ii) Let $\mathfrak{g}=\mathfrak{gl}(N)$ or $\mathfrak{su}(N)$, $(X,Y)=\kappa\,\mathrm{tr}(XY)$ with a constant κ, so that $(-,-)^{-1}=\kappa^{-1}\sum_A T_A\otimes T^A$ for dual bases, and for bosonic outputs expand*

$$\mathrm{ad}_{Y_r}\cdots\mathrm{ad}_{Y_1}(Z)=\sum_{S\subseteq\{1,\dots,r\}}(-1)^{r-|S|}\Big(\prod_{j\in S}^{\leftarrow}Y_j\Big)\,Z\,\Big(\prod_{j\notin S}^{\rightarrow}Y_j\Big),$$

the first product in decreasing and the second in increasing order of j. Replacing the colour factors of $X_a\otimes X_b$ by $(1\otimes\mathrm{ad}_{Y_r}\cdots\mathrm{ad}_{Y_1})(-,-)^{-1}$ maps, for products M_1, M_2 of the other colour factors (for $\mathfrak{gl}(N)$, and for $\mathfrak{su}(N)$ when $r\geq1$),

$$\mathrm{tr}(M_1X_aX_b)\mapsto\kappa^{-1}(-1)^rN\,\mathrm{tr}(M_1Y_1\cdots Y_r)+\kappa^{-1}\sum_{S\neq\emptyset}(-1)^{r-|S|}\,\mathrm{tr}\Big(\prod_{j\in S}^{\leftarrow}Y_j\Big)\,\mathrm{tr}\Big(M_1\prod_{j\notin S}^{\rightarrow}Y_j\Big),$$

$$\mathrm{tr}(M_1X_aM_2X_b)\mapsto\kappa^{-1}\sum_S(-1)^{r-|S|}\,\mathrm{tr}\Big(M_2\prod_{j\in S}^{\leftarrow}Y_j\Big)\,\mathrm{tr}\Big(M_1\prod_{j\notin S}^{\rightarrow}Y_j\Big),$$

$$\mathrm{tr}(M_1X_a)\,\mathrm{tr}(M_2X_b)\mapsto\kappa^{-1}\sum_S(-1)^{r-|S|}\,\mathrm{tr}\Big(M_1\prod_{j\notin S}^{\rightarrow}Y_j\,M_2\prod_{j\in S}^{\leftarrow}Y_j\Big).$$

The pair (X_1, X_n) *of* $\operatorname{tr}(X_1X_2\cdots X_n)$ *is the second case with* $M_1 = 1$, $M_2 = X_2\cdots X_{n-1}$, *and its leading term is* $+\kappa^{-1}N\operatorname{tr}(X_2\cdots X_{n-1}Y_r\cdots Y_1)$, *the first case read from* X_b. *For* $r = 0$ *and* $\mathfrak{su}(N)$: $\operatorname{tr}(M_1X_aX_b) \mapsto \kappa^{-1}(N - \frac{1}{N})\operatorname{tr}M_1$, $\operatorname{tr}(M_1X_aM_2X_b) \mapsto \kappa^{-1}\big(\operatorname{tr}M_1\operatorname{tr}M_2 - \frac{1}{N}\operatorname{tr}(M_1M_2)\big)$ *and* $\operatorname{tr}(M_1X_a)\operatorname{tr}(M_2X_b) \mapsto \kappa^{-1}\big(\operatorname{tr}(M_1M_2) - \frac{1}{N}\operatorname{tr}M_1\operatorname{tr}M_2\big)$.

- *(iii) For odd letters and outputs the same formulas hold with the graded commutator* $[Y, Z] = YZ - (-1)^{|Y||Z|}ZY$ *in the expansion, the Koszul sign of* (10.4) *for the letters, and* $\operatorname{tr}(ZM) = (-1)^{|Z||M|}\operatorname{tr}(MZ)$ *for cyclic moves.*

- *(iv) For* $\mathfrak{su}(N)$, $\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y) = 2N\operatorname{tr}(XY)$, *so* $\frac{\hbar N}{16\pi^2\kappa}$ *is half the unit: a term of the pair kernel equal to a* $\frac{\hbar}{16\pi^2}$ *times* $(1\otimes\operatorname{ad}_{Y_r}\cdots\operatorname{ad}_{Y_1})(-,-)^{-1}$, $r \geq 1$, *contributes to an adjacent pair of a single trace* $(-1)^r\frac{a}{2}$ *units times the single trace with the pair replaced by* $Y_1\cdots Y_r$, *plus terms with more traces or lower powers of* N.

Proof. (i) Every component of P has the colour factor $(-,-)^{-1}$. A cubic vertex has the colour form $(Z_1, [Z_2, Z_3])$ (Definition 1.13); with output Y and cycle slots Z_2, Z_3 it is $(\operatorname{ad}_Y Z_2, Z_3)$, and the quartic vertex (5.2) gives $\pm(Z, \operatorname{ad}_Y\operatorname{ad}_{Y'}Z')$ or $\pm(Z, \operatorname{ad}_{[Y,Y']}Z')$. Composing along the chain from X_a, the vertex nearest X_a acts first, and the invariance of $(-,-)$ moves the word to either factor. (ii) $[Y_r, [\cdots[Y_1, Z]]]$ expands into the sum over S, Y_j multiplying from the left for $j \in S$ and from the right with a sign otherwise. With $\sum_A (T_A)_{ij}(T^A)_{kl} = \delta_{il}\delta_{jk}$ for $\mathfrak{gl}(N)$, $\sum_A \operatorname{tr}(PT_AQT^A) = \operatorname{tr}P\operatorname{tr}Q$ and $\sum_A \operatorname{tr}(PT_A)\operatorname{tr}(QT^A) = \operatorname{tr}(PQ)$, which give the three rules; in the first rule $S = \emptyset$ gives $\operatorname{tr}1 = N$, and for the pair (X_1, X_n) the complement of S empty gives $\operatorname{tr}1$ and $\operatorname{tr}(M_2Y_r\cdots Y_1)$ with the sign $+1$. For $\mathfrak{su}(N)$ the completeness relation acquires $-\frac{1}{N}1\otimes 1$, which drops out for $r \geq 1$ because the word annihilates 1. (iii) Koszul rule. (iv) The unit is $\frac{2N\hbar}{16\pi^2\kappa}$. □

10.3 Census of the cycles and the output bound

The components of λ in $S_+\otimes W$ and $S_-\otimes W^*$ are written $\lambda|_{S_+\otimes W}$ and $\lambda|_{S_-\otimes W^*}$, those of λ^* in $S_+\otimes W^*$ and $S_-\otimes W$ are written $\lambda^*|_{S_+\otimes W^*}$ and $\lambda^*|_{S_-\otimes W}$ (Proposition 1.2).

Lemma 10.9 (Active letters)**.** *A letter of* M *on the cycle of a graph of* (10.1) *lies in a slot with a nonzero component of* $P(\varepsilon, T)$: *the active slots are* $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, λ *of both chiralities,* c *and* $A^*|_V$, *and the letters of* $A^*|_{W_{\mathbb{R}}^\perp}$, λ^* *and* c^* *are spectators of every pair kernel. The cycle leaves an active letter along one of the edges*

$$A|_V - A|_V, \qquad A|_{W_{\mathbb{R}}^\perp} - A|_{W_{\mathbb{R}}^\perp}, \qquad \lambda|_{S_+\otimes W} - \lambda|_{S_-\otimes W^*}, \qquad c - A^*|_V.$$

Proof. A half-edge of M on the cycle is contracted with a component of $P(\varepsilon, T)$ (Proposition 3.17), and $\rho(p)$ exchanges the four-dimensional chiralities (Proposition 1.2). □

Definition 10.10 (Chains)**.** A chain from an active letter X_a to an active letter X_b is a sequence of $k \geq 0$ vertices of I, each with an entry slot and an exit slot, such that the edge of Lemma 10.9 leaving the slot of X_a ends at the entry slot of the first vertex, consecutive vertices are joined by nonzero components of $P(\varepsilon, T)$, and the exit slot of the last vertex is joined to the slot of X_b; for $k = 0$ the two slots are joined directly. The remaining slots of the vertices are the outputs, and their fields, read off in the order of the chain, form the leg set of the chain. The chains are the Taylor components of the dressed propagator of Definition 10.3; a chain and its reverse are the same cycle.

Lemma 10.11 (Superficial degree)**.** *A cycle with edges* $P(\varepsilon, T)$ *through the letters* X_a, X_b, *whose outputs have total Weyl weight* w *(fields only), has superficial degree* $w_{X_a} + w_{X_b} - w$.

Proof. Every vertex of I has Weyl weight 4, fields plus derivatives, and an edge between slots of Weyl weights w_1, w_2 has numerator degree $w_1 + w_2 - 2$ (Proposition 3.17). With k vertices and $k+1$ edges the numerator degree is the sum over the vertices of 4 minus the weights of their slots, plus the sum over the edges of $w_1 + w_2 - 2$, plus the derivatives of the letters; the internal half-edges cancel, the letters appear in edges and the outputs in vertices, which gives $4k - 2(k+1) + w_{X_a} + w_{X_b} - w$. The superficial degree is this plus $4 - 2(k+1)$ (Lemma 4.6). □

Proposition 10.12 (Output bound of the counterterm)**.** *In the pair kernel $\partial_\delta I^{CT}[\varepsilon](X_a X_b)$:*

- *(i) the coefficient of ε^{-m} is local of Weyl weight $w_{X_a} + w_{X_b} - 2m$ in the outputs, and the coefficient of $\log\varepsilon$ is local of Weyl weight $w_{X_a} + w_{X_b}$;*
- *(ii) a chain whose outputs have total Weyl weight w contributes ε^{-m} for $1 \le m \le \frac{1}{2}(w_{X_a} + w_{X_b} - w)$ and $\log\varepsilon$ for $w \le w_{X_a} + w_{X_b}$; for $w > w_{X_a} + w_{X_b}$ it is convergent and contributes nothing to the counterterm;*
- *(iii) every output slot has Weyl weight plus degree at least 1, with equality exactly for c and A, and the outputs carry the pair degree, so the number of output fields is at most*

$$w + (\text{degree of } X_a) + (\text{degree of } X_b) \;\le\; w_{X_a} + w_{X_b} + (\text{degree of } X_a) + (\text{degree of } X_b),$$

  and the number k of vertices of the chain is at most the number of output fields;
- *(iv) for each pair of slots and each pair weight finitely many leg sets occur; their numbers for pair weights up to 7 are in Table 10.3.*

Proof. (i), (ii): Lemma 10.11 and Lemma 4.6: the coefficient of ε^{-m} has momentum degree equal to the superficial degree minus $2m$, the coefficient of $\log\varepsilon$ momentum degree equal to the superficial degree, the output weight is w plus the momentum degree, and both coefficients vanish for negative superficial degree. (iii): the Weyl weight plus the degree is $1, 1, \frac{3}{2}, 2, \frac{3}{2}, 2$ for c, A, λ, A^*, λ^*, c^*; vertices and edges have degree 0, so the degrees of the outputs sum to the pair degree; each vertex has at least three slots, two of them on the cycle. (iv): finitely many leg sets satisfy (iii) at a given pair weight. □

Proposition 10.13 (Census)**.**

- *(i) Every chain conserves the degree, the fermion parity and the action of the centre $\mathbb{Z}/4$ of $SL(4,\mathbb{C})$, $W \mapsto iW$.*
- *(ii) At the pair weight without derivatives the coefficient of $\log\varepsilon$ of the pair kernel has its outputs in the leg sets of Table 10.3, and the coefficient of ε^{-1} in the leg sets of its last column; the empty leg set, the term $k = 0$, enters for even pair weight only, at m equal to half the pair weight.*
- *(iii) $k = 0$ occurs exactly for the pairs $(A|_V, A|_V)$, $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, $(\lambda|_{S_+\otimes W}, \lambda|_{S_-\otimes W^*})$ and $(c, A^*|_V)$; $k = 1$ occurs for all pairs except $(A|_{W_{\mathbb{R}}^\perp}, c)$, (λ, c) and (c, c), whose shortest chains have $k = 2$.*
- *(iv) The coefficient of $\log\varepsilon$ of $\partial_\delta I^{CT}[\varepsilon](A|_{W_{\mathbb{R}}^\perp} A|_{W_{\mathbb{R}}^\perp})$ vanishes on the leg set $A|_V$, and its two chains with the leg set $A|_V\, A|_V$, the triangle and the quartic bubble, cancel.*

Proof. (i) Each vertex of I and each component of $P(\varepsilon, T)$ conserves these charges. (ii) Enumeration of the chains with the vertex list of Definition 1.13 and the edges of Lemma 10.9, with the superficial degree of Lemma 10.11; the empty leg set is Lemma 10.6. (iii) A chain with one vertex from $A|_{W_{\mathbb{R}}^\perp}$ to c needs a vertex with the slots $A|_{W_{\mathbb{R}}^\perp}$ and $A^*|_V$, from λ to c one with the slots λ and $A^*|_V$, from c to c one with two slots $A^*|_V$; I has no such vertex. (iv) Proposition 10.29: the coefficients of the two chains are 2 and -2 times the same structure. □

pair	pair weight	pair degree	$\log\varepsilon$ leg sets	ε^{-1} leg sets
$(A\|_V, A\|_V)$	2	0	$A\|_V$, $A\|_V^2$, $A\|_{W_{\mathbb{R}}^\perp}^2$	$\emptyset$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	2	0	$A\|_V\, A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	–
$(A\|_V, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	0	$A\|_V\, \lambda\|_{S_+\otimes W}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda\|_{S_-\otimes W^*}$, $c\,\lambda^*\|_{S_-\otimes W}$, $\lambda\|_{S_+\otimes W}$	–
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	0	$A\|_V\, \lambda\|_{S_-\otimes W^*}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda\|_{S_+\otimes W}$, $c\,\lambda^*\|_{S_+\otimes W^*}$, $\lambda\|_{S_-\otimes W^*}$	–
$(A\|_V, c)$	1	1	$c\,A\|_V$, c	–
$(A\|_V, A^*\|_V)$	4	-1	$A\|_V\, A^*\|_V$, $A\|_{W_{\mathbb{R}}^\perp}\, A^*\|_{W_{\mathbb{R}}^\perp}$, $A^*\|_V$, $c\,c^*$, $\lambda\|_{S_-\otimes W^*}\,\lambda^*\|_{S_-\otimes W}$, $\lambda\|_{S_+\otimes W}\,\lambda^*\|_{S_+\otimes W^*}$	–
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	2	0	$A\|_V$, $A\|_V^2$, $A\|_{W_{\mathbb{R}}^\perp}^2$	$\emptyset$
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	0	$A\|_V\, \lambda\|_{S_-\otimes W^*}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda\|_{S_+\otimes W}$, $\lambda\|_{S_-\otimes W^*}$	–
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	0	$A\|_V\, \lambda\|_{S_+\otimes W}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda\|_{S_-\otimes W^*}$, $\lambda\|_{S_+\otimes W}$	–
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	1	1	$c\,A\|_{W_{\mathbb{R}}^\perp}$	–
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	4	-1	$A\|_V\, A^*\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}\, A^*\|_V$, $A^*\|_{W_{\mathbb{R}}^\perp}$, $\lambda\|_{S_-\otimes W^*}\,\lambda^*\|_{S_+\otimes W^*}$, $\lambda\|_{S_+\otimes W}\,\lambda^*\|_{S_-\otimes W}$	–
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	3	0	$A\|_V^2\, A\|_{W_{\mathbb{R}}^\perp}$, $A\|_V\, A\|_{W_{\mathbb{R}}^\perp}^2$, $A\|_{W_{\mathbb{R}}^\perp}^3$, $(\lambda\|_{S_-\otimes W^*})^2$, $(\lambda\|_{S_+\otimes W})^2$	$A\|_{W_{\mathbb{R}}^\perp}$
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	3	0	$A\|_V$, $A\|_V^2$, $A\|_V^3$, $A\|_V\, A\|_{W_{\mathbb{R}}^\perp}^2$, $A\|_{W_{\mathbb{R}}^\perp}^2$, $\lambda\|_{S_+\otimes W}\,\lambda\|_{S_-\otimes W^*}$	$A\|_V$
$(\lambda\|_{S_+\otimes W}, c)$	$\frac{3}{2}$	1	$c\,\lambda\|_{S_+\otimes W}$	–
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	$\frac{3}{2}$	-1	$A\|_V^2\, \lambda^*\|_{S_-\otimes W}$, $A\|_V\, A\|_{W_{\mathbb{R}}^\perp}\, \lambda^*\|_{S_+\otimes W^*}$, $A\|_V\, \lambda^*\|_{S_-\otimes W}$, $A\|_{W_{\mathbb{R}}^\perp}^2\, \lambda^*\|_{S_-\otimes W}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda^*\|_{S_+\otimes W^*}$, $\lambda\|_{S_+\otimes W}\, A^*\|_V$, $\lambda\|_{S_-\otimes W^*}\, A^*\|_{W_{\mathbb{R}}^\perp}$, $\lambda^*\|_{S_-\otimes W}$	$\lambda^*\|_{S_-\otimes W}$
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	3	0	$A\|_V^2\, A\|_{W_{\mathbb{R}}^\perp}$, $A\|_V\, A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}^3$, $(\lambda\|_{S_-\otimes W^*})^2$, $(\lambda\|_{S_+\otimes W})^2$	$A\|_{W_{\mathbb{R}}^\perp}$
$(\lambda\|_{S_-\otimes W^*}, c)$	$\frac{3}{2}$	1	$c\,\lambda\|_{S_-\otimes W^*}$	–
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	$\frac{3}{2}$	-1	$A\|_V^2\, \lambda^*\|_{S_+\otimes W^*}$, $A\|_V\, A\|_{W_{\mathbb{R}}^\perp}\, \lambda^*\|_{S_-\otimes W}$, $A\|_V\, \lambda^*\|_{S_+\otimes W^*}$, $A\|_{W_{\mathbb{R}}^\perp}^2\, \lambda^*\|_{S_+\otimes W^*}$, $A\|_{W_{\mathbb{R}}^\perp}\, \lambda^*\|_{S_-\otimes W}$, $\lambda\|_{S_-\otimes W^*}\, A^*\|_V$, $\lambda\|_{S_+\otimes W}\, A^*\|_{W_{\mathbb{R}}^\perp}$, $\lambda^*\|_{S_+\otimes W^*}$	$\lambda^*\|_{S_+\otimes W^*}$
(c, c)	0	2	c^2	–
$(c, A^*\|_V)$	3	0	$A\|_V$, $A\|_V^2$, $A\|_V^3$, $c\,A^*\|_V$	$A\|_V$
$(A^*\|_V, A^*\|_V)$	6	-2	$A\|_V^2\, c^*$, $A\|_V\, c^*$, $A\|_V\, A^*\|_{S_+\otimes W^*}\,\lambda^*\|_{S_-\otimes W}$, $A\|_{W_{\mathbb{R}}^\perp}\,(\lambda^*\|_{S_-\otimes W})^2$, $A\|_{W_{\mathbb{R}}^\perp}\,(\lambda^*\|_{S_+\otimes W^*})^2$, $(A^*\|_V)^2$, $(A^*\|_{W_{\mathbb{R}}^\perp})^2$, c^*, $\lambda^*\|_{S_+\otimes W^*}\,\lambda^*\|_{S_-\otimes W}$	c^*

The leg sets of the pair kernels at the pair weight without derivatives, for the 21 unordered pairs of active slots; the outputs carry the pair degree. $\emptyset$ is the term $k = 0$ of Lemma 10.6. The last column lists the leg sets admitted by the slot structure for the coefficient of ε^{-1}.

pair	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$
$(A\|_V, A\|_V)$	3 (2)	7 (3)	18 (4)	34 (5)	67 (6)	121 (7)		
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	2 (2)	7 (3)	16 (4)	32 (5)	59 (6)	114 (7)		
$(A\|_V, \lambda\|_{S_+\otimes W})$	4 (2)	9 (3)	20 (4)	42 (5)	79 (6)			
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	4 (2)	9 (3)	20 (4)	42 (5)	79 (6)			
$(A\|_V, c)$	2 (2)	4 (3)	8 (4)	19 (5)	35 (6)	68 (7)	122 (8)	
$(A\|_V, A^*\|_V)$	6 (2)	15 (3)	35 (4)	73 (5)				
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	3 (2)	6 (3)	13 (4)	24 (5)	48 (6)	89 (7)		
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	3 (2)	7 (3)	16 (4)	35 (5)	68 (6)			
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	3 (2)	7 (3)	16 (4)	35 (5)	68 (6)			
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	1 (2)	2 (3)	7 (4)	16 (5)	32 (6)	59 (7)	114 (8)	
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	5 (2)	13 (3)	29 (4)	67 (5)				
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	6 (3)	12 (4)	24 (5)	44 (6)	84 (7)			
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	6 (3)	14 (4)	26 (5)	51 (6)	94 (7)			
$(\lambda\|_{S_+\otimes W}, c)$	1 (2)	4 (3)	9 (4)	20 (5)	42 (6)	79 (7)		
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	8 (3)	21 (4)	46 (5)					
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	6 (3)	12 (4)	24 (5)	44 (6)	84 (7)			
$(\lambda\|_{S_-\otimes W^*}, c)$	1 (2)	4 (3)	9 (4)	20 (5)	42 (6)	79 (7)		
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	8 (3)	21 (4)	46 (5)					
(c, c)	1 (2)	2 (3)	4 (4)	8 (5)	19 (6)	35 (7)	68 (8)	122 (9)
$(c, A^*\|_V)$	4 (3)	10 (4)	20 (5)	41 (6)	80 (7)			
$(A^*\|_V, A^*\|_V)$	9 (3)	22 (4)						

The number of leg sets of the coefficient of $\log\varepsilon$ of the pair kernels and, in brackets, the largest number of output fields among them, for n derivatives on the two letters, up to pair weight 7.

Remark 10.14 (Reading of the bound; two loops)**.** At the pair weight without derivatives the bound of Proposition 10.12 (iii) is attained except for $(A|_V, A^*|_V)$ and $(A|_{W_{\mathbb{R}}^\perp}, A^*|_V)$, with bound 3 and largest number 2, and $(A^*|_V, A^*|_V)$, with bound 4 and largest number 3. A ghost output has Weyl weight 0 and degree $+1$, compensated by an antifield output or by the pair degree. At two loops the counterterm of an operator consists of connected cycles through at most three letters and of the factorised terms of the figure-eight graph with M at the junction, which after the one-loop subtractions leave minus the product of the singular parts of its two one-loop subgraphs: a product of two pair kernels acting on

four letters. The counting of (iii) applies to each connected kernel with the pair weight replaced by the weight of its letters.

10.4 Covariance: the BRST relations of the pair kernels

Lemma 10.15 (Derivation rule)**.** *For an odd derivation d of the product of functionals with $d\Delta_0+\Delta_0 d=0$, in particular $d=Q$, $\{I,-\}$ and $Q+\{I,-\}$,*

$$d\{f,g\}=-\{df,g\}-(-1)^{|f|}\{f,dg\},\qquad \textit{that is}\qquad [d,\{f,-\}]=-\{df,-\}.$$

Proof. Insert $\{f,g\}=\Delta_0(fg)-(\Delta_0 f)g-(-1)^{|f|}f\Delta_0 g$ (Definition 1.6) and $d(fg)=(df)g+(-1)^{|f|}f\,dg$, and move d through Δ_0. For Q, $[\partial_{K_0},Q]=\partial_{(Q\otimes 1+1\otimes Q)K_0}=0$ (Proposition 1.11); $\{I,-\}$ is a derivation anticommuting with Δ_0 since $\Delta_0 I=0$ (proof of Theorem 4.30). For odd f the identity reads $s\{f,g\}-\{f,sg\}=-\{sf,g\}$, the identity of Proposition 8.19 (a). □

Theorem 10.16 (BRST covariance of the logarithmic coefficient)**.** *For every local functional M,*

$$\begin{aligned}&\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big((Q+\{I,-\})M\big)\big)\\ &\quad=(Q+\{I,-\})\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](M)\big)+\big\{\textit{coefficient of } \log\varepsilon \textit{ in } I^{CT}[\varepsilon],\,M\big\}.\end{aligned}\tag{10.5}$$

Equivalently, the map

$$M\longmapsto \big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](M)\big)-\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),\,M\Big\}$$

commutes with $Q+\{I,-\}$, and for the operators of Theorem 6.11 twice this map, applied to O, is the dilatation coefficient of that theorem times O.

Proof. Take $T=\infty$ and $J=I-I^{CT}[\varepsilon]+\delta M(\varepsilon)$. By Proposition 4.26, $Q\,W(P(\varepsilon,\infty),J)$ is the flow of the defect vertex $QJ+\frac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J$, and it has a limit as $\varepsilon\to 0$ at generic momenta (Proposition 5.3; Q multiplies the leg functions by polynomials). Its δ-linear part at order $\hbar$ consists of the following graphs.

- The tree vertices $-Q\,\partial_\delta I^{CT}[\varepsilon](M)-\{I,\partial_\delta I^{CT}[\varepsilon](M)\}_\varepsilon-\{I^{CT}[\varepsilon],M\}_\varepsilon$. The bracket $\{-,-\}_\varepsilon$ is $\{-,-\}$ plus the bracket built from $K_\varepsilon-K_0=(e^{-\varepsilon H}-1)K_0$; since $\varepsilon^{-m}(e^{-\varepsilon H}-1)=\sum_{j\ge1}\frac{(-H)^j}{j!}\varepsilon^{j-m}$ consists of pure powers and $\log\varepsilon\,(e^{-\varepsilon H}-1)\to 0$ at a generic momentum, the coefficient of $\log\varepsilon$ of these vertices is minus $(Q+\{I,-\})$ of the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](M)$ minus the bracket of the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ with M. Their part of order ε^0 is the finite local vertex

  $$-\sum_{m\ge1}\frac{(-1)^m}{m!}\Big(\big\{I,\text{coefficient of }\varepsilon^{-m}\text{ in }\partial_\delta I^{CT}[\varepsilon](M)\big\}_{H^m}+\big\{\text{coefficient of }\varepsilon^{-m}\text{ in }I^{CT}[\varepsilon],M\big\}_{H^m}\Big),\tag{10.6}$$

  with $\{-,-\}_{H^m}$ the bracket built from $(H^m\otimes 1)K_0$ in place of K_0; it is the coefficient of ε^0 of the counterterm partners of the graphs with the special edge on a bridge, and with these graphs it tends to 0 (Lemma 12.37).

- The one-loop graphs with the vertex $QM+\{I,M\}_\varepsilon$, which is $(Q+\{I,-\})M$ plus the special-edge part of $\{I,M\}_\varepsilon$. The cycles through $(Q+\{I,-\})M$ have the singular part $\partial_\delta I^{CT}[\varepsilon]\big((Q+\{I,-\})M\big)$; the graphs with this vertex on a tree pair with the trees of $-I^{CT}[\varepsilon]$ and have limits (Theorem 10.4 (ii)). The cycles through a special edge have no $\log\varepsilon$ (Lemma 10.24), and a special edge on a bridge multiplies a singular part by $e^{-\varepsilon|p|^2}-1$, which gives pure powers as above.

- $\hbar\Delta_\varepsilon M$: contractions through K_ε, whose scalar factor $(4\pi\varepsilon)^{-2}e^{-|x|^2/4\varepsilon}$ has Taylor coefficients that are single powers of ε; no $\log\varepsilon$.

- The δ-independent defect: at order $\hbar^0$ the special edge of $\frac{1}{2}\{I,I\}_\varepsilon$ on a cycle through M, without $\log\varepsilon$, or on a bridge; at order $\hbar$ the defect of the action joined to M by trees, which has a limit.

The sum has a limit, so the local vertex formed by the coefficients of $\log\varepsilon$ vanishes, the tree-level flow being injective (Theorem 8.15 (i)); this is (10.5). By Theorem 4.22 the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ is the unit times $(Q+\{I,-\})$ of the field-counting functional, which is odd, and Lemma 10.15 gives $\{(Q+\{I,-\})F,M\} = -(Q+\{I,-\})\{F,M\} + \{F,(Q+\{I,-\})M\}$ for the field-counting functional F; inserted into (10.5) this is the commutation. The last statement is (6.8): its right side is $2\big((E+w_O)O\big)[\infty]$ plus 2 times the image of O under the map. □

For a letter X and a constant tensor ζ of the dual of its fibre and colour space, $(Q+\{I,-\})\int\omega\,(\zeta,X) = \int\omega\,(\zeta,\text{a polynomial in letters})$, the image of the letter, by Proposition 1.11 and the brackets of the proof of Proposition 9.4 (iii). For $M=\int\omega\,X_aX_b$, $(Q+\{I,-\})M$ is $\int\omega$ of the image of X_a times X_b plus $(-1)^{|X_a|}X_a$ times the image of X_b.

Corollary 10.17 (Ward identities of the pair kernels)**.**

- *(W1) For every single letter $N=\int\omega\,(\zeta,X)$, the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]\big((Q+\{I,-\})N\big)$, the sum over the pairs of letters of the image of X of their logarithmic pair kernels, equals $\{$coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon],N\}$.*

- *(W2) For every pair of letters, the sum of the logarithmic pair kernels of the pairs formed by one letter of the image of X_a with X_b, the other letters of the image being spectators, plus $(-1)^{|X_a|}$ times the same sum for X_a and the image of X_b, equals $(Q+\{I,-\})$ applied to the outputs of the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](X_aX_b)$.*

(W1) and (W2) for all letters are equivalent to (10.5).

Proof. For a single letter N the pair kernels are absent, so the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](N)$ vanishes and (10.5) is (W1). For $M=\int\omega\,X_aX_b$ the pairs of the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]\big((Q+\{I,-\})M\big)$ inside the image of X_a or of X_b give $\{$coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon],M\}$ by (W1), the bracket being a derivation of degree 1, and the pairs across the two images give the left side of (W2); $(Q+\{I,-\})$ of the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](M)$ is the right side of (W2). For n letters the spectators sort in the same way, by (10.4). □

Proposition 10.18 (The one-body identities letter by letter)**.** *For a single letter N, $\{$coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon],N\}$ is the unit times $\{S,N\} = (Q+\{I,-\})N$ with every term of S weighted by its multiplier under the field-counting functional:*

$-\frac{1}{4}(dA,dA):\ 4,\quad \frac{1}{2}(\lambda,\not{d}\lambda):\ 8,\quad (dc,A^*):\ -1,\quad$ *cubic gauge term*$:\ 6,\quad \frac{1}{2}(\lambda,(\not{d}_A-\not{d})\lambda):\ 10,\quad ([A,c],A^*):\ 1,$

Hence (W1) reads, with the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](QN)$ equal to 0, QN being a single letter:

- *for $X=A|_{W^\perp_{\mathbb{R}}}$, λ, c: the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](\{I,N\})$ is 1 unit times $\{I,N\}$, from the pairs $(A|_{W^\perp_{\mathbb{R}}},c)$, (λ,c), (c,c), and $QN=0$;*

- *for $X=A|_V$: it is the unit times $\{I,N\}-QN$, from the pair $(A|_V,c)$, with $QN=-\int\omega\,(\zeta,dc)$;*

- *for $X=A^*|_{W^\perp_{\mathbb{R}}}$: it is the unit times $4QN$ plus 6, 8, 10 and 1 times the brackets of N with the cubic gauge term, the quartic term, $\frac{1}{2}\int(\lambda,(\not{d}_A-\not{d})\lambda)$ and $\int([A,c],A^*)$;*

- *for $X=A^*|_V$ the same with the multipliers $4,6,8,10,1$ and without (dc,A^*), for $X=\lambda^*$ with $8,10,1$ on the terms containing λ, and for $X=c^*$ with $-1,1,1,1$ on the terms containing c.*

Proof. By Lemma 4.21 and the proof of Theorem 4.22, the bracket of the unit times $(Q + \{I, -\})$ of the field-counting functional with N contracts the letter of N with the partner slot in each term of S, multiplied by the weighted count of that term. QN is the contraction with the free part of S and $\{I, N\}$ the contraction with I. □

Proposition 10.19 (Triangularity of (W2) and the independent data)**.** *Grade the logarithmic pair kernels by the number r of output fields, and split the image of a letter into QX, a single letter, and $\{I, X\}$, of at least two letters. On the active letters Q acts by $A|_V \mapsto -dc$ and $A^*|_V \mapsto$ a letter $A|_V$ with two derivatives (Proposition 1.11), and annihilates c, $A|_{W_{\mathbb{R}}^\perp}$ and λ. The Q-depth of an active letter is the number of applications of Q that reach a letter annihilated by Q: 0 for c, $A|_{W_{\mathbb{R}}^\perp}$, λ, 1 for $A|_V$ and 2 for $A^*|_V$; that of a pair is the sum.*

- *(i) The part of (W2) with r output fields reads*

  $Q\big($*part with r outputs of the pair* $(X_a, X_b)\big) - \big($*part with r outputs of* $(QX_a, X_b)\big) - (-1)^{|X_a|}\big($*part with r outputs*

  with the source built from the parts with fewer than r outputs: $\{I, -\}$ on their outputs, and the pair kernels of the pairs with one letter of $\{I, X_a\}$ or of $\{I, X_b\}$, the other letters of these being spectators. The pairs (QX_a, X_b) and (X_a, QX_b) have Q-depth one less: the system is triangular in r and the Q-depth.

- *(ii) For a pair of Q-depth 0 the equation is $Q($part with r outputs$) = ($source$)$, and its homogeneous solutions are Q-closed in the outputs. On a leg set of field outputs, of c, $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$ and λ, Q substitutes for an output $A|_V$ of momentum p and polarisation a the ghost with the factor $-i(p, a)$, different outputs giving different leg sets; so a Q-closed kernel is annihilated by $a \mapsto p$ in every leg $A|_V$, and by the exactness of the Koszul complex $(\wedge^\bullet V^\vee \otimes \mathbb{C}[p], \iota_p)$ in positive degree it depends on every output $A|_V$ through $p \wedge a$, that is through $dA|_V$.*

- *(iii) For a pair of Q-depth at least 1 the homogeneous solutions depend on the outputs $A|_V$ through more than $dA|_V$: a homogeneous kernel of the Q-depth-0 pair $(\partial^\alpha c, X_b)$ enters, through the term with QX_a, the equation of $(\partial^\beta A|_V, X_b)$. The pair $(A|_V, A|_{W_{\mathbb{R}}^\perp})$ at pair weight 2 has on the leg set $A|_V\, A|_{W_{\mathbb{R}}^\perp}$ the logarithmic kernel $-\frac{1}{2}(a_1, b_1)(a_2, b_2)$, the sum of its two chains in Proposition 10.29, which depends on the output $A|_V$ through more than $dA|_V$.*

- *(iv) Hence the data that (W2) leaves free, given the parts with fewer outputs, are at each r: the Q-closed parts of the kernels of the pairs of Q-depth 0, the pairs $(\partial^\alpha c, X_b)$ with derivatives of the ghost, on which (W1) fixes the combinations that enter the image of $\partial^\alpha X$, for example the sum of the pair kernels of $(\partial A|_{W_{\mathbb{R}}^\perp}, c)$ and $(A|_{W_{\mathbb{R}}^\perp}, \partial c)$, and the Q-closed parts of the kernels of the pairs of Q-depth at least 1. The covariant completion of the outputs, $d \mapsto d_A$ and $dA \mapsto F_A$, is fixed by (i) from the parts with fewer outputs.*

Proof. (i) is the part of (W2) with r output fields: $\{I, -\}$ on a kernel adds one or two fields; in the pair kernels of $(\{I, X_a\}, X_b)$ the letters of $\{I, X_a\}$ other than the paired one are spectators, so the kernel part has fewer outputs; QX is a single letter of the next lower Q-depth (Proposition 1.11). (ii) For letters of Q-depth 0, $QX_a = QX_b = 0$. On field outputs, Q of a functional linear in an output $A|_V$ of polarisation a substitutes $a \mapsto -ip$ times the ghost; vanishing for all ghosts gives a dependence $(a, h(p))$ with $(p, h(p)) = 0$, and $h = \iota_p g$ with g antisymmetric, so the dependence is $g(p, a)$. Antifield outputs are moved by Q into other leg sets, $A^* \to A$, $\lambda^* \to \lambda$, $c^* \to A^*$, and couple the leg sets; (ii) is stated for field outputs. (iii) The kernel $-\frac{1}{2}(a_1, b_1)(a_2, b_2)$ changes under $b_1 \mapsto p_1$ to $-\frac{1}{2}(a_1, p_1)(a_2, b_2) \neq 0$. (iv) Induction on r and the Q-depth with (i)–(iii). □

Proposition 10.20 (Instances of the Ward identities)**.** *With the images $A|_V \mapsto -dc-[A|_V, c]$, $A|_{W_{\mathbb{R}}^\perp} \mapsto -[A|_{W_{\mathbb{R}}^\perp}, c]$, $c \mapsto \frac{1}{2}[c,c]$, $\lambda \mapsto [c,\lambda]$ of the field letters (Proposition 1.11 and the proof of Proposition 9.4 (iii)), written for $\mathfrak{g} = \mathfrak{su}(N)$ as $-dc-(A|_V c - cA|_V)$, $-(A|_{W_{\mathbb{R}}^\perp} c - cA|_{W_{\mathbb{R}}^\perp})$, cc, $c\lambda+\lambda c$:*

- *(i) the pair kernels satisfy (W1), with the brackets of Proposition 10.18, for the field letters $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, λ, c, and for the antifield letters $A^*|_{W_{\mathbb{R}}^\perp}$, $A^*|_V$, λ^*, c^* on their leg sets with several field outputs;*
- *(ii)* (10.5) *holds for $M = \int \omega\, \mathrm{tr}(\zeta X_a X_b)$, ζ a constant matrix, in the planar reading of Lemma 10.8 (iv), on every leg set of at most three field outputs whose identity involves no image of an antifield output, for the pairs of field letters $(A|_V, A|_V)$, $(A|_V, A|_{W_{\mathbb{R}}^\perp})$, $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, $(A|_V, c)$, $(A|_{W_{\mathbb{R}}^\perp}, c)$, (c,c), $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$, (λ,λ) in the three chirality combinations and (λ, c) at the pair weight without derivatives, and for a set of instances with one and two derivatives on the two letters that contains all leg sets of $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, $(A|_V, c)$, $(A|_{W_{\mathbb{R}}^\perp}, c)$ and (c,c) with one derivative;*

Proof. By Corollary 10.17 each instance is an identity between finite sums of pair kernels, which are exact rational numbers times polynomials in the output momenta and fibre structures (Theorem 10.22); both sides are evaluated exactly with the pair kernels of Sections 10.5 and 10.6 in explicit realisations of the fibres (Remark 10.51). □

10.5 Closed forms: simplex integrals and generating functions

Definition 10.21 (Cycle data)**.** For a chain with k vertices the cycle has $k+1$ edges $e_0, \dots, e_k$, with e_0 leaving X_a, e_k ending at X_b and e_j joining the j-th to the $(j+1)$-th vertex. Let q_j be the sum of the incoming momenta of the outputs of the first j vertices, $q_0 = 0$, so that e_j carries $l + q_j$ for the loop momentum l and the letters have the incoming momenta $-l$ and $l + q_k$. The numerator is the polynomial in l and the output momenta formed by the vertex functions, the polynomial factors of the edges, the fibre elements and the colour factors (Lemma 4.6); the chain numerator is the numerator without the factors of the two letters. On the simplex $\{\alpha \in [0,1]^{k+1} : \sum_j \alpha_j = 1\}$, with the Lebesgue measure $d\alpha$ of $(\alpha_1, \dots, \alpha_k)$, the centre of the cycle is $-\sum_j \alpha_j q_j$ and its momentum form is

$$\sum_j \alpha_j |q_j|^2 - \Big|\sum_j \alpha_j q_j\Big|^2 = \tfrac{1}{2}\sum_{i,j} \alpha_i \alpha_j |q_i - q_j|^2.$$

For $m \geq 0$ the m-th simplex integrand is

$$\sum_{j \geq m+k-1} \frac{\big((\sum_\mu \partial_{l_\mu}^2)^j \text{ numerator}\big)(\text{centre})}{j!\, 4^j} \, \frac{(-\text{momentum form})^{\,j+1-k-m}}{(j+1-k-m)!},$$

a polynomial in α and the output momenta. For each edge e the region of the simplex where α_e is the smallest of the α's has volume $\frac{1}{k+1}$ of the simplex.

Theorem 10.22 (Singular coefficients as simplex integrals)**.** *The value of the cycle with edges $P(\varepsilon, \infty)$,*

$$\int_{[\varepsilon,\infty)^{k+1}} \prod_j dt_j \int \frac{d^4 l}{(2\pi)^4}\, (\textit{numerator}) \prod_j e^{-t_j |l+q_j|^2},$$

has, at generic output momenta, the coefficients

$$\begin{aligned} 16\pi^2\big(\textit{coefficient of } \varepsilon^{-m}\big) &= \frac{1}{m}\int \big(\min_e \alpha_e\big)^m \big(m\textit{-th simplex integrand}\big)\, d\alpha \qquad (m \geq 1), \\ 16\pi^2\big(\textit{coefficient of } \log\varepsilon\big) &= -\int \big(0\textit{-th simplex integrand}\big)\, d\alpha. \end{aligned} \tag{10.7}$$

The first integral is the sum over the edges e of the integrals of α_e^m times the integrand over the region where α_e is the smallest, the graph with e labelled by $-\varepsilon L_\varepsilon$ of Proposition 4.2. Every coefficient is a polynomial in the output momenta with rational coefficients, by $\int \prod_e \alpha_e^{n_e}\, d\alpha = \prod_e n_e!/(\sum_e n_e + k)!$; every coefficient of a pair kernel is a rational multiple of $\frac{\hbar}{16\pi^2}$ times its colour and fibre factors.

Proof. $\sum_j t_j|l+q_j|^2 = \tau|l-\text{centre}|^2 + \tau\cdot(\text{momentum form})$ with $\tau = \sum_j t_j$ and $\alpha_j = t_j/\tau$. The Gaussian integral (Lemma 4.4, iterated) is

$$\int \frac{d^4l}{(2\pi)^4}\,(\text{numerator})\,e^{-\tau|l-\text{centre}|^2} = \frac{1}{16\pi^2\tau^2}\sum_j \frac{\big((\sum_\mu \partial_{l_\mu}^2)^j\,\text{numerator}\big)(\text{centre})}{j!\,(4\tau)^j}.$$

With $\prod_j dt_j = \tau^k d\tau\, d\alpha$ and the cutoff $\tau \geq \varepsilon/\min_e \alpha_e$, the expansion of $e^{-\tau\cdot(\text{momentum form})}$ gives powers τ^{-1-n}, and $\int_{\varepsilon/\min_e\alpha_e}^1 \tau^{-1-n}d\tau$ is $\frac{1}{n}\big(\varepsilon^{-n}(\min_e\alpha_e)^n - 1\big)$ for $n \geq 1$ and $-\log\varepsilon + \log\min_e\alpha_e$ for $n=0$; the region $\tau \geq 1$ has a limit. Collecting the terms with $n=m$ gives (10.7). For the boundary form put $t_f = \varepsilon u_f$ with $u_e = 1$, $u_f \geq 1$; $u \mapsto \alpha = u/\sum_f u_f$ maps $[1,\infty)^k$ onto the region where α_e is the smallest, with $d\alpha = (\sum_f u_f)^{-k-1}du$ and $\sum_f u_f = 1/\alpha_e$. These regions are simplices with rational vertices. □

Proposition 10.23 (Shape independence of the logarithm)**.** *For a Schwinger domain $\{\tau \geq \varepsilon\cdot(\text{cutoff shape})\}$, the cutoff shape a positive function on the interior of the simplex with integrable logarithm ($1/\min_e\alpha_e$ for the cutoff on every edge, 1 for a cutoff on the total Schwinger time),*

$$16\pi^2\big(\text{coefficient of }\varepsilon^{-m}\big) = \frac{1}{m}\int (\text{cutoff shape})^{-m}\big(m\text{-th simplex integrand}\big)\,d\alpha, \qquad 16\pi^2\big(\text{coefficient of }\log\varepsilon\big) = -\int$$
$$16\pi^2\big(\text{coefficient of }\varepsilon^0\big) \ni -\int \big(0\text{-th simplex integrand}\big)\log(\text{cutoff shape})\,d\alpha.$$

The coefficient of $\log\varepsilon$ is independent of the cutoff shape; the coefficients of ε^{-m} and the finite part depend on it. In units of $\frac{1}{16\pi^2}$, for $k=1$ and the numerator $|l|^4$ the cutoff on every edge gives $\frac{|p|^2}{8}\varepsilon^{-1} + \frac{1}{4}\varepsilon^{-2}$ and the cutoff on the total time $|p|^2\varepsilon^{-1} + 3\varepsilon^{-2}$, with coefficient 0 of $\log\varepsilon$ in both; for the triangle with the numerator $|l|^4$ the coefficients of ε^{-1} are $\frac{1}{3}$ and 3, and those of $\log\varepsilon$ agree.

Proof. $\int_{\varepsilon\cdot\text{shape}}^1 \tau^{-1-n}d\tau$ is $\frac{1}{n}\big((\varepsilon\cdot\text{shape})^{-n} - 1\big)$ for $n \geq 1$ and $-\log\varepsilon - \log(\text{shape})$ for $n=0$; the other regions have limits independent of the shape. The examples are exact evaluations of these integrals. □

Lemma 10.24 (Cycles with one special edge)**.** *Number the edges so that e_0 carries $K_\varepsilon - K_0 = -\int_0^\varepsilon (H\otimes 1)K_t\,dt$ (Proposition 4.27): $t_0 \in [0,\varepsilon]$, with the factor $-|l+q_0|^2e^{-t_0|l+q_0|^2}$ times the fibre factor of K_0 included in the numerator, and the other edges $P(\varepsilon,\infty)$. Then the value has no $\log\varepsilon$, and*

$$16\pi^2\big(\text{coefficient of }\varepsilon^{-m}\big) = \frac{1}{m}\int_{\alpha_0\ \text{smallest}} \Big[\big(\min_{f\neq 0}\alpha_f\big)^m - \alpha_0^m\Big]\big(m\text{-th simplex integrand}\big)\,d\alpha, \qquad 16\pi^2\big(\text{coefficient of }\varepsilon^0\big)$$

the remainder tending to 0. For $k=1$, the numerator $-|l|^2$ and e_0 carrying l, the other edge $l-p$, $16\pi^2$ times the value is $-\frac{1}{2\varepsilon} - \frac{3}{8}|p|^2$ plus a function tending to 0, equal to $16\pi^2$ times the direct evaluation of $\int \frac{d^4l}{(2\pi)^4}\big(e^{-\varepsilon|l|^2}-1\big)\frac{e^{-\varepsilon|l-p|^2}}{|l-p|^2}$.

Proof. The domain $\{t_0 \leq \varepsilon,\ t_f \geq \varepsilon\ (f\neq 0)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq0}\alpha_f$ and $1/\min_f\alpha_f$, which differ on the region where α_0 is the smallest; Proposition 10.23 gives the coefficients, the two terms $\log\varepsilon$ cancelling. The convergent terms differ by $O(\varepsilon\log\varepsilon)$ and the region $\tau\geq 1$ by $O(\varepsilon)$. In the example the first simplex integrand is -2 and the 0-th is $|p|^2(1-\alpha_0)(3\alpha_0-1)$, and $\int_0^{1/2}(1-a)(3a-1)\log\frac{a}{1-a}\,da = \frac{3}{8}$. □

Proposition 10.25 (Generating functions)**.** *For the numerator $e^{i(y_b-y_a,l)}$, a formal power series in y_b-y_a, the m-th simplex integrand is*

$$(-1)^{m+k-1}\,e^{i(y_b-y_a,\,centre)}\Big(\frac{|y_b-y_a|}{2\sqrt{momentum\ form}}\Big)^{m+k-1}I_{m+k-1}\Big(|y_b-y_a|\sqrt{momentum\ form}\Big)=(-1)^{m+k-1}e^{i(y_b-y_a,\,cent} \tag{10.8}$$

with I_n the modified Bessel function, and 0 for $m+k-1<0$. For the numerator (chain numerator) $e^{i(y_b-y_a,l)}$ the integrand is the chain numerator, with l replaced by $-i$ times the derivative in y_b-y_a, applied to (10.8). With the letters at the points $x+y_a$ and $x+y_b$, $\varphi_a(x+y_a)\varphi_b(x+y_b)=\sum_{\alpha,\beta}\frac{y_a^\alpha y_b^\beta}{\alpha!\beta!}\partial^\alpha\varphi_a\,\partial^\beta\varphi_b(x)$, the generating functions of the singular coefficients of the pair kernels are

$$\sum_{\alpha,\beta}\frac{y_a^\alpha y_b^\beta}{\alpha!\beta!}\big(coefficient\ of\ \varepsilon^{-m}\ in\ \partial_\delta I^{CT}[\varepsilon](\partial^\alpha\varphi_a\,\partial^\beta\varphi_b)\big)=\frac{\hbar}{16\pi^2 m}\,e^{i(y_b,q_k)}\int\big(\min_e\alpha_e\big)^m\big(m\text{-}th\ simplex\ integrand\big)\,d\alpha,$$

$$\sum_{\alpha,\beta}\frac{y_a^\alpha y_b^\beta}{\alpha!\beta!}\big(coefficient\ of\ \log\varepsilon\ in\ \partial_\delta I^{CT}[\varepsilon](\partial^\alpha\varphi_a\,\partial^\beta\varphi_b)\big)=-\frac{\hbar}{16\pi^2}\,e^{i(y_b,q_k)}\int\big(0\text{-}th\ simplex\ integrand\big)\,d\alpha,$$

summed over the chains, the integrands taken for the generating numerator, the simplex variable α of the integrals being independent of the summation multi-indices α,β; in position space iq_j is the derivative acting on the product of the outputs at x.

Proof. On plane waves the letters at $x+y_a$, $x+y_b$ carry $e^{-i(y_a,l)+i(y_b,l+q_k)}=e^{i(y_b-y_a,l)}e^{i(y_b,q_k)}$. $(\sum_\mu\partial_{l_\mu}^2)^j e^{i(y,l)}=(-|y|^2)^j e^{i(y,l)}$, so the j-th term of the simplex integrand is $e^{i(y,\mathrm{centre})}(-|y|^2/4)^j/j!$ times the power of the momentum form, and the sum over $j=i+m+k-1$ is the series of (10.8), which is $(|y|^2/4)^n(z/2)^{-n}I_n(z)$ with $y=y_b-y_a$ and $z=|y|\sqrt{\text{momentum form}}$, $n=m+k-1$. Each order in y is a finite sum, and the derivatives in y commute with the operations in l. □

Remark 10.26 (The chain numerator)**.** The chain numerator is the product along the chain of the edge numerators and the momentum factors of the vertices:

factor	degree in l	form
edge $A-A$	0	$2(-,-)^{-1}\otimes$ (inverse metric of V_{10})
edge $c-A^*\vert_V$	1	$-i(l+q_j)\,(-,-)^{-1}$
edge $\lambda-\lambda$	1	$i\rho(l+q_j)\,(-,-)^{-1}$
vertices $(A\vert_V)^3$, $A\vert_V(A\vert_{W_{\mathbb{R}}^\perp})^2$	at most 1	one leg momentum: $l+q_{j-1}$, $l+q_j$ or that of the output
vertices $\lambda\lambda A$, $A\,c\,A^*$, $c\,\lambda\,\lambda^*$, $c\,c\,c^*$, quartic	0	constant

It is constant exactly for the chains whose edges are $A-A$ and whose vertices carry no momentum factor on the cycle legs: the quartic bubbles ($k=1$), and the cubic gauge vertices with the momentum factor on the output.

Proposition 10.27 (Explicit forms for $k=0,1,2$)**.**

- *(i) $k=0$: the centre and the momentum form vanish, the m-th integrand is $(-1)^{m-1}(|y_b-y_a|^2/4)^{m-1}/(m-1)!$ and the 0-th is 0; the generating function of the term $k=0$ is the Taylor series of $P(\varepsilon,\infty)$ at the separation y_b-y_a, $16\pi^2$ times the scalar factor at y_b-y_a being $\sum_m\varepsilon^{-m}(-1)^{m+1}(|y_b-y_a|^2/4)^{m-1}/m!$; no $\log\varepsilon$.*

- *(ii) $k=1$, with $\alpha=\alpha_1$, the centre $-\alpha q_1$ and the momentum form $\alpha(1-\alpha)|q_1|^2$:*

$$16\pi^2\big(coefficient\ of\ \log\varepsilon\big)=-e^{i(y_b,q_1)}\,(chain\ numerator)\int_0^1 e^{-i\alpha(y_b-y_a,q_1)}I_0\Big(|y_b-y_a||q_1|\sqrt{\alpha(1-\alpha)}\Big)\,d\alpha,$$

$$16\pi^2\big(coefficient\ of\ \varepsilon^{-m}\big)=\frac{(-1)^m}{m}\,e^{i(y_b,q_1)}\,(chain\ numerator)\int_0^1\min(\alpha,1-\alpha)^m\,e^{-i\alpha(y_b-y_a,q_1)}\Big(\frac{|y_b-y_a|}{2|q_1|\sqrt{\alpha(1-\alpha)}}$$

with l replaced by $-i$ times the derivative in $y_b - y_a$ in the chain numerator. For a constant chain numerator (Remark 10.26), in closed form in the numbers of derivatives,

$$16\pi^2\big(\textit{coefficient of } \log\varepsilon\big) = -\sum_{n_1,n_2,j\geq 0} \big(i(y_a,q_1)\big)^{n_1}\big(i(y_b,q_1)\big)^{n_2}\,|y_b-y_a|^{2j}\Big(\frac{|q_1|^2}{4}\Big)^j \frac{(n_1+j)!\,(n_2+j)!}{n_1!\,n_2!\,(j!)^2\,(n_1+n_2+2j+1)} \tag{10.9}$$

times the chain numerator: the pair $\varphi_a(x+y_a)\varphi_b(x+y_b)$ is replaced by

$$-\frac{\hbar}{16\pi^2}\sum_{n_1,n_2,j}\frac{(n_1+j)!\,(n_2+j)!}{n_1!\,n_2!\,(j!)^2\,(n_1+n_2+2j+1)!}\,(y_a,\partial)^{n_1}(y_b,\partial)^{n_2}|y_b-y_a|^{2j}\big(-\tfrac14\partial^2\big)^j$$

acting on the product of the outputs, times the colour and fibre factors of the chain, $\partial^2=\sum_\mu\partial_\mu^2$.

- *(iii) $k=2$, with the centre $-\alpha_1q_1-\alpha_2q_2$ and the momentum form $\alpha_0\alpha_1|q_1|^2+\alpha_0\alpha_2|q_2|^2+\alpha_1\alpha_2|q_1-q_2|^2$:*

$$16\pi^2\big(\textit{coefficient of } \log\varepsilon\big) = e^{i(y_b,q_2)}\,(\textit{chain numerator})\int e^{i(y_b-y_a,\,\textit{centre})}\sum_{i\geq0}\frac{(|y_b-y_a|^2/4)^{i+1}(\textit{momentum form})^i}{i!\,(i+1)!}$$

$$16\pi^2\big(\textit{coefficient of } \varepsilon^{-m}\big) = \frac{(-1)^{m+1}}{m}\,e^{i(y_b,q_2)}\,(\textit{chain numerator})\int \big(\min_e\alpha_e\big)^m e^{i(y_b-y_a,\,\textit{centre})}\sum_{i\geq0}\frac{(|y_b-y_a|^2/4)^{i+}}{i!\,(}$$

with $\int\alpha_0^{n_0}\alpha_1^{n_1}\alpha_2^{n_2}\,d\alpha=\frac{n_0!\,n_1!\,n_2!}{(n_0+n_1+n_2+2)!}$.

- *(iv) For general k,* (10.8) *applies, with k at most the number of output fields (Proposition 10.12 (iii)).*

Proof. Specialise (10.8). For (10.9), $e^{i(y_b,q_1)-i\alpha(y_b-y_a,q_1)}=e^{i(1-\alpha)(y_b,q_1)+i\alpha(y_a,q_1)}$; multiply by $I_0=\sum_j\big(|y_b-y_a|^2|q_1|^2\alpha(1-\alpha)/4\big)^j/(j!)^2$ and use $\int_0^1\alpha^{n_1+j}(1-\alpha)^{n_2+j}d\alpha=\frac{(n_1+j)!\,(n_2+j)!}{(n_1+n_2+2j+1)!}$. In position space iq_1 is the derivative of the product of the outputs, so $|q_1|^2$ is $-\partial^2$. □

Example 10.28 (The scalar bilinears in closed form)**.** For the quartic bubble (e) of Proposition 5.14 the chain numerator is constant, and (10.9) at $n_1=n_2=j=0$ gives -1, the coefficient $-\frac{1}{16\pi^2}$ of Lemma 5.10. With one derivative on each letter the coefficient of $y_a^\mu y_b^\nu$ comes from $(n_1,n_2,j)=(1,1,0)$ and $(0,0,1)$ and is $\frac{q_{1\mu}q_{1\nu}}{6}+\frac{\delta_{\mu\nu}|q_1|^2}{12}$: the pair $(\partial_\mu\varphi_a,\partial_\nu\varphi_b)$ of a quartic bubble is replaced by $\frac{\hbar}{16\pi^2}\log\varepsilon\,\big(-\frac16\partial_\mu\partial_\nu-\frac{1}{12}\delta_{\mu\nu}\partial^2\big)$ of the product of its two outputs, times its colour and fibre factor. For the triangle (f) the top part of the numerator is a constant times $|l|^2$; the 0-th simplex integrand is twice that constant, and minus its integral over the two-dimensional simplex, of volume $\frac12$, is minus the constant, the coefficient of Lemma 5.10.

10.6 The pair kernels at the lowest weight

Proposition 10.29 (Bosonic pair kernels without derivatives)**.** *At the pair weight without derivatives, the singular coefficients of the pair kernels of two bosonic-slot letters ($A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, c, $A^*|_V$) on their bosonic leg sets are those of Tables 10.6–10.6. A row is a chain with the outputs $Y_1,\dots,Y_r$ in the order of the chain from the first letter; its entry is the coefficient of $\frac{\hbar}{16\pi^2}\log\varepsilon$, or of $\frac{\hbar}{16\pi^2}\varepsilon^{-1}$ where marked, times the colour factor $(1\otimes\mathrm{ad}_{Y_r}\cdots\mathrm{ad}_{Y_1})(-,-)^{-1}$ (Lemma 10.8 (i)), given as the plane-wave value of Definition 1.13 on letters with the fibre elements a_1, a_2 and outputs with the fibre elements $b_1,\dots,b_r$ and the incoming momenta $p_1,\dots,p_r$; $(-,-)$ is the metric of V_{10}, which vanishes between V and $W_{\mathbb{R}}^\perp$, the ghost and c^* carry no fibre element, and ip_j is the derivative of the j-th output. Every entry is in $\mathbb{Q}[i]$. The pairs with one or two letters λ have the leg sets of Table 10.3; their kernels on the planar sector of Proposition 10.33 are given there.*

Proof. Theorem 10.22 and Proposition 10.27, with the chains of Proposition 10.13, the vertex functions of Definition 1.13 and the components of Proposition 3.17, the sign of Lemma 10.5 for the odd letters c and $A^*|_V$, and the colour factor of Lemma 10.8 (i). Each entry is the unique combination of the invariant structures of $O(4) \times O(6)$ in the fibre elements and the output momenta that reproduces the exact values on integer plane waves in an explicit realisation (Remark 10.51). □

pair	outputs	chain	coefficient
$(A\|_V, A\|_V)$	none	$k = 0$, ε^{-1}	$2(a_1, a_2)$
$(A\|_V, A\|_V)$	$A\|_V$	one cubic vertex	$3i\big[(a_2, b_1)(a_1, p_1) - (a_1, b_1)(a_2, p_1)\big]$
$(A\|_V, A\|_V)$	$A\|_V\, A\|_V$	two cubic vertices	$\frac{5}{2}(a_1, a_2)(b_1, b_2) + (a_1, b_1)(a_2, b_2) - 2(a_1, b_2)(a_2, b_1)$
$(A\|_V, A\|_V)$	$A\|_V\, A\|_V$	quartic vertex	$-2(a_1, a_2)(b_1, b_2) - 2(a_1, b_1)(a_2, b_2) + 4(a_1, b_2)(a_2, b_1)$
$(A\|_V, A\|_V)$	$A\|_{W_{\mathbb{R}}^\perp} A\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$\frac{1}{2}(a_1, a_2)(b_1, b_2)$
$(A\|_V, A\|_V)$	$A\|_{W_{\mathbb{R}}^\perp} A\|_{W_{\mathbb{R}}^\perp}$	quartic vertex	$-2(a_1, a_2)(b_1, b_2)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_V\, A\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$\frac{3}{2}(a_1, b_1)(a_2, b_2)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_V\, A\|_{W_{\mathbb{R}}^\perp}$	quartic vertex	$-2(a_1, b_1)(a_2, b_2)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp} A\|_V$	two cubic vertices	$-(a_1, b_2)(a_2, b_1)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp} A\|_V$	quartic vertex	$4(a_1, b_2)(a_2, b_1)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp}$	one cubic vertex	$3i(a_1, p_1)(a_2, b_1)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	none	$k = 0$, ε^{-1}	$2(a_1, a_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_V\, A\|_V$	two cubic vertices	$2(a_1, a_2)(b_1, b_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_V\, A\|_V$	quartic vertex	$-2(a_1, a_2)(b_1, b_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp} A\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$2(a_1, b_1)(a_2, b_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp} A\|_{W_{\mathbb{R}}^\perp}$	quartic vertex	$-2(a_1, a_2)(b_1, b_2) - 2(a_1, b_1)(a_2, b_2) + 4(a_1, b_2)(a_2, b_1)$

The pair kernels of the gauge field and the scalars at pair weight 2. The chain of $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ with the output $A|_V$ vanishes.

pair	outputs	chain	coefficient
$(A\|_V, c)$	$A\|_V\, c$	two cubic vertices	$\frac{3}{2}(a_1, b_1)$
$(A\|_V, c)$	$c\, A\|_V$	two cubic vertices	$-\frac{1}{2}(a_1, b_2)$
$(A\|_V, c)$	c	one cubic vertex	$i(a_1, p_1)$
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	$A\|_{W_{\mathbb{R}}^\perp} c$	two cubic vertices	$2(a_1, b_1)$
(c, c)	$c\, c$	two cubic vertices	2
$(A\|_V, A^*\|_V)$	$A\|_V\, A^*\|_V$	two cubic vertices	$\frac{1}{2}(a_1, a_2)(b_1, b_2) + \frac{1}{2}(a_1, b_1)(a_2, b_2) - (a_1, b_2)(a_2, b_1)$
$(A\|_V, A^*\|_V)$	$A^*\|_V\, A\|_V$	two cubic vertices	$-\frac{1}{2}(a_1, b_1)(a_2, b_2)$
$(A\|_V, A^*\|_V)$	$A\|_{W_{\mathbb{R}}^\perp} A^*\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$\frac{1}{2}(a_1, a_2)(b_1, b_2)$
$(A\|_V, A^*\|_V)$	$A^*\|_V$	one cubic vertex	$i(a_1, b_1)(a_2, p_1)$
$(A\|_V, A^*\|_V)$	$c\, c^*$	two cubic vertices	$-\frac{1}{2}(a_1, a_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	$A\|_V\, A^*\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$-(a_2, b_1)(a_1, b_2)$
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	$A^*\|_{W_{\mathbb{R}}^\perp} A\|_V$	two cubic vertices	$-\frac{1}{2}(a_2, b_2)(a_1, b_1)$
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	$A\|_{W_{\mathbb{R}}^\perp} A^*\|_V$	two cubic vertices	$\frac{1}{2}(a_2, b_2)(a_1, b_1)$
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	$A^*\|_{W_{\mathbb{R}}^\perp}$	one cubic vertex	$i(a_2, p_1)(a_1, b_1)$

The pair kernels with a ghost letter or a letter $A^*|_V$ and a field letter, at the pair weights 1, 0 and 4.

pair	outputs	chain	coefficient
$(c, A^*\|_V)$	$A\|_V$	one cubic vertex, ε^{-1}	$-\frac{1}{8}(a_2, b_1)$
$(c, A^*\|_V)$	$A\|_V$	one cubic vertex	$-\frac{1}{12}(a_2, b_1)\|p_1\|^2 - \frac{1}{6}(a_2, p_1)(b_1, p_1)$
$(c, A^*\|_V)$	$A\|_V\, A\|_V$	two cubic vertices	$\frac{i}{12}(b_1, b_2)(a_2, p_1 + 2p_2) - \frac{i}{12}(a_2, b_2)(b_1, 2p_1 + p_2) + \frac{i}{12}(a_2, b_1)(b_2, p_1 - p_2)$
$(c, A^*\|_V)$	$A\|_V\, A\|_V\, A\|_V$	three cubic vertices	$-\frac{1}{24}\big[(a_2, b_1)(b_2, b_3) + (a_2, b_2)(b_1, b_3) + (a_2, b_3)(b_1, b_2)\big]$
$(c, A^*\|_V)$	$c\, A^*\|_V$	two cubic vertices	$\frac{1}{2}(a_2, b_2)$
$(A^*\|_V, A^*\|_V)$	c^*	one cubic vertex, ε^{-1}	$-\frac{1}{8}(a_1, a_2)$
$(A^*\|_V, A^*\|_V)$	c^*	one cubic vertex	$-\frac{1}{12}(a_1, a_2)\|p_1\|^2 - \frac{1}{6}(a_1, p_1)(a_2, p_1)$
$(A^*\|_V, A^*\|_V)$	$A\|_V\, c^*$	two cubic vertices	$\frac{i}{12}(a_1, b_1)(a_2, p_1 + 2p_2) - \frac{i}{12}(a_2, b_1)(a_1, 2p_1 + p_2) + \frac{i}{12}(a_1, a_2)(b_1, p_1 - p_2)$
$(A^*\|_V, A^*\|_V)$	$c^* A\|_V$	two cubic vertices	$\frac{i}{12}(a_1, b_2)(a_2, p_1 + 2p_2) - \frac{i}{12}(a_2, b_2)(a_1, 2p_1 + p_2) + \frac{i}{12}(a_1, a_2)(b_2, p_1 - p_2)$
$(A^*\|_V, A^*\|_V)$	$A^*\|_V A^*\|_V$	two cubic vertices	$\frac{1}{2}(a_1, a_2)(b_1, b_2)$
$(A^*\|_V, A^*\|_V)$	$A^*\|_{W_{\mathbb{R}}^\perp} A^*\|_{W_{\mathbb{R}}^\perp}$	two cubic vertices	$\frac{1}{2}(a_1, a_2)(b_1, b_2)$
$(A^*\|_V, A^*\|_V)$	$A\|_V\, A\|_V\, c^*$	three cubic vertices	$-\frac{1}{24}\big[(a_1, a_2)(b_1, b_2) + (a_1, b_1)(a_2, b_2) + (a_1, b_2)(a_2, b_1)\big]$
$(A^*\|_V, A^*\|_V)$	$A\|_V\, c^* A\|_V$	three cubic vertices	$-\frac{1}{24}\big[(a_1, a_2)(b_1, b_3) + (a_1, b_1)(a_2, b_3) + (a_1, b_3)(a_2, b_1)\big]$
$(A^*\|_V, A^*\|_V)$	$c^* A\|_V\, A\|_V$	three cubic vertices	$-\frac{1}{24}\big[(a_1, a_2)(b_2, b_3) + (a_1, b_2)(a_2, b_3) + (a_1, b_3)(a_2, b_2)\big]$

The pair kernels with two letters of the ghost line, at the pair weights 3 and 6.

Corollary 10.30 (The scalar bilinears graph by graph)**.** *For $O = \int \omega\, (\iota_{w_1} A, \iota_{w_2} A)$, (10.4) has the single pair $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ with $a_1 = w_1$, $a_2 = w_2$, and the rows of Table 10.6 are the graphs of Proposition 5.14: the term $k = 0$ is (a), the vanishing chain with the output $A|_V$ is (b), the quartic and two-cubic chains with the outputs $A|_V\, A|_V$ are (c) and (d), with the coefficients -2 and 2 of $(w_1, w_2) \int \omega\, (A|_V, A|_V)$, and those with the outputs $A|_{W_{\mathbb{R}}^\perp} A|_{W_{\mathbb{R}}^\perp}$ are (e) and (f), with $2\,O - 2(w_1, w_2) \int \omega\, (A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ and $2\,O$, in units times $\log \varepsilon$. The sum is Theorem 5.18.*

Proof. The two colour factors of the letters are paired by $(-,-)$; evaluated on $(1 \otimes \mathrm{ad}_{Y_2} \mathrm{ad}_{Y_1})(-,-)^{-1}$ this gives $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{Y_2} \mathrm{ad}_{Y_1})$, and $\frac{\hbar}{16\pi^2} \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{Y_2} \mathrm{ad}_{Y_1})$ is the unit times (Y_1, Y_2) (Lemma 5.7). The fibre structures give $(a_1, a_2)(b_1, b_2) \mapsto (w_1, w_2)$ times the pairing of the two outputs, and $(a_1, b_1)(a_2, b_2)$, $(a_1, b_2)(a_2, b_1) \mapsto O$, by the symmetry of the pairing. The chain with the output $A|_V$ carries $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{Y_1}) = 0$ and vanishes identically. Example 10.7 gives (a). □

10.7 Planar densities and spectra

Throughout this section $\mathfrak{g} = \mathfrak{su}(N)$ at leading order in N, O is a single trace of letters, every adjacent pair of the trace, the pair of the last and the first letter included, receives the leading term of Lemma 10.8 (ii), and the bracket with the unit times the field-counting functional is distributed over the adjacent pairs, half of the weight of each letter to each of its two pairs. The map

$$O \longmapsto \big(\text{coefficient of } \log \varepsilon \text{ in } \partial_\delta I^{CT}[\varepsilon](O)\big) - \frac{\hbar\, \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X, Y)} \Big\{ 2 \int (A, A^*) - 4 \int (\lambda^*, \lambda) - \int (c^*, c),\, O \Big\}$$

of Theorem 10.16 is then a sum over the adjacent pairs of a two-site density; twice it is the dilatation coefficient of Theorem 6.11.

Proposition 10.31 (The planar scalar density)**.** *On single traces of letters $A|_{W_{\mathbb{R}}^\perp}$ without derivatives, on the part with outputs $A|_{W_{\mathbb{R}}^\perp}$, the coefficient of $\log \varepsilon$ of the pair kernel of an adjacent pair acts as*

$$\mathrm{tr}(\cdots \iota_{w_1} A\, \iota_{w_2} A \cdots) \longmapsto \frac{\hbar\, \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X, Y)} \Big[2\, \mathrm{tr}(\cdots \iota_{w_2} A\, \iota_{w_1} A \cdots) - (w_1, w_2) \sum_K \mathrm{tr}(\cdots \iota_{e_K} A\, \iota_{e_K} A \cdots) \Big],$$

e_K an orthonormal basis of $W_{\mathbb{R}}^\perp$, and the two-site density of the map above is -2 units times

$$\mathrm{tr}(\cdots \iota_{w_1} A\, \iota_{w_2} A \cdots) - \mathrm{tr}(\cdots \iota_{w_2} A\, \iota_{w_1} A \cdots) + \tfrac{1}{2}(w_1, w_2) \sum_K \mathrm{tr}(\cdots \iota_{e_K} A\, \iota_{e_K} A \cdots).$$

On the outputs $A|_V\, A|_V$ the two chains cancel. On the bilinears the map is 0 on the twenty traceless ones and -12 units on the trace.

Proof. By Table 10.6 the two chains with the outputs $A|_{W_{\mathbb{R}}^{\perp}} A|_{W_{\mathbb{R}}^{\perp}}$ sum to $-2(a_1,a_2)(b_1,b_2)+4(a_1,b_2)(a_2,b_1)$, and by Lemma 10.8 (iv) with $r=2$ an adjacent pair receives half of this, in units, with the outputs Y_1Y_2 in the order of the chain; $(a_1,b_2)(a_2,b_1)$ places the fibre element of the second letter on Y_1. The field-counting functional gives 2 per letter, 1 to each of its pairs. The two chains with the outputs $A|_V\, A|_V$ have the coefficients 2 and -2. On the trace of two letters both adjacent pairs act; the trace part of $\sum_K e_K \otimes e_K$ has (e_K, e_K) summed to 6, which gives $-2\cdot 3\cdot 2 = -12$ on the trace and 0 on the symmetric traceless part. □

Proposition 10.32 (The $\mathfrak{su}(2)$ sector)**.** *Let $w, w' \in W_{\mathbb{R}}^{\perp} \otimes \mathbb{C}$ linearly independent with $(w,w) = (w',w') = (w,w') = 0$, and consider the single traces of L letters, $L-M$ of them $\iota_w A$ and M of them $\iota_{w'} A$. The map above is -2 units times $\sum_l$ of one minus the exchange of the letters l and $l+1$, the Hamiltonian of the Heisenberg chain; on the cyclic states the eigenvalues of $\sum_l(1-\text{exchange})$ are:*

(L,M)	*eigenvalues of $\sum_l(1-\text{exchange})$ on the cyclic states*
$M \le 1$	0
$(4,2)$	0, 6
$(5,2)$	0, 4
$(6,2)$	0, $5 \pm \sqrt{5}$
$(6,3)$	0, 6, $5 \pm \sqrt{5}$
$(7,2)$	0, 2, 6
$(7,3)$	0, 2, 5, 5, 6
$(8,2)$	0, $4 - 4\cos\frac{2\pi n}{7}$ *($n=1,2,3$)*
$(8,3)$	0, 6, 4, 4, $4 - 4\cos\frac{2\pi n}{7}$ *($n=1,2,3$)*
$(8,4)$	0, 6, 4, 4, $4 - 4\cos\frac{2\pi n}{7}$ *($n=1,2,3$), the three roots of the cubic with the coefficients* $1, -20, 116, -200$

Proof. For these fibre elements the trace term of Proposition 10.31 vanishes on every adjacent pair, since $(w,w) = (w',w') = (w,w') = 0$; what remains is 2 times the exchange minus 2 times the identity. The spectra are those of the cyclic sector of $\sum_l(1-\text{exchange})$, computed exactly for $L \le 8$ and all M; the cubic $\prod_{n=1}^{3}\big(\,\cdot -4 + 4\cos\frac{2\pi n}{7}\big)$ has the coefficients $1, -14, 56, -56$. □

Proposition 10.33 (The $\mathfrak{su}(1|1)$ sector)**.** *Choose an orthonormal basis $e_1, \dots, e_6$ of $W_{\mathbb{R}}^{\perp}$ and the Cartan subalgebra of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ rotating the planes (e_1,e_2), (e_3,e_4), (e_5,e_6). Let $Z = \iota_{e_5+ie_6} A$, of weights $(0,0,1)$, and let ψ, in this proposition a letter and elsewhere in the chapter the finite part of the operators of Definition 8.11, be the component of $\lambda|_{S_+\otimes W}$ along a fixed element of S_+ and the weight vector of W of weights $(\frac12,\frac12,\frac12)$. For adjacent pairs of single traces of the letters Z and ψ, on the outputs Z and ψ, the coefficients of $\log\varepsilon$ of the pair kernels act as*

$$(Z,\psi) \mapsto 1\cdot(Z\psi) + 2\cdot(\psi Z), \qquad (Z,Z) \mapsto 2\cdot(ZZ), \qquad (\psi,\psi) \mapsto 0,$$

in units, with (Y_1Y_2) the single trace with the pair replaced by Y_1Y_2. The two-site density of the map above is -2 units times one minus the graded exchange, which sends $Z\psi$ to ψZ, ZZ to ZZ and $\psi\psi$ to $-\psi\psi$.

Proof. The kernels are exact evaluations of the chains of the pairs $(A|_{W_{\mathbb{R}}^{\perp}}, \lambda)$, $(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$ and (λ,λ) on these components. The field-counting functional gives 2 for Z and 4 for ψ, half to each pair: $1+2=3$ on (Z,ψ), 2 on (Z,Z), 4 on (ψ,ψ). Hence the density is $(1-3)(Z\psi)+2(\psi Z)$, $(2-2)(ZZ)$ and $-4(\psi\psi)$ units, which is -2 units times $(Z\psi-\psi Z)$, 0 and $2\psi\psi$. □

Proposition 10.34 (Twist two)**.** *Let $n \in V\otimes\mathbb{C}$ with $(n,n)=0$, let $\iota_n d_{A|_V}$ be the covariant derivative along n, and let $Z = \iota_w A$ with $w \in W_{\mathbb{R}}^{\perp}\otimes\mathbb{C}$, $(w,w)=0$. On the span of the single traces $\operatorname{tr}\big((\iota_n d_{A|_V})^k Z\,(\iota_n d_{A|_V})^{S'-k} Z\big)$, $0 \le k \le S'$, whose densities contain the letters $A|_V$ of the covariant derivatives, the map above has the eigenvalues*

$$-8\,h(S)\ \textit{units}, \qquad h(S) = \sum_{j=1}^{S}\frac{1}{j}, \qquad S \le S'\ \textit{even}: \qquad 0,\ -12,\ -\tfrac{50}{3},\ -\tfrac{98}{5},\ -\tfrac{761}{35} \quad (S=0,2,4,6,8),$$

for $S' \le 8$. The two-field part receives the pair kernels of the pairs (Z,Z) with two outputs and of the pairs $(A|_V, Z)$ from the covariant derivatives with one output. In the basis $(\iota_n d)^k Z/k!$ of the two sites, its matrix is -2 units times $H'_{12}+H'_{21}$, with H'_{12} the two-site Hamiltonian of [B03c, (3.14)] (N. Beisert, The complete one-loop dilatation operator of $N=4$ super Yang–Mills theory, hep-th/0307015),

$$\frac{(\iota_n d)^k Z\,(\iota_n d)^{S'-k} Z}{k!\,(S'-k)!} \longmapsto \sum_{k'} \Big(\delta_{kk'}\big(h(k)+h(S'-k)\big) - \frac{1-\delta_{kk'}}{|k-k'|}\Big) \frac{(\iota_n d)^{k'} Z\,(\iota_n d)^{S'-k'} Z}{k'!\,(S'-k')!}.$$

Proof. The pair kernels with derivatives on both letters are the closed forms of Proposition 10.27 for the quartic bubble and Proposition 10.25 for the chains with a gauge-field edge; the outputs depend on the output momenta through (n,p) alone, because every derivative is contracted with n and $(n,n)=0$. The pairs $(A|_V, Z)$ of the covariant derivatives give the outputs Z with derivatives that complete the two-field part; without them the eigenvalues at $S'=2$ are 0 and $-\frac{8}{3}$ units. The matrix is computed exactly on the basis and diagonalised. □

Corollary 10.35 (Comparison with the one-loop dilatation operator)**.** *On the families of Propositions 10.31–10.34, twice the map above, the dilatation coefficient of Theorem 6.11, has the spectrum of $\frac{g^2N}{8\pi^2}\sum_l H_{l,l+1}$, with the two-site Hamiltonian $H_{12}=2h(J_{12})$ of [B03c, (1.1)–(1.3)], the unit being $-\frac{g^2N}{32\pi^2}$: $\frac{g^2N}{8\pi^2}$ times the spectrum of $\sum_l(1-\text{exchange}+\frac{1}{2}\,\text{trace})$ on the scalars, of the Heisenberg Hamiltonian on the $\mathfrak{su}(2)$ sector and of $\sum_l(1-\text{graded exchange})$ on the $\mathfrak{su}(1|1)$ sector, $\frac{g^2N}{2\pi^2}h(S)$ on twist two [B03c, (2.31)], and $\frac{3g^2N}{4\pi^2}$ for the trace bilinear, the Konishi value of Remark 6.13. The map is built from coefficients of $\log\varepsilon$ alone, which are independent of the cutoff shape (Proposition 10.23), of every finite counterterm and of the finite parts of the operators.*

Proof. $2\cdot(-2)$ units $=\frac{g^2N}{8\pi^2}$ and $2\cdot(-8h(S))$ units $=\frac{g^2N}{2\pi^2}h(S)$. On these families H_{12} reduces to $1-\text{exchange}+\frac{1}{2}\,\text{trace}$ on the scalars, to the exchange form on the $\mathfrak{su}(2)$ and $\mathfrak{su}(1|1)$ sectors and to H'_{12} on twist two [B03c, (3.14)], whose eigenvalues on two sites are $2h(S)$ for even S. Theorem 10.16 identifies twice the map with the dilatation coefficient. □

10.8 The dilatation and translation components

The components along Δ and P_b are defined by the master formula of Theorem 11.44 (iii): the sum over the pairs of letters of the coefficients of ε^0 of their defect cycles, minus the bracket with the arity-one finite counterterm. This section evaluates the same flow a second time, through the scaling of the regulator and the composite kernel, and identifies the pieces of the second form; the agreement of the two forms is a theorem of the construction (Corollary 11.56). Throughout, $v=2E$, $v.$ is the operator of Lemma 9.16, $D_1(O)=\hbar L_1(O)^{(1)}(\Delta)$ and $L(O)^{(1)}(\Delta)=\{S^{(1)}(\Delta),O\}+D_1(O)$ (Definition 8.16). By Example 2.13 and the proof of Lemma 9.16 (i), $S^{(1)}(\Delta)$ is the sum of its parts in the sectors of $A|_{W_{\mathbb{R}}^{\perp}}$, of λ and of $(A|_V, c)$,

$$\int \big(2E(A|_{W_{\mathbb{R}}^{\perp}})+2A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}}\big), \qquad -\int \big(2E(\lambda)+3\lambda,\lambda^*\big), \qquad \int \big(2E(A|_V)+2A|_V, A^*|_V\big)-\int \big(2E(c),c^*\big),$$

the Q-closed directions of Theorem 9.26 (iii). The bracket of each part with a field letter X of its sector is $v.X=2(E+w_X)X$, and with the field letters of the other sectors it is 0; the sum $S^{(1)}(\Delta)$ is annihilated by Q and by $\{I,-\}$ (Lemma 9.16 (iii)).

Definition 10.36 (Hypotheses and the composite term along Δ)**.** The hypotheses along Δ are:

- (i) $\hbar F_1^{(0)}$ solves the arity-zero part of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ and contains the finite counterterm of Theorem 4.32, and $\hbar F_1^{(1)}(\Delta)$ solves (9.10), whose right side is the sum of the coefficients of ε^0 of the defect cycles along Δ (Proposition 9.25 (ii)), equal to 4 times the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ plus the composite part (Proposition 9.25 (iii)); so $\hbar F_1^{(1)}(\Delta)$ minus 4 units times the field-counting functional has the composite part as its image under $Q + \{I, -\}$. Its quadratic part is (9.16): the translation-invariant solution of Theorem 9.26 (iii), whose part beyond 4 units times the field-counting functional is $\big[-\frac{2}{3}\int(A, A^*) + \frac{1}{3}\int(\lambda^*, \lambda) + \frac{4}{3}\int(c^*, c)\big]$ units, plus the Q-closed directions: the three sector parts of $S^{(1)}(\Delta)$, with the coefficients θ, θ_λ, θ_V units, θ the coefficient of (9.17), and $Q\int(\iota_x A|_V, c^*)$, $Q\int(\rho(x)\lambda^*, \lambda^*)$. The arity-two equations along (Δ, P_b) and (Δ, ξ) fix $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ (Theorem 9.28); the statements below are written for general values, which is how the free directions enter.
- (ii) O is homogeneous, $(Q + \{I, -\})O = 0$, and $L_1(O)^{(0)} = 0$ for a finite part ψ of the operators that is linear in ω, translation invariant and preserves the Weyl weight; such a ψ removes the outputs of Remark 5.6.
- (iii) The local representative $L(O)^{(1)}(\Delta)$ exists (Definition 8.14); given (i), this holds for every local functional O (Theorem 11.44 (v)).

The composite graphs of O are the graphs of the linearised flow (9.5) in which the composite kernel $2(P \star v.P \star \otimes 1)K_0$, a line from c to $A|_V$ (Proposition 9.18 (ii)), closes two legs of the tree-level flow of O: either the cycle passes through two letters X_a, X_b of O, the composite cycles through the pair, or it closes on the tree attached to one letter, which gives the renormalised composite two-point functional of Remark 9.27 on a leg of O. Their singular parts are the arity-one counterterm of O along Δ and the quadratic part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ (Proposition 9.25 (i)). The composite term of O is the local functional whose tree-level flow at scale T is

$$\lim_{\varepsilon\to 0} Q\big(\text{composite graphs of } O \text{ minus their counterterm trees}\big)[T] - \big(\text{tree insertion of the composite part into } O[T]\big),$$

and the composite term of the pair (X_a, X_b) is the image under Q of the renormalised composite cycles through the pair.

Lemma 10.37 (Finite parts)**.** *Under $\psi \mapsto \psi + \psi'$ (Definition 8.11), L_1 changes by $[s, \psi']$ (Proposition 8.19 (b)): the arity-zero part by $[Q + \{I, -\}, \psi']$ and the component along Δ by $[\{S^{(1)}(\Delta), -\}, \psi'] = [v., \psi']$. For ψ' linear in ω, translation invariant and preserving the Weyl weight, $[v., \psi'] = 0$, and D_1 is unchanged.*

Proof. $\psi'(\int\omega\, D) = \int\omega\,\psi'(D)$ with $\psi'(D)$ built from the density D at the same point, of the same weight and commuting with ∂; after integration by parts against ω, $v.\int\omega\, D = \int\omega\, 2(x\cdot\partial + w_D)D$ (Lemma 9.16 (i)), and ψ' commutes with $x\cdot\partial$ and with the multiplication by $2w_D$. The first statements are Proposition 8.19 (b) and the proof of Proposition 9.36. □

Theorem 10.38 (Decomposition of D_1)**.** *Under the hypotheses of Definition 10.36, at every $T \in (0, \infty]$, the component $D_1(O)$ of the master formula of Theorem 11.44 (iii) is*

$$D_1(O) = 4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](O)\big) - \big\{\hbar F_1^{(1)}(\Delta), O\big\} + \big(\textit{composite term of } O\big). \quad (10.10)$$

In terms of the letters of $O = \int\omega\, X_1\cdots X_n$, with the Koszul signs of (10.4)*,*

$$D_1(O) = \sum_{a<b} \pm\int \omega\cdots\Big[4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](X_aX_b)\big) + \big(\textit{composite term of } (X_a, X_b)\big)\Big]\cdots$$
$$+ \sum_a \int \omega\cdots\Big[-4\,\textit{units}\cdot(\textit{weight of } X_a)\,X_a - \big\{\hbar F_1^{(1)}(\Delta) - 4\,\textit{units}\cdot(\textit{field-counting functional}), X_a\big\} + \big\{\textit{rer}$$

the weight of a letter being its multiplier under the field-counting functional (Lemma 6.7). On the letters $A|_{W_{\mathbb{R}}^{\perp}}$ *the quadratic part of the renormalised composite two-point functional is local,* $\big[\frac{5}{3}\int(2E(A|_{W_{\mathbb{R}}^{\perp}}) + 2A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}}) - \frac{2}{3}\int(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})\big]$ *units (Remark 9.27), and for the solution* (9.17) *the one-output part of the one-body term of* $X = \partial^{\alpha}A|_{W_{\mathbb{R}}^{\perp}}$ *is*

$$\big[-8X + \big(\tfrac{5}{3} - \theta\big)\, v.X\big] \text{ units.}$$

On the letters $A|_V$, λ, c *that functional contains* $\log|p|^2$ *of the momentum, and the sum of the composite terms of the pairs and of the legs is local. For the translation-invariant solution of Theorem 9.26 (iii) the parts of the one-body terms without the composite two-point functional are:*

letter X	-4 *units* $\cdot$ (*weight of* X) $X - \{\hbar F_1^{(1)}(\Delta) - 4$ *units* $\cdot$ (*field-counting functional*)$, X\}$
$\partial^{\alpha}A|_{W_{\mathbb{R}}^{\perp}}$	$-\frac{22}{3}\,X$ *units*
$\partial^{\alpha}A|_{V}$	$-\frac{22}{3}\,X$ *units*
$\partial^{\alpha}\lambda$	$-\frac{47}{3}\,X$ *units*
$\partial^{\alpha}c$	$-\frac{8}{3}\,X$ *units*

A Q*-closed direction* G *added to* $\hbar F_1^{(1)}(\Delta)$ *adds* $-\{G, X\}$*: a units times the sector part of* $S^{(1)}(\Delta)$ *containing* X *adds* $-a\, v.X$ *units, and* $Q\int(\iota_x A|_V, c^*)$, $Q\int(\rho(x)\lambda^*, \lambda^*)$ *add minus their brackets with the letters* $A|_V$, c *and* λ.

Proof. At $T = \infty$, $d_{\infty} = d_{CE} + Q$, and d_{CE} vanishes on arity zero; by (ii) and (iii) the arity-one component along Δ of $d_{\infty}O[\infty] = L(O)[\infty]$ is Q of the arity-one part of $O[\infty]$ along Δ, which is $\big(\{S^{(1)}(\Delta), O\} + D_1(O)\big)[\infty]$; that arity-one part is the $\varepsilon \to 0$ limit of the δ-part of the linearised flow, at $J = I - I^{CT}[\varepsilon] - \hbar F_1^{(0)} + \delta\, O(\varepsilon)$, of the arity-one vertex of Definition 9.23 minus δ times the arity-one counterterm of O. The graph sums converge at $T = \infty$ at generic momenta, a composite line being of order $|p|^{-3}$ at small momentum and the one-loop integrals with external legs at generic momenta converging (Propositions 8.25, 9.18 (iv)); by Theorem 8.15 (ii) the local representative is the same at every finite T.

(1) The vertex $S^{(1)}(\Delta)$. By (9.7) at $T = \infty$, Q of its linearised flow is $\{S^{(1)}(\Delta), W\}_{\varepsilon}$ minus the terms with QW. The special-edge part of $\{S^{(1)}(\Delta), W\}_{\varepsilon}$ on external legs carries $e^{-\varepsilon|p|^2} - 1 \to 0$. By (9.9), $\{S^{(1)}(\Delta), W\}$ is the linearised flow of $(v. - 4\varepsilon\frac{d}{d\varepsilon})J$ plus $4\varepsilon\frac{d}{d\varepsilon}W$, which tends to 0; $v.I = v.\hbar F_1^{(0)} = 0$, minus $(v. - 4\varepsilon\frac{d}{d\varepsilon})I^{CT}[\varepsilon]$ is 4 times the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$, and by (10.2) $(v. - 4\varepsilon\frac{d}{d\varepsilon})O(\varepsilon)$ is $(v.O)(\varepsilon)$ plus 4 times the coefficient of $\log\varepsilon$ in $\partial_{\delta}I^{CT}[\varepsilon](O)$. The δ-part tends to $(v.O)[\infty]$ plus 4 times that coefficient at scale ∞ plus 4 times the tree insertion of the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ into $O[\infty]$ (Corollary 9.22). In the terms with QW, QW is the flow of the arity-zero defect; by (i) its limit vanishes and by (ii) its δ-part too, and Q of the composite graphs remains, since $-\frac{\hbar}{2}\partial(QW_{\text{tree}}) = Q(\frac{\hbar}{2}\partial W_{\text{tree}})$ for the operator ∂ of the composite kernel (Proposition 9.18 (iii)).

(2) The vertex $-\hbar F_1^{(1)}(\Delta)$ enters by tree insertion, and by Lemma 6.8 with $(Q + \{I, -\})O = 0$, $-Q$ of its tree insertion into O is minus the tree insertion of $(Q + \{I, -\})\hbar F_1^{(1)}(\Delta)$, which is 4 times the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ plus the composite part, minus $\{\hbar F_1^{(1)}(\Delta), O\}[\infty]$; the first term cancels the last term of (1).

(3) The counterterm vertices remove the singular parts of the composite graphs (Proposition 9.25 (i), Definition 8.5); their images under Q complete Q of the composite graphs to Q of the renormalised ones.

Collecting, $\big(\{S^{(1)}(\Delta), O\} + D_1(O)\big)[\infty]$ is $(v.O)[\infty]$ plus 4 times the coefficient of $\log\varepsilon$ in $\partial_{\delta}I^{CT}[\varepsilon](O)$, minus $\{\hbar F_1^{(1)}(\Delta), O\}$, plus the composite term, all at scale ∞, with $v.O = \{S^{(1)}(\Delta), O\}$; by (iii) and Theorem 8.15 (i) this is (10.10), and the local representative is the one given by the master formula, both being the unique local functional with this tree-level flow.

Letters. Four times the coefficient of $\log\varepsilon$ is two-body by Theorem 10.4; the brackets with $\hbar F_1^{(1)}(\Delta)$ and with the field-counting functional are derivations; composite graphs of the first kind use two letters, of the second kind one. For the second kind the identity of Lemma 6.8 is used for tree insertions at generic momenta, where its proof, the derivation property of Q and $QP\star+P\star Q=1$ on the tree edges at nonzero momentum, applies to the nonlocal composite two-point functional and gives its bracket with O, nonlocal unless the functional is local in the sector of the letter. On $X=\partial^\alpha A|_{W_{\mathbb{R}}^\perp}$: $\{\int(A,A^*),X\}=X$, the bracket of the sector part with X is $v.X$, and $-8X$ from the weight 2, $+\frac{2}{3}X$ from the translation-invariant solution, $-\theta\, v.X$ from (9.17) and $\frac{5}{3}v.X-\frac{2}{3}X$ from Remark 9.27 sum to $-8X+(\frac{5}{3}-\theta)v.X$, in units. The table: the weights 2, 4, 1 of A, λ, c, and $\{\int(A,A^*),X\}=X$, $\{\int(\lambda^*,\lambda),\lambda\}=-\lambda$, $\{\int(c^*,c),c\}=-c$ (Lemma 4.21): $-8+\frac{2}{3}=-\frac{22}{3}$, $-16+\frac{1}{3}=-\frac{47}{3}$, $-4+\frac{4}{3}=-\frac{8}{3}$. □

Proposition 10.39 (Output bound of the composite term)**.** *Under the hypotheses of Definition 10.36, the composite term of a pair (X_a,X_b) is a local functional of the outputs of Weyl weight $w_{X_a}+w_{X_b}$, the position counted at weight -1, at most linear in the position, of the pair degree and of the fermion parity $|X_a|+|X_b|$; its number of output fields is at most*

$$w_{X_a}+w_{X_b}+1+(\textit{degree of } X_a)+(\textit{degree of } X_b).$$

A composite cycle through the pair contributes only if its outputs before the application of Q have total Weyl weight at most $w_{X_a}+w_{X_b}+3$. At the pair weight without derivatives the composite cycles with the fewest vertices have the leg sets, before Q:

pair	*leg sets of the composite cycles, number of vertices in brackets*
$(A\|_{W_{\mathbb{R}}^\perp},A\|_{W_{\mathbb{R}}^\perp})$	$A\|_{W_{\mathbb{R}}^\perp}A^*\|_{W_{\mathbb{R}}^\perp}$ (2), $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}A^*\|_{W_{\mathbb{R}}^\perp}$ (2), $A\|^2_{W_{\mathbb{R}}^\perp}A^*\|_V$ (3)
$(A\|_V,A\|_{W_{\mathbb{R}}^\perp})$	$A^*\|_{W_{\mathbb{R}}^\perp}$ (1); $A\|_V\,A^*\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}A^*\|_V$, $\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_+\otimes W^*}$, $\lambda\|_{S_+\otimes W}\lambda^*\|_{S_-\otimes W}$, $A\|^2_V A^*\|_{W_{\mathbb{R}}^\perp}$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}A^*\|_V$, $A\|^2_{W_{\mathbb{R}}^\perp}A^*\|_{W_{\mathbb{R}}^\perp}$ (2)
$(A\|_V,A\|_V)$	$A^*\|_V$ (1); $A\|_V\,A^*\|_V$, $A\|_{W_{\mathbb{R}}^\perp}A^*\|_{W_{\mathbb{R}}^\perp}$, $c\,c^*$, $\lambda\|_{S_+\otimes W}\lambda^*\|_{S_+\otimes W^*}$, $\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_-\otimes W}$, $A\|^2_V A^*\|_V$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}A^*\|_{W_{\mathbb{R}}^\perp}$, $A\|^2_{W_{\mathbb{R}}^\perp}A^*\|_V$ (2)
$(\lambda\|_{S_+\otimes W},\lambda\|_{S_-\otimes W^*})$	$\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_-\otimes W}$, $\lambda\|_{S_+\otimes W}\lambda^*\|_{S_+\otimes W^*}$ (2)*; leg sets of three outputs* (3)
$(A\|_{W_{\mathbb{R}}^\perp},c)$	$A\|_V\,A\|_{W_{\mathbb{R}}^\perp}$ (1); $A\|^2_V A\|_{W_{\mathbb{R}}^\perp}$, $c\,A^*\|_{W_{\mathbb{R}}^\perp}$ (2)
(c,c)	$A\|_V\,c$, $A\|^2_V c$ (2); $c^2A^*\|_V$ (3)

Proof. The composite kernel contains $v.$ once, so the graphs and their images under Q are at most linear in the position. Its component $c\otimes A|_V$ obeys the weight rule of Proposition 3.17 with $0+1-4=-3$, the order $|p|^{-3}$ of Proposition 9.18 (iv), and has degree -1; Q has degree $+1$ and preserves the Weyl weight, and Lemma 10.11 gives the total weight $w_{X_a}+w_{X_b}$ of the image, the position at weight -1. Q changes one output into a field of lower Weyl weight with derivatives, two for A^* and one for λ^*, c^* and $A|_V$, and locality of the result requires its momentum degree, which is $w_{X_a}+w_{X_b}$ plus the number of positions minus the output weight plus that number of derivatives, to be nonnegative: the output weight is at most $w_{X_a}+w_{X_b}+3$. With Weyl weight plus degree at least 1 for every output, the number of output fields is at most the field weight plus the pair degree, and the field weight is at most $w_{X_a}+w_{X_b}+1$. The table is the enumeration of Proposition 10.13 with the composite kernel in place of one edge. □

Proposition 10.40 (The planar scalar sector)**.** *Let $\mathfrak{g}=\mathfrak{su}(N)$ at leading order in N and O a single trace whose letters are $A|_{W_{\mathbb{R}}^\perp}$ without derivatives, with its covariant completion. On the part with the outputs $A|_{W_{\mathbb{R}}^\perp}$ alone, with the inputs Theorem 9.30, Proposition 9.31, Theorem 9.35 and Remark 9.27:*

- *(i) for every adjacent pair, modulo terms whose sum over the cyclically adjacent pairs of every trace vanishes, the composite term of the pair is $-v.(X_aX_b)$ units along Δ, and $-\partial_b(X_aX_b)$ units along the translation P_b;*
- *(ii) the composite term of O minus the bracket of O with $\hbar F_1^{(1)}(\Delta)$ minus 4 units times the field-counting functional is $-(\frac{1}{3}+\theta)$ units times $v.O$, so that, with the map of Theorem 10.16,*

$$D_1(O)=4\big(\textit{the map}\big)(O)-\big(\tfrac{1}{3}+\theta\big)\textit{ units}\cdot\{S^{(1)}(\Delta),O\},\qquad P_1(O)=-\big(\tfrac{1}{3}+\theta_P\big)\textit{ units}\cdot\{S^{(1)}(P_b),O\},$$

and on an adjacent pair the two-site density of D_1 is

$$\operatorname{tr}(\cdots \iota_{w_1} A\, \iota_{w_2} A \cdots) \mapsto -8\ \textit{units}\Big[\operatorname{tr}(\cdots \iota_{w_1} A\, \iota_{w_2} A \cdots) - \operatorname{tr}(\cdots \iota_{w_2} A\, \iota_{w_1} A \cdots) + \tfrac{1}{2}(w_1, w_2) \sum_K \operatorname{tr}(\cdots \iota_{e_K} A\, \iota_{e_K} A \cdots$$

- *(iii) at the value $\theta = -\frac{1}{3}$ of Theorem 9.28 D_1 is 4 times the map of Theorem 10.16, and its spectrum is twice that of the one-loop dilatation coefficients on this sector, $\frac{g^2 N}{4\pi^2}$ times the spectrum of $\sum_l (1 - \textit{exchange} + \frac{1}{2}\,\textit{trace})$; with the finite part $\hbar\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$, $a = -(\frac{1}{3} + \theta_P)$ units (Proposition 9.36), $P_1 = 0$ and D_1 is unchanged.*

Proof. (a) Composite cycles. By Proposition 10.39 the composite cycles through $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ whose image under Q has the two outputs $A|_{W_{\mathbb{R}}^\perp}$ are the two triangles formed by the vertex $A|_V (A|_{W_{\mathbb{R}}^\perp})^2$, entered at $A|_{W_{\mathbb{R}}^\perp}$, left at $A|_V$, with the output $A|_{W_{\mathbb{R}}^\perp}$, and the vertex $([A|_{W_{\mathbb{R}}^\perp}, c], A^*|_{W_{\mathbb{R}}^\perp})$, entered at c, left at $A|_{W_{\mathbb{R}}^\perp}$, with the output $A^*|_{W_{\mathbb{R}}^\perp}$, joined by the composite line in either order; a cycle with two outputs has two cubic vertices and the composite line between them, since the line ends on no letter $A|_{W_{\mathbb{R}}^\perp}$. Both vertices act as the identity on the $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ index along the cycle, so each triangle maps $(\iota_{w_1} A, \iota_{w_2} A)$ to outputs with the fibre elements w_1, w_2 in the order of the letters. The leg sets $A|_V\, A|_{W_{\mathbb{R}}^\perp} A^*|_{W_{\mathbb{R}}^\perp}$ and $A|^2_{W_{\mathbb{R}}^\perp} A^*|_V$ give outputs with a factor $A|_V$, the covariant completion of $x \cdot \partial$, outside this part.

(b) Form. A local Spin(4)-invariant density of weight 2, the position at weight -1, at most linear in the position, with the outputs $\iota_{w_1} A\, \iota_{w_2} A$ and no further derivatives, is $b_1\, \iota_{w_1} A\, \iota_{w_2} A + b_2\, (x \cdot \partial\, \iota_{w_1} A)\, \iota_{w_2} A + b_3\, \iota_{w_1} A\, (x \cdot \partial\, \iota_{w_2} A)$ in the planar adjacent form of Lemma 10.8 (iv), with numbers b_1, b_2, b_3.

(c) Normalisation. For the bilinear $\int \omega\, \operatorname{tr}(\iota_{w_1} A\, \iota_{w_2} A)$ the single pair is adjacent on both sides: by Lemma 10.8 (ii) with $M_1 = 1$ and $r = 2$ the terms $S = \emptyset$ and $S = \{1, 2\}$ give $N \operatorname{tr}(Y_1 Y_2)$ and $N \operatorname{tr}(Y_2 Y_1)$, and the others vanish for traceless outputs, so the bilinear receives the density on (X_1, X_2) and on (X_2, X_1): $2b_1 O + (b_2 + b_3) \sum_l x \cdot \partial X_l$. Proposition 9.31 gives -2 units times $v.O = \sum_l (2x \cdot \partial + 2) X_l$ for the traceless and the trace parts alike (the vertices act as the identity on $\mathfrak{so}(W_{\mathbb{R}}^\perp)$, table of Theorem 9.30): $b_1 = -4$ and $b_2 + b_3 = -4$ units.

(d) Length L. Summing over the L cyclically adjacent pairs gives $b_1 L\, O + (b_2 + b_3) \sum_l x \cdot \partial X_l = -2\, v.O$ units; the part $\frac{b_2 - b_3}{2} \big[(x \cdot \partial X_a) X_b - X_a (x \cdot \partial X_b)\big]$ sums to zero, each letter receiving $x \cdot \partial$ once with each sign. So the composite term of the pair is $-v.(X_a X_b)$ units modulo such terms.

(e) One body. By Theorem 10.38 each letter receives $(\frac{5}{3} - \theta) v.X$ units beyond $-8X$; with (d), the composite term of O minus the bracket with $\hbar F_1^{(1)}(\Delta)$ minus 4 units times the field-counting functional is $(\frac{5}{3} - \theta - 2) v.O$ units. For the density, 4 times the map of Theorem 10.16 per pair is -8 units times the bracket of Proposition 10.31, and $v.O$ is half the sum over the adjacent pairs of $v.(X_a X_b)$.

(f) Translations. By Proposition 10.42 and Theorem 9.35, the composite line along P_b carries no position, the density is $\nu_2 (\partial_b X_a) X_b + \nu_3 X_a \partial_b X_b$ with numbers ν_2, ν_3, the bilinear (the composite triangle along P_b summed over the two orderings, -2 units times $\{S^{(1)}(P_b), O\}$) gives $\nu_2 + \nu_3 = -2$ units, and the one-body part is $(\frac{5}{3} - \theta_P) \partial_b X$ units; (d) gives $P_1 = -(\frac{1}{3} + \theta_P)$ units times $\partial_b O$. (iii): Theorem 9.28, Corollary 10.35 and Proposition 9.36, the change along translations being $\frac{a}{2} \lambda_{P_b} \{S^{(1)}(P_b), -\}$ with $\lambda_{P_b} = -2$, and zero along Δ. □

Remark 10.41 (The input of the planar scalar sector)**.** Proposition 10.40 derives the sector from the bilinear values of Chapter 9. Its input is the table of Theorem 9.30, the master formula on the bilinears, which determines the two coefficients 8 of the composite triangle of Proposition 9.31 (Remark 9.32); the composite terms of the pairs enter through that table.

Proposition 10.42 (The translation component)**.** *Under the hypotheses of Definition 10.36 with P_b in place of Δ, the component $P_1(O)$ of the master formula of Theorem 11.44 (iii) is*

$$P_1(O) = -\big\{\hbar F_1^{(1)}(P_b), O\big\} + \big(\textit{composite term of } O \textit{ along } P_b\big),$$

the composite term along P_b defined as along Δ with the composite kernel $2\big(\partial_b(P(\varepsilon,\infty)\star)^2\otimes 1\big)K_0$ (Theorem 9.35), a line from c to $A|_V$ of momentum degree -2 without explicit position. There is no term with the coefficient of $\log\varepsilon$, since ∂_b commutes with $P(\varepsilon,\infty)\star$. Its two-body part is the image under Q of the renormalised composite cycles through a pair along P_b, local of Weyl weight $w_{X_a}+w_{X_b}+1$. On the part with two legs $A|_{W_{\mathbb{R}}^{\perp}}$ of the scalar bilinears $P_1(O)=-(\frac{1}{3}+\theta_P)$ units times $\{S^{(1)}(P_b),O\}$ (Theorem 9.35), which vanishes in the translation-covariant scheme, and on the planar scalar sector Proposition 10.40 applies.

Proof. As for Theorem 10.38, with ∂_b in place of v.: (9.9) holds without the scaling term (Section 9.5), so step (1) gives $(\partial_b O)[\infty]$ plus the terms with QW, and the arity-one equation along P_b has no term with the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$. □

Proposition 10.43 (Spectrum)**.** *Under the hypotheses of Definition 10.36, on the cohomology of $\Lambda^{(0)}=s+\hbar L_1^{(0)}$ in a fixed classical Weyl weight:*

- *(i) the map of Theorem 10.16 induces an operator, and twice it is the one-loop dilatation coefficient (Theorem 6.11), independent of the Q-closed directions of $\hbar F_1^{(1)}(\Delta)$, of ψ and of the finite counterterms (Theorem 8.20 (iii));*
- *(ii) D_1 is 4 times the map plus the composite term of O minus the bracket of O with $\hbar F_1^{(1)}(\Delta)$ minus 4 units times the field-counting functional, and this difference is the image under Q of a non-local functional of degree -1, the renormalised composite graphs of O minus the tree insertion of $\hbar F_1^{(1)}(\Delta)-4$ units times the field-counting functional into O. The difference depends on $\hbar F_1^{(1)}(\Delta)$ through the bracket alone: a sector part of $S^{(1)}(\Delta)$ with the coefficient a units in $\hbar F_1^{(1)}(\Delta)$ contributes $-a$ units times its bracket, and a term $(Q+\{I,-\})G$ contributes $(Q+\{I,-\})\{G,O\}$, which is zero on cohomology;*
- *(iii) on the part with the outputs $A|_{W_{\mathbb{R}}^{\perp}}$ alone of the planar scalar sector the difference is $-(\frac{1}{3}+\theta)$ units times $\{S^{(1)}(\Delta),-\}$ (Proposition 10.40).*

Proof. (i) Theorem 10.16 and Theorem 6.11. (ii) From the proof of Theorem 10.38 and the local form of Lemma 6.8: for $(Q+\{I,-\})O=0$, minus the tree insertion of the image under $Q+\{I,-\}$ of a local G is minus Q of the tree insertion of G plus $\{G,O\}[\infty]$. The Q-closed directions enter through the bracket with $\hbar F_1^{(1)}(\Delta)$, and for $(Q+\{I,-\})O=0$ Lemma 10.15 gives $-\{(Q+\{I,-\})G,O\}=(Q+\{I,-\})\{G,O\}$. (iii) Proposition 10.40. □

Corollary 10.44 (The dilatation component from the pair kernels)**.** *Let a family of gauge-invariant cocycles stable under Λ satisfy the hypotheses of Definition 10.36, with the solution $\hbar F_1^{(1)}(\Delta)$ whose quadratic part is* (9.16) *at the computed values $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$ of Theorem 9.28, the translation-invariant solution of Theorem 9.26 (iii) minus $\frac{1}{3}$ units times $S^{(1)}(\Delta)$. If on its cohomology the composite term of O minus the bracket of O with the translation-invariant solution minus 4 units times the field-counting functional induces $-\frac{1}{3}$ units times $\{S^{(1)}(\Delta),-\}$, then the spectrum of D_1 on the family is 4 times that of the map of Theorem 10.16: twice the one-loop anomalous dimensions. The condition holds on the part with the outputs $A|_{W_{\mathbb{R}}^{\perp}}$ alone of the scalar bilinears (Theorem 9.30) and of the planar scalar sector (Proposition 10.40), on the cohomology of the scalar bilinears when their parts with the outputs $(A|_V,A|_V)$ and $(c,A^*|_V)$ lie in the image of $\Lambda^{(0)}$ (Corollary 9.34 (iii)), and on the families of Theorem 11.38, where it is the pair of identities (b), (c) of Definition 11.30 (Theorem 11.32 (i)).*

Proof. By Proposition 10.43 (ii) the term $-\frac{1}{3}$ units times $S^{(1)}(\Delta)$ of the solution contributes $+\frac{1}{3}$ units times $\{S^{(1)}(\Delta),-\}$, which cancels the operator of the condition. $S^{(1)}(\Delta)$ is annihilated by Q and by $\{I,-\}$ (Lemma 9.16 (iii)), so this solution solves (9.10) at every order in the fields; Theorem 9.28 identifies it as the solution compatible with the arity-two equations on two legs. □

10.9 The components along every generator

For x in a homogeneous basis of $\mathfrak{psl}(4|4)$ let λ_x be its ad_Δ eigenvalue, as in Proposition 9.36: -2 for P_b, $+2$ for K_b, -1 on Σ, $+1$ on Σ^* and 0 for Δ, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ (Remark 2.6). The degree in the position of the vector field or the conformal Killing spinor of x (Definitions 2.1, 2.2) is 0 for P_b, Σ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, 1 for Δ, $\mathfrak{so}(V)$ and Σ^*, and 2 for K_b. The one-loop components Q_1, S_1, D_1, K_1, P_1, M_1, R_1 are the values $\hbar L_1(O)^{(1)}(x)$ of Definition 8.16.

Proposition 10.45 (Structure of the one-loop components)**.** *Let $\hbar F_1^{(1)}(x)$ solve the arity-one equation along x, and let O satisfy (ii) and (iii) of Definition 10.36 along x. Then*

$$\hbar L_1(O)^{(1)}(x) = \sum_{a<b} \pm \int \omega \cdots \big(\textit{two-body kernel of } (X_a, X_b) \textit{ along } x\big) \cdots + \sum_a \int \omega \cdots \big(\textit{one-body kernel of } X_a \textit{ along } x\big)$$

with the Koszul signs of (10.4)*, the two-body kernel depending on the pair and the one-body kernel on the letter alone. The two-body kernel of (X_a, X_b) along x is the coefficient of ε^0 of the defect cycles of the pair, the cycles through the two letters with exactly one special edge $K_\varepsilon - K_0$ and exactly one vertex $S^{(1)}(x)$, placed on a line (for even x the composite line of Definition 9.17 with the vector field of x in place of v), at the special edge, or as a cubic vertex (Definition 11.43); the one-body kernel of a letter X is $-\{\hbar F_1^{(1)}(x), X\}$. This is the master formula of Theorem 11.44 (iii). For $x = \Delta$ and $x = P_b$, Theorem 10.38 and Proposition 10.42 evaluate the same flow a second time, through the scaling of the regulator and the composite kernel.*

Proof. By Lemma 8.4 and Definition 8.14 the component along x of $L(O)[T]$ is read from the $\varepsilon \to 0$ limit of the graphs with one defect vertex linear in δ and in x. At order $\hbar$ such a graph has one cycle or none; a cycle passes through O along two half-edges, two letters, and a tree meets O in one letter per edge. Lemma 4.7, as in Theorem 10.4 (i), separates the cycle through O from the trees attached to it, which carry the fields of the outputs. The local representative and Theorem 8.15 (i) transfer the sum to local functionals. □

Lemma 10.46 (Active letters at arity one)**.** *The special edge has the components of K_0: $A \otimes A^*$, $\lambda \otimes \lambda^*$, $c \otimes c^*$ and their transposes. For $x \in \Sigma$ the quadratic part of $S^{(1)}(x)$ joins λ to A^* and A to λ^* (proof of Proposition 8.25), and for even x it joins a field to its antifield, through $(v.A, A^*)$, $(v.\lambda, \lambda^*)$ and $(v.c, c^*)$. Hence every letter, $A^*|_{W_{\mathbb{R}}^{\perp}}$, λ^* and c^* included, is active at arity one, and the cycles have edges $P(\varepsilon, T)$, at most one special edge or composite line, vertices of I and at most one vertex $S^{(1)}(x)$.*

Proof. Proposition 4.27, the proof of Proposition 8.25, and Proposition 9.18 (ii). □

Proposition 10.47 (Output bound along x)**.** *Under (iii) of Definition 10.36 along x, the two-body kernel of (X_a, X_b) along x is a local functional of Weyl weight $w_{X_a} + w_{X_b} - \frac{1}{2}\lambda_x$, the position counted at weight -1, a polynomial in the position of degree at most that of x, of the pair degree and of the fermion parity $|X_a| + |X_b| + |x|$; its number of output fields is at most*

$$w_{X_a} + w_{X_b} - \tfrac{1}{2}\lambda_x + (\textit{degree of } x \textit{ in the position}) + (\textit{degree of } X_a) + (\textit{degree of } X_b).$$

For each pair of slots and each pair weight finitely many leg sets enter.

Proof. $\{S^{(1)}(x), -\}$ shifts the Weyl weight by $-\frac{1}{2}\lambda_x$, since $[\Delta, x] = \lambda_x x$, and preserves the degree. The defect insertions are built from $S^{(1)}(x)$, I, K_0 and $P(\varepsilon, T)$, each of which obeys the weight rule of Proposition 3.17 with the position at weight -1, and contain the position at most to the degree of the vector field or the Killing spinor of x. Locality and the bound Weyl weight plus degree at least 1 for every output give the bound as in Proposition 10.39. □

Remark 10.48 (Scheme dependence)**.** (a) $\hbar F_1^{(1)}(x)$ enters the one-body kernel alone, through $-\{\hbar F_1^{(1)}(x), -\}$. (b) $\hbar F_1^{(0)}$ enters through the arity-one equation, which contains $\{\hbar F_1^{(0)}, S^{(1)}(x)\}$, and therefore through $\hbar F_1^{(1)}(x)$ in the one-body kernel. (c) A finite part ψ' of the operators changes L_1 by $[s, \psi']$ (Proposition 8.19 (b)), and the component along x by $[\{S^{(1)}(x), -\}, \psi']$; for $\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$ by $\frac{a}{2}\lambda_x\{S^{(1)}(x), -\}$ (Proposition 9.36). (d) The cutoff shape enters the finite parts of the special-edge cycles (Proposition 10.23, Lemma 10.24). The finite vertex (10.6), the coefficient of ε^0 of the special-edge parts of the counterterm brackets, is cancelled in the components by the graphs with the special edge on a bridge: a bridge and its counterterm partner give $e^{-\varepsilon|p|^2} - 1$ times a renormalised subgraph, which tends to 0 (Theorem 11.44 (ii)); the two-body kernels are the coefficients of ε^0 of the defect cycles alone, in arity one and in arity zero. The spectrum of D_1 on cohomology, at the coefficients of $\hbar F_1^{(1)}(\Delta)$ fixed by the arity-two equations (Theorem 9.28), is independent of (a)–(d) (Theorem 8.20 (iii)).

Remark 10.49 (The data that complete the one-loop components)**.** The following data are given by the graphs of this chapter and are the next computations. The composite terms of the pairs beyond the planar scalar sector, first on the twist-two pairs $\big((\iota_n d)^k Z, (\iota_n d)^{k'} Z\big)$ and $(A|_V, Z)$ and on a pair of gaugino letters, are the special-edge cycles of Theorem 9.30 through the pair, with one special edge and the composite line or $S^{(1)}(\Delta)$ on an edge at a letter; with them the condition of Corollary 10.44 becomes a computation on these families, carried out in Theorem 11.38. The coefficients of the three sector parts of $S^{(1)}(\Delta)$ in $\hbar F_1^{(1)}(\Delta)$ are fixed to $-\frac{1}{3}$ by the two-leg arity-two equations along (Δ, P_b) and (Δ, ξ) (Theorem 9.28); the cubic part of (9.10), in which the bracket of I with each sector part is nonzero while $\{I, S^{(1)}(\Delta)\} = 0$, is then a consistency condition on three legs. The two-body kernels of Q_1, S_1 and K_1 are the defect cycles of Theorem 11.44, with the active letters of Lemma 10.46 and the bound of Proposition 10.47; along $\xi \in \Sigma$ a two-body kernel has Weyl weight $w_{X_a} + w_{X_b} + \frac{1}{2}$, as in the second-order operator on pairs of letters of [BK, (3.7)]. The identity (10.5) with the images of antifield outputs and of antifield letters, on the pairs $(A|_V, \lambda)$, (λ, c) with derivatives, $(c, A^*|_V)$ and the targets of $(A|_V, A|_V)$ with three outputs, extends Proposition 10.20, and Proposition 10.19 (ii) for antifield outputs, which couple the leg sets under Q, requires the same images. Field-strength families, the general fermion families and the identification of the basis with Beisert's oscillator states beyond two sites complete Corollary 10.35.

Remark 10.50 (Two-component forms)**.** With the components of Definition 8.1:

- Letters. $\partial^\alpha c$, $\partial^\alpha A_m$, $\partial^\alpha A_{ij}$, $\partial^\alpha \lambda_\alpha{}^i$, $\partial^\alpha \bar\lambda_{\dot\alpha i}$, $\partial^\alpha A^*_m$, $\partial^\alpha A^*_{ij}$, $\partial^\alpha \lambda^{*\alpha}{}_i$, $\partial^\alpha \bar\lambda^{*\dot\alpha i}$, $\partial^\alpha c^*$, of Weyl weights $0, 1, 1, \frac{3}{2}, \frac{3}{2}, 3, 3, \frac{5}{2}, \frac{5}{2}, 4$ plus $|\alpha|$; $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$ are $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, and $\iota_w A = \frac{1}{4} w^{ij} A_{ij}$, $(u, w) = \frac{1}{4} u^{ij} w_{ij}$ on $W^\perp_{\mathbb{R}}$.
- Decomposition. (10.4) is the same sum over the pairs, the pair kernel being $\hbar\,\mathrm{Sing}_\varepsilon$ of the contraction $\langle X_a(x) X_b(y)\rangle$ in the background of the fields at $y = x$, built from the propagator components of Proposition 8.2,

  $$\langle A_m A_n\rangle = 2(-,-)^{-1}\delta_{mn}, \quad \langle A_{ij} A_{kl}\rangle = 4(-,-)^{-1}\varepsilon_{ijkl}, \quad \langle c\, A^*_m\rangle = \langle A^*_m\, c\rangle = -(-,-)^{-1} i p_m, \quad \langle \lambda_\alpha{}^i \bar\lambda_{\dot\beta j}\rangle = -(-,$$

  times $\frac{e^{-\varepsilon|p|^2} - e^{-T|p|^2}}{|p|^2}$, and from the vertices of the action of Proposition 8.2; the Yukawa vertices $\frac{1}{2}(\lambda_\alpha{}^j, \varepsilon^{\alpha\beta}[A_{ji}, \lambda_\beta{}^i])$ and $-\frac{1}{2}(\bar\lambda_{\dot\alpha j}, \varepsilon^{\dot\alpha\dot\beta}[A^{ji}, \bar\lambda_{\dot\beta i}])$ are the vertices $\lambda|_{S_+\otimes W}\lambda|_{S_+\otimes W} A|_{W^\perp_{\mathbb{R}}}$ and $\lambda|_{S_-\otimes W^*}\lambda|_{S_-\otimes W^*} A|_{W^\perp_{\mathbb{R}}}$. The four edges of Lemma 10.9 are $\langle A_m A_n\rangle$, $\langle A_{ij} A_{kl}\rangle$, $\langle \lambda_\alpha{}^i \bar\lambda_{\dot\beta j}\rangle$ and $\langle c\, A^*_m\rangle$; a gaugino edge enters the generating function of Proposition 10.25 as $i\sigma^m_{\alpha\dot\beta}$ times the derivative in $(y_b - y_a)^m$ plus the shift by q_j.
- Field counting. The field-counting functional is $2\int\big[(A_m, A^*_m) + \frac{1}{4}(A^{ij}, A^*_{ij})\big] - 4\int\big[(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big] - \int(c^*, c)$; its bracket multiplies c, A_m, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ by $1, 2, 2, 4, 4$ and the antifields by

$-1,-2,-2,-4,-4$. The Q-depths are 0 for c, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$, 1 for A_m and 2 for A^*_m; for $X=A_m$, (W1) contains the one-field output $-QN=\int\omega\,(\zeta_m,\partial_m c)$ with the coefficient 1 unit, from the pair (A_m,c) of $\{I,N\}$, and the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon]\big(\frac14\int\omega\,(\zeta^{ij},[A_{ij},c])\big)$ is 1 unit times itself.

- Tables. In Tables 10.6–10.6, $(a_1,a_2)=a_{1m}a_{2m}$ for fibre elements in V and $\frac14 a_1^{ij}a_{2ij}$ for fibre elements in $W^\perp_{\mathbb R}$, and $(a,p)=a_m p_m$. The scalar bilinear is $\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)=\frac{1}{16}\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})$, its counterterm $\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\big[\frac14\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})-\frac18 w_1^{ij}w_{2ij}\int\omega\,(A^{kl},A_{kl})\big]+\frac{\hbar\dim\mathfrak g\,w_1^{ij}w_{2ij}}{32\pi^2\varepsilon}\int\omega$.

- Planar densities. On adjacent scalars of a single trace, in units,

$$(A_{ij},A_{kl})\mapsto 2A_{kl}A_{ij}-\tfrac12\varepsilon_{ijkl}A^{mn}A_{mn},\qquad (A_I,A_J)\mapsto 2A_JA_I-\delta_{IJ}A_KA_K,$$

  with $A^{ij}A_{ij}=4A_IA_I$; the two-site density of the map of Theorem 10.16 is -2 units times $1-$ exchange $+\frac12$ trace on the $\mathfrak{so}(6)$ indices. On the $\mathfrak{su}(1|1)$ sector $Z=A_5+iA_6$, ψ the component $\lambda_\alpha{}^i$ of the $\mathfrak{so}(6)$ weights $(\frac12,\frac12,\frac12)$, and $(Z,\psi)\mapsto(Z\psi)+2(\psi Z)$ units. On twist two $Z=A_1+iA_2$, $n=(1,i,0,0)$ and $\iota_n d_{A|_V}=n^mD_m$, $D_m=\partial_m+[A_m,-]$.

- Dilatation. $\{S^{(1)}(\Delta),O\}=2(E+w_O)O$ with $E=x^m\partial_m$, and the three sector parts of $S^{(1)}(\Delta)$ are $\frac14\int\big((2E+2)A^{ij},A^*_{ij}\big)$, $-\int\big[\big((2E+3)\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i\big)-\big((2E+3)\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i}\big)\big]$ and $\int\big((2E+2)A_m,A^*_m\big)-\int\big(2E(c),c^*\big)$ (Remark 9.29). The part of $\hbar F_1^{(1)}(\Delta)$ beyond 4 units times the field-counting functional is, for the translation-invariant solution, $\big\{-\frac23\int\big[(A_m,A^*_m)+\frac14(A^{ij},A^*_{ij})\big]+\frac13\int\big[(\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i)-(\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i})\big]+\frac43\int(c^*,c)\big\}$ units. The one-body term of $\partial^\alpha A_{ij}$ for the solution (9.17) is $\big[-8+(\frac53-\theta)(2E+2+2|\alpha|)\big]\partial^\alpha A_{ij}$ units, and on an adjacent pair of the planar scalar sector

$$D_1:\ (A_{ij},A_{kl})\mapsto\Big[8A_{kl}A_{ij}-2\varepsilon_{ijkl}A^{mn}A_{mn}-8A_{ij}A_{kl}-\tfrac12\big(\tfrac13+\theta\big)\big(\big((2E+2)A_{ij}\big)A_{kl}+A_{ij}\big((2E+2)A_{kl}\big)\big)\Big]\ \text{units},$$

  so that on $\mathrm{tr}(A_{I_1}\cdots A_{I_L})$, $D_1=-8$ units times $\sum_l(1-\text{exchange}+\frac12\,\text{trace})_{l,l+1}$ minus $(\frac13+\theta)$ units times $2(E+L)$. For the Konishi operator $\frac14\int\omega\,(A^{ij},A_{ij})$ at the value $\theta=-\frac13$ of Theorem 9.28, $D_1=-48$ units times O, which is $\frac{3g^2N}{2\pi^2}O$ for $\mathfrak{su}(N)$, twice the Konishi anomalous dimension $\frac{3g^2N}{4\pi^2}$.

- Weinberg. On the Dirac column $(\lambda_\alpha{}^i,\bar\lambda^{\dot\alpha}{}_i)$, $\rho(v)=\gamma^\mu_W v_\mu$ gives Weinberg's form of each line, and with $A_m=-igA^a_mT^a$, $A_I=-igA^a_IT^a$, $\lambda=-\frac{ig}{\sqrt2}\lambda^aT^a$, $\mathrm{tr}(T^aT^b)=\frac12\delta^{ab}$, the unit is $-\frac{g^2N}{32\pi^2}$.

Remark 10.51 (Method of computation)**.** Each chain is evaluated exactly: the Gaussian loop integral and the simplex formulas of Theorem 10.22, or the generating functions of Proposition 10.25, reduce it to Dirichlet integrals of polynomials over the simplex and over its regions, with the fibre algebra in explicit realisations of V_{10} and Σ over $\mathbb Q(i)$ and the colour words reduced by Lemma 10.8. A second evaluation expands the numerator in momentum space, fixes the Schwinger time of one edge at the boundary face of Proposition 4.2 and integrates the others; it gives the same entries at the pair weight without derivatives, at one derivative with at most two outputs, and at two derivatives with at most two outputs for the pairs of the letters $A|_V$ and $A|_{W^\perp_{\mathbb R}}$. The index-free entries of Tables 10.6–10.6 are the combinations of invariant structures that reproduce every component, and the spectra of Section 10.7 are exact diagonalisations. Every integral is a four-dimensional Schwinger integral with the cutoff ε on every edge.

Chapter 11

Composite kernels and the odd two-body kernels

In the construction every one-loop component of a local operator along a generator x of $\mathfrak{psl}(4|4)$ is read from the defect flow of Lemma 8.4: it is the sum over the pairs of letters of the coefficients of ε^0 of the defect cycles, cycles through the two letters with one special edge and one vertex $S^{(1)}(x)$, minus the bracket of the operator with the arity-one finite counterterm $\hbar F_1^{(1)}(x)$, which solves the arity-one equation of the action with all its free directions. This master formula (Theorem 11.44) needs an argument: the graphs with the special edge on a bridge have finite limits, which cancel exactly against the special-edge parts of their counterterm partners. The chapter first treats the composite term of Chapter 10, the second evaluation of D_1 and P_1 through the scaling of the regulator, in which $S^{(1)}(\Delta)$ or $S^{(1)}(P_b)$ sits on the cycle as a line from a ghost leg to a gauge leg (Theorem 10.38, Proposition 10.42); on the letters $A|_V$, λ, c the renormalised composite two-point functional contains $\log|p|^2$ of the momentum (Remark 9.27), and the locality of the composite term is proved: Q maps the composite graphs of a cocycle to cycles with one special edge, and the composite term is a sum over the pairs of letters of the composite pair kernels. These kernels are simplex integrals, evaluated on twist two, the $\mathfrak{su}(1|1)$ sector, the fermion and gluon $\mathfrak{sl}(2)$ sectors and two mixed families; with the sector coefficients $-\frac{1}{3}$ units of $\hbar F_1^{(1)}(\Delta)$ computed from the action-level arity-two equations (Theorem 9.28), D_1 is four times the map of Theorem 10.16, with spectrum twice the one-loop anomalous dimensions, and P_1 vanishes on cohomology in the translation-covariant scheme; the operator-level arity-two equation reproduces the value $-\frac{1}{3}$ family by family. The last section derives the master formula along every generator, computes the action-level obstructions and solutions along the odd generators and along K_b with their free directions and the arity-two equations that relate them, assembles Q_1, S_1, D_1, P_1 and K_1 on the scalar bilinears as representatives together with their scheme-independent content, and compares with the one-loop differential of the holomorphic twist at the level of classes. Chapter 12 states the construction for every generator and every arity together (Theorems 12.38, 12.31 and 12.56).

The unit is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, equal to $-\frac{g^2N}{32\pi^2}$ for $\mathfrak{g}=\mathfrak{su}(N)$ (Remark 6.13). Edges carry $P(\varepsilon,T)$ of Definition 8.3; the special edge is $K_\varepsilon - K_0 = -\int_0^\varepsilon (H\otimes 1)K_t\,dt$, the factor $e^{-\varepsilon|p|^2}-1$ times the components of K_0 (Proposition 4.27). Every component below is independent of the endpoint $T\in(0,\infty]$ of the transfer (Theorem 8.15 (ii)) and is evaluated at $T=\infty$, where the graph sums converge at generic momenta (Proposition 8.25); along K_b it is taken at a finite T (Definition 8.26). On arity zero $s = Q+\{I,-\}$ (Theorem 8.8). Letters, the pair weight, the pair degree and the Koszul signs are those of Definition 10.1 and (10.4). The field-counting functional is $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$, whose bracket multiplies the coordinate functionals of c, A, λ, A^*, λ^*, c^* by 1, 2, 4, -2, -4, -1 (Lemma 6.7), and the map of Theorem 10.16 is the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](O)$ minus the unit times the bracket of O with it (Section 10.7). Along Δ, $v=2E$ and $v.$ is the operator of Lemma 9.16, so that

$\{S^{(1)}(\Delta), M\} = v.M$; along P_b, $v.$ is ∂_b on every slot (Proposition 7.6); for $x \in \{\Delta, P_b\}$, $v_x.$ is the operator of the vector field of x (Definition 7.4). The two-component form of Definition 8.1 of every main formula is in Remark 11.68.

11.1 The composite term as a sum over pairs

Definition 11.1 (Square kernel and composite kernels)**.** For $0 < \varepsilon < T \leq \infty$ and $P = P(\varepsilon, T)$, the square kernel is $\big((P\star)^2 \otimes 1\big)K_0$, and the composite kernel along $x \in \{\Delta, P_b\}$ is

$$\big(P \star v_x. \otimes 1 + 1 \otimes P \star v_x.\big)P,$$

the two legs of $S^{(1)}(x)$ joined, in both orders, to two edges P (Definition 9.17, Theorem 9.35). The sector parts of $S^{(1)}(P_b)$ are

$$\int \big(\partial_b(A|_{W_{\mathbb{R}}^\perp}), A^*|_{W_{\mathbb{R}}^\perp}\big), \qquad -\int (\partial_b\lambda, \lambda^*), \qquad \int \big(\partial_b(A|_V), A^*|_V\big) - \int (\partial_b c, c^*);$$

their sum is $S^{(1)}(P_b)$, and each sector part of $S^{(1)}(\Delta)$ of Section 10.8 is $2\int x^\mu$ times the density of the sector part of $S^{(1)}(P_{e_\mu})$ of the same sector, summed over μ, plus a translation-invariant functional, with e_μ an orthonormal basis of V and x^μ the coordinates.

Lemma 11.2 (Structure of the composite kernels)**.**

- *(i) The composite kernel along x is $2\big(P \star v_x.\, P \star \otimes 1\big)K_0 = (v_x. \otimes 1 - 1 \otimes v_x.)$ applied to the square kernel.*
- *(ii) $(v_x. \otimes 1 + 1 \otimes v_x.)$ applied to the square kernel is $-4\big(\varepsilon\partial_\varepsilon + T\partial_T\big)$ of the square kernel for $x = \Delta$ and 0 for $x = P_b$. Hence the composite kernel along Δ is $2(v. \otimes 1)$ of the square kernel plus $4(\varepsilon\partial_\varepsilon + T\partial_T)$ of it, and the composite kernel along P_b is $2(\partial_b \otimes 1)$ of the square kernel.*
- *(iii) The square kernel and the composite kernels have components in $c \otimes A|_V$ and $A|_V \otimes c$ alone: as operators, $(P\star)^2$ and $P \star v_x.\, P\star$ map the slot of $A^*|_V$ to the slot of c and the slot of c^* to the slot of $A|_V$, and vanish on the other slots.*
- *(iv) In a graph the composite kernel is the vertex $S^{(1)}(x)$ with both legs on edges P. Its term $\int (v_x.A|_V, A^*|_V)$ contributes, with the leg $A|_V$ on the component $A \otimes A$ of P and the leg $A^*|_V$ on the component $A^*|_V \otimes c$. Along Δ this internal vertex is*

$$\int \big((2E+2)A|_V, A^*|_V\big) = 2\int x^\mu \big(\partial_\mu(A|_V), A^*|_V\big) + 2\int \big(A|_V, A^*|_V\big),$$

along P_b it is $\int (\partial_b(A|_V), A^*|_V)$. The components of P lie in $A \otimes A$, $\lambda \otimes \lambda$, $c \otimes A^*|_V$ and $A^*|_V \otimes c$ (Proposition 3.17), so the terms of $S^{(1)}(x)$ in the sectors of $A|_{W_{\mathbb{R}}^\perp}$, λ and c, whose antifield legs are $A^*|_{W_{\mathbb{R}}^\perp}$, λ^*, c^*, enter a cycle through the special edge.

- *(v) $(Q \otimes 1 + 1 \otimes Q)$ annihilates the composite kernel; the operator ∂ of the composite kernel is odd and anticommutes with Q.*
- *(vi) At $T = \infty$ the component of the square kernel is $(Q^{GF})^2$ on the slot of $A^*|_V$, which is $2d^*$ with values in the slot of c (Proposition 9.18 (ii)), times $(-,-)^{-1}$ and*

$$\int_\varepsilon^\infty \int_\varepsilon^\infty e^{-(s'+s'')|p|^2}\, ds'\, ds'' = e^{-2\varepsilon|p|^2}|p|^{-4}:$$

a doubled edge with two Schwinger times $s', s'' \in [\varepsilon, \infty)$, of momentum degree -3, which obeys the weight rule of Proposition 3.17 for the slots c (weight 0) and $A|_V$ (weight 1).

Proof. (i) The first equality is Proposition 9.18 (i) along Δ; along P_b the same computation holds, ∂_b preserving the pairing and commuting with Q^{GF} and H (Proposition 7.6). For the second, $2P\star\, v_x.\, P\star = v_x.(P\star)^2+(P\star)^2 v_x.$, along Δ because $[v., P\star]$ is Q^{GF} times a function of H and commutes with $P\star$ (proof of Proposition 9.18 (i)), along P_b because ∂_b commutes with $P\star$. For the even operator $X = (P\star)^2$, $(X\, v_x.\otimes 1)K_0 = (X\otimes 1)(v_x.\otimes 1)K_0 = -(1\otimes v_x.)(X\otimes 1)K_0$ by Lemma 9.15, $v_x.$ being even and pairing preserving.

(ii) Since $(v_x.\otimes 1 + 1\otimes v_x.)K_0 = 0$, the left side is $([v_x., X]\otimes 1)K_0$. Along Δ, $X = (Q^{GF})^2 g(H)^2$ with $g(H) = \int_\varepsilon^T e^{-tH}dt$, $[v., Q^{GF}] = -4Q^{GF}$ and $[v., f(H)] = -4Hf'(H)$ (Proposition 7.10). From $Hg'(H) = Te^{-TH} - \varepsilon e^{-\varepsilon H} - g = (\varepsilon\partial_\varepsilon + T\partial_T)g - g$,

$$[v., X] = -8X - 4(Q^{GF})^2 H\partial_H\big(g^2\big) = -8X - 4\big((\varepsilon\partial_\varepsilon + T\partial_T)X - 2X\big) = -4\big(\varepsilon\partial_\varepsilon + T\partial_T\big)X.$$

Along P_b, $[\partial_b, X] = 0$. Insert into (i).

(iii) $(Q^{GF})^2$ vanishes on the slots of c, A, λ, λ^*; on A^* and c^* it is $2d^*(A^*|_V)$ in the slot of c and $-2dc^*$ in the slot of $A|_V$ (Proposition 9.18 (ii)); $v_x.$ preserves the slots, and the components $A^*\otimes A$ and $c^*\otimes c$ of K_0 go to $c\otimes A|_V$ and $A|_V\otimes c$.

(iv) Applied to $A^*\otimes A$, the kernel $(P\star\, v_x.\, P\star\otimes 1)K_0$ is the component $A^*\to A$ of Q^{GF}, then $v_x.$ on the slot of A, then the component $A\to c$ of Q^{GF}: the vertex $\int(v_x.A|_V, A^*|_V)$ between an $A\otimes A$ and an $A^*|_V\otimes c$ component, the factor 2 counting the two orientations. On $A|_V$, $L_{2E}A|_V = 2x^\mu\partial_\mu(A|_V) + 2A|_V$ (Definition 7.4).

(v) Proposition 9.18 (iii) along Δ; along P_b, $QP\star + P\star Q = e^{-\varepsilon H} - e^{-TH}$ commutes with ∂_b, which commutes with Q, so $Q(P\star\,\partial_b\, P\star) = (P\star\,\partial_b\, P\star)Q$. The identity $[\partial_\phi, Q] = \partial_{(Q\otimes 1+1\otimes Q)\phi}$ of Lemma 9.15 gives the anticommutation; the composite kernel has degree -1.

(vi) Proposition 9.18 (iv) and $\big(\int_\varepsilon^\infty e^{-s|p|^2}ds\big)^2 = e^{-2\varepsilon|p|^2}|p|^{-4}$. □

Definition 11.3 (Composite graphs and the arity-one counterterm of an operator)**.** Let M be a local functional, δ a square-zero parameter of degree $-|M|$, $x\in\{\Delta, P_b\}$ and $P = P(\varepsilon,\infty)$. The composite graphs of M along x are the δ-linear part of

$$\tfrac{\hbar}{2}\,\partial_{\text{composite kernel along } x}\, W(P, I+\delta M)_{\text{tree}}:$$

the composite kernel closes two legs of a tree of vertices I and one vertex M. The cycle so formed either passes through two letters X_a, X_b of M, a composite cycle through the pair (X_a, X_b) with trees of I on its outputs, or passes through vertices of I alone and is joined to M by a tree through one letter, a composite loop of I on a leg of M. The arity-one counterterm of M along x is $\hbar\,\mathrm{Sing}_\varepsilon$ of the amputated composite cycles through the pairs, summed over the pairs with the Koszul signs of (10.4). The singular part of the composite loops of I is $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ (Proposition 9.25 (i)); its quadratic part along Δ is the unit times $\log\varepsilon$ times $Q\big(\int(\iota_x A|_V, c^*) - \frac{1}{2}\int(\rho(x)\lambda^*, \lambda^*)\big)$, with no term in ε^{-1} (Theorem 9.26 (i)).

Proposition 11.4 (The image of the composite graphs under Q)**.** *For every local functional M, $x\in\{\Delta, P_b\}$ and $\varepsilon>0$, at $T=\infty$,*

$$Q\big(\textit{composite graphs of } M \textit{ along } x\big) = -\tfrac{\hbar}{2}\,\partial_{\textit{composite kernel along } x}\Big[\textit{tree-level flow, } \delta\textit{-linear, with one vertex } sM \textit{ or one } s$$

the special edge joining a vertex I to a letter of M, from $\{I, M\}_\varepsilon - \{I, M\}$, or two vertices I, from $\frac{1}{2}(\{I, I\}_\varepsilon - \{I, I\})$. Since $\Delta_\varepsilon - \Delta_0 = -\partial_{K_\varepsilon - K_0}$ (Definition 1.6, Proposition 4.27), $\{f, g\}_\varepsilon - \{f, g\}$ is minus f and g joined by one edge $K_\varepsilon - K_0$; for $sM = 0$ the right side is $\frac{\hbar}{2}\,\partial_{\textit{composite kernel along } x}$ of the trees with exactly one edge $K_\varepsilon - K_0$.

Proof. By Lemma 11.2 (v), Q anticommutes with ∂ of the composite kernel. By Proposition 4.26 at order $\hbar^0$, $Q\,W(P, J)_{\text{tree}}$ is the tree-level flow of $QJ + \frac{1}{2}\{J, J\}_\varepsilon$; the term $\hbar\Delta_\varepsilon J$ is of order $\hbar$, and its

composite closure has two cycles. For $J = I + \delta M$ the classical master equation $QI + \frac{1}{2}\{I, I\} = 0$ (Proposition 1.14) leaves $\frac{1}{2}(\{I, I\}_\varepsilon - \{I, I\})$ in the δ-independent part and $sM + (\{I, M\}_\varepsilon - \{I, M\})$ in the δ-linear part. □

Definition 11.5 (Special-edge composite cycles and the composite pair kernel)**.** For two letters X_a, X_b of any slots and $x \in \{\Delta, P_b\}$, the special-edge composite cycles of (X_a, X_b) along x are

$$\tfrac{\hbar}{2}\,\partial_{\text{composite kernel along } x}\,\partial_{K_\varepsilon - K_0}\big(\text{amputated chains from } X_a \text{ to } X_b\big)\,:$$

the chains of Definition 10.10 with $k \geq 1$ vertices of I in which one edge e_c is the composite kernel along x, one other edge e_σ is the special edge, with the components of K_0 (each field slot with its antifield slot), which may end at a letter of any slot, and every remaining edge is $P(\varepsilon, \infty)$. By Proposition 11.4 they are minus $\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the chains with one bracket-defect vertex. A chain is read with the signs of Lemma 10.5 along the chain from X_a, times

$$(-1)^{n(<\sigma)+n(<c)+[c<\sigma]},$$

where $n(< j)$ is the number of odd outputs at the vertices before the edge e_j, and $[c < \sigma]$ is 1 if e_c precedes e_σ and 0 otherwise; the outputs are left open as in Definition 10.3. The composite pair kernel of (X_a, X_b) along x is the coefficient of ε^0 of its special-edge composite cycles.

By Proposition 11.4, for $sM = 0$ the image under Q of the composite cycles through the pairs and of the composite loops on the legs is the sum of the special-edge composite cycles with trees of I on their outputs, together with the composite closures in which the special edge lies on a tree; Theorem 11.18 identifies the composite term of a cocycle with the sum over its pairs of letters of the composite pair kernels.

Proposition 11.6 (Reachability and census)**.**

- *(i) The composite kernel joins a slot c to a slot $A|_V$. The letters c and $A|_V$ reach it directly; the letters $A|_{W_\mathbb{R}^\perp}$, λ and $A^*|_V$ reach it through one vertex, $A|_V(A|_{W_\mathbb{R}^\perp})^2$ entered at $A|_{W_\mathbb{R}^\perp}$, $\lambda|_{S_+\otimes W}\lambda|_{S_-\otimes W^*}A|_V$ entered at λ, $([A|_V, c], A^*|_V)$ entered at c, each left at $A|_V$; the letters $A^*|_{W_\mathbb{R}^\perp}$, λ^*, c^*, on which P and the composite kernel have no component, reach it through the special edge, to a slot $A|_{W_\mathbb{R}^\perp}$, λ, c.*

- *(ii) A chain of Definition 11.5 exists for 45 of the 55 unordered pairs of the ten slots; the other ten are the pairs of two letters among $A^*|_{W_\mathbb{R}^\perp}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, which would need two special edges. Every chain conserves the degree, the fermion parity and the centre $\mathbb{Z}/4$ of $SL(4, \mathbb{C})$, and its outputs carry the pair degree; the composite kernel has degree -1 and the special edge $+1$.*

- *(iii) Every chain has $k \geq 1$ vertices. A chain with $k = 1$ has the composite kernel at one letter and the special edge at the other, so $k = 1$ occurs exactly for the pairs with a letter $A|_V$ and for the pairs $(c, A^*|_V)$, $(c, A^*|_{W_\mathbb{R}^\perp})$, (c, λ^*), (c, c^*); the other pairs, the field pairs $(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$, $(A|_{W_\mathbb{R}^\perp}, \lambda)$, (λ, λ) among them, start at $k = 2$.*

- *(iv) At the pair weight without derivatives the leg sets of the ten pairs of field letters are those of Table 11.1; the numbers of leg sets with $n \leq 3$ derivatives on the two letters, for the 21 pairs of the active slots of Lemma 10.9, are those of Table 11.1.*

Proof. (i) Lemma 11.2 (iii), Lemma 10.9 and the vertices of Definition 1.13. (ii), (iv) Enumeration of the chains with one composite kernel and one special edge, the vertices of I and the components of P and K_0, up to six vertices and output weight four above the pair weight; every chain is checked for degree, parity and the centre. A pair of two letters among $A^*|_{W_\mathbb{R}^\perp}$, λ^*, c^* needs an edge with a component on each letter; K_0 alone has one, and a chain has one special edge. (iii) A chain with $k = 0$ would join the two letters by the composite kernel and the special edge at once; the rest is read off from the enumeration. □

pair	pair weight	leg sets of the special-edge composite cycles, fewest vertices in brackets; †: position part along Δ and part along P_b
$(A\|_V, A\|_V)$	2	$A\|_V$ (1), $A\|_V^2$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2$ (2), $A\|_V^{3\dagger}$ (2), $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^{2\dagger}$ (2), $A^*\|_V c^\dagger$ (2), $\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (2)
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	2	$A\|_{W_{\mathbb{R}}^\perp}$ (1), $A\|_V A\|_{W_{\mathbb{R}}^\perp}$ (2), $A\|_V^2 A\|_{W_{\mathbb{R}}^\perp}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^{3\dagger}$ (2), $A^*\|_{W_{\mathbb{R}}^\perp} c^\dagger$ (2), $(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (2), $(\lambda\|_{S_+\otimes W})^{2\dagger}$ (2)
$(A\|_V, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	$\lambda\|_{S_+\otimes W}$ (1), $A\|_V \lambda\|_{S_+\otimes W}$ (2), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}$ (2), $c\lambda^*\|_{S_-\otimes W}$ (2), $A\|_V^2\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}{}^\dagger$ (3), $A\|_V c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	$\lambda\|_{S_-\otimes W^*}$ (1), $A\|_V \lambda\|_{S_-\otimes W^*}$ (2), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}$ (2), $c\lambda^*\|_{S_+\otimes W^*}$ (2), $A\|_V^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_V c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	2	$(A\|_{W_{\mathbb{R}}^\perp})^2$ (2), $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^{2\dagger}$ (2)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	$A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	$A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	3	$(\lambda\|_{S_+\otimes W})^2$ (2), $A\|_V(\lambda\|_{S_+\otimes W})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_+\otimes W}\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	3	$\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}$ (2), $A\|_V \lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}(\lambda\|_{S_+\otimes W})^{2\dagger}$ (3), $c\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_+\otimes W}\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	3	$(\lambda\|_{S_-\otimes W^*})^2$ (2), $A\|_V(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)

The leg sets of the special-edge composite cycles through the ten pairs of field letters at the pair weight without derivatives. The leg sets without † enter the translation-invariant part along Δ as well.

pair	$n=0$	$n=1$	$n=2$	$n=3$
$(A\|_V, A\|_V)$	7 (3)	18 (7)	34 (18)	67 (34)
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	7 (2)	16 (7)	32 (16)	59 (32)
$(A\|_V, \lambda\|_{S_+\otimes W})$	9 (4)	20 (9)	42 (20)	79 (42)
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	9 (4)	20 (9)	42 (20)	79 (42)
$(A\|_V, c)$	4 (2)	8 (4)	19 (8)	35 (19)
$(A\|_V, A^*\|_V)$	15 (6)	35 (15)	73 (35)	142 (73)
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	2 (1)	7 (2)	16 (7)	38 (16)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	4 (1)	12 (4)	30 (12)	62 (30)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	4 (1)	12 (4)	30 (12)	62 (30)
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	2 (1)	7 (2)	16 (7)	32 (16)
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	10 (2)	26 (10)	64 (26)	127 (64)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	4 (1)	12 (4)	28 (12)	63 (28)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	6 (1)	15 (6)	36 (15)	75 (36)
$(\lambda\|_{S_+\otimes W}, c)$	4 (1)	9 (4)	20 (9)	42 (20)
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	17 (5)	42 (17)	88 (42)	173 (88)
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	4 (1)	12 (4)	28 (12)	63 (28)
$(\lambda\|_{S_-\otimes W^*}, c)$	4 (1)	9 (4)	20 (9)	42 (20)
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	17 (5)	42 (17)	88 (42)	173 (88)
(c, c)	2 (1)	4 (2)	8 (4)	19 (8)
$(c, A^*\|_V)$	18 (7)	34 (18)	67 (34)	121 (67)
$(A^*\|_V, A^*\|_V)$	18 (5)	52 (18)	114 (52)	235 (114)

The number of leg sets of the special-edge composite cycles entering the part along P_b and, in brackets, the translation-invariant part along Δ, for n derivatives on the two letters.

Proposition 11.7 (Output bound)**.**

- *(i) A chain of Definition 11.5 with outputs of total Weyl weight w has superficial degree $w_{X_a} + w_{X_b} - w$ before the insertion factor of $S^{(1)}(x)$; the factor ∂_b adds one, and the translation-invariant part along Δ adds none.*

- *(ii) The composite pair kernel along P_b is a translation-invariant local functional of Weyl weight $w_{X_a} + w_{X_b} + 1$; the composite pair kernel along Δ is local of Weyl weight $w_{X_a} + w_{X_b}$, the position counted at weight -1, at most linear in the position. Both have the pair degree and the parity $|X_a| + |X_b|$.*

- *(iii) Chains with $w \le w_{X_a} + w_{X_b} + 1$ contribute to the kernel along P_b and to the part along Δ linear in the position, chains with $w \le w_{X_a} + w_{X_b}$ to the translation-invariant part along Δ; a chain above these bounds has a vanishing limit as $\varepsilon \to 0$.*

- *(iv) The number of output fields is at most $w_{X_a} + w_{X_b} + 1$ plus the pair degree for both kernels; the coefficients of ε^{-m} of the special-edge composite cycles have Weyl weight $w_{X_a} + w_{X_b} - 2m$ along Δ and $w_{X_a} + w_{X_b} + 1 - 2m$ along P_b.*

The same bounds hold for the composite cycles through the pair and for the arity-one counterterm, of degree one less than the pair degree: the coefficient of $\log\varepsilon$ *of the arity-one counterterm of* X_aX_b *has Weyl weight* $w_{X_a}+w_{X_b}$ *along* Δ *and* $w_{X_a}+w_{X_b}+1$ *along* P_b.

Proof. (i) Every edge obeys the weight rule w_1+w_2-4: P (Proposition 3.17), K_0 (a field and its antifield have the weights w and $4-w$, momentum degree 0) and the composite kernel (Lemma 11.2 (vi)); the count of Lemma 10.11 applies. (ii) The coefficient of ε^0 is a polynomial in the output momenta of degree equal to the superficial degree (Theorem 11.11); along Δ the position enters once, through the factor $2x^\mu$ of Lemma 11.2 (iv). Degree and parity: Proposition 11.6 (ii). (iii) A polynomial of negative degree vanishes, and so does the limit of a chain of negative degree with a special edge (Theorem 11.11). (iv) Weyl weight plus degree is at least 1 for every output (Proposition 10.12), so the number of outputs is at most w plus the pair degree. The statements on the arity-one counterterm follow in the same way from Lemma 4.6. □

11.2 Closed forms and locality

Definition 11.8 (Position part and translation-invariant part)**.** A local functional is in canonical form if it is written $\int\omega\,D$ with $\omega\in\Omega^4_c(\mathbb{R}^4)$ and D a polynomial in letters, every derivative on a field. A map M of local functionals, linear in ω, of the form

$$M\Big(\int\omega\,D\Big)=\int\omega\,\big(x^\mu M_\mu(D)+M_0(D)\big),$$

with M_μ, M_0 maps of densities at one point and x^μ the coordinates of V in an orthonormal basis, has the position part $b^\mu M_\mu$ along $b\in V$ and the translation-invariant part M_0. A local functional $G=\int\big(x^\mu g_\mu+g_0\big)$ at most linear in the position has the position part $\int b^\mu g_\mu$ along b and the translation-invariant part $\int g_0$. On plane waves, with the momenta of the fields independent and ω absorbing the total momentum, a graph is $\alpha\,\delta(\cdot)+\beta^\mu\partial_\mu\delta(\cdot)$, and M_0 corresponds to α, M_μ to $i\beta^\mu$.

Lemma 11.9 (Translations and the canonical form)**.** *M is well defined, independently of integrations by parts against ω, exactly when every M_μ commutes with the derivatives and $[M_0,\partial_\mu]=M_\mu$. Then $[M,\{S^{(1)}(P_b),-\}]$ is the map $\int\omega\,D\mapsto\int\omega\,b^\mu M_\mu(D)$. For $M=v.=\{S^{(1)}(\Delta),-\}$ the position part is $2\partial_b$ and the translation-invariant part is the multiplication by $2w_D$; $\{S^{(1)}(P_b),-\}$ has position part 0.*

Proof. $M(\int\omega\,\partial_\mu D)$ and $M(-\int(\partial_\mu\omega)D)=\int\omega\big(M_\mu(D)+x^\nu\partial_\mu M_\nu(D)+\partial_\mu M_0(D)\big)$ agree for all ω exactly under the two conditions. Then $M(\int\omega\,\partial_bD)-\{S^{(1)}(P_b),M(\int\omega\,D)\}=\int\omega\,\big(M_0(\partial_bD)-\partial_bM_0(D)\big)=\int\omega\,b^\mu M_\mu(D)$, since $\{S^{(1)}(P_b),-\}$ acts by ∂_b on the fields and leaves the position and ω untouched. Lemma 9.16 (i) gives $v.\int\omega\,D=\int\omega\,2(x\cdot\partial+w_D)D$. □

Proposition 11.10 (Position decomposition)**.**

- *(i) For every pair of letters, the position part along b of the composite pair kernel along Δ is twice the composite pair kernel along P_b. For every local functional the position part of its arity-one counterterm along Δ is twice its arity-one counterterm along P_b; on cocycles the position part of the composite term along Δ is twice the composite term along P_b; and the position part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ along b is $2I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(P_b)$. In particular the quadratic part of the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(P_b)$ is $\frac12$ unit times $Q\big(\int(\iota_bA|_V,c^*)-\frac12\int(\rho(b)\lambda^*,\lambda^*)\big)$, and the quadratic part of the arity-one obstruction along P_b vanishes.*
- *(ii) The translation-invariant part of the composite pair kernel along Δ is computed with the internal vertex $\int\big(2(x-x_O)^\mu\partial_\mu(A|_V)+2A|_V,A^*|_V\big)$, x_O the point of the density, and pair by pair its commutator with $\{S^{(1)}(P_b),-\}$ is twice the composite pair kernel along P_b.*

Proof. (i) By Lemma 11.2 (iv) the internal vertex along Δ is $2\int x^\mu s_\mu + \int s_0$, with $\int s_\mu$ the internal vertex along P_{e_μ} and $s_0 = 2(A|_V, A^*|_V)$; insert $x^\mu = x_O^\mu + (x - x_O)^\mu$. On plane waves the factor x^μ is $-i\partial/\partial r^\mu$ of $e^{i(r,x)}$, with the momentum r injected at the internal vertex and absorbed by ω; its action on the momentum delta function is the position part, $-i$ times the graph with $\int s_\mu$ at $r = 0$, which is $2 \cdot (-i)$ times the graph along P_{e_μ}; its action on the integrand is translation invariant. The same holds graph by graph for the composite cycles, their singular parts, the counterterm trees and the composite loops of I. Theorem 9.26 (i) gives the quadratic part along Δ, which is Q of $\int x^\mu$ times a density with Q mapping letters to letters; Theorem 9.26 (ii) gives a quadratic obstruction along Δ without explicit position. (ii) The first statement is the decomposition of (i); the second is Lemma 11.9 applied to the composite pair kernel along Δ, whose position part is given by (i). □

Theorem 11.11 (Special-edge composite cycles as simplex integrals)**.** *For a chain of Definition 11.5 with k vertices of I and edges $e_0, \dots, e_k$, e_0 at X_a, let e_c be the composite kernel and e_σ, $\sigma \neq c$, the special edge. The cycle has $k+2$ Schwinger times: $t_j \in [\varepsilon, \infty)$ for $j \notin \{c, \sigma\}$, $t_\sigma \in [0, \varepsilon]$, and $t'_c, t''_c \in [\varepsilon, \infty)$ for the two segments of the composite kernel (Lemma 11.2 (vi)), which meet at the internal vertex of $S^{(1)}(x)$. The edge e_j carries $l + q_j$ as in Definition 10.21, both segments carry $l + q_c$, and along Δ a momentum r is injected at the internal vertex and carried by the second segment and by $e_{c+1}, \dots, e_k$ to X_b and ω. The numerator is the product of the vertex functions, the edge factors of Definition 10.21, the special-edge factor $-|l + q_\sigma|^2$ times the fibre factor of K_0, the segment factors $2(-,-)^{-1}$ times the inverse metric and $-i(l + q_c)(-,-)^{-1}$, and the insertion: $i(l + q_c)_b$ along P_b; 2 for $2\int (A|_V, A^*|_V)$ and $2i(l + q_c)_\mu$ for $2\int x^\mu(\partial_\mu(A|_V), A^*|_V)$ along Δ. With the simplex of dimension $k+1$ of the coordinates $\alpha = (\alpha_j)_{j\neq c} \cup (\alpha'_c, \alpha''_c)$, its centre, momentum form and m-th simplex integrand as in Definition 10.21 with $k+1$ in place of k, at generic output momenta and $T = \infty$,*

$$
\begin{aligned}
16\pi^2\big(\textit{coefficient of } \varepsilon^{-m}\big) &= \frac{1}{m}\int_{\alpha_\sigma \ \textit{smallest}} \Big[\big(\min_{f\neq\sigma} \alpha_f\big)^m - \alpha_\sigma^m\Big]\big(m\textit{-th simplex integrand}\big)\, d\alpha \qquad (m \geq 1),\\
16\pi^2\big(\textit{coefficient of } \varepsilon^{0}\big) &= \int_{\alpha_\sigma \ \textit{smallest}} \big(0\textit{-th simplex integrand}\big)\, \log \frac{\min_{f\neq\sigma} \alpha_f}{\alpha_\sigma}\, d\alpha,
\end{aligned}
\tag{11.1}
$$

the coefficient of $\log \varepsilon$ is 0, and the remainder tends to 0. Each coefficient is a polynomial in the output momenta and in r, of degree the superficial degree minus $2m$, and the limit vanishes for negative superficial degree. The coefficients of ε^0 are integrals of polynomials times logarithms of ratios of affine functions over rational subsimplices; for $k = 1$ they lie in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$ (Proposition 11.13).

Proof. The domain $\{t_\sigma \leq \varepsilon,\ t_f \geq \varepsilon\ (f \neq \sigma)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma} \alpha_f$ and $1/\min_f \alpha_f$, which differ on the region where α_σ is the smallest coordinate. Proposition 10.23 applies to each with $k+2$ edges, the Schwinger measure being $\tau^{k+1} d\tau\, d\alpha$, and the difference is taken as in Lemma 10.24: the two terms in $\log \varepsilon$ cancel, and the finite parts differ by $-\int F \log(\alpha_\sigma/\min_{f\neq\sigma} \alpha_f)$ over that region, F the 0-th simplex integrand. The region $\tau \geq 1$ converges at $T = \infty$: the composite kernel is of order $|p|^{-3}$ at small momentum (Proposition 9.18 (iv)) and the other edges carry distinct generic momenta (proof of Theorem 10.4 (ii)), so the integrand is integrable and the measure of $\{\alpha_\sigma \leq \varepsilon/\tau\}$ is of order ε; the injected momentum enters polynomially at each order in r. The degree statement is Proposition 11.7 (i). □

Proposition 11.12 (Generating functions in the point splitting)**.** *With the letters at $x + y_a$ and $x + y_b$, $y = y_b - y_a$, the m-th simplex integrand for the numerator (chain numerator) $e^{i(y,l)}$ is the chain numerator with l replaced by $-i\partial_y$, applied to*

$$
(-1)^{m+k} e^{i(y,\textit{centre})} \sum_{i\geq 0} \frac{(|y|^2/4)^{i+m+k}(\textit{momentum form})^i}{i!\,(i+m+k)!} = (-1)^{m+k} e^{i(y,\textit{centre})} \Big(\frac{|y|}{2\sqrt{\textit{momentum form}}}\Big)^{m+k} I_{m+k}\big(|y|\sqrt{m}
$$

and, summed over the chains with the signs of Definition 11.5,

$$\sum_{\alpha,\beta}\frac{y_a^\alpha y_b^\beta}{\alpha!\,\beta!}\big(\textit{composite pair kernel of } (\partial^\alpha\varphi_a,\partial^\beta\varphi_b) \textit{ along } x\big)=\frac{\hbar}{16\pi^2}\,e^{i(y_b,q_k)}\int_{\alpha_\sigma\ \textit{smallest}}\big(0\textit{-th simplex integrand}\big)\log\frac{\min}{}$$

with q_k including r along Δ; the coefficients of ε^{-m} have the same generating function with the weights of (11.1)*. In position space iq_j is the derivative of the product of the outputs at x.*

Proof. As Proposition 10.25, with $k+2$ edges: the power τ^{-1-m} of the Gaussian-integrated integrand times τ^{k+1} fixes the index shift $m+k$, and Theorem 11.11 replaces the weight $(\min_e \alpha_e)^m$ by the weights of the region where α_σ is the smallest. □

Proposition 11.13 (One vertex in closed form)**.** *Let $k=1$, $\beta:=\alpha_\sigma$, and let γ' and $\gamma''=1-\beta-\gamma'$ be the coordinates of the two segments. At $r=0$ the centre $-\beta q_\sigma-(1-\beta)q_c$ and the momentum form $\beta(1-\beta)|q_c-q_\sigma|^2$ depend on β alone, the region where β is the smallest coordinate requires $\beta\le\frac13$, and*

$$16\pi^2\big(\textit{composite pair kernel}\big)=\int_0^{1/3}\big(0\textit{-th simplex integrand}\big)(\beta)\Big[(1-\beta)\log\frac{1-\beta}{2\beta}-(1-3\beta)\Big]d\beta,$$

$$16\pi^2\big(\textit{coefficient of }\varepsilon^{-m}\big)=\int_0^{1/3}\big(m\textit{-th simplex integrand}\big)(\beta)\,\frac{2}{m}\Big[\frac{\big(\frac{1-\beta}{2}\big)^{m+1}-\beta^{m+1}}{m+1}-\beta^m\Big(\frac{1-\beta}{2}-\beta\Big)\Big]d\beta.$$

The part linear in r carries one factor γ'' in the simplex integrand, and $\int\gamma''(\cdots)\,d\gamma'=\frac{1-\beta}{2}\int(\cdots)\,d\gamma'$ by the symmetry $\gamma'\leftrightarrow\gamma''$ of the region and of the weights. The coefficients are integrals of $\beta^a(1-\beta)^b$ times these weights over $[0,\frac13]$, in $\mathbb{Q}+\mathbb{Q}\log2+\mathbb{Q}\log3$.

Proof. On the region, $\min_{f\neq\sigma}\alpha_f=\min(\gamma',\gamma'')$ and $\gamma'\in[\beta,1-2\beta]$. By the symmetry about $\gamma'=\frac{1-\beta}{2}$,

$$\int\log\frac{\min(\gamma',\gamma'')}{\beta}\,d\gamma'=2\int_\beta^{(1-\beta)/2}\log\frac{\gamma'}{\beta}\,d\gamma'=(1-\beta)\log\frac{1-\beta}{2\beta}-(1-3\beta),$$

and $\frac1m\int\big(\min(\gamma',\gamma'')^m-\beta^m\big)d\gamma'$ is the second weight. At $r=0$ both segments carry $l+q_c$, so the Gaussian data depend on β alone; along Δ the derivative in r of $\sum_j t_j|l+q_j|^2$ is $2t_c''(l+q_c)\cdot$, linear in γ''. The integrals $\int_0^{1/3}\beta^a(1-\beta)^b\log\frac{1-\beta}{2\beta}\,d\beta$ are rational combinations of 1, $\log2$, $\log3$, the endpoint $\beta=\frac13$ giving $\log\frac23$. □

Definition 11.14 (Composite loops of the action)**.** For $x\in\{\Delta,P_b\}$ and $P=P(\varepsilon,T)$, the composite loops of I along x, written $\Xi_x(\varepsilon)$, are the amputated one-particle-irreducible cycles of vertices of I and edges P in which one edge is the composite kernel along x, all other legs open: $\frac{\hbar}{2}\,\partial_{\text{composite kernel along }x}W(P,I)_{\text{tree}}$ is the tree-level flow of $\Xi_x(\varepsilon)$. $\Xi_x^\sigma(\varepsilon)$ is the sum of the same loops in which one edge of the cycle other than the composite kernel is the special edge, with the sign of Proposition 11.4, minus the loops with one edge $K_\varepsilon-K_0$. The renormalised composite functional is $\Xi_x^{\text{ren}}:=\lim_{\varepsilon\to0}\big(\Xi_x(\varepsilon)-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)\big)$, a functional of degree -1 at generic momenta, whose quadratic part along Δ is the renormalised composite two-point functional of Remark 9.27. The composite graphs of a local functional O (Definition 11.3) are of the first type when the cycle passes through two letters of O and of the second type when it passes through vertices of I alone; the second type is the tree insertion of $\Xi_x(\varepsilon)$ into O, written $\mathrm{tr}(\Xi_x,O)$.

Lemma 11.15 (Images of the composite loops)**.** *For every $\varepsilon>0$, as functionals at generic momenta, $s\,\Xi_x(\varepsilon)=\Xi_x^\sigma(\varepsilon)$. Consequently Ξ_x^σ has no term in $\log\varepsilon$ and no poles, $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)=\log\varepsilon\,L^{(1)}(x)$ with $L^{(1)}(x)$ an s-closed local functional, and the composite part along x is $s\,\Xi_x^{\text{ren}}$, the coefficient of ε^0 of Ξ_x^σ, a local functional.*

Proof. Compute Q of the tree-level flow of Ξ_x in two ways. First, $\frac{\hbar}{2}\,\partial_{\text{composite kernel}} W(P,I)_{\text{tree}}$ is that flow, and ∂ of the composite kernel anticommutes with Q (Lemma 11.2 (v)); by Proposition 4.26 at order $\hbar^0$ and the classical master equation, Q of it is $-\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the tree-level flow of $\frac{1}{2}\{I,I\}_\varepsilon - \frac{1}{2}\{I,I\}$, the composite closures of the trees of I with one special edge (Proposition 11.4 with $M=0$). Sort them by the position of the special edge: on the cycle closed by the composite kernel, which gives the tree-level flow of Ξ_x^σ, or on a tree, between two vertices of I or between Ξ_x and a vertex of I. Second, with Ξ_x as a vertex, Proposition 4.26 applied to $I + \delta\,\Xi_x$, δ of degree $+1$, gives the tree-level flow of $s\,\Xi_x$ plus the tree-level flows with the special-edge parts of $\{I,\Xi_x\}_\varepsilon$ and of $\frac{1}{2}\{I,I\}_\varepsilon$ on tree edges. The graphs with the special edge on a tree are the same in both evaluations, with the same signs, the sign rule of Proposition 11.4 being the Koszul rule of the second evaluation. Hence the tree-level flows of $s\,\Xi_x$ and of Ξ_x^σ agree, and the tree-level flow is injective on functionals graded by the number of legs (the argument of Theorem 8.15 (i)). Ξ_x^σ is a sum of cycles with one special edge and the composite kernel, which by the argument of Theorem 11.11 are $\sum_m \varepsilon^{-m} a_m + a_0 + o(1)$ without $\log\varepsilon$. The poles of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ vanish by the weight count of Proposition 9.25 (i), and along P_b by the same count at Weyl weight $5-2m$; so $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) = \log\varepsilon\, L^{(1)}(x)$, and the coefficient of $\log\varepsilon$ in $s\,\Xi_x = s\,I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) + s\,\Xi_x^{\text{ren}} + o(1)$ gives $s\,L^{(1)}(x) = 0$, while the poles of Ξ_x^σ vanish. The coefficient of ε^0 gives the last statement. □

Lemma 11.16 (Special edges on trees of the composite graphs)**.** *Let O be a local functional. The composite graphs of the first type with the special edge on a tree, at an output of the cycle or at a spectator letter of O, together with the special-edge parts of the counterterm trees of the arity-one counterterm of O, are $e^{-\varepsilon|p|^2}-1$ at a generic momentum p times renormalised composite cycles, and tend to* 0*. The composite graphs of the second type with the special edge on a tree tend to* 0*, and so do the special-edge parts of the counterterm trees of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$.*

Proof. A special edge on a tree edge of momentum p factorises: the graph is $(e^{-\varepsilon|p|^2}-1)$ times the composite cycle through the pair with its trees; the counterterm vertex joined by the same special edge gives $(e^{-\varepsilon|p|^2}-1)$ times minus its singular part, so the sum is $(e^{-\varepsilon|p|^2}-1)$ times the renormalised composite cycle, which has a limit at generic momenta (Proposition 9.25 (i)). At a spectator letter of O the same holds with the special-edge parts of $\{I,O\}_\varepsilon$ and of the bracket of I with the arity-one counterterm of O. The graphs of the second type are $(e^{-\varepsilon|p|^2}-1)$ times $\Xi_x(\varepsilon)$, which is $\log\varepsilon\, L^{(1)}(x)$ plus a convergent functional (Lemma 11.15), and $(e^{-\varepsilon|p|^2}-1)\log\varepsilon \to 0$; the counterterm trees carry $(e^{-\varepsilon|p|^2}-1)\log\varepsilon$. □

Proposition 11.17 (The image of the composite graphs of each type)**.** *For every local functional O and every $\varepsilon > 0$, with (A) the special-edge composite cycles of the pairs of letters of O with trees of I on their outputs (Definition 11.5), (B) the composite graphs of the first type with the special edge on a tree, (D) those of the second type with the special edge on a tree, and $\epsilon_\Xi \in \{\pm 1\}$ the sign with which the bracket $\{\Xi_x, O\}$ enters $Q\,\mathrm{tr}(\Xi_x, O)$,*

$$Q\big(\textit{composite graphs of the first type of } O\big) = \text{(A)} + \text{(B)} - \epsilon_\Xi\{\Xi_x(\varepsilon), O\} - \big(\textit{composite graphs of } sO\big) \mp \mathrm{tr}(\Xi_x, sO),$$
$$Q\,\mathrm{tr}(\Xi_x, O) = \mathrm{tr}(s\,\Xi_x, O) + \epsilon_\Xi\{\Xi_x, O\} \pm \mathrm{tr}(\Xi_x, sO) + \text{(D)},$$

the terms with sO being present only for $sO \neq 0$.

Proof. By Proposition 11.4 the image under Q of all composite graphs of O is $-\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the trees with one vertex sO or one special edge; the trees with one vertex sO closed by the composite kernel are the composite graphs of sO, of both types. The special edge lies on the cycle through two letters (A), on a tree of a graph of the first type (B), on the cycle of a composite loop of I, which gives $\mathrm{tr}(\Xi_x^\sigma, O)$, or on a tree of a graph of the second type (D). The image of the second type

is computed with Ξ_x as a vertex, by Proposition 4.26 applied to $I + \delta_1\Xi_x + \delta_2 O$, the part linear in δ_1 and δ_2: the vertex $s\,\Xi_x$ with O, the vertex sO with Ξ_x, the direct contraction $\{\Xi_x, O\}$, and the special edges on trees, which are (D). Subtract: $\mathrm{tr}(s\,\Xi_x, O) = \mathrm{tr}(\Xi_x^\sigma, O)$ by Lemma 11.15, so the loops with the special edge on the cycle cancel, (D) cancels, and $\pm\mathrm{tr}(\Xi_x, sO)$ moves to the first line with the opposite sign. □

Theorem 11.18 (The composite term is two-body and local)**.** *Let $O = \int \omega\, X_1 \cdots X_n$ be a local functional with $sO = 0$, $x \in \{\Delta, P_b\}$, $T = \infty$, generic momenta, and $\hbar F_1^{(0)}$ as in (i) of Definition 10.36.*

- *(i) The special-edge composite cycles of (X_a, X_b) along x are $\sum_{m\geq 1}\varepsilon^{-m}$ times local coefficients, plus the composite pair kernel, plus a remainder tending to 0; they contain no $\log\varepsilon$.*
- *(ii) s annihilates the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is $\log\varepsilon$ times that coefficient in every sector.*
- *(iii) (Arity-one BRST covariance.) s of the coefficient of $\log\varepsilon$ in the arity-one counterterm of O along x is minus the bracket of O with the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and the sum over the pairs, with the Koszul signs of (10.4), of the coefficients of ε^{-m} of the special-edge composite cycles is s of the coefficient of ε^{-m} in the arity-one counterterm of O.*
- *(iv) The composite term of O along Δ (Definition 10.36) and along P_b (Proposition 10.42) is*

$$\big(\textit{composite term of } O \textit{ along } x\big) = \sum_{a<b}(-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1 \cdots X_{a-1}\, \big(\textit{composite pair kernel of } (X_a, X \tag{11.2}$$

  a local two-body functional with no one-body term.

- *(v) Hence the composite terms of the pairs and of the legs of Theorem 10.38, summed, equal the sum over the pairs of the composite pair kernels along Δ: the nonlocal parts of the brackets of the letters $A|_V$, λ, c with the renormalised composite two-point functional, $\log|p|^2$ of the leg momentum and $(p,a)(p,b)|p|^{-2}$ on $A|_V$ (Remark 9.27), cancel against those of the images under Q of the renormalised composite cycles through the pairs containing these letters.*
- *(vi) For $sO \neq 0$ the right side of (11.2) acquires the renormalised composite graphs of sO, minus $\frac{\hbar}{2}\partial$ of the composite kernel applied to the trees with one vertex sO; these contain the nonlocal bracket with the renormalised composite two-point functional on the letters of sO.*

Proof. (i) Theorem 11.11 applied to each chain of Definition 11.5. (ii) Lemma 11.15.

(iii), (iv) The composite term is the limit of Q of the composite graphs minus their counterterm trees, minus the tree insertion of the composite part. Write the counterterm trees as $-\mathrm{tr}(\mathrm{ct}_x^{(1)}(O)[\varepsilon])$, with $\mathrm{ct}_x^{(1)}(O)[\varepsilon] = \sum_{m\geq 1}\varepsilon^{-m}\mathrm{ct}_{x,m}^{(1)}(O) + \log\varepsilon\,\mathrm{lc}_x^{(1)}(O)$ the arity-one counterterm of O along x, and $-\mathrm{tr}(I^{CT,(1)}, O)$, with $I^{CT,(1)}$ short for $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, an odd vertex. By Proposition 4.26 and Lemma 6.8, with $sO = 0$,

$$\begin{aligned} Q\big(-\mathrm{tr}(\mathrm{ct}_x^{(1)}(O)[\varepsilon])\big) &= -\mathrm{tr}\big(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]\big) + (\text{special edges}),\\ Q\big(-\mathrm{tr}(I^{CT,(1)}, O)\big) &= -\mathrm{tr}\big(s\,I^{CT,(1)}, O\big) - \epsilon_\Xi\{I^{CT,(1)}, O\} + (\text{special edges}), \end{aligned}$$

the sign of the bracket being that for Ξ_x, both vertices having degree -1 and being joined to O by a tree. By Proposition 11.17 the sum of the images is

$$(\mathrm{A}) + (\mathrm{B}) + (\mathrm{D}) + (\text{special edges}) + \mathrm{tr}\big(s(\Xi_x - I^{CT,(1)}), O\big) - \epsilon_\Xi\{I^{CT,(1)}, O\} - \mathrm{tr}\big(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]\big),$$

the brackets $\pm\epsilon_\Xi\{\Xi_x, O\}$ of the two types cancelling. By Lemma 11.16, (B), (D) and the special edges tend to 0. By Lemma 11.15, $\mathrm{tr}(s(\Xi_x - I^{CT,(1)}), O)$ tends to $\mathrm{tr}(s\,\Xi_x^{\mathrm{ren}}, O)$, the tree insertion of the composite part, which is subtracted in the definition. What remains is (A) minus $\epsilon_\Xi \log\varepsilon\{L^{(1)}(x), O\}$ minus $\mathrm{tr}(s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon])$. The left side has a limit, the renormalised composite graphs converging and Q multiplying plane-wave values by polynomials in the momenta, and (A) has no $\log\varepsilon$ by (i): the coefficient of $\log\varepsilon$ gives $\mathrm{tr}\big(s\,\mathrm{lc}_x^{(1)}(O) + \epsilon_\Xi\{L^{(1)}(x), O\}\big) = 0$, and the coefficient of ε^{-m}, with $a_m(X_a, X_b)$ the coefficient of ε^{-m} of the special-edge composite cycles of the pair, gives $\mathrm{tr}\big(\sum_{a<b} \pm a_m(X_a, X_b) - s\,\mathrm{ct}_{x,m}^{(1)}(O)\big) = 0$; the trees with these local vertices are the same at every ε, and the injectivity of the tree-level flow gives (iii). At order ε^0 the trees on the outputs of (A) converge to trees with edges $P(0,T)$, and the limit is the tree-level flow of the sum over the pairs of the composite pair kernels; Theorem 8.15 (i) gives (11.2).

(v) By Proposition 11.17, the image of the first type contains $-\epsilon_\Xi\{\Xi_x, O\}$ and that of the second type $+\epsilon_\Xi\{\Xi_x, O\}$; the counterterm trees replace Ξ_x by Ξ_x^{ren} in the second and add $-\epsilon_\Xi \log\varepsilon\{L^{(1)}(x), O\}$, which (iii) removes from the first. The renormalised composite loops on a letter X give the bracket of Ξ_x^{ren} with X, whose quadratic part along Δ is the functional of Remark 9.27, local on $A|_{W_{\mathbb{R}}^\perp}$ and containing $\log|p|^2$ on $A|_V$, λ, c. The form of Theorem 10.38 splits the same sum into the images of the renormalised composite cycles through the pairs, the composite terms of the pairs, and the brackets with Ξ_x^{ren} on the letters together with the tree insertion of $s\,\Xi_x^{\mathrm{ren}}$, the composite terms of the legs; the brackets cancel between the two, and the sum is local.

(vi) For $sO \neq 0$, Proposition 11.4 keeps the trees with one vertex sO, whose closures by the composite kernel are the composite graphs of sO of both types (Proposition 11.17); with their counterterm trees they converge to the renormalised composite graphs of sO, nonlocal through the brackets with Ξ_x^{ren} on the letters of sO. □

Remark 11.19 (What the locality uses and what it fixes)**.** The argument uses $sO = 0$ exactly, the classical master equation, the anticommutation of Q with ∂ of the composite kernel, which rests on $[v_x\cdot, Q] = 0$ and $QP\star + P\star Q = e^{-\varepsilon H}$ (Lemma 11.2 (v)), the absence of $\log\varepsilon$ in cycles with one special edge, a property of the cutoff on every edge (Theorem 11.11), the absence of a term ε^{-1} in the composite loops of I (Theorem 11.18 (ii)), and generic momenta at $T = \infty$ for the convergence of the cycles and of the composite loops, which are of order $|p|^{-3}$ (Proposition 9.18 (iv)), or any finite T (Theorem 8.15 (ii)); (i) of Definition 10.36 enters through the terms with QW of the proof of Theorem 10.38, which are outside the composite term. For a functional of the fields $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, λ alone, $sO = 0$ is the gauge invariance of its density. Statement (iv) fixes the one-body part of the composite term: a term t units times $v.X_a$ on every letter, added to the right side of (11.2), would replace each sector coefficient θ by $\theta + t$ in Sections 11.3–11.5 and leave every spectrum unchanged; the operator-level value $\theta = -\frac{1}{3}$ of Proposition 11.33 is the value in the normalisation of (11.2), and it agrees with the action-level value of Theorem 9.28. The steps of the proof are Lemma 11.15 for the composite loops of the action, Lemma 11.16 for the special edges on trees and Proposition 11.17 for the cancellation of the brackets $\pm\epsilon_\Xi\{\Xi_x, O\}$ between the two types, forced by Proposition 11.4, whose right side contains no bracket; the sign ϵ_Ξ never needs to be evaluated. The composite term enters the second evaluation of D_1 and P_1; the components themselves are given by the master formula of Theorem 11.44, which holds along every generator. Statement (iv) is the input of Corollary 11.20, Propositions 11.21 and 11.35 and Sections 11.3–11.5; its consequences are evaluated on the bilinears, where the single chain of Example 11.22 reproduces the value of the route of the special edges, on the planar scalar sector, and through the spectra of Table 11.5, which a common one-body term leaves unchanged.

Corollary 11.20 (Local two-body form of D_1 and P_1)**.** *Under the hypotheses of Definition 10.36, and*

of the same definition with P_b in place of Δ, for $O = \int \omega\, X_1 \cdots X_n$, with the Koszul signs of (10.4),

$$D_1(O) = \sum_{a<b} \pm \int \omega \cdots \Big[4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](X_a X_b)\big) + \big(\textit{composite pair kernel of } (X_a, X_b) \textit{ along } \Delta\big)\Big] \cdots$$
$$+ \sum_a \int \omega \cdots \Big[-4\,\textit{units} \cdot (\textit{weight of } X_a)\, X_a - \big\{\hbar F_1^{(1)}(\Delta) - 4\,\textit{units} \cdot (\textit{field-counting functional}), X_a\big\}\Big] \cdots,$$

$$P_1(O) = \sum_{a<b} \pm \int \omega \cdots \big(\textit{composite pair kernel of } (X_a, X_b) \textit{ along } P_b\big) \cdots - \sum_a \int \omega \cdots \big\{\hbar F_1^{(1)}(P_b), X_a\big\} \cdots,$$

every term local; D_1 minus four times the map of Theorem 10.16 is the sum over the pairs of the composite pair kernels along Δ minus the bracket of O with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional.

Proof. Insert (11.2) into Theorem 10.38 and Proposition 10.42; four times the coefficient of $\log\varepsilon$ is two-body by Theorem 10.4 (iii), and the brackets with the field-counting functional, with $\hbar F_1^{(1)}(\Delta)$ and with $\hbar F_1^{(1)}(P_b)$ are derivations. □

Proposition 11.21 (The route of the special edges)**.** *For O as in Corollary 11.20, the limit as $\varepsilon \to 0$ of the cycles through the pairs of letters of O with one special edge and one vertex $S^{(1)}(\Delta)$ anywhere on the cycle, as the composite kernel or with one leg on the special edge, minus the trees of s of the arity-one counterterm of O, the graphs of Lemma 8.4 used in Theorem 9.30, is the sum over the pairs of four times the coefficient of $\log\varepsilon$ of the pair kernel plus the composite pair kernel along Δ.*

Proof. Both routes compute the local representative $D_1(O)$, which is unique (Theorem 8.15 (i)). In the route of the special edges the one-body terms are the trees with $-\{\hbar F_1^{(1)}(\Delta), O\}$, the arity-one defect of the action joined to O vanishing by (9.10) (proof of Theorem 9.30); in Corollary 11.20 the one-body terms are -4 units times the weights minus the bracket with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional, which is $-\{\hbar F_1^{(1)}(\Delta), -\}$. The two-body sums therefore agree. □

Example 11.22 (The scalar bilinears)**.** For $O = \int \omega\, (\iota_{w_1} A, \iota_{w_2} A)$, on the part with two legs $A|_{W_{\mathbb{R}}^\perp}$, Theorem 9.30 and Remark 9.32 give the value $4\,(\text{coefficient of } \log\varepsilon \text{ in } \partial_\delta I^{CT}[\varepsilon](O)) - \frac{4}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ units of the route of the special edges, so by Proposition 11.21

$$\big(\text{composite pair kernel of } (A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}) \text{ along } \Delta\big)\Big|_{\text{outputs } A|_{W_{\mathbb{R}}^\perp} A|_{W_{\mathbb{R}}^\perp}} = \Big[-\tfrac{1}{3}\, v.O - \tfrac{4}{3}\, O\Big] \text{ units},$$

for the traceless and the trace parts alike. It is the single chain of Table 11.1,

$$A|_{W_{\mathbb{R}}^\perp} \xrightarrow{K_\varepsilon - K_0} \big([A|_{W_{\mathbb{R}}^\perp}, c], A^*|_{W_{\mathbb{R}}^\perp}\big) \xrightarrow{\text{composite kernel}} A|_V (A|_{W_{\mathbb{R}}^\perp})^2 \xrightarrow{P} A|_{W_{\mathbb{R}}^\perp}.$$

Its position part along b is $-\frac{2}{3}$ units times ∂_b, so the composite pair kernel along P_b is $-\frac{1}{3}$ units times $\{S^{(1)}(P_b), O\}$, in agreement with Theorem 9.35, the composite triangle and the composite two-point functional along P_b giving $-2 + \frac{5}{3}$; its translation-invariant part is $-\frac{2}{3} \cdot 2\, O - \frac{4}{3}\, O = -\frac{8}{3}\, O$ units.

11.3 The free directions and the arity-two equation

Definition 11.23 (Finite counterterms along Δ and P_b)**.** Write the solutions of the arity-one equations along Δ and along P_b as

$$\begin{aligned}\hbar F_1^{(1)}(\Delta) &= 4\,\text{units}\cdot(\text{field-counting functional}) + \Big[-\tfrac{2}{3}\int(A,A^*)+\tfrac{1}{3}\int(\lambda^*,\lambda)+\tfrac{4}{3}\int(c^*,c)\Big]\,\text{units}\\ &\quad+\theta\,\text{units}\int\big(2E(A|_{W_\mathbb{R}^\perp})+2A|_{W_\mathbb{R}^\perp},A^*|_{W_\mathbb{R}^\perp}\big)-\theta_\lambda\,\text{units}\int\big(2E(\lambda)+3\lambda,\lambda^*\big)\\ &\quad+\theta_V\,\text{units}\Big[\int\big(2E(A|_V)+2A|_V,A^*|_V\big)-\int\big(2E(c),c^*\big)\Big]+\text{units}\cdot Q\Big(\kappa_V\int(\iota_xA|_V,c^*)+\kappa_\lambda\int(\rho(x)\lambda^*,\lambda\\ \hbar F_1^{(1)}(P_b) &= \theta_P\,\text{units}\int\big(\partial_b(A|_{W_\mathbb{R}^\perp}),A^*|_{W_\mathbb{R}^\perp}\big)-\theta_{P,\lambda}\,\text{units}\int(\partial_b\lambda,\lambda^*)+\theta_{P,V}\,\text{units}\Big[\int\big(\partial_b(A|_V),A^*|_V\big)-\int(\partial_bc,c^*)\Big]\\ &\quad+Q\big(\text{a combination of }\textstyle\int(\iota_bA|_V,c^*)\text{ and }\int(\rho(b)\lambda^*,\lambda^*)\big)+(\text{cubic and higher}),\end{aligned}$$

with numbers $\theta,\theta_\lambda,\theta_V,\kappa_V,\kappa_\lambda,\theta_P,\theta_{P,\lambda},\theta_{P,V}$; θ is the coefficient of (9.17) and θ_P that of Theorem 9.35. The three terms with θ, θ_λ, θ_V are the sector parts of $S^{(1)}(\Delta)$ in the sectors of $A|_{W_\mathbb{R}^\perp}$, of λ and of $(A|_V,c)$ times these numbers, and those with θ_P, $\theta_{P,\lambda}$, $\theta_{P,V}$ the sector parts of $S^{(1)}(P_b)$ (Definition 11.1). The scheme is translation covariant if the position part along b of $\hbar F_1^{(1)}(\Delta)$ is $2\hbar F_1^{(1)}(P_b)$; then $\theta_P=\theta$, $\theta_{P,\lambda}=\theta_\lambda$, $\theta_{P,V}=\theta_V$. The arity-two equations along (Δ,P_b) and (Δ,ξ) fix $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$ (Theorem 9.28), and those along (ξ,P_b) give $\theta_P=\theta_{P,\lambda}=\theta_{P,V}$ (Theorem 11.51).

Proposition 11.24 (Completeness of the quadratic directions)**.** *Let $\mathcal{F}_{-1}$ and $\mathcal{F}_{-2}$ be the quadratic local functionals of degree -1 and -2, of Weyl weight 4 with the position at weight -1, at most linear in the position and invariant under* $\mathrm{Spin}(4)\times\mathrm{Spin}(6)$*. On the parity-even subspaces of Theorem 9.26 (iii), invariant under the reflection of one direction of V together with one direction of $W_\mathbb{R}^\perp$,*

$$\frac{\ker\big(Q|_{\mathcal{F}_{-1}}\big)}{Q\,\mathcal{F}_{-2}}=\mathrm{span}\big\{\textit{the three sector parts of }S^{(1)}(\Delta)\big\},\qquad Q\,\mathcal{F}_{-2}=\mathrm{span}\Big\{Q\int(\iota_xA|_V,c^*),\,Q\int(\rho(x)\lambda^*,\lambda^*)\Big\}.$$

Without the reflection the quotient has one further class, $\int(\lambda^*,\rho(e_1e_2e_3e_4)\lambda)$, with $e_1,\dots,e_4$ an oriented orthonormal basis of V.

Proof. By Schur's lemma, $\mathcal{F}_{-1}$ is spanned by $\int(A^*|_V,D\,A|_V)$ with the structures D built from 1, $x\cdot\partial$, $x_\mu\partial_\nu$, $x_\nu\partial_\mu$ on the vector index and $\epsilon_{\mu\nu\kappa\tau}x^\kappa\partial^\tau$; by $\int(A^*|_{W_\mathbb{R}^\perp},D\,A|_{W_\mathbb{R}^\perp})$ with $D\in\{1,x\cdot\partial\}$; by $\int(\lambda^*,D\lambda)$ with $D\in\{1,\ x\cdot\partial,\ \rho(x)\rho(\partial)\}$ times $\{1,\rho(e_1e_2e_3e_4)\}$; and by $\int(c^*,Dc)$ with $D\in\{1,x\cdot\partial\}$. $\mathcal{F}_{-2}$ is spanned by $\int x^\mu(A_\mu,c^*)$, $\int(\lambda^*,\rho(x)\lambda^*)$ and $\int(\lambda^*,\rho(e_1e_2e_3e_4)\rho(x)\lambda^*)$. The structures with ϵ or with one factor $\rho(e_1e_2e_3e_4)$ change sign under the reflection. With Q of Proposition 1.11, the ranks on the plane-wave normal forms, computed exactly, are, in the sectors of $A|_{W_\mathbb{R}^\perp}$, of λ and of $(A|_V,c)$: $\dim\mathcal{F}_{-1}=2,3,6$, $\dim\ker Q=1,2,2$, $\dim Q\mathcal{F}_{-2}=0,1,1$ for the reflection-invariant subspaces. The kernel is spanned by the sector part of $A|_{W_\mathbb{R}^\perp}$; by the sector part of λ and $Q\int(\rho(x)\lambda^*,\lambda^*)$; by the sector part of $(A|_V,c)$ and $Q\int(\iota_xA|_V,c^*)$. Without the reflection, $\dim\mathcal{F}_{-1}=15$, the ϵ-structure is not Q-closed, $\int(\lambda^*,\rho(e_1e_2e_3e_4)\lambda)$ is closed and not exact, and $\int(\lambda^*,\rho(e_1e_2e_3e_4)\rho(x)\lambda^*)=0$. □

Proposition 11.25 (What the arity-one equations fix)**.** • *(i) Along Δ the quadratic part of* (9.10) *fixes the translation-invariant part $\big[-\frac{2}{3}\int(A,A^*)+\frac{1}{3}\int(\lambda^*,\lambda)+\frac{4}{3}\int(c^*,c)\big]$ units (Theorem 9.26 (iii)) and leaves θ, θ_λ, θ_V, κ_V, κ_λ free. Along P_b the quadratic part of the arity-one obstruction vanishes (Proposition 11.10 (i)), so the quadratic part of $\hbar F_1^{(1)}(P_b)$ is Q-closed and every coefficient of Definition 11.23 along P_b is free in arity one, the arity-two equations relating them to the coefficients along Σ (Theorem 11.51); the translation-covariant choice, $\hbar F_1^{(1)}(P_b)$ equal to half the position part along b of $\hbar F_1^{(1)}(\Delta)$, solves the equation along P_b at every order in the fields.*

- *(ii) Two solutions along Δ differ by an s-closed local functional of degree -1 and Weyl weight 4. The common shift of θ, θ_λ, θ_V by t adds t units times $S^{(1)}(\Delta)$, which is s-closed (Lemma 9.16 (iii)); the directions $Q\int(\iota_x A|_V, c^*)$ and $Q\int(\rho(x)\lambda^*, \lambda^*)$ complete to $s\int(\iota_x A|_V, c^*)$ and $s\int(\rho(x)\lambda^*, \lambda^*)$, and an s-exact direction sG contributes $s\{G, O\}$ to $D_1(O)$ on cocycles, zero on cohomology.*

- *(iii) Since $\{I, S^{(1)}(\Delta)\} = 0$ while the brackets of I with the sector parts are nonzero, the cubic part of* (9.10) *depends on the sector coefficients through $(\theta - \theta_V)$ times the bracket of the cubic part of I with the sector part of $A|_{W_{\mathbb{R}}^\perp}$ plus $(\theta_\lambda - \theta_V)$ times its bracket with the sector part of λ. It is solvable exactly when this sum is congruent to the cubic part of the composite part minus the bracket of the cubic part of I with the translation-invariant part, modulo the images under Q of the cubic local functionals of degree -1. If the classes of the two brackets modulo the images under Q of the cubic local functionals of degree -1, Weyl weight 4 and degree at most one in the position are linearly independent, the cubic equation fixes $\theta - \theta_V$ and $\theta_\lambda - \theta_V$.*

Proof. (i) Theorem 9.26 (iii) along Δ, Proposition 11.10 (i) along P_b. For the covariant choice: s maps letters to letters and carries no position, so it commutes with taking the position part, and the position part of the arity-one obstruction along Δ is twice the obstruction along P_b by Proposition 11.10 (i), the term 4 units times s of the field-counting functional having no position part. (ii) Theorem 8.8 (iii) restricted to arity one along Δ, and Lemma 10.15: $\{sG, O\} = -s\{G, O\}$ for $sO = 0$ and G of degree -2. (iii) s of the sector terms is the sum of θ, θ_λ, θ_V units times the brackets of I with the sector parts, whose sum over the three sectors vanishes; at cubic order s of the solution is Q of its cubic part plus the bracket of the cubic part of I with its quadratic part. □

Theorem 11.26 (The position part of D_1 is exact)**.** *Assume a one-loop quantisation through arity two along Δ and P_b, that is,* (8.6) *modulo $\hbar^2$ in its components of arity at most two along these generators, and local representatives through arity two along (Δ, P_b) on a space $\mathfrak{F}$ of local functionals stable under Λ, with P_1 translation covariant, $P_1(\int\omega\, D) = \int\omega\, P_1(D)$ for a map P_1 of densities commuting with the derivatives; this holds when $\hbar F_1^{(1)}(P_b)$ is translation invariant. Then for every $O = \int\omega\, D \in \mathfrak{F}$ with $sO = 0$*

$$\big(\textit{position part of } D_1 \textit{ along } b\big)(O) = \big[D_1, \{S^{(1)}(P_b), -\}\big](O) = \pm\, s\Big(\hbar L_1(O)^{(2)}(\Delta, P_b)\Big)\,:$$

the position part of D_1 is s-exact on cocycles, with the arity-two component of O as primitive. With the composite pair kernels this gives an operator-level determination of the sector coefficients along Δ (Proposition 11.33), which reproduces the values of the action-level arity-two equations (Theorem 9.28).

Proof. By Corollary 8.17 (ii), $[\Lambda^{(1)}(\Delta), \Lambda^{(1)}(P_b)] + \Lambda^{(1)}([\Delta, P_b])$ is a graded commutator of $\Lambda^{(0)}$ with an operator built from $\Lambda^{(2)}(\Delta, P_b)$. With $[\Delta, P_b] = -2P_b$ (Remark 2.6) and $S^{(2)}(\Delta, P_b) = 0$ (Definition 2.15), the order-$\hbar^0$ part of that operator vanishes and the order-$\hbar$ part of the identity reads

$$\big[D_1, \{S^{(1)}(P_b), -\}\big] + \big[v., P_1\big] - 2P_1 = \big[s, H_1\big],$$

with H_1 built from $\hbar L_1^{(2)}(\Delta, P_b)$. For translation-covariant P_1, $v.(P_1 O) = \int\omega\, 2(x\cdot\partial + w_D + 1)P_1(D)$ and $P_1(v.O) = \int\omega\,\big(2x^\mu P_1(\partial_\mu D) + 2w_D P_1(D)\big)$, so $[v., P_1] = 2P_1$. Lemma 11.9 identifies $[D_1, \{S^{(1)}(P_b), -\}]$ with the position part, and on a cocycle $[s, H_1]O = s(H_1 O)$. □

Lemma 11.27 (Position part of a bracket)**.** *For a local functional $G = \int(x^\mu g_\mu + g_0)$, the position part along b of $\{G, -\}$ is $\{\int b^\mu g_\mu, -\}$.*

Proof. The bracket contracts a slot of g_μ at y with a letter $\partial^\alpha\varphi$ of D at x through K_0, which gives $\partial_x^\alpha(x^\mu\cdots)$; the term with no derivative on x^μ is x^μ times the bracket with $\int g_\mu$, and the others carry no position. □

Corollary 11.28 (The composite term along P_b and the translation component)**.** *Under the hypotheses of Theorem 11.26 and of Definition 10.36, on every cocycle $O \in \mathfrak{F}$:*

- *(i) twice the composite term of O along P_b minus the bracket of O with the position part along b of $\hbar F_1^{(1)}(\Delta)$ is s-exact; this position part is*

  $$2\theta \text{ units} \int \big(\partial_b(A|_{W_{\mathbb{R}}^\perp}), A^*|_{W_{\mathbb{R}}^\perp}\big) - 2\theta_\lambda \text{ units} \int (\partial_b\lambda, \lambda^*) + 2\theta_V \text{ units} \Big[\int \big(\partial_b(A|_V), A^*|_V\big) - \int (\partial_b c, c^*) \Big] + \text{units} \cdot Q\Big(\kappa_V$$

  plus the position parts of the higher terms;

- *(ii) if $\theta = \theta_\lambda = \theta_V$, the composite term along P_b induces θ units times $\{S^{(1)}(P_b), -\}$ on cohomology on the parts with the fewest fields of the cocycles of $\mathfrak{F}$, on which $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part, and on every family in which $\{S^{(1)}(P_b), O\}$ is not exact for some cocycle the composite term along P_b determines θ;*

- *(iii) in the translation-covariant scheme the position part of D_1 is $2P_1$, and the arity-two equation along (Δ, P_b) is equivalent to $P_1 \cong 0$ on cohomology: $P_1(O) = \pm\frac{1}{2}\, s\big(\hbar L_1(O)^{(2)}(\Delta, P_b)\big)$;*

- *(iv) if $\theta = \theta_\lambda = \theta_V$ and $\theta_P = \theta_{P,\lambda} = \theta_{P,V}$, then P_1 induces $(\theta - \theta_P)$ units times $\{S^{(1)}(P_b), -\}$ on cohomology, and the finite part of the operators $\hbar\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$ (Definition 8.11; the letter ψ of Section 11.5 is a different object), $a = (\theta - \theta_P)$ units, of Proposition 9.36 gives $P_1 \cong 0$ on every family at once, leaves D_1, M_1, R_1 unchanged and changes the components along K_b, Σ and Σ^* by a, $-\frac{a}{2}$ and $\frac{a}{2}$ times the classical transformation. If the differences of the coefficients along P_b and along Δ depend on the sector, a multiple of $\{S^{(1)}(\Delta), -\}$ removes P_1 on the families with letters of one sector, and the translation-covariant scheme removes it on every family.*

Proof. (i) By Corollary 11.20 and Proposition 11.10 (i), the position part of D_1 is twice the sum over the pairs of the composite pair kernels along P_b minus the bracket with the position part of $\hbar F_1^{(1)}(\Delta)$, by Lemma 11.27; the coefficients of $\log\varepsilon$ and the bracket with the field-counting functional commute with the translations. Theorem 11.26 gives exactness, and the position part of each sector part of $S^{(1)}(\Delta)$ is twice the sector part of $S^{(1)}(P_b)$ (Definition 11.1). (ii) For equal coefficients the θ-terms are 2θ units times $S^{(1)}(P_b)$, which is s-closed, and the other directions are s-exact after completion, with $\{sG, O\} = -s\{G, O\}$. (iii) In the covariant scheme the position part of $\hbar F_1^{(1)}(\Delta)$ is $2\hbar F_1^{(1)}(P_b)$, and P_1 is the sum of the composite pair kernels along P_b minus the bracket with $\hbar F_1^{(1)}(P_b)$ (Corollary 11.20). (iv) By (ii), $P_1 \cong \theta$ units times $\{S^{(1)}(P_b), -\}$ minus θ_P units times the same; Proposition 9.36 with $\lambda_{P_b} = -2$. For sector-dependent differences P_1 induces the sum over the sectors of the differences times the brackets with the sector parts of $S^{(1)}(P_b)$, plus the completions, which is a multiple of $\{S^{(1)}(P_b), -\}$ on the cocycles with letters of one sector. □

11.4 The dilatation identities in kernel form

Corollary 10.44 requires that on cohomology the composite term minus the bracket with the translation-invariant finite counterterm beyond the field-counting functional induce $-\frac{1}{3}$ units times $\{S^{(1)}(\Delta), -\}$, the operator cancelled by the sector coefficients $-\frac{1}{3}$ of Theorem 9.28. By Corollary 11.20 this is a statement about the composite pair kernels, which splits into a statement along P_b and a statement about the translation-invariant part.

Lemma 11.29 (One-body terms)**.** *For $\theta = \theta_\lambda = \theta_V$, $\kappa_V = \kappa_\lambda = 0$ and the quadratic part of $\hbar F_1^{(1)}(\Delta)$ of Definition 11.23, the translation-invariant part of $-\big\{\hbar F_1^{(1)}(\Delta) - 4 \text{ units} \cdot (\text{field-counting functional}), X\big\}$ on a field letter $X = \partial^\alpha\varphi$ is $(c_\varphi - 2\theta\, w_X)$ units times X, with $c_\varphi = \frac{2}{3}$ for $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$, $\frac{1}{3}$ for λ and*

$\frac{4}{3}$ *for* c*, and its position part along* b *is* -2θ *units times* $\partial_b X$*. At* $\theta = -\frac{1}{3}$ *the translation-invariant part is* $\frac{2}{3}(2+|\alpha|)$ *units times* X *for every field slot and for* c*.*

Proof. $\{\int(A,A^*),X\} = X$, $\{\int(\lambda^*,\lambda),\lambda\} = -\lambda$ and $\{\int(c^*,c),c\} = -c$ (Lemma 4.21), and the bracket of the sector part with a letter of its sector is $v.X$, with translation-invariant part $2w_X X$ and position part $2\partial_b X$ (Lemma 11.9). At $\theta = -\frac{1}{3}$: $\frac{2}{3}+\frac{2}{3}(1+|\alpha|)$, $\frac{1}{3}+\frac{2}{3}(\frac{3}{2}+|\alpha|)$ and $\frac{4}{3}+\frac{2}{3}|\alpha|$ are all $\frac{2}{3}(2+|\alpha|)$. □

Definition 11.30 (The identities in kernel form)**.** Let $\mathfrak{F}$ be a family of gauge-invariant cocycles of field letters, preserved by D_1 and contained in a space of local functionals stable under Λ, satisfying (ii) and (iii) of Definition 10.36. The letter count is the derivation that multiplies a monomial $\int \omega\, X_1 \cdots X_n$ of letters $X_a = \partial^{\alpha_a}\varphi_a$, $\varphi_a \in \{A|_V, A|_{W_{\mathbb{R}}^\perp}, \lambda\}$, by $\sum_a (2+|\alpha_a|)$; it is defined on canonical forms, and its commutator with $\{S^{(1)}(P_b),-\}$ is $\{S^{(1)}(P_b),-\}$. The three identities on $\mathfrak{F}$ are:

- (a) the cubic part of (9.10) forces $\theta = \theta_\lambda = \theta_V$;
- (b) the sum over the pairs, with the Koszul signs of (10.4), of the composite pair kernels along P_b induces $-\frac{1}{3}$ units times $\{S^{(1)}(P_b),-\}$ on the cohomology of $\mathfrak{F}$;
- (c) the sum over the pairs of the translation-invariant parts of the composite pair kernels along Δ induces $-\frac{2}{3}$ units times the letter count on the cohomology of $\mathfrak{F}$.

Remark 11.31 (Dependences among the identities)**.** (1) By Proposition 11.10 (ii), pair by pair the commutator of the translation-invariant part of the composite pair kernel along Δ with $\{S^{(1)}(P_b),-\}$ is twice the composite pair kernel along P_b, and the commutator of the letter count with $\{S^{(1)}(P_b),-\}$ is $\{S^{(1)}(P_b),-\}$; hence (c) on $\mathfrak{F}$ and on $\{S^{(1)}(P_b),\mathfrak{F}\}$ gives (b) along b, and the content of (b) beyond (c) lies in the directions b with $\{S^{(1)}(P_b),\mathfrak{F}\}$ outside $\mathfrak{F}$. (2) Identity (a) follows from (b) on three families with letters of one sector each, under the hypotheses of Theorem 11.26 (Proposition 11.33); the cubic arity-one equation is then a consistency condition. (3) Identities (b) and (c) are identities between graph sums at the regulator of Definition 8.3; the numbers $-\frac{1}{3}$ and $-\frac{2}{3}$ are finite parts of special-edge cycles, which depend on the cutoff shape (Remark 10.48 (d), Proposition 10.23). The statement independent of the scheme is the spectrum of D_1 at the coefficients fixed by the arity-two equation (Remark 10.48).

Theorem 11.32 (Consequences of the identities)**.** *Assume (a), (b) and (c) of Definition 11.30 on* $\mathfrak{F}$*.*

- *(i) On the parts with the fewest fields of the cocycles of* $\mathfrak{F}$*, on which* $\hbar F_1^{(1)}(\Delta)$ *acts through its quadratic part,* D_1 *minus four times the map of Theorem 10.16 induces* $-(\frac{1}{3}+\theta)$ *units times* $\{S^{(1)}(\Delta),-\}$ *on cohomology: the assumption of Corollary 10.44, for the solution with sector coefficient* θ*.*
- *(ii) Under the hypotheses of Theorem 11.26, and if* $\{S^{(1)}(P_b),O\}$ *is not exact for some cocycle* $O \in \mathfrak{F}$*,* $\theta = -\frac{1}{3}$*, and* D_1 *induces four times the map of Theorem 10.16; its spectrum on* $\mathfrak{F}$ *in a fixed classical Weyl weight is twice that of the one-loop dilatation coefficients,* -8 *units times the eigenvalues of* $\sum_l H_{l,l+1}$ *of [B03c, (1.1)–(1.3)] on the planar families of Corollary 10.35 and of Table 11.5.*
- *(iii) Under the hypotheses of (ii), in the translation-covariant scheme* $P_1 \cong 0$ *on* $\mathfrak{F}$*; otherwise Corollary 11.28 (iv) applies.*

Conversely, if D_1 *minus four times the map induces* $-(\frac{1}{3}+\theta)$ *units times* $\{S^{(1)}(\Delta),-\}$ *for equal sector coefficients, then (b) and (c) hold on* $\mathfrak{F}$*.*

Proof. (i) By Corollary 11.20, D_1 minus four times the map is the sum of the composite pair kernels along Δ minus the bracket with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional. Its position part along b is, by (b) and Lemma 11.29, $2 \cdot (-\frac{1}{3})\partial_b - 2\theta\,\partial_b = -2(\frac{1}{3} + \theta)\partial_b$ units. Its translation-invariant part is, by (c) and Lemma 11.29, $-\frac{2}{3}$ units times the letter count plus $\sum_a (c_{\varphi_a} - 2\theta w_{X_a})$ units, which is $-2(\frac{1}{3} + \theta)w_O$ units, by $\sum_a (c_{\varphi_a} + \frac{2}{3} w_{X_a}) = \frac{2}{3}\sum_a (2 + |\alpha_a|)$. Together they give $-(\frac{1}{3} + \theta)(2x \cdot \partial + 2w_O)$ units, which is $-(\frac{1}{3} + \theta)$ units times v. (Lemma 11.9). (ii) By Theorem 11.26 the position part of D_1, which is $-2(\frac{1}{3} + \theta)$ units times ∂_b on cohomology, is exact, and $\partial_b O$ is not exact for that cocycle, so $\theta = -\frac{1}{3}$; then Theorem 10.16 and Corollary 10.35. (iii) Corollary 11.28 (iii). The converse reverses the two computations of (i). □

Proposition 11.33 (The arity-two equation fixes each sector coefficient)**.** *Assume the hypotheses of Theorem 11.26 and of Definition 10.36. Let $\mathfrak{F}_\perp$, $\mathfrak{F}_\lambda$, $\mathfrak{F}_V$ be families of gauge-invariant cocycles whose two-field parts have letters of the sector of $A|_{W_{\mathbb{R}}^\perp}$, of λ and of $A|_V$ alone, each containing a cocycle O for which $\{S^{(1)}(P_b), O\}$ is not exact. If (b) holds along b on $\mathfrak{F}_\perp$, then $\theta = -\frac{1}{3}$; if it holds along b on $\mathfrak{F}_\lambda$, then $\theta_\lambda = -\frac{1}{3}$; if it holds along b on $\mathfrak{F}_V$, then $\theta_V = -\frac{1}{3}$. In particular (b) on the three families gives $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, the equality asserted in (a), and the cubic part of* (9.10) *admits the solution $\theta - \theta_V = \theta_\lambda - \theta_V = 0$. These are the values computed from the action-level arity-two equations along (Δ, P_b) and (Δ, ξ) (Theorem 9.28); on each family the operator-level determination is a cross-check of that computation.*

Proof. By Corollary 11.20, Proposition 11.10 (i) and Lemma 11.27, the position part of D_1 is twice the sum of the composite pair kernels along P_b minus the bracket with the position part of $\hbar F_1^{(1)}(\Delta)$. On the two-field part $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part; the bracket of the sector part of the sector of X has the position part $2\partial_b X$ and the other sector parts have bracket 0 with X, and the κ-directions are s-exact (Proposition 11.25 (ii)). With (b), on the cohomology of the family of the sector with coefficient $\theta_\star \in \{\theta, \theta_\lambda, \theta_V\}$,

$$\big(\text{position part of } D_1 \text{ along } b\big) \cong -\tfrac{2}{3}\,\text{units} \cdot \partial_b - 2\theta_\star\,\text{units} \cdot \partial_b = -2\big(\tfrac{1}{3} + \theta_\star\big)\,\text{units} \cdot \partial_b.$$

The left side is s-exact on cocycles (Theorem 11.26) and $\partial_b O$ is not exact, so $\theta_\star = -\frac{1}{3}$. The last statement is Proposition 11.25 (iii); the existence of the quantisation is a hypothesis of Theorem 11.26. □

Corollary 11.34 (The dilatation component without the cubic equation)**.** *Under the hypotheses of Proposition 11.33, if (b) holds on $\mathfrak{F}_\perp$, $\mathfrak{F}_\lambda$ and $\mathfrak{F}_V$, and (b) and (c) hold on $\mathfrak{F}$, then on the cohomology of $\mathfrak{F}$, D_1 induces four times the map of Theorem 10.16, with spectrum -8 units times the eigenvalues of $\sum_l H_{l,l+1}$ on the planar families of Corollary 10.35 and of Table 11.5, and $P_1 \cong 0$ in the translation-covariant scheme.*

Proof. Proposition 11.33 gives $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, and Theorem 11.32 (i)–(iii) applies. □

Proposition 11.35 (The planar scalar sector)**.** *On the part with the outputs $A|_{W_{\mathbb{R}}^\perp}$ of the planar scalar sector (Proposition 10.40), (b) and (c) hold as identities modulo terms whose sum over the cyclically adjacent pairs of every trace vanishes: per adjacent pair the composite pair kernel along Δ is $\big[-\frac{1}{6}v.(X_aX_b) - \frac{2}{3}X_aX_b\big]$ units and along P_b it is $-\frac{1}{6}$ units times $\partial_b(X_aX_b)$, so that on $O = \operatorname{tr}(\iota_{w_1}A \cdots \iota_{w_L}A)$ the composite term along Δ is $-\big(\frac{1}{3}v. + \frac{2}{3}L\big)O$ units, the composite term along P_b is $-\frac{1}{3}$ units times $\partial_b O$, and the sum of the translation-invariant parts is $-\frac{4}{3}L$ units, $-\frac{2}{3}$ units times the letter count. On the bilinears these are exact identities (Example 11.22).*

Proof. By Proposition 10.40 (i), (e), the composite term of an adjacent pair is $-v.(X_aX_b)$ units, and the bracket with the renormalised composite two-point functional on a letter is $(\frac{5}{3}v.X - \frac{2}{3}X)$ units (Remark 9.27); by Theorem 11.18 (v) their sum is the sum of the composite pair kernels, and the one-body term

distributed over the two adjacent pairs of each letter gives the density per pair. For L letters without derivatives the letter count is $2L$ and the translation-invariant part of $v.$ is $2L$: $-\frac{1}{3}\cdot 2L - \frac{2}{3}L = -\frac{4}{3}L$. Along P_b: Theorem 9.35, $-2+\frac{5}{3} = -\frac{1}{3}$. □

11.5 The composite kernels on the families

Throughout this section $\mathfrak{g} = \mathfrak{su}(N)$ at leading order in N and O is a single trace, with the conventions of Section 10.7. The projection to cohomology on the two-field part is the evaluation on the free equations of motion, $|p|^2 = 0$ on the scalar legs, the Weyl equation on the gaugino legs and transverse on-shell gauge legs, which removes the s-exact terms $s(\text{antifield}\times\text{field})$; total derivatives are kept, as descendants through ω. The letters are written in the components of Definition 8.1.

Definition 11.36 (Test families)**.** Let $n_m := \sigma^m_{11} = (0,0,1,i)$, a null vector, and $D_n := n_m(\partial_m + [A_m,-])$, the covariant derivative $\iota_n d_{A|_V}$ along n; let $Z = \iota_w A$ with $w \in W_{\mathbb{R}}^{\perp}\otimes\mathbb{C}$, $(w,w)=0$; let $\psi := \lambda_1{}^i$ for a fixed i, the component $\alpha=1$ of $\lambda|_{S_+\otimes W}$; and let F_{11} be the component $\alpha=\beta=1$ of $(\sigma^m\bar\sigma^n)_\alpha{}^\gamma\varepsilon_{\gamma\beta}F_{mn}$, $F_{mn}=\partial_m A_n - \partial_n A_m + [A_m,A_n]$, whose two-field part is a component of $dA|_V$ and whose three-field part is a component of $[A|_V\wedge A|_V]$. The test families are the spans of the following single traces and their gauge completions:

- the scalars: $\mathrm{tr}(\iota_{w_1}A\cdots\iota_{w_L}A)$, the bilinears, and the $\mathfrak{su}(2)$ sector of Proposition 10.32 for $L\le 8$;
- twist two: $\mathrm{tr}\left(D_n^k Z\, D_n^{S'-k}Z\right)$, $0\le k\le S'$ (Proposition 10.34);
- the $\mathfrak{su}(1|1)$ sector: single traces of the letters Z and ψ of Proposition 10.33, $L\le 6$;
- the fermion $\mathfrak{sl}(2)$ sector: $\mathrm{tr}\left(D_n^k\psi\, D_n^{S'-k}\psi\right)$;
- two gaugino components: $\mathrm{tr}\left(D_n^k\psi^{(3)}\, D_n^{S'-k}\psi^{(4)}\right)$ with $\psi^{(3)}=\lambda_1{}^3$, $\psi^{(4)}=\lambda_1{}^4$, together with $\mathrm{tr}\left(D_n^k Z'\, D_n^{S'-k}F_{11}\right)$, Z' the scalar letter with the R-symmetry weights of $\psi^{(3)}\psi^{(4)}$, with which the family closes;
- the gluon $\mathfrak{sl}(2)$ sector: $\mathrm{tr}\left(D_n^k F_{11}\, D_n^{S'-k}F_{11}\right)$;
- the mixed gluon–gaugino family: $\mathrm{tr}\left(D_n^k F_{11}\, D_n^{S'-k}\psi\right)$.

Twist two, the fermion and the gluon $\mathfrak{sl}(2)$ sectors are families of Proposition 11.33 for the sectors of $A|_{W_{\mathbb{R}}^{\perp}}$, of λ and of $A|_V$.

The two-field part of a gauge-invariant operator with covariant derivatives receives the pairs of its two leading letters with two outputs and the pairs of a letter $A|_V$ from a covariant derivative, or from the three-field part of a field strength, with another letter, with one output (Proposition 10.34). For the composite pair kernels these are the leg sets $A|_{W_{\mathbb{R}}^{\perp}}$ of $(A|_V, A|_{W_{\mathbb{R}}^{\perp}})$, λ of $(A|_V,\lambda)$ and $A|_V$ of $(A|_V,A|_V)$ of Table 11.1, all with $k=1$, with derivatives.

Proposition 11.37 (Spectra from the harmonic action)**.** *On the two-site graded cyclic states, single traces of two letters with S' derivatives D_n in total, the eigenvalues of $\sum_l H_{l,l+1}$ of [B03c, (1.1)–(1.3)], computed from the harmonic action [B03c, (4.27)–(4.29)] on the oscillator letters [B03c, (4.9)], are:*

- *(i) scalars $D_n^k Z$: $4h(S)$, $S\le S'$ even [B03c, (2.31)];*
- *(ii) gluons $D_n^k F_{11}$: $4h(S+2)$, $S\le S'$ even, so that* $\mathrm{tr}(F_{11}F_{11})$ *has the Konishi value* 6*;*
- *(iii) one gaugino component $D_n^k\psi$: $4h(S+1)$, $S\le S'$ odd;*

- *(iv) two gaugino components $\psi^{(3)}$, $\psi^{(4)}$ with $Z'F_{11}$: $0, 6$ at $S' = 0$; $0, 6, 6, 6$ at $S' = 1$; $0, 6, 6, 6, 6, \frac{25}{3}$ at $S' = 2$; 0, four times 6 and three times $\frac{25}{3}$ at $S' = 3$; 0, four times 6, four times $\frac{25}{3}$ and $\frac{49}{5}$ at $S' = 4$;*

- *(v) (F_{11}, ψ): 6; $6, 6$; $6, 6, \frac{25}{3}$; $6, 6, \frac{25}{3}, \frac{25}{3}$; $6, 6, \frac{25}{3}, \frac{25}{3}, \frac{49}{5}$ at $S' = 0, \ldots, 4$.*

On the $\mathfrak{su}(1|1)$ sector the two-site Hamiltonian is one minus the graded exchange (Proposition 10.33), with the cyclic spectra of Table 11.5.

Proof. Exact evaluation of [B03c, (4.27)–(4.29)], with the harmonic numbers of the coefficients $c_{n,0,0} = h(\frac{n}{2})$, on bases of the families closed under H_{12}, with the graded exchange for H_{21} and the projection to graded cyclic states, and exact diagonalisation. The same evaluation gives [B03c, (2.31)] on twist two and the values 0, twenty times, and 6 on the scalar bilinears, which fixes the central charge per site, the signs of the fermionic oscillators and the graded cyclicity. □

(L, M)	eigenvalues on the graded cyclic states of $\mathrm{tr}(Z^{L-M}\psi^M)$
$M \leq 1$	0
$(3,2)$, $(3,3)$	6
$(4,2)$, $(4,3)$	4
$(5,2)$, $(5,3)$	$5 \pm \sqrt{5}$
$(5,4)$, $(5,5)$	10
$(6,2)$	2, 6
$(6,3)$	2, 6, 8, 8
$(6,4)$	8, 8, 8
$(6,5)$	8

The eigenvalues of $\sum_l H_{l,l+1}$ on the $\mathfrak{su}(1|1)$ sector, $L \leq 6$; the sectors $(2,2)$, $(4,4)$, $(6,6)$ have no graded cyclic state.

Theorem 11.38 (The composite pair kernels on the families)**.** • *(i) (The two-site density.) For every ordered adjacent pair (X_a, X_b) of letters among $A|_{W^\perp_{\mathbb{R}}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ without derivatives, on the free shell, and for two letters $A|_{W^\perp_{\mathbb{R}}}$ also off the shell and for all their components, the planar composite pair kernels have, on the same word X_aX_b, the translation-invariant part $-\frac{4}{3}$ units times X_aX_b along Δ and the value $-\frac{1}{6}$ units times $\partial_b(X_aX_b)$ along P_b; on the exchanged word and on the trace they vanish.*

- *(ii) On the parts with the fewest fields of the families of Definition 11.36, the two-field parts for two letters, on the free shell, the sum over the pairs of the translation-invariant parts of the composite pair kernels along Δ is the diagonal operator of Table 11.5, which is $-\frac{2}{3}$ units times the letter count, and the sum of the composite pair kernels along n is $-\frac{1}{3}$ units times ∂_n; on twist two for $S' \leq 4$ the sum along every $b \in V$ is $-\frac{1}{3}$ units times ∂_b.*

- *(iii) Hence (b) along n and (c) of Definition 11.30 hold on each family in the ranges of Table 11.5, and (b) along every b holds on twist two for $S' \leq 4$.*

Proof. Exact evaluation of the special-edge composite cycles by Theorem 11.11, Propositions 11.12 and 11.13, with the colour words reduced to the planar adjacent form of Lemma 10.8 (iv), on the leg sets of Table 11.1 with the derivatives of the families, the outputs evaluated on the free shell (Remark 11.40). On the families with derivatives the outputs depend on the output momenta through (n, p), every derivative being contracted with n and $(n, n) = 0$; the sums are diagonal in the bases of the families. (iii) is (i), (ii) and the values of the letter count: $2L$ on L letters without derivatives, $4 + S'$ on twist two and on two gaugino letters, $5 + S'$ on $Z'F_{11}$ and on (F_{11}, ψ), $6 + S'$ on two gluon letters, a letter F_{11} counting 3 through its two-field part $dA|_V$. □

family	range	sum of the translation-invariant parts along Δ	spectrum of D_1 at $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$, in units
scalars	bilinears; $\mathfrak{su}(2)$, $L\le 8$	$-\frac{4}{3}L$	0, -48; -8 times the Heisenberg spectra
twist two	$S'\le 6$	$-\frac{2}{3}(4+S')$	0, -48, $-\frac{200}{3}$, $-\frac{392}{5}$: $-32h(S)$, $S\le S'$ even
$\mathfrak{su}(1\|1)$	$L\le 6$	$-\frac{4}{3}L$	-8 times the $\mathfrak{su}(1\|1)$ spectra
fermion $\mathfrak{sl}(2)$	$S'\le 5$	$-\frac{2}{3}(4+S')$	-48, $-\frac{200}{3}$, $-\frac{392}{5}$: $-32h(S+1)$, $S\le S'$ odd
$\psi^{(3)}\psi^{(4)}$ with $Z'F_{11}$	$S'\le 2$	$-\frac{2}{3}(4+S')$ on $\psi^{(3)}\psi^{(4)}$, $-\frac{2}{3}(5+S')$ on $Z'F_{11}$	$0,-48$; $0,-48,-48,-48$; 0, four times -48, $-\frac{200}{3}$
gluon $\mathfrak{sl}(2)$	$S'\le 2$	$-\frac{2}{3}(6+S')$	-48, $-\frac{200}{3}$: $-32h(S+2)$, $S\le S'$ even
(F_{11},ψ)	$S'\le 4$	$-\frac{2}{3}(5+S')$	at $S'=4$: -48, -48, $-\frac{200}{3}$, $-\frac{200}{3}$, $-\frac{392}{5}$

The composite pair kernels and the spectrum of D_1 on the families: the third column is identity (c), the fourth is -8 units times the eigenvalues of Proposition 11.37, of Proposition 10.32 on the $\mathfrak{su}(2)$ sector and of Table 11.5 on the $\mathfrak{su}(1|1)$ sector.

Corollary 11.39 (D_1 and P_1 on the families)**.** *Under the hypotheses of Definition 10.36, on the cohomology of the part with the fewest fields of every family of Definition 11.36, with every free direction of Definition 11.23,*

$$D_1 \cong 4\,(\textit{the map}) - \sum_a \left(\theta_{X_a}+\tfrac{1}{3}\right) \textit{units}\cdot v.X_a, \qquad P_1 \cong -\sum_a \left(\theta_{P,X_a}+\tfrac{1}{3}\right) \textit{units}\cdot \partial_b X_a,$$

with the map of Theorem 10.16, the position part of D_1 and the component P_1 taken along $b=n$, and along every $b\in V$ on twist two for $S'\le 4$ and on the letters without derivatives (Theorem 11.38), where θ_{X_a} is θ, θ_λ or θ_V and θ_{P,X_a} is θ_P, $\theta_{P,\lambda}$ or $\theta_{P,V}$ according to the sector of the letter X_a, $v.X_a$ and $\partial_b X_a$ act on the letter X_a alone, and the directions $Q\int(\iota_x A|_V, c^*)$, $Q\int(\rho(x)\lambda^*,\lambda^*)$ contribute s-exact terms. At the values $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$ of Theorem 9.28, which twist two, the fermion and the gluon $\mathfrak{sl}(2)$ sectors reproduce under the hypotheses of Theorem 11.26 (Proposition 11.33), D_1 induces four times the map of Theorem 10.16 on every family, with the spectra of Table 11.5, -8 units times the eigenvalues of $\sum_l H_{l,l+1}$, which is twice the one-loop anomalous dimensions; and $P_1\cong 0$ in the translation-covariant scheme, while for equal coefficients θ_P along P_b, $P_1\cong(\theta-\theta_P)$ units times ∂_b with $\theta=-\frac{1}{3}$.

Proof. Corollary 11.20 and Theorem 11.38: on the two-field part $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part, the bracket of the sector part of the sector of X_a with X_a being $v.X_a$ and the others vanishing. As in the proof of Theorem 11.32 (i), letter by letter, the position part of the composite pair kernels and of the bracket is $-2(\frac{1}{3}+\theta_{X_a})\partial_b$ units and the translation-invariant part is $-2(\frac{1}{3}+\theta_{X_a})w_{X_a}$ units; along P_b the composite pair kernels give $-\frac{1}{3}$ units times ∂_b and the bracket with $\hbar F_1^{(1)}(P_b)$ gives $-\theta_{P,X_a}$ units times $\partial_b X_a$. The κ-directions are s-exact (Proposition 11.25 (ii)). The rest is Proposition 11.33, Theorem 11.32 and Corollary 11.28 (iii), (iv). □

Remark 11.40 (Method of computation)**.** Each special-edge composite cycle is evaluated exactly: the Gaussian loop integral and the integrals of (11.1) over the ordered subsimplices of the region where α_σ is the smallest coordinate, the letter derivatives through Proposition 11.12, the position through the injected momentum r (Proposition 11.10), the chains with $k=1$ through Proposition 11.13, the fibre algebra in explicit realisations of V_{10} and Σ over $\mathbb{Q}(i)$, and the colour words by Lemma 10.8, with its planar adjacent rule also checked in realisations of $\mathfrak{su}(2)$, $\mathfrak{su}(3)$ and $\mathfrak{gl}(3)$. In this evaluation every chain carries the sign of Definition 11.5, the factor $(-1)^{[c<\sigma]}$ included, and the families are evaluated at generic null leg momenta. A second evaluation works in momentum space with the kernel form of Lemma 11.2 (ii), $v.$ acting on the Fourier components as an Euler operator plus constants, Feynman parameters for the edges P and the special Schwinger time integrated last, applies the sign of Definition 11.5 as written, and evaluates the families on on-shell legs with momenta proportional to one null vector $\bar n$, $(n,\bar n)\neq 0$, which determine the translation-invariant parts and the parts along n (proof of Theorem 11.38). The two evaluations give the same off-shell entries with at most two outputs and no letter derivatives, for the pairs of bosonic letters and for the pairs with a gaugino letter or output, whose values lie in $\mathbb{Q}+\mathbb{Q}\log 2+\mathbb{Q}\log 3$ for $k\le 2$, and the same family matrices on the scalars, on twist two for $S'\le 4$, on the $\mathfrak{su}(1|1)$ sector, on the fermion $\mathfrak{sl}(2)$ sector for $S'\le 5$ and on the gluon $\mathfrak{sl}(2)$ sector

for $S' \leq 2$; the first evaluation extends twist two to $S' \leq 6$, and the second evaluation gives the two families with F_{11} and a gaugino letter. A third evaluation, a Taylor expansion in momentum space at generic null leg momenta, gives the sums on twist two for $S' \leq 4$ along a generic b. Every integral is a four-dimensional Schwinger integral with the cutoff ε on every edge.

11.6 The two-body kernels of Q_1, S_1 and K_1

For x in a homogeneous basis of $\mathfrak{psl}(4|4)$, λ_x is its ad_Δ eigenvalue, -1 on Σ, $+1$ on Σ^* and $+2$ for K_b, and the degree of x in the position is 0, 1 and 2 (Section 10.9). The two-body kernel of a pair of letters along x is the kernel of Proposition 10.45; this section computes it along the supercharges, the special supercharges and the special conformal fields. Along Σ the vertex $S^{(1)}(\xi)$ meets the gauge-fixing operator in a new way: the commutator $[\xi., Q^{GF}]$ joins the gaugino to the ghost, and $S^{(1)}(\xi)$ sits on a cycle as a line from λ to c whose image under Q is nonzero, in contrast to the composite lines along Δ and P_b. The section shows that the component along x of every local functional is nevertheless a sum over its pairs of letters of local two-body kernels, the finite parts of the cycles with one vertex $S^{(1)}(x)$ and one special edge, plus a one-body term; writes these kernels as simplex integrals, independent of the endpoint along K_b; computes the action-level obstructions and solutions along the odd generators, the rotations and K_b on two legs with all their free directions, and the arity-two equations that relate the free coefficients; assembles Q_1, S_1, D_1, P_1, M_1, R_1 and K_1 on the scalar bilinears, where the two-body part of Q_1 on the traceless part is $-2\log\frac{3}{2}$ units times the classical transformation, as representatives with their scheme-independent content; and evaluates the pair entries of the twisted differential along a rank-one supercharge. Edges carry $P(\varepsilon,\infty)$ along Σ and Σ^* (Proposition 8.25) and $P(\varepsilon,T)$ at a finite T along K_b (Definition 8.26). In this section ψ denotes the finite part of the operators of Definition 8.11, and $L := \log\frac{3}{2}$; with its superscript, $L^{(1)}(x)$ is the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ (Lemma 11.15).

Definition 11.41 (The terms of $S^{(1)}(x)$)**.** For $\xi \in S_+ \otimes W$ the functional $S^{(1)}(\xi)$ of (2.6) is the sum of the seven terms of Table 11.6: the quadratic terms $T_1, \dots, T_4$ and the three cubic terms T_5, with $F_A = dA + \frac{1}{2}[A \wedge A]$ reduced to V and $dA|_V$, $dA|_{W_{\mathbb{R}}^\perp}$ the parts of dA in $\wedge^2 V$ and in $V \wedge W_{\mathbb{R}}^\perp$; T_3 and T_4 carry one derivative on the leg A. For $\xi \in S_- \otimes W^*$ the table holds with $\lambda|_{S_+\otimes W}$ and $\lambda|_{S_-\otimes W^*}$ exchanged and $\lambda^*|_{S_+\otimes W^*}$ and $\lambda^*|_{S_-\otimes W}$ exchanged. The quadratic part of $S^{(1)}(\xi)$ is $S_2^V(\xi) + S_2^\perp(\xi)$, with $S_2^V(\xi) := T_1 + T_3$, the part with the legs $(\lambda, A^*|_V)$ and $(A|_V, \lambda^*)$, and $S_2^\perp(\xi) := T_2 + T_4$, the part with the legs $(\lambda, A^*|_{W_{\mathbb{R}}^\perp})$ and $(A|_{W_{\mathbb{R}}^\perp}, \lambda^*)$. For $\eta \in \Sigma^*$ and an orthonormal basis e_μ of V, (2.7) reads

$$S^{(1)}(\eta) = \sum_\mu \int x^\mu s_\mu + \int s_0, \qquad \int s_\mu := S^{(1)}\big(\rho(e_\mu)\eta\big), \qquad s_0 := -\big(\rho(A|_{W_{\mathbb{R}}^\perp})\eta, \lambda^*\big); \tag{11.3}$$

for $\eta \in S_+ \otimes W^*$ the spinors $\rho(e_\mu)\eta$ lie in $S_- \otimes W^*$ and s_0 has the legs $A|_{W_{\mathbb{R}}^\perp}$, $\lambda^*|_{S_+\otimes W^*}$, and for $\eta \in S_- \otimes W$ they lie in $S_+ \otimes W$ and s_0 has the legs $A|_{W_{\mathbb{R}}^\perp}$, $\lambda^*|_{S_-\otimes W}$. For an even generator with vector field v, $S^{(1)}(x) = \int \big((v.A, A^*) - (v.\lambda, \lambda^*) - (v.c, c^*)\big)$ (Definition 7.4).

term	functional, $\int$ dvol understood	legs
T_1	$-\big(\Gamma(\xi,\lambda)\|_V, A^*\|_V\big)$	$\lambda\|_{S_-\otimes W^*}$, $A^*\|_V$
T_2	$-\big(\Gamma(\xi,\lambda)\|_{W_{\mathbb{R}}^\perp}, A^*\|_{W_{\mathbb{R}}^\perp}\big)$	$\lambda\|_{S_+\otimes W}$, $A^*\|_{W_{\mathbb{R}}^\perp}$
T_3	$\frac{1}{2}\big(\rho(dA\|_V)\xi, \lambda^*\big)$	$A\|_V$, $\lambda^*\|_{S_+\otimes W^*}$
T_4	$\frac{1}{2}\big(\rho(dA\|_{W_{\mathbb{R}}^\perp})\xi, \lambda^*\big)$	$A\|_{W_{\mathbb{R}}^\perp}$, $\lambda^*\|_{S_-\otimes W}$
T_5	$\frac{1}{2}\big(\rho(\frac{1}{2}[A\wedge A])\xi, \lambda^*\big)$	$(A\|_V, A\|_V, \lambda^*\|_{S_+\otimes W^*})$, $(A\|_V, A\|_{W_{\mathbb{R}}^\perp}, \lambda^*\|_{S_-\otimes W})$, $(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp}, \lambda^*\|_{S_+\otimes W^*})$

The terms of $S^{(1)}(\xi)$ for $\xi \in S_+ \otimes W$ and their legs; their two-component forms are in Remark 11.68.

Lemma 11.42 (The generators on the regulator and their lines)**.** *For an odd generator x, $x.$ is the odd operator on $\mathscr{E}$ that induces the bracket with the quadratic part of $S^{(1)}(x)$ (Proposition 7.11); for an even one it is $v.$ of Definition 7.4. Let $g(H) := \int_\varepsilon^T e^{-tH}dt$, so that $P\star = Q^{GF}g(H)$ and $QP\star + P\star Q = e^{-\varepsilon H} - e^{-TH} =: E(H)$ (Proposition 9.18 (iii)).*

- *(i) For $\xi \in \Sigma$, $\xi.$ maps the slot of λ to the slot of A by $\lambda \mapsto +\Gamma(\xi,\lambda)$, the slot of A to the slot of λ by $A \mapsto +\frac{1}{2}\rho(dA)\xi$, the slots of A^* and of λ^* to each other by the transposed arrows, and vanishes on the slots of c and c^*. In particular*

$$\Big\{S^{(1)}(\xi), \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)\Big\} = \int \omega\,\Big[\big(\iota_{w_1}A, (\rho(w_2)\xi,\lambda)\big) + \big(\iota_{w_2}A, (\rho(w_1)\xi,\lambda)\big)\Big].$$

- *(ii) $[\xi., Q] = 0$, $[\xi., H] = 0$, and $C_\xi := [\xi., Q^{GF}]$ is an even operator of degree -1 with constant coefficients and of order one, with four components, each nonzero for $\xi \neq 0$: $\lambda \mapsto -d^*\big(\Gamma(\xi,\lambda)|_V\big)$ in the slot of c, on a plane wave $e^{i(p,x)}$ the map $\lambda \mapsto +i\big(p, \Gamma(\xi,\lambda)|_V\big)$; a component from $A^*|_V$ to λ, a nonzero multiple of $A^*|_V \mapsto (p, A^*|_V)\,\xi$; and their transposes from λ^* to $A|_V$ and from c^* to λ^*. For $\xi \in S_+ \otimes W$ these are $\lambda|_{S_-\otimes W^*} \to c$, $A^*|_V \to \lambda|_{S_+\otimes W}$, $\lambda^*|_{S_+\otimes W^*} \to A|_V$ and $c^* \to \lambda^*|_{S_-\otimes W}$. The variation kernel $(\xi.\otimes 1 + 1\otimes\xi.)P(\varepsilon,T) = (C_\xi\, g(H)\otimes 1)K_0$ has its components in $\lambda\otimes A|_V$, $c\otimes\lambda^*$ and their transposes.*

- *(iii) The term T_1 with both legs on edges P, its leg λ on the component $\lambda\otimes\lambda$ and its leg $A^*|_V$ on the component $A^*|_V\otimes c$, is the composite line along ξ, with the kernel $2(P\star\,\xi.\,P\star\otimes 1)K_0$, whose components lie in $c\otimes\lambda$ and $\lambda\otimes c$. For $\xi \in S_+\otimes W$ and $T = \infty$ it is $-2\,\xi\, e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $c\otimes\lambda|_{S_+\otimes W}$ and $+2\,\xi\, e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $\lambda|_{S_+\otimes W}\otimes c$, and*

$$P\star\,\xi.\,P\star = C_\xi\, Q^{GF}g(H)^2 - \xi.\,(Q^{GF})^2 g(H)^2, \qquad \big[Q, P\star\,\xi.\,P\star\big] = E(H)\,C_\xi\,g(H) \neq 0.$$

  The terms of $S^{(1)}(\xi)$ with a leg $A^*|_{W_\mathbb{R}^\perp}$ or λ^* enter a cycle with that leg on the special edge, and a term T_5 also with its leg λ^* as an output.

- *(iv) For $\eta \in \Sigma^*$, with $\xi_\mu := \rho(e_\mu)\eta$,*

$$\eta. = \sum_\mu x^\mu\,\xi_\mu. + \eta^0. + Y_\eta, \qquad [\eta., Q] = 0, \qquad [\eta., H] = 2\sum_\mu \xi_\mu.\,\partial_\mu,$$

  where $\eta^0.$ is the constant operator of order zero induced by s_0, with the arrows $A|_{W_\mathbb{R}^\perp} \to \lambda$ and $\lambda^* \to A^*|_{W_\mathbb{R}^\perp}$, and Y_η is the constant operator of order zero from λ^* to A^* produced when the derivative of T_3, T_4 in $\sum_\mu \int x^\mu\big(S_2^V(\xi_\mu) + S_2^\perp(\xi_\mu)\big)$ is moved off A, $\partial_\nu(x^\mu\,\cdot\,) = x^\mu\partial_\nu + \delta^\mu_\nu$. On a cycle $S^{(1)}(\eta)$ forms the line of (iii) at the spinor $\rho(x)\eta$, linear in the position, and s_0 enters with its leg λ^* on the special edge.

- *(v) For K_b, with $v(x) = (x,x)b - 2(b,x)x$ and $\operatorname{div} v = -8(b,x)$, on a slot of Weyl weight w with the rotation generators $\Sigma^{\rho\nu}$, and with $\sigma^\rho(D) := [D, x^\rho]$ for a constant-coefficient operator D,*

$$[v., Q^{GF}] = 4(b,x)\,Q^{GF} + C_K, \qquad C_K := 2\,\sigma^\rho(Q^{GF})\big(w\,b_\rho - b_\nu\Sigma^{\rho\nu}\big),$$
$$[v., H] = 4(b,x)\,H - 4(w-1)(b,\partial) + 4b_\nu\Sigma^{\rho\nu}\partial_\rho,$$

  C_K of order zero with constant coefficients, zero on $A^*|_{W_\mathbb{R}^\perp}$, with $C_K A|_V = -4(b, A|_V)$ in the slot of c. $S^{(1)}(K_b)$ forms on a cycle the composite line from c to $A|_V$ with the kernel $P\star\, v.\,P\star = v.(P\star)^2 + [P\star, v.]P\star$, quadratic in the position, at a finite T (Definition 8.26).

In each case every letter, $A^*|_{W_\mathbb{R}^\perp}$, λ^* and c^* included, is active (Lemma 10.46).

Proof. (i) The terms $-(\Gamma(\xi,\lambda),A^*)$ and $\frac12(\rho(dA)\xi,\lambda^*)$ of (2.6) pair the slots (λ,A^*) and (A,λ^*); the bracket of a quadratic functional with a coordinate functional is the coordinate functional composed with the corresponding arrow (proof of Lemma 9.16 (i)), and each pairing term gives one arrow between field slots and the transposed arrow between antifield slots (proof of Proposition 7.11). In the reading of Proposition 1.11 the term (dc,A^*) of (1.2) gives $c\mapsto -dc$; the term $-(\Gamma(\xi,\lambda),A^*)$, whose letter λ is odd like c, gives $\lambda\mapsto+\Gamma(\xi,\lambda)$, and $[\xi.,Q]=0$ on the slot of A, the part of polynomial degree two of $\{S,S^{(1)}(\xi)\}=0$, fixes $A\mapsto+\frac12\rho(dA)\xi$. With $\iota_w\Gamma(\xi,\lambda)=(\rho(w)\xi,\lambda)$ [ESW, (28)] the classical component on the bilinear is the first functional of Corollary 7.14.

(ii) $[\xi.,Q]=0$ and $[\xi.,H]=0$ are Proposition 7.11. Q^{GF} maps $A^*\mapsto -2A^*$ into the slot of A, $A\mapsto -d^*(A|_V)$ into the slot of c, $c^*\mapsto dc^*$ into the slot of $A^*|_V$ and $\lambda^*\mapsto -\not{\partial}\lambda^*$ into the slot of λ, and vanishes on c and λ (Chapter 3). Hence $\xi.Q^{GF}$ has the arrows $A^*\to A\to\lambda$, $c^*\to A^*|_V\to\lambda^*$, $\lambda^*\to\lambda\to A$, and $Q^{GF}\xi.$ the arrows $\lambda\to A\to c$, $A^*\to\lambda^*\to\lambda$, $\lambda^*\to A^*\to A$; each composite has order one. On λ, $C_\xi\lambda=Q^{GF}\Gamma(\xi,\lambda)=-d^*\big(\Gamma(\xi,\lambda)|_V\big)$, and $d^*=-\iota_\partial$ is $-i\iota_p$ on plane waves. On $A^*=a\,e^{i(p,x)}$ the two composites to λ are opposite multiples of $\rho(p)\rho(a)\xi$ plus, through $\rho(q(p\wedge a))=\rho(p)\rho(a)-(p,a)$ [ESW, (29)] in the first, a multiple of $(p,a)\xi$; for $a\in W_{\mathbb R}^\perp$ the two cancel. The components from λ^* and c^* are the transposes, since $\xi.$ preserves the pairing and Q^{GF} has the adjointness of Proposition 3.8, so $(C_\xi\otimes1)K_0$ is graded symmetric. $\Gamma(\xi,-)|_V$ is nonzero on one chirality of λ for $\xi\neq0$, and the component from A^* to λ is nonzero by Proposition 7.11. The variation kernel is the third item of Proposition 7.11 with T in place of ∞.

(iii) As in Proposition 9.18 (i), with the signs of Lemma 9.15 for the odd $\xi.$, both orders of the two legs give $(P\star\,\xi.\,P\star\otimes1)K_0$. Since $[\xi.,g(H)]=0$, $[\xi.,P\star]=C_\xi g(H)$, which gives the first identity; with $[\xi.,Q]=0$,

$$Q\,(P\star\xi.P\star)=E\,\xi.P\star-P\star Q\,\xi.P\star=E\,\xi.P\star+P\star\xi.E-P\star\xi.P\star Q,$$

so $[Q,P\star\xi.P\star]=E\,(\xi.P\star+P\star\xi.)=E\,C_\xi g$. The components follow the arrows $\lambda^*\to\lambda\to A\to c$ and $c^*\to A^*|_V\to\lambda^*\to\lambda$. On a plane wave the composite on c^* is a multiple of $\rho(p)\rho(p)\xi=|p|^2\xi$ times $g(H)^2=e^{-2\varepsilon|p|^2}|p|^{-4}$ at $T=\infty$, a scalar line of momentum degree -2; the multiple, -2 on $c\otimes\lambda$, is the value of both evaluations of Remark 11.66. A leg $A^*|_{W_{\mathbb R}^\perp}$ or λ^* has no component of P (Proposition 3.17).

(iv) By (11.3) the field arrows of $\int x^\mu S_2(\xi_\mu)$, S_2 the quadratic part, are x^μ times those of $\xi_\mu.$; its antifield arrows, the transposes, are obtained by integrating by parts the derivative of T_3, T_4 against $x^\mu\lambda^*$, which gives x^μ times the antifield arrows of $\xi_\mu.$ plus Y_η. For the odd Q, $[x^\mu\xi_\mu.,Q]=x^\mu[\xi_\mu.,Q]+[Q,x^\mu]\xi_\mu.$, and the vanishing of $[\eta.,Q]$ is the part of polynomial degree two of $\{S,S^{(1)}(\eta)\}=0$ (Lemma 2.11); it requires Y_η. H is $-\partial^2$ on every component, $[x^\mu,-\partial^2]=2\partial_\mu$, and $\eta^0.$, Y_η and $\xi_\mu.$ have constant coefficients.

(v) For a conformal Killing field, $\partial_\mu v_\nu=\frac14(\operatorname{div}v)\delta_{\mu\nu}+\omega_{\mu\nu}$ with ω antisymmetric and $v.=v^\mu\partial_\mu+\frac12\omega_{\mu\nu}\Sigma^{\mu\nu}+\frac w4\operatorname{div}v$; the transport, spin and weight terms and the rotation covariance of a constant-coefficient operator D of weight -2 give

$$\begin{aligned}[v.,D]&=-\tfrac12(\operatorname{div}v)D-\tfrac w4\sigma^\rho(D)\,\partial_\rho(\operatorname{div}v)-\tfrac12\sigma^\rho(D)(\partial_\rho\omega_{\mu\nu})\Sigma^{\mu\nu},\\ [v.,H]&=-\tfrac12(\operatorname{div}v)H+\tfrac{w-1}2(\partial_\rho\operatorname{div}v)\partial_\rho+(\partial_\rho\omega_{\mu\nu})\Sigma^{\mu\nu}\partial_\rho,\end{aligned}$$

with $\partial^2v_\mu=-\frac12\partial_\mu\operatorname{div}v$. For K_b, $\omega_{\mu\nu}=2(x_\mu b_\nu-x_\nu b_\mu)$ and $\partial_\rho\omega_{\mu\nu}=2(\delta_{\rho\mu}b_\nu-\delta_{\rho\nu}b_\mu)$. On $A^*|_{W_{\mathbb R}^\perp}$, $Q^{GF}=-2$ has order zero; on $A|_V$, $\sigma^\rho(Q^{GF})Y=Y_\rho$, $w=1$ and $\sigma^\rho(Q^{GF})\Sigma^{\rho\nu}A=3A_\nu$, so $C_KA|_V=2(b,A|_V)-6(b,A|_V)$. The line is Lemma 11.2 (iii), (iv) with this v. □

Definition 11.43 (Defect cycles and two-body kernels)**.** Let x be a generator, X_a, X_b two letters of any slots and $0<\varepsilon<T\le\infty$.

- (i) A defect cycle of (X_a,X_b) along x is a chain $X_a-e_0-V_1-e_1-\cdots-V_n-e_n-X_b$ of Definition 10.10 whose vertices are vertices of I and exactly one vertex V_s given by a term of $S^{(1)}(x)$, entered

and left through two of its legs, the third leg of a cubic term being an output, and whose edges are components of $P(\varepsilon,T)$ and exactly one special edge $e_\sigma = K_\varepsilon - K_0$, with the components of K_0. The vertex V_s is placed on a line, with both legs on edges P; at the special edge, with one leg on e_σ; or cubic, a term T_5 with two legs on the cycle.

- (ii) $\mathcal{D}_x(X_a,X_b;\varepsilon,T)$ is the sum of the defect cycles of the pair, amputated, with the outputs left open as in Definition 10.3: the δ-linear graphs of arity one along x of the flow of the defect vertex of Lemma 8.4, applied to $e^{(J+\delta M(\varepsilon))/\hbar}$ with $M=\int\omega\,X_aX_b$, whose cycle is a defect cycle of the pair. A chain read from X_a carries, beyond the signs of Lemma 10.5, the factor

$$-(-1)^{n(<\sigma)+|S^{(1)}(x)|\,(n(<s)+[s<\sigma])},$$

with $n(<j)$ the number of odd outputs at the vertices before the position j, $[s<\sigma]=1$ if V_s precedes e_σ and 0 otherwise, and $|S^{(1)}(x)|$ the total parity of $S^{(1)}(x)$, 0 for odd x and 1 for even x; the leading sign is that of the special edge (Proposition 11.4), and for even x the factor is that of Definition 11.5 times this sign.

- (iii) Let $\mathrm{ct}_m(M)$ and $\mathrm{lc}(M)$ be the coefficients of ε^{-m} and of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](M)$, and $\mathrm{ct}^{(1)}_{x,m}(M)$ and $\mathrm{lc}^{(1)}_x(M)$ those of the arity-one counterterm of M along x, the $\hbar\,\mathrm{Sing}_\varepsilon$ of the cycles through two letters of M with one vertex $S^{(1)}(x)$ on the cycle and no special edge (Definition 11.3 for $x\in\{\Delta,P_b\}$). The finite special vertex of the pair is

$$\mathcal{F}_x(X_a,X_b) := -\sum_{m\geq 1}\frac{(-1)^m}{m!}\Big(\big\{I,\mathrm{ct}^{(1)}_{x,m}(X_aX_b)\big\}_{H^m}+\big\{S^{(1)}(x),\mathrm{ct}_m(X_aX_b)\big\}_{H^m}\Big),$$

with $\{-,-\}_{H^m}$ the bracket of (10.6): the coefficient of ε^0 of the special-edge parts of the counterterm brackets $-\{I,\mathrm{ct}^{(1)}_x(X_aX_b)[\varepsilon]\}_\varepsilon$ and $-\{S^{(1)}(x),\partial_\delta I^{CT}[\varepsilon](X_aX_b)\}_\varepsilon$. It is cancelled in the components by the graphs with the special edge on a bridge (Theorem 11.44 (ii)); the two-body kernel is the coefficient of ε^0 of the defect cycles alone.

- (iv) The two-body kernel of (X_a,X_b) along x is

$$\Upsilon_x(X_a,X_b) := [\varepsilon^0]\,\mathcal{D}_x(X_a,X_b;\varepsilon,T),$$

with $[\varepsilon^0]$ the coefficient of ε^0 in the expansion of Theorem 11.45: a local density of the outputs at the point of the pair, with values in the tensor product of the fibres and colour spaces of the two letters.

- (v) In arity zero the special-edge cycles of the pair are the chains of Definition 10.10 with vertices of I, edges $P(\varepsilon,T)$ and exactly one special edge, and the arity-zero two-body kernel $\Upsilon_0(X_a,X_b)$ is the coefficient of ε^0 of their sum.

For $x\in\{\Delta,P_b\}$ the defect cycles with V_s on a line are the special-edge composite cycles of Definition 11.5.

Theorem 11.44 (Two-body form of the components along every generator)**.** *Let x be a generator and $O=\int\omega\,X_1\cdots X_n$ a monomial local functional, gauge-variant ones included. Let $J=I_{\mathfrak{psl}(4|4)}-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]-\hbar F_1$ with $\hbar F_1$ a one-loop quantisation through arity one along x, that is,* (8.6) *modulo $\hbar^2$ in arity zero and in arity one along x, equivalently the first line of* (9.12)*, and let $\psi=0$. The endpoint is $T=\infty$ along Σ, Σ^*, Δ, P_b and the rotations, and any $T\in(0,\infty)$ along K_b.*

- *(i) The δ-linear graphs of arity one along x and of order $\hbar$ of the flow of the defect vertex of Lemma 8.4, applied to $e^{(J+\delta O(\varepsilon))/\hbar}$, form the classes: (F) the one-loop graphs of the vertex $\delta\{S^{(1)}(x),O\}$ and of the vertex $\delta(Q+\{I,-\})O$ with a separate vertex $S^{(1)}(x)$; (C2) the defect cycles of the pairs of letters of O, with trees of I on the outputs; (C0) the cycles through two letters of O with the special edge or a contraction through K_ε, with $S^{(1)}(x)$ on a tree; (Br) the graphs with the special edge on a tree; (Ac) the graphs whose cycle passes through vertices of the action alone; (Ct) the trees with one counterterm vertex of order $\hbar$.*

- *(ii) (F) is the one-loop part of the arity-one flow of sO, and (C0) with the arity-zero counterterm vertices of (Ct) tends to the trees of $\hbar L_1(O)^{(0)}$ with one vertex $S^{(1)}(x)$, both on the left side of (8.7). A graph of (Br) separates at its special edge, of generic momentum p, a one-loop subgraph G; when G is divergent the bare graph has the finite limit $\sum_m \frac{(-1)^m}{m!}|p|^{2m}a_m(G)$, $a_m(G)$ the coefficients of ε^{-m} of G, and its counterterm partner, the special-edge part of the counterterm bracket across the same edge, has the opposite limit: the sum is $(e^{-\varepsilon|p|^2}-1)(G-\operatorname{Sing}G)$ and tends to 0. The partners are the special-edge parts of $-\{I,\operatorname{ct}_x^{(1)}(O)[\varepsilon]\}_\varepsilon$, $-\{S^{(1)}(x),\partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon$ and $-\{I^{CT}[\varepsilon],O\}_\varepsilon$, whose coefficients of ε^0 are the finite special vertices of Definition 11.43 (iii) and the vacuum-pole terms of (10.6); they are used up here. (Ac) with the defect vertices of the action tends to 0 by the master equation in arity zero and in arity one along x.*

- *(iii) (Master formula.) With the Koszul signs of (10.4),*

$$\hbar L_1(O)^{(1)}(x) = \sum_{1\le a<b\le n} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1\cdots X_{a-1}\,\Upsilon_x(X_a,X_b)\,X_{a+1}\cdots\widehat{X_b}\cdots X_n - \big\{\hbar F_1^{(1)}(x),O\big\}, \tag{11.4}$$

the two-body kernels $\Upsilon_x=[\varepsilon^0]\mathcal{D}_x$ of Definition 11.43 summed over the unordered pairs of letters, and a one-body term, the bracket with the arity-one finite counterterm along x. In arity zero, with the same signs,

$$\hbar L_1(O)^{(0)} = \sum_{a<b} \pm \int \omega\cdots\Upsilon_0(X_a,X_b)\cdots - \big\{\hbar F_1^{(0)},O\big\}. \tag{11.5}$$

Every number of a component is a coefficient of ε^0 of a defect cycle or a coefficient of the action-side solution $\hbar F_1$; these formulas are the definition of the components in the construction.

- *(iv) The coefficients of $\log\varepsilon$ and of ε^{-m} cancel. The defect cycles carry no $\log\varepsilon$ (Theorem 11.45), and the first cancellation is the logarithmic covariance*

$$\operatorname{lc}\big(\{S^{(1)}(x),O\}\big) - \big\{S^{(1)}(x),\operatorname{lc}(O)\big\} = s\operatorname{lc}_x^{(1)}(O) - \operatorname{lc}_x^{(1)}(sO) + \big\{L^{(1)}(x),O\big\}, \tag{11.6}$$

with $L^{(1)}(x)$ the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, which has no term ε^{-m} along Σ, Σ^*, Δ, P_b and K_b. The second is the pole identity, for $m\ge 1$,

$$\sum_{a<b} \pm[\varepsilon^{-m}]\,\mathcal{D}_x(X_a,X_b) = s\operatorname{ct}_{x,m}^{(1)}(O) + \big\{S^{(1)}(x),\operatorname{ct}_m(O)\big\} - \operatorname{ct}_m\big(\{S^{(1)}(x),O\}\big) - \operatorname{ct}_{x,m}^{(1)}(sO), \tag{11.7}$$

the special-edge parts of the counterterm brackets, with all their powers of ε, having been grouped with the bridges in (ii); for a single pair $O=\int\omega\,X_aX_b$ the right side contains the poles of the pairs of the cubic monomials of $\{S^{(1)}(x),O\}$ and of sO times their third letter.

- *(v) (11.4) holds along K_b at every $T\in(0,\infty)$ with the same right side. Given the master equation through arity one along x, the local representative of $L(O)$ exists through arity one along x for every local functional O: (iii) of Definition 10.36 holds along every generator.*

Proof. (i) At order $\hbar$ a graph of the flow has one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Proposition 3.35); the part of arity one along x and linear in δ places x on one vertex and δ on one vertex. Expand the defect vertex in δ, in the arity and in $\hbar$, with $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ and $O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$. The classical master equation (2.4) leaves at order $\hbar^0$ the vertices $\delta\{S^{(1)}(x), O\}$ and $\delta(Q + \{I, -\})O$ and the special-edge parts of $\{S^{(1)}(x), O\}_\varepsilon$, $\{I, O\}_\varepsilon$, $\{I, S^{(1)}(x)\}_\varepsilon$, $\frac{1}{2}\{I, I\}_\varepsilon$, together with $\delta\hbar\Delta_\varepsilon O$; $\hbar\Delta_\varepsilon S^{(1)}(x)$ vanishes, its quadratic terms pairing a field with the antifield of another field and its cubic terms carrying λ^* and no λ. A cycle through a vertex of the first two kinds is in (F). A cycle with a special edge passes through O along two letters (Theorem 10.4 (i)) or avoids O, and the positions of $S^{(1)}(x)$ and of the special edge sort it into (C2), (C0), (Br) or (Ac).

(ii) By Theorem 8.15 (iv) the left side of (8.7) in arity one along x is $\big(L(O)^{(1)}(x)\big)[T] + \big(L(O)^{(0)}\big)[T]^{(1)}(x)$ with $L(O) = sO + \hbar L_1(O)$; the one-loop part of $(sO)[T]^{(1)}(x)$ consists of the graphs (F) and their counterterm trees. For (C0), the arity-zero analysis of Theorem 10.4 and of the present argument at arity zero applies with a tree carrying $S^{(1)}(x)$ attached at generic momenta. (Br): the special edge on a tree edge of momentum p multiplies the subgraph G it separates by $e^{-\varepsilon|p|^2} - 1 = \sum_{j\geq 1}(-\varepsilon|p|^2)^j/j!$. For $G = \sum_{m\geq 1}\varepsilon^{-m}a_m + a_0 + o(1)$, a cycle through two letters of O with $S^{(1)}(x)$ on it or a cycle of the action such as the self-energy with the pole of I^{CT}, the terms $j = m$ give the finite limit stated. The counterterm vertex of G joined by the same special edge is $-(e^{-\varepsilon|p|^2} - 1)\sum_m \varepsilon^{-m}a_m$; its coefficient of ε^0 is $-\sum_m \frac{(-1)^m}{m!}|p|^{2m}a_m$, which in the bracket notation of (10.6) is $-\frac{(-1)^m}{m!}\{f, g_m\}_{H^m}$ with $(f, g_m) = (I, \mathrm{ct}^{(1)}_{x,m}(O))$, $(S^{(1)}(x), \mathrm{ct}_m(O))$ or (coefficient of ε^{-m} in $I^{CT}[\varepsilon], O)$. The sum of a bridge and its partner is $(e^{-\varepsilon|p|^2} - 1)$ times the renormalised subgraph, which has a limit (Definition 8.5), and tends to 0; the partners with $\log\varepsilon$ carry $(e^{-\varepsilon|p|^2} - 1)\log\varepsilon \to 0$ (Theorem 8.8 (ii), Proposition 4.27). Along K_b at finite T the subgraphs carry at fixed ε the term $\frac{3}{8\pi^2}\hbar\log T$ times a tree with the same factor (remark after Theorem 8.27), which also tends to 0. (Ac): the cycle and its counterterm vertices form the flow of the defect of the action in arity zero or in arity one along x, joined to O by a tree at generic momenta, whose limit is the tree-level flow of $\hbar\,\mathrm{Ob} - s(\hbar F_1)$ in these arities (Definition 8.7), zero by the master equation.

(iii) What remains is (C2) with the arity-one vertices of (Ct) without special edge, against the tree of $\hbar L_1(O)^{(1)}(x)$ and the counterterm trees of $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x), O\})$ and $-\mathrm{ct}^{(1)}_x(sO)$ on the left. The vertex $-\{\hbar F_1^{(1)}(x), O\}_\varepsilon$ gives the one-body term, its special-edge part tending to 0. Each defect cycle is a sum of poles, $[\varepsilon^0]\mathcal{D}_x$ and a vanishing remainder (Theorem 11.45); the trees on its outputs converge, and Theorem 8.15 (i) identifies the limit with the tree-level flow of (11.4). The sum over the pairs with the Koszul signs is that of Theorem 10.4 (iii), each defect cycle meeting O in exactly two letters. In arity zero the same argument with the special-edge cycles of the pairs, the vertex $-\{\hbar F_1^{(0)}, O\}_\varepsilon$ and the bridges through the self-energy cycles grouped with the special-edge parts of $-\{I^{CT}[\varepsilon], O\}_\varepsilon$ gives (11.5).

(iv) The left side has a limit; collecting the coefficients of $\log\varepsilon$ and of ε^{-m} of (C2), of the brackets without special edge and of the counterterm trees moved from the left side gives (11.6) and (11.7), the special-edge parts of the counterterm brackets cancelling against the bridges at every power of ε by (ii). A coefficient of ε^{-m} in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is a colour-singlet local functional of degree -1 and total Weyl weight $4 - \frac{1}{2}\lambda_x - 2m$, the position at weight -1 and of degree p at most that of x (Proposition 10.47); a colour singlet of degree -1 contains an antifield of weight at least $\frac{5}{2}$ and a second adjoint field of weight at least 1, or c^* and c, so its density has weight at least $\frac{7}{2}$ and the total weight is at least $\frac{7}{2} - p$. For $m \geq 1$ the total weight is at most $\frac{5}{2}$, $\frac{3}{2}$, 2, 3, 1 along Σ, Σ^*, Δ, P_b, K_b, below $\frac{7}{2}$, $\frac{5}{2}$, $\frac{5}{2}$, $\frac{7}{2}$, $\frac{3}{2}$. Read for one pair, the cubic terms of $\{S^{(1)}(x), O\}$ and of sO have pairs with the empty leg set of Theorem 10.4 (iv), whose poles stand on the right with the third letter.

(v) The argument uses sO through (F), which cancels between the two sides, and through the counterterm trees. Along K_b at a fixed finite T the edges $P(\varepsilon, T)$ are bounded uniformly in T and decay

as $e^{-|y|^2/4T}$ (Definition 8.3); Υ_{K_b} is independent of T by Theorem 11.45 (iii), and Theorem 8.27 gives the independence of the local representative. The classes (C2) and (Ct) have local limits, so Theorem 8.15 (i) gives the local representative. □

Theorem 11.45 (Defect cycles as simplex integrals)**.** *For a defect cycle with n vertices and the edges $e_0, \dots, e_n$, let q_j, the centre, the momentum form and the m-th simplex integrand F_m be those of Definition 10.21 with n in place of k, the numerator containing the function of the term of $S^{(1)}(x)$ at V_s (Table 11.6, Definition 7.4) and the special-edge factor $-|l+q_\sigma|^2$ times the fibre factor of K_0; along η and K_b a momentum r is injected at V_s and carried by $e_s, \dots, e_n$ to X_b and ω. Let D be the superficial degree of Proposition 11.47.*

- *(i) At generic output momenta and $T=\infty$, for $x \in \Sigma \cup \Sigma^* \cup \{\Delta, P_b\} \cup \mathfrak{so}(V)$, the value of the cycle is $\sum_{m\ge 1} \varepsilon^{-m} a_m + a_0 + o(1)$ with*

$$\begin{aligned} 16\pi^2 a_m &= \frac{1}{m} \int_{\alpha_\sigma \ smallest} \Big[\big(\min_{f\neq\sigma} \alpha_f \big)^m - \alpha_\sigma^m \Big] F_m \, d\alpha \qquad (m \ge 1), \\ 16\pi^2 a_0 &= \int_{\alpha_\sigma \ smallest} F_0 \, \log \frac{\min_{f\neq\sigma} \alpha_f}{\alpha_\sigma} \, d\alpha, \end{aligned} \tag{11.8}$$

  and no term in $\log\varepsilon$; a_m is a polynomial of degree $D-2m$ in the output momenta and in r, and $a_0 = 0$ for $D<0$.

- *(ii) Along $\eta \in \Sigma^*$, at the point x_O of the pair, $\Upsilon_\eta = \sum_\mu x_O^\mu \, \Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta$: the position part along b is the kernel along $\rho(b)\eta \in \Sigma$, and the translation-invariant part Υ^0_η is $-i\sum_\mu \partial_{r_\mu}$ at $r=0$ of (11.8) with $S^{(1)}(\rho(e_\mu)\eta)$ at V_s, plus (11.8) with s_0 at V_s.*

- *(iii) Along K_b, with $z = x - x_O$,*

$$v(x) = v(x_O) + \big(-2(b,x_O)z + \omega_{x_O} z \big) + v(z), \qquad \omega_{x_O} z := 2\big((x_O, z) b - (b,z) x_O \big),$$

  and correspondingly

$$\Upsilon_{K_b} = v(x_O)^\mu \, \Upsilon_{P_{e_\mu}} - (b, x_O) \, \Upsilon^0_\Delta + \Upsilon^0_{M_{\omega_{x_O}}} + \Upsilon^0_{K_b}, \tag{11.9}$$

  with Υ^0_y the translation-invariant part along the generator y centred at x_O, the rotation $M_{\omega_{x_O}}$ having the vector field $\omega_{x_O} z$; the monomials z^μ and $z^\mu z^\nu$ of the Killing data act as $-i\partial_{r_\mu}$ and $-\partial_{r_\mu}\partial_{r_\nu}$ at $r=0$. At every finite T the coefficients of the expansion in ε are given by (11.8), and $T\partial_T$ of every defect cycle is $O(\varepsilon)$: Υ_{K_b} is independent of T.

- *(iv) For a cycle with $n_t = n+1$ edges and a constant integrand F_0 ($D=0$), $16\pi^2 a_0 = F_0 \log \frac{n_t}{n_t-1} / (n_t - 1)!$, so that $16\pi^2 a_0 = \frac{1}{2} F_0 \log \frac{3}{2}$ for $n_t = 3$ and $\frac{1}{6} F_0 \log \frac{4}{3}$ for $n_t = 4$; for a constant F_1 ($D=2$), $16\pi^2 a_1 = F_1 / \big(n_t^2 (n_t-1)(n_t-1)! \big)$. For $n=1$, the vertex V_s at the special edge, with $\alpha := \alpha_\sigma \in [0, \frac{1}{2}]$,*

$$16\pi^2 a_0 = \int_0^{1/2} F_0(\alpha) \log \frac{1-\alpha}{\alpha} \, d\alpha, \qquad 16\pi^2 a_m = \frac{1}{m} \int_0^{1/2} \big((1-\alpha)^m - \alpha^m \big) F_m(\alpha) \, d\alpha,$$

  in $\mathbb{Q} + \mathbb{Q}\log 2$ for polynomial numerators; for $n=2$ with V_s on a line between a letter or a vertex of I and the special edge, Proposition 11.13 applies, with values in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$.

Proof. (i) The domain $\{t_\sigma \le \varepsilon,\ t_f \ge \varepsilon \ (f \neq \sigma)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma}\alpha_f$ and $1/\min_f \alpha_f$, which differ on the region where α_σ is the smallest coordinate; Proposition 10.23 gives for each shape the poles, the same coefficient of $\log\varepsilon$ and the finite part, and the difference

is (11.8), as in Lemma 10.24. The region $\tau \geq 1$ converges at $T = \infty$: the edges carry distinct generic momenta (proof of Theorem 10.4 (ii)), the composite line along ξ is of order $|p|^{-2}$ at small momentum (Lemma 11.42 (iii)), the composite kernels along Δ, P_b and $\mathfrak{so}(V)$ of order $|p|^{-3}$ (Proposition 9.18 (iv)), and along η the line carries the spinor $\rho(b)\eta$ in the position part and one derivative in r in the translation-invariant part; the measure of $\{\alpha_\sigma \leq \varepsilon/\tau\}$ is of order ε. The coefficient of τ^{-1-m} after the Gaussian integral is homogeneous of degree $D-2m$. (ii) By (11.3), insert $x^\mu = x_O^\mu + (x-x_O)^\mu$ and argue as in Proposition 11.10 (i): on plane waves the factor $(x-x_O)^\mu$ is $-i\partial/\partial r_\mu$ of the momentum injected at V_s and absorbed by ω, which commutes with the Gaussian integral and the simplex integration; the same holds for the special edges, the counterterm vertices and the finite special vertices. (iii) Expand $(x_O+z, x_O+z)b - 2(b, x_O+z)(x_O+z)$; $S^{(1)}$ of Definition 7.4 is linear in v with its weight and spin terms, and $\operatorname{div} v(x) = -8(b, x_O) - 8(b, z)$, $\operatorname{div}(2z) = 8$, $\operatorname{div}(\omega_{x_O} z) = 0$. At a finite T the region $\tau \geq \tau_0$ is compact in τ, so the expansion in ε is that of the region of small τ. $T\partial_T$ replaces one edge e by TL_T, of scalar factor $Te^{-T|l+q_e|^2}$, which confines the loop momentum, while the special edge carries $e^{-\varepsilon|l+q_\sigma|^2} - 1 = O\big(\varepsilon(1+|l|)^2\big)$; the polynomial dependence on the position enters through derivatives in r, which preserve these bounds, and the finite special vertices are built from the coefficients ct_m, $\mathrm{ct}^{(1)}_{x,m}$, independent of T (Proposition 8.6). (iv) For a constant integrand the integrals of (11.8) over the region where α_σ is the smallest coordinate are elementary; for $n = 1$ the simplex is $[0,1]$, the region is $\alpha_\sigma \leq \frac{1}{2}$ and $\min_{f\neq\sigma} \alpha_f = 1 - \alpha_\sigma$, and $\int_0^{1/2} \alpha^a (1-\alpha)^b \log\frac{1-\alpha}{\alpha}\, d\alpha \in \mathbb{Q} + \mathbb{Q}\log 2$. Proposition 11.13 uses that the two segments adjacent to V_s carry the same momentum at $r = 0$ and that the special edge is the third edge. □

Example 11.46 (The cubic term at the special edge)**.** The defect cycles with $n = 1$ and an output have a cubic term of $S^{(1)}(\xi)$ at the special edge. For $\xi \in S_+ \otimes W$ and the pair $(A|_{W^\perp_{\mathbb{R}}}, \lambda|_{S_+\otimes W})$, the cycle

$$A|_{W^\perp_{\mathbb{R}}} \xrightarrow{P} T_5\big(A|_{W^\perp_{\mathbb{R}}}, A|_{W^\perp_{\mathbb{R}}}, \lambda^*|_{S_+\otimes W^*}\big) \xrightarrow{K_\varepsilon - K_0} \lambda|_{S_+\otimes W},$$

with the output $A|_{W^\perp_{\mathbb{R}}}$ of momentum q, has by Theorem 11.45 (iv) and Lemma 10.24 the value

$$\frac{\hbar}{16\pi^2}\Big(-\frac{1}{2\varepsilon} - \frac{3}{8}|q|^2\Big)$$

times the value of T_5 on the fibre elements and colour factors of the letters and of the output, with the fibre factor $2(-,-)^{-1}$ of the edge $A \otimes A$ and the inverse fibre pairing of (λ, λ^*) on the special edge. Its contribution to $\Upsilon_\xi\big(A|_{W^\perp_{\mathbb{R}}}, \lambda|_{S_+\otimes W}\big)$ on the leg set $A|_{W^\perp_{\mathbb{R}}}$ is $\frac{3}{8}\frac{\hbar}{16\pi^2}$ times ∂^2 of the output times that factor, and its pole enters (11.7) against ct_1 of the pair $(\lambda|_{S_+\otimes W}, \lambda|_{S_+\otimes W})$ of the classical transformation $\{S^{(1)}(\xi), A|_{W^\perp_{\mathbb{R}}}\} = \Gamma(\xi,\lambda)|_{W^\perp_{\mathbb{R}}}$, whose coefficient of ε^{-1} has the leg set $A|_{W^\perp_{\mathbb{R}}}$ (Table 10.3).

Proposition 11.47 (Output bounds along Σ, Σ^* and K_b)**.** *A defect cycle of (X_a, X_b) along x whose outputs have total Weyl weight w, fields and derivatives, has the superficial degree*

$$D = w_{X_a} + w_{X_b} + \delta - w,$$

with $\delta = \frac{1}{2}$ along $\xi \in \Sigma$ and for the position part along η, $\delta = -\frac{1}{2}$ for Υ^0_η, and $\delta = 1, 0, -1$ for the parts of Υ_{K_b} multiplying $v(x_O)$, the first derivatives of v at x_O and its second derivatives. Hence:

- *(i) along $\xi \in \Sigma$: $\Upsilon_\xi(X_a, X_b)$ is translation invariant, of Weyl weight $w_{X_a} + w_{X_b} + \frac{1}{2}$, of the pair degree and of parity $|X_a| + |X_b| + 1$, with at most $w_{X_a} + w_{X_b} + \frac{1}{2}$ plus the pair degree output fields, as for the second-order operator on pairs of letters of [BK, (3.7)];*
- *(ii) along $\eta \in \Sigma^*$: $\Upsilon_\eta = \sum_\mu x^\mu\, \Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta$, of parity $|X_a| + |X_b| + 1$, with Υ^0_η translation invariant of Weyl weight $w_{X_a} + w_{X_b} - \frac{1}{2}$ and at most $w_{X_a} + w_{X_b} - \frac{1}{2}$ plus the pair degree output fields;*

- *(iii) along K_b: Υ_{K_b} is a polynomial of degree at most two in the position, of Weyl weight $w_{X_a} + w_{X_b} - 1$ with the position at weight -1, of the pair degree and of parity $|X_a| + |X_b|$, with at most $w_{X_a} + w_{X_b} + 1$ plus the pair degree output fields; $\Upsilon^0_{K_b}$ has Weyl weight $w_{X_a} + w_{X_b} - 1$.*

The same bounds hold for the counterterm cycles, whose outputs carry the pair degree minus one, and for the finite special vertices, which are supported on the leg sets with $D \geq 2$ and cancel against the bridges in the components (Theorem 11.44 (ii)). These are the bounds of Proposition 10.47 with $\lambda_x = -1, +1, +2$, valid without (iii) of Definition 10.36 by Theorem 11.44 (v).

Proof. Lemma 10.11 with one vertex of weight $4+\delta$: every vertex of I has weight 4, the term of $S^{(1)}(x)$ has weight $4 + \delta$ on its fields and derivatives once the Killing data carry the weights of Proposition 7.13 ($-\frac{1}{2}$ for a constant spinor, -1, 0, $+1$ for the value and the first and second derivatives of a vector field, the position at weight -1), every edge P has numerator degree $w_1 + w_2 - 2$, and the special edge, $-|l+q|^2$ over a Schwinger interval $[0, \varepsilon]$, has degree $0 = w + (4 - w) - 4$. The degree of the coefficient of ε^0 is Theorem 11.45 (i). $S^{(1)}(x)$ has degree -1 and total parity 0 for odd x and 1 for even x, the special edge degree $+1$ and parity 1. Weyl weight plus degree is at least 1 for every field and antifield, with equality for c and A (Proposition 10.12), which bounds the number of outputs. The position parts are Theorem 11.45 (ii), (iii). On a leg set of degree D the term of $\mathcal{F}_x$ with ct_m or $\mathrm{ct}^{(1)}_{x,m}$ contains a coefficient of ε^{-m}, of degree $D - 2m \geq 0$ in the momenta, and $m \geq 1$. □

Proposition 11.48 (Census of the defect cycles)**.**

- *(i) Every defect cycle conserves the degree and the centre $\mathbb{Z}/4$ of $SL(4,\mathbb{C})$, $W \mapsto iW$, once x carries its charge, $+1$ on $S_+ \otimes W$ and on $S_- \otimes W$, -1 on $S_- \otimes W^*$ and on $S_+ \otimes W^*$, 0 on the even part; it shifts the total parity by that of $S^{(1)}(x)$ plus one.*

- *(ii) Along every generator, defect cycles exist for 45 of the 55 unordered pairs of the ten slots; the other ten are the pairs of two letters among $A^*|_{W^\perp_{\mathbb{R}}}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, each of which needs the special edge.*

- *(iii) A defect cycle without vertices of I has V_s at the special edge, letter–P–V_s–special edge–letter; a cycle without outputs is homogeneous in ε of degree $-\frac{D}{2}$ (Lemma 10.6), so its coefficient of ε^0 is nonzero at $D = 0$ alone.*

- *(iv) Along $\xi \in S_+ \otimes W$ the leg sets through the pairs of field and ghost letters without derivatives are those of Table 11.6; the numbers of leg sets with $n \leq 3$ derivatives on the two letters, along $\xi \in S_+ \otimes W$ and for the translation-invariant parts along $\eta \in S_+ \otimes W^*$ and along K_b, are those of Table 11.6. Along $S_- \otimes W^*$ and $S_- \otimes W$ the tables hold with the chiralities exchanged, and for the position parts with δ of Proposition 11.47.*

- *(v) Along $\xi \in S_+ \otimes W$ the pair (c, c) has the empty set of leg sets at the pair weight without derivatives, and the pairs of c with $A|_V$, $A|_{W^\perp_{\mathbb{R}}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ have leg sets with one ghost output.*

Proof. (i) Each vertex of I, each term of $S^{(1)}(x)$ and each component of P and of K_0 carries these charges (Table 11.6, Definition 1.13, Proposition 3.17). (ii)–(v) Exhaustive enumeration of the chains with up to six vertices on the cycle, exactly one term of $S^{(1)}(x)$ in each of its placements and exactly one special edge, with output weights up to $w_{X_a} + w_{X_b} + 3 + \delta$; every chain is checked for degree, parity and the centre, and its superficial degree against the count of Lemma 10.11. A letter $A^*|_{W^\perp_{\mathbb{R}}}$, λ^* or c^* has no component of P and lies on the special edge, of which a defect cycle has one. □

pair	output weight at most	leg sets along $\xi \in S_+ \otimes W$: fewest vertices of I; number of chains
$(A\|_V, A\|_V)$	5/2	$\lambda\|_{S_-\otimes W^*}\ (1;2)$, $A\|_V\,\lambda\|_{S_-\otimes W^*}\ (1;8)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_+\otimes W}\ (1;7)$, $c\,\lambda^*\|_{S_+\otimes W^*}\ (1;4)$
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	5/2	$\lambda\|_{S_+\otimes W}\ (1;4)$, $A\|_V\,\lambda\|_{S_+\otimes W}\ (1;15)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_-\otimes W^*}\ (1;15)$, $c\,\lambda^*\|_{S_-\otimes W}\ (1;6)$
$(A\|_V, \lambda\|_{S_+\otimes W})$	3	$\emptyset\ (0;2)$, $A\|_V\ (0;6)$, $A\|_V^2\ (1;13)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\ (1;11)$, $A\|_V^3\ (1;25)$, $A\|_V\,(A\|_{W_{\mathbb{R}}^\perp})^2\ (1;55)$, $c\,A^*\|_V\ (2;9)$, $\lambda\|_{S_+\otimes W}\,\lambda\|_{S_-\otimes W^*}\ (2;15)$
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	3	$A\|_{W_{\mathbb{R}}^\perp}\ (0;5)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\ (1;19)$, $A\|_V^2\,A\|_{W_{\mathbb{R}}^\perp}\ (1;51)$, $(A\|_{W_{\mathbb{R}}^\perp})^3\ (1;19)$, $c\,A^*\|_{W_{\mathbb{R}}^\perp}\ (2;2)$, $(\lambda\|_{S_-\otimes W^*})^2\ (2;7)$, $(\lambda\|_{S_+\otimes W})^2\ (2;6)$
$(A\|_V, c)$	3/2	$\lambda\|_{S_-\otimes W^*}\,c\ (2;10)$
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	5/2	$\lambda\|_{S_-\otimes W^*}\ (1;2)$, $A\|_V\,\lambda\|_{S_-\otimes W^*}\ (1;7)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_+\otimes W}\ (1;8)$, $c\,\lambda^*\|_{S_+\otimes W^*}\ (1;2)$
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	3	$A\|_{W_{\mathbb{R}}^\perp}\ (0;6)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\ (1;22)$, $A\|_V^2\,A\|_{W_{\mathbb{R}}^\perp}\ (1;59)$, $(A\|_{W_{\mathbb{R}}^\perp})^3\ (1;21)$, $c\,A^*\|_{W_{\mathbb{R}}^\perp}\ (2;7)$, $(\lambda\|_{S_-\otimes W^*})^2\ (2;7)$, $(\lambda\|_{S_+\otimes W})^2\ (2;8)$
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	3	$\emptyset\ (0;2)$, $A\|_V\ (0;5)$, $A\|_V^2\ (1;10)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\ (1;11)$, $A\|_V^3\ (1;18)$, $A\|_V\,(A\|_{W_{\mathbb{R}}^\perp})^2\ (1;52)$, $\lambda\|_{S_+\otimes W}\,\lambda\|_{S_-\otimes W^*}\ (2;13)$
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	3/2	$\lambda\|_{S_+\otimes W}\,c\ (2;9)$
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	7/2	$\lambda\|_{S_+\otimes W}\ (1;3)$, $A\|_V\,\lambda\|_{S_+\otimes W}\ (1;11)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_-\otimes W^*}\ (1;9)$, $c\,\lambda^*\|_{S_-\otimes W}\ (2;7)$, $A\|_V^2\,\lambda\|_{S_+\otimes W}\ (2;27)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_-\otimes W^*}\ (2;42)$, $A\|_V\,c\,\lambda^*\|_{S_-\otimes W}\ (2;30)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\,\lambda\|_{S_+\otimes W}\ (2;21)$, $A\|_{W_{\mathbb{R}}^\perp}\,c\,\lambda^*\|_{S_+\otimes W^*}\ (3;9)$
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	7/2	$\lambda\|_{S_-\otimes W^*}\ (1;5)$, $A\|_V\,\lambda\|_{S_-\otimes W^*}\ (1;18)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_+\otimes W}\ (1;17)$, $c\,\lambda^*\|_{S_+\otimes W^*}\ (2;7)$, $A\|_V^2\,\lambda\|_{S_-\otimes W^*}\ (2;44)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_+\otimes W}\ (2;79)$, $A\|_V\,c\,\lambda^*\|_{S_+\otimes W^*}\ (2;30)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\,\lambda\|_{S_-\otimes W^*}\ (2;40)$, $A\|_{W_{\mathbb{R}}^\perp}\,c\,\lambda^*\|_{S_-\otimes W}\ (2;21)$
$(\lambda\|_{S_+\otimes W}, c)$	2	$c\ (1;5)$, $A\|_V\,c\ (1;17)$, $A\|_V^2\,c\ (2;39)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\,c\ (2;18)$
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	7/2	$\lambda\|_{S_+\otimes W}\ (1;2)$, $A\|_V\,\lambda\|_{S_+\otimes W}\ (1;7)$, $A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_-\otimes W^*}\ (1;8)$, $A\|_V^2\,\lambda\|_{S_+\otimes W}\ (2;17)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\,\lambda\|_{S_-\otimes W^*}\ (2;37)$, $(A\|_{W_{\mathbb{R}}^\perp})^2\,\lambda\|_{S_+\otimes W}\ (2;19)$, $A\|_{W_{\mathbb{R}}^\perp}\,c\,\lambda^*\|_{S_+\otimes W^*}\ (2;12)$
$(\lambda\|_{S_-\otimes W^*}, c)$	2	$A\|_{W_{\mathbb{R}}^\perp}\,c\ (1;10)$, $A\|_V\,A\|_{W_{\mathbb{R}}^\perp}\,c\ (2;44)$
(c, c)	1/2	–

The leg sets of the defect cycles along $\xi \in S_+ \otimes W$ through the pairs of field and ghost letters without derivatives, with the fewest vertices of I and the number of chains; along $\xi \in S_- \otimes W^*$ the chiralities of λ and λ^* are exchanged. A leg set enters Υ_ξ when its output weight is at most the bound of the second column, and the empty leg set $\emptyset$ enters the poles.

pair	ξ, $n=0$	1	2	3	Υ^0_η, $n=0$	1	2	3	$\Upsilon^0_{K_b}$, $n=0$	1	2	3
$(A\|_V, A\|_V)$	4	9	20	43	1	4	9	20	2	4	8	19
$(A\|_V, A\|_{W^\perp_{\mathbb{R}}})$	4	9	20	43	1	4	9	20	1	2	7	16
$(A\|_V, \lambda\|_{S_+\otimes W})$	8	19	35	68	2	7	16	33	1	4	9	20
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	7	16	33	61	4	8	19	35	1	4	9	20
$(A\|_V, c)$	1	4	9	20	0	1	4	9	1	2	4	8
$(A\|_V, A^*\|_V)$	8	22	49	97	3	8	22	49	1	6	15	35
$(A\|_V, A^*\|_{W^\perp_{\mathbb{R}}})$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_V, \lambda^*\|_{S_+\otimes W^*})$	3	11	26	62	1	6	15	35	1	3	8	21
$(A\|_V, \lambda^*\|_{S_-\otimes W})$	6	15	35	73	0	3	11	26	1	3	8	21
$(A\|_V, c^*)$	3	13	34	80	0	3	13	34	1	3	10	25
$(A\|_{W^\perp_{\mathbb{R}}}, A\|_{W^\perp_{\mathbb{R}}})$	4	9	19	39	1	4	9	19	2	4	7	14
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda\|_{S_+\otimes W})$	7	16	33	61	4	7	16	30	1	3	7	16
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda\|_{S_-\otimes W^*})$	7	16	30	60	2	7	16	33	1	3	7	16
$(A\|_{W^\perp_{\mathbb{R}}}, c)$	1	4	9	20	0	1	4	9	0	1	2	7
$(A\|_{W^\perp_{\mathbb{R}}}, A^*\|_V)$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_{W^\perp_{\mathbb{R}}}, A^*\|_{W^\perp_{\mathbb{R}}})$	7	18	40	81	3	7	18	40	0	1	5	16
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda^*\|_{S_+\otimes W^*})$	3	10	26	59	1	5	14	31	0	1	4	14
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda^*\|_{S_-\otimes W})$	5	14	31	70	0	3	10	26	0	1	4	14
$(A\|_{W^\perp_{\mathbb{R}}}, c^*)$	3	13	34	80	0	3	13	34	0	1	4	19
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	9	20	42	79	3	7	16	35	2	6	12	24
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	9	20	43	82	4	9	20	43	4	7	15	27
$(\lambda\|_{S_+\otimes W}, c)$	4	8	19	35	1	2	7	16	0	1	4	9
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	15	35	73	143	5	14	31	70	3	8	21	46
$(\lambda\|_{S_+\otimes W}, A^*\|_{W^\perp_{\mathbb{R}}})$	14	31	70	135	3	10	26	59	1	4	14	35
$(\lambda\|_{S_+\otimes W}, \lambda^*\|_{S_+\otimes W^*})$	8	21	47	88	3	8	21	47	0	2	7	19
$(\lambda\|_{S_+\otimes W}, \lambda^*\|_{S_-\otimes W})$	8	21	46	86	1	3	11	29	0	1	5	14
$(\lambda\|_{S_+\otimes W}, c^*)$	10	25	62	116	2	5	19	49	0	2	10	29
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	7	16	35	68	4	9	20	42	2	6	12	24
$(\lambda\|_{S_-\otimes W^*}, c)$	2	7	16	33	2	4	8	19	0	1	4	9
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	14	31	70	135	6	15	35	73	3	8	21	46
$(\lambda\|_{S_-\otimes W^*}, A^*\|_{W^\perp_{\mathbb{R}}})$	10	26	59	123	5	14	31	70	1	4	14	35
$(\lambda\|_{S_-\otimes W^*}, \lambda^*\|_{S_+\otimes W^*})$	3	11	29	60	3	8	21	46	0	1	5	14
$(\lambda\|_{S_-\otimes W^*}, \lambda^*\|_{S_-\otimes W})$	8	21	47	88	3	8	21	47	0	2	7	19
$(\lambda\|_{S_-\otimes W^*}, c^*)$	5	19	49	101	3	10	25	62	0	2	10	29
(c, c)	0	1	4	9	0	0	1	4	0	1	2	4
$(c, A^*\|_V)$	9	20	43	82	4	9	20	43	4	8	19	35
$(c, A^*\|_{W^\perp_{\mathbb{R}}})$	9	20	43	82	4	9	20	43	2	7	16	32
$(c, \lambda^*\|_{S_+\otimes W^*})$	6	14	30	53	4	8	19	35	1	4	9	20
$(c, \lambda^*\|_{S_-\otimes W})$	8	19	35	64	2	6	14	30	1	4	9	20
(c, c^*)	8	22	49	91	3	8	22	49	1	6	15	35
$(A^*\|_V, A^*\|_V)$	13	34	80	169	3	13	34	80	3	10	25	61
$(A^*\|_V, A^*\|_{W^\perp_{\mathbb{R}}})$	13	34	80	169	3	13	34	80	1	4	19	49
$(A^*\|_V, \lambda^*\|_{S_+\otimes W^*})$	5	19	49	101	3	10	25	62	0	2	10	29
$(A^*\|_V, \lambda^*\|_{S_-\otimes W})$	10	25	62	116	2	5	19	49	0	2	10	29
$(A^*\|_V, c^*)$	4	15	46	101	1	4	15	46	0	1	7	24

The number of leg sets of the defect cycles entering Υ_ξ, $\xi \in S_+ \otimes W$, the translation-invariant part Υ^0_η, $\eta \in S_+ \otimes W^*$, and the translation-invariant part $\Upsilon^0_{K_b}$, for n derivatives on the two letters and up to six vertices on the cycle, on the 45 pairs of slots with defect cycles; the other ten pairs, of two letters among $A^*|_{W^\perp_\mathbb{R}}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, are those of Proposition 11.48 (ii).

Proposition 11.49 (The quadratic free directions along Σ and Σ^*).

- *(i) The quadratic local functionals of degree -1, of Weyl weight $\frac{9}{2}$ in the fields and derivatives, linear in $\xi \in S_+ \otimes W$, translation invariant and covariant under* $\mathrm{Spin}(4) \times \mathrm{Spin}(6) \times \mathfrak{g}$*, are spanned by T_1, T_2, T_3, T_4 and $\int \big((d^*A|_V)\xi, \lambda^*\big)$; their Q-closed combinations are spanned by $S_2^V(\xi)$ and $S_2^\perp(\xi)$, and the Q-exact ones are* 0.

- *(ii) The arity-one equation along ξ, the arity-one part along ξ of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ (Theorem 8.8 (ii)), is*

$$(Q + \{I, -\})\,\hbar F_1^{(1)}(\xi) + \big\{S^{(1)}(\xi), \hbar F_1^{(0)}\big\} = \hbar\,\mathrm{Ob}^{(1)}(\xi). \tag{11.10}$$

  At fixed $\hbar F_1^{(0)}$ two solutions differ, in the quadratic sector, by θ_V^ξ units times $S_2^V(\xi)$ plus $\theta_\perp^\xi$ units times $S_2^\perp(\xi)$, and the cubic part of (11.10) *then contains $\theta_V^\xi\{I_3, S_2^V(\xi)\} + \theta_\perp^\xi\{I_3, S_2^\perp(\xi)\}$ units, I_3 the cubic part of I. For $\theta_V^\xi = \theta_\perp^\xi =: \theta^\xi$ the direction extends to θ^ξ units times $S^{(1)}(\xi)$, the arity-one value of G_B (Definition 8.9) and of the field redefinition by the R-rotation $S^{(1)}(R_\xi)$, $[R_\xi, \xi] = \xi$ (Theorem 8.23). The cubic part fixes $\theta_V^\xi - \theta_\perp^\xi$ when the class of $\{I_3, S_2^V(\xi)\}$ modulo the images under Q of the cubic local functionals of degree -1 is nonzero, as for the sector coefficients along Δ (Proposition 11.25 (iii)).*

- *(iii) The one-body kernel of a letter X along ξ is $-\{\hbar F_1^{(1)}(\xi), X\}$; the free directions add $-\theta_V^\xi$ units times the arrows of $S_2^V(\xi)$ and $-\theta_\perp^\xi$ units times those of $S_2^\perp(\xi)$, which are $\{S_2^V(\xi), A|_V\} = \Gamma(\xi,\lambda)|_V$ and $\{S_2^\perp(\xi), A|_{W^\perp_\mathbb{R}}\} = \Gamma(\xi,\lambda)|_{W^\perp_\mathbb{R}}$ on the gauge field and the scalars and $\frac{1}{2}\rho(dA|_V)\xi$ and $\frac{1}{2}\rho(dA|_{W^\perp_\mathbb{R}})\xi$ on λ (Lemma 11.42 (i)).*

- *(iv) Along Σ^*, with the position included at weight -1 and η at weight $\frac{1}{2}$, the Q-closed quadratic directions of degree -1 and Weyl weight* 4*, linear in η, are spanned by $\sum_\mu \int x^\mu S_2^V(\rho(e_\mu)\eta)$ and $\sum_\mu \int x^\mu S_2^\perp(\rho(e_\mu)\eta) + \int s_0$, whose sum is the quadratic part of $S^{(1)}(\eta)$.*

The same holds along $S_- \otimes W^*$ with the chiralities exchanged.

Proof. (i) The field contents of degree -1 and field weight $\frac{9}{2}$ linear in a constant spinor are (λ, A^*) without derivatives and (A, λ^*) with one derivative, since c and c^* would need a further weight $\frac{1}{2}$; the covariant pairings are $\Gamma(\xi,\lambda)$ with A^* in its parts along V and $W^\perp_\mathbb{R}$, and $\rho(\partial A)\xi$ with λ^*, ∂A being $dA|_V$, $d^*A|_V$ or the mixed part $dA|_{W^\perp_\mathbb{R}}$, the volume element acting by a sign on each chirality. On functionals Q substitutes $A \mapsto -dc$, $A^* \mapsto -\frac{1}{2}d^*dA$, $\lambda^* \mapsto \not{d}\lambda$, $c^* \mapsto d^*(A^*|_V)$ and annihilates c and λ (Proposition 1.11). $Q\int((d^*A|_V)\xi, \lambda^*)$ contains $-\int((d^*dc)\xi, \lambda^*)$, produced by no other structure, so its coefficient vanishes in a closed combination; Q preserves the split into the parts along V and along $W^\perp_\mathbb{R}$, and $S_2^V(\xi)$, $S_2^\perp(\xi)$ are closed, being the two parts of the quadratic part of $\{S, S^{(1)}(\xi)\} = 0$. A Q-exact functional is Q of a functional of degree -2 and field weight $\frac{9}{2}$, and two antifields have weight at least 5, while c^* needs a further field of weight $\frac{1}{2}$. Both evaluations of Remark 11.66 give the same ranks on plane-wave normal forms. (ii) Theorem 8.8 (iii) restricted to arity one along ξ at fixed arity-zero part. The cubic part of $s\big(\theta_V^\xi S_2^V + \theta_\perp^\xi S_2^\perp\big)$ is the bracket with I_3, and $\{I_3, S_2^V(\xi) + S_2^\perp(\xi)\} = -QS_3(\xi)$ with $S_3(\xi)$ the cubic part of $S^{(1)}(\xi)$, the cubic part of $\{S, S^{(1)}(\xi)\} = 0$; $G_B^{(1)}(\xi) = S^{(1)}(B\xi) = S^{(1)}(\xi)$ for $\xi \in S_+ \otimes W$. (iii) Theorem 11.44 (iii) and Lemma 11.42 (i). (iv) The same enumeration with $\int x^\mu$ times the structures of (i) at $\rho(e_\mu)\eta$, $\int s_0$ and $\int(\rho(A|_V)\eta, \lambda^*)$, and the exact ranks of Q on plane-wave normal forms; s_0 enters the closed direction together with the part along $W^\perp_\mathbb{R}$. □

Proposition 11.50 (Two-leg obstructions and solutions along the odd generators, the rotations and K_b)**.** *Let $\hbar F_1^{(0)}$ contain the finite counterterm $\frac{27}{16}$ units times $\int(d^*A|_V, d^*A|_V)$ of Theorem 4.32, write $L = \log\frac{3}{2}$, and, for $\xi \in S_+ \otimes W$, $\Gamma_V := \Gamma(\xi,\lambda)|_V$, $\Gamma_\perp := \Gamma(\xi,\lambda)|_{W_\mathbb{R}^\perp}$,*

$$g_1 = \int(\Gamma_V, d^*d\,A|_V), \qquad g_2 = \int(\Gamma_V, dd^*A|_V), \qquad g_3 = \int(\Gamma_\perp, d^*d\,A|_{W_\mathbb{R}^\perp}), \qquad g_4 = \int\big(d^*dc, (\xi,\lambda^*)\big), \qquad T_0 = \int$$

The obstructions are the sums of the coefficients of ε^0 of the defect cycles with two legs along x, as in Proposition 9.25 (ii), with the placements of Lemma 11.42.

- *(i) Along ξ, for both chiralities: $QT_1 = -QT_3 = -\frac{1}{2}g_1$, $QT_2 = -QT_4 = -\frac{1}{2}g_3$, $QT_0 = -(g_2+g_4)$; the cohomology of Q on the quadratic functionals of degree 0, Weyl weight $\frac{9}{2}$, linear in ξ and covariant is zero, and*

$$\hbar\,\mathrm{Ob}^{(1)}(\xi) = \textit{units}\cdot\Big[\big(\tfrac{79}{24} - \tfrac{1}{2}L\big)(g_1+g_3) + \tfrac{27}{8}\,g_2 + 0\cdot g_4\Big].$$

  The arity-one equation (11.10) *is solvable on two legs, with the solutions*

$$\hbar F_1^{(1)}(\xi) = \big(L - \tfrac{79}{12}\big)\,\textit{units}\cdot(T_1+T_2) + \theta_V^\xi\,\textit{units}\cdot S_2^V(\xi) + \theta_\perp^\xi\,\textit{units}\cdot S_2^\perp(\xi),$$

  the part $\frac{27}{8}g_2$ being the bracket $\{\hbar F_1^{(0)}, S^{(1)}(\xi)\}$; the field-counting directions of $\hbar F_1^{(0)}$ (Theorem 9.11) shift this by brackets with $S^{(1)}(\xi)$ (Proposition 11.52 (i)). On two legs the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\xi)$ is $\{\mathcal{N}, S^{(1)}(\xi)\}$ units, $\mathcal{N} = 2\int(A,A^*) - 4\int(\lambda^*,\lambda) - \int(c^*,c)$ the field-counting functional: $2(T_1+T_2-T_3-T_4)$ units.

- *(ii) Along $\eta \in \Sigma^*$, for both chiralities, with position degree at most one, writing $u_0 := \int(\rho(A|_V)\eta, \lambda^*)$ and s_0 of* (11.3)*,*

$$\hbar\,\mathrm{Ob}^{(1)}(\eta) = \sum_\mu x^\mu\,\hbar\,\mathrm{Ob}^{(1)}(\rho(e_\mu)\eta) + \textit{units}\cdot\Big[-\tfrac{5}{2}\int(\eta,\lambda)\,d^*A|_V - \tfrac{5}{2}\int(\eta,\rho(dA|_V)\lambda) + \big(3L-\tfrac{3}{2}\big)\int(\eta,\rho(dA|_{W_\mathbb{R}^\perp})\lambda) +$$

  the cohomology of Q at this weight is one-dimensional and the residual has no component along it, and the solutions are

$$\hbar F_1^{(1)}(\eta) = \sum_\mu x^\mu\big(L-\tfrac{79}{12}\big)\,\textit{units}\cdot(T_1+T_2)(\rho(e_\mu)\eta) - \big(3L-\tfrac{3}{2}\big)\,\textit{units}\cdot s_0 - \tfrac{5}{2}\,\textit{units}\cdot u_0 + \theta_V^\eta\,\textit{units}\cdot S^V(\eta) + \theta_\perp^\eta\,\textit{units}\cdot S^\perp(\eta)$$

  with $S^V(\eta) = \sum_\mu\int x^\mu S_2^V(\rho(e_\mu)\eta)$ and $S^\perp(\eta) = \sum_\mu\int x^\mu S_2^\perp(\rho(e_\mu)\eta) + \int s_0$ (Proposition 11.49 (iv)).

- *(iii) Along $M_\omega \in \mathfrak{so}(V)$, with $\rho_\mu = \rho(e_\mu)$ and $S^{(1)}(M_\omega)_{\mathrm{spin}} := \int\omega_{\mu\nu}(A_\mu, A_\nu^*) + \int\big(\frac{1}{4}\omega_{\mu\nu}\rho_\mu\rho_\nu\lambda,\lambda^*\big)$: $\hbar\,\mathrm{Ob}^{(1)}(M_\omega) = \frac{1}{3}$ units times $Q\,S^{(1)}(M_\omega)_{\mathrm{spin}}$, 0 on $(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$, and $\hbar F_1^{(1)}(M_\omega) = \frac{1}{3}$ units times $S^{(1)}(M_\omega)_{\mathrm{spin}}$ plus the three sector parts of $S^{(1)}(M_\omega)$ with coefficients $\theta_\perp^M, \theta_\lambda^M, \theta_V^M$ units.*

- *(iv) Along $R \in \mathfrak{so}(W_\mathbb{R}^\perp)$, $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(R) = 0$ and $\mathrm{Ob}^{(1)}(R) = 0$: R. commutes with Q, Q^{GF} and H, preserves the pairing, annihilates I, $I^{CT}[\varepsilon]$ and $\hbar F_1^{(0)}$, and $S^{(1)}(R)$ forms no line; the solutions are $\theta_\perp^R$, θ_λ^R units times the sector parts of $S^{(1)}(R)$.*

- *(v) Along K_b, at a finite endpoint T and with position degree at most two,*

$$\hbar F_1^{(1)}(K_b) = \textit{units}\int(b,x)\Big\{-\tfrac{22}{3}(A,A^*) + 16(\lambda,\lambda^*) + \tfrac{8}{3}(c,c^*)\Big\} + \tfrac{2}{3}\,\textit{units}\int\big\{(\iota_bA, \iota_xA^*) - (\iota_xA, \iota_bA^*)\big\}$$
$$- \tfrac{1}{3}\,\textit{units}\int b_\mu x_\nu\big(\lambda, \rho_\mu\rho_\nu\lambda^*\big) + \textit{units}\cdot\sum_d\theta_d^K\,S^{(1)}(K_b)_d + \sum_{j=1}^4\kappa_j^K\,G_j^K,$$

with the sector parts $S^{(1)}(K_b)_d$ and four Q-closed quadratic functionals $G_1^K,\dots,G_4^K$ of degree -1, the second-order symmetries of the free fields built from the conformal Killing tensors, which are neither exact nor of the form $S^{(1)}(z)$ on two legs; the cohomology of Q is four-dimensional in degree 0 and seven-dimensional in degree -1, and the residual has no component in degree 0. The part linear in x is $-(b,x)$ times the translation-invariant solution along Δ plus the spin term of the rotation part of the derivative of the vector field, $16-\frac{1}{3}=\frac{47}{3}$ on λ.

Proof. (i) The census of Proposition 11.49 (i) and its degree-0 analogue: $\xi\otimes\lambda|_{S_-\otimes W^*}$ contains the $SU(4)$-singlet in $V\otimes\mathbf{1}$ alone, so every bilinear in $(\xi,\lambda|_{S_-\otimes W^*})$ is a component of Γ_V, paired with $A|_V$ and two derivatives in d^*d and dd^*; $\xi\otimes\lambda|_{S_+\otimes W}$ pairs with $A|_{W_{\mathbb{R}}^\perp}$ through $\Gamma_\perp$; (ξ,λ^*) pairs with c and two derivatives. Q substitutes $A^*\mapsto-\frac{1}{2}d^*dA$, $\lambda^*\mapsto\not{d}\lambda$, $A|_V\mapsto-dc$ (Proposition 1.11), with $(\xi,\not{d}\lambda)=-d^*\Gamma_V$; $Qg_1=Qg_3=0$ and Qg_2, Qg_4 are proportional and nonzero, so the closed functionals of degree 0 are spanned by g_1, g_3, g_2+g_4, which is the image of Q. The obstruction is the sum of the coefficients of ε^0 of the defect cycles with two legs, evaluated in two independent ways; its components along g_2, g_4 are fixed by the Wess–Zumino identity of Lemma 9.24, $Q\,\mathrm{Ob}^{(1)}(\xi)=-\{\mathrm{Ob}^{(0)},S^{(1)}(\xi)\}$ with the two-leg arity-zero obstruction $-\frac{27}{8}$ units times $\int(d^*dc,d^*A|_V)$ (Theorem 4.30). The bracket $\{\int(d^*A|_V,d^*A|_V),T_1\}$ is $2g_2$ up to the sign fixed by the Wess–Zumino identity, so the residual $\mathrm{Ob}^{(1)}(\xi)-\{F_1^{(0)},S^{(1)}(\xi)\}$ is $(\frac{79}{24}-\frac{1}{2}L)(g_1+g_3)$ units, Q of the stated particular solution; the free directions are Proposition 11.49 (i). The coefficient of $\log\varepsilon$ is the singular part of the composite and cubic cycles along ξ; its difference with $\{\mathcal{N},S^{(1)}(\xi)\}$ units is Q-closed on two legs, the coefficient of $\log\varepsilon$ of the counterterm cochain being s-closed and its arity-zero part Q of $\mathcal{N}$ units (Theorem 4.22, Lemma 9.24), hence a combination of $S_2^V(\xi)$ and $S_2^\perp(\xi)$, whose two coefficients are computed to vanish; and $\{\mathcal{N},-\}$ multiplies each arrow $f\to f'$ of $S^{(1)}(\xi)$ by the difference of the weights 2, 4, -2, -4 of A, λ, A^*, λ^*. (ii) The position part along b of every defect cycle along η is the cycle along $\rho(b)\eta$ (Theorem 11.45 (ii)); Q is block triangular for the split into position parts and translation-invariant parts, and the closed directions are those of Proposition 11.49 (iv); the intrinsic residual is computed on a complete basis of the structures $(\eta,\lambda)\,d^*A|_V$, $(\eta,\rho(dA|_V)\lambda)$, $(\eta,\rho(dA|_{W_{\mathbb{R}}^\perp})\lambda)$, $(\rho(dc)\eta,\lambda^*)$ and their images, in both evaluations. (iii) M_ω. commutes with Q, Q^{GF} and H; in the leg sets $(A|_V,A|_V)$, $(c,A^*|_V)$ and (λ,λ) the closed functionals of degree 0 linear in ω are spanned by $Q\int\omega_{\mu\nu}(A_\mu,A_\nu^*)$ and $Q\int(\frac{1}{4}\omega_{\mu\nu}\rho_\mu\rho_\nu\lambda,\lambda^*)$, the structure $\int(\lambda,\omega_{\mu\nu}\rho_{[\mu}\rho_\nu\rho_{\rho]}\partial_\rho\lambda)$ being a total derivative, so the cohomology vanishes there, and the defect cycles give the multiple $\frac{1}{3}$ in both sectors. (iv) The graphs with one vertex $S^{(1)}(R)$ are $R.$ applied to the legs of the flow, and the singular part of the one-loop part is removed by $I^{CT}[\varepsilon]$; $[Q,D_R]$ is the derivation by $(e^{-\varepsilon H}-e^{-TH})R.$, whose action tends to $R.$ of the renormalised flow, zero on the R-invariant action. (v) The sector bound of Proposition 10.47 with $\lambda_{K_b}=2$ and position degree two gives the leg sets $(c,A^*|_V)$, $(A|_V,A|_V)$, $(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$ and (λ,λ); the cohomology is computed on complete bases in both evaluations, and the defect cycles are evaluated at a finite T, their coefficients of ε^0 being independent of T (Theorem 11.45 (iii)). The part linear in x is the position decomposition (11.9) applied to the solution. □

Theorem 11.51 (The arity-two equations on two legs and the uniform coefficients)**.** *With the solutions of Theorem 9.26 (iii), Theorem 9.35 (i) and Proposition 11.50, write θ_d^Δ for $\theta,\theta_\lambda,\theta_V$, θ_d^P for $\theta_P,\theta_{P,\lambda},\theta_{P,V}$, and $\theta_d^{\xi\pm}$, $\theta_d^{\eta\pm}$ for the coefficients of Proposition 11.50 along $\xi_\pm\in S_+\otimes W$, $S_-\otimes W^*$ and $\eta_+\in S_+\otimes W^*$, $\eta_-\in S_-\otimes W$, with $d\in\{V,\perp\}$ along the odd generators and $L=\log\frac{3}{2}$. The classes of the arity-two residuals of Lemma 9.24 on two legs, the obstructions being the sums of the coefficients of ε^0 of the cycles with one special edge and two vertices $S^{(1)}(x)$, $S^{(1)}(y)$ or one vertex $S^{(2)}(x,y)$, vanish exactly under the conditions of Table 11.51.*

pair	*condition*
$(\Delta,P_b),\ (\Delta,\xi_\pm)$	$\theta^\Delta_\perp=\theta^\Delta_\lambda=\theta^\Delta_V=-\frac{1}{3}$
(ξ_+,ξ_-)	$\theta^P_d=\theta^{\xi_+}_V+\theta^{\xi_-}_V-\frac{27}{4}-2L$ *for every* d, $\theta^{\xi_+}_V+\theta^{\xi_-}_V=\theta^{\xi_+}_\perp+\theta^{\xi_-}_\perp$
$(\xi_\pm,\xi'_\pm)$	$\theta^{\xi_\pm}_V=\theta^{\xi_\pm}_\perp$
$(\xi_\pm,P_b)$	$\theta^P_V=\theta^P_\perp=\theta^P_\lambda$
(ξ_+,η_-)	$\theta^{\xi_+}_V-\theta^{\xi_+}_\perp=\theta^{\eta_-}_V-\theta^{\eta_-}_\perp$
(ξ_+,η_+)	$\theta^R_\perp=\theta^R_\lambda=:\tau$, $\theta^\Delta_d=\theta^M_d=\tau-\frac{1}{3}$, $\theta^{\xi_+}_d+\theta^{\eta_+}_d=\tau+\frac{77}{12}+2L$ $(d=V,\perp)$

The conditions under which the classes of the two-leg arity-two residuals vanish, $L=\log\frac{3}{2}$.

On $(\xi_\pm,P_b)$ *the obstruction is* $\frac{1}{4}\log\frac{4}{3}$ *times a* Q*-exact functional, and the arity-two generating cochain of two odd generators is* $I+cS^{(1)}(\xi)+c'S^{(1)}(\xi')-cc'S^{(2)}(\xi,\xi')$ *for odd parameters* c, c'. *Consequently:*

- *(i) every sector coefficient along* P_b, Σ, Σ^*, $\mathfrak{so}(V)$ *and* $\mathfrak{so}(W^\perp_{\mathbb{R}})$ *is uniform,* $\theta^z_d=:t_z$, *and*

$$t_\Delta=-\tfrac{1}{3},\qquad t_M=t_\Delta,\qquad t_R=t_\Delta+\tfrac{1}{3}=0,\qquad t_{\xi_+}+t_{\xi_-}-t_P=\tfrac{27}{4}+2\log\tfrac{3}{2},\qquad t_{\xi_+}+t_{\eta_+}-t_\Delta=\tfrac{27}{4}+2\log\tfrac{3}{2};\tag{11.11}$$

- *(ii) the solutions form an affine space over the inner direction* $t_z\mapsto t_z+c\lambda_z$, $\lambda_z=-1,+1,-2,+2,0$ *on* Σ, Σ^*, P_b, K_b *and the even rotations and* Δ, *generated by the field redefinition* $\phi^{(0)}=c\,\text{units}\cdot S^{(1)}(\Delta)$, *and the outer direction* $t_{\xi_\pm}\mapsto t_{\xi_\pm}\pm a$, $t_{\eta_\pm}\mapsto t_{\eta_\pm}\mp a$, *the direction* G_B *of Definition 8.9; the relations* (11.11) *and the sector uniformity are invariant under both, and they are the scheme-independent content of the computed pairs on the action side;*

- *(iii) in the translation-covariant scheme of Definition 11.23,* $t_P=t_\Delta=-\frac{1}{3}$, *hence* $t_{\xi_+}+t_{\xi_-}=\frac{77}{12}+2\log\frac{3}{2}$ *and* $t_{\eta_+}=t_{\xi_-}$*;*

- *(iv) the individual* $t_{\xi_\pm}$, $t_{\eta_\pm}$, t_P, *the coefficients* θ^K_d, κ^K_j, κ_V, κ_λ *and every particular part with* $\log\frac{3}{2}$ *are scheme artefacts.*

The relations (11.11) *are the one-loop parts of* $[\Delta,P_b]=-2P_b$, $[\Delta,\xi]=-\xi$, $[\xi,P_b]=0$, $[\xi_+,\xi_-]=P_{\Gamma(\xi_+,\xi_-)|_V}$ *and of the bracket* $[\xi,\eta]$ *with its components along* Δ, $\mathfrak{so}(V)$ *and* $\mathfrak{so}(W^\perp_{\mathbb{R}})$ *(Proposition 2.4): with the corrected generators* $S^{(1)}(z)+\hbar t_z\,\text{units}\cdot S^{(1)}(z)$ *the arity-two equation along* (x,y) *states that the order-*$\hbar$ *part of* $(-1)^{|x|}\{S^{(1)}(x),S^{(1)}(y)\}+S^{(1)}([x,y])$ *closes up to* $\Lambda^{(0)}$*-exact terms once the computed class of the obstruction is included.*

Proof. In (9.12) the free directions enter through $F^{(1)}_1([x,y])$ and the brackets $\{F^{(1)}_1(x),S^{(1)}(y)\}$, $\{S^{(1)}(x),F^{(1)}_1(y)\}$; for uniform coefficients, $(-1)^{|x|}\{S^{(1)}(x),S^{(1)}(y)\}=-S^{(1)}([x,y])-\Lambda^{(0)}S^{(2)}(x,y)$ (Lemma 2.11) gives the change $(t_x+t_y-t_{[x,y]})$ units times $S^{(1)}([x,y])$ modulo exact terms, $t_{[x,y]}S^{(1)}([x,y])$ meaning the sum over the components of $[x,y]$ in the blocks. On two legs the sector parts enter separately through the brackets $\{S^V_2(\xi),S_2(\xi')\}$, $\{S_2(\xi),S^V_2(\xi')\}$ and their analogues, whose classes in the cohomology of Q are combinations of the classes of the sector parts of $S^{(1)}([x,y])$ by the Fierz identities of the arrows $\lambda\to A|_V\to\lambda$ and $\lambda\to A|_{W^\perp_{\mathbb{R}}}\to\lambda$. The obstructions are the coefficients of ε^0 of the cycles of the statement, evaluated in two independent ways, and their classes are computed on complete bases of the two-leg functionals with the quantum numbers of the pair, the generating cochain of two odd generators having the sign forced by the classical master equation, $QS^{(2)}(\xi,\xi')+S^{(1)}(P_{\Gamma(\xi,\xi')|_V})-\{S^{(1)}(\xi),S^{(1)}(\xi')\}=0$ on two legs; the vanishing of the residual classes gives the table, the rows along Δ being Theorem 9.28. (i) $(\xi_\pm,\xi'_\pm)$ and $(\xi_\pm,P_b)$ give the uniformity along Σ and P_b, then (ξ_+,η_-) the uniformity along Σ^*, and (ξ_+,η_+) with $\theta^\Delta_d=-\frac{1}{3}$ gives $\tau=0$, $t_M=t_\Delta$, $t_R=0$ and $t_{\xi_+}+t_{\eta_+}=\frac{77}{12}+2L=t_\Delta+\frac{27}{4}+2L$; (ξ_+,ξ_-) gives the fourth relation. (ii) The field redefinition generated by $\phi^{(0)}$ changes $F^{(1)}_1(x)$ by $\{\phi^{(0)},S^{(1)}(x)\}=c\lambda_x$ units times $S^{(1)}(x)$ modulo exact terms (Lemma 2.11 with $S^{(2)}(\Delta,x)=0$), and G_B is an s-closed cochain whose arity-one value along $\xi_\pm$ is $\pm S^{(1)}(\xi_\pm)$ and along $\eta_\pm$ is $\mp S^{(1)}(\eta_\pm)$ (Theorem 8.8 (iv));

both preserve the solvability of every equation, λ and B being derivations of $\mathfrak{psl}(4|4)$. The left sides of (11.11) are invariant: the inner direction shifts $t_{\xi_\pm}$, $t_{\eta_\pm}$, t_P by $-c$, $+c$, $-2c$ and leaves t_Δ, t_M, t_R fixed, and the outer direction shifts $t_{\xi_\pm}$ by $\pm a$ and $t_{\eta_\pm}$ by $\mp a$. (iii) The position part of $\hbar F_1^{(1)}(\Delta)$ along b is $2\hbar F_1^{(1)}(P_b)$, so $t_P = t_\Delta$, and (11.11). (iv) Under the inner and outer directions the individual coefficients move. □

Proposition 11.52 (Renormalisation freedom of Q_1, S_1 and K_1)**.** *Under the changes of Proposition 8.19 the components change as follows; $\{S^{(1)}(x), -\}$ is the classical transformation along x.*

- *(i) A field redefinition $F_{\mathfrak{psl}(4|4)} \mapsto F_{\mathfrak{psl}(4|4)} + \hbar s\phi$ changes the component along x by $-\{(s\phi)^{(1)}(x), -\}$, a derivation of the letters: it changes the one-body kernels and leaves the two-body kernels unchanged. For the field countings $\phi^{(0)} = \sum_f y_f \int (f, f^*)$ the change of Q_1 multiplies each arrow $f \to f'$ of the classical transformation by $y_{f'} - y_f$; for $\phi^{(0)} = t\, S^{(1)}(\Delta)$ it is a multiple of the classical transformation proportional to λ_x.*

- *(ii) The coupling direction leaves Q_1, S_1, K_1 unchanged. The outer direction G_B changes Q_1 along $S_+ \otimes W$ by $-a$ and along $S_- \otimes W^*$ by $+a$ times the classical transformation, S_1 along $S_+ \otimes W^*$ by $+a$ and along $S_- \otimes W$ by $-a$ times it, and leaves K_1, D_1, P_1, M_1, R_1 unchanged.*

- *(iii) The free directions of $\hbar F_1^{(1)}(x)$ change the one-body kernels: along Σ by Proposition 11.49 (iii); along Δ and P_b through the sector coefficients of Definition 11.23, which enter D_1, P_1 and, through P_1, the relations of Proposition 11.59 and Corollary 11.60; along K_b through the s-closed directions of degree -1 quadratic in the Killing data. $\hbar F_1^{(0)}$ enters the one-body kernels through* (11.10) *and through the trees with $S^{(1)}(x)$ (Remark 10.48 (a), (b)).*

- *(iv) A finite part ψ' of the operators changes the component along x by $[\{S^{(1)}(x), -\}, \psi']$, on O by $\{S^{(1)}(x), \psi'(O)\} - \psi'(\{S^{(1)}(x), O\})$. For $\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$ this is $\frac{a}{2}\lambda_x\{S^{(1)}(x), -\}$: $+a$, $-\frac{a}{2}$, $+\frac{a}{2}$ times the classical transformation along K_b, Σ, Σ^* (Proposition 9.36), a change of the one-body kernels. A finite part ψ' that vanishes on O changes the component of O along x by $-\psi'(\{S^{(1)}(x), O\})$: the finite parts of the operators of the Weyl weight of $\{S^{(1)}(x), O\}$ enter the two-body kernels, as the finite part of Lemma 11.53 on the trace bilinear and the shift of the Konishi entry in Remark 11.63.*

- *(v) The cutoff shape enters the finite parts of the defect cycles (Proposition 10.23, Lemma 10.24), and of the special-edge cycles of $L_1^{(0)}$; since the counterterms are unique up to finite local terms [C07, Thm B], a change of the shape is a change of $\hbar F_1$ and of ψ, a combination of (i)–(iv).*

- *(vi) At the values $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ of Theorem 9.28, in the translation-covariant scheme, $P_1 \cong 0$ on the families of Definition 11.36 (Corollary 11.39), and the condition $P_1 = 0$ of Definition 8.35 holds there with $\psi' = 0$: the components along K_b, Σ and Σ^* are those of the scheme. For equal coefficients $\theta_P \neq \theta$ along P_b, the finite part with $a = (\theta - \theta_P)$ units gives $P_1 \cong 0$ and changes the components along K_b, Σ, Σ^* by a, $-\frac{a}{2}$, $\frac{a}{2}$ times the classical transformation (Corollary 11.28 (iv)).*

- *(vii) Under (i), (iv) and the coupling direction the representation of $\mathfrak{psl}(4|4)$ on the cohomology of $\Lambda^{(0)}$ changes by an isomorphism, under G_B by the automorphism $e^{-a\hbar B}$, and the spectrum of D_1 in a fixed classical Weyl weight is invariant (Theorem 8.20); the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ, its kernel, image and rank are invariant under (i), (ii) and (iv) (Theorem 8.23).*

Proof. (i), (ii) Proposition 8.19 (a), (c); the bracket with a local functional is a derivation in its second argument, so it acts letter by letter; for $\phi^{(0)} = t\, S^{(1)}(\Delta)$, $\{S^{(1)}(x), S^{(1)}(\Delta)\}$ is $S^{(1)}$ of $\pm[x, \Delta] = \mp\lambda_x x$

by Lemma 2.11, with $S^{(2)}(x,\Delta)=0$ (Definition 2.15). (iii) Proposition 11.49, Definition 11.23, Remark 10.48. (iv) Proposition 8.19 (b) and Proposition 9.36, with $[\{S^{(1)}(x),-\},\psi'](O)=\{S^{(1)}(x),\psi'(O)\}-\psi'(\{S^{(1)}(x),O\})$. (v) Remark 10.48 (d); the finite counterterm and the finite parts of the operators absorb the change. (vi) Corollaries 11.39 and 11.28 (iv). (vii) Theorems 8.20 and 8.23. □

Lemma 11.53 (The trace bilinear in a cocycle scheme)**.** *Let $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, $\xi\in S_+\otimes W$ or $\xi\in S_-\otimes W^*$, and let $\mathcal{B}_1,\dots,\mathcal{B}_4$ be the four functionals of Corollary 7.14,*

$$\mathcal{B}_1:=\{S^{(1)}(\xi),O\},\qquad \mathcal{B}_2:=(w_1,w_2)\int\omega\,\big(\rho(A|_{W_{\mathbb{R}}^\perp})\xi,\lambda\big),$$
$$\mathcal{B}_3:=(w_1,w_2)\int\omega\,\big(\rho(A|_V)\xi,\lambda\big),\qquad \mathcal{B}_4:=(w_1,w_2)\int\omega\,\big(c,(\xi,\lambda^*)\big),$$

$\mathcal{B}_1$ being the first functional listed there (Lemma 11.42 (i)). Write the two-body part of $Q_1(O)$ as $a_1\mathcal{B}_1+a_2\mathcal{B}_2+a_3\mathcal{B}_3+a_4\mathcal{B}_4$.

- *(i) If $L_1^{(0)}=0$ on O and on $\mathcal{B}_1$, then $a_3=a_4=0$, and the one-body terms of $Q_1(O)$ are multiples of $\mathcal{B}_1$.*
- *(ii) If $L_1^{(0)}(O)=2\kappa_O(w_1,w_2)\int\omega\,(dc,A|_V)$ and the part of $L_1^{(0)}(\mathcal{B}_1)$ with the two letters c, λ is $-\kappa_{\mathcal{B}}\,s\mathcal{B}_3$, the finite part $\psi(O)=\kappa_O(w_1,w_2)\int\omega\,(A|_V,A|_V)$, $\psi(\mathcal{B}_1)=\kappa_{\mathcal{B}}\mathcal{B}_3$ removes both and changes $Q_1(O)$ by $(2\kappa_O-\kappa_{\mathcal{B}})\mathcal{B}_3$. At $\psi=0$ the part with the letters c, λ of (R1) of Proposition 11.61 is the identity $a_3+2\kappa_O-\kappa_{\mathcal{B}}=0$.*

Proof. (i) By (R1), $s\,Q_1(O)=0$. $s\mathcal{B}_1=0$, since $[s,\{S^{(1)}(\xi),-\}]=\{\{S,S^{(1)}(\xi)\},-\}=0$ at tree level and $sO=0$; $s\mathcal{B}_2=0$, since Q annihilates the coordinate functionals of $A|_{W_{\mathbb{R}}^\perp}$ and λ and the cubic parts of s on them are the gauge transformation with parameter c, which leaves the invariant pairing fixed. In the sector with the two letters c, λ, $s\mathcal{B}_3=-(w_1,w_2)\int\omega\,(\rho(dc)\xi,\lambda)$ and $s\mathcal{B}_4=(w_1,w_2)\int\omega\,(c,(\xi,\not{d}\lambda))$, whose densities carry the derivative on different letters and are linearly independent for arbitrary ω; hence $a_3=a_4=0$. The one-body terms are $-\{\hbar F_1^{(1)}(\xi),O\}$; a term of $\hbar F_1^{(1)}(\xi)$ acting on the letters $A|_{W_{\mathbb{R}}^\perp}$ of O has the leg $A^*|_{W_{\mathbb{R}}^\perp}$, and by Proposition 11.49 (i) it is a multiple of T_2, which gives a multiple of $\mathcal{B}_1$. (ii) $Q\int\omega\,(A|_V,A|_V)=-2\int\omega\,(dc,A|_V)$ (Remark 5.6), and the cubic part of $s\int\omega\,(A|_V,A|_V)$ vanishes by the invariance of $(-,-)$; L_1 changes by $[s,\psi]$ (Lemma 10.37), the component along ξ by $[\{S^{(1)}(\xi),-\},\psi]$ (Proposition 8.19 (b)), and $\{S^{(1)}(\xi),\int\omega\,(A|_V,A|_V)\}=2\int\omega\,(\Gamma(\xi,\lambda)|_V,A|_V)$ with $(\Gamma(\xi,\lambda)|_V,a)=(\rho(a)\xi,\lambda)$ [ESW, (28)]. In the shifted scheme (i) gives $a_3+2\kappa_O-\kappa_{\mathcal{B}}=0$; at $\psi=0$ the part with the letters c, λ of (R1), $s\,Q_1(O)+L_1^{(0)}(\mathcal{B}_1)+\{S^{(1)}(\xi),L_1^{(0)}(O)\}=0$, is the same equation, since there $s\,Q_1(O)=a_3\,s\mathcal{B}_3$, $L_1^{(0)}(\mathcal{B}_1)=-\kappa_{\mathcal{B}}\,s\mathcal{B}_3$ and $\{S^{(1)}(\xi),L_1^{(0)}(O)\}=2\kappa_O\,s\mathcal{B}_3$. □

Theorem 11.54 (Q_1 on the scalar bilinears)**.** *Let $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, $\xi\in S_+\otimes W$ or $\xi\in S_-\otimes W^*$, and $\mathcal{B}_1,\dots,\mathcal{B}_4$ as in Lemma 11.53. At $\psi=0$:*

- *(i) the two-body part of $Q_1(O)$ is*

$$-2L\,\mathcal{B}_1+4L\,\mathcal{B}_2+4L\,\mathcal{B}_3+0\cdot\mathcal{B}_4\ \textit{units},\qquad L=\log\tfrac{3}{2},\tag{11.12}$$

  without poles and the same for both chiralities; its trace part is $4L$ units times $(w_1,w_2)\int\omega\,(\rho(A)\xi,\lambda)$, with the whole connection $A=A|_V+A|_{W_{\mathbb{R}}^\perp}$;
- *(ii) on the traceless part, $(w_1,w_2)=0$,*

$$Q_1(O)=c_Q\,\{S^{(1)}(\xi),O\},\qquad c_Q=-2\log\tfrac{3}{2}-t_\perp\ \textit{units},$$

with $t_\perp$ units the coefficient of T_2 in $\hbar F_1^{(1)}(\xi)$, which the two-leg solution of (11.10) *gives as $t_\perp = \log\frac{3}{2} - \frac{79}{12} + \theta_\perp^\xi$ (Proposition 11.50 (i)), so that $c_Q = \frac{79}{12} - 3\log\frac{3}{2} - \theta_\perp^\xi$ units; c_Q moves by -1 times the change of the free direction $\theta_\perp^\xi$, by $\mp a$ under G_B along $S_+ \otimes W$ and $S_- \otimes W^*$, by $-\frac{a}{2}$ under $\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$, and under the field countings as in Proposition 11.52 (i);*

- *(iii) $L_1^{(0)}(O) = -(2+4L)$ units times $(w_1, w_2)\int\omega\,(dc, A|_V)$, and the part of $L_1^{(0)}(\mathcal{B}_1)$ with the letters c, λ is 2 units times $s\mathcal{B}_3$; Lemma 11.53 (ii) applies with $\kappa_O = -(1+2L)$ units and $\kappa_\mathcal{B} = -2$ units, and in the cocycle scheme*

$$\psi(O) = -(1+2L)\,\textit{units}\cdot(w_1,w_2)\int\omega\,(A|_V, A|_V), \qquad \psi(\mathcal{B}_1) = -2\,\textit{units}\cdot\mathcal{B}_3,$$

the two-body part of $Q_1(O)$ is $-2L\,\mathcal{B}_1 + 4L\,\mathcal{B}_2$ units, $a_3 = 4L + 2\kappa_O - \kappa_\mathcal{B} = 0$;

- *(iv) at $\psi = 0$, for $\xi \in S_+ \otimes W$, the part with the letters c, $\lambda|_{S_-\otimes W^*}$ of (R1) is the identity*

$$\begin{aligned} &s\,Q_1(O) + L_1^{(0)}\big(\{S^{(1)}(\xi), O\}\big) + \big\{S^{(1)}(\xi), L_1^{(0)}(O)\big\} \\ &\qquad = \big(-4L - 2 + (2+4L)\big)\,\textit{units}\cdot(w_1,w_2)\int\omega\,\big(\rho(dc)\xi, \lambda\big) = 0; \end{aligned}$$

- *(v) at $\psi = 0$, with the uniform coefficient $t_\xi = \theta_V^\xi = \theta_\perp^\xi$ forced by the arity-two equations along $(\xi_\pm, \xi'_\pm)$ (Theorem 11.51), $Q_1(O) = \big(\frac{79}{12} - 3L - t_\xi\big)\mathcal{B}_1 + 4L\,\mathcal{B}_2 + 4L\,\mathcal{B}_3$ units: the two-body part* (11.12) *and the one-body part $-\big(L - \frac{79}{12} + t_\xi\big)\mathcal{B}_1$ units.*

Proof. (i) By Theorem 11.44 (iii) the two-body part is Υ_ξ on the single pair $(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ of O, whose leg sets are those of Table 11.6: $\lambda|_{S_-\otimes W^*}$, $A|_V\,\lambda|_{S_-\otimes W^*}$, $A|_{W_\mathbb{R}^\perp}\lambda|_{S_+\otimes W}$ and $c\,\lambda^*|_{S_+\otimes W^*}$ for $\xi \in S_+\otimes W$. The single output Y has the colour factor $(1\otimes \mathrm{ad}_Y)(-,-)^{-1}$, which the colour pairing of O contracts to $\mathrm{tr}_\mathfrak{g}\,\mathrm{ad}_Y = 0$. The other three leg sets have $D = 0$, so they carry no poles and no finite special vertex (Proposition 11.47), and they give $\mathcal{B}_1$, $\mathcal{B}_2$ (outputs $A|_{W_\mathbb{R}^\perp}\lambda$), $\mathcal{B}_3$ (outputs $A|_V\lambda$) and $\mathcal{B}_4$ (outputs $c\,\lambda^*$). Their chains have three edges and, at $D = 0$, constant integrands, so each chain gives a rational multiple of $\frac{1}{2}\log\frac{3}{2}$ (Theorem 11.45 (iv)); the exact evaluation of the chains in all placements, in both ways of Remark 11.66, gives (11.12), with the same coefficient for $\mathcal{B}_2$ and $\mathcal{B}_3$. (ii) By Corollary 7.14 the functionals at $(w_1, w_2) = 0$ are the multiples of $\mathcal{B}_1$; the one-body term is $-t_\perp\mathcal{B}_1$ units (proof of Lemma 11.53 (i)); the changes are Propositions 11.49 and 11.52. (iii) $L_1^{(0)}$ is given by the arity-zero two-body kernels, the cycles with one special edge of the proof of Theorem 10.16, of the pairs $(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ of O and $(A|_{W_\mathbb{R}^\perp}, \lambda)$ of $\mathcal{B}_1$, evaluated in the same way; then Lemma 11.53 (ii). (v) By (i) and (ii), T_2 being the only term of the solution of Proposition 11.50 (i) with the leg $A^*|_{W_\mathbb{R}^\perp}$. (iv) With $s\mathcal{B}_3 = -(w_1,w_2)\int\omega\,(\rho(dc)\xi, \lambda)$ in this sector, the three terms are $4L\,s\mathcal{B}_3$, $2\,s\mathcal{B}_3$ and $2\kappa_O\,s\mathcal{B}_3 = -(2+4L)\,s\mathcal{B}_3$ units (proof of Lemma 11.53 (ii)). □

Proposition 11.55 (S_1 and K_1 on the scalar bilinears)**.** *Let $O = \int\omega\,(\iota_{w_1}A, \iota_{w_2}A)$ and $\psi = 0$.*

- *(i) $\Upsilon_\eta^0(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ has the single leg set λ, which vanishes on the colour singlet, and s_0 has zero bracket with a letter $A|_{W_\mathbb{R}^\perp}$. Hence the two-body part of $S_1(O)$ at the point x of the density is the two-body part of $Q_1(O)$ along the Killing spinor $\rho(x)\eta$, given by* (11.12) *along $S_-\otimes W^*$ for $\eta \in S_+\otimes W^*$ and along $S_+\otimes W$ for $\eta \in S_-\otimes W$; the term linear in $\not{d}(\rho(x)\eta) = 4\eta$ is absent (Corollary 7.15). With the solution of Proposition 11.50 (ii), whose position part is $x^\mu\hbar F_1^{(1)}(\rho(e_\mu)\eta)$ with the uniform coefficient t_η and whose terms s_0, u_0 have no leg A^*, the component is*

$$S_1(O) = \big(\tfrac{79}{12} - 3\log\tfrac{3}{2} - t_\eta\big)\,\textit{units}\cdot\{S^{(1)}(\eta), O\} + x^\mu\Big(4\log\tfrac{3}{2}\,\mathcal{B}_2 + 4\log\tfrac{3}{2}\,\mathcal{B}_3\Big)\big(\rho(e_\mu)\eta\big)\ \textit{units},$$

at the point x of the density, with $\mathcal{B}_2$, $\mathcal{B}_3$ of Lemma 11.53 at the spinor $\rho(e_\mu)\eta$; its intrinsic part vanishes.

- *(ii) $\Upsilon^0_{K_b}(A|_{W^\perp_{\mathbb{R}}}, A|_{W^\perp_{\mathbb{R}}})$ has the leg sets $\emptyset$, of superficial degree 1, and $A|_V$, both zero on the colour singlet, and $\Upsilon^0_{M_\omega}$ vanishes there. By (11.9) the two-body part of $K_1(O)$ is $v(x)^\mu\,\Upsilon_{P_{e_\mu}} - (b,x)\,\Upsilon^0_\Delta$, and the one-body part is the bracket with the solution of Proposition 11.50 (v), whose terms with the leg $A^*|_{W^\perp_{\mathbb{R}}}$ are the part $-\frac{22}{3}$ units times $\int(b,x)(A,A^*)$, the sector part $\theta^K_\perp$ units times $S^{(1)}(K_b)_\perp$ and the directions G^K_j. On the part with two legs $A|_{W^\perp_{\mathbb{R}}}$, with $\Upsilon^0_\Delta = \frac{40}{3}O$, $-\frac{104}{3}O$ units at the point of the density and $\Upsilon_{P_{e_\mu}} = -\frac{1}{3}$ units times $\partial_\mu O$ (Corollary 11.56),*

$$K_1(O) = -\big(\tfrac{1}{3} + \theta^K_\perp\big)\ \textit{units}\cdot\{S^{(1)}(K_b), O\} + \begin{cases} 0 & \textit{(traceless part)},\\ 48\ \textit{units}\cdot(b,x)\,O & \textit{(trace)}, \end{cases} \; - \sum_{j=1}^{4} \kappa^K_j \{G^K_j, O\},$$

$$\{S^{(1)}(K_b), O\} = v^\mu \partial_\mu O - 4(b,x)\,O,$$

so that at $\theta^K_\perp = -\frac{1}{3}$ and $\kappa^K_j\{G^K_j, O\} = 0$, $K_1(O) = -(b,x)D_1(O)$: 0 on the traceless bilinears and 48 units times $(b,x)\,O$ on the trace, the relation of Proposition 8.36 with $\kappa = -24$ units. The coefficients θ^K_d and κ^K_j enter the arity-two equations along $(K_b, P_{b'})$ and (η, η') at a finite endpoint.

Proof. (i) Table 11.6 and the evaluation of the leg set: a single output Y gives $\mathrm{tr}_{\mathfrak{g}}\,\mathrm{ad}_Y = 0$ on the colour singlet; s_0 has the legs $A|_{W^\perp_{\mathbb{R}}}$, λ^*, and its bracket with a letter needs the leg $A^*|_{W^\perp_{\mathbb{R}}}$. Theorem 11.45 (ii) gives the position part. (ii) The leg set $A|_V$ has a single output; the empty leg set is homogeneous in ε of degree $-\frac{1}{2}$ (Lemma 10.6). Theorem 11.45 (iii) gives the two-body parts; by Corollary 7.16 the term in the second derivatives of v and the term in the antisymmetric part of dv vanish on a scalar bilinear. The one-body part: $-\{-\frac{22}{3}\,\text{units}\int(b,x)(A,A^*), O\} = \frac{44}{3}(b,x)\,O$ units, the rotation and spin terms of the solution acting on $A|_V$ and λ alone, and $-\theta^K_\perp$ units times $\{S^{(1)}(K_b), O\}$. With $v(x)^\mu\partial_\mu O$ and $\{S^{(1)}(K_b), O\} = v^\mu\partial_\mu O + \frac{2}{4}(\mathrm{div}\,v)O$ (Definition 7.4), $-\frac{1}{3}v^\mu\partial_\mu O - \frac{40}{3}(b,x)O + \frac{44}{3}(b,x)O = -\frac{1}{3}\{S^{(1)}(K_b), O\}$ on the traceless part, and on the trace $-(b,x)\Upsilon^0_\Delta$ adds $(\frac{104}{3} + \frac{40}{3})(b,x)O = 48(b,x)O$, the one-body part being the same. Proposition 8.36: $K_1(O) = -2(b,x)\kappa\,O$ for $D_1(O) = 2\kappa\,O$. □

Corollary 11.56 (The dilatation and the translations by the defect cycles)**.** *On every local functional O with $sO = 0$, with the Koszul signs of (10.4),*

$$\sum_{a<b} \pm\Upsilon_\Delta(X_a, X_b) = \sum_{a<b} \pm\big(4\,\mathrm{lc}(X_aX_b) + \textit{composite pair kernel along } \Delta\big),$$
$$\sum_{a<b} \pm\Upsilon_{P_b}(X_a, X_b) = \sum_{a<b} \pm\big(\textit{composite pair kernel along } P_b\big),$$

and the one-body terms of the two forms agree. On the part with two legs $A|_{W^\perp_{\mathbb{R}}}$ of the scalar bilinears, at the point of the density, Υ^0_Δ is $\frac{40}{3}$ units times O on the traceless part and $-\frac{104}{3}$ units times O on the trace, and Υ_{P_b} is $-\frac{1}{3}$ units times $\partial_b O$; with the position part $2x^\mu\Upsilon_{P_{e_\mu}}$ this is $\frac{44}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ and $-\frac{100}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ units, the first row of the table of Theorem 9.30, and the value $-\frac{1}{3}$ of Theorem 9.35.

Proof. Both forms compute the unique local representative (Theorem 8.15 (i)): for Δ this is Proposition 11.21, whose route of the special edges is (11.4), and for P_b the same argument applies with Corollary 11.20 and the one-body term $-\{\hbar F^{(1)}_1(P_b), -\}$ in both forms. On the bilinears, the defect cycles with $S^{(1)}(\Delta)$ and $S^{(1)}(P_b)$ in all their placements, evaluated by Theorem 11.45, give the stated values, with $\{S^{(1)}(\Delta), O\} = 2x^\mu\partial_\mu O + 4O$ at the point of the density: $\frac{40}{3} + \frac{4}{3} = \frac{44}{3}$ and $-\frac{104}{3} + \frac{4}{3} = -\frac{100}{3}$. □

Proposition 11.57 (The dilatation and the translations on the families from the defect cycles)**.** *With the master formula* (11.4) *along* Δ *and* P_n*, the sector coefficients* $-\frac{1}{3}$ *of Theorem 9.28 and the families and kinematics of Definition 11.36, on the parts with the fewest fields:*

- *(i) on the* $\mathfrak{su}(2)$ *sector,* $L \leq 8$*,* $D_1 = -8\,\text{units} \cdot \sum_l (1 - P_{l,l+1})$*,* $P_{l,l+1}$ *the exchange of the letters at the sites* l*,* $l+1$*, which is* $\frac{g^2 N}{4\pi^2}$ *times the Heisenberg Hamiltonian; on the* $\mathfrak{su}(1|1)$ *sector,* $L \leq 6$*, on all twenty-two graded cyclic sectors,* $D_1 = -8\,\text{units} \cdot \sum_l (1 - \Pi_{l,l+1})$ *with the graded exchange* $\Pi_{l,l+1}$*;*
- *(ii) on twist two,* $S' \leq 6$*, the spectrum of* D_1 *is* $\{-32\,h(j)\ \text{units} : j \leq S'\ \text{even}\}$*, and on the fermion* $\mathfrak{sl}(2)$ *sector,* $S' \leq 5$*, it is* $\{-32\,h(j+1)\ \text{units} : j \leq S'\ \text{odd}\}$*;*
- *(iii) on the family of two gaugino components with* $Z'F_{11}$*,* $S' \leq 3$*, the eigenvalues are* 0*,* -48*,* $-\frac{200}{3}$ *units, on the gluon* $\mathfrak{sl}(2)$ *sector,* $S' \leq 2$*,* -48*,* $-\frac{200}{3}$ *units, and on* (F_{11}, ψ)*,* $S' \leq 4$*,* -48*,* $-\frac{200}{3}$*,* $-\frac{392}{5}$ *units:* -8 *units times the eigenvalues of Proposition 11.37;*
- *(iv) the sum over the pairs of* Υ_{P_n} *is* $-\frac{1}{3}$ *units times* ∂_n *on twist two and on the fermion* $\mathfrak{sl}(2)$ *sector, and on the families of (iii), and* $D_1(\partial_n O) = \partial_n D_1(O)$ *on twist two, the fermion* $\mathfrak{sl}(2)$ *sector, the family of two gaugino components with* $Z'F_{11}$ *and the family* (F_{11}, ψ)*, as an identity of matrices from* S' *to* $S'+1$*;*
- *(v) pair by pair,* $\Upsilon_\Delta(X_a, X_b) = 4\,\mathrm{lc}(X_a X_b) + \big(\text{composite pair kernel along } \Delta\big)$*,* Υ_{P_b} *and* Υ_{M_ω} *equal to the composite pair kernels along* P_b *and* M_ω*, and* $\Upsilon_R = 0$*, on every pair of slots with defect cycles and every leg set with at most one letter derivative on which both evaluations are complete, and with at most two letter derivatives in the simplex evaluation.*

The spectra of (i)–(iii) are those of Table 11.5, obtained here without the composite term.

Proof. The two-body kernels are the coefficients of ε^0 of the defect cycles of the pairs of letters with $S^{(1)}(\Delta)$ or $S^{(1)}(P_n)$ at the special edge or on a line, evaluated exactly by Theorem 11.45 with the colour words reduced to the planar adjacent form of Lemma 10.8 (iv), and the one-body terms are the brackets with (9.16) and with $\hbar F_1^{(1)}(P_n)$; the operators are diagonalised exactly on bases of the families closed under them, with the projection to cohomology of Remark 11.40. Items (i), (ii), (iv) on twist two and the fermion sector and (v) at one letter derivative are obtained in both evaluations of Remark 11.66; the families of (iii), the matrix identity of (iv) on them and (v) at two letter derivatives in the simplex evaluation. Item (v) summed over the pairs of a cocycle is Corollary 11.56. □

Corollary 11.58 (The components on the scalar bilinears: representatives and invariant content)**.** *Let* $O = \int \omega\,(\iota_{w_1} A, \iota_{w_2} A)$*,* $\psi = 0$*,* $L = \log\frac{3}{2}$*,* $\mathcal{B}_1, \ldots, \mathcal{B}_4$ *as in Lemma 11.53, and let the action-side solutions be those of Theorem 9.26 (iii), Theorem 9.35 (i) and Proposition 11.50, with the uniform coefficients* t_Δ*,* t_P*,* t_M*,* t_R*,* $t_{\xi_\pm}$*,* $t_{\eta_\pm}$ *of Theorem 11.51. The representatives of the components given by the master formula on the part with two legs* $A|_{W_{\mathbb{R}}^\perp}$ *are, in units,*

$$Q_1(O) = \big(\tfrac{79}{12} - 3L - t_\xi\big)\mathcal{B}_1 + 4L\,\mathcal{B}_2 + 4L\,\mathcal{B}_3 + 0\cdot\mathcal{B}_4, \qquad \hbar L_1(O)^{(0)} = -(2+4L)\,(w_1, w_2)\int \omega\,(dc, A|_V),$$

$$S_1(O) = \big(\tfrac{79}{12} - 3L - t_\eta\big)\{S^{(1)}(\eta), O\} + x_O^\mu\big(4L\,\mathcal{B}_2 + 4L\,\mathcal{B}_3\big)\big(\rho(e_\mu)\eta\big),$$

$$D_1(O) = -\big(4t_\Delta + \tfrac{4}{3}\big)\,O \ \textit{(traceless)}, \qquad D_1(O) = -\big(4t_\Delta + \tfrac{148}{3}\big)\,O \ \textit{(trace)}, \quad \textit{at the point } x_O \textit{ of the density},$$

$$P_1(O) = -\big(\tfrac{1}{3} + t_P\big)\,\partial_b O, \qquad M_1(O) = -\big(\tfrac{1}{3} + t_M\big)\,\{S^{(1)}(M_\omega), O\}, \qquad R_1(O) = 0,$$

$$K_1(O) = -\big(\tfrac{1}{3} + \theta_\perp^K\big)\{S^{(1)}(K_b), O\} + \begin{cases} 0 & \textit{(traceless)}, \\ 48\,(b,x)\,O & \textit{(trace)}, \end{cases} - \sum_{j=1}^{4} \kappa_j^K \big\{G_j^K, O\big\},$$

with t_ξ, t_η the coefficients of the chirality of ξ, η, the spinor $\rho(e_\mu)\eta$ in $\mathcal{B}_2$, $\mathcal{B}_3$ in place of ξ, and the intrinsic parts of S_1 and K_1 equal to 0. Hence:

- *(i) the scheme-independent content is $D_1 = 0$ on the twenty traceless bilinears and $D_1 = -48$ units times O on the trace, $\frac{3g^2N}{2\pi^2}O$ for $\mathfrak{g} = \mathfrak{su}(N)$, twice the Konishi anomalous dimension: these are the eigenvalues of the induced operator on the cohomology of $\Lambda^{(0)}$, at the value $t_\Delta = -\frac{1}{3}$ fixed by the arity-two equations, invariant under the inner and outer directions and under the finite parts ψ' (Theorem 8.20 (iii));*

- *(ii) as functions of the action-side free directions at $\psi = 0$, with c_Q, c_S the coefficients of $\mathcal{B}_1$ and of $\{S^{(1)}(\eta), O\}$ and $p_1 = -(\frac{1}{3} + t_P)$, on the traceless bilinears*

$$c_Q(\xi_+) + c_Q(\xi_-) - p_1 = c_Q(\xi_+) + c_S(\eta_+) - \tfrac{1}{4}D_1 = \tfrac{27}{4} - 8\log\tfrac{3}{2},$$

independent of t_Δ, t_P, t_ξ, t_η and of the inner and outer directions by (11.11); a finite part ψ' of the operators shifts c_Q, c_S and p_1 by the coboundaries of Proposition 11.52 (iv), with weight-dependent coefficients on the levels of Weyl weight 2, $\frac{5}{2}$ and 3, so these combinations become scheme-independent together with the one-loop components on the weight-$\frac{5}{2}$ level that enter (R2) and (R3) of Proposition 11.61;

- *(iii) P_1, M_1 and R_1 vanish at the values $t_P = t_M = t_\Delta = -\frac{1}{3}$ of the translation-covariant scheme, and $L_1^{(0)}(O)$ is removed by the finite part $\psi(O) = -(1 + 2L)$ units times $(w_1, w_2)\int \omega\,(A|_V, A|_V)$ (Theorem 11.54 (iii));*

- *(iv) the scheme artefacts are c_Q and c_S separately, every coefficient with $\log\frac{3}{2}$, P_1, $L_1^{(0)}(O)$, M_1, R_1, the one-body part of K_1, which depends on $\theta_\perp^K$ and the four directions κ_j^K left free on two legs, and the individual two-body kernels.*

Proof. The two-body parts are Theorem 11.54 (i), Proposition 11.55 and Corollary 11.56: $\Upsilon^0_\Delta = \frac{40}{3}O$, $-\frac{104}{3}O$, $\Upsilon_{P_b} = -\frac{1}{3}\partial_b O$, $\Upsilon^0_{M_\omega} = 0$, $\Upsilon_R = 0$, the intrinsic parts along η and K_b vanishing on the colour singlet. The one-body parts are the brackets of O with the solutions; on O only the terms with the leg $A^*|_{W^\perp_\mathbb{R}}$ act. Along ξ, T_2 is the only such term of $\hbar F_1^{(1)}(\xi)$ and $\{T_2, O\} = \mathcal{B}_1$, which gives $-(L - \frac{79}{12} + t_\xi)\mathcal{B}_1$; along η the position part of the solution is $x^\mu \hbar F_1^{(1)}(\rho(e_\mu)\eta)$ with the uniform t_η, and s_0, u_0 have no leg A^*; along Δ, $-\{\frac{22}{3}\int(A, A^*), O\} = -\frac{44}{3}O$ and $-\{t_\Delta S^{(1)}(\Delta)_\perp, O\} = -4t_\Delta O$ at the point of the density; along P_b, M_ω, R the sector parts in the sector of $A|_{W^\perp_\mathbb{R}}$, with $\{S^{(1)}(M_\omega), O\} = (\omega x)^\mu \partial_\mu O$ on a scalar bilinear and the position part $(\omega x_O)^\mu \Upsilon_{P_{e_\mu}}$ of Υ_{M_ω}; along K_b, the part linear in x of Proposition 11.50 (v) gives $\frac{44}{3}(b, x)O$, which with $-(b, x)\Upsilon^0_\Delta$ and $v(x)^\mu \Upsilon_{P_{e_\mu}}$ is $-\frac{1}{3}\{S^{(1)}(K_b), O\}$ on the traceless part and the same plus $48(b, x)O$ on the trace, $\{S^{(1)}(K_b), O\} = v^\mu \partial_\mu O - 4(b, x)O$. (i) Theorem 9.30 and Theorem 11.51. (ii) $c_Q(\xi_+) + c_Q(\xi_-) - p_1 = \frac{79}{6} - 6L + \frac{1}{3} - (t_{\xi_+} + t_{\xi_-} - t_P)$ and $c_Q(\xi_+) + c_S(\eta_+) - \frac{1}{4}D_1 = \frac{79}{6} - 6L + \frac{1}{3} - (t_{\xi_+} + t_{\eta_+} - t_\Delta)$, both $\frac{27}{2} - 6L - \frac{27}{4} - 2L$ by (11.11). (iii) Theorem 11.51 (i), (iii). (iv) Theorem 11.51 (ii), (iv) and Proposition 11.52. □

Proposition 11.59 (The special supercharges and the supercharges)**.** *Let $\eta \in \Sigma^*$ and $b \in V$, so that $\rho(b)\eta \in \Sigma$.*

- *(i) Graph by graph, the position part along b of the two-body kernel of (X_a, X_b) along η is the two-body kernel of (X_a, X_b) along $\rho(b)\eta$; its translation-invariant part is computed with $S^{(1)}(\eta)$ at the spinor $\rho(x - x_O)\eta$, x_O the point of the density, together with the term $-\int(\rho(A|_{W^\perp_\mathbb{R}})\eta, \lambda^*)$.*

- *(ii) Under the hypotheses of Proposition 7.13 and Corollary 8.17, on cohomology, the position part along b of S_1 along η is Q_1 along $\rho(b)\eta$ minus the commutator of $\{S^{(1)}(\eta), -\}$ with P_1.*

- *(iii) On a family on which $P_1 \cong 0$, in particular on the families of Definition 11.36 in the translation-covariant scheme at the values $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ of Theorem 9.28, the position part of S_1 along η is Q_1 along $\rho(b)\eta$ on cohomology, and S_1 along η is fixed by Q_1 along Σ up to the term of Proposition 7.13 linear in $\not{A}(\rho(x)\eta) = 4\eta$. On the scalar bilinears this term is absent (Corollary 7.15), and S_1 along η is, on cohomology, the combination of the functionals of Corollary 7.15 with the coefficients of Q_1 along Σ in Corollary 7.14, given by Theorem 11.54 and Proposition 11.55 (i); the component S_1 computed from its own defect cycles and from the action-side solution along η (Proposition 11.55 (i)) satisfies this relation.*

- *(iv) In a reflection-compatible scheme, on the family formed by a traceless scalar bilinear O at the origin and its descendants $\{S^{(1)}(\xi), O\}$, on which $Q_1 = c_Q\{S^{(1)}(\xi), -\}$ along the unit generator Q^i_α (Theorem 11.54 (ii)) and $S_1 = c_S\{S^{(1)}(\eta), -\}$ along $S^\alpha{}_i$, Corollary 8.41 reads*

$$c_S = \overline{c_Q} + g_{5/2} - g_2,$$

with g_w the eigenvalue of $\mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1$ in Weyl weight w on the family; the outer direction changes c_Q by $-a$ and c_S by $+a$ and preserves the relation for $\bar{a} = -a$.

Proof. (i) By (11.3), $S^{(1)}(\eta) = \int x^\mu s_\mu + \int s_0$, where $\int s_\mu$ is $S^{(1)}$ along the constant spinor $\rho(e_\mu)\eta \in \Sigma$. Insert $x^\mu = x^\mu_O + (x - x_O)^\mu$ and argue as in the proof of Proposition 11.10 (i): on plane waves the factor x^μ is $-i\partial/\partial r^\mu$ of the momentum injected at the vertex, and its action on the momentum delta function is $-i$ times the graph with $\int s_\mu$; the same holds for the special edges, the vertices $S^{(1)}(\eta)$ on the cycle and the counterterm vertices, which are the defect insertions of Proposition 10.45 (Theorem 11.45 (ii)). (ii) Corollary 8.17 (ii) at (η, P_b): by Proposition 2.4, $[P_b, \eta] = \rho(b)\eta$, so $[\eta, P_b] = -\rho(b)\eta$, and $S^{(2)}(\eta, P_b) = 0$ (Definition 2.15). At order $\hbar$ the identity states that the commutator of S_1 along η with $\{S^{(1)}(P_b), -\}$, plus the commutator of $\{S^{(1)}(\eta), -\}$ with P_1, minus Q_1 along $\rho(b)\eta$, is a graded commutator with $\Lambda^{(0)}$, which induces 0 on cohomology; at order $\hbar^0$ the same identity is the statement of (i) for the classical transformation, which fixes the signs. By Proposition 10.47, S_1 along η is at most linear in the position, and Lemma 11.9 identifies the first commutator with its position part. (iii) By Proposition 7.13 the representative of S_1 along η at a point x_0 is a term linear in the value $\rho(x_0)\eta$ plus a term linear in 4η; the first has the position part given by its value at $\rho(b)\eta$, and as η and b vary, $\rho(b)\eta$ spans Σ, so by (ii) it is the representative of Q_1 evaluated at $\rho(x_0)\eta$ on cohomology. (iv) Write $X_0 + \hbar X_1$ for the matrix of $-\Lambda^{(1)}(Q^i_\alpha)$ and $Y_0 + \hbar Y_1$ for that of $-\Lambda^{(1)}(-S^\alpha{}_i)$ (Theorem 8.31 (i)): $\hbar X_1 = c_Q X_0$ and $\hbar Y_1 = c_S Y_0$, and Y_0 maps Weyl weight $\frac{5}{2}$ to Weyl weight 2. Corollary 8.41 gives $\hbar Y_1 = \overline{c_Q}\, X_0^{\mathrm{adj}} + \hbar[Y_0, \mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1] = \big(\overline{c_Q} + g_{5/2} - g_2\big)Y_0$ on the descendants, with $Y_0 = X_0^{\mathrm{adj}}$. The outer direction: Proposition 11.52 (ii) and Corollary 8.41. □

Corollary 11.60 (K_1 in the translation-covariant scheme)**.** *Under the hypotheses of Theorems 8.40 and 11.26, on a family of Definition 11.36 at the origin of one length, in the translation-covariant scheme at the values $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ of Theorem 9.28, K_1 along $K_{\bar{b}}$ equals $\hbar\big[\{S^{(1)}(K_{\bar{b}}), -\}, \mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1\big]$, the statement $K = P^\dagger$ of Corollary 8.41. For equal coefficients θ_P along P_b, K_1 along $K_{\bar{b}}$ is $(\theta - \theta_P)$ units times $\{S^{(1)}(K_{\bar{b}}), -\}$ plus the same commutator, with the radial Gram matrix of Definition 8.39 in that scheme. On a scalar bilinear at the origin both sides vanish: the defect cycles give $K_1(O) = 0$ there (Proposition 11.55 (ii)), and the commutator vanishes on a primary at the origin on which $\mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1$ is diagonal. On its descendants $\partial_\mu O$ the relation equates the intrinsic kernels $\Upsilon^0_{K_b}$ of the descendant pairs with the difference of the two-point normalisations in Weyl weights $w_O + 1$ and w_O times the classical transformation. The component K_1 itself is computed from its own defect cycles at a finite endpoint and from the action-side solution along K_b (Proposition 11.55 (ii)); this corollary is a cross-check of it through the inner product and enters no number.*

Proof. Corollary 11.39 gives $P_1 \cong 0$, resp. $P_1 \cong (\theta - \theta_P)$ units times $\{S^{(1)}(P_b), -\}$ with a real coefficient; Corollary 8.41 gives the two relations. At the origin $v(0) = 0$ and $(b, 0) = 0$, so the formula of

Proposition 11.55 (ii) vanishes; $\{S^{(1)}(K_{\bar b}),-\}$ lowers the Weyl weight by one and annihilates a primary at the origin. □

Proposition 11.61 (The relations at order $\hbar$)**.** *Write $\Lambda^{(1)}(x)=\Lambda_0^{(1)}(x)+\hbar\Lambda_1^{(1)}(x)$ with $\Lambda_0^{(1)}(x)=\pm\{S^{(1)}(x),-\}$ (Theorem 8.15 (iv)), so that $\hbar\Lambda_1^{(1)}(x)O$ is the component of O along x, and $\Lambda^{(0)}=s+\hbar L_1^{(0)}$. Let $\mathfrak{F}$ be a family of colour-singlet local functionals of degree 0, stable under Λ and under the classical transformations, with $L_1^{(0)}=0$ on $\mathfrak{F}$ for a finite part ψ as in Definition 10.36 (ii), and let $O\in\mathfrak{F}$ with $sO=0$.*

- *(R1) $s\big(\hbar\Lambda_1^{(1)}(x)O\big)=0$ for every generator x; along ξ, $s\,Q_1(O)=0$, an identity of local functionals.*
- *(R2) For $\xi,\xi'\in\Sigma$ with $b=\Gamma(\xi,\xi')|_V$, $[\Lambda_1^{(1)}(\xi),\Lambda_0^{(1)}(\xi')]O+[\Lambda_0^{(1)}(\xi),\Lambda_1^{(1)}(\xi')]O+\Lambda_1^{(1)}(P_b)O=\Lambda^{(0)}(\chi)$, with χ local of degree -1, built from the arity-two component of O along (ξ,ξ'); for $\xi,\xi'\in S_+\otimes W$, $b=0$.*
- *(R3) For $\xi\in\Sigma$ and $\eta\in\Sigma^*$ with $[\xi,\eta]=\frac12(\xi,\eta)\Delta+M_{\xi\eta}+R_{\xi\eta}$ (Proposition 2.4), $M_{\xi\eta}\in\mathfrak{so}(V)$, $R_{\xi\eta}\in\mathfrak{so}(W_{\mathbb{R}}^{\perp})$,*

  $$[\Lambda_1^{(1)}(\xi),\Lambda_0^{(1)}(\eta)]O+[\Lambda_0^{(1)}(\xi),\Lambda_1^{(1)}(\eta)]O+\tfrac12(\xi,\eta)\,D_1(O)+M_1(O)+R_1(O)=\Lambda^{(0)}(\chi'),$$

  the components M_1, R_1 taken along $M_{\xi\eta}$, $R_{\xi\eta}$.
- *(R4) For a rank-one ξ, $Q_0^\xi Q_1^\xi+Q_1^\xi Q_0^\xi=0$ on every local functional (Theorem 8.23).*
- *(R5) With $[K_b,P_{b'}]=(b,b')\Delta+M_{bb'}$ (Remark 7.17), $[\Lambda_1^{(1)}(K_b),\Lambda_0^{(1)}(P_{b'})]+[\Lambda_0^{(1)}(K_b),\Lambda_1^{(1)}(P_{b'})]+(b,b')\hbar\Lambda_1^{(1)}(\Delta)+\hbar\Lambda_1^{(1)}(M_{bb'})\cong 0$ on cohomology; with Theorem 11.45 (iii) it states that the parts of K_1 multiplying the position are those of* (11.9).
- *(R6) For $\eta_1,\eta_2\in\Sigma^*$ with $[\eta_1,\eta_2]=-K_b$, $b=\Gamma(\eta_1,\eta_2)|_V$, at a finite T, K_1 along K_b is, on cohomology, $[\Lambda_1^{(1)}(\eta_1),\Lambda_0^{(1)}(\eta_2)]+[\Lambda_0^{(1)}(\eta_1),\Lambda_1^{(1)}(\eta_2)]$.*

In (R2), (R3), (R5) and (R6) the right sides vanish on cohomology. For O without letters λ and c, such as the scalar bilinears and the families of scalars, the parts of order $\hbar^0$ of χ, χ' vanish, the right sides are s-exact, and their parts with two field letters and no antifield vanish on free on-shell plane waves, $|p|^2=0$ on the scalars and the gauge field, $\rho(p)\lambda=0$ and $(p,a)=0$ on the polarisations a of $A|_V$. On the scalar bilinears at $\theta=-\frac13$, (R3) contains $D_1(O)=0$ on the traceless part and -48 units times O on the trace (Corollary 9.34), the Konishi relation between Q_1, S_1 and the anomalous dimension, and (R2) with $\xi\in S_+\otimes W$, $\xi'\in S_-\otimes W^*$ contains $P_1(O)=-(\frac13+\theta_P)$ units times $\partial_b O$ (Theorem 9.35).

Proof. Corollary 8.17 (i) at order $\hbar$ reads $[s,\Lambda_1^{(1)}(x)]+[L_1^{(0)},\Lambda_0^{(1)}(x)]=0$; on O with $sO=0$, $L_1^{(0)}O=0$ and $L_1^{(0)}\{S^{(1)}(x),O\}=0$ (stability of $\mathfrak{F}$) this is (R1). Corollary 8.17 (ii) at order $\hbar$ for the pairs (ξ,ξ'), (ξ,η), $(K_b,P_{b'})$, (η_1,η_2), with the brackets of Proposition 2.4 and Remark 7.17, has the right side $[\Lambda^{(0)},H]$ with H built from $\Lambda^{(2)}$, which on O with $\Lambda^{(0)}O=0$ is $\Lambda^{(0)}(HO)$; the part of order $\hbar^0$ of H is $\pm\{S^{(2)}(x_1,x_2),-\}$, and $S^{(2)}$ has the legs (λ^*,λ^*) and (A,c^*) (Definition 2.15), whose bracket with a functional of the fields A alone vanishes. (R4) is Theorem 8.23. For the vanishing on shell: the part of $s\chi$ with two field letters is $Q\chi_2+\{I_3,\chi_1\}$, with χ_k the part of χ with k letters; χ_1 is a single adjoint letter contracted with a constant tensor, zero for a colour singlet; χ_2 of degree -1 has one antifield and one field, or c^* and c, and Q substitutes $A^*\mapsto-\frac12 d^*dA$, $\lambda^*\mapsto\not d\lambda$, $c^*\mapsto d^*(A^*|_V)$, $A\mapsto -dc$ (Proposition 1.11), so the part of $Q\chi_2$ without antifields and ghosts is a free field equation times a field. The values on the bilinears are Corollary 9.34 and Theorem 9.35, with $R_1=0$ on them (Remark 7.9). □

Proposition 11.62 (The chiral-primary bilinears along a rank-one supercharge)**.** *Let $\xi \in S_+ \otimes W$ be of rank one, ${\xi_\alpha}^i = u_\alpha z^i$, and let $w_1, w_2 \in W_{\mathbb{R}}^{\perp} \otimes \mathbb{C}$ satisfy $\rho(w_1)\xi = \rho(w_2)\xi = 0$. Then $(w_1, w_2) = 0$, and under the hypotheses of Proposition 7.13 the component along ξ of $O = \int \omega\, (\iota_{w_1} A, \iota_{w_2} A)$ vanishes at every order at which they hold: its classical part $\{S^{(1)}(\xi), O\}$ and $Q_1(O)$ are 0, in every scheme in which the counterterms are equivariant for translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ and $\mathfrak{g}$. At one loop the order-$\hbar$ part of the twisted differential $Q_0^\xi + \hbar Q_1^\xi$ (Definition 8.22) vanishes on O identically: the arity-one component is $Q_1(O) = 0$, and the arity-two component along (ξ, ξ) has no output with two letters. This holds for every free direction of the one-loop quantisation, for every finite part ψ and independently of the normalisation of the transport map Φ of Theorem 11.65; through Φ, O is the untwisted form of $\operatorname{tr} \gamma^{(I}\gamma^{J)}$, and the statement is the case $n = 2$ of [BK, Sec. 3.4.1, (3.77)] at the level of classes.*

Proof. $\rho(w_1)\rho(w_2) + \rho(w_2)\rho(w_1) = 2(w_1, w_2)$ applied to $\xi \neq 0$ gives $(w_1, w_2) = 0$. By Corollary 7.14, for $(w_1, w_2) = 0$ the component is a multiple of $\int \omega \left[(\iota_{w_1} A, (\rho(w_2)\xi, \lambda)) + (\iota_{w_2} A, (\rho(w_1)\xi, \lambda))\right]$, which vanishes; the classical component is the same functional (proof of Corollary 7.14). In the evaluation of Theorem 11.54 the two-body part on these bilinears, $-2L\,\mathcal{B}_1 + 4L(\mathcal{B}_2 + \mathcal{B}_3)$ units, vanishes term by term, with $\mathcal{B}_1 = \{S^{(1)}(\xi), O\} = 0$ and $(w_1, w_2) = 0$, and the one-body term is a multiple of $\mathcal{B}_1$; by Corollary 11.58 the same holds with the computed solution and every free direction, the coefficients of $\mathcal{B}_1, \dots, \mathcal{B}_4$ multiplying functionals that vanish here, and $\hbar L_1(O)^{(0)}$ vanishes with (w_1, w_2). The component along (ξ, ξ) of a functional of degree 0 and Weyl weight 2 has degree -1 and Weyl weight 3 (Proposition 7.13, applied once for each argument), so an output with two letters contains an antifield, of Weyl weight at least $\frac{5}{2}$, and a field of weight at least 1, of total weight at least $\frac{7}{2} > 3$; the one-body term $-\{\hbar F_1^{(2)}(\xi, \xi), O\}$ vanishes, a quadratic functional of degree -2 and Weyl weight 5 having the legs (λ^*, λ^*) or (A, c^*) and no leg A^*. Φ maps classes to classes, so a vanishing differential stays zero under every normalisation. □

Remark 11.63 (The pair entries of the holomorphic twist)**.** The twisted one-loop differential is $\hbar Q_1^\xi(O) = \hbar L_1(O)^{(0)} + Q_1(O) - \frac{1}{2}\hbar L_1(O)^{(2)}(\xi, \xi) - \cdots$ (Definition 8.22), so its pair rule [BK, (3.7)–(3.8)] corresponds to the two-body kernels of Q_1 along ξ and of the arity-two component along (ξ, ξ). By Proposition 7.13, applied once for each argument, a component of arity k along ξ of an operator of Weyl weight w and degree d has representatives of Weyl weight $w + \frac{k}{2}$ and degree $d + 1 - k$. [BK, (3.51)] relates the holomorphic letters to the fields up to field redefinitions; with ${\xi_\alpha}^i = u_\alpha z^i$, $u' \in S_+$ with $u'^\alpha u_\alpha = 1$, and the representatives $\gamma \leftrightarrow A_{ij} z^j$ (Weyl weight 1, degree 0), $\beta \leftrightarrow u'^\alpha {\lambda_\alpha}^i$ modulo z^i $(\frac{3}{2}, 0)$, $\partial_{\dot\alpha} c \leftrightarrow \bar\lambda_{\dot\alpha i} z^i$ $(\frac{3}{2}, 0)$, $c \leftrightarrow c$ $(0, 1)$, $b \leftrightarrow u'^\alpha F_{mn} (\sigma^m \bar\sigma^n)_\alpha{}^\beta u'_\beta$ $(2, 0)$ or a component of λ^* $(\frac{5}{2}, -1)$, and $\partial_{\dot\alpha} \leftrightarrow u'^\alpha \sigma^m_{\alpha\dot\alpha} D_m$ $(1, 0)$, the pair entries of [BK, (3.64)–(3.69)] lie in the following arities:

entry	Weyl weight and degree of the pair	of the output	arity
$b\,c$ [BK, (3.64)]	$(2, 1)$	$(3, 0)$	2
$b\,b$ [BK, (3.65)]	$(4, 0)$	$(\frac{9}{2}, 0)$	1
$\beta\,\gamma$ [BK, (3.66)]	$(\frac{5}{2}, 0)$	$(3, 0)$	1
$\beta\,\beta$ [BK, (3.67)]	$(3, 0)$	$(\frac{7}{2}, 0)$	1
$b\,\gamma$ [BK, (3.68)]	$(3, 0)$	$(\frac{7}{2}, 0)$	1
$\beta\,b$ [BK, (3.69)]	$(\frac{7}{2}, 0)$	$(4, 0)$	1

The arities of the one-loop pair entries of the holomorphic twist, with b represented by the self-dual field strength; with b represented by a component of λ^* the entry $b\,c$ has arity 1.

The entry $\beta\gamma$, the Konishi entry, is the two-body kernel of Q_1 along ξ on the pair $(\lambda|_{S_+ \otimes W}, A|_{W_{\mathbb{R}}^{\perp}})$ with two outputs $\lambda|_{S_- \otimes W^*}$: the coefficient of $\int \omega\, (\bar\lambda_{\dot\alpha i} z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ in Q_1 of $\int \omega\, (u'^\alpha {\lambda_\alpha}^i, A_{ij} z^j)$. The classical component of this pair along ξ is a number κ times a cubic functional of the scalars, and a finite part of the operators of Weyl weight 3 that adds $y \int \omega\, (\bar\lambda_{\dot\alpha i} z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ to that cubic functional changes the entry by $-\kappa y$ (Proposition 11.52 (iv)), the two functionals having the same Weyl weight and quantum

numbers. The pair entries are therefore compared through the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ (Theorem 8.23), transported to the holomorphic letters through [ESW, Thm 10.9] (remark after Theorem 8.23). Proposition 11.62 is the exact untwisted form, for two letters, of [BK, (3.77)], $Q_1(\gamma^I\gamma^J)=0$: for $\rho(w_a)\xi=0$, $\iota_{w_a}A$ is a combination of the components $A_{ij}z^j$, the letters γ. Theorem 11.64 gives the arity-one entries at $\psi=0$, and Theorem 11.65 states what their comparison with [BK] tests.

Theorem 11.64 (The pair entries along a rank-one supercharge)**.** *Let $\xi=u\otimes z\in S_+\otimes W$ be of rank one, with u', the twisted letters and their representatives of Remark 11.63, and let $\psi=0$. Normalise the letters as the components along the weight vectors of a Cartan subalgebra of $\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^\perp)$ for which u, u' and z are weight vectors, read the outputs on the dual weight vectors, take b as the component $F(d_1,d_2)$ of the field strength on the two null vectors d_1, d_2 of $V\otimes\mathbb{C}$ along which $\partial_{\dot\alpha}$ differentiates, and expand the output momenta in the null coordinates; this fixes each entry up to one nonzero factor for each letter and each output. Reduce modulo $\operatorname{Im}Q_0^\xi$, total derivatives kept: the derivatives $u^\alpha\sigma^m_{\alpha\dot\alpha}\partial_m$ are Q_0^ξ-exact on the letters, and so is, up to terms with antifields, the holomorphic Dirac component of a derivative on an output $\partial_{\dot\alpha}c$. Then the arity-one pair entries of Q_1 are those of Table 11.6, and:*

- *(i) the entry $\beta\gamma$, the kernel of $(\lambda|_{S_+\otimes W},A|_{W_{\mathbb{R}}^\perp})$ on the leg set $(\lambda|_{S_-\otimes W^*})^2$, vanishes for every $\xi\in S_+\otimes W$, the placements on a line and at the special edge cancelling chain by chain; the one-body terms and $L_1^{(0)}$ have outputs of other Weyl weights or with a spectator letter, so the Konishi entry of Remark 11.63 is 0 at $\psi=0$; the entry bc with b represented by (ξ,λ^*) and the entry $\gamma\gamma$ vanish as well;*
- *(ii) the entries $\beta\beta$ and $\beta b\to\gamma\gamma$ carry the colour and derivative structure of [BK, (3.67)] and of the last term of [BK, (3.69)]: the output γ^K, resp. the outputs γ^J, γ^K, with the indices complementary to those of the letters β, the same value in the two colour orders, and, for $\beta b\to\gamma\gamma$, the antisymmetric combination of the two holomorphic derivatives;*
- *(iii) the entry $b\gamma$ has different values in the two colour orders, each with a term in $\log\frac{3}{2}$, and the entry $\beta b\to\beta\,\partial c$ is nonzero in one colour order;*
- *(iv) every derivative in the entries is holomorphic.*

Statements (i)–(iv) are independent of the normalisation factors; the values of the individual entries depend on the scheme (Proposition 11.52 (iv), Remark 11.63).

entry	pair → outputs	order (Y_1,Y_2)	order (Y_2,Y_1)	Dirac components
$\beta\gamma$ [BK, (3.66)]	$(\beta_I,\gamma^J)\to\partial c\,\partial c$	0	0	0, 0
bc [BK, (3.64)], $b\leftrightarrow(\xi,\lambda^*)$	$(c,b)\to\partial c\,\partial c$	0	0	0, 0
$\beta\beta$ [BK, (3.67)]	$(\beta_I,\beta_J)\to\gamma^K\,\partial c$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}-2L$, $\frac{1}{3}-2L$
$b\gamma$ [BK, (3.68)]	$(b,\gamma^I)\to\gamma^I\,\partial c$	$\frac{64}{3}-128L$	$32+64L$	$-\frac{64}{3}-128L$, $-\frac{32}{3}+64L$
$\beta b\to\beta\,\partial c$ [BK, (3.69)]	$(\beta_I,b)\to\beta_I\,\partial c$	$-\frac{64}{3}$	0	0, 0
$\beta b\to\gamma\gamma$ [BK, (3.69)]	$(\beta_I,b)\to\gamma^J\gamma^K$	$\frac{8}{3}\big(\partial_2\gamma^J\partial_1\gamma^K-\partial_1\gamma^J\partial_2\gamma^K\big)$	the same	
$\gamma\gamma$ [BK, (3.77)]	$(\gamma^I,\gamma^J)\to$ twisted letters	0	0	

The arity-one pair entries of Q_1 along $\xi=u\otimes z$ at $\psi=0$, reduced modulo $\operatorname{Im}Q_0^\xi$, in the normalisation of Theorem 11.64: coefficients of $\frac{\hbar}{16\pi^2}$ times the colour factor $(1\otimes\mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$ of the colour order (Y_1,Y_2), Y_1 the first output listed, with $L=\log\frac{3}{2}$. The third and fourth columns give the coefficient of the holomorphic derivative on the output other than ∂c, and for $\beta b\to\gamma\gamma$ the derivative structure, ∂_1 and ∂_2 being the derivatives along d_1 and d_2; the last column gives, in the two orders, the coefficient of the holomorphic Dirac component on the output ∂c, which is Q_0^ξ-exact up to terms with antifields; K is complementary to I,J, and J,K to I.

Proof. The entries are the kernels on the leg sets $(\lambda|_{S_-\otimes W^*})^2$ of $(\lambda|_{S_+\otimes W},A|_{W_{\mathbb{R}}^\perp})$; $A|_{W_{\mathbb{R}}^\perp}\lambda|_{S_-\otimes W^*}$ of $(\lambda|_{S_+\otimes W},\lambda|_{S_+\otimes W})$ and of $(A|_V,A|_{W_{\mathbb{R}}^\perp})$ with one derivative on $A|_V$; $\lambda|_{S_+\otimes W}\lambda|_{S_-\otimes W^*}$ and $(A|_{W_{\mathbb{R}}^\perp})^2$ of

$(\lambda|_{S_+\otimes W}, A|_V)$ with one derivative on $A|_V$; and $(\lambda|_{S_-\otimes W^*})^2$ of $(c, \lambda^*|_{S_+\otimes W^*})$ (Tables 11.6 and 11.6); all have superficial degree 0, so they carry no poles and no finite special vertex (Proposition 11.47), and Theorem 11.45 evaluates them exactly, with the letter vectors contracted exactly in the weight basis (Remark 11.66). The colour factors are those of Lemma 10.8 with the output of a term T_5 read as an output of a cubic vertex. (i) The vanishing of $\beta\gamma$ for general ξ and of bc holds in both evaluations. The outputs $(\lambda|_{S_-\otimes W^*})^2$ have Weyl weight 3, while $L_1^{(0)}$ preserves the Weyl weight $\frac{5}{2}$ of the Konishi descendant, a one-body term keeps the other letter as a spectator, and the arity-two kernels have outputs of degree -1; hence the coefficient of $\int \omega\,(\bar\lambda_{\dot\alpha i} z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ in $\hbar Q_1^\xi$ of the Konishi descendant is the entry $\beta\gamma$. $\gamma\gamma$ is Proposition 11.62. (ii)–(iv) are read off the table. Rescaling a weight vector rescales its letter by the inverse factor and its output by the factor, which multiplies an entry by one product of factors, the same for the two colour orders, for the reduced and the Dirac coefficients, and for the rational part and the part in $\log\frac{3}{2}$; the vanishing, the equality of the two orders, the indices and the derivative structure are therefore independent of the normalisation. □

Theorem 11.65 (What the comparison with the holomorphic twist tests)**.** *Let $\xi = u \otimes z$ be of rank one, let Φ be the quasi-isomorphism from the cohomology of Q_0^ξ on local operators to that of the free holomorphic differential on the letters c, γ, β, b, $\partial_{\dot\alpha}$ of [BK], transported through [ESW, Thm 10.9] (remark after Theorem 8.23), and let D be the difference, read through Φ, of the one-loop quantisation $r_\xi(\hbar F_1)$ of the twisted theory restricted from the equivariant one and the holomorphic quantisation of [BK, (2.42), footnote 22] and [W18, Thm 3.4] (B. R. Williams, Renormalization for holomorphic field theories, arXiv:1809.02661). If $D \in \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$, then $\Phi \circ [Q_1^\xi] \circ \Phi^{-1}$ is the class of the one-loop differential of [BK, (3.7), (3.64)–(3.69)], and:*

- *(a) the one-body terms of Q_1^ξ on the twisted letters are $a\,Q_0^\xi + [Q_0^\xi, N]$ for a number a and a derivation N of the letters, zero on cohomology;*
- *(b) on the pairs from $\{c, \gamma\}$ and on (β, c) the two-body kernels lie, on the twisted letters, in $\operatorname{Im} Q_0^\xi$ plus terms cancelled on cohomology by the one-body terms, as in [BK, (3.11)]; on (γ, γ) they vanish (Proposition 11.62);*
- *(c) on the pairs $\beta\gamma$, $\beta\beta$, $b\gamma$, βb, bb and bc the kernels, reduced modulo $\operatorname{Im} Q_0^\xi$ as in Theorem 11.64, have the colour factors $(1\otimes \mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$, which are $\frac{1}{2} f_{ACD} f_{BCE}$ of [BK] up to normalisation, and holomorphic derivatives $\partial_{\dot\alpha} = u'^\alpha \sigma^m_{\alpha\dot\alpha}\partial_m$ on the outputs;*
- *(d) their coefficients satisfy, modulo the finite parts of Proposition 11.52 (iv), the relations [BK, (3.59)–(3.63)], which are the order-$\hbar$ part of Corollary 8.17 (ii) at (ξ, R) for the elements R of the tree-level commutant $\mathfrak{psl}(3|3)$ of ξ, restricted by r_ξ;*
- *(e) every statement of the holomorphic twist about $[Q_1]$ on classes of several letters is the image under Φ of $[Q_1^\xi]$, computed by the pair rule from the kernels of (c), with the normalisation of $\hbar$ relative to the unit fixed by Φ.*

For a general D the two class maps differ by $[\{D, -\}]$, and (a)–(e) compute that class. On the chiral bilinears the comparison is complete at the level of classes: by Proposition 11.62 the order-$\hbar$ twisted differential vanishes on them for every free direction of the equivariant quantisation and every normalisation of Φ, which is the case $n = 2$ of [BK, (3.77)].

Proof. Two quantisations of the twisted theory differing by a Q_0^ξ-closed D of degree 0 have class maps differing by $[\{D, -\}]$, which vanishes for $D \in \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$, since $\{S^\xi, -\} = Q_0^\xi$ and $\{Q_0^\xi \chi, -\} = \pm[Q_0^\xi, \{\chi, -\}]$ (Theorem 8.23 and the remark after it). (a) The one-body terms are brackets with local functionals (Theorem 11.44 (iii)), and the condition on D makes them aQ_0^ξ plus commutators with

Q_0^{ξ} on the twisted letters. (b), (c) The pair rule [BK, (3.7)–(3.11)], the colour factors of Lemma 10.8 with two outputs, and the exactness of the antiholomorphic derivatives, $\{Q, \bar{Q}^1_{\dot\alpha}\} = P_{-\dot\alpha}$ [BK, (2.5)]. (d) Corollary 8.17 (ii) at (ξ, R) with $[\xi, R] = 0$, at order $\hbar$, mapped by r_ξ (Lemma 8.21). (e) $[Q_1^{\xi}]$ is determined by its values on representatives, polynomials in the letters, through the pair rule. □

Remark 11.66 (Method of computation)**.** Each defect cycle is evaluated exactly as in Remark 11.40: the Gaussian loop integral and the integrals of (11.8) over the ordered subsimplices of the region where α_σ is the smallest coordinate, the letter derivatives through Proposition 11.12 with $n + 1$ edges, the position along η and K_b through the injected momentum r, the fibre algebra in an explicit realisation of V_{10} and Σ over $\mathbb{Q}(i)$ with the vertices $S^{(1)}(x)$ of Chapter 2 and the sign of Definition 11.43 (ii), letter vectors with fractional components contracted exactly, and the colour words by Lemma 10.8; the twisted letters are built from the weights of a Cartan subalgebra. A second evaluation works in momentum space with an independently written realisation of $S^{(1)}(x)$ in the components of Definition 8.1, the special edge as the factor $e^{-\varepsilon|p|^2} - 1$, Feynman parameters for the other edges and numerical integration at several values of ε with thirty digits, extrapolated to ε^0 with the poles fitted; along K_b it evaluates the cycles at the endpoints $T = 1, 4, 16$. The two evaluations give the same operators C_ξ, Y_η, C_K and the same arrows of Lemma 11.42, the same composite line along ξ, the same free directions of Proposition 11.49, the same kernels Υ_ξ on every leg set with at most two outputs of the pairs of field and ghost letters without derivatives and with one letter derivative, the same coefficients of Theorem 11.54 for both chiralities, the same entries $\beta\gamma$, $\beta\beta$ and bc, and the same translation-invariant kernels along η and K_b where both are evaluated; both reproduce Example 11.46, Proposition 11.62 and the value along P_b of Corollary 11.56. The first evaluation gives the values along Δ of Corollary 11.56, extends Υ_ξ to three letter derivatives, and gives the entries of Table 11.6 and the identity (iv) of Theorem 11.54; the second gives the closed forms of Theorem 11.45 (iv) for $n_t = 2, \dots, 6$, the coefficients of ε^{-m} of the defect cycles on the pairs of gaugino letters and on the pair of Example 11.46, against the right side of (11.7), the position identity of Theorem 11.45 (ii), and, along K_b, $T\partial_T$ of the defect cycles of the intrinsic vertex $S^{(1)}(K_b)$ on $(A|_V, A|_V)$ with the output $A|_V$, with the vertex on a line and at the special edge, decreasing at least as ε at the three endpoints, as Theorem 11.45 (iii) states. The action-level obstructions of Proposition 11.50 and the arity-two classes of Theorem 11.51 are computed in the same two ways, on cycles of vertices of the action with one special edge, and the cohomology of Q on complete bases of the two-leg functionals fixed by representation counts. Every integral is a four-dimensional Schwinger integral with the cutoff ε on every edge.

Remark 11.67 (The data that complete the one-loop components)**.** The next computations are the following. On the action side: the arity-two equations along $(K_b, P_{b'})$ and (η, η') at a finite endpoint, which contain the coefficients θ_d^K and κ_j^K of Proposition 11.50 (v) and give the one-loop parts of $[K, P] = D + M$ and $\{S, S\} = K$, and along $(\xi_-, \eta_\pm)$; the three- and four-leg parts of the arity-one equations, which decide whether the sector parts of the generators extend to symmetries of the interacting theory, fix the differences of the sector coefficients (Proposition 11.25 (iii), Proposition 11.49 (ii)) and the couplings κ_k of Theorem 9.11; and the arity-three component of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$. On the operator side: Q_1, S_1, M_1, K_1 and P_1 along generic directions on the families of Definition 11.36; the arity-two operator kernels, with which the relations (R2)–(R6) of Proposition 11.61 become identities of local functionals before the projection to cohomology; and on the scalar bilinears the one-loop components on the level of Weyl weight $\frac{5}{2}$, with which (R2) and (R3), the Konishi relation between Q_1, S_1 and $D_1 = -48$ units, make the combinations of Corollary 11.58 (ii) scheme independent. The class-level comparison of Theorem 11.65 beyond the chiral bilinears uses the transport map Φ built from [ESW, Thm 10.9] with the normalisation of the letters b, β, γ, $\partial_{\dot\alpha} c$ of [BK], the condition on D, the arity-two kernels along (ξ, ξ), which give the entry bc with b represented by the self-dual field strength and the entry bb on the pair $(A|_V, A|_V)$ with two derivatives, and the relations [BK, (3.59)–(3.63)] applied to the entries of Table 11.6. The inner-product relations of Proposition 11.59 (iv) and Corollary

11.60 use Gram_1 from the two-point graphs of Definition 8.37.

Remark 11.68 (Two-component forms)**.** With the components of Definition 8.1:

- Letters. $A|_V$, $A|_{W_{\mathbb{R}}^{\perp}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ are A_m, A_{ij} (the $\mathbf{6}$ of $SU(4)_R$), $\lambda_\alpha{}^i$ (undotted, $\mathbf{4}$), $\bar\lambda_{\dot\alpha i}$ (dotted, $\bar{\mathbf{4}}$), with the antifields A_m^*, A_{ij}^*, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^* and the degrees $1,0,0,0,0,-1,-1,-1,-1,-2$ for $c, A_m, A_{ij}, \lambda_\alpha{}^i, \bar\lambda_{\dot\alpha i}, A_m^*, A_{ij}^*, \lambda^{*\alpha}{}_i, \bar\lambda^{*\dot\alpha i}, c^*$; $(\lambda,\lambda^*) = (\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})$ and $(u,w) = \frac14 u^{ij}w_{ij}$ on $W_{\mathbb{R}}^{\perp}$.

- Composite kernels. The internal vertex along Δ is $\int\big((2E+2)A_m, A_m^*\big) = 2\int x_n(\partial_n A_m, A_m^*) + 2\int(A_m, A_m^*)$ with $E = x_n\partial_n$, along P_b it is $\int(b_n\partial_n A_m, A_m^*)$; it sits between $\langle A_m A_n\rangle = 2(-,-)^{-1}\delta_{mn}$ and $\langle A_m^*\, c\rangle = -(-,-)^{-1} ip_m$, each times $e^{-\varepsilon|p|^2}|p|^{-2}$ at $T=\infty$, and the insertion factors are 2, $2i(l+q_c)_n$ and $ib_n(l+q_c)_n$. The special edge carries $e^{-\varepsilon|p|^2}-1$ times $(-,-)^{-1}$ and the inverse fibre pairing of (A_m, A_m^*), $\frac14(A^{ij}, A_{ij}^*)$, (λ,λ^*) and (c,c^*): on $(\lambda_\alpha{}^i, \lambda^{*\beta}{}_j)$ the factor $-|l+q_\sigma|^2\delta_\alpha^\beta\delta_j^i(-,-)^{-1}$, on $(\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\beta j})$ the opposite sign.

- Bilinears. For $O = \frac{1}{16}\int\omega\,(w_1^{ij}A_{ij}, w_2^{kl}A_{kl})$ the composite pair kernel along Δ is

$$\Big[-\tfrac13\big((2E+2)A_{ij}\big)A_{kl} - \tfrac13 A_{ij}\big((2E+2)A_{kl}\big) - \tfrac43 A_{ij}A_{kl}\Big] \text{ units},$$

  and along P_b it is $-\frac13$ units times $b_m\partial_m(A_{ij}A_{kl})$, with the same contraction.

- Two-site density. On adjacent pairs (A_{ij}, A_{kl}), (Z,ψ), (ψ, Z), (ψ,ψ) of a single trace the composite pair kernels give $-\frac43$ units (translation-invariant part along Δ) and $-\frac16$ units times $b_m\partial_m$ (along P_b) on the same word, and 0 on the exchanged word.

- Free directions. The sector parts of $S^{(1)}(\Delta)$ are $\frac14\int\big((2E+2)A^{ij}, A_{ij}^*\big)$, $-\int\big[\big((2E+3)\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i\big) - \big((2E+3)\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}\big)\big]$ and $\int\big((2E+2)A_m, A_m^*\big) - \int\big(2E(c), c^*\big)$; those of $S^{(1)}(P_b)$ are $\frac14\int(b_m\partial_m A^{ij}, A_{ij}^*)$, $-\int\big[(b_m\partial_m\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (b_m\partial_m\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big]$ and $\int(b_m\partial_m A_n, A_n^*) - \int(b_m\partial_m c, c^*)$. The exact directions are $Q\int x_m(A_m, c^*)$ and $Q\int(\rho(x)\lambda^*, \lambda^*) = -2i\,Q\int x_m\big(\lambda^{*\alpha}{}_i, \sigma^m_{\alpha\dot\alpha}\bar\lambda^{*\dot\alpha i}\big)$. The further class of Proposition 11.24 without the reflection is a multiple of $\int\big[(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big]$, the rotation $\lambda_\alpha{}^i \mapsto e^{i\varphi}\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}\mapsto e^{-i\varphi}\bar\lambda_{\dot\alpha i}$ of the free gaugino.

- Letter count. It counts 2 for each letter A_m, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ and 1 for each derivative ∂_m on it; (b) reads: the composite pair kernels along P_b sum to $-\frac13$ units times $b_m\partial_m$ on cohomology.

- Families and spectra. $n_m = \sigma^m_{1\dot1}$, $D_n = n_m(\partial_m + [A_m, -])$, $\psi = \lambda_1{}^i$, and F_{11} the self-dual field strength with both undotted indices 1. In Srednicki's normalisation, $A_m = -igA_m^aT^a$, $A_I = -igA_I^aT^a$, $\lambda = -\frac{ig}{\sqrt2}\lambda^aT^a$, $\mathrm{tr}(T^aT^b) = \frac12\delta^{ab}$, the unit is $-\frac{g^2N}{32\pi^2}$ and $-\frac13$ units is $\frac{g^2N}{96\pi^2}$. On the families $D_1 = -8$ units times the eigenvalues E of $\sum_l H_{l,l+1}$, which is $\frac{g^2N}{4\pi^2}E = 2\gamma$ with the one-loop anomalous dimension $\gamma = \frac{g^2N}{8\pi^2}E$: $\gamma = \frac{g^2N}{2\pi^2}h(S)$ on twist two, $\gamma = \frac{3g^2N}{4\pi^2}$ for $\mathrm{tr}(F_{11}F_{11})$ and the trace bilinear, the Konishi multiplet.

- Odd generators. $Q_1(O) = \xi_\alpha{}^i\,Q_1(O)^\alpha{}_i + \bar\xi_{\dot\alpha i}\,\bar Q_1(O)^{\dot\alpha i}$ and $S_1(O) = \eta^\alpha{}_i\,S_1(O)_\alpha{}^i + \bar\eta^{\dot\alpha i}\,\bar S_1(O)_{\dot\alpha i}$ (Definition 8.16); the Killing spinor of η has the components $(\rho(x)\eta)_\alpha{}^i = -i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$ and $(\rho(x)\eta)_{\dot\alpha i} = i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ (8.1), so on a family with $P_1\cong 0$ the position part along b of S_1 along η is, on cohomology, Q_1 along $\xi_\alpha{}^i = -i\,b_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\xi_{\dot\alpha i} = i\,b_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$. For $\xi_\alpha{}^i = u_\alpha z^i$, $(\rho(w)\xi)^\alpha{}_j = \varepsilon^{\alpha\beta}w_{ji}u_\beta z^i$, so $\rho(w)\xi = 0$ exactly when $w_{ij}z^j = 0$, and then $\iota_w A = \frac14 w^{ij}A_{ij}$ is a combination of the components $A_{ij}z^j$: Proposition 11.62 is the vanishing of Q_1 on $\int\omega\,(m_1^iA_{ij}z^j, m_2^kA_{kl}z^l)$, $m_1, m_2\in W$.

- Terms of $S^{(1)}(\xi)$. For $\xi = (\xi_\alpha{}^i, 0)$, with $(\xi^j\lambda^i) := \xi^{\alpha j}\lambda_\alpha{}^i$ and $F^0_{mn} = \partial_m A_n - \partial_n A_m$: $T_1 = -i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i, A^*_m\big)$, $T_2 = \frac12\varepsilon_{jikl}\big((\xi^j\lambda^i), A^{*kl}\big)$, $T_3 = -\frac14\big(F^0_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i, \lambda^{*\alpha}{}_i\big)$, $T_4 = -\frac{i}{2}\big(\xi^{\alpha j}\sigma^m_{\alpha\dot\alpha}\partial_m A_{ij}, \bar\lambda^{*\dot\alpha i}\big)$ and T_5 the three terms $-\frac14\big([A_m, A_n](\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i, \lambda^{*\alpha}{}_i\big)$, $-\frac{i}{2}\big(\xi^{\alpha j}\sigma^m_{\alpha\dot\alpha}[A_m, A_{ij}], \bar\lambda^{*\dot\alpha i}\big)$, $-\frac14\big([A^{ik}, A_{kj}]\xi_\alpha{}^j, \lambda^{*\alpha}{}_i\big)$; for $\xi_\alpha{}^i = u_\alpha z^i$ these are the terms of the remark after Theorem 8.23. The terms with $\bar\xi_{\dot\alpha i}$ follow by $\lambda_\alpha{}^i \leftrightarrow \bar\lambda_{\dot\alpha i}$, $\lambda^{*\alpha}{}_i \leftrightarrow \bar\lambda^{*\dot\alpha i}$, $\sigma \leftrightarrow \bar\sigma$. The intrinsic term along η is $s_0 = \big(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \lambda^{*\alpha}{}_k\big) - \big(A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}, \bar\lambda^{*\dot\beta k}\big)$, and $S^{(1)}(K_b) = \int\big\{(v.A_m, A^*_m) + \frac14(v.A^{ij}, A^*_{ij}) - (v.\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (v.\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (v.c, c^*)\big\}$ with $v^m = x^2 b^m - 2(b\cdot x)x^m$, $v.A_m = v^n\partial_n A_m + A_n\partial_m v^n$, $v.A_{ij} = v^n\partial_n A_{ij} - 2(b\cdot x)A_{ij}$, $v.\lambda_\alpha{}^i = v^n\partial_n\lambda_\alpha{}^i - \frac12(x_m b_n - x_n b_m)(\sigma^m\bar\sigma^n)_\alpha{}^\beta\lambda_\beta{}^i - 3(b\cdot x)\lambda_\alpha{}^i$, $v.c = v^n\partial_n c$.

- Arrows and lines. $\xi.$ maps $A_{kl} \mapsto \Gamma_{kl}(\xi,\lambda) = -2\varepsilon_{jikl}(\xi^j\lambda^i)$ and $A_m \mapsto \Gamma^m(\xi,\lambda) = i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$. On a plane wave C_ξ maps λ to c by $\lambda \mapsto -p_m\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$. The composite line along $\xi_\alpha{}^i$ joins a leg $\lambda_\alpha{}^i$ to a leg c through $\langle\lambda_\alpha{}^i\bar\lambda_{\dot\beta j}\rangle$, the term T_1 and $\langle A^*_m\, c\rangle$; with $\sigma^k_{\alpha\dot\beta}\bar\sigma^{m\dot\beta\gamma}p_k p_m = -|p|^2\delta^\gamma_\alpha$ it is $-2\,\xi_\alpha{}^i\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $c\otimes\lambda_\alpha{}^i$ and $+2\,\xi_\alpha{}^i\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $\lambda_\alpha{}^i\otimes c$. Along η, $(\rho(e_m)\eta)_\alpha{}^i = -i\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$ and $(\rho(e_m)\eta)_{\dot\alpha i} = i\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$; along K_b, $\operatorname{div} v = -8\,b\cdot x$, $\Sigma^{\mu\nu}$ acts on A_m as $(\Sigma^{\mu\nu}A)_m = \delta_{m\mu}A_\nu - \delta_{m\nu}A_\mu$ and on $\lambda_\alpha{}^i$ through $-\frac14(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)$, and C_K maps A_m to $-4b_m A_m$ in the slot of c.

- Bilinears along ξ. For $O = \frac{1}{16}\int\omega\,(w_1^{ij}A_{ij}, w_2^{kl}A_{kl})$,

$$\mathcal{B}_1 = \tfrac14\int\omega\,\big[(w_1^{kl}A_{kl}, w_{2\,ji}\xi^{\alpha i}\lambda_\alpha{}^j) + (1\leftrightarrow 2)\big], \qquad \mathcal{B}_2 = \tfrac14 w_1^{ij}w_{2\,ij}\int\omega\,(A_{ji}, \xi^{\alpha i}\lambda_\alpha{}^j),$$

  $\mathcal{B}_3 = \frac14 w_1^{ij}w_{2\,ij}\int\omega\,i\big(\bar\lambda_{\dot\alpha i}, A_m\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i\big)$ and $\mathcal{B}_4 = \frac14 w_1^{ij}w_{2\,ij}\int\omega\,(c, \xi_\alpha{}^i\lambda^{*\alpha}{}_i)$; Theorem 11.54 reads, in units $-\frac{g^2N}{32\pi^2}$,

$$Q_1(O)\big|_{\text{two-body}} = -2\log\tfrac32\,\mathcal{B}_1 + 4\log\tfrac32\,(\mathcal{B}_2 + \mathcal{B}_3),$$
$$L_1^{(0)}(O) = -\big(2 + 4\log\tfrac32\big)\tfrac14 w_1^{ij}w_{2\,ij}\int\omega\,(\partial_m c, A_m),$$

  and the cocycle finite part is $\psi(O) = -\big(1 + 2\log\frac32\big)\frac14 w_1^{ij}w_{2\,ij}\int\omega\,(A_m, A_m)$ units, $\psi(\mathcal{B}_1) = -2\,\mathcal{B}_3$ units. On the twenty traceless bilinears the transformation along Q^i_α is $1 + c_Q$ times the classical one, with c_Q of Theorem 11.54 (ii).

- Special supercharges and special conformal fields on the bilinears. The two-body part of $S_1(O)$ along $\eta^\alpha{}_i$ is that of $\bar Q_1(O)$ along $\bar\xi_{\dot\alpha i} = i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, and along $\bar\eta^{\dot\alpha i}$ that of $Q_1(O)$ along $\xi_\alpha{}^i = -i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$. On the part with two scalar legs, $K_1(O) = -\big(\frac13 + \theta\big)\big(v^m\partial_m O - 4(b\cdot x)O\big)$ units on the traceless bilinears and the same plus $48\,(b\cdot x)\,O$ units on the trace $\frac14\int\omega\,(A^{ij}, A_{ij})$.

- Twist. With $z = e_4$: $\gamma^I = A_{I4}$, $\beta_I = u'^\alpha\lambda_\alpha{}^I$ $(I = 1,2,3)$, $\partial_{\dot\alpha}c = \bar\lambda_{\dot\alpha 4}$, $b = u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$ or $(\xi,\lambda^*) = u_\alpha\lambda^{*\alpha}{}_4$, and $\partial_{\dot\alpha} = u'^\alpha\sigma^m_{\alpha\dot\alpha}D_m$, so that $\partial_2\gamma^J\partial_1\gamma^K - \partial_1\gamma^J\partial_2\gamma^K$ of Table 11.6 is a multiple of $\partial_{\dot\alpha}\gamma^J\partial^{\dot\alpha}\gamma^K$. The Konishi entry is the coefficient of $\int\omega\,(\bar\lambda_{\dot\alpha 4}, \bar\lambda^{\dot\alpha}{}_4)$ in $Q_1\big(\int\omega\,(u'^\alpha\lambda_\alpha{}^i, A_{i4})\big)$, 0 at $\psi = 0$; [BK, (3.66)] is $Q_1\big((\beta_I)^A(\gamma^J)^B\big) = \frac12\delta^J_I f_{ACD}f_{BCE}\,\partial_{\dot\alpha}c^D\partial^{\dot\alpha}c^E$, and the entries $\beta\beta$, $b\gamma$, βb of [BK, (3.67)–(3.69)] have the same factor $\frac12 f_{ACD}f_{BCE}$ in both terms.

- Action-side data along Σ and Σ^*. For $\xi = (\xi_\alpha{}^i, 0)$, with $\Gamma^m_V = i\,\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i$ and $(\Gamma_\perp)_{kl} = 2\varepsilon_{jikl}\,\xi_\alpha{}^j\varepsilon^{\alpha\beta}\lambda_\beta{}^i$, the functionals of Proposition 11.50 are $g_1 = \int d^4x\,(\Gamma^n_V, -\partial^2 A_n + \partial_n\partial_m A_m)$, $g_2 = -\int d^4x\,(\Gamma^n_V, \partial_n\partial_m A_m)$, $g_3 = -\frac14\int d^4x\,((\Gamma_\perp)_{kl}, \partial^2 A^{kl})$, $g_4 = -\int d^4x\,(\partial^2 c, \xi_\alpha{}^i\lambda^{*\alpha}{}_i)$ and $T_0 = -\int d^4x\,(\partial_m A_m, \xi_\alpha{}^i\lambda^{*\alpha}{}_i)$, and the solution along ξ is $\big[(\log\frac32 - \frac{79}{12})(T_1+T_2) + \theta^\xi_V(T_1+T_3) + \theta^\xi_\perp(T_2+T_4)\big]$ units with the terms of the bullet above. For $\eta = (\eta^\alpha{}_i, 0)$, $s_0 = \int d^4x\,(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \lambda^{*\alpha}{}_k)$ and $u_0 = -i\int d^4x\,(A_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}, \bar\lambda^{*\dot\alpha i})$. Along M_ω and K_b the products $\rho_\mu\rho_\nu$ of the spin terms are $\sigma^\mu\bar\sigma^\nu$ on the undotted and $\bar\sigma^\mu\sigma^\nu$ on the dotted components, and $(\iota_b A, \iota_x A^*) = (b_m A_m, x_n A^*_n)$.

- Relations among the coefficients. (11.11) reads $t_Q + t_{\bar{Q}} - t_P = t_Q + t_S - t_\Delta = \frac{27}{4} + 2\log\frac{3}{2}$ for the coefficients along Q^i_α ($\xi_\alpha{}^i$), $\bar{Q}_{\dot{\alpha}i}$ ($\bar{\xi}_{\dot{\alpha}i}$), $S^\alpha{}_i$ ($\eta^\alpha{}_i$) and P_m, with $t_\Delta = t_M = -\frac{1}{3}$ and $t_R = 0$; the outer direction moves t_Q, $t_{\bar{S}}$ by $+a$ and $t_{\bar{Q}}$, t_S by $-a$.

- Components on the bilinears. In units $-\frac{g^2N}{32\pi^2}$, $Q_1(O) = \big(\frac{79}{12} - 3\log\frac{3}{2} - t_Q\big)\mathcal{B}_1 + 4\log\frac{3}{2}\,(\mathcal{B}_2 + \mathcal{B}_3)$ along $\xi_\alpha{}^i$, $S_1(O) = \big(\frac{79}{12} - 3\log\frac{3}{2} - t_S\big)\{S^{(1)}(\eta), O\}$ plus the two-body terms $4\log\frac{3}{2}\,(\mathcal{B}_2 + \mathcal{B}_3)$ at the spinor $\bar{\xi}_{\dot{\alpha}i} = i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot{\alpha}}$ along $\eta^\alpha{}_i$, $D_1 = 0$ and -48 units, $P_1(O) = -(\frac{1}{3} + t_P)\,b_m\partial_m O$, $M_1(O) = -(\frac{1}{3} + t_M)\,\omega_{mn}x_n\partial_m O$, $R_1 = 0$, and $K_1(O) = -(\frac{1}{3} + \theta^K_\perp)\big(v^m\partial_m O - 4(b\cdot x)O\big)$ plus $48\,(b\cdot x)\,O$ on the trace and the brackets with G^K_j; the invariant combination of Corollary 11.58 (ii) is $\frac{27}{4} - 8\log\frac{3}{2}$ units.

- Weinberg. On the Dirac column $(\lambda_\alpha{}^i, \bar{\lambda}^{\dot{\alpha}}{}_i)$, $\rho(v) = \gamma^\mu_W v_\mu$ gives Weinberg's form of each line, as in Remark 10.50.

Chapter 12

The one-loop equivariant structure from the quantum master equation

Chapters 9–11 computed the one-loop components of local operators from two-body kernels, and several inputs of that computation were identities read off the structure of the regulator: the dilatation component as four times the logarithmic coefficient, the coefficient $\theta = -\frac{1}{3}$ from the arity-two component of $L \circ L = 0$, the special supercharges from the supercharges by the position decomposition, $K = P^{\dagger}$, and the comparison with the holomorphic twist. This chapter derives every one-loop component from the equivariant quantum master equation at scale ε: the obstruction of Definition 8.7 along each generator is computed from the graphs with one special edge, the equation $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ is solved arity by arity with all its free directions, and the components on local operators follow from the defect flow of Lemma 8.4 through the local representative of Definition 8.14. The construction needs new input at three places: along the supercharges the composite line has a nonzero image under Q, so the obstruction is a finite sum of regulator graphs computed graph by graph; the free directions of the arity-one solution are fixed by the arity-two equations, which therefore carry the coefficient θ; and the finite counterterm vertices of Chapters 10 and 11 cancel against the bridges, so that the two-body kernels are the finite parts of the defect cycles themselves. The former shortcuts become theorems of the construction or cross-checks, and every component is stated as a representative together with its class, the part that is independent of the scheme.

The unit is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$, equal to $-\frac{g^2 N}{32\pi^2}$ for $\mathfrak{g} = \mathfrak{su}(N)$ (Remark 6.13), and a number followed by "units" is that number times the unit. Edges carry $P(\varepsilon, T)$ of Definition 8.3; the special edge is $K_\varepsilon - K_0 = -\int_0^\varepsilon (H \otimes 1) K_t\, dt$, the factor $e^{-\varepsilon|p|^2} - 1$ times the components of K_0 (Proposition 4.27). The endpoint T is the endpoint of the transfer of [C07]; every component below is independent of it (Theorem 8.15 (ii)), and it is taken as $T = \infty$ where the graph sums converge at generic momenta (Proposition 8.25) and finite along K_b and along the pairs $(K_b, P_{b'})$ and (η, η') with $\Gamma(\eta, \eta')|_V \neq 0$ (Definition 8.26). The bracket is that of Definition 1.6, and $s = d_{CE} + \{S_{\mathfrak{psl}(4|4)}, -\}$ on cochains (Theorem 8.8). We write $\mathcal{N} := 2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$ for the field-counting functional of Chapter 6, whose bracket multiplies the coordinate functionals of c, A, λ, A^*, λ^*, c^* by 1, 2, 4, -2, -4, -1 (Lemma 6.7). Every number of this chapter is computed by the two independent evaluations of Remark 12.69, unless a statement says otherwise; the two-component forms of Definition 8.1 of the main formulas are collected in Remark 12.72.

12.1 The component equations of the quantum master equation

Definition 12.1 (Generators, their data and the sector parts)**.** For a homogeneous generator x of $\mathfrak{psl}(4|4)$ let λ_x be its ad_Δ eigenvalue (Remark 2.6), p_x the degree of its Killing data in the position and

$|x|$ its parity. The functional $S^{(1)}(x)$ of Definition 2.12 has degree -1 and Weyl weight $4-\frac{1}{2}\lambda_x$, the position counted at weight -1. We write $\xi_+ \in S_+ \otimes W$ and $\xi_- \in S_- \otimes W^*$ for the two chiralities of $\xi \in \Sigma$, $\eta_+ \in S_+ \otimes W^*$ and $\eta_- \in S_- \otimes W$ for those of $\eta \in \Sigma^*$, M_ω for the rotation of V with $\omega \in \mathfrak{so}(V)$ and R_r for the rotation of $W^\perp_{\mathbb{R}}$ with $r \in \mathfrak{so}(W^\perp_{\mathbb{R}})$, acting on λ by its spin lift.

x	λ_x	p_x	$\lvert x\rvert$	terms of $S^{(1)}(x)$	endpoint
Δ, $v=2E$	0	1	even	quadratic	$T=\infty$
P_b, $v=b$	-2	0	even	quadratic	$T=\infty$
M_ω, $v=\omega x$	0	1	even	quadratic	$T=\infty$
R_r	0	0	even	quadratic, on $A\lvert_{W^\perp_{\mathbb{R}}}$ and λ	$T=\infty$
$\xi \in \Sigma$	-1	0	odd	$T_1,\dots,T_4$ quadratic, T_5 cubic	$T=\infty$
$\eta \in \Sigma^*$	1	1	odd	$\sum_\mu x^\mu S^{(1)}(\rho(e_\mu)\eta)+\int s_0$	$T=\infty$
K_b, $v=(x,x)b-2(b,x)x$	2	2	even	quadratic	$T<\infty$

The generators and their data; $T_1,\dots,T_5$ are the terms of Table 11.6 and $s_0=-(\rho(A\lvert_{W^\perp_{\mathbb{R}}})\eta,\lambda^*)$ is the term of (11.3).

For even x with vector field v the sector parts of $S^{(1)}(x)$ are

$$S^{(1)}(x)_\perp := \int \big(v.A\lvert_{W^\perp_{\mathbb{R}}}, A^*\lvert_{W^\perp_{\mathbb{R}}}\big), \qquad S^{(1)}(x)_\lambda := -\int (v.\lambda,\lambda^*), \qquad S^{(1)}(x)_V := \int \big(v.A\lvert_V, A^*\lvert_V\big) - \int (v.c,c^*),$$

each Q-closed, $v.$ commuting with Q (Propositions 7.6, 7.10); along R_r the sector V is absent. For $\xi \in \Sigma$ the sector parts are $S_2^V(\xi)=T_1+T_3$ and $S_2^\perp(\xi)=T_2+T_4$ (Definition 11.41); for $\eta \in \Sigma^*$ they are

$$S^V(\eta) := \sum_\mu \int x^\mu S_2^V(\rho(e_\mu)\eta), \qquad S^\perp(\eta) := \sum_\mu \int x^\mu S_2^\perp(\rho(e_\mu)\eta) + \int s_0,$$

whose sum is the quadratic part of $S^{(1)}(\eta)$ (Proposition 11.49 (iv)). The index d of a sector runs over $\{\perp,\lambda,V\}$ for even x and over $\{V,\perp\}$ for odd x. The quadratic part of a solution $\hbar F_1^{(1)}(x)$ contains $\sum_d \theta_d^x$ units times the sector parts; $\theta_\perp^\Delta$, θ_λ^Δ, θ_V^Δ are θ, θ_λ, θ_V of Definition 11.23 and $\theta_d^{P_b}$ are θ_P, $\theta_{P,\lambda}$, $\theta_{P,V}$. When the θ_d^x are equal, their common value t_x is the uniform coefficient: the coefficient, in units, of $S^{(1)}(x)$ with its cubic part.

Definition 12.2 (Components of cochains, of the obstruction and of the logarithmic coefficient)**.** A local cochain f of $C^\bullet(\mathfrak{psl}(4|4),\mathscr{O}_l(\mathscr{E}))$ has components $f^{(k)}(x_1,\dots,x_k)$ (Definition 2.7), normalised as in Lemma 2.11: the arity-one component of $d_{CE}f+\frac{1}{2}\{f,f\}$ at $f=S_{\mathfrak{psl}(4|4)}$ is $\{S,S^{(1)}(x)\}$, and its arity-two component is $\{S,S^{(2)}(x,y)\}+S^{(1)}([x,y])+(-1)^{|x|}\{S^{(1)}(x),S^{(1)}(y)\}$. $\mathrm{Ob}^{(k)}$ and $\hbar F_1^{(k)}$ are the components of the one-loop obstruction of Definition 8.7 and of the finite counterterm $\hbar F_1$ of Definition 8.5 in this normalisation. We write $I^{CT,(k)}$ for the components of the counterterm cochain $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$, and L for its coefficient of $\log\varepsilon$, with components $L^{(0)}$, $L^{(1)}(x)$, $L^{(2)}(x,y)$; $L^{(0)}$ is the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ and $L^{(1)}(x)$ that in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, as in Theorem 11.44 (iv).

Lemma 12.3 (The derivation rule of the bracket)**.** *Let $\{f,g\}_{\mathrm{std}}$ be the odd bracket with $\Delta_0(fg)=\Delta_0(f)g+(-1)^{|f|}f\Delta_0(g)+(-1)^{|f|}\{f,g\}_{\mathrm{std}}$. Then the bracket of Definition 1.6 is $\{f,g\}=(-1)^{|f|}\{f,g\}_{\mathrm{std}}$, and every derivation $\mathcal{D}$ of degree 1 of $\{,\}_{\mathrm{std}}$, in particular Q, $\{I,-\}$, $Q+\{I,-\}$ and $\{F,-\}$ for F of degree 0, satisfies*

$$\mathcal{D}\{a,b\} = -\{\mathcal{D}a,b\} - (-1)^{|a|}\{a,\mathcal{D}b\}. \tag{12.1}$$

Proof. Definition 1.6 gives $\{f,g\}=\Delta_0(fg)-\Delta_0(f)g-(-1)^{|f|}f\Delta_0(g)$, which is $(-1)^{|f|}\{f,g\}_{\mathrm{std}}$. A derivation of degree 1 satisfies $\mathcal{D}\{a,b\}_{\mathrm{std}}=\{\mathcal{D}a,b\}_{\mathrm{std}}+(-1)^{|a|+1}\{a,\mathcal{D}b\}_{\mathrm{std}}$; multiplying by $(-1)^{|a|}$ and using $|\mathcal{D}a|=|a|+1$,

$$\mathcal{D}\{a,b\} = (-1)^{|a|}(-1)^{|a|+1}\{\mathcal{D}a,b\} - \{a,\mathcal{D}b\}_{\mathrm{std}} = -\{\mathcal{D}a,b\} - (-1)^{|a|}\{a,\mathcal{D}b\}.$$

□

Lemma 12.4 (Components of the defect vertex of the action)**.** *Let $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \hbar F_1$ and let $\mathfrak{d} := (d_{CE}+Q)J + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J$ be the defect vertex of Lemma 8.4. Write $\{f,g\}_\varepsilon = \{f,g\} + \{f,g\}^\sigma_\varepsilon$, the second term being minus f and g joined by one special edge, and $I^{CT} := I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$. Modulo $\hbar^2$ and constants,*

$$
\begin{aligned}
\mathfrak{d}^{(0)} &= \tfrac{1}{2}\{I,I\}^\sigma_\varepsilon - Q\, I^{CT,(0)} - \{I, I^{CT,(0)}\}_\varepsilon - \hbar\big(QF_1^{(0)} + \{I, F_1^{(0)}\}_\varepsilon\big),\\
\mathfrak{d}^{(1)}(x) &= \{I, S^{(1)}(x)\}^\sigma_\varepsilon + \hbar\Delta_\varepsilon S^{(1)}(x) - Q\, I^{CT,(1)}(x) - \{I, I^{CT,(1)}(x)\}_\varepsilon - \{I^{CT,(0)}, S^{(1)}(x)\}_\varepsilon\\
&\quad - \hbar\big(QF_1^{(1)}(x) + \{I, F_1^{(1)}(x)\}_\varepsilon + \{F_1^{(0)}, S^{(1)}(x)\}_\varepsilon\big),\\
\mathfrak{d}^{(2)}(x,y) &= (-1)^{|x|}\{S^{(1)}(x), S^{(1)}(y)\}^\sigma_\varepsilon + \{I, S^{(2)}(x,y)\}^\sigma_\varepsilon - \Big[Q\, I^{CT,(2)}(x,y) + \{I, I^{CT,(2)}(x,y)\}_\varepsilon + I^{CT,(1)}([x,y])\\
&\quad + (-1)^{|x|}\big(\{I^{CT,(1)}(x), S^{(1)}(y)\}_\varepsilon + \{S^{(1)}(x), I^{CT,(1)}(y)\}_\varepsilon\big) + \{I^{CT,(0)}, S^{(2)}(x,y)\}_\varepsilon\Big] - \hbar\big[\textit{the same with } F_1 \textit{ i}
\end{aligned}
\tag{12.2}
$$

with $\hbar\Delta_\varepsilon I = 0$, $\hbar\Delta_\varepsilon S^{(1)}(x)$ a constant for even x and 0 for odd x, and $\hbar\Delta_\varepsilon S^{(2)}(x,y) = 0$.

Proof. The part of order $\hbar^0$ without special edge is $d_{CE} I_{\mathfrak{psl}(4|4)} + \frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\}$, which vanishes by (2.4); its arity components are those of Lemma 2.11, and the remainder is the special-edge part of each bracket. The terms linear in I^{CT} and in F_1 are the linearisation of the same component formulas at $S_{\mathfrak{psl}(4|4)}$ (Lemma 12.6 (i)), with the bracket at scale ε. Δ_ε contracts a field with its own antifield inside one monomial: in I it gives $\operatorname{tr}\operatorname{ad} = 0$ (proof of Theorem 4.30); in $S^{(1)}(x)$ for even x the terms $(v.e, e^*)$ give the field-independent supertrace of $v.\,e^{-\varepsilon H}$; the quadratic terms of $S^{(1)}(\xi)$ and $S^{(1)}(\eta)$ pair λ with A^* and A with λ^*, the cubic terms have the legs A, A, λ^*, and s_0 the legs $A|_{W^\perp_{\mathbb{R}}}$, λ^*; the terms of $S^{(2)}$ in (2.8) have the legs (λ^*,λ^*) and (A, c^*). □

Definition 12.5 (Arity components of the obstruction and residuals)**.** With $F_1 = 0$, Ob is the local cochain of degree 1 of Definition 8.7: its tree-level flow with edges $P(0,T)$ is the coefficient of $\hbar$ of the $\varepsilon \to 0$ limit of the connected graphs with one vertex $\mathfrak{d}$, the other vertices from J and edges $P(\varepsilon, T)$. In the normalisation of Definition 12.2 the equation $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ of Theorem 8.8 (ii) reads, arity by arity,

$$
\begin{aligned}
&(Q + \{I,-\})\hbar F_1^{(0)} = \hbar\,\mathrm{Ob}^{(0)}, \qquad (Q + \{I,-\})\hbar F_1^{(1)}(x) + \big\{\hbar F_1^{(0)}, S^{(1)}(x)\big\} = \hbar\,\mathrm{Ob}^{(1)}(x),\\
&(Q + \{I,-\})\hbar F_1^{(2)}(x,y) + \hbar F_1^{(1)}([x,y]) + (-1)^{|x|}\Big(\big\{\hbar F_1^{(1)}(x), S^{(1)}(y)\big\} + \big\{S^{(1)}(x), \hbar F_1^{(1)}(y)\big\}\Big) + \big\{\hbar F_1^{(0)}, S^{(2)}(x,y)\big\} =
\end{aligned}
\tag{12.3}
$$

Given a solution of the first equation, the arity-one residual is $\mathcal{R}^{(1)}(x) := \hbar\,\mathrm{Ob}^{(1)}(x) - \{\hbar F_1^{(0)}, S^{(1)}(x)\}$; given also a solution of the second, the arity-two residual $\mathcal{R}^{(2)}(x,y)$ is $\hbar\,\mathrm{Ob}^{(2)}(x,y)$ minus the terms of the third equation that contain $F_1^{(1)}$ and $F_1^{(0)}$. $\mathrm{Ob}^{(1)}(x)$ is local of degree 0 and Weyl weight $4 - \frac{1}{2}\lambda_x$, $\mathrm{Ob}^{(2)}(x,y)$ of degree -1 and weight $4 - \frac{1}{2}(\lambda_x + \lambda_y)$, and $\hbar F_1^{(1)}(x)$, $\hbar F_1^{(2)}(x,y)$ have degrees -1, -2 and the same weights. The arity-one equation with $\hbar F_1^{(0)}$ in the form of (11.10) is the second equation, $\{\hbar F_1^{(0)}, S^{(1)}(x)\} = \{S^{(1)}(x), \hbar F_1^{(0)}\}$ for $F_1^{(0)}$ even of degree 0.

Lemma 12.6 (Component formulas of s and the Wess–Zumino identities)**.**

- *(i) For a cochain F of degree 0, $(sF)^{(0)} = (Q + \{I,-\})F^{(0)}$, $(sF)^{(1)}(x) = (Q + \{I,-\})F^{(1)}(x) + \{F^{(0)}, S^{(1)}(x)\}$, and $(sF)^{(2)}(x,y)$ is the left side of the third equation of (12.3) with F in place of $\hbar F_1$.*
- *(ii) For a cochain G of degree 1, $(sG)^{(1)}(x) = (Q + \{I,-\})G^{(1)}(x) + \{G^{(0)}, S^{(1)}(x)\}$ and*

  $(sG)^{(2)}(x,y) = (Q+\{I,-\})G^{(2)}(x,y) - G^{(1)}([x,y]) + (-1)^{|x|}\{G^{(1)}(x), S^{(1)}(y)\} - \{S^{(1)}(x), G^{(1)}(y)\} + \{G^{(0)}, S^{(2)}(x,$

- *(iii) (Wess–Zumino.) Since* $s\,\mathrm{Ob}=0$ *(Theorem 8.8 (i)),*

$$(Q+\{I,-\})\mathrm{Ob}^{(0)}=0,\qquad (Q+\{I,-\})\mathrm{Ob}^{(1)}(x)=-\big\{\mathrm{Ob}^{(0)},S^{(1)}(x)\big\},$$
$$(Q+\{I,-\})\mathrm{Ob}^{(2)}(x,y)=\mathrm{Ob}^{(1)}([x,y])-(-1)^{|x|}\big\{\mathrm{Ob}^{(1)}(x),S^{(1)}(y)\big\}+\big\{S^{(1)}(x),\mathrm{Ob}^{(1)}(y)\big\}-\big\{\mathrm{Ob}^{(0)},S^{(2)}(x,y)\big\}\tag{12.4}$$

and the residuals are closed: $(Q+\{I,-\})\mathcal{R}^{(1)}(x)=0$ *when the arity-zero equation holds, and* $(Q+\{I,-\})\mathcal{R}^{(2)}(x,y)=0$ *when the equations of arity zero and one hold.*

Proof. Lemma 9.24 and its proof, with $\Lambda^{(0)}=Q+\{I,-\}$ and $S(x)=S^{(1)}(x)$; the components of arity zero are $(sF)^{(0)}=(Q+\{I,-\})F^{(0)}$ and $(s\,\mathrm{Ob})^{(0)}=0$, and the closedness of $\mathcal{R}^{(1)}(x)$ is the arity-one component of $s\big(\hbar\,\mathrm{Ob}-s(\hbar F_1)\big)=0$ when the arity-zero component of $\hbar\,\mathrm{Ob}-s(\hbar F_1)$ vanishes. □

Proposition 12.7 (Pole structure of the counterterm cochain)**.** *At one loop*

$$I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]=\tfrac{9}{4}\,\varepsilon^{-1}\int \mathrm{dvol}\,(A,A)\ \textit{units}+\log\varepsilon\cdot L,\tag{12.5}$$

with L *a local cochain of degree* 0*: the components of arity one and two have no pole.*

Proof. The coefficient of ε^{-m} of $I^{CT,(k)}(x_1,\dots,x_k)$ is a colour-singlet local functional of degree $-k$, total Weyl weight $4-\frac12\sum_i\lambda_{x_i}-2m$ and position degree at most $\sum_i p_{x_i}$ (Proposition 10.47). A colour singlet of degree -1 has an antifield and a second adjoint field, field weight at least 4, or $\frac72$ with the legs (A,λ^*) when the generator supplies a spinor; one of degree -2 has field weight at least 5, or $\frac{11}{2}$ with an odd number of spinor legs. The total weight is at least the field weight minus the position degree, and for $m\geq1$ the inequality fails in every case:

$$\Delta:\ 4-2m<3,\quad P_b:\ 5-2m<4,\quad M_\omega:\ 4-2m<3,\quad R_r:\ 4-2m<4,\quad \Sigma:\ \tfrac92-2m<\tfrac72,\quad \Sigma^*:\ \tfrac72-2m<\tfrac52$$
$$(\Delta,P_b):\ 5-2m<4,\quad (K_b,P_{b'}):\ 4-2m<3,\quad (\xi,\xi'):\ 5-2m<5,\quad (\xi,\eta):\ 4-2m<4,$$
$$(\xi,P_b):\ \tfrac{11}{2}-2m<\tfrac{11}{2},\quad (\eta,\eta'):\ 3-2m<3,\quad (\Delta,\xi):\ \tfrac92-2m<\tfrac92.$$

In arity zero the weight $4-2m$ with $m\geq1$ leaves the legs (A,A) at weight 2, the term of Theorem 4.16. One loop gives integer powers of ε and the first power of $\log\varepsilon$ (Lemma 4.6). □

Proposition 12.8 (The logarithmic coefficient is s-closed)**.** $sL=0$*. In components,* $(Q+\{I,-\})L^{(0)}=0$*,* $(Q+\{I,-\})L^{(1)}(x)+\{L^{(0)},S^{(1)}(x)\}=0$*, and the arity-two formula of Lemma 12.6 (i) applied to* L *vanishes. With* $L^{(0)}=(Q+\{I,-\})\mathcal{N}$ *units (Theorem 4.22),*

$$L^{(1)}(x)-\big\{\mathcal{N},S^{(1)}(x)\big\}\ \textit{units}\tag{12.6}$$

is $(Q+\{I,-\})$*-closed of degree* -1*.*

Proof. By [C07, Thm B] and Definition 8.5 the flow $W(P(\varepsilon,T),I_{\mathfrak{psl}(4|4)}-I^{CT})$ has a limit; Q acts on plane-wave values by polynomial substitutions, so the flow of $\mathfrak{d}$ has a limit (Lemma 8.4). In this flow at order $\hbar$ the coefficient of $\log\varepsilon$ comes from the trees of the counterterm vertices of (12.2): their parts without special edge give $-\log\varepsilon\,(sL)$, their special-edge parts carry $(e^{-\varepsilon|p|^2}-1)\log\varepsilon\to0$; the cycles with a special edge have no $\log\varepsilon$ (Lemma 12.10), and the subgraphs of the reducible graphs are of these two kinds. The tree-level flow is injective on local cochains (Theorem 8.15 (i)), so $sL=0$. For the last statement, (12.1) with $(Q+\{I,-\})S^{(1)}(x)=0$ gives $(Q+\{I,-\})\{\mathcal{N},S^{(1)}(x)\}=-\{(Q+\{I,-\})\mathcal{N},S^{(1)}(x)\}$, which is $-\{L^{(0)},S^{(1)}(x)\}=(Q+\{I,-\})L^{(1)}(x)$ in units. □

12.2 The arity-one obstruction from the defect cycles

Lemma 12.9 (Placements of a vertex $S^{(1)}(x)$ on a cycle)**.** *The propagator has the components $A\otimes A$, $c\otimes A^*|_V$, $A^*|_V\otimes c$ and $\lambda\otimes\lambda$ (Proposition 3.17), and a special edge joins a field slot to its antifield slot. On a cycle with exactly one special edge a vertex $S^{(1)}(x)$ occurs in the placements of the following table and in no other.*

x	*on a line*	*at the special edge*	*cubic*
Δ, P_b, M_ω, K_b	$\int(v.A\|_V, A^*\|_V)$, *a line from c to $A\|_V$*	*the five slot pairs*	*none*
R_r	*none*	$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_{W_{\mathbb{R}}^\perp})$, (λ,λ^*)	*none*
ξ	T_1, *a line from λ to c*	$T_1,\dots,T_4$	T_5
η	$x^\mu T_1(\rho(e_\mu)\eta)$	$x^\mu T_1,\dots,x^\mu T_4$ *at $\rho(e_\mu)\eta$, and s_0*	$x^\mu T_5(\rho(e_\mu)\eta)$

The placements of the terms of $S^{(1)}(x)$ on a cycle with one special edge. At the special edge one leg of the term lies on $K_\varepsilon - K_0$ and the other on an edge P; a cubic term has two legs A on the cycle and its third leg external.

A vertex $S^{(2)}(x,y)$ has no line placement; its legs λ^*, c^* lie on the special edge or are external.

Proof. A vertex on a cycle uses two of its legs on cycle edges. The legs λ^*, c^* and $A^*|_{W_{\mathbb{R}}^\perp}$ have no component of P, so they lie on the special edge or are external; $A^*|_V$ lies on $c\otimes A^*|_V$. For even x the term with the leg $A^*|_V$ is $\int(v.A|_V, A^*|_V)$, and $v.$ vanishes on $A|_V$ and c along R_r; for ξ it is T_1 (Table 11.6); along η the terms are those of (11.3). For $S^{(2)}$ the legs are those of (2.8). □

Lemma 12.10 (Special-edge lemma)**.** *Let $\mathcal{C}$ be a cycle all of whose vertices lie on it, with external legs at generic momenta, n_t edges, one special edge e_σ and the other edges $P(\varepsilon,T)$, possibly one composite line, with vertices from I and from $S^{(1)}(x)$ and $S^{(1)}(y)$ or $S^{(2)}(x,y)$; along η, Δ, M_ω and K_b the position factors at the vertex $S^{(1)}(x)$ are realised by derivatives in a momentum r injected there, at $r=0$. Let D be its superficial degree and F_m the m-th simplex integrand of Definition 10.21. At $T=\infty$*

$$\mathcal{C}(\varepsilon)=\sum_{m\ge 1}\varepsilon^{-m}a_m+a_0+o(1),\qquad 16\pi^2 a_m=\frac{1}{m}\int_{\alpha_\sigma\ \text{smallest}}\Big[\big(\min_{f\neq\sigma}\alpha_f\big)^m-\alpha_\sigma^m\Big]F_m\,d\alpha,\qquad 16\pi^2 a_0=\int_{\alpha_\sigma\ \text{smallest}}F_0\log \tag{12.7}$$

with no term in $\log\varepsilon$; a_m is a polynomial of degree $D-2m$ in the external momenta and in r, and a_0 a polynomial of degree D, zero for $D<0$. At finite T the coefficients are the same and $T\partial_T\mathcal{C}(\varepsilon)=O(\varepsilon)$ at fixed T.

Proof. As in Theorems 11.11 and 11.45: the Schwinger domain $\{t_\sigma\le\varepsilon\le t_f\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma}\alpha_f$ and $1/\min_f\alpha_f$, which differ on the region where α_σ is the smallest coordinate. Proposition 10.23 applies to each; the coefficients of $\log\varepsilon$ cancel and the finite parts differ by the stated integral. The region of large total Schwinger time converges at $T=\infty$ because the external momenta are generic and a composite line is of order $|p|^{-3}$ (Proposition 9.18 (iv)); the injected momentum enters polynomially. At finite T, $T\partial_T$ replaces one edge by $(Q^{GF}\otimes 1)K_T$, which confines the loop momentum to $|l|^2\lesssim T^{-1}$, where the special edge is $O(\varepsilon|l|^2)$; the position factors grow at most as $T^{p_x/2}$ and are compensated by the decay of the other edges (Theorem 8.27). □

Theorem 12.11 (Graph classes and the local formula for the arity-one obstruction)**.** *Fix a generator x and $n\ge 2$. At order $\hbar$, with $F_1=0$, the part with n legs of the arity-one component along x of the flow of $\mathfrak{d}$ is the sum of the following classes of connected graphs with edges $P(\varepsilon,T)$, one vertex from* (12.2) *and one vertex carrying x:*

- *(C) irreducible defect cycles: every vertex on the cycle, exactly one special edge, exactly one vertex $S^{(1)}(x)$ in a placement of Table 12.9, the other vertices cubic or quartic vertices of I with their external legs; the special edge joins $S^{(1)}(x)$ to I (placement at the special edge) or two vertices of I (line and cubic placements);*

- *(Rd) reducible graphs: a subgraph of type (C) with fewer legs, or a one-loop subgraph of the arity-zero defect, joined by edges $P(\varepsilon,T)$ to trees of vertices of I and $S^{(1)}(x)$;*
- *(Br) graphs whose special edge is a bridge;*
- *(Ct) trees with one counterterm vertex of* (12.2) *of order $\hbar$, with the parts without and with special edge of each bracket;*
- *(K) the constant $\hbar\Delta_\varepsilon S^{(1)}(x)$.*

As $\varepsilon\to 0$, (Br) with the special-edge parts of (Ct) tends to 0; (Rd) tends to the trees of the components of Ob *with fewer legs; the $\log\varepsilon$ terms of (Ct) cancel identically (Proposition 12.8) and (C) has none; the poles satisfy*

$$\sum_{\mathcal{C}\in(C),\,n=2} a_1(\mathcal{C}) = \tfrac{9}{4}\Big\{\int (A,A), S^{(1)}(x)\Big\} \text{ units}, \qquad \sum_{\mathcal{C}\in(C),\,n} a_m(\mathcal{C}) = 0 \quad (m\geq 2 \text{ or } n\geq 3), \tag{12.8}$$

and

$$\hbar\,\mathrm{Ob}^{(1)}(x)\big|_{n \text{ legs}} = \sum_{\mathcal{C}\in(C),\,n \text{ legs}} a_0(\mathcal{C}), \tag{12.9}$$

a local functional independent of T. With an arity-zero finite counterterm included the same graphs give $\mathcal{R}^{(1)}(x)$.

Proof. At order $\hbar$ a connected graph has one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Proposition 3.35); arity one places x on one vertex. The marked vertices of order $\hbar^0$ in $\mathfrak{d}^{(1)}(x)$ and $\mathfrak{d}^{(0)}$ are special-edge parts, so the special edge lies on the cycle, on a bridge, or on a tree attached to the cycle, which again makes it a bridge. A graph with the special edge on the cycle is the cycle with trees attached; if a tree is attached through an edge the graph is reducible, otherwise it is in (C). (Br): removing the bridge separates a one-loop subgraph at a generic momentum p; with the counterterm trees of the subgraph and the special-edge part of the counterterm bracket across the same edge, the edge carries $e^{-\varepsilon|p|^2}-1$ times the renormalised subgraph, which has a limit (Definition 8.5), so the sum tends to 0; along η and K_b the derivatives in the injected momentum act on $e^{-\varepsilon|p+r|^2}-1$ and on a function of class C^2 at $p\neq 0$. (Rd): the subgraph equals the corresponding component of the obstruction plus $o(1)$ after its counterterm trees are included (induction on the number of legs; the arity-zero two-leg subgraphs are Theorem 4.30). (Ct): by Proposition 12.7 the only pole is $-\varepsilon^{-1}\frac{9}{4}\{\int(A,A),S^{(1)}(x)\}$ units in $-\{I^{CT,(0)},S^{(1)}(x)\}_\varepsilon$, which has two legs, $\{\int(A,A),T_5\}=0$ since T_5 has no leg A^*. The flow has a limit (proof of Proposition 12.8), so the poles of (C) cancel this term, which is (12.8), and Lemma 12.10 gives the finite parts. The tree-level flow of the arity-one part of Ob with n legs is $\mathrm{Ob}^{(1)}(x)|_n$ plus trees of the components with fewer legs, the limits of (Rd); this gives (12.9), and T-independence is Definition 8.7 with Lemma 12.10. □

Proposition 12.12 (Composite lines and the commutators with the regulator)**.** *Let $x.$ be the operator of Lemma 11.42 (v. of Definition 7.4 for even x), $P\star = Q^{GF}g(H)$, $g(H)=\int_\varepsilon^T e^{-tH}dt$ and $E(H)=e^{-\varepsilon H}-e^{-TH}$, so that $QP\star + P\star Q = E(H)$. The line placement of $S^{(1)}(x)$ is the composite kernel $2(P\star\, x.\, P\star\otimes 1)K_0$, with*

$$P\star\, x.\, P\star = x.\,(P\star)^2 - [x., Q^{GF}]\, g(H)\, P\star - Q^{GF}[x., g(H)]\, P\star, \tag{12.10}$$

and the variation kernel $\mathcal{V}_x := (x.\otimes 1 + 1\otimes x.)P(\varepsilon,T)$ is $([x.,P\star]\otimes 1)K_0$. Generator by generator:

- *(i) P_b, M_ω, R_r: the commutators with Q^{GF} and H vanish, $\mathcal{V}_x=0$, the kernel is $2(x.(P\star)^2\otimes 1)K_0$, zero along R_r, and $[Q, P\star x.P\star]=0$;*

- *(ii)* Δ*:* $[v.,Q^{GF}]=-4Q^{GF}$, $[v.,H]=-4H$, $\mathcal{V}_\Delta=-4(\varepsilon\partial_\varepsilon+T\partial_T)P(\varepsilon,T)$, *and* $[Q,P\star v.P\star]=0$ *(Proposition 9.18);*

- *(iii)* ξ*:* $C_\xi:=[\xi.,Q^{GF}]$, $[\xi.,H]=0$, $P\star\xi.P\star=C_\xi Q^{GF}g^2-\xi.(Q^{GF})^2g^2$, $\mathcal{V}_\xi=(C_\xi g(H)\otimes 1)K_0$, *and* $[Q,P\star\xi.P\star]=E(H)C_\xi g(H)\neq 0$ *(Lemma 11.42 (ii), (iii));*

- *(iv)* η*:* $\eta.=\sum_\mu x^\mu\xi_\mu.+\eta^0.+Y_\eta$ *with* $\xi_\mu=\rho(e_\mu)\eta$, $[\eta.,H]=2\sum_\mu\xi_\mu.\partial_\mu$ *and*

  $$[\eta.,P\star]=\sum_\mu x^\mu C_{\xi_\mu}g(H)+\sum_\mu[x^\mu,Q^{GF}]\,\xi_\mu.\,g(H)+[\eta^0.+Y_\eta,Q^{GF}]\,g(H)+2\,Q^{GF}\sum_\mu\xi_\mu.\,\partial_\mu\,g'(H)$$

  (Lemma 11.42 (iv));

- *(v)* K_b*:* $[v.,Q^{GF}]=4(b,x)Q^{GF}+C_K$, $[v.,H]=4(b,x)H-4(w-1)(b,\partial)+4b_\nu\Sigma^{\rho\nu}\partial_\rho$ *on a slot of Weyl weight* w *(Lemma 11.42 (v)), and* $[Q,P\star v.P\star]=[v.,P\star]E(H)-[v.,E(H)]P\star\neq 0$.

Proof. $P\star x.P\star=x.P\star P\star+[P\star,x.]P\star$ with $[P\star,x.]=-[x.,Q^{GF}]g-Q^{GF}[x.,g]$, graded. For $x.$ preserving the pairing, $(1\otimes x.)K_0=-(-1)^{|x|}(x.\otimes 1)K_0$ up to the Koszul sign of Lemma 9.15, which gives $\mathcal{V}_x$. For even $x.$ with $[x.,Q]=0$, $Q\,P\star x.P\star=E\,x.P\star-P\star x.E+P\star x.P\star Q$ by $QP\star+P\star Q=E$; for P_b, M_ω, R_r and Δ the first two terms cancel, $x.$ commuting with E and $P\star$ up to the scaling of (ii), and for K_b they equal $[v.,P\star]E-[v.,E]P\star$, using $EP\star=P\star E$. The commutators are those of the cited statements; for η, $[x^\mu\xi_\mu.,Q^{GF}]=x^\mu C_{\xi_\mu}+[x^\mu,Q^{GF}]\xi_\mu.$ and $[\eta.,f(H)]=2\sum_\mu\xi_\mu.\partial_\mu f'(H)$. □

Theorem 12.13 (Second evaluation: Q of the renormalised arity-one amplitudes)**.** *Let* $W^{(1)}(x)$ *be the arity-one component along* x *of* $W(P(\varepsilon,T),I_{\mathfrak{psl}(4|4)}-I^{CT})$*, the linearised flow of* $S^{(1)}(x)-I^{CT,(1)}(x)$ *at* $I-I^{CT,(0)}$*, and* $W^{(1)}_{\mathrm{ren}}(x)$ *the* $\varepsilon\to 0$ *limit of its coefficient of* $\hbar$ *at generic momenta.*

- *(i) At* $T=\infty$ *the tree-level flow of* $\hbar\,\mathrm{Ob}^{(1)}(x)$ *is* $Q\,W^{(1)}_{\mathrm{ren}}(x)$*, the action of* Q *on plane-wave values commuting with the limit; at finite* T *add* $\{W^{(0)},W^{(1)}(x)\}_T+\hbar\Delta_T W^{(1)}(x)$. $\hbar\,\mathrm{Ob}^{(1)}(x)|_n$ *is the part of* $Q\,W^{(1)}_{\mathrm{ren}}(x)$ *with* n *legs minus the trees of the components with fewer legs.*

- *(ii) (Conjugation.) With* D_x *the derivation induced by* $P\star x.$, $D^{(3)}_x$ *the first-order operator given by the cubic term of* $S^{(1)}(x)$ *with one leg* A *joined by* P *to the argument,* $\partial^{(3)}_x$ *the second-order operator with both legs* A *joined by* P*, and* $\mathcal{K}_x=2(P\star x.P\star\otimes 1)K_0$*,*

  $$e^{\hbar\partial_P}\,S^{(1)}(x)\,e^{-\hbar\partial_P}=S^{(1)}(x)+\hbar\big(D_x+D^{(3)}_x\big)+\tfrac{\hbar^2}{2}\big(\partial_{\mathcal{K}_x}+\partial^{(3)}_x\big),\tag{12.11}$$

  and the series terminates; the composite cycles $\frac{1}{2}\partial_{\mathcal{K}_x}W_0$ *and the cubic cycles* $\frac{1}{2}\partial^{(3)}_x W_0$ *carry the singular part* $I^{CT,(1)}(x)$*.*

- *(iii)* R_r*:* $I^{CT,(1)}(R_r)=0$ *and* $\mathrm{Ob}^{(1)}(R_r)=0$.

- *(iv)* P_b*,* M_ω*:* $\hbar\,\mathrm{Ob}^{(1)}(x)$ *is* Q *of the renormalised composite cycles minus* $(Q+\{I,-\})I^{CT,(1)}(x)$*, the composite part along* x*.*

- *(v)* Δ*:* $\hbar\,\mathrm{Ob}^{(1)}(\Delta)=4L^{(0)}+$ (*composite part along* Δ) *in every sector in which the arity-zero obstruction is removed.*

- *(vi)* ξ*:* $\hbar\,\mathrm{Ob}^{(1)}(\xi)$ *is* Q *of the renormalised composite and cubic cycles minus* $(Q+\{I,-\})I^{CT,(1)}(\xi)$*, plus the one-loop graphs of* W *with one edge of the cycle replaced by* $\mathcal{V}_\xi$*.*

Proof. (i) Lemma 8.4 at arity one; d_{CE} vanishes on arity zero. (ii) ad_{∂_P} applied to the multiplication by the quadratic part gives D_x (Lemma 9.16 (iv), whose proof applies to every quadratic $S^{(1)}(x)$ whose legs have no common component of P); applied twice it gives $\partial_{\mathcal{K}_x}$ by (9.3), a constant-coefficient operator commuting with ∂_P. For the cubic term, $\partial_P T_5 = 0$, the contraction of its two legs A giving $[X_a, X^a] = 0$; $\mathrm{ad}_{\partial_P} T_5 = D_x^{(3)}$, $\mathrm{ad}^2_{\partial_P} T_5 = \partial_x^{(3)}$, whose coefficient λ^* has no component of P, so $\mathrm{ad}^3_{\partial_P} T_5 = 0$. (iii) $v.$ commutes with Q, Q^{GF}, H, preserves the pairing, annihilates I and the $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$-invariant counterterms, and $\mathcal{K}_{R_r} = 0$; the one-loop part of the linearised flow is $D_x W_1$, whose singular part is that of W_1 with $P \star v.$ on a leg, removed by $I^{CT,(0)}$. Then $QW^{(1)} = [Q, D_x]W_1 + D_x Q W_1$; $[Q, D_x]$ is the derivation by $E(H)v.$, whose action on W_1 tends to $v.W_{1,\mathrm{ren}} = 0$, and $D_x Q W_1$ is a tree of $\mathrm{Ob}^{(0)}$ with $S^{(1)}(x)$. (iv) The same with $\mathcal{K}_x \neq 0$ and $[Q, \mathcal{K}_x] = 0$ (Proposition 12.12 (i)). (v) Proposition 9.25 (ii), whose proof uses (9.7) and (9.9), both identities of the flow; Lemma 12.41 below gives the same statement from Theorem 12.40. (vi) $[Q, P \star \xi.] = E(H)\xi.$ and $[Q, \mathcal{K}_\xi] = 2(EC_\xi g \otimes 1)K_0$ (Proposition 12.12 (iii)). □

Proposition 12.14 (Position parts at the level of the action)**.** *For a local functional G with polynomial dependence on the position, the position part along b is $-\{S^{(1)}(P_b), G\}$ (Lemma 11.9). Graph by graph, with the momentum injection of Lemma 12.10, the position part along b of a cycle along Δ is twice the same cycle along P_b, along M_ω the cycle along $P_{\omega b}$, along η the cycle along $\rho(b)\eta \in \Sigma$, and along K_b the cycle along the generator with the Killing field $\partial_{b'} v$, which is $(b, b')\Delta$ plus the rotation $x \mapsto 2(b, x)b' - 2(b', x)b$ (Remark 7.17). Hence*

$$-\{S^{(1)}(P_b), \mathrm{Ob}^{(1)}(\Delta)\} = 2\,\mathrm{Ob}^{(1)}(P_b), \qquad -\{S^{(1)}(P_b), \mathrm{Ob}^{(1)}(\eta)\} = \mathrm{Ob}^{(1)}(\rho(b)\eta), \qquad -\{S^{(1)}(P_b), \mathrm{Ob}^{(1)}(M_\omega)\} = \mathrm{Ob}^{(} \tag{12.12}$$

with the analogous identities along K_b, and the same for $I^{CT,(1)}$ and $L^{(1)}$. The intrinsic part of $\mathrm{Ob}^{(1)}(x)$ is $\mathrm{Ob}^{(1)}(x)$ minus a lift $\int x^\mu o_\mu$ of its position parts, translation invariant and fixed by the canonical densities of Definition 11.8.

Proof. As Proposition 11.10 (i): the factor x^μ at the vertex $S^{(1)}(x)$ is $-i\partial/\partial r_\mu$ of $e^{i(r,x)}$; its action on the momentum delta function is the position part, $-i$ times the graph with the vertex differentiated along e_μ. The derivative of the Killing data of Δ along e_μ is $2e_\mu$, that of η is $\rho(e_\mu)\eta$, that of M_ω is ωe_μ and that of K_b is $\partial_\mu v$. The special edge, the edges P and the vertices of I are translation invariant, and the obstruction is the sum (12.9). Finally $-\{S^{(1)}(P_b), \int x^\mu o_\mu\} = \int b^\mu o_\mu$. □

Theorem 12.15 (Sector bound)**.** $\mathrm{Ob}^{(1)}(x)$ *is a colour-singlet local functional of degree* 0*, Weyl weight* $4 - \frac{1}{2}\lambda_x$*, degree at most* p_x *in the position, linear in the Killing data of* x*, of fermion parity* $|x|$*, equivariant for* $\mathrm{Spin}(4) \times \mathrm{Spin}(6)$ *and the adjoint action of* $\mathfrak{g}$*, and invariant under the reflection of one direction of* V *together with one of* $W_{\mathbb{R}}^{\perp}$ *(Proposition 9.3). A leg set with field weight* w_f *and* D *derivatives occurs at position degree* p *only if* $D = 4 - \frac{1}{2}\lambda_x + p - w_f \geq 0$*, the numbers* $n_\pm$ *of* $S_\pm$*-indices and* n_V *of* V*-indices (legs* $A|_V$*,* $A^*|_V$*, derivatives, positions and the vectors* b*,* ω *of the data) satisfy* $n_+ + n_V \equiv n_- + n_V \equiv 0$ *modulo* 2*, and the* $SU(4)$ *charges (*W*: +1,* W^**: −1,* $W_{\mathbb{R}}^{\perp}$*: 2) add to* 0 *modulo* 4*. The same rules hold for* $\mathrm{Ob}^{(2)}(x, y)$ *with degree* -1 *and weight* $4 - \frac{1}{2}(\lambda_x + \lambda_y)$*, and for* $\hbar F_1^{(1)}$*,* $\hbar F_1^{(2)}$ *with degrees* -1*,* -2*. At position degree* 0 *the number of legs of* $\mathrm{Ob}^{(1)}(x)$ *is at most* 4 *along* Δ*,* M_ω*,* R_r *and* Σ*, at most* 5 *along* P_b *(the leg sets* $(A|_V)^5$*,* $(A|_V)^3(A|_{W_{\mathbb{R}}^{\perp}})^2$*,* $A|_V(A|_{W_{\mathbb{R}}^{\perp}})^4$ *without derivatives) and at most* 3 *along* Σ^**; the intrinsic part along* K_b *has two legs; the arity-two obstruction has at most three legs at position degree* 0*, four along* (ξ, P_b)*, and none at position degree* 0 *along* (η, η')*. The two-leg leg sets at position degree* 0 *are those of Table 12.2.*

x	weight	$\mathrm{Ob}^{(1)}(x)$, two legs	$\hbar F_1^{(1)}(x)$, two legs
Δ, M_ω, R_r	4	$(c, A^*\|_V)_1$, $(A\|_V, A\|_V)_2$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A\|_{W_{\mathbb{R}}^{\perp}})_2$, $(\lambda_+, \lambda_-)_1$	$(c, c^*)_0$, $(A\|_V, A^*\|_V)_0$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A^*\|_{W_{\mathbb{R}}^{\perp}})_0$, $(\lambda_\pm, \lambda_\pm^*)_0$
P_b	5	$(c, A^*\|_V)_2$, $(A\|_V, A\|_V)_3$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A\|_{W_{\mathbb{R}}^{\perp}})_3$, $(\lambda_+, \lambda_-)_2$	$(c, c^*)_1$, $(A\|_V, A^*\|_V)_1$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A^*\|_{W_{\mathbb{R}}^{\perp}})_1$, $(\lambda_\pm, \lambda_\pm^*)_1$
ξ_+	$\frac{9}{2}$	$(c, \lambda_+^*)_2$, $(A\|_V, \lambda_-)_2$, $(A\|_{W_{\mathbb{R}}^{\perp}}, \lambda_+)_2$	$(A\|_V, \lambda_+^*)_1$, $(A\|_{W_{\mathbb{R}}^{\perp}}, \lambda_-^*)_1$, $(\lambda_+, A^*\|_{W_{\mathbb{R}}^{\perp}})_0$, $(\lambda_-, A^*\|_V)_0$
η_+	$\frac{7}{2}$	$(c, \lambda_-^*)_1$, $(A\|_V, \lambda_+)_1$, $(A\|_{W_{\mathbb{R}}^{\perp}}, \lambda_-)_1$	$(A\|_V, \lambda_-^*)_0$, $(A\|_{W_{\mathbb{R}}^{\perp}}, \lambda_+^*)_0$
K_b	3	$(c, A^*\|_V)_0$, $(A\|_V, A\|_V)_1$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A\|_{W_{\mathbb{R}}^{\perp}})_1$, $(\lambda_+, \lambda_-)_0$	none at $p = 0$; at $p = 1$: $(c, c^*)_0$, $(A\|_V, A^*\|_V)_0$, $(A\|_{W_{\mathbb{R}}^{\perp}}, A^*\|_{W_{\mathbb{R}}^{\perp}})_0$, $(\lambda_\pm, \lambda_\pm^*)_0$

The two-leg leg sets at position degree 0 allowed by Theorem 12.15, $(\cdots)_D$ with D derivatives; here λ_+, λ_- are the components of λ in $S_+ \otimes W$ and $S_- \otimes W^*$, and λ^*_+, λ^*_- those of λ^* in $S_+ \otimes W^*$ and $S_- \otimes W$. For ξ_-, η_- exchange the chiralities. The parts of higher position degree are fixed by Proposition 12.14.

Proof. Weight and degree: Definition 12.5. A local functional with D derivatives has the weight of its legs plus D minus the position degree. The centres of $SU(2) \times SU(2)$ and of $SU(4)$ act on each leg and datum by the stated signs and charges, and an invariant exists exactly when they act trivially. A one-loop graph with external legs in the adjoint has at least two of them. Equivariance is Propositions 7.6 and 7.13. The leg sets are the enumeration of the multisets of at most six legs; along K_b the cubic leg sets of the intrinsic part pair a symmetric kinematic factor with the antisymmetric colour factor, or are parity odd, and vanish. □

Proposition 12.16 (Endpoint)**.** *The components* $\mathrm{Ob}^{(1)}(x)$, $\mathrm{Ob}^{(2)}(x,y)$, $L^{(k)}$ *and the solutions* $\hbar F_1^{(k)}$ *are independent of* T. *Along* K_b, $(K_b, P_{b'})$ *and* (η, η') *with* $\Gamma(\eta,\eta')|_V \neq 0$ *the defect cycles are evaluated at a finite* T, *with the same limit at every* T; *the coefficient* $\frac{3}{8\pi^2}\hbar \log T$ *of the composite insertions along* K_b *(Proposition 8.25) sits in the flows and in the trees of the observables, and the obstruction is free of it.*

Proof. Definition 8.7 and Proposition 8.6 give the T-independence of the singular parts and of the obstruction, Lemma 12.10 that of the finite parts of the cycles, and Proposition 8.25 the convergence at $T = \infty$ along the other generators. By Theorem 8.27, the coefficient of $\log T$ in a defect graph carries a special edge on a tree at a generic momentum and is $O(\varepsilon)$. □

12.3 The arity-one equation: solutions and free directions

Theorem 12.17 (The arity-one equation)**.** *Let* $\hbar F_1^{(0)}$ *solve the arity-zero equation of* (12.3) *and let* x *be a generator.*

- *(i)* $\mathcal{R}^{(1)}(x)$ *is* $(Q + \{I, -\})$*-closed, of degree* 0, *Weyl weight* $4 - \frac{1}{2}\lambda_x$ *and position degree at most* p_x, *with the covariance of Theorem 12.15.*

- *(ii) The equation* $(Q + \{I, -\})\hbar F_1^{(1)}(x) = \mathcal{R}^{(1)}(x)$ *has a local solution exactly when the class* $[\mathcal{R}^{(1)}(x)]$ *vanishes in the cohomology* $H^0(Q + \{I, -\})$ *of the local functionals with these quantum numbers.*

- *(iii) Two solutions differ by a* $(Q + \{I, -\})$*-closed local functional of degree* -1 *with the same quantum numbers, and modulo exact ones by a class of* $H^{-1}(Q + \{I, -\})$.

- *(iv) Replacing* $\hbar F_1^{(0)}$ *by* $\hbar F_1^{(0)} + G^{(0)}$ *with* $(Q + \{I, -\})G^{(0)} = 0$ *replaces* $\mathcal{R}^{(1)}(x)$ *by* $\mathcal{R}^{(1)}(x) - \{G^{(0)}, S^{(1)}(x)\}$; *for* $G^{(0)} = (Q + \{I, -\})\phi^{(0)}$ *the change is absorbed by* $\hbar F_1^{(1)}(x) \mapsto \hbar F_1^{(1)}(x) + \{\phi^{(0)}, S^{(1)}(x)\}$, *the field redefinition generated by* $\phi^{(0)}$ *(Proposition 8.19 (a)).*

- *(v) In* $H^0(Q + \{I, -\})$,

$$[\mathcal{R}^{(1)}([x,y])] = (-1)^{|x|}[\{\mathcal{R}^{(1)}(x), S^{(1)}(y)\}] - [\{S^{(1)}(x), \mathcal{R}^{(1)}(y)\}], \tag{12.13}$$

  the Chevalley–Eilenberg one-cocycle condition of $\mathfrak{psl}(4|4)$ *with values in* $H^0(Q + \{I, -\})$, *the action being the bracket with* $S^{(1)}(y)$; *the change (iv) is the coboundary of* $[G^{(0)}]$, *and the obstruction to solving through arity one, for some arity-zero solution, is the class of* $x \mapsto [\mathcal{R}^{(1)}(x)]$ *in* $H^1(\mathfrak{psl}(4|4); H^0(Q + \{I, -\}))$.

Proof. (i) Lemma 12.6 (iii). (ii), (iii) Definition of the cohomology of local functionals. (iv) $\{(Q+\{I,-\})\phi^{(0)},S^{(1)}(x)\}=-(Q+\{I,-\})\{\phi^{(0)},S^{(1)}(x)\}$ by (12.1). (v) The arity-two identity of Lemma 12.6 (ii) for $G=\hbar\,\mathrm{Ob}-s(\hbar F_1)$, whose arity-zero component vanishes, reads $(Q+\{I,-\})G^{(2)}(x,y)=G^{(1)}([x,y])-(-1)^{|x|}\{G^{(1)}(x),S^{(1)}(y)\}+\{S^{(1)}(x),G^{(1)}(y)\}$ with $G^{(1)}=\mathcal{R}^{(1)}$; in cohomology the left side vanishes. For $\mathcal{R}^{(1)}(x)=-\{G^{(0)},S^{(1)}(x)\}$ the identity follows from $(-1)^{|x|}\{S^{(1)}(x),S^{(1)}(y)\}=-S^{(1)}([x,y])-(Q+\{I,-\})S^{(2)}(x,y)$ (Lemma 2.11) and (12.1) for $\{G^{(0)},-\}$, the term $\{G^{(0)},(Q+\{I,-\})S^{(2)}(x,y)\}$ being exact. □

On two legs the bracket with I raises the number of legs, so the equation reads $Q\,\hbar F_1^{(1)}(x)|_2=\mathcal{R}^{(1)}(x)|_2$, and the relevant cohomology is that of Q on quadratic functionals.

Proposition 12.18 (Two legs along Σ)**.** *Let $\xi\in S_+\otimes W$; for $\xi\in S_-\otimes W^*$ exchange the chiralities. With $\Gamma_V:=\Gamma(\xi,\lambda)|_V$, $\Gamma_\perp:=\Gamma(\xi,\lambda)|_{W^\perp_{\mathbb{R}}}$ and (ξ,λ^*) the $\mathfrak{g}$-valued pairings, define*

$$g_1:=\int(\Gamma_V,d^*d\,A|_V),\quad g_2:=\int(\Gamma_V,dd^*A|_V),\quad g_3:=\int(\Gamma_\perp,d^*d\,A|_{W^\perp_{\mathbb{R}}}),\quad g_4:=\int\big(d^*dc,(\xi,\lambda^*)\big),\quad T_0:=\int\big(d^*A|$$

- *(i) The quadratic covariant local functionals of degree -1 and Weyl weight $\frac{9}{2}$, linear in ξ, are spanned by T_0,T_1,T_2,T_3,T_4; those of degree 0 by g_1,g_2,g_3,g_4.*
- *(ii) $QT_1=-QT_3=-\frac12 g_1$, $QT_2=-QT_4=-\frac12 g_3$ and $QT_0=-(g_2+g_4)$. The kernel of Q at degree -1 is spanned by $S_2^V(\xi)=T_1+T_3$ and $S_2^\perp(\xi)=T_2+T_4$, and no nonzero combination of them is Q-exact.*
- *(iii) The Q-closed functionals of degree 0 are spanned by g_1, g_3 and g_2+g_4, which is the image of Q: the cohomology $H^0(Q)$ of these quadratic functionals is zero.*
- *(iv) The two-leg part of the arity-one equation along ξ is solvable for every arity-zero solution: with $\mathcal{R}^{(1)}(\xi)|_2=r_1g_1+r_3g_3+r_0(g_2+g_4)$,*

$$\hbar F_1^{(1)}(\xi)|_2=-2r_1T_1-2r_3T_2-r_0T_0+\theta_V^\xi\,S_2^V(\xi)\ \textit{units}+\theta_\perp^\xi\,S_2^\perp(\xi)\ \textit{units},\tag{12.14}$$

  with free θ_V^ξ, $\theta_\perp^\xi$. The coefficient r_0 of the gauge-fixing structure receives the finite counterterm $\frac{27}{16}\int(d^*A|_V,d^*A|_V)$ units through $\{\int(d^*A|_V,d^*A|_V),T_1\}=2g_2$, and the parts g_2, g_4 of $\hbar\,\mathrm{Ob}^{(1)}(\xi)|_2$.
- *(v) On two legs the Wess–Zumino identity is $Q\,\mathrm{Ob}^{(1)}(\xi)|_2=-\{\mathrm{Ob}^{(0)}|_2,T_1\}$, with $\hbar\,\mathrm{Ob}^{(0)}|_2=-\frac{27}{8}\int(d^*dc,d^*A|_V)$ units (Theorem 4.30): it fixes the components of $\mathrm{Ob}^{(1)}(\xi)|_2$ along g_2, g_4 outside the closed combination.*
- *(vi) The field-counting directions $(Q+\{I,-\})\mathcal{N}_f$ of $\hbar F_1^{(0)}$ (Theorem 9.11) shift $\hbar F_1^{(1)}(\xi)|_2$ by $\{\mathcal{N}_f,S_2^{(1)}(\xi)\}$, which multiplies each arrow $f\to f'$ of the classical transformation by the difference of the counting weights (Theorem 12.17 (iv)).*

Proof. (i) $\xi\otimes\lambda|_{S_-\otimes W^*}\in(S_+\otimes S_-)\otimes(W\otimes W^*)$ contains the $SU(4)$ singlet in $V\otimes\mathbf{1}$, so every bilinear in $(\xi,\lambda|_{S_-\otimes W^*})$ is a component of Γ_V; with $A|_V$ and two derivatives, after integration by parts onto $A|_V$, the contractions are $\Gamma_V\cdot d^*dA|_V$ and $\Gamma_V\cdot dd^*A|_V$. $\xi\otimes\lambda|_{S_+\otimes W}\in(\mathbf{1}\oplus\wedge^2_+)\otimes(\mathbf{6}\oplus\mathbf{10})$ pairs with $A|_{W^\perp_{\mathbb{R}}}$ through $\Gamma_\perp$, and $\xi\otimes\lambda^*|_{S_+\otimes W^*}$ contains the scalar (ξ,λ^*), paired with c and two derivatives. At degree -1, (λ,A^*) without derivatives gives T_1, T_2, and (A,λ^*) with one derivative gives T_0, T_3, T_4 (Proposition 11.49 (i)). (ii) Q substitutes $A^*\mapsto-\frac12 d^*dA$, $\lambda^*\mapsto\not{d}\lambda$, $A|_V\mapsto-dc$ (Proposition 1.11): $T_1\mapsto-\frac12\Gamma_V\cdot d^*dA|_V$, $T_2\mapsto-\frac12\Gamma_\perp\cdot d^*dA|_{W^\perp_{\mathbb{R}}}$; in T_0 the substitution in $A|_V$ gives $-g_4$ and that in λ^* gives $(\xi,\not{d}\lambda)\,d^*A|_V$ with $(\xi,\not{d}\lambda)=-d^*\Gamma_V$, hence $-g_2$. S_2^V and $S_2^\perp$ are closed, the two sector

parts of the quadratic part of $\{S, S^{(1)}(\xi)\} = 0$. The images g_1, g_3, $g_2 + g_4$ are independent, so the kernel is two-dimensional; a Q-exact functional of degree -1 would be Q of a quadratic functional of degree -2 and weight $\frac{9}{2}$, and these are zero (Proposition 11.49 (i)). (iii) $Qg_1 = Qg_3 = 0$, and $Qg_2 = -\int(\Gamma_V, dd^*dc) = -Qg_4$, nonzero; the closed space is three-dimensional and equals the image. (iv) The residual is closed (Lemma 12.6 (iii)); (ii) and (iii) give the solution, and $\{\int(d^*A|_V, d^*A|_V), T_1\}$ contracts $A|_V$ with $A^*|_V$. (v) Lemma 12.6 (iii) on two legs. (vi) The quadratic part of $(Q + \{I, -\})\mathcal{N}_f$ is $Q\mathcal{N}_f$, and $\{Q\mathcal{N}_f, S_2^{(1)}(\xi)\} = -Q\{\mathcal{N}_f, S_2^{(1)}(\xi)\}$ by (12.1). □

Proposition 12.19 (Two legs along the even generators, Σ^* and K_b).

- *(i) Δ. The Q-closed quadratic functionals of degree -1, Weyl weight 4, position degree at most one and reflection-even are spanned, modulo Q-exact ones, by the three sector parts of $S^{(1)}(\Delta)$; the exact ones by $Q\int(\iota_x A|_V, c^*)$ and $Q\int(\rho(x)\lambda^*, \lambda^*)$ (Proposition 11.24). $H^0(Q)$ vanishes on the two-leg target.*

- *(ii) P_b. The translation-invariant quadratic closed functionals of degree -1, weight 5, linear in b, are spanned modulo exact ones by the three sector parts of $S^{(1)}(P_b)$, the position parts of those of (i); the exact ones by $Q\int(\iota_b A|_V, c^*)$ and $Q\int(\rho(b)\lambda^*, \lambda^*)$.*

- *(iii) M_ω, R_r. The sector parts are closed. Along M_ω, in the leg sets $(A|_V, A|_V)_2$, $(c, A^*|_V)_1$ the closed functionals of degree 0 are spanned by $Q\int\omega_{\mu\nu}(A_\mu, A^*_\nu)$, and in the leg set $(\lambda, \lambda)_1$ by $Q\int(\frac{1}{4}\omega_{\mu\nu}\rho_\mu\rho_\nu\lambda, \lambda^*)$, the structure $\int(\lambda, \omega_{\mu\nu}\rho_{[\mu}\rho_\nu\rho_{\kappa]}\partial_\kappa\lambda)$ being a total derivative: $H^0(Q) = 0$ there, and the spin term*

$$S^{(1)}(M_\omega)_{\text{spin}} := \int \omega_{\mu\nu}(A_\mu, A^*_\nu) + \int \left(\tfrac{1}{4}\omega_{\mu\nu}\rho_\mu\rho_\nu\lambda, \lambda^*\right)$$

  of $\hbar F_1^{(1)}(M_\omega)$ is fixed by $\mathrm{Ob}^{(1)}(M_\omega)|_2$. Along R_r, $\hbar F_1^{(1)}(R_r) = 0$ solves the equation (Theorem 12.13 (iii)).

- *(iv) $\eta \in S_+ \otimes W^*$. The translation-invariant quadratic functionals of degree -1 and weight $\frac{7}{2}$ are spanned by $\int(\rho(A|_V)\eta, \lambda^*)$ and $\int s_0$; Q is block triangular for the split into position parts and intrinsic parts; its kernel on the full degree-(-1) space is spanned by $S^V(\eta)$ and $S^\perp(\eta)$; the cohomology $H^0(Q)$ on the two-leg degree-0 functionals of weight $\frac{7}{2}$ with position degree at most one is one-dimensional, a class in the sector $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$.*

- *(v) K_b. The sector parts $S^{(1)}(K_b)_d$ are closed; with position degree at most two, $H^{-1}(Q)$ and $H^0(Q)$ on two legs have dimensions 7 and 4: H^{-1} is spanned by the three classes $[S^{(1)}(K_b)_d]$ and four second-order symmetries of the free fields, written $G_1^K, \dots, G_4^K$, closed and outside the span of the $S^{(1)}(z)$. $\{\mathrm{Ob}^{(0)}|_2, S^{(1)}(K_b)\}$ and $\{\hbar F_1^{(0)}, S^{(1)}(K_b)\}$ are nonzero, $\int(d^*dc, d^*A|_V)$ being outside the special conformal invariants.*

Proof. (i) Proposition 11.24 and Theorem 9.26 (iii). (ii) The position part along b commutes with Q (Lemma 11.9) and maps the closed and exact directions of (i) to those listed. (iii) $v.$ commutes with Q; in the vector sector $Q\int\omega_{\mu\nu}(\partial_\mu A_\nu, d^*A|_V)$ and $Q\int\omega_{\mu\nu}(\partial_\mu c, A^*_\nu)$ are proportional and nonzero, so the closed subspace of these two is one-dimensional and contains $Q\int\omega_{\mu\nu}(A_\mu, A^*_\nu) \neq 0$; in the gaugino sector Γ is symmetric. (iv), (v) The leg sets of Table 12.2 and their position-dependent completions, with the exact ranks of Q on plane-wave normal forms; Q commutes with $-\{S^{(1)}(P_b), -\}$, which gives the triangular form. $\{\int(d^*dc, d^*A|_V), S^{(1)}(K_b)\}$ is the variation of a functional outside the special conformal invariants. □

Theorem 12.20 (The arity-one obstructions on two legs)**.** *On two legs the arity-one obstructions are*

$$\begin{aligned}
\hbar\,\mathrm{Ob}^{(0)}|_2 &= -\tfrac{27}{8}\int (d^*dc, d^*A|_V)\ \textit{units},\\
\hbar\,\mathrm{Ob}^{(1)}(\Delta)|_2 &= \Big[-\tfrac{11}{3}\int (dA, dA) + \tfrac{47}{3}\int (\lambda,\, \not{d}\lambda) - \tfrac{14}{3}\int (dc, A^*)\Big]\ \textit{units},\\
\hbar\,\mathrm{Ob}^{(1)}(P_b)|_2 &= 0, \qquad \hbar\,\mathrm{Ob}^{(1)}(R_r)|_2 = 0, \qquad \hbar\,\mathrm{Ob}^{(1)}(M_\omega)|_2 = \tfrac{1}{3}\,Q\,S^{(1)}(M_\omega)_{\mathrm{spin}}\ \textit{units},\\
\hbar\,\mathrm{Ob}^{(1)}(\xi)|_2 &= \Big[\big(\tfrac{79}{24} - \tfrac{1}{2}\log\tfrac{3}{2}\big)(g_1+g_3) + \tfrac{27}{8}\,g_2 + 0\cdot g_4\Big]\ \textit{units},\\
\hbar\,\mathrm{Ob}^{(1)}(\eta)|_2 &= \sum_\mu x^\mu\,\hbar\,\mathrm{Ob}^{(1)}(\rho(e_\mu)\eta)|_2 + \Big[-\tfrac{5}{2}\int (\eta,\lambda)\,d^*A|_V - \tfrac{5}{2}\int (\eta, \rho(dA|_V)\lambda)\\
&\qquad + \big(-\tfrac{3}{2} + 3\log\tfrac{3}{2}\big)\int (\eta, \rho(dA|_{W_{\mathbb{R}}^\perp})\lambda) + \tfrac{5}{2}\int (\rho(dc)\eta, \lambda^*)\Big]\ \textit{units} + Q(\cdot),
\end{aligned} \tag{12.15}$$

the same for both chiralities of ξ and η, the intrinsic part along η in the canonical density of Definition 11.8. Along Δ the coefficient of $\int (d^*A|_V, d^*A|_V)$ and of every functional with an explicit factor x is 0; along K_b the obstruction lies in the span of the canonical data with position degree at most two. In every case the poles of the defect cycles sum to $\frac{9}{4}\{\int (A,A), S^{(1)}(x)\}$ units, as (12.8) *requires; along Δ this is $-9\int (A,A)$ units and along ξ it is $\frac{9}{2}\big[\int (\Gamma_V, A|_V) + \int (\Gamma_\perp, A|_{W_{\mathbb{R}}^\perp})\big]$ units. The logarithmic coefficient along ξ is $L^{(1)}(\xi)|_2 = \{\mathcal{N}, S_2^{(1)}(\xi)\}$ units $= 2(T_1 + T_2 - T_3 - T_4)$ units, so the closed combination* (12.6) *vanishes on two legs along ξ. The Wess–Zumino identities* (12.4) *hold on every entry.*

Proof. Theorem 4.30 in arity zero, Theorem 9.26 (ii) along Δ, Theorem 9.35 (i) along P_b, and Proposition 11.50 (i)–(iv) along ξ, η, M_ω and R_r, each obstruction being the sum (12.9) over the cycles of Theorem 12.11 on the bases of Propositions 12.18 and 12.19; the position part along η is Proposition 12.14. $L^{(1)}(\xi)|_2$ is the coefficient of $\log\varepsilon$ of the composite and cubic cycles, and $\{\mathcal{N}, -\}$ multiplies the arrows of T_1, T_2 ($\lambda \to A^*$) by $-2+4=2$ and those of T_3, T_4 ($A \to \lambda^*$) by $2-4=-2$. □

Theorem 12.21 (Two-leg solutions with all free directions)**.** *On two legs the arity-zero solution is $\hbar F_1^{(0)}|_2 = \frac{27}{16}\int (d^*A|_V, d^*A|_V)$ units plus the s-closed arity-zero directions, and the arity-one solutions*

are

$$\begin{aligned}
\hbar F_1^{(1)}(\Delta)|_2 &= \Big[\tfrac{22}{3}\int (A,A^*) - \tfrac{47}{3}\int(\lambda^*,\lambda) - \tfrac{8}{3}\int(c^*,c)\Big]\ \textit{units} + \sum_d \theta_d^{\Delta}\, S^{(1)}(\Delta)_d\ \textit{units}\\
&\quad + \kappa_V\, Q\int(\iota_x A|_V, c^*)\ \textit{units} + \kappa_\lambda\, Q\int(\rho(x)\lambda^*,\lambda^*)\ \textit{units},\\
\hbar F_1^{(1)}(P_b)|_2 &= \sum_d \theta_d^{P_b}\, S^{(1)}(P_b)_d\ \textit{units} + (\textit{the parts along } b \textit{ of the } \kappa \textit{ directions}),\\
\hbar F_1^{(1)}(M_\omega)|_2 &= \tfrac{1}{3}\, S^{(1)}(M_\omega)_{\mathrm{spin}}\ \textit{units} + \sum_d \theta_d^{M}\, S^{(1)}(M_\omega)_d\ \textit{units}, \qquad \hbar F_1^{(1)}(R_r)|_2 = \sum_d \theta_d^{R}\, S^{(1)}(R_r)_d\ \textit{units},\\
\hbar F_1^{(1)}(\xi)|_2 &= \big(\log\tfrac{3}{2} - \tfrac{79}{12}\big)(T_1+T_2)\ \textit{units} + \theta_V^{\xi}\, S_2^V(\xi)\ \textit{units} + \theta_\perp^{\xi}\, S_2^{\perp}(\xi)\ \textit{units},\\
\hbar F_1^{(1)}(\eta)|_2 &= \sum_\mu x^\mu\big(\log\tfrac{3}{2} - \tfrac{79}{12}\big)(T_1+T_2)(\rho(e_\mu)\eta)\ \textit{units} - \big(3\log\tfrac{3}{2} - \tfrac{3}{2}\big)\int s_0\ \textit{units} - \tfrac{5}{2}\int(\rho(A|_V)\eta,\lambda^*)\ \textit{units}\\
&\quad + \theta_V^{\eta}\, S^V(\eta)\ \textit{units} + \theta_\perp^{\eta}\, S^{\perp}(\eta)\ \textit{units},\\
\hbar F_1^{(1)}(K_b)|_2 &= \int(b,x)\Big\{-\tfrac{22}{3}(A,A^*) + 16(\lambda,\lambda^*) + \tfrac{8}{3}(c,c^*)\Big\}\ \textit{units} + \tfrac{2}{3}\int\big\{(\iota_b A)(\iota_x A^*) - (\iota_x A)(\iota_b A^*)\big\}\ \textit{units}\\
&\quad - \tfrac{1}{3}\int b_\mu x_\nu(\lambda,\rho_\mu\rho_\nu\lambda^*)\ \textit{units} + \sum_d \theta_d^{K}\, S^{(1)}(K_b)_d\ \textit{units} + \sum_{j=1}^{4}\kappa_j^K G_j^K,
\end{aligned} \tag{12.16}$$

with the free directions displayed: the sector coefficients θ_d^x, the exact directions κ_V, κ_λ, the four closed directions κ_j^K along K_b, and, along ξ, the shifts of Proposition 12.18 (vi). Q of the first bracket of $\hbar F_1^{(1)}(\Delta)|_2$ is $\hbar\,\mathrm{Ob}^{(1)}(\Delta)|_2$. The part of $\hbar F_1^{(1)}(K_b)|_2$ linear in x is $-(b,x)$ times the particular part along Δ plus the spin term of the rotation part of $\partial_{b'}v$, $16 - \frac{1}{3} = \frac{47}{3}$ on the gaugino. On two legs the residual has no component along $H^0(Q)$ in each case: $H^0(Q)$ vanishes along Δ, P_b, M_ω, R_r, Σ, and along Σ^* and K_b the computed residual is orthogonal to its classes.

Proof. Theorem 9.26 (iii) along Δ, Theorem 9.35 (i) along P_b, and Proposition 11.50 along M_ω, R_r, ξ, η and K_b, with the free directions of Propositions 12.18 and 12.19. Along ξ, (12.14) with $r_0 = 0$, the coefficient $\frac{27}{8}$ of g_2 in the obstruction being the contribution $\frac{27}{16}\cdot 2$ of the finite counterterm, and $-2r_1 = -2r_3 = \log\frac{3}{2} - \frac{79}{12}$. □

Theorem 12.22 (The cubic part fixes the sector differences)**.** *Write the quadratic part of a solution as $\hbar F_1^{(1)}(x)|_2 = F_2^{\mathrm{part}}(x) + \sum_d \theta_d^x S^{(1)}(x)_d$ units + exact terms, with I_3 the cubic part of I. The part of the arity-one equation with three legs is*

$$Q\,\hbar F_1^{(1)}(x)|_3 + \{I_3, \hbar F_1^{(1)}(x)|_2\} = \mathcal{R}^{(1)}(x)|_3, \qquad \mathcal{R}^{(1)}(x)|_3 = \hbar\,\mathrm{Ob}^{(1)}(x)|_3 - \{\hbar F_1^{(0)}|_2, S_3^{(1)}(x)\} - \{\hbar F_1^{(0)}|_3, S_2^{(1)}(x)\}.$$

- *(i) Each $\{I_3, S^{(1)}(x)_d\}$ is Q-closed and $\sum_d\{I_3, S^{(1)}(x)_d\} = -QS_3^{(1)}(x)$; the equation depends on the θ_d^x through $\sum_{d\neq d_0}(\theta_d^x - \theta_{d_0}^x)[\{I_3, S^{(1)}(x)_d\}]$ in the cohomology $H^0(Q)_3$ of the cubic functionals of degree 0, weight $4 - \frac{1}{2}\lambda_x$ and position degree at most p_x.*
- *(ii) It is solvable exactly when $[\mathcal{R}^{(1)}(x)|_3 - \{I_3, F_2^{\mathrm{part}}(x)\}] = \sum_{d\neq d_0}(\theta_d^x - \theta_{d_0}^x)[\{I_3, S^{(1)}(x)_d\}]$ units in $H^0(Q)_3$.*
- *(iii) The classes $[\{I_3, S^{(1)}(x)_d\}]$, $d \neq d_0$, are linearly independent exactly when no combination $\sum_d c_d S^{(1)}(x)_d$ other than the multiples of $S_2^{(1)}(x)$ extends to a $(Q + \{I, -\})$-closed functional through three legs; then the cubic equation fixes the differences $\theta_d^x - \theta_{d_0}^x$.*

Proof. On three legs $(Q+\{I,-\})$ applied to a functional with two and three legs is Q of the cubic part plus the bracket of I_3 with the quadratic part. (i) $QI_3=0$ and $QS^{(1)}(x)_d=0$ give $Q\{I_3,S^{(1)}(x)_d\}=0$ by (12.1); the cubic part of $\{S,S^{(1)}(x)\}=0$ is $QS_3^{(1)}(x)+\{I_3,S_2^{(1)}(x)\}=0$. (ii) Definition of $H^0(Q)_3$; the exact quadratic directions contribute $\{I_3,Q\psi\}=\pm Q\{I_3,\psi\}$. (iii) A relation $\sum_d c_d[\{I_3,S^{(1)}(x)_d\}]=0$ means $\{I_3,\sum_d c_d S^{(1)}(x)_d\}=-QY_3$, that is, $(Q+\{I,-\})(\sum_d c_d S^{(1)}(x)_d+Y_3)=0$ through three legs; $\sum_d S^{(1)}(x)_d$ always extends, by $Y_3=S_3^{(1)}(x)$. □

Proposition 12.23 (The arity-one equations along Σ constrain the arity-zero couplings)**.** *Let $\mathcal{H}_4$ be $H^0(Q+\{I,-\})$ on the colour-singlet local functionals of degree* 0 *and weight* 4*, invariant under* $\mathrm{Spin}(4)\times\mathrm{Spin}(6)$ *and the reflection: the classical couplings modulo field redefinitions, containing the class of S and the classes of Theorem 9.11 that extend to closed functionals. For $\xi\in\Sigma$ let $\delta_\xi[G]:=[\{G,S^{(1)}(\xi)\}]$. The arity-one equations along Σ are solvable for some arity-zero solution exactly when $[\mathcal{R}^{(1)}(\xi)]$ lies in the image of δ_ξ for every ξ, one $[G]$ serving all ξ; the arity-zero freedom left after them is $\bigcap_\xi\ker\delta_\xi$, which contains $[S]$. Along Δ, P_b, M_ω, R_r and K_b the map $[G]\mapsto[\{G,S^{(1)}(x)\}]$ vanishes on $\mathcal{H}_4$.*

Proof. Theorem 12.17 (iv), (v). For even x, $\{G,S^{(1)}(x)\}=\mp v.G$ (Lemma 9.16 (i), Definition 7.4), zero for a translation- and rotation-invariant G of weight 4; along K_b the gauge-invariant functionals of weight 4 of A and λ are conformally invariant with the weights 1, $\frac{3}{2}$ (Proposition 2.14), and the antifield terms of weight 4 because S is. □

Definition 12.24 (The classical symmetry condition)**.** The classical symmetry condition at the weight of x (resp. of a pair (x,y)) is: every $(Q+\{I,-\})$-closed colour-singlet local functional of degree -1, of Weyl weight $4-\frac{1}{2}\lambda_x$ (resp. $4-\frac{1}{2}(\lambda_x+\lambda_y)$), position degree at most p_x (resp. p_x+p_y), linear in the Killing data and equivariant as in Theorem 12.15, is a linear combination of the functionals $S^{(1)}(z)$, $z\in\mathfrak{psl}(4|4)$, with these quantum numbers, plus a $(Q+\{I,-\})$-exact functional. It states that the global symmetries of the interacting classical theory at these weights are the superconformal ones. On two legs $H^{-1}(Q)$ is larger, containing the sector parts $S^{(1)}(z)_d$ and, at the weights with two powers of the position, second-order symmetries of the free fields built from conformal Killing tensors; the condition concerns their extensions through three legs, which Theorem 12.22 (iii) tests.

Theorem 12.25 (Solution space at arity one)**.** *Assume the classical symmetry condition at the weights of the generators and that the arity-one equations are solvable. Then for every generator x the solutions are $\hbar F_1^{(1)}(x)=F^{\mathrm{part}}(x)+t_x\,S^{(1)}(x)$ units $+(Q+\{I,-\})(\mathit{local})$, with one number t_x, the sector coefficients being equal to t_x up to the fixed differences of Theorem 12.22 contained in $F^{\mathrm{part}}(x)$. In a scheme equivariant for translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, t_x depends on the block of x alone: t_Δ, t_P, t_M, t_R, t_{ξ_+}, t_{ξ_-}, t_{η_+}, t_{η_-}, t_K.*

Proof. Theorem 12.17 (iii) and the condition of Definition 12.24: the functional $S^{(1)}(z)$ with the weight, position degree and Killing data of x that is equivariant is $S^{(1)}(x)$ itself. Equivariance under the compact groups makes t_x constant on each irreducible block. □

Remark 12.26 (The free directions and the equations that fix them)**.** The differences of the sector coefficients are fixed by the cubic part of the arity-one equation (Theorem 12.22) and, along Σ, Σ^*, P_b, Δ, M_ω, R_r, already by the two-leg parts of the arity-two equations (Theorem 12.31); the uniform parts by the arity-two equations of the next section:

x	closed quadratic directions of $\hbar F_1^{(1)}(x)$	uniform part fixed by
Δ	$S^{(1)}(\Delta)_d$; $Q\int(\iota_x A\vert_V, c^*)$, $Q\int(\rho(x)\lambda^*, \lambda^*)$	(Δ, P_b), (Δ, ξ), (ξ, η)
P_b	$S^{(1)}(P_b)_d$; $Q\int(\iota_b A\vert_V, c^*)$, $Q\int(\rho(b)\lambda^*, \lambda^*)$	up to the inner direction
M_ω	$S^{(1)}(M_\omega)_d$	$(K_b, P_{b'})$, (ξ, η)
R_r	$S^{(1)}(R_r)_\perp$, $S^{(1)}(R_r)_\lambda$	(ξ, η)
Σ	$S_2^V(\xi)$, $S_2^\perp(\xi)$; difference fixed by $(\xi_\pm, \xi'_\pm)$	up to the inner and outer directions
Σ^*	$S^V(\eta)$, $S^\perp(\eta)$; difference fixed by (ξ_+, η_-)	up to the inner and outer directions
K_b	$S^{(1)}(K_b)_d$, $G_1^K, \dots, G_4^K$	$(K_b, P_{b'})$, (η_+, η_-), up to the inner direction

The free directions of the arity-one solutions and the equations that fix them; the inner and outer directions are those of Theorem 12.29 (ii).

12.4 Arity two: the coefficients and the one-loop algebra

Proposition 12.27 (Graph classes at arity two)**.** *Fix generators x, y and $n \geq 2$. At order $\hbar$, with $F_1 = 0$, the part with n legs of the arity-two component along (x, y) of the flow of $\mathfrak{d}$ consists of: (C2) irreducible cycles with exactly one special edge and either two vertices $S^{(1)}(x)$, $S^{(1)}(y)$ or one vertex $S^{(2)}(x, y)$ on the cycle, the other vertices from I; (Rd2) reducible graphs whose subgraphs are cycles of type (C) along x or y, of type (C2) with fewer legs, or of the arity-zero defect, joined by edges $P(\varepsilon, T)$ to trees; (Br2) bridges; (Ct2) the counterterm trees of $\mathfrak{d}^{(2)}(x, y)$, of $\mathfrak{d}^{(1)}$ with a separate $S^{(1)}(y)$ or $S^{(1)}(x)$, and of $\mathfrak{d}^{(0)}$ with two separate vertices. Two vertices $S^{(1)}(x)$, $S^{(1)}(y)$ in the line placement are adjacent through an $A\vert_V \otimes A\vert_V$ edge when both are even (the chain c–$A\vert_V$–$A\vert_V$–c, a double composite line) or through a $\lambda \otimes \lambda$ edge when both are odd, and never when one is even and one odd. As $\varepsilon \to 0$, (Br2) with the special-edge parts of (Ct2) tends to 0, (Rd2) tends to the trees of the components of* Ob *of lower arity or with fewer legs, the $\log \varepsilon$ terms of (Ct2) cancel, (Ct2) has no pole, $\{\int(A, A), S^{(2)}(x, y)\}$ being 0, and*

$$\sum_{\mathcal{C} \in (C2),\, n} a_m(\mathcal{C}) = 0 \ (m \geq 1), \qquad \hbar\, \mathrm{Ob}^{(2)}(x, y)\big|_{n \text{ legs}} = \sum_{\mathcal{C} \in (C2),\, n \text{ legs}} a_0(\mathcal{C}). \tag{12.17}$$

For two odd generators the arity-two generating cochain is $I + c\, S^{(1)}(\xi) + c' S^{(1)}(\xi') - cc' S^{(2)}(\xi, \xi')$ with odd parameters c, c', the sign forced by the classical master equation in the form $Q S_2^{(2)}(\xi, \xi') + S_2^{(1)}(P_{\Gamma(\xi,\xi')\vert_V}) - \{S_2^{(1)}(\xi), S_2^{(1)}(\xi')\} = 0$.

Proof. As Theorem 12.11, with Lemma 12.4 for $\mathfrak{d}^{(2)}$: its marked vertices of order $\hbar^0$ are the special-edge parts of $(-1)^{\vert x\vert}\{S^{(1)}(x), S^{(1)}(y)\}_\varepsilon$ and $\{I, S^{(2)}(x, y)\}_\varepsilon$, and the marked vertices of lower arity enter with the remaining generators on separate vertices. $\int(A, A)$ has no antifield and $S^{(2)}$ no leg A^*. The adjacency rules follow from the components of P (Proposition 3.17) and the legs of Table 12.9. The generating cochain: the arity-two component of $d_{CE} f + \frac{1}{2}\{f, f\}$ at the two odd parameters, with Lemma 2.11. □

Theorem 12.28 (The arity-two equation and its cocycle)**.** *Let $\hbar F_1^{(0)}$, $\hbar F_1^{(1)}$ solve the equations of arity zero and one.*

- *(i) $\mathcal{R}^{(2)}(x, y)$ is $(Q + \{I, -\})$-closed of degree -1, Weyl weight $4 - \frac{1}{2}(\lambda_x + \lambda_y)$ and position degree at most $p_x + p_y$; the arity-two equation along (x, y) is solvable exactly when $[\mathcal{R}^{(2)}(x, y)] = 0$ in $H^{-1}(Q + \{I, -\})$.*
- *(ii) Replacing $\hbar F_1^{(1)}$ by $\hbar F_1^{(1)} + \varphi$ with $(Q + \{I, -\})\varphi(x) = 0$ for all x replaces $\mathcal{R}^{(2)}(x, y)$ by $\mathcal{R}^{(2)}(x, y) - \varphi([x, y]) - (-1)^{\vert x\vert}\big(\{\varphi(x), S^{(1)}(y)\} + \{S^{(1)}(x), \varphi(y)\}\big)$, minus the Chevalley–Eilenberg coboundary of $[\varphi]$ with values in $H^{-1}(Q + \{I, -\})$; exact φ change $\mathcal{R}^{(2)}$ by exact terms.*

- *(iii)* $\{\hbar F_1^{(1)}(x), \hbar F_1^{(1)}(y)\}$ *is of order* $\hbar^2$*; at one loop* $\hbar F_1^{(1)}$ *enters linearly.* $(x,y) \mapsto [\mathcal{R}^{(2)}(x,y)]$ *is a two-cocycle of* $\mathfrak{psl}(4|4)$ *with values in* $H^{-1}(Q+\{I,-\})$*, and solvability through arity two, for some* $\hbar F_1^{(1)}$*, is the vanishing of its class in* $H^2(\mathfrak{psl}(4|4); H^{-1}(Q+\{I,-\}))$*.*

- *(iv) Under the classical symmetry condition at the weights of the pairs,* $H^{-1}(Q+\{I,-\})$ *there is spanned by the classes* $[S^{(1)}(z)]$*, the adjoint module, and the uniform directions* $\varphi(x) = t_x S^{(1)}(x)$ *units change the class by*

$$[\mathcal{R}^{(2)}(x,y)] \longmapsto [\mathcal{R}^{(2)}(x,y)] + \big(t_x + t_y - t_{[x,y]}\big)[S^{(1)}([x,y])] \text{ units}, \tag{12.18}$$

where $t_{[x,y]} S^{(1)}([x,y])$ *is* $\sum_z t_z S^{(1)}(z)$ *over the components* z *of* $[x,y]$ *in the blocks of Theorem 12.25.*

Proof. (i) Lemma 12.6 (iii). (ii) Definition 12.5; for $\varphi = (Q+\{I,-\})\psi$ the brackets are exact by (12.1). (iii) The arity-three component of s on a degree-1 cochain G with $G^{(0)} = G^{(1)} = 0$ is $(Q+\{I,-\})G^{(3)}$ plus a signed Chevalley–Eilenberg differential of $G^{(2)}$, the signs forced as in Lemma 12.6 (ii) by $s^2 = 0$ on $G = sF^{(2)}$; in H^{-1} the first term vanishes, and $S^{(3)} = 0$. (iv) $(-1)^{|x|}\{S^{(1)}(x), S^{(1)}(y)\} = -S^{(1)}([x,y]) - (Q+\{I,-\})S^{(2)}(x,y)$ (Lemma 2.11), so the change in (ii) for $\varphi = tS^{(1)}$ units is $-t_{[x,y]}S^{(1)}([x,y]) + (t_x + t_y)\big(S^{(1)}([x,y]) + (Q+\{I,-\})S^{(2)}(x,y)\big)$ units. □

Theorem 12.29 (What the arity-two equations fix)**.** *Assume the classical symmetry condition, the solvability of the arity-one equations, and write* $[\mathcal{R}^{(2)}(x,y)]_{t=0} = [S^{(1)}(\varrho(x,y))]$ *units with* $\varrho(x,y) \in \mathfrak{psl}(4|4)$ *of weight* $\lambda_x + \lambda_y$*, the class at* $t_x = 0$ *for every* x*. The arity-two equations along all pairs are solvable exactly when*

$$\varrho(x,y) = t\cdot[x,y] - [t\cdot x, y] - [x, t\cdot y] \qquad \text{for all } x, y, \qquad t\cdot z := t_z z, \tag{12.19}$$

that is, when the cocycle ϱ *with values in the adjoint module is the coboundary of the diagonal one-cochain* $t\cdot$*. Pair by pair:*

pair	*bracket*	*equation*
(Δ, P_b)	$-2P_b$	$\varrho = 2t_\Delta P_b$
(Δ, ξ)	$-\xi$	$\varrho = t_\Delta\, \xi$
(ξ_+, ξ_-)	$P_{\Gamma(\xi_+,\xi_-)\vert_V}$	$\varrho = (t_P - t_{\xi_+} - t_{\xi_-})P_\Gamma$
(ξ_+, η_+)	*components along* Δ, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$	$\varrho_z = (t_z - t_{\xi_+} - t_{\eta_+})\, z$
(η_+, η_-)	$K_{\Gamma(\eta_+,\eta_-)\vert_V}$	$\varrho = (t_K - t_{\eta_+} - t_{\eta_-})K_\Gamma$
$(K_b, P_{b'})$	$(b,b')\Delta + M_{bb'}$	$\varrho_\Delta = (t_\Delta - t_K - t_P)(b,b')\Delta$, $\varrho_M = (t_M - t_K - t_P)M_{bb'}$
(M_ω, P_b), (M_ω, ξ), (R_r, ξ)	$[M_\omega, P_b]$, $[M_\omega, \xi]$, $[R_r, \xi]$	$\varrho = -t_M[M_\omega, P_b]$, $-t_M[M_\omega, \xi]$, $-t_R[R_r, \xi]$
(Δ, η), (Δ, K_b)	η, $2K_b$	$\varrho = -t_\Delta \eta$, $-2t_\Delta K_b$
(P_b, η), (K_b, ξ)	$\rho(b)\eta$, $\rho(b)\xi$	$\varrho = (t_\xi - t_P - t_\eta)\rho(b)\eta$, $(t_\eta - t_K - t_\xi)\rho(b)\xi$
(ξ_+, ξ'_+), (ξ_+, η_-), (ξ, P_b), (η_+, η'_+), $(P_b, P_{b'})$	0	$\varrho = 0$

The arity-two equations of Theorem 12.29, and likewise with the chiralities exchanged; in the rows (P_b, η) *and* (K_b, ξ) *the image has the opposite chirality and* t_ξ*,* t_η *are the coefficients of the image block.*

Consequently:

- *(i)* t_Δ*,* t_M *and* t_R *are fixed,* t_Δ *overdetermined by* (Δ, P_b)*,* $(\Delta, \xi_\pm)$ *and the* Δ*-components of* $(\xi_\pm, \eta_\pm)$*;*

- *(ii) the solutions* $(t_P, t_{\xi_\pm}, t_{\eta_\pm}, t_K)$ *form an affine plane over the two diagonal derivations of* $\mathfrak{psl}(4|4)$*: the inner one,* $t_z \mapsto t_z + c\lambda_z$*, generated by the field redefinition* $\phi^{(0)} = c\, S^{(1)}(\Delta)$ *units, and the outer one,* $t_z \mapsto t_z + aB_z$*, the direction* G_B *of Definition 8.9;*

- *(iii) the remaining relations are consistency conditions, consequences of the cocycle condition of Theorem 12.28 (iii);*

- *(iv) in a translation-covariant scheme $t_P = t_\Delta$, the inner direction being excluded since it shifts t_P by $c\lambda_{P_b} = -2c$ and t_Δ by 0; with the computed classes of Theorem 12.31, $t_{\eta_+} = t_{\xi_-}$, a relation invariant under the outer direction;*

- *(v) on the commuting pairs $\varrho = 0$ holds automatically, $\mathfrak{psl}(4|4)$ having no element of the required weight and type, and the computed classes there test the classical symmetry condition; on two legs the sector coefficients enter separately, the $[S^{(1)}(z)_d]$ being independent in $H^{-1}(Q)$, and the two-leg parts of the commuting-pair equations impose sector uniformity.*

Proof. (12.18): the equation along (x, y) is $[S^{(1)}(\varrho(x, y))] + (t_x + t_y - t_{[x,y]})[S^{(1)}([x, y])] = 0$ in units, which is (12.19) componentwise. The brackets are those of Proposition 2.4, Remark 2.6 and Remark 7.17; $[\xi_+, \eta_-] = 0$ because $S_+ \otimes S_- \otimes W \otimes W$ contains no element of $\mathfrak{so}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^{\perp}) \oplus \mathbb{C}\Delta$. On the six unknowns $t_P, t_{\xi_\pm}, t_{\eta_\pm}, t_K$ the equations from $(\xi_\pm, \eta_\pm)$, (ξ_+, ξ_-), (η_+, η_-), $(K_b, P_{b'})$ have rank four, the sum of the first four being the fifth up to the values of ϱ; the homogeneous solutions are $t_z = c\lambda_z$ and $t_z = aB_z$, whose diagonal maps are derivations ($\lambda_{[x,y]} = \lambda_x + \lambda_y$, and B is a derivation). The inner direction: $(s\phi)^{(1)}(x) = \{\phi^{(0)}, S^{(1)}(x)\} = c\{S^{(1)}(\Delta), S^{(1)}(x)\} \equiv c\lambda_x S^{(1)}(x)$ units modulo exact terms; the outer one: Proposition 8.19 (c). □

Theorem 12.30 (The sector coefficients along Δ from the arity-two equation)**.** *Let $\hbar F_1^{(1)}(\Delta)$ and $\hbar F_1^{(1)}(P_b)$ be the solutions of Theorem 12.21. On two legs the arity-two equation along (Δ, P_b) reads*

$$Q\,\hbar F_1^{(2)}(\Delta, P_b)\big|_2 + 2 \sum_{d \in \{\perp, \lambda, V\}} \theta_d^{\Delta}\, S^{(1)}(P_b)_d \text{ units} = \hbar\,\mathrm{Ob}^{(2)}(\Delta, P_b)\big|_2 + Q(\text{local}), \tag{12.20}$$

independently of the $\theta_d^{P_b}$. The classes $[S^{(1)}(P_b)_d]$ are independent in $H^{-1}(Q)$ on two legs; hence, with o_d the coefficient of $[S^{(1)}(P_b)_d]$ in $[\hbar\,\mathrm{Ob}^{(2)}(\Delta, P_b)|_2]$, the equation is solvable on two legs exactly when this class lies in the span of the $[S^{(1)}(P_b)_d]$, and then $\theta_d^{\Delta} = \frac{1}{2} o_d$ in units.

Proof. Theorem 9.28 (i) and its proof, in the notation θ_d^{Δ}, $\theta_d^{P_b}$ of Definition 12.1. □

Theorem 12.31 (The computed arity-two equations on two legs)**.** *With the solutions of Theorem 12.21, the classes of the arity-two residuals on two legs vanish exactly under the conditions of Table 12.4. The obstructions are*

$$\begin{aligned}
\hbar\,\mathrm{Ob}^{(2)}(\Delta, P_b)|_2 = \Big[&-\tfrac{2}{3}\int (\partial_b A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}}) + \tfrac{2}{3}\int (\partial_b \lambda, \lambda^*) + \big(-\tfrac{3}{2} + \log\tfrac{4}{3}\big)\int (\rho(b)\not{\partial}\lambda, \lambda^*) \\
&- \tfrac{2}{3}\int (\partial_b A|_V, A^*|_V) + \big(\tfrac{3}{2} - \log\tfrac{4}{3}\big)\int (\partial_\mu(\iota_b A|_V), A_\mu^*) + \big(-\tfrac{5}{6} + \log\tfrac{4}{3}\big)\int (\partial_b c, c^*)\Big] \text{ units}, \\
\hbar\,\mathrm{Ob}^{(2)}(\Delta, \xi)|_2 = \Big[8(T_1 + T_2) - \tfrac{26}{3}(T_3 + T_4) + \log\tfrac{4}{3}\, Q(\psi_1 + \psi_2)\Big] \text{ units}, \qquad &\psi_1 := \int x^\mu \big(\Gamma(\xi, \lambda)^\mu, c^*\big), \quad \psi_2 := \int x^\mu
\end{aligned} \tag{12.21}$$

with $o_d = -\frac{2}{3}$ units in each sector, no position part and no pole along (Δ, P_b); $\hbar F_1^{(2)}(\Delta, \xi)|_2 = \log\frac{4}{3}(\psi_1 + \psi_2)$ units plus closed directions of degree -2; $\hbar\,\mathrm{Ob}^{(2)}(\xi, P_b)|_2$ is $\frac{1}{4}\log\frac{4}{3}$ times a Q-exact functional; every two-leg arity-two pole sum vanishes. Consequently

$$t_\Delta = -\tfrac{1}{3}, \qquad t_M = t_\Delta, \qquad t_R = t_\Delta + \tfrac{1}{3}, \qquad t_{\xi_+} + t_{\xi_-} - t_P = \tfrac{27}{4} + 2\log\tfrac{3}{2}, \qquad t_{\xi_+} + t_{\eta_+} - t_\Delta = \tfrac{27}{4} + 2\log\tfrac{3}{2}, \tag{12.22}$$

and the sector coefficients of t_P, t_ξ, t_η, t_M, t_R are uniform. The value $t_\Delta = -\frac{1}{3}$ is obtained twice, from (Δ, P_b) and from $(\Delta, \xi_\pm)$.

pair	condition
(Δ, P_b)	$\theta^{\Delta}_{\perp} = \theta^{\Delta}_{\lambda} = \theta^{\Delta}_{V} = -\frac{1}{3}$
$(\Delta, \xi_{\pm})$	$\theta^{\Delta}_{\perp} = \theta^{\Delta}_{\lambda} = \theta^{\Delta}_{V} = -\frac{1}{3}$, θ^{ξ}_{d} free
(ξ_+, ξ_-)	$\theta^{P_b}_{d} = \theta^{\xi_+}_{V} + \theta^{\xi_-}_{V} - \frac{27}{4} - 2\log\frac{3}{2}$ for every d, and $\theta^{\xi_+}_{V} + \theta^{\xi_-}_{V} = \theta^{\xi_+}_{\perp} + \theta^{\xi_-}_{\perp}$
$(\xi_{\pm}, \xi'_{\pm})$	$\theta^{\xi_{\pm}}_{V} = \theta^{\xi_{\pm}}_{\perp}$
$(\xi_{\pm}, P_b)$	$\theta^{P_b}_{V} = \theta^{P_b}_{\perp} = \theta^{P_b}_{\lambda}$
(ξ_+, η_-)	$\theta^{\xi}_{V} - \theta^{\xi}_{\perp} = \theta^{\eta}_{V} - \theta^{\eta}_{\perp}$
(ξ_+, η_+)	$\theta^{R}_{\perp} = \theta^{R}_{\lambda} =: \tau$, $\theta^{\Delta}_{d} = \theta^{M}_{d} = \tau - \frac{1}{3}$, $\theta^{\xi}_{d} + \theta^{\eta}_{d} = \tau + \frac{77}{12} + 2\log\frac{3}{2}$ $(d = V, \perp)$

The conditions under which the two-leg arity-two equations are solvable, in the parametrisation of (12.16).

Proof. Theorem 9.28 for the rows along Δ, with Theorem 12.30 and $o_d = -\frac{2}{3}$ units, and Theorem 11.51 for the other rows and for (12.22); the obstructions (12.21) are the sums (12.17) over the cycles of Proposition 12.27. □

Corollary 12.32 (Scheme-independent content on the action side)**.** *The inner direction ($t_z \mapsto t_z + c\lambda_z$, $\lambda_z = -1, +1, -2, +2$ on Σ, Σ^*, P_b, K_b and 0 on Δ, M_ω, R_r) and the outer direction ($t_{\xi_\pm} \mapsto t_{\xi_\pm} \pm a$, $t_{\eta_\pm} \mapsto t_{\eta_\pm} \mp a$) leave invariant exactly*

$$t_\Delta = -\tfrac{1}{3}, \qquad t_M - t_\Delta = 0, \qquad t_R - t_\Delta = \tfrac{1}{3}, \qquad t_{\xi_+} + t_{\xi_-} - t_P = \tfrac{27}{4} + 2\log\tfrac{3}{2}, \qquad t_{\xi_+} + t_{\eta_+} - t_\Delta = \tfrac{27}{4} + 2\log\tfrac{3}{2},$$

together with the sector uniformity of t_P, t_ξ, t_η; these are the one-loop parts of $[D,P] = -2P$, $[D,Q] = -Q$, $[Q,P] = 0$, $[Q,Q] = P$ and $\{Q,S\} = D + M + R$ in the form of Theorem 12.33. In a translation-covariant scheme $t_P = t_\Delta = -\frac{1}{3}$, hence $t_{\xi_+} + t_{\xi_-} = \frac{77}{12} + 2\log\frac{3}{2}$ and $t_{\eta_+} = t_{\xi_-}$. The individual $t_{\xi_\pm}$, $t_{\eta_\pm}$, t_P, t_K, the coefficients θ^K_d, κ^K_j, κ_V, κ_λ, the particular parts with $\log\frac{3}{2}$ and every representative $\hbar F_1^{(k)}$ depend on the scheme.

Proof. Theorem 12.31 and Theorem 12.29 (ii): each combination has zero weight under λ and under B; for instance $t_{\xi_+} + t_{\xi_-} - t_P$ changes by $c(-1-1+2)$ and $a(1-1)$, and $t_{\xi_+} + t_{\eta_+} - t_\Delta$ by $c(-1+1)$ and $a(1-1)$. The translation-covariant scheme is Theorem 12.29 (iv). □

Theorem 12.33 (One-loop algebra relations at the level of the action)**.** *Assume the classical symmetry condition and the arity-one equations. For each pair (x, y) the following are equivalent: (a) the arity-two equation along (x, y) is solvable; (b) $[\mathcal{R}^{(2)}(x, y)] = 0$, that is, (12.19) holds at (x, y); (c) with the one-loop corrected generators $S_\hbar(z) := S^{(1)}(z) + t_z\, S^{(1)}(z)$ units, $z \in \{x, y, [x, y]\}$, the part of order $\hbar$ of $(-1)^{|x|}\{S_\hbar(x), S_\hbar(y)\} + S_\hbar([x, y])$ plus $S^{(1)}(\varrho(x, y))$ units is $(Q + \{I, -\})$-exact. The relations are the one-loop corrections of $[Q,Q] = P$ along (ξ, ξ'), of $\{Q,S\} = D + M + R$ along (ξ, η), of $[K,P] = D + M$ along $(K_b, P_{b'})$, of $[D,Q] = -Q$ along (Δ, ξ), of $[D,P] = -2P$ along (Δ, P_b) and of $\{S,S\} = K$ along (η, η'). Computed on two legs: $[D,P]$, $[D,Q]$, $[Q,P]$, $[Q,Q]$, and $\{Q,S\}$ along $(\xi_+, \eta_\pm)$, by Theorem 12.31.*

Proof. (a)⇔(b) is Theorem 12.28 (i). (b)⇔(c): with $(-1)^{|x|}\{S^{(1)}(x), S^{(1)}(y)\} = -S^{(1)}([x, y]) - (Q + \{I, -\})S^{(2)}(x, y)$, the part of order $\hbar$ is $(t_x + t_y - t_{[x,y]})S^{(1)}([x, y])$ units modulo exact terms, and (12.18) applies. □

Remark 12.34 (The coefficient θ as a cross-check)**.** Theorem 9.33 derives $\theta = -\frac{1}{3}$ from the arity-two component of $L \circ L = 0$ on the scalar bilinears, under the hypothesis that the quantisation and the local representatives exist through arity two, and Corollary 11.28 with Proposition 11.33 does the same family by family for the three sector coefficients. In the present construction $\theta^{\Delta}_d = -\frac{1}{3}$ is computed from the arity-two obstruction along (Δ, P_b) and again along $(\Delta, \xi_\pm)$ (Theorem 12.31), and agrees with these derivations.

12.5 The components on local operators from the defect flow

In this section $O = \int \omega\, X_1 \cdots X_n$ is a monomial local functional, gauge-variant ones included, δ a square-zero parameter of degree $-|O|$, $O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$ (Definition 8.11 with $\psi = 0$), and $\hbar F_1$ a solution of (12.3) through arity one along the generators considered. As in Definition 11.43 (iii), $\mathrm{ct}_m(M)$ and $\mathrm{lc}(M)$ are the coefficients of ε^{-m} and $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](M)$, and $\mathrm{ct}^{(1)}_{x,m}(M)$, $\mathrm{lc}^{(1)}_x(M)$ those of the arity-one counterterm of M along x. For a two-body kernel κ we write

$$\sum_{a<b} \pm\kappa(X_a, X_b) := \sum_{1\le a<b\le n} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1 \cdots X_{a-1}\, \kappa(X_a, X_b)\, X_{a+1} \cdots \widehat{X_b} \cdots X_n, \tag{12.23}$$

the sum over the unordered pairs with the Koszul signs of (10.4).

Definition 12.35 (Defect cycles of a pair and two-body kernels)**.** For a generator x, two letters X_a, X_b of any slots and $0 < \varepsilon < T \le \infty$, the defect cycles of the pair along x and their sum $\mathcal{D}_x(X_a, X_b; \varepsilon, T)$, amputated with the outputs open and read with the sign of Definition 11.43 (ii), are those of Definition 11.43 (i), (ii): chains from X_a to X_b with vertices of I, exactly one vertex given by a term of $S^{(1)}(x)$ in a placement of Table 12.9, edges $P(\varepsilon, T)$ and exactly one special edge. The two-body kernel of the pair along x is

$$\Upsilon_x(X_a, X_b) := [\varepsilon^0]\, \mathcal{D}_x(X_a, X_b; \varepsilon, T), \tag{12.24}$$

the coefficient of ε^0 of the expansion of Lemma 12.10. In arity zero, $\Upsilon_0(X_a, X_b)$ is the coefficient of ε^0 of the chains from X_a to X_b with vertices of I, edges $P(\varepsilon, T)$ and exactly one special edge. For even x acting on $A|_V$ the defect cycles with the vertex on a line are the special-edge composite cycles of Definition 11.5, and their coefficient of ε^0 is the composite pair kernel, written $\mathcal{C}_x(X_a, X_b)$.

Lemma 12.36 (The defect vertex of an operator)**.** *Modulo $\hbar^2$ and constants, the defect vertex $\mathfrak{d}_O := (d_{CE} + Q)(J + \delta O(\varepsilon)) + \frac{1}{2}\{J + \delta O(\varepsilon), J + \delta O(\varepsilon)\}_\varepsilon + \hbar\Delta_\varepsilon(J + \delta O(\varepsilon))$ has the parts: (a) the defect vertex of the action, independent of δ; (b) in arity zero, δ times*

$$sO + \{I, O\}^\sigma_\varepsilon + \hbar\Delta_\varepsilon O - s\,\partial_\delta I^{CT}[\varepsilon](O) - \{I, \partial_\delta I^{CT}[\varepsilon](O)\}^\sigma_\varepsilon - \{I^{CT}[\varepsilon] + \hbar F_1^{(0)}, O\}_\varepsilon;$$

(c) in arity one along x, δ times

$$\pm\{S^{(1)}(x), O\} \pm \{S^{(1)}(x), O\}^\sigma_\varepsilon - s\,\mathrm{ct}^{(1)}_x(O)[\varepsilon] - \{I, \mathrm{ct}^{(1)}_x(O)[\varepsilon]\}^\sigma_\varepsilon \mp \{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon - \big\{I^{CT,(1)}(x) + \hbar F_1^{(1)}(x), O\big\}_\varepsilon,$$

with the signs of the cochain bracket of Definition 2.8 on an arity-zero argument (Theorem 8.15 (iv)), all $+$ for $|O| = 0$, and $\mathrm{ct}^{(1)}_x(O)[\varepsilon] = \sum_{m\ge1} \varepsilon^{-m}\mathrm{ct}^{(1)}_{x,m}(O) + \log\varepsilon\,\mathrm{lc}^{(1)}_x(O)$.

Proof. Expand in δ, in the arity and in $\hbar$, with $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$. The classical master equation (2.4) removes the brackets without special edge in (a); d_{CE} vanishes on arity zero, so the parts of arity zero and one of $(d_{CE} + Q)(\delta O) + \{I_{\mathfrak{psl}(4|4)}, \delta O\}$ are $\delta\, sO$ and $\pm\delta\{S^{(1)}(x), O\}$. $\hbar\Delta_\varepsilon$ of the arity-one counterterms and of $\hbar F_1$, and the brackets of two counterterms, are of order $\hbar^2$; $\hbar\Delta_\varepsilon S^{(1)}(x)$ is a constant (Lemma 12.4). □

Lemma 12.37 (Bridges and their counterterm partners)**.** *Let G be a subgraph at a generic momentum p, joined to the rest of a graph by a special edge on a tree, with $G = \sum_{m\ge1} \varepsilon^{-m} a_m + a_0 + o(1)$ and singular part $\mathrm{Sing}\, G = \sum_{m\ge1} \varepsilon^{-m} a_m$ (plus a multiple of $\log\varepsilon$). The bare bridge has the finite limit $\sum_{m\ge1} \frac{(-1)^m}{m!}|p|^{2m} a_m$. Its counterterm partner, the special-edge part of the counterterm bracket across the same edge, is $-(e^{-\varepsilon|p|^2} - 1)\,\mathrm{Sing}\, G$, whose coefficient of ε^0 is the negative of that limit; for G a cycle through two letters of M with $S^{(1)}(x)$ on it, or a cycle through two letters of M in arity zero, these*

coefficients are the finite special vertex of Definition 11.43 (iii) and the finite vertex (10.6)*. The sum of the bridge and its partner is*

$$(e^{-\varepsilon|p|^2}-1)\big(G-\operatorname{Sing}G\big)\longrightarrow 0. \tag{12.25}$$

Proof. $e^{-\varepsilon|p|^2}-1=\sum_{j\geq 1}(-\varepsilon|p|^2)^j/j!$ at a generic momentum; multiplying $\varepsilon^{-m}a_m$ by it gives the poles $j<m$, the finite term $j=m$ and terms that vanish for $j>m$; a term $\log\varepsilon$ gives $(e^{-\varepsilon|p|^2}-1)\log\varepsilon\to 0$. The counterterm vertex across the edge carries $-\operatorname{Sing}G$, so the partner is $-(e^{-\varepsilon|p|^2}-1)\operatorname{Sing}G$, and in the sum the convergent graph $G-\operatorname{Sing}G$ is multiplied by a factor tending to 0. The coefficients $-\frac{(-1)^m}{m!}\{f,g_m\}_{H^m}$ of the brackets defining Definition 11.43 (iii) and (10.6) are the ε^0 coefficients of the partners with $f=I$, $g_m=\mathrm{ct}^{(1)}_{x,m}$ or ct_m, and $f=S^{(1)}(x)$, $g_m=\mathrm{ct}_m$. □

Theorem 12.38 (Every component from the defect flow)**.** *Let x be a generator, $O=\int\omega\,X_1\cdots X_n$ of degree 0, $\psi=0$, and $\hbar F_1$ a solution of* (12.3) *in arity zero and in arity one along x. Take $T=\infty$ for $x\neq K_b$ and any $T\in(0,\infty)$ for $x=K_b$, and write the left side of* (8.7) *in arity one along x as $\big(L(O)^{(1)}(x)\big)[T]+\big(L(O)^{(0)}\big)[T]^{(1)}(x)$, with $L(O)^{(1)}(x)=\{S^{(1)}(x),O\}+\hbar L_1(O)^{(1)}(x)$ and $L(O)^{(0)}=sO+\hbar L_1(O)^{(0)}$ (Theorem 8.15 (iv)).*

- *(i) The δ-linear graphs of arity one along x and of order $\hbar$ of the flow of $\mathfrak{d}_O$ form the classes: (F) one-loop graphs with the vertex $\pm\delta\{S^{(1)}(x),O\}$, or with the vertex $\delta\,sO$ and a vertex $S^{(1)}(x)$; (C2) the defect cycles of the pairs of letters of O with trees of I on the outputs; (C0) cycles through two letters of O with one special edge or one contraction through K_ε, the vertex $S^{(1)}(x)$ on a tree; (Br) graphs with a special edge on a tree; (Ac) graphs whose cycle passes through vertices of J alone; (Ct) trees with one vertex of order $\hbar$ of (b) or (c) of Lemma 12.36.*

- *(ii) As $\varepsilon\to 0$: (F) with the counterterm trees of $\{S^{(1)}(x),O\}$ and of sO gives the one-loop parts of $(\{S^{(1)}(x),O\})[T]$ and of $(sO)[T]^{(1)}(x)$; (C0) with the arity-zero vertices of (Ct) tends to $\big(\hbar L_1(O)^{(0)}\big)[T]^{(1)}(x)$; (Br) with its counterterm partners in (Ct) tends to* 0 *by Lemma 12.37, the partners being used up there; (Ac) tends to* 0 *by the arity-zero and arity-one equations.*

- *(iii) (Master formula.)*

$$\hbar L_1(O)^{(1)}(x)=\sum_{a<b}\pm\Upsilon_x(X_a,X_b)-\big\{\hbar F_1^{(1)}(x),O\big\},\qquad \hbar L_1(O)^{(0)}=\sum_{a<b}\pm\Upsilon_0(X_a,X_b)-\big\{\hbar F_1^{(0)},O\big\}. \tag{12.26}$$

- *(iv) The defect cycles contain no $\log\varepsilon$; the coefficient of $\log\varepsilon$ cancels by* (11.6)*, and the coefficients of ε^{-m} by* (11.7)*, the poles of the bridge partners being grouped with the bridges in (ii); $I^{CT,(1)}(x)$ has no pole along every generator.*

- *(v) Along K_b the right side of* (12.26) *is the same local functional at every finite T; along the other generators at every $T\in(0,\infty]$.*

Proof. By Lemma 8.4, $(d_{CE}+Q+\hbar\Delta_T)e^{\hbar\partial_{P(\varepsilon,T)}}e^{(J+\delta O(\varepsilon))/\hbar}=e^{\hbar\partial_{P(\varepsilon,T)}}\big(\hbar^{-1}\mathfrak{d}_O\,e^{(J+\delta O(\varepsilon))/\hbar}\big)$, so the left side of (8.7) at scale T is the δ-linear part of the $\varepsilon\to 0$ limit of the connected graphs with one vertex $\mathfrak{d}_O$, the others from $J+\delta O(\varepsilon)$ and edges $P(\varepsilon,T)$ (Lemma 8.13, Proposition 4.26), and by Definition 8.14 it equals $L(O)[T]$.

(i) At order $\hbar$ a connected graph has one cycle and vertices of order $\hbar^0$, or one vertex of order $\hbar$ and no cycle (Proposition 3.35); one vertex carries δ and one carries x. A cycle passes through O along exactly two letters or avoids it (Theorem 10.4 (i)); the positions of the special edge and of $S^{(1)}(x)$ relative to the cycle sort the graph into (C2), (C0), (Br) or (Ac), the vertices $\pm\delta\{S^{(1)}(x),O\}$ and $\delta\,sO$ into (F), and the trees with a vertex of order $\hbar$ are (Ct).

(ii) (F): graph by graph the one-loop graphs of the observables $\{S^{(1)}(x), O\}$ and sO at scale T; the differences of their counterterm trees with the vertices of (b), (c) are local and singular and enter (iv). (C0): Theorem 10.16 (proof), with a tree carrying $S^{(1)}(x)$ attached at generic momenta, and the arity-zero part of the present argument. (Br): Lemma 12.37; the subgraph is a cycle through two letters of O with $S^{(1)}(x)$ on it, a cycle through two letters in arity zero, or a cycle of J alone such as the self-energy with the pole of I^{CT}. Along K_b at finite T the subgraphs carry at fixed ε the term $\frac{3}{8\pi^2}\hbar\log T$ times a tree with the same factor (Theorem 8.27), which also tends to 0. (Ac): a cycle of J alone, with its counterterm and finite vertices, is a graph of the flow of the defect of the action attached to O by a tree; its limit is the tree-level flow of $\hbar\,\mathrm{Ob} - s(\hbar F_1)$ (Theorem 8.8 (ii)), zero in arity zero and in arity one along x.

(iii) What remains is (C2) and the arity-one vertices of (Ct) without their bridge partners: $-s\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]$, $\mp\{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}$, $-\{I^{CT,(1)}(x), O\}$, $-\{\hbar F_1^{(1)}(x), O\}_\varepsilon$, with the counterterm trees $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x), O\})$ and $-\mathrm{ct}_x^{(1)}(sO)$ moved from the left side. Each defect cycle is $\sum_{m\geq 1}\varepsilon^{-m}a_m + a_0 + o(1)$ with local coefficients and no $\log\varepsilon$ (Lemma 12.10, Theorem 11.45); the trees on its outputs converge to trees with edges $P(0,T)$. $-\{\hbar F_1^{(1)}(x), O\}_\varepsilon$ gives $-\{\hbar F_1^{(1)}(x), O\}$, its special-edge part tending to 0. After the cancellations of (iv) the sum converges to the tree flow of the right side of (12.26), and Theorem 8.15 (i) identifies the local functionals; the sum over the pairs with the Koszul signs is Theorem 10.4 (iii). In arity zero the same argument with the vertices (b) gives the second formula; the finite part of $\{I^{CT}[\varepsilon], O\}_\varepsilon^\sigma$ is the partner of the bridges through the self-energy cycles and cancels with them.

(iv) The left side has a limit, so the coefficients of $\log\varepsilon$ and of ε^{-m} of the collected vertices vanish, which with the injectivity of the tree flow gives (11.6) and (11.7) in the stated form. A coefficient of ε^{-m} of $I^{CT,(1)}(x)$ has degree -1 and total Weyl weight $4 - \frac{1}{2}\lambda_x - 2m$ with position degree at most p_x; a colour singlet of degree -1 has total weight at least $\frac{7}{2} - p_x$, while for $m \geq 1$ the total weight is at most $2, 3, 2, 2, \frac{5}{2}, \frac{3}{2}, 1$ along $\Delta, P_b, M_\omega, R_r, \Sigma, \Sigma^*, K_b$, below $\frac{5}{2}, \frac{7}{2}, \frac{5}{2}, \frac{7}{2}, \frac{7}{2}, \frac{5}{2}, \frac{3}{2}$ (Proposition 12.7).

(v) Lemma 12.10 and Theorem 11.45 (iii); $\hbar F_1^{(1)}(K_b)$ is independent of T (Proposition 12.16); along the other generators Proposition 8.25 and Theorem 8.15 (ii). □

Remark 12.39 (The finite special vertices in Chapters 10 and 11)**.** Lemma 12.37 pairs the finite special vertex of Definition 11.43 (iii) and the finite vertex (10.6) with the bare bridges: both are the coefficients of ε^0 of the counterterm partners of the bridges, and they cancel against the bridges. The two-body kernels of the components are therefore the coefficients of ε^0 of the defect cycles alone, (12.24), and the pole identity holds in the form of Theorem 12.38 (iv). The kernel (12.24) is that of Definition 11.43 (iv); on the leg sets with superficial degree $D < 2$ the defect cycles have no poles and the finite special vertex vanishes (Proposition 11.47), and the composite pair kernels of Definition 11.5 are coefficients of ε^0 of cycles. On the computed leg sets with $D \geq 2$ the pair form of Remark 12.44 holds with (12.24), in both evaluations of Remark 12.69.

Every number of a component is a coefficient of ε^0 of a defect cycle or a coefficient of the action-side solution: the master formula uses Lemma 8.4, Definition 8.14, the injectivity of the tree flow, the classical master equation, the equations (12.3) through arity one along x, the simplex form of the defect cycles and generic momenta. The identities below are proved from it.

Theorem 12.40 (Ward identity of the flow along an even generator)**.** *Let x be even, J even and at least cubic modulo $\hbar$, $P = P(\varepsilon, T)$ and $W = W(P, J)$. With $\{S^{(1)}(x), -\} = x.$ on functionals (Definition 7.4),*

$$\{S^{(1)}(x), W\} = \partial_\delta W\big(P, J + \delta\{S^{(1)}(x), J\}\big) - \hbar\,\partial_{\mathcal{V}_x} W - \tfrac{1}{2}\big(\partial_{\mathcal{V}_x}(W^2) - 2W\,\partial_{\mathcal{V}_x} W\big), \tag{12.27}$$

the last two terms being the graphs of W with one edge P replaced by the variation kernel $\mathcal{V}_x$. In particular:

- *(i)* $\{S^{(1)}(\Delta),W\}=\partial_\delta W(P,J+\delta\{S^{(1)}(\Delta),J\})+4(\varepsilon\partial_\varepsilon+T\partial_T)W$*, the derivatives acting on* P*, which is* (9.9)*;*
- *(ii) for* $x\in\{P_b,M_\omega,R_r\}$*,* $\{S^{(1)}(x),W\}=\partial_\delta W(P,J+\delta\{S^{(1)}(x),J\})$ *at every* ε *and* T*, which is* (7.1)*.*

Proof. By Lemma 9.15, $[\partial_\phi,x.]=\partial_{(x.\otimes 1+1\otimes x.)\phi}$ for even $x.$, so $x.\,\partial_P=\partial_P\,x.-\partial_{\mathcal{V}_x}$; the operators ∂_ϕ commute, hence $x.\,e^{\hbar\partial_P}=e^{\hbar\partial_P}(x.-\hbar\partial_{\mathcal{V}_x})$. Apply this to $e^{J/\hbar}$: $x.\,e^{W/\hbar}=\hbar^{-1}(x.W)e^{W/\hbar}$, $e^{\hbar\partial_P}(x.e^{J/\hbar})=\hbar^{-1}\partial_\delta W(P,J+\delta\,x.J)e^{W/\hbar}$, and $\partial_{\mathcal{V}_x}e^{W/\hbar}=\big(\hbar^{-1}\partial_{\mathcal{V}_x}W+\frac{1}{2}\hbar^{-2}(\partial_{\mathcal{V}_x}(W^2)-2W\partial_{\mathcal{V}_x}W)\big)e^{W/\hbar}$ (Definition 3.21); multiply by $\hbar e^{-W/\hbar}$. (i) and (ii) are Proposition 12.12 (ii) and (i). □

Lemma 12.41 (The arity-one obstruction along an even generator)**.** *For* $x\in\{\Delta,P_b,M_\omega,R_r\}$*, with* $\hbar F_1^{(0)}$ *a solution of the arity-zero equation invariant under* x*,* $I^{CT,(1)}(x)$ *is the singular part of the composite loops of* I*, and*

$$\hbar\,\mathrm{Ob}^{(1)}(x)=\mathcal{V}_x^{\mathrm{act}}+(\textit{composite part along } x),\qquad \mathcal{V}_\Delta^{\mathrm{act}}=4L^{(0)}=4\,s\mathcal{N}\textit{ units},\qquad \mathcal{V}_{P_b}^{\mathrm{act}}=\mathcal{V}_{M_\omega}^{\mathrm{act}}=\mathcal{V}_{R_r}^{\mathrm{act}}=0,\tag{12.28}$$

the composite part along x *being the local functional whose tree flow is* $\lim_{\varepsilon\to0}\big(Q(\textit{composite loops of } I)-\textit{tree flow of } s\,I^{CT,(1)}(x)\big)$*; it is* 0 *along* R_r*, so* $\mathrm{Ob}^{(1)}(R_r)=0$ *and* $I^{CT,(1)}(R_r)=0$*.*

Proof. As Proposition 9.25, with Theorem 12.40 in place of (9.5), (9.7) and (9.9). By Theorem 12.13 (ii) the linearised flow of $S^{(1)}(x)$ is the vertex, the vertex joined by one edge to a leg of W, and the composite closures; Q of it is $\{S^{(1)}(x),W\}_\varepsilon$ minus the terms with QW, using $QP\star x.+P\star x.Q=E(H)x.$ and $\{S^{(1)}(x),F\}_t=(e^{-tH}x.)F$ (Lemma 9.16 (ii)). The special-edge part on legs tends to 0, and by (12.27) $\{S^{(1)}(x),W\}$ is the linearised flow of $x.J$ plus the variation term. For Δ, $x.I=x.\hbar F_1^{(0)}=0$, $(v.-4\varepsilon\frac{d}{d\varepsilon})I^{CT}[\varepsilon]=-4L^{(0)}$ (proof of Corollary 9.22), and $4\varepsilon\partial_\varepsilon W\to0$, which gives $\mathcal{V}_\Delta^{\mathrm{act}}$. For P_b, M_ω, R_r, $x.I=0$, $x.I^{CT}[\varepsilon]=0$ by the covariance of the counterterms (Proposition 10.2 (iv)), $x.\hbar F_1^{(0)}=0$, and the variation kernel vanishes. The terms with QW tend to 0 by the arity-zero equation, except Q of the composite loops, Q anticommuting with ∂ of the composite line (Proposition 12.12). Along R_r the composite line is absent (Table 12.9). □

Theorem 12.42 (The conjugation evaluation along Δ, P_b, M_ω, R_r)**.** *Let* $x\in\{\Delta,P_b,M_\omega,R_r\}$*,* $\hbar F_1^{(0)}$ *invariant under* x*, and* O *homogeneous with* $sO=0$ *and* $\hbar L_1(O)^{(0)}=0$ *for a finite part* ψ *of the operators that is linear in* ω*, translation invariant, weight preserving and covariant under* $\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^{\perp})$*. The composite term of* O *along* x *is the local functional whose tree flow is the limit of* Q *of the composite graphs of* O *minus their counterterm trees, minus the tree insertion of the composite part into* O *(Definition 10.36 along* Δ*). Then at every* $T\in(0,\infty]$

$$\hbar L_1(O)^{(1)}(x)=\mathcal{V}_x(O)+(\textit{composite term of } O \textit{ along } x)-\big\{\hbar F_1^{(1)}(x),O\big\},\qquad \mathcal{V}_\Delta(O)=4\,\mathrm{lc}(O),\quad \mathcal{V}_{P_b}=\mathcal{V}_{M_\omega}=\mathcal{V}_{R_r}=\tag{12.29}$$

Proof. At $T=\infty$, $d_\infty=d_{CE}+Q$, and the arity-one component along x of $d_\infty O[\infty]$ is Q of the arity-one part of $O[\infty]$, the limit of the δ-part of the linearised flow at $J+\delta O(\varepsilon)$ of $S^{(1)}(x)-I^{CT,(1)}(x)-\hbar F_1^{(1)}(x)-\delta\,\mathrm{ct}_x^{(1)}(O)[\varepsilon]$. (1) By Theorem 12.40 the vertex $S^{(1)}(x)$ gives the linearised flow of $x.J$ plus the variation term; the δ-part of $x.J$ is $x.O(\varepsilon)$. For Δ, $(v.-4\varepsilon\frac{d}{d\varepsilon})O(\varepsilon)=(v.O)(\varepsilon)+4\,\mathrm{lc}(O)$ by (10.2), and the δ-independent part gives 4 times the tree insertion of $L^{(0)}$ into $O[\infty]$; for P_b, M_ω, R_r, $x.\partial_\delta I^{CT}[\varepsilon](O)=\partial_\delta I^{CT}[\varepsilon](x.O)$ (Proposition 10.2 (iv)) and $x.\psi=\psi x.$. The terms with QW vanish in the limit by the arity-zero equation, $sO=0$ and $L_1(O)^{(0)}=0$, except Q of the composite graphs. (2) The vertex $-\hbar F_1^{(1)}(x)$ enters by tree insertion; by Lemma 6.8 with $sO=0$, $-Q$ of its tree insertion is minus the tree insertion of $s\hbar F_1^{(1)}(x)=\hbar\,\mathrm{Ob}^{(1)}(x)$, which by Lemma 12.41 is $\mathcal{V}_x^{\mathrm{act}}$ plus

the composite part, minus $\{\hbar F_1^{(1)}(x), O\}[\infty]$; the first cancels the δ-independent term of (1) along Δ. (3) The vertices $-I^{CT,(1)}(x)$ and $-\delta\,\mathrm{ct}_x^{(1)}(O)$ are the counterterm trees of the composite graphs. With $\{S^{(1)}(x), O\} = x.O$, Theorem 8.15 (i) gives (12.29); along R_r the composite graphs vanish, and Theorem 8.15 (ii) gives every finite T. □

Theorem 12.43 (Dilatation, translations and rotations from the defect flow)**.** *Under the hypotheses of Theorem 12.42, for every cocycle $O = \int \omega\, X_1 \cdots X_n$:*

$$\begin{aligned} D_1(O) &= 4\big(\mathrm{lc}(O) - \{\mathcal{N}, O\}\ \textit{units}\big) + \sum_{a<b} \pm\mathcal{C}_\Delta(X_a, X_b) - \big\{\hbar F_1^{(1)}(\Delta) - 4\mathcal{N}\ \textit{units}, O\big\}, \\ P_1(O) &= \sum_{a<b} \pm\mathcal{C}_{P_b}(X_a, X_b) - \big\{\hbar F_1^{(1)}(P_b), O\big\}, \qquad M_1(O) = \sum_{a<b} \pm\mathcal{C}_{M_\omega}(X_a, X_b) - \big\{\hbar F_1^{(1)}(M_\omega), O\big\}, \\ R_1(O) &= -\big\{\hbar F_1^{(1)}(R_r), O\big\}, \end{aligned} \tag{12.30}$$

and, summed over the pairs of the cocycle,

$$\sum_{a<b} \pm\Upsilon_\Delta = \sum_{a<b} \pm\big(4\,\mathrm{lc}(X_aX_b) + \mathcal{C}_\Delta(X_a, X_b)\big), \quad \sum_{a<b} \pm\Upsilon_{P_b} = \sum_{a<b} \pm\mathcal{C}_{P_b}, \quad \sum_{a<b} \pm\Upsilon_{M_\omega} = \sum_{a<b} \pm\mathcal{C}_{M_\omega}, \quad \sum_{a<b} \pm\Upsilon_{R_r} = 0. \tag{12.31}$$

Proof. Theorems 12.38 (iii) and 12.42 compute the same local representative, which is unique (Theorem 8.15 (i)), and the one-body terms $-\{\hbar F_1^{(1)}(x), O\}$ are the same in both. By Theorem 12.53 (iv), with Remark 12.54 for M_ω, the composite term of a cocycle is $\sum_{a<b} \pm\mathcal{C}_x(X_a, X_b)$. For Δ, $4\,\mathrm{lc}(O) = \sum_{a<b} \pm 4\,\mathrm{lc}(X_aX_b)$ (Theorem 10.4 (iii)), and the map $M \mapsto \mathrm{lc}(M) - \{\mathcal{N}, M\}$ units is that of Theorem 10.16. □

The factor four and the logarithmic coefficient in (12.30) are outputs of the theorem, an identity between two evaluations of one defect flow: (12.26), which defines the numbers, and the evaluation through the regulator variation of Theorem 12.40, a property of $P(\varepsilon, T)$ proved from its definition through (9.8) and (10.2), the latter following from $\sigma_\lambda P(\varepsilon, \infty) = P(\lambda^2\varepsilon, \infty)$ (Proposition 6.3).

Remark 12.44 (The pair form)**.** The pair form is the statement, for every pair of letters of any slots with any derivatives, that $\Upsilon_\Delta = 4\,\mathrm{lc} + \mathcal{C}_\Delta$, $\Upsilon_{P_b} = \mathcal{C}_{P_b}$, $\Upsilon_{M_\omega} = \mathcal{C}_{M_\omega}$ and $\Upsilon_{R_r} = 0$. The pair kernels are graded symmetric, and the difference $\Upsilon_x - \mathcal{V}_x - \mathcal{C}_x$ is a two-body kernel whose sum over the pairs of every cocycle vanishes by (12.31). On the pair $(A|_{W_\mathbb{R}^\perp}, A|_{W_\mathbb{R}^\perp})$ the pair form follows from (12.31), the scalar bilinears being cocycles with this single pair: there $\Upsilon^0_\Delta = \frac{40}{3}O$ (traceless) and $-\frac{104}{3}O$ (trace) units, with $4\,\mathrm{lc} = 16O$ and $-32O$ units and $\mathcal{C}^0_\Delta = -\frac{8}{3}O$ units (Example 11.22), and $\Upsilon_{P_b} = \mathcal{C}_{P_b} = -\frac{1}{3}\partial_b O$ units (Corollary 11.56); along R_r graded symmetry and $\mathfrak{so}(W_\mathbb{R}^\perp)$-covariance leave the form $a\big((R_r.A|_{W_\mathbb{R}^\perp})A|_{W_\mathbb{R}^\perp} + A|_{W_\mathbb{R}^\perp}(R_r.A|_{W_\mathbb{R}^\perp})\big)$, which the cocycle identity on the traceless bilinears sets to 0. On every complete row of the tables of Remark 12.69 the pair form holds entry by entry, in both evaluations at one letter derivative and in the first at two. For a pair whose functional $\int \omega\, X_aX_b$ is not s-closed, Theorem 12.42 gives the pair form once the terms in $s\int \omega\, X_aX_b$ of the two evaluations, Theorem 12.53 (vi) in the second and the classes (F), (C0) in the first, are shown to agree.

Definition 12.45 (Centred Killing data)**.** For a point x_O and $z := x - x_O$ the Killing data with explicit position decompose as

$$\Delta:\ 2x = 2x_O + 2z, \qquad M_\omega:\ \omega x = \omega x_O + \omega z, \qquad \eta:\ \rho(x)\eta = \rho(x_O)\eta + \rho(z)\eta, \qquad K_b:\ v(x) = v(x_O) + \big(-2(b, x_O)z + \omega_{x_O}z\big)$$

with $\omega_{x_O}z := 2\big((x_O, z)b - (b, z)x_O\big)$. For y with Killing data of positive degree, $\Upsilon^0_y(X_a, X_b)$ is the two-body kernel at the point x_O of the pair computed with the Killing data centred at x_O; on plane waves, with the momentum r injected at the vertex and absorbed by ω, the monomials z^μ, $z^\mu z^\nu$ act as $-i\partial_{r_\mu}$ and $-\partial_{r_\mu}\partial_{r_\nu}$ at $r = 0$.

Theorem 12.46 (Position decompositions). • *(i) For every pair of letters at the point x_O, graph by graph,*

$$\begin{aligned}
&\Upsilon_\Delta = 2x_O^\mu\,\Upsilon_{P_{e_\mu}} + \Upsilon^0_\Delta, \qquad \Upsilon_{M_\omega} = (\omega x_O)^\mu\,\Upsilon_{P_{e_\mu}} + \Upsilon^0_{M_\omega}, \qquad \Upsilon_\eta = \sum_\mu x_O^\mu\,\Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta,\\
&\Upsilon_{K_b} = v(x_O)^\mu\,\Upsilon_{P_{e_\mu}} - (b,x_O)\,\Upsilon^0_\Delta + \Upsilon^0_{M_{\omega_{x_O}}} + \Upsilon^0_{K_b},
\end{aligned} \tag{12.32}$$

and the same decompositions hold for every coefficient $\mathrm{ct}^{(1)}_{x,m}$, $\mathrm{lc}^{(1)}_x$ and for $I^{CT,(1)}(x)$; Υ^0_η is computed with $S^{(1)}(\eta)$ at the spinor $\rho(z)\eta$ together with s_0, the term Y_η of η. belonging to it.

- *(ii) If $\hbar F_1$ solves the arity-two equation along (y,P_b) for $y\in\{\Delta,M_\omega,\eta,K_b\}$, then, with $S^{(2)}(y,P_b)=0$, the position part along b of $\hbar F_1^{(1)}(y)$ is*

$$\pm\hbar F_1^{(1)}([y,P_b]) \pm \big\{S^{(1)}(y),\hbar F_1^{(1)}(P_b)\big\} \pm s\big(\hbar F_1^{(2)}(y,P_b)\big) \mp \hbar\,\mathrm{Ob}^{(2)}(y,P_b), \tag{12.33}$$

with $[\Delta,P_b]=-2P_b$, $[M_\omega,P_b]=-P_{\omega b}$, $[\eta,P_b]=-\rho(b)\eta$ and $[K_b,P_{b'}]=(b,b')\Delta+M_{bb'}$; in a scheme with $\hbar F_1^{(1)}(P_b)=0$ and an s-exact arity-two obstruction along (y,P_b) the one-body term $-\{\hbar F_1^{(1)}(y),O\}$ decomposes as (12.32) up to s-exact brackets, zero on cohomology (Lemma 10.15).

- *(iii) If the local representatives exist through arity two along (y,P_b) on a family stable under Λ (Corollary 8.17), then on its cocycles, with $\Lambda^{(1)}(x)=\Lambda_0^{(1)}(x)+\hbar\Lambda_1^{(1)}(x)$ as in Proposition 11.61,*

$$\big[\Lambda_1^{(1)}(y),\Lambda_0^{(1)}(P_b)\big] + \big[\Lambda_0^{(1)}(y),\Lambda_1^{(1)}(P_b)\big] + \Lambda_1^{(1)}([y,P_b]) \cong 0 \tag{12.34}$$

modulo $\Lambda^{(0)}$-exact terms: the position part of D_1 vanishes on cohomology when P_1 is translation covariant; the position part of S_1 along η is, on cohomology, Q_1 along $\rho(b)\eta$ up to the commutator of $\{S^{(1)}(\eta),-\}$ with P_1; the position part of K_1 along b' is, on cohomology, the component along $(b,b')\Delta+M_{bb'}$ up to the commutator of $\{S^{(1)}(K_b),-\}$ with P_1.

Proof. (i) $S^{(1)}(y)$ is linear in the Killing data: for even y through $v.=v^\mu\partial_\mu+\frac12\omega_{\mu\nu}\Sigma^{\mu\nu}+\frac{w}{4}\operatorname{div}v$, whose terms are linear in v, dv and $\operatorname{div}v$; for η through (11.3). Insert Definition 12.45: on plane waves $x^\mu=x_O^\mu+z^\mu$ acts as x_O^μ on the delta function of the total momentum absorbed by ω and as $-i\partial_{r_\mu}$ on the integrand (Proposition 11.10 (i)); the constant part $v(x_O)$ of the field of K_b is a translation, its linear part is $-(b,x_O)$ times the dilatation field $2z$ plus the rotation $\omega_{x_O}z$, with $\operatorname{div}v(x)=-8(b,x_O)-8(b,z)$, $\operatorname{div}(2z)=8$, $\operatorname{div}(\omega_{x_O}z)=0$, and the quadratic part $v(z)$ is K_b centred at x_O. The decomposition commutes with the loop integral, the simplex integration, the special edge and the expansion in ε (Theorem 11.45 (ii), (iii)), and the counterterms are built linearly from $S^{(1)}(y)$. (ii) The arity-two component of $s(\hbar F_1)=\hbar\,\mathrm{Ob}$ at (y,P_b) has the terms $\pm\hbar F_1^{(1)}([y,P_b])$, $\pm\{S^{(1)}(y),\hbar F_1^{(1)}(P_b)\}\pm\{S^{(1)}(P_b),\hbar F_1^{(1)}(y)\}$ and $(Q+\{I,-\})\hbar F_1^{(2)}(y,P_b)$; $\{S^{(1)}(P_b),G\}$ for $G=\int(x^\mu g_\mu+g_0)$ is minus $\int b^\mu g_\mu$, the position part. (iii) Corollary 8.17 (ii) at (y,P_b) with $S^{(2)}(y,P_b)=0$, whose order-$\hbar$ part is a graded commutator with $\Lambda^{(0)}$; Lemma 11.9 identifies $[\mathcal{M},\{S^{(1)}(P_b),-\}]$ with the position part of a map $\mathcal{M}$ polynomial in the position. □

Corollary 12.47 (Special supercharges and special conformal fields). *For $O=\int\omega\,X_1\cdots X_n$ at the point x_O of its density, with the superscript 0 denoting the parts centred at x_O,*

$$\begin{aligned}
&S_1(O) = Q_1(O)\big|_{\rho(x_O)\eta} + S_1^0(O), \qquad S_1^0(O) := \sum_{a<b}\pm\Upsilon^0_\eta(X_a,X_b) - \big\{\hbar F_1^{(1)}(\eta)^0,O\big\},\\
&K_1(O) = v(x_O)^\mu P_1(O)\big|_{P_{e_\mu}} - (b,x_O)\,D_1^0(O) + M_1^0(O)\big|_{\omega_{x_O}} + K_1^0(O),
\end{aligned} \tag{12.35}$$

when the one-body terms decompose as in Theorem 12.46 (ii), and otherwise up to the bracket with the right side of (12.33) *minus its first term. On the scalar bilinears* $\Upsilon^0_\eta(A|_{W^\perp_{\mathbb{R}}}, A|_{W^\perp_{\mathbb{R}}})$ *vanishes on the colour singlet and* $\{\int s_0, A|_{W^\perp_{\mathbb{R}}}\} = 0$*, so the two-body part of* S_1 *is that of* Q_1 *along* $\rho(x_O)\eta$*;* $\Upsilon^0_{K_b}(A|_{W^\perp_{\mathbb{R}}}, A|_{W^\perp_{\mathbb{R}}})$ *and the rotation part vanish there (Proposition 11.55).*

Proof. Insert (12.32) into (12.26) for $y = \eta$, K_b, with the decompositions of the one-body terms; the components along P_{e_μ}, $\rho(e_\mu)\eta$, Δ and $M_{\omega_{x_O}}$ are those of (12.26) along these generators. □

Proposition 12.48 (The endpoint along K_b)**.** *Along* K_b *at every finite* T*: every defect cycle has* $T\partial_T = O(\varepsilon)$*, so* Υ_{K_b} *is independent of* T*; the graphs of the classes (Br) and (Ac) carry at fixed* ε *the term* $\frac{3}{8\pi^2}\hbar \log T$ *times a tree with the factor* $e^{-\varepsilon|p|^2} - 1$ *at a generic momentum, or with its gradient* $-2\varepsilon p\, e^{-\varepsilon|p|^2}$ *along* (η_1, η_2)*, and tend to* 0*; hence* $K_1(O)$ *is independent of* T*, and the observables keep the term* $\frac{3}{8\pi^2}\hbar \log T$ *of their order-*$\hbar^0$ *part generated by the flow (Theorem 8.27), whose image under* Q *vanishes on gauge-invariant* O*.*

Proof. Theorem 12.38 (v). The coefficient of $\log T$ of a graph with $S^{(1)}(K_b)$ on its cycle is $\frac{3}{8\pi^2}\hbar$ times the tree obtained by cutting the insertion out of the cycle with the freed legs on the constant modes (Proposition 8.25); in (Br) and (Ac) this tree contains the special edge at a generic momentum. Theorem 8.27. □

12.6 Locality of the composite term

Throughout this section $x \in \{\Delta, P_b, M_\omega\}$, so that the composite line $\mathsf{L}_x := (P \star x. \otimes 1 + 1 \otimes P \star x.)P$ is Q-closed (Proposition 12.12), $T = \infty$ with generic momenta or any finite T, and $P = P(\varepsilon, T)$. The composite term enters the second evaluation (Theorem 12.42) and the identities of Theorem 12.43; the components themselves are given by Theorem 12.38 along every generator. Lemma 12.50, Lemma 12.51, Proposition 12.52 and Theorem 12.53 are Lemma 11.15, Lemma 11.16, Proposition 11.17 and Theorem 11.18, proved there for Δ and P_b, together with the rotations of Remark 12.54.

Definition 12.49 (Composite loops of the action)**.** $\Xi_x(\varepsilon)$ is the sum of the amputated one-particle-irreducible composite loops of I along x: cycles of vertices of I and edges P in which one edge is L_x, all other legs open, so that $\frac{\hbar}{2}\partial_{\mathsf{L}_x} W(P, I)_{\text{tree}}$ is the tree flow of $\Xi_x(\varepsilon)$. $\Xi^\sigma_x(\varepsilon)$ is minus the sum of the same loops with one edge of the cycle other than L_x replaced by the special edge, with the sign of Proposition 11.4. The renormalised composite functional is $\Xi^{\text{ren}}_x := \lim_{\varepsilon\to 0}\big(\Xi_x(\varepsilon) - I^{CT,(1)}(x)\big)$, a functional of degree -1 at generic momenta, whose quadratic part along Δ is the renormalised composite two-point functional of Remark 9.27. The composite graphs of a local functional O are of type I when the cycle passes through two letters of O and of type II when it passes through vertices of I alone; type II is the tree insertion of $\Xi_x(\varepsilon)$ into O, written $\mathrm{tr}(\Xi_x, O)$.

Lemma 12.50 (The image of the composite loops under s)**.** *For every* $\varepsilon > 0$*, as functionals at generic momenta,* $s\,\Xi_x(\varepsilon) = \Xi^\sigma_x(\varepsilon)$*. Consequently: (i)* Ξ^σ_x *has no term in* $\log\varepsilon$ *and no poles, and* $I^{CT,(1)}(x) = \log\varepsilon \cdot L^{(1)}(x)$ *with* $sL^{(1)}(x) = 0$*; (ii) the composite part along* x *of Lemma 12.41 is* $s\,\Xi^{\text{ren}}_x = [\varepsilon^0]\Xi^\sigma_x$*, a local functional, although* Ξ^{ren}_x *is nonlocal on the letters* $A|_V$*,* λ*,* c*.*

Proof. Lemma 11.15 and its proof, which use of the composite line only $[Q, P \star x.P\star] = 0$, true along P_b, Δ and M_ω (Proposition 12.12 (i), (ii)); the absence of poles of $I^{CT,(1)}(x)$ is Proposition 12.7. □

Lemma 12.51 (Special edges on trees)**.** *Let* O *be a local functional. (B) The composite graphs of type I with the special edge on a tree, at an output of the cycle or at a spectator letter of* O*, together with the special-edge parts of the counterterm trees of* $\mathrm{ct}^{(1)}_x(O)$*, are* $e^{-\varepsilon|p|^2} - 1$ *at a generic momentum* p *times renormalised composite cycles, and tend to* 0*. (D) The composite graphs of type II with the special edge on a tree, and the special-edge parts of the counterterm trees of* $I^{CT,(1)}(x)$*, tend to* 0*.*

Proof. Lemma 11.16 and its proof; (B) is Lemma 12.37 with G the composite cycle through the pair with its trees. □

Proposition 12.52 (The image of the composite cycles through the pairs)**.** *For every local functional O and every $\varepsilon > 0$,*

$$Q(\textit{type I composite graphs of } O) = (\mathrm{A}) + (\mathrm{B}) - \epsilon_\Xi\{\Xi_x(\varepsilon), O\} - (\textit{composite graphs of } sO) \mp \operatorname{tr}(\Xi_x, sO),$$

where (A) are the special-edge composite cycles of the pairs of letters of O with trees of I on their outputs (Definition 11.5), (B) the type I graphs with the special edge on a tree, $\epsilon_\Xi \in \{\pm 1\}$ the sign with which $\{\Xi_x, O\}$ enters $Q\operatorname{tr}(\Xi_x, O)$, and the last two terms are present for $sO \neq 0$; moreover

$$Q\operatorname{tr}(\Xi_x, O) = \operatorname{tr}(s\Xi_x, O) + \epsilon_\Xi\{\Xi_x, O\} \pm \operatorname{tr}(\Xi_x, sO) + (\mathrm{D}),$$

with (D) the type II graphs with the special edge on a tree.

Proof. Proposition 11.17 and its proof, with Lemma 12.50 in place of Lemma 11.15. □

Theorem 12.53 (The composite term is two-body and local)**.** *Let $O = \int \omega\, X_1 \cdots X_n$ be a monomial local functional and $x \in \{\Delta, P_b, M_\omega\}$.*

- *(i) The special-edge composite cycles of every pair are $\sum_{m\geq 1} \varepsilon^{-m}$ times local coefficients, plus $\mathcal{C}_x(X_a, X_b)$, plus a remainder tending to 0; they contain no $\log\varepsilon$.*
- *(ii) $I^{CT,(1)}(x) = \log\varepsilon\, L^{(1)}(x)$ with $sL^{(1)}(x) = 0$, and the composite part along x is $s\,\Xi_x^{\mathrm{ren}} = [\varepsilon^0]\Xi_x^\sigma$.*
- *(iii) (Arity-one BRST covariance.) If $sO = 0$: $s\,\mathrm{lc}_x^{(1)}(O) = -\epsilon_\Xi\{L^{(1)}(x), O\}$, and $\sum_{a<b} \pm[\varepsilon^{-m}](\textit{special-edge com}$ $s\,\mathrm{ct}_{x,m}^{(1)}(O)$ for $m \geq 1$.*
- *(iv) If $sO = 0$, the composite term of O along x exists and*

$$(\textit{composite term of } O \textit{ along } x) = \sum_{a<b} \pm\mathcal{C}_x(X_a, X_b), \tag{12.36}$$

  a local two-body functional with no one-body term.

- *(v) (Cancellation of the nonlocal one-body terms.) If $sO = 0$, the image under Q of the renormalised type I graphs contains $-\epsilon_\Xi\{\Xi_x^{\mathrm{ren}}, O\}$ and that of the renormalised type II graphs $+\epsilon_\Xi\{\Xi_x^{\mathrm{ren}}, O\}$; both are nonlocal one-body terms on the letters $A|_V$, λ, c (the factors $\log|p|^2$ and, on $A|_V$, $(p,a)(p,b)|p|^{-2}$ of Remark 9.27) and local on $A|_{W_{\mathbb{R}}^\perp}$, and they cancel in the composite term. In the form of Theorem 10.38, the composite terms of the pairs plus those of the legs equal $\sum_{a<b} \pm\mathcal{C}_x(X_a, X_b)$.*
- *(vi) (Gauge-variant operators.) For $sO \neq 0$ the right side of (12.36) acquires minus the limit of the renormalised composite graphs of sO of both types minus their counterterm trees: the nonlocal brackets $\{\Xi_x^{\mathrm{ren}}, -\}$ on the letters of sO and the Q-preimages of the composite cycles through the pairs of letters of sO. The components of O are local and two-body by Theorem 12.38.*

Proof. Theorem 11.18 (i)–(vi) and its proof, with Lemmas 12.50, 12.51 and Proposition 12.52 in place of Lemmas 11.15, 11.16 and Proposition 11.17; along M_ω every step applies by Remark 12.54. □

Remark 12.54 (Rotations, and the generators with a non-closed line)**.** For M_ω the composite line is the term $\int(\omega x\cdot\partial A|_V+\omega A|_V,A^*|_V)$ joined to two edges, Q-closed because M_ω. commutes with Q^{GF} and H; every step above applies, and by Theorem 12.46 (i) $\mathcal{C}_{M_\omega}=(\omega x_O)^\mu\mathcal{C}_{P_{e_\mu}}+\mathcal{C}^0_{M_\omega}$, the intrinsic part computed with the spin term $\frac12\omega_{\mu\nu}\Sigma^{\mu\nu}$ and the transport $\omega z\cdot\partial$ centred at x_O. Along K_b, ξ and η the composite line has a nonzero image under Q (Proposition 12.12), Proposition 11.4 acquires the terms of that image, and the components along these generators are local and two-body by Theorem 12.38. The proof of Theorem 12.53 uses $sO=0$ exactly (or keeps sO, (vi)), the classical master equation, $[x.,Q]=0$ with $[Q,P\star x.P\star]=0$, the injectivity of the tree flow on functionals graded by legs, the absence of $\log\varepsilon$ on cycles with one special edge and the weight count that excludes poles of $I^{CT,(1)}(x)$; the sign ϵ_Ξ enters through the cancellation of the two brackets and is never evaluated.

12.7 Representatives and classes on the families

Definition 12.55 (Representative and class of a component)**.** Let $\mathfrak{F}$ be a family of colour-singlet local functionals of degree 0, stable under Λ and under the classical transformations along a subalgebra $\mathfrak{h}\subset\mathfrak{psl}(4|4)$, with $\hbar L_1^{(0)}=0$ on $\mathfrak{F}$ for a finite part ψ (Definition 10.36 (ii)), and let H be the cohomology of s on $\mathfrak{F}$. Write $\Lambda^{(1)}(x)=\Lambda_0^{(1)}(x)+\hbar\Lambda_1^{(1)}(x)$ with $\Lambda_0^{(1)}(x)=\pm\{S^{(1)}(x),-\}$ (Proposition 11.61). The representative of the component along x on $O\in\mathfrak{F}$ is the local functional $\hbar\Lambda_1^{(1)}(x)O=\hbar L_1(O)^{(1)}(x)$ of (12.26); its class is the map $\delta(x)$ induced on H by $-\hbar\Lambda_1^{(1)}(x)$, and $\rho_0(x)$ is the map induced by $-\Lambda_0^{(1)}(x)$. A change of the finite part of the operators or a field redefinition changes δ by the coboundary $x\mapsto[\rho_0(x),\phi]$ with $\phi\in\mathrm{End}(H)$ preserving the classical Weyl weight, the coupling direction leaves δ unchanged, and the outer direction G_B changes it by $-a\,\rho_0\circ B$ (Theorem 12.63 (iv)). The scheme-independent class of the components is the class of δ in $H^1(\mathfrak{h};\mathrm{End}(H))$ modulo $\mathbb{C}\,\rho_0\circ B$; it contains the spectrum of $\delta(\Delta)$ on each classical weight space of H.

Theorem 12.56 (The components on the scalar bilinears)**.** *Let $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ with $w_1,w_2\in W_{\mathbb{R}}^\perp\otimes\mathbb{C}$, $\psi=0$, $\xi\in\Sigma$ of either chirality, and $\mathcal{B}_1,\dots,\mathcal{B}_4$ the functionals of Lemma 11.53: $\mathcal{B}_1=\{S^{(1)}(\xi),O\}$, $\mathcal{B}_2=(w_1,w_2)\int\omega\,(\rho(A|_{W_{\mathbb{R}}^\perp})\xi,\lambda)$, $\mathcal{B}_3=(w_1,w_2)\int\omega\,(\rho(A|_V)\xi,\lambda)$, $\mathcal{B}_4=(w_1,w_2)\int\omega\,(c,(\xi,\lambda^*))$. With the uniform coefficients of Theorem 12.31, the representatives are*

$$
\begin{aligned}
&Q_1(O)=\Big[\big(\tfrac{79}{12}-3\log\tfrac32-t_\xi\big)\mathcal{B}_1+4\log\tfrac32\,\mathcal{B}_2+4\log\tfrac32\,\mathcal{B}_3+0\cdot\mathcal{B}_4\Big]\ \textit{units},\qquad \hbar L_1(O)^{(0)}=-\big(2+4\log\tfrac32\big)(w_1,w_2)\int\omega\\
&S_1(O)=\big(\tfrac{79}{12}-3\log\tfrac32-t_\eta\big)\{S^{(1)}(\eta),O\}\ \textit{units}+x_O^\mu\Big[4\log\tfrac32\,\mathcal{B}_2+4\log\tfrac32\,\mathcal{B}_3\Big](\rho(e_\mu)\eta)\ \textit{units},\\
&D_1(O)=\big(-4t_\Delta-\tfrac43\big)O\ \textit{units (traceless)},\qquad \big(-4t_\Delta-\tfrac{148}{3}\big)O\ \textit{units (trace)},\qquad \textit{at the point of the density},\\
&P_1(O)=-\big(\tfrac13+t_P\big)\,\partial_bO\ \textit{units},\qquad M_1(O)=-\big(\tfrac13+t_M\big)\{S^{(1)}(M_\omega),O\}\ \textit{units},\qquad R_1(O)=0,\\
&K_1(O)=v(x_O)^\mu P_1(O)\big|_{P_{e_\mu}}-(b,x_O)\,D_1^0(O)+M_1^0(O)\big|_{\omega_{x_O}}+K_1^0(O),
\end{aligned}
\tag{12.37}
$$

with the two-body part of K_1^0 equal to 0 and its one-body part

$$
K_1(O)\big|_{\textit{one-body}}=\tfrac{44}{3}\int\omega\,(b,x)(\iota_{w_1}A,\iota_{w_2}A)\ \textit{units}-\theta_\perp^K\big\{S^{(1)}(K_b)_\perp,O\big\}\ \textit{units}-\sum_{j=1}^4\kappa_j^K\{G_j^K,O\},
$$

whose part linear in x is $-(b,x_O)$ times the one-body part $-\frac{44}{3}O$ units of D_1^0. The coefficient of $\mathcal{B}_1$ in Q_1 is the two-body $-2\log\frac32$ plus the one-body $-\big(\log\frac32-\frac{79}{12}+t_\xi\big)$; the intrinsic part of S_1 vanishes. At $t_\Delta=t_M=-\frac13$: $D_1=0$ on the traceless bilinears, $D_1=-48\,O$ units $=\frac{3g^2N}{2\pi^2}O$ on the trace, and $M_1=0$.

Proof. Theorem 12.38 with the two-body kernels of the pair $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ and the one-body terms from Theorem 12.21; on O the terms of $\hbar F_1^{(1)}(x)$ with a leg $A^*|_{W_{\mathbb{R}}^\perp}$ act and the others give 0. Along ξ: of $\hbar F_1^{(1)}(\xi)|_2 = (\log\frac{3}{2} - \frac{79}{12})(T_1+T_2) + t_\xi(S_2^V + S_2^\perp)(\xi)$ units the terms with that leg are T_2 in both, and $\{T_2, O\} = \{S^{(1)}(\xi), O\} = \mathcal{B}_1$, the arrow $A|_{W_{\mathbb{R}}^\perp} \to \lambda$ of $S^{(1)}(\xi)$ being T_2 alone; the two-body part is (11.12) (leg sets with $D=0$, Remark 12.39) and $L_1^{(0)}(O)$ is Theorem 11.54 (iii). Along η the position part of the solution is $x^\mu \hbar F_1^{(1)}(\rho(e_\mu)\eta)$ with the uniform t_η (Theorem 12.31), s_0 and $\int(\rho(A|_V)\eta, \lambda^*)$ have no leg A^*, and Corollary 12.47 gives the two-body part. Along Δ, $-\{\frac{22}{3}\int(A, A^*), O\} = -\frac{44}{3}O$ and $-\{t_\Delta S^{(1)}(\Delta)_\perp, O\} = -4t_\Delta O$ units at the point, and the two-body sums are $\frac{40}{3}O$ and $-\frac{104}{3}O$ units (Remark 12.44). Along P_b, $\Upsilon_{P_b} = -\frac{1}{3}\partial_b O$ units and $-\{t_P S^{(1)}(P_b)_\perp, O\} = -t_P\partial_b O$ units. Along M_ω, $\Upsilon^0_{M_\omega} = 0$ on the pair, the position part is $(\omega x_O)^\mu \Upsilon_{P_{e_\mu}}$ (Theorem 12.46), the spin term of $\hbar F_1^{(1)}(M_\omega)$ has no leg $A^*|_{W_{\mathbb{R}}^\perp}$, and $\{S^{(1)}(M_\omega), O\}$ is the transport $(\omega x)\cdot\partial O$. Along R_r, $\Upsilon_{R_r} = 0$ and $t_R = 0$. Along K_b, Corollary 12.47 with the x-linear part of $\hbar F_1^{(1)}(K_b)|_2$ (Theorem 12.21). The values at $t_\Delta = -\frac{1}{3}$: $-\frac{4}{3} + \frac{4}{3} = 0$ and $-\frac{148}{3} + \frac{4}{3} = -48$, with the unit $-\frac{g^2 N}{32\pi^2}$. □

Corollary 12.57 (Classes on the scalar bilinears).

- *(i) The spectrum of $\delta(\Delta)$ on the scalar bilinears is 0 on the twenty traceless bilinears and -48 units $= \frac{3g^2N}{2\pi^2}$ on the trace, the Konishi value $2\gamma_K$ with $\gamma_K = \frac{3g^2N}{4\pi^2}$: it is independent of the scheme, $t_\Delta = -\frac{1}{3}$ being fixed by the arity-two equations (Theorem 12.31).*

- *(ii) On the traceless bilinears $\delta(\xi)$ is $c_Q\,\rho_0(\xi)$ with $c_Q = \frac{79}{12} - 3\log\frac{3}{2} - t_\xi$ units, $\delta(\eta)$ is $c_S\,\rho_0(\eta)$ with $c_S = \frac{79}{12} - 3\log\frac{3}{2} - t_\eta$ units, and $\delta(P_b) = p_1\rho_0(P_b)$ with $p_1 = -(\frac{1}{3} + t_P)$ units. The outer direction shifts $c_Q(\xi_\pm)$ by $\mp a$, and a finite part of the operators shifts c_Q, c_S, p_1 by coboundaries; beyond $\delta(\Delta)$, the scheme-independent content of these components on the scalar bilinears is contained in the relations (R2), (R3) of Corollary 12.64.*

- *(iii) At $\psi = 0$, as functions of the action-side free directions,*

$$c_Q(\xi_+) + c_Q(\xi_-) - p_1 = c_Q(\xi_+) + c_S(\eta_+) - \tfrac{1}{4}D_1\big|_{traceless} = \tfrac{27}{4} - 8\log\tfrac{3}{2} \text{ units}, \tag{12.38}$$

  independent of t_Δ, t_P, t_ξ, t_η and of the inner and outer directions, by (12.22)*. A finite part of the operators shifts c_Q, c_S and p_1 by coboundaries, so these combinations depend on the operator scheme; the relations (R2), (R3) of Corollary 12.64 on the scalar bilinears contain them together with the components on the descendants of Weyl weight $\frac{5}{2}$ and the arity-two kernels.*

- *(iv) The coefficients c_Q, c_S, every coefficient with $\log\frac{3}{2}$, P_1, $L_1^{(0)}(O)$ (removed by $\psi(O) = -(1 + 2\log\frac{3}{2})(w_1, w_2)\int\omega\,(A|_V, A|_V)$ units), M_1, R_1 and the individual pair kernels depend on the scheme.*

Proof. (i) Theorem 12.56 and Theorem 8.20. (ii) Theorem 12.56, Theorem 12.63 (iv) and Proposition 11.52 (ii), (iv). (iii) $2\cdot\frac{79}{12} - 6\log\frac{3}{2} - (t_{\xi_+} + t_{\xi_-}) + \frac{1}{3} + t_P = \frac{81}{12} - 8\log\frac{3}{2}$ by (12.22), and $2\cdot\frac{79}{12} - 6\log\frac{3}{2} - (t_{\xi_+} + t_{\eta_+}) + \frac{1}{3} + t_\Delta$ is the same. (iv) Theorem 11.54 (iii), Corollary 11.28 and Theorem 12.63 (iv). □

Proposition 12.58 (The twisted differential on the chiral bilinears). *Let ${\xi_\alpha}^i = u_\alpha z^i$ be of rank one and w_1, w_2 with $\rho(w_1)\xi = \rho(w_2)\xi = 0$, so that $O = \int\omega\,(\iota_{w_1}A, \iota_{w_2}A)$ is a combination of the bilinears $\int\omega\,(A_{ij}z^j, A_{kl}z^l)$ (Proposition 11.62). Then the order-$\hbar$ part of the twisted differential vanishes on O: $\hbar Q_1^\xi(O) = \hbar L_1(O)^{(0)} + Q_1(O) - \frac{1}{2}\hbar L_1(O)^{(2)}(\xi,\xi) - \cdots = 0$ (Definition 8.22), for every value of the action-side free directions, of ψ and of the outer and inner directions. For every quasi-isomorphism Φ of Theorem 11.65, whatever its normalisation, the class $\Phi\circ[Q_1^\xi]\circ\Phi^{-1}$ vanishes on the image of $[O]$, the classes of the two-letter chiral operators $\operatorname{tr}\gamma^{(I}\gamma^{J)}$.*

Proof. $\mathcal{B}_1 = \{S^{(1)}(\xi), O\}$ contains $\iota_{w_a}\Gamma(\xi,\lambda) = (\rho(w_a)\xi,\lambda) = 0$ [ESW, (28)], and $(w_1,w_2) = 0$ (Proposition 11.62) removes $\mathcal{B}_2$, $\mathcal{B}_3$, $\mathcal{B}_4$ and $L_1^{(0)}(O)$; Theorem 12.56 holds for every ξ, being linear in ξ, and its one-body term is a multiple of $\mathcal{B}_1$ for every value of t_ξ. The (ξ,ξ) component of a functional of degree 0 and weight 2 has degree -1 and weight 3, while a colour-singlet output of degree -1 has at least two letters, an antifield of weight at least $\frac{5}{2}$ and a field of weight at least 1, total at least $\frac{7}{2} > 3$; the component of arity $k \geq 3$ has degree $1-k \leq -2$ and weight $2+\frac{k}{2}$, below the weight 5 of the outputs of degree -2, and vanishes likewise; the one-body term $-\{\hbar F_1^{(2)}(\xi,\xi), O\}$ vanishes, $\hbar F_1^{(2)}(\xi,\xi)$ having the legs (λ^*,λ^*) (Table 12.2 and Theorem 12.15). Under the scheme changes Q_1^ξ changes by a commutator with Q_0^ξ or by a multiple of Q_0^ξ (Theorem 8.23), both zero on the class of O, and Φ is linear. □

Remark 12.59 (The comparison with the holomorphic twist)**.** Proposition 12.58 is the case $n = 2$ of [BK, Sec. 3.4.1, (3.77)], $Q_1(\gamma^I\gamma^J) = 0$, at the level of classes. The full comparison of $[Q_1^\xi]$ modulo $\operatorname{Im} Q_0^\xi + \mathbb{C}\, S^\xi$ with [BK, (3.7), (3.59)–(3.70)] is the statement of Theorem 11.65; its inputs are the transport map Φ of [ESW, Thm 10.9] with the normalisation of [BK] for the letters b, β, γ, $\partial_{\dot\alpha}c$, the arity-one entries of Table 11.6, which are kernels of the form (12.24) (their leg sets have $D = 0$), the arity-two kernels along (ξ,ξ) for the entries $b\,c$ and $b\,b$ with b represented by the field strength, and the finite parts ψ, t_ξ, a that move the colour-order-dependent terms in $\log\frac{3}{2}$ of the individual entries into $\operatorname{Im} Q_0^\xi$. The comparison is a cross-check: no number of the components is taken from it.

Proposition 12.60 (The dilatation component on the families)**.** *On the families of Definition 11.36, at $t_\Delta = -\frac{1}{3}$ and ψ as in Definition 10.36 (ii), $\delta(\Delta)$ has the spectra of Table 12.7, independent of the scheme; with the unit $-\frac{g^2N}{32\pi^2}$, $D_1 = -8$ units times the eigenvalue E of $\sum_l H_{l,l+1}$ is $\frac{g^2N}{4\pi^2}E = 2\gamma$. On twist two and the fermion $\mathfrak{sl}(2)$ sector, and in the first evaluation of Remark 12.69 on the three families with gaugino or field-strength letters, $\sum_{a<b} \pm\Upsilon_{P_n} = -\frac{1}{3}\partial_n$ units, so $P_1 = -(\frac{1}{3} + t_P)\partial_n$ units; on twist two, the fermion sector, the family with two gaugino components and the mixed family the dilatation commutes with the classical translation along the null direction, $D_1^{(S'+1)}\partial_n = \partial_n D_1^{(S')}$, as an exact matrix identity. On every complete row of the single-pair tables of Remark 12.69 the pair form of Remark 12.44 holds.*

family	range	spectrum of D_1 in units
$\mathfrak{su}(2)$ sector $\operatorname{tr}(Z^{L-M}X^M)$	$L \leq 8$	$-8\sum_l(1-P_{l,l+1})$, the Heisenberg spectrum
$\mathfrak{su}(1\|1)$ sector $\operatorname{tr}(Z^{L-M}\psi^M)$	$L \leq 6$	$-8\sum_l(1-\Pi_{l,l+1})$, graded exchange Π; Table 11.5 times -8
twist two $\operatorname{tr}(D_n^k Z\, D_n^{S'-k} Z)$	$S' \leq 6$	$-32h(j)$, $j \leq S'$ even: $0, -48, -\frac{200}{3}, -\frac{392}{5}$
fermion $\operatorname{tr}(D_n^k\psi\, D_n^{S'-k}\psi)$	$S' \leq 5$	$-32h(j+1)$, $j \leq S'$ odd
two gaugino components with $Z'F_{11}$	$S' \leq 3$	$0, -48;\ 0, (-48)^3;\ 0, (-48)^4, -\frac{200}{3};\ 0, (-48)^4, (-\frac{200}{3})^3$
gluon $\operatorname{tr}(D_n^k F_{11}\, D_n^{S'-k} F_{11})$	$S' \leq 2$	$-48;\ -48;\ -\frac{200}{3}, -48$
mixed $\operatorname{tr}(D_n^k F_{11}\, D_n^{S'-k}\psi)$	$S' \leq 4$	$-48;\ (-48)^2;\ -\frac{200}{3}, (-48)^2;\ (-\frac{200}{3})^2, (-48)^2;\ -\frac{392}{5}, (-\frac{200}{3})^2, (-48)^2$

The spectra of D_1 on the families, h the harmonic number and a power the multiplicity; the entries separated by semicolons are $S' = 0, 1, \ldots$. The first four rows are computed by both evaluations of Remark 12.69, the last three by the first, with the eigenvalues recomputed independently.

Proof. Theorem 12.38 along Δ and P_n: the two-body kernels are the coefficients of ε^0 of the defect cycles of the pairs of letters of the families, evaluated through the two-field parts (Proposition 10.34 and the remark before Proposition 11.37), and the one-body term is the bracket with $\hbar F_1^{(1)}(\Delta)$ at $t_\Delta = -\frac{1}{3}$ (Proposition 11.57). Theorem 12.43 with the composite pair kernels of Theorem 11.38 gives the same spectra, a second evaluation. Scheme independence is Theorem 8.20 (iii), t_Δ being fixed by Theorem 12.31. The translation identity is (R7) of Corollary 12.64 along P_n. □

Remark 12.61 (The spectra as a cross-check)**.** The spectra of Table 12.7 are -8 units times the eigenvalues of Proposition 11.37, of Proposition 10.32 and of Table 11.5, the one-loop dilatation operator

of [B03c]: $\gamma = \frac{g^2N}{2\pi^2}h(S)$ on twist two, $\gamma = \frac{3g^2N}{4\pi^2}$ on $\mathrm{tr}(F_{11}F_{11})$ and the trace bilinear, the Konishi multiplet. The agreement is a cross-check of the construction; the numbers come from (12.26).

12.8 The one-loop algebra on operators

Proposition 12.62 (The arity-two component from the defect flow)**.** *Let $\hbar F_1$ solve* (12.3) *through arity two along (x,y), with the local representatives through arity one along x, y, $[x,y]$. For O of degree* 0*, with T finite if x or y is K_b or if $(x,y)=(\eta_1,\eta_2)$ with $\Gamma(\eta_1,\eta_2)|_V \neq 0$, and $T=\infty$ otherwise,*

$$\hbar L_1(O)^{(2)}(x,y) = \sum_{a<b} \pm \Upsilon_{x,y}(X_a,X_b) - \big\{\hbar F_1^{(2)}(x,y),O\big\} + \big(\textit{trees of } \hbar F_1^{(1)}(x), \hbar F_1^{(1)}(y) \textit{ with } S^{(1)}(y), S^{(1)}(x)\big),$$

with $\Upsilon_{x,y}$ the coefficient of ε^0 of the cycles through the pair with one special edge and either two vertices $S^{(1)}(x)$, $S^{(1)}(y)$ or one vertex $S^{(2)}(x,y)$; the bridges cancel their counterterm partners as in Lemma 12.37.

Proof. The proof of Theorem 12.38 applies in arity two: the defect vertex has the arity-two parts $\pm\{S^{(2)}(x,y),O\}$, the special-edge parts of the products of two arity-one brackets, $-\{\hbar F_1^{(2)}(x,y),O\}_\varepsilon$, $-\{I^{CT,(2)}(x,y),O\}_\varepsilon$ and the arity-two counterterm of O; the classes are the same with two arity-one vertices or one arity-two vertex, and (Ac) vanishes by the arity-two equation. The trees with $\hbar F_1^{(1)}(x)$ and $S^{(1)}(y)$ attached to O are the arity-two parts of the products $\Lambda\Lambda$ in the local representative (Theorem 8.15 (iv)). □

Theorem 12.63 (The one-loop algebra)**.** *Let $\mathfrak{F}$, H, ρ_0, δ be as in Definition 12.55.*

- *(i) If the quantisation holds through arity one along x and the local representatives exist, then $s\big(\hbar\Lambda_1^{(1)}(x)O\big)=0$ for every $O\in\mathfrak{F}$ with $sO=0$ (R1).*
- *(ii) If moreover the quantisation holds through arity two along (x,y) and the local representatives exist through arity two, then on the cocycles of $\mathfrak{F}$*

$$\big[\Lambda_1^{(1)}(x),\Lambda_0^{(1)}(y)\big]O + \big[\Lambda_0^{(1)}(x),\Lambda_1^{(1)}(y)\big]O + \Lambda_1^{(1)}([x,y])O = \Lambda^{(0)}\big(\chi_{x,y}(O)\big), \tag{12.39}$$

  with $\chi_{x,y}(O)$ local of degree -1, built from $\hbar L_1(O)^{(2)}(x,y)$ and $\{S^{(2)}(x,y),O\}$ (Corollary 8.17 (ii)).
- *(iii) Under (ii) for all pairs in $\mathfrak{h}$, $\delta([x,y])=[\delta(x),\rho_0(y)]+[\rho_0(x),\delta(y)]$, graded commutators: δ is a one-cocycle of $\mathfrak{h}$ with values in* $\mathrm{End}(H)$*.*
- *(iv) A finite part ψ' or a field redefinition changes δ by the coboundary $x\mapsto[\rho_0(x),\psi'_H]$; the coupling direction leaves δ unchanged; G_B changes it by $-a\,\rho_0\circ B$, a cocycle that on $\mathfrak{psl}(4|4)$ is not a coboundary.*

Proof. (i), (ii) Theorem 8.15 (iv), $L\circ L=0$, at arity one and two on an arity-zero input at order $\hbar$, with Corollary 8.17 and Proposition 11.61; the right side of (ii) is $[\Lambda^{(0)},H_{x,y}]O=\Lambda^{(0)}(H_{x,y}O)$ for $\Lambda^{(0)}O=0$. (iii) On cohomology $\Lambda^{(0)}$-exact terms vanish; ρ_0 is a representation by the order-$\hbar^0$ identity, the term $S^{(2)}$ acting as 0 on cohomology, and the order-$\hbar$ part of $[\rho(x),\rho(y)]=\rho([x,y])$ with $\rho=\rho_0+\delta$ is the cocycle identity. (iv) Proposition 8.19: conjugation by $1+\hbar\psi'$ changes ρ by $[\rho_0,\psi'_H]$ at order $\hbar$; Theorem 8.20 (ii) for G_B, and B is outer (Proposition 8.10). □

Corollary 12.64 (The relations on cohomology)**.** *Under the hypotheses of Theorem 12.63, on cohomology:*

- *(R2) for $\xi,\xi'\in\Sigma$, $[\delta(\xi),\rho_0(\xi')]+[\rho_0(\xi),\delta(\xi')]=\delta(P_{\Gamma(\xi,\xi')|_V})$, with right side 0 for $\xi,\xi'\in S_+\otimes W$;*
- *(R3) for $\xi\in\Sigma$, $\eta\in\Sigma^*$, $[\delta(\xi),\rho_0(\eta)]+[\rho_0(\xi),\delta(\eta)]=\frac{1}{2}(\xi,\eta)\delta(\Delta)+\delta(M_{\xi\eta})+\delta(R_{\xi\eta})$ (Proposition 2.4);*
- *(R4) for a rank-one ξ, $\{\delta(\xi),\rho_0(\xi)\}=0$, the cohomological form of $Q_0^\xi Q_1^\xi+Q_1^\xi Q_0^\xi=0$ (Theorem 8.23);*
- *(R5) $[\delta(K_b),\rho_0(P_{b'})]+[\rho_0(K_b),\delta(P_{b'})]=(b,b')\delta(\Delta)+\delta(M_{bb'})$;*
- *(R6) for $\eta_1,\eta_2\in\Sigma^*$, $[\delta(\eta_1),\rho_0(\eta_2)]+[\rho_0(\eta_1),\delta(\eta_2)]=\delta([\eta_1,\eta_2])$, a special conformal field for $\Gamma(\eta_1,\eta_2)|_V\neq 0$;*
- *(R7) on a family of operators at one point, graded by the classical Weyl weight and closed under the classical action of the generators considered, $[\delta(\Delta),\rho_0(x)]=0$: the one-loop dilatation commutes with the classical action and is constant on the classical multiplets of the family;*
- *(R8) if the family is finite-dimensional and stable under the classical rotations, the restriction of δ to $\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ is a coboundary, and a finite part on the family makes $M_1\cong 0$ and $R_1\cong 0$.*

(R7) along the null translation P_n holds as an exact matrix identity on twist two, the fermion $\mathfrak{sl}(2)$ sector, the family with two gaugino components and the mixed family, and on the $\mathfrak{su}(2)$ sector.

Proof. (R2)–(R6): Theorem 12.63 (iii) with the brackets of Proposition 2.4 and Remark 7.17. (R7): $[\Delta,x]=\lambda_x x$, so $\lambda_x\delta(x)=[\delta(\Delta),\rho_0(x)]+[\rho_0(\Delta),\delta(x)]$; at a point $\rho_0(\Delta)$ acts on the classical Weyl weight w as a fixed multiple of w, and $\delta(x)$ maps weight w to $w-\frac{1}{2}\lambda_x$, as $\rho_0(x)$ does (Proposition 10.47), so $[\rho_0(\Delta),\delta(x)]=\lambda_x\delta(x)$ and the remaining term vanishes. (R8): $\mathfrak{so}(4,\mathbb{C})\oplus\mathfrak{so}(6,\mathbb{C})$ is semisimple and $\mathrm{End}(H)$ finite-dimensional, so the first cohomology vanishes (Whitehead's first lemma), and a coboundary $x\mapsto[\rho_0(x),\phi]$ is removed by the finite part $-\phi$. The last statement: Proposition 12.60. □

Proposition 12.65 (Free directions fixed by the algebra on operators)**.** *Suppose that on the cohomology of a family the free directions of $\hbar F_1^{(1)}(x)$ act as $-t_x$ units times $\rho_0(x)$, with t_x constant on the blocks P, K, Δ, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, $S_+\otimes W$, $S_-\otimes W^*$, $S_+\otimes W^*$, $S_-\otimes W$; along Σ, Σ^*, P_b, Δ, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ this holds on every family, the sector coefficients being uniform (Theorem 12.31). If (R2)–(R7) hold for two choices of these numbers, their difference t satisfies $t_{[x,y]}=t_x+t_y$ whenever $\rho_0([x,y])\neq 0$ on the cohomology, and $t=aB+\frac{c}{2}\lambda$: the outer direction and the finite part $\frac{c}{2}\{S^{(1)}(\Delta),-\}$ (Proposition 9.36). In particular the difference vanishes on Δ, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$.*

Proof. With $\delta_t(x)=-t_x\rho_0(x)$ the cocycle identity reads $-t_{[x,y]}\rho_0([x,y])=-(t_x+t_y)\rho_0([x,y])$. From $[\Delta,x]=\lambda_x x$ with $\lambda_x\neq 0$, $t_\Delta=0$. Write α, β for t on $S_+\otimes W$ and $S_-\otimes W^*$; the brackets $[\Sigma,\Sigma^*]\ni\Delta$ of equal chirality give $t=-\alpha$, $-\beta$ on $S_+\otimes W^*$, $S_-\otimes W$ and $t=0$ on the rotations they produce; $[S_+\otimes W,S_-\otimes W^*]\ni P$ gives $t_P=\alpha+\beta$; $[P_b,\eta]=\rho(b)\eta$ and $[K_b,\xi]=\rho(b)\xi$ are consistent with $t_K=-(\alpha+\beta)$ and $[K,P]\ni\Delta$. The two parameters are $a=\frac{1}{2}(\alpha-\beta)$ and $c=-(\alpha+\beta)$, with $B=\pm 1$ (Definition 8.9) and $\lambda=-1,+1,-2,+2$ on Σ,Σ^*,P,K. □

Lemma 12.66 (Ward identity of the two-point cochain from the defect flow)**.** *Let O_1, O_2 be gauge-invariant local functionals of degree 0 with disjoint supports and local representatives with $L(O_a)^{(0)}=0$ modulo $\hbar^2$, and let $\langle\,,\rangle_T$ be the arity-zero two-point function at scale T (Definition 8.37). In arity one along a generator Y, modulo $O(\hbar^2)$,*

$$\big\langle\Lambda^{(1)}(Y)O_1,O_2\big\rangle_T+(-1)^{|Y||O_1|}\big\langle O_1,\Lambda^{(1)}(Y)O_2\big\rangle_T=B_T(Y;O_1,O_2),\tag{12.40}$$

where B_T is $\hbar\Delta_T$ of the quadratic part at the zero field of the arity-one two-point cochain. $B_\infty=0$, and along Σ, Σ^*, Δ, the translations and the rotations (12.40) *holds at $T=\infty$ with zero right side.*

Proof. By Lemmas 8.4 and 8.13 applied to $J + t_1 O_1(\varepsilon) + t_2 O_2(\varepsilon)$, the $t_1 t_2$-part of d_T of the effective functional at scale T is the flow of the special-edge part of $\{t_1 O_1(\varepsilon), t_2 O_2(\varepsilon)\}_\varepsilon$, of order $e^{-|x_1-x_2|^2/4\varepsilon}$ at separated supports, together with the flows with $t_a L(O_a)$ in place of $t_a O_a$. At the zero field Q and d_{CE} give 0, $\{F, G\}_T$ needs one-leg parts of both arguments, which vanish for colour singlets and for the effective interaction, and $\hbar\Delta_T$ of the $t_1 t_2$-part leaves B_T; $\Delta_\infty = 0$ (Definition 3.21). □

Remark 12.67 ($K = P^\dagger$ as a cross-check)**.** Assume (H1) a reflection-compatible one-loop quantisation through arity one along P_n, K_n and the odd generators considered and through arity two along (P_n, K_n) (Theorem 8.33 (iii)); (H2) local representatives on a family $\mathfrak{F}$ of gauge-invariant cocycles at the origin of one length, stable under $\Lambda^{(1)}(P_n)$, $\Lambda^{(1)}(K_n)$ and the classical generators, with $L_1^{(0)} = 0$ for a reflection-compatible finite part; (H3) $\lim_{T\to\infty} B_T(K_n; O_1, O_2) = 0$ on the descendants of $\mathfrak{F}$ entering the radial inner product. Lemma 12.66 gives the Ward identity of Theorem 8.40 along every generator for which the two-point cochain converges, and along K_n through (H3), K_1 being independent of T (Proposition 12.48). Theorem 8.40 and Corollary 8.41 with $P_b^\dagger = -K_{\bar b}$ (Theorem 8.31 (i)) then give, on $\mathfrak{F}$ with the radial inner product of Definition 8.39,

$$K_1 \text{ along } K_{\bar b} = -\big(P_1 \text{ along } P_b\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(K_{\bar b}), -\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],$$

and the same form for S_1 against Q_1 and for D_1; in a scheme with $P_1 = 0$ on $\mathfrak{F}$ (Definition 8.35) the first term is absent. In the simplest graph of $B_T(K_n)$ the antifield leg A^* of $S^{(1)}(K_n)$ at y, joined by $P \star v$. to O_1, is contracted by K_T with a letter of O_2, an integrand of order $|y|^{-1}$, and $\int_{|y|^2 \lesssim T} |y|^{-1} T^{-2} d^4y = O(T^{-1/2})$. The identity is a cross-check of K_1, which (12.26) computes from its own defect cycles at a finite endpoint.

12.9 Former shortcuts, method and continuation

Remark 12.68 (The former shortcuts)**.**

- The coefficient $\theta = -\frac{1}{3}$ is computed from the construction twice, from (Δ, P_b) and from $(\Delta, \xi_\pm)$ (Theorem 12.31); Theorem 9.33 and Corollary 11.28 are cross-checks (Remark 12.34).

- "D_1 is four times the logarithmic coefficient" is Theorem 12.43, with Theorem 12.40 (i) and Lemma 12.41; on the action side, $\hbar\,\mathrm{Ob}^{(1)}(\Delta) = 4L^{(0)}+$ (composite part) is Theorem 12.13 (v). On two legs it holds placement class by placement class: the defect cycles at the special edge give $[-4, 16, -4]$ units and the line placement $[\frac{1}{3}, -\frac{1}{3}, -\frac{2}{3}]$ units on $\int(dA, dA)$, $\int(\lambda, \not{d}\lambda)$, $\int(dc, A^*)$, whose sum is (12.15) along Δ.

- The covariance of the regulator along P_b, M_ω, R_r is Proposition 12.12 (i) and Theorem 12.40 (ii); $\mathrm{Ob}^{(1)}(P_b)|_2 = 0$ and $\mathrm{Ob}^{(1)}(R_r) = 0$ are also computed by direct cycles (Theorem 12.20).

- S_1 from Q_1 by the position decomposition is Theorem 12.46 and Corollary 12.47; S_1 is computed from its own defect cycles along η and from the action-side solution along η, whose position part is $x^\mu \hbar F_1^{(1)}(\rho(e_\mu)\eta)$ by computation.

- $K = P^\dagger$ is Remark 12.67, a cross-check; K_1 is computed by (12.26) at a finite endpoint.

- The twist is Proposition 12.58 and Remark 12.59, a cross-check.

Remark 12.69 (Method of computation)**.** Every number is computed by two independent evaluations, except where a statement names one of them. The first evaluates each defect cycle in the scaled

Schwinger form: with $t_e = \varepsilon\tau_e$, $\tau_\sigma \in [0,1]$ for the special edge and $\tau_e \in [1,\infty)$ for the other edges (both segments of a composite line included), the coefficient of ε^0 is a finite sum of the integrals

$$J(n_t;\beta;N) = \int_0^1 d\tau_\sigma \int_{[1,\infty)^{n_t-1}} \prod_{e\neq\sigma} d\tau_e\, \tau^\beta \Big(\sum_e \tau_e\Big)^{-N},$$

τ^β a monomial, convergent for $N > |\beta| + n_t - 1$, evaluated exactly in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$ for $n_t \le 3$ (Proposition 12.70); letter derivatives through the generating functions of Proposition 11.12, the position through the injected momentum r by exact interpolation on integer r, the fibre algebra in an explicit realisation of $V \oplus W_{\mathbb{R}}^{\perp}$ and Σ over $\mathbb{Q}(i)$, colour in the realisations $\mathfrak{su}(2)$, $\mathfrak{su}(3)$ and $\mathfrak{gl}(3)$ by Lemma 10.8, and exact linear algebra on bases fixed by representation counts. The second computes the renormalised amplitudes $W^{(1)}_{\mathrm{ren}}(x)$, $W^{(2)}_{\mathrm{ren}}(x,y)$ and the defect cycles in momentum space with an independently written two-component realisation of $S^{(1)}(x)$ (Definition 8.1), Feynman parameters for the edges P, the special edge as the factor $e^{-\varepsilon|p|^2} - 1$, exact integrals where they close and otherwise thirty-digit numerics at several ε, extrapolated to ε^0 with the poles fitted, and along K_b the endpoints $T = 1, 4, 16$; it applies Q to $W^{(1)}_{\mathrm{ren}}(x)$ (Theorem 12.13) and evaluates the second route of Theorem 12.42. The two evaluations agree on every entry of Theorems 12.20, 12.21, 12.31, 12.56 and of the first four rows of Table 12.7, and check the Wess–Zumino identities (12.4) on every computed entry, the pole identities (12.8), (12.17) and the position identities (12.12); the single-pair tables cover the 45 pairs of slots with defect cycles with at most two letter derivatives in the first evaluation and one in the second, on the rows on which the second is complete. The overall sign of the defect cycles is normalised in each evaluation to the arity-zero obstruction of Theorem 4.30 and to the cubic obstruction of Section 9.1.

Proposition 12.70 (Closed forms)**.** *With F_m the m-th simplex integrand of* (12.7)*:*

- *(i) for $n_t = 2$, $16\pi^2 a_0 = \int_0^{1/2} F_0(\alpha)\log\frac{1-\alpha}{\alpha}\,d\alpha$ with $\alpha = \alpha_\sigma$, which is $F_0\log 2$ for constant F_0;*
- *(ii) for constant F_0 ($D = 0$) and any n_t, $16\pi^2 a_0 = \frac{F_0}{(n_t-1)!}\log\frac{n_t}{n_t-1}$, so $\frac{1}{2}F_0\log\frac{3}{2}$ for $n_t = 3$; for constant F_1 ($D = 2$), $16\pi^2 a_1 = \frac{F_1}{n_t^2(n_t-1)(n_t-1)!}$;*
- *(iii) for one vertex of I and a composite line, $16\pi^2 a_0 = \int_0^{1/3} F_0(\beta)\big[(1-\beta)\log\frac{1-\beta}{2\beta} - (1-3\beta)\big]d\beta$, in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$;*
- *(iv) triangles of cubic vertices with the special edge have values in $\mathbb{Q} + \mathbb{Q}\log\frac{3}{2}$.*

Proof. (i) For $n_t = 2$ the region where α_σ is the smaller coordinate is $\alpha \le \frac{1}{2}$, with $\min_{f\neq\sigma}\alpha_f = 1-\alpha$, and $\int_0^{1/2}\log\frac{1-\alpha}{\alpha}d\alpha = \log 2$. (ii) By symmetry the region where α_σ is smallest is $1/n_t$ of the simplex, of volume $1/(n_t-1)!$, and the logarithmic average over it is $\log\frac{n_t}{n_t-1}$ (Theorem 11.45 (iv)). (iii) Proposition 11.13. (iv) Theorem 9.9. □

Remark 12.71 (The computations that complete the construction)**.** The construction is complete through the two-leg parts of arity one along every generator and of arity two along (Δ, P_b), $(\Delta, \xi_\pm)$, (ξ_+, ξ_-), $(\xi_\pm, \xi'_\pm)$, $(\xi_\pm, P_b)$, $(\xi_+, \eta_\pm)$, and through the components on the scalar bilinears and the spectra of the families. The next computations, each a finite evaluation of the classes of graphs above, are:

- on the action side, the two-leg arity-two equations along $(K_b, P_{b'})$, (η, η') and $(\xi_-, \eta_\pm)$, which fix t_K and give the one-loop forms of $[K, P] = D + M$ and $\{S, S\} = K$; the three- and four-leg parts of the arity-one equations, which fix the differences of the sector coefficients through Theorem 12.22, the four directions G^K_j, the free classes in the **15** of $\mathfrak{su}(4)$ at the weight of (ξ_+, ξ_-), the arity-zero couplings of Proposition 12.23 and the five-leg sector along P_b, and test the classical symmetry condition of Definition 12.24; the arity-three component of s written term by term;

- on the operator side, the identities of Lemma 12.50 and Proposition 12.52 as computed identities on the composite loops with two and three legs and on the scalar bilinears, (11.6) and the full pole identity on the line rows; Q_1, S_1, M_1, K_1 and P_1 along generic b on the families at generic kinematics; the arity-two operator kernels of Proposition 12.62 and the two-sided relations (R2)–(R6); the second evaluation on the families with two gaugino components, the gluon sector and the mixed family; Gram_1 from the two-point graphs for Remark 12.67; the overall sign of the defect cycles from the sign rule of Definition 11.43 (ii);

- for the twist, the transport map Φ with the normalisation of [BK] and the (ξ,ξ) kernels of the entries $b\,c$ and $b\,b$ (Remark 12.59).

With the arity-two kernels and the components on the weight-$\frac{5}{2}$ descendants, (R3) on the trace bilinear becomes the Konishi relation between Q_1, S_1 and $D_1 = -48$ units, a scheme-independent test of the combinations (12.38).

Remark 12.72 (Two-component forms)**.** With Definition 8.1, $\sigma^m_{\alpha\dot\alpha} = (\vec\sigma, i)$, $\bar\sigma^{m\dot\alpha\alpha} = (-\vec\sigma, i)$, $\varepsilon^{12} = \varepsilon^{\dot1\dot2} = +1$, $A_m = -igA^a_m T^a$, $A_I = -igA^a_I T^a$, $A_{ij} = A_I \Sigma^I_{ij}$, $A^{ij} = \frac12 \varepsilon^{ijkl} A_{kl}$, $\lambda = -\frac{ig}{\sqrt2}\lambda^a T^a$, $\mathrm{tr}(T^a T^b) = \frac12 \delta^{ab}$, the unit $-\frac{g^2 N}{32\pi^2}$, and $(-,-)$ the pairing of $\mathfrak{g}$ around the components:

- Pairings. $\int (A, A^*) = \int d^4x \{(A_m, A^*_m) + \frac14 (A_{ij}, A^{*ij})\}$, $\int (\lambda^*, \lambda) = \int d^4x \{(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\}$, so $\mathcal{N} = \int d^4x \{2(A_m, A^*_m) + \frac12 (A_{ij}, A^{*ij}) - 4(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + 4(\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (c^*, c)\}$, with the order of the odd factors of Lemma 6.7.

- Supercharge data. For $\xi = (\xi_\alpha{}^i, 0) \in S_+ \otimes W$: $\Gamma^m_V = i\,\bar\lambda_{\dot\alpha i} \bar\sigma^{m\dot\alpha\alpha} \xi_\alpha{}^i$, $(\Gamma_\perp)_{kl} = 2\varepsilon_{jikl}\, \xi_\alpha{}^j \varepsilon^{\alpha\beta} \lambda_\beta{}^i$, $(\xi, \lambda^*) = \xi_\alpha{}^i \lambda^{*\alpha}{}_i$, and

  $$T_0 = -\int d^4x \,\big(\partial_m A_m, \xi_\alpha{}^i \lambda^{*\alpha}{}_i\big), \quad T_1 = -i \int d^4x \,\big(\bar\lambda_{\dot\alpha i} \bar\sigma^{m\dot\alpha\alpha} \xi_\alpha{}^i, A^*_m\big), \quad T_2 = -\tfrac12 \int d^4x\, \varepsilon_{jikl} \big(\xi_\alpha{}^j \varepsilon^{\alpha\beta} \lambda_\beta{}^i, A^{*kl}\big),$$

  $$T_3 = -\tfrac14 \int d^4x \,\big((\partial_m A_n - \partial_n A_m) \sigma^m_{\alpha\dot\alpha} \bar\sigma^{n\dot\alpha\beta} \xi_\beta{}^i, \lambda^{*\alpha}{}_i\big), \quad T_4 = -\tfrac{i}{2} \int d^4x \,\big(\varepsilon^{\alpha\beta} \xi_\beta{}^j \sigma^m_{\alpha\dot\alpha} \partial_m A_{ij}, \bar\lambda^{*\dot\alpha i}\big),$$

  $g_1 = \int d^4x \,(\Gamma^n_V, -\partial^2 A_n + \partial_n \partial_m A_m)$, $g_2 = -\int d^4x \,(\Gamma^n_V, \partial_n \partial_m A_m)$, $g_3 = -\frac14 \int d^4x \,((\Gamma_\perp)_{kl}, \partial^2 A^{kl})$, $g_4 = -\int d^4x \,(\partial^2 c, \xi_\alpha{}^i \lambda^{*\alpha}{}_i)$. The terms with $\bar\xi_{\dot\alpha i}$ follow by $\lambda_\alpha{}^i \leftrightarrow \bar\lambda_{\dot\alpha i}$, $\lambda^{*\alpha}{}_i \leftrightarrow \bar\lambda^{*\dot\alpha i}$, $\sigma \leftrightarrow \bar\sigma$.

- Special supercharge data. For $\eta = (\eta^\alpha{}_i, 0)$: $(\rho(e_m)\eta)_{\dot\alpha i} = i\,\eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}$, and for $\eta = (0, \bar\eta^{\dot\alpha i})$: $(\rho(e_m)\eta)_\alpha{}^i = -i\,\sigma^m_{\alpha\dot\alpha} \bar\eta^{\dot\alpha i}$ (8.1); $s_0 = \int d^4x \,(A^{kj} \varepsilon_{\alpha\beta} \eta^\beta{}_j, \lambda^{*\alpha}{}_k)$ and $\int (\rho(A|_V)\eta, \lambda^*) = -i \int d^4x \,(A_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}, \bar\lambda^{*\dot\alpha i})$.

- Sector parts. $S^{(1)}(\Delta)_\perp = \frac14 \int d^4x \,((2x_n\partial_n + 2) A_{ij}, A^{*ij})$, $S^{(1)}(\Delta)_\lambda = -\int d^4x \{((2x_n\partial_n + 3)\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - ((2x_n\partial_n + 3)\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\}$, $S^{(1)}(\Delta)_V = \int d^4x \{((2x_n\partial_n + 2) A_m, A^*_m) - (2x_n\partial_n c, c^*)\}$, and $S^{(1)}(P_b)_d$ the same with $2x_n\partial_n + 2w$ replaced by $b_n\partial_n$; $S^{(1)}(M_\omega)_{\mathrm{spin}} = \int d^4x \{\omega_{mn}(A_m, A^*_n) + \frac14 \omega_{mn}[(\lambda^{*\alpha}{}_i (\sigma^m \bar\sigma^n)_\alpha{}^\beta \lambda_\beta{}^i) -$ (the same on the dotted components with $\bar\sigma^m \sigma^n$)]$\}$.

- Obstructions. $\hbar\,\mathrm{Ob}^{(0)}|_2 = -\frac{27}{8} \int d^4x \,(\partial^2 c, \partial_m A_m)$ units, $\hbar F^{(0)}_1|_2 = \frac{27}{16} \int d^4x \,(\partial_m A_m, \partial_n A_n)$ units, and

  $$\hbar\,\mathrm{Ob}^{(1)}(\Delta)|_2 = \int d^4x \Big\{ -\tfrac{11}{3} \Big[\tfrac12 (F^{\mathrm{lin}}_{mn}, F^{\mathrm{lin}}_{mn}) + \tfrac14 (\partial_m A^{ij}, \partial_m A_{ij}) \Big] + \tfrac{94}{3}\, i \big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha} \partial_m \lambda_\alpha{}^i\big) - \tfrac{14}{3} (\partial_m c, A^*_m) \Big\} \text{ units},$$

  $F^{\mathrm{lin}}_{mn} = \partial_m A_n - \partial_n A_m$; along ξ the obstruction is (12.15) with $g_1, \dots, g_4$ above.

- Solutions. $\hbar F^{(1)}_1(\Delta)|_2 = \int d^4x \{\frac{22}{3}[(A_m, A^*_m) + \frac14 (A_{ij}, A^{*ij})] - \frac{47}{3}[(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})] - \frac83 (c^*, c)\}$ units plus the free directions; $\hbar F^{(1)}_1(\xi)|_2 = (\log\frac32 - \frac{79}{12})(T_1 + T_2)$ units $+ t_\xi (T_1 + T_2 + T_3 + T_4)$ units; along K_b, $\int (\iota_b A)(\iota_x A^*) = \int d^4x \,(b_m A_m, x_n A^*_n)$ and $\rho_\mu \rho_\nu$ is $\sigma^\mu \bar\sigma^\nu$ on the undotted and $\bar\sigma^\mu \sigma^\nu$ on the dotted components.

- Arity two. $S^{(2)}(\xi,\xi') = \int\{\frac{1}{4}(\Gamma(\xi,\xi'),\Gamma(\lambda^*,\lambda^*)) - \frac{1}{2}(\xi,\lambda^*)(\xi',\lambda^*) - (\iota_{\Gamma(\xi,\xi')}A, c^*)\}$ with $\Gamma(\xi,\bar\xi')|^m_V = i\,\bar\xi'_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i$ for $\xi \in S_+\otimes W$, $\bar\xi' \in S_-\otimes W^*$, and $(\Gamma(\xi,\xi')|_\perp)_{kl} = 2\varepsilon_{jikl}\,\xi_\alpha{}^j\varepsilon^{\alpha\beta}\xi'^{\,i}_\beta$ for $\xi,\xi' \in S_+\otimes W$; $\psi_1 = \int d^4x\, x_m(\Gamma^m_V, c^*)$, $\psi_2 = \int d^4x\, x_m(A^*_m, \xi_\alpha{}^i\lambda^{*\alpha}{}_i)$.

- Bilinears. For $O = \frac{1}{16}\int\omega\, w_1^{ij}w_2^{kl}(A_{ij},A_{kl})$: $\mathcal{B}_1 = \frac{1}{4}\int\omega\,[(w_1^{kl}A_{kl}, w_{2\,ji}\xi^{\alpha i}\lambda_\alpha{}^j) + (1 \leftrightarrow 2)]$, $\mathcal{B}_2 = \frac{1}{4}w_1^{ij}w_{2\,ij}\int\omega\,(A_{ji},\xi^{\alpha i}\lambda_\alpha{}^j)$, $\mathcal{B}_3 = \frac{1}{4}w_1^{ij}w_{2\,ij}\int\omega\, i(\bar\lambda_{\dot\alpha i}, A_m\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i)$, $\mathcal{B}_4 = \frac{1}{4}w_1^{ij}w_{2\,ij}\int\omega\,(c,\xi_\alpha{}^i\lambda^{*\alpha}{}_i)$; Theorem 12.56 then reads $Q_1(O) = (\frac{79}{12} - 3\log\frac{3}{2} - t_\xi)\mathcal{B}_1 + 4\log\frac{3}{2}(\mathcal{B}_2+\mathcal{B}_3)$ in units $-\frac{g^2N}{32\pi^2}$, and on the traceless bilinears the transformation along Q^i_α is $1+c_Q$ times the classical one.

- Generators and directions. B is $+1$ on $\xi_\alpha{}^i$, $\bar\eta^{\dot\alpha i}$ and -1 on $\bar\xi_{\dot\alpha i}$, $\eta^\alpha{}_i$ (Definition 8.9); the residual freedom $t = aB + \frac{c}{2}\lambda$ multiplies the one-loop actions of Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$, P_m, K_m by $1-(a-\frac{c}{2})$, $1-(-a-\frac{c}{2})$, $1-(-a+\frac{c}{2})$, $1-(a+\frac{c}{2})$, $1+c$, $1-c$ at one-loop order; $Q_1(O) = \xi_\alpha{}^i Q_1(O)^\alpha{}_i + \bar\xi_{\dot\alpha i}\bar Q_1(O)^{\dot\alpha i}$ and the same with $\eta^\alpha{}_i$, $\bar\eta^{\dot\alpha i}$ along $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$.

- Position decompositions. The position part of S_1 along $S^\alpha{}_i$ at x_O is Q_1 along $\bar\xi_{\dot\alpha i} = i\,x_{O,m}\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, and along $\bar S^{\dot\alpha i}$ it is Q_1 along $\xi_\alpha{}^i = -i\,x_{O,m}\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$; $\Upsilon_{K_b} = v^m(x_O)\Upsilon_{P_m} - (b\cdot x_O)\Upsilon^0_\Delta + \Upsilon^0_{M_{\omega_{x_O}}} + \Upsilon^0_{K_b}$ with $v^m(x_O) = x_O^2 b^m - 2(b\cdot x_O)x_O^m$ and $(\omega_{x_O})^m{}_n = 2(b^m x_{O,n} - x_O^m b_n)$.

- Weinberg. On the Dirac column $(\lambda_\alpha{}^i, \bar\lambda^{\dot\alpha}{}_i)$, $\rho(v) = \gamma^\mu_W v_\mu$ gives Weinberg's form of each line (Proposition 8.2).